

Data Book

Bayesian Macro-Regime Mapping for TAA

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Series count: 307

Table of Contents

1. [FRED API	] Output / Production	Real Gross Domestic Product
2. [FRED API	] Output / Production	Chicago Fed National Activity Index
3. [FRED API	] Output / Production	Industrial Production: Total Index
4. [FRED API	] Output / Production	Industrial Production: Manufacturing
5. [FRED API	] Output / Production	Nonfarm Business Sector: Labor Productivity (Output per Hour)
6. [FRED API	] Consumption	Real Personal Consumption Expenditures
7. [FRED API	] Consumption	Retail Trade
8. [FRED API	] Consumption	Personal Consumption Expenditures
9. [FRED API	] Consumption	Personal Consumption Expenditures: Durable Goods
10. [FRED API	] Consumption	Personal Consumption Expenditures: Services
11. [FRED API	] Income	Personal Income
12. [FRED API	] Income	Real Personal Income
13. [FRED API	] Income	Disposable Personal Income
14. [FRED API	] Income	Gross Domestic Income: Wages and Salaries
15. [FRED API	] Orders / Inventories	Total Business Inventories
16. [FRED API	] Orders / Inventories	Retailers: Inventories to Sales Ratio
17. [FRED API	] Orders / Inventories	Wholesale Inventories to Sales Ratio
18. [FRED API	] Orders / Inventories	Retail Inventories
19. [FRED API	] Orders / Inventories	Manufacturers Total Inventories
20. [FRED API	] Labor market	Unemployment Rate
21. [FRED API	] Labor market	Employment Level
22. [FRED API	] Labor market	U-6 Unemployment Rate
23. [FRED API	] Labor market	Long-term Unemployment
24. [FRED API	] Wages / labor costs	Employment Cost Index
25. [FRED API	] Wages / labor costs	Unit Labor Costs
26. [FRED API	] Wages / labor costs	Real Hourly Compensation
27. [FRED API	] Inflation / Prices	CPI: All Items: Total for United States
28. [FRED API	] Inflation / Prices	CPI Less Food and Energy
29. [FRED API	] Inflation / Prices	CPI Durables
30. [FRED API	] Inflation / Prices	CPI Services
31. [FRED API	] Inflation / Prices	CPI Rent of Primary Residence
32. [FRED API	] Inflation / Prices	CPI Medical Care
33. [FRED API	] Inflation / Prices	CPI Transportation
34. [FRED API	] Inflation / Prices	CPI Food
35. [FRED API	] Inflation / Prices	CPI Energy
36. [FRED API	] Inflation / Prices	PCE Less Food, Energy, Housing
37. [FRED API	] Inflation / Prices	PPI: All Commodities
38. [FRED API	] Inflation / Prices	PPI Finished Goods
39. [FRED API	] Inflation / Prices	Trimmed Mean PCE Inflation Rate
40. [FRED API	] Inflation / Prices	1-Year Expected Inflation
41. [FRED API	] Inflation / Prices	5-Year Expected Inflation
42. [FRED API	] Inflation / Prices	10-Year Expected Inflation
43. [FRED API	] Inflation / Prices	University of Michigan: Inflation Expectation
44. [FRED API	] Inflation / Prices	5-Year Breakeven Inflation Rate
45. [FRED API	] Inflation / Prices	10-Year Breakeven Inflation Rate
46. [FRED API	] Inflation / Prices	5-Year, 5-Year Forward Inflation Expectation Rate
47. [FRED API	] Inflation / Prices	Equity Volatility: Inflation
48. [FRED API	] Selected export	Asset Growth YoY
49. [FRED API	] Selected export	Assets, All Commercial Banks
50. [FRED API	] Selected export	Assets-to-Deposits Ratio
51. [FRED API	] Selected export	Commercial and Industrial Loans Monthly
52. [FRED API	] Selected export	Commercial and Industrial Loans YoY
53. [FRED API	] Selected export	Consumer Loans YoY
54. [FRED API	] Selected export	Credit Cards YoY
55. [FRED API	] Selected export	Deposit Growth YoY
56. [FRED API	] Selected export	Liabilities, All Commercial Banks
57. [FRED API	] Selected export	Liability Growth YoY
58. [FRED API	] Selected export	Loan-to-Deposit Ratio
59. [FRED API	] Selected export	Loan Growth YoY
60. [FRED API	] Selected export	Loans and Leases in Bank Credit Monthly
61. [FRED API	] Selected export	Revolving Consumer Credit Monthly
62. [FRED API	] Selected export	Revolving Consumer Credit YoY
63. [FRED API	] Selected export	Total Consumer Credit Monthly
64. [FRED API	] Selected export	Total Consumer Credit YoY

Table of Contents (cont.)

65. [FRED API	] Selected export	Total Bank Credit Monthly
66. [FRED API	] Selected export	Total Bank Credit YoY
67. [Derived	] Selected export	B Minus BB OAS
68. [Derived	] Selected export	BAA Minus AAA
69. [Derived	] Selected export	BB Minus BBB OAS
70. [Derived	] Selected export	BBB Minus AAA OAS
71. [Derived	] Selected export	CCC Minus B OAS
72. [Derived	] Selected export	High Yield Minus Investment Grade
73. [Derived	] Selected export	Belly Slope
74. [Derived	] Selected export	20Y Breakeven Inflation
75. [Derived	] Selected export	30Y Breakeven Inflation
76. [Derived	] Selected export	Curve Curvature
77. [Derived	] Selected export	Inversion Dummy 10Y Minus 2Y
78. [Derived	] Selected export	Inversion Dummy 10Y Minus 3M
79. [Derived	] Selected export	Long-End Slope
80. [Derived	] Selected export	Real Slope 10Y Minus 5Y
81. [Derived	] Selected export	Real Slope 30Y Minus 10Y
82. [Derived	] Selected export	Short-End Slope
83. [Derived	] Selected export	Slope 10Y Minus 2Y
84. [Derived	] Selected export	Slope 10Y Minus 3M
85. [Derived	] Selected export	Slope 30Y Minus 10Y
86. [Derived	] Selected export	Slope 5Y Minus 2Y
87. [Derived	] Selected export	Slope 5Y Minus 3M
88. [Derived	] Selected export	30Y Mortgage Rate Minus 10Y Treasury Yield
89. [WRDS	] Selected export	CRSP Curve Curvature
90. [WRDS	] Selected export	CRSP Slope 10Y Minus 2Y
91. [WRDS	] Selected export	CRSP Slope 10Y Minus 3M
92. [WRDS	] Selected export	CRSP Slope 2Y Minus 1Y
93. [WRDS	] Selected export	CRSP Slope 30Y Minus 10Y
94. [WRDS	] Selected export	CRSP Slope 5Y Minus 2Y
95. [WRDS	] Selected export	CRSP Slope 5Y Minus 3M
96. [WRDS	] Selected export	CRSP 10Y Treasury Yield
97. [WRDS	] Selected export	CRSP 1M Treasury Yield
98. [WRDS	] Selected export	CRSP 1Y Treasury Yield
99. [WRDS	] Selected export	CRSP 20Y Treasury Yield
100. [WRDS	] Selected export	CRSP 2Y Treasury Yield
101. [WRDS	] Selected export	CRSP 30Y Treasury Yield
102. [WRDS	] Selected export	CRSP 3M Treasury Yield
103. [WRDS	] Selected export	CRSP 5Y Treasury Yield
104. [WRDS	] Selected export	CRSP 7Y Treasury Yield
105. [WRDS	] Selected export	CRSP Curve Curvature First Difference
106. [WRDS	] Selected export	CRSP Curve Curvature 12M Rolling Volatility
107. [WRDS	] Selected export	CRSP Curve Curvature 3M Rolling Volatility
108. [WRDS	] Selected export	CRSP Curve Curvature 6M Rolling Volatility
109. [WRDS	] Selected export	CRSP Slope 10Y Minus 2Y First Difference
110. [WRDS	] Selected export	CRSP Slope 10Y Minus 2Y 12M Rolling Volatility
111. [WRDS	] Selected export	CRSP Slope 10Y Minus 2Y 3M Rolling Volatility
112. [WRDS	] Selected export	CRSP Slope 10Y Minus 2Y 6M Rolling Volatility
113. [WRDS	] Selected export	CRSP Slope 10Y Minus 3M First Difference
114. [WRDS	] Selected export	CRSP Slope 10Y Minus 3M 12M Rolling Volatility
115. [WRDS	] Selected export	CRSP Slope 10Y Minus 3M 3M Rolling Volatility
116. [WRDS	] Selected export	CRSP Slope 10Y Minus 3M 6M Rolling Volatility
117. [WRDS	] Selected export	CRSP Slope 2Y Minus 1Y First Difference
118. [WRDS	] Selected export	CRSP Slope 2Y Minus 1Y 12M Rolling Volatility
119. [WRDS	] Selected export	CRSP Slope 2Y Minus 1Y 3M Rolling Volatility
120. [WRDS	] Selected export	CRSP Slope 2Y Minus 1Y 6M Rolling Volatility
121. [WRDS	] Selected export	CRSP Slope 30Y Minus 10Y First Difference
122. [WRDS	] Selected export	CRSP Slope 30Y Minus 10Y 12M Rolling Volatility
123. [WRDS	] Selected export	CRSP Slope 30Y Minus 10Y 3M Rolling Volatility
124. [WRDS	] Selected export	CRSP Slope 30Y Minus 10Y 6M Rolling Volatility
125. [WRDS	] Selected export	CRSP Slope 5Y Minus 2Y First Difference
126. [WRDS	] Selected export	CRSP Slope 5Y Minus 2Y 12M Rolling Volatility
127. [WRDS	] Selected export	CRSP Slope 5Y Minus 2Y 3M Rolling Volatility
128. [WRDS	] Selected export	CRSP Slope 5Y Minus 2Y 6M Rolling Volatility

Table of Contents (cont.)

129.	[WRDS	] Selected export	CRSP Slope 5Y Minus 3M First Difference
130.	[WRDS	] Selected export	CRSP Slope 5Y Minus 3M 12M Rolling Volatility
131.	[WRDS	] Selected export	CRSP Slope 5Y Minus 3M 3M Rolling Volatility
132.	[WRDS	] Selected export	CRSP Slope 5Y Minus 3M 6M Rolling Volatility
133.	[WRDS	] Selected export	CRSP 10Y Treasury Yield First Difference
134.	[WRDS	] Selected export	CRSP 10Y Treasury Yield 12M Rolling Volatility
135.	[WRDS	] Selected export	CRSP 10Y Treasury Yield 3M Rolling Volatility
136.	[WRDS	] Selected export	CRSP 10Y Treasury Yield 6M Rolling Volatility
137.	[WRDS	] Selected export	CRSP 1M Treasury Yield First Difference
138.	[WRDS	] Selected export	CRSP 1M Treasury Yield 12M Rolling Volatility
139.	[WRDS	] Selected export	CRSP 1M Treasury Yield 3M Rolling Volatility
140.	[WRDS	] Selected export	CRSP 1M Treasury Yield 6M Rolling Volatility
141.	[WRDS	] Selected export	CRSP 1Y Treasury Yield First Difference
142.	[WRDS	] Selected export	CRSP 1Y Treasury Yield 12M Rolling Volatility
143.	[WRDS	] Selected export	CRSP 1Y Treasury Yield 3M Rolling Volatility
144.	[WRDS	] Selected export	CRSP 1Y Treasury Yield 6M Rolling Volatility
145.	[WRDS	] Selected export	CRSP 20Y Treasury Yield First Difference
146.	[WRDS	] Selected export	CRSP 20Y Treasury Yield 12M Rolling Volatility
147.	[WRDS	] Selected export	CRSP 20Y Treasury Yield 3M Rolling Volatility
148.	[WRDS	] Selected export	CRSP 20Y Treasury Yield 6M Rolling Volatility
149.	[WRDS	] Selected export	CRSP 2Y Treasury Yield First Difference
150.	[WRDS	] Selected export	CRSP 2Y Treasury Yield 12M Rolling Volatility
151.	[WRDS	] Selected export	CRSP 2Y Treasury Yield 3M Rolling Volatility
152.	[WRDS	] Selected export	CRSP 2Y Treasury Yield 6M Rolling Volatility
153.	[WRDS	] Selected export	CRSP 30Y Treasury Yield First Difference
154.	[WRDS	] Selected export	CRSP 30Y Treasury Yield 12M Rolling Volatility
155.	[WRDS	] Selected export	CRSP 30Y Treasury Yield 3M Rolling Volatility
156.	[WRDS	] Selected export	CRSP 30Y Treasury Yield 6M Rolling Volatility
157.	[WRDS	] Selected export	CRSP 3M Treasury Yield First Difference
158.	[WRDS	] Selected export	CRSP 3M Treasury Yield 12M Rolling Volatility
159.	[WRDS	] Selected export	CRSP 3M Treasury Yield 3M Rolling Volatility
160.	[WRDS	] Selected export	CRSP 3M Treasury Yield 6M Rolling Volatility
161.	[WRDS	] Selected export	CRSP 5Y Treasury Yield First Difference
162.	[WRDS	] Selected export	CRSP 5Y Treasury Yield 12M Rolling Volatility
163.	[WRDS	] Selected export	CRSP 5Y Treasury Yield 3M Rolling Volatility
164.	[WRDS	] Selected export	CRSP 5Y Treasury Yield 6M Rolling Volatility
165.	[WRDS	] Selected export	CRSP 7Y Treasury Yield First Difference
166.	[WRDS	] Selected export	CRSP 7Y Treasury Yield 12M Rolling Volatility
167.	[WRDS	] Selected export	CRSP 7Y Treasury Yield 3M Rolling Volatility
168.	[WRDS	] Selected export	CRSP 7Y Treasury Yield 6M Rolling Volatility
169.	[FRED API	] Selected export	1Y Treasury Yield
170.	[FRED API	] Selected export	1M Treasury Yield
171.	[FRED API	] Selected export	2Y Treasury Yield
172.	[FRED API	] Selected export	20Y Treasury Yield
173.	[FRED API	] Selected export	3Y Treasury Yield
174.	[FRED API	] Selected export	30Y Treasury Yield
175.	[FRED API	] Selected export	3M Treasury Yield
176.	[FRED API	] Selected export	5Y Treasury Yield
177.	[FRED API	] Selected export	6M Treasury Yield
178.	[FRED API	] Selected export	7Y Treasury Yield
179.	[Derived	] Selected export	Commercial and Industrial Loans YoY
180.	[Derived	] Selected export	Case-Shiller National Home Price Index YoY
181.	[Derived	] Selected export	Housing Starts YoY
182.	[Derived	] Selected export	New One Family Houses Sold YoY
183.	[Derived	] Selected export	M1 YoY
184.	[Derived	] Selected export	M2 YoY
185.	[Derived	] Selected export	Owners' Equivalent Rent YoY
186.	[Derived	] Selected export	Building Permits YoY
187.	[Derived	] Selected export	Rent of Primary Residence YoY
188.	[Derived	] Selected export	Revolving Consumer Credit YoY
189.	[Derived	] Selected export	Total Consumer Credit YoY
190.	[Derived	] Selected export	Total Bank Credit YoY
191.	[Derived	] Selected export	Total Reserves YoY
192.	[Derived	] Selected export	FHFA US House Price Index YoY

Table of Contents (cont.)

193. [Derived	] Selected export	Fed Total Assets YoY
194. [FRED-MD	] Sentiment / Surveys	UMich Consumer Sentiment
195. [FRED API	] Inflation / Prices	University of Michigan: Inflation Expectation
196. [FRED API	] Sentiment / Surveys	Conf Board Consumer Confidence (OECD proxy)
197. [FRED-MD	] Sentiment / Surveys	ISM Manufacturing PMI Composite
198. [FRED-MD	] Sentiment / Surveys	ISM Mfg New Orders
199. [FRED-MD	] Sentiment / Surveys	ISM Mfg Production
200. [FRED-MD	] Sentiment / Surveys	ISM Mfg Supplier Deliveries
201. [FRED-MD	] Sentiment / Surveys	ISM Mfg Inventories
202. [FRED-MD	] Sentiment / Surveys	ISM Mfg Employment
203. [FRED-MD	] Sentiment / Surveys	ISM Mfg Prices
204. [FRED API	] Sentiment / Surveys	Philly/Empire State Mfg
205. [FRED API	] Inflation / Prices	5-Year Breakeven Inflation Rate
206. [FRED API	] Inflation / Prices	10-Year Breakeven Inflation Rate
207. [FRED API	] Inflation / Prices	5-Year, 5-Year Forward Inflation Expectation Rate
208. [FRED-MD	] Volatility / Stress	CBOE VIX
209. [FRED-MD	] Volatility / Stress	10Y Treasury Yield
210. [FRED API	] Volatility / Stress	St. Louis Fed Financial Stress Index
211. [FRED API	] Volatility / Stress	Chicago Fed NFCI
212. [FRED API	] Volatility / Stress	Chicago Fed Adjusted NFCI
213. [FRED API	] Volatility / Stress	Kansas City Fed Financial Stress
214. [FRED API	] Volatility / Stress	Economic Policy Uncertainty (Monthly)
215. [FRED-MD	] Cross-Asset	S&P 500 Price Index
216. [FRED-MD	] Cross-Asset	S&P 500 Dividend Yield
217. [FRED-MD	] Cross-Asset	S&P 500 PE Ratio
218. [WRDS	] Cross-Asset	S&P 500 Total Return (monthly)
219. [Computed	] Cross-Asset	CRSP Value-Weighted Market Return
220. [FRED API	] Cross-Asset	Nasdaq Composite
221. [WRDS	] Cross-Asset	Russell 2000 (IWM ETF)
222. [Computed	] Cross-Asset	S&P 500 Realized Vol (20d)
223. [Computed	] Cross-Asset	S&P 500 Realized Vol (60d)
224. [Computed	] Cross-Asset	Sector Return Dispersion
225. [Computed	] Cross-Asset	Defensive vs Cyclical Rel Perf
226. [FRED API	] Cross-Asset	ICE BofA US Corp Total Return Index
227. [WRDS	] Cross-Asset	iShares 7-10Y Treasury ETF
228. [WRDS	] Cross-Asset	iShares 20+Y Treasury ETF
229. [WRDS	] Cross-Asset	iShares US Agg Bond ETF
230. [Computed	] Cross-Asset	10Y Treasury Approx Total Return
231. [Computed	] Cross-Asset	10Y Treasury Drawdown
232. [Computed	] Cross-Asset	10Y Treasury Momentum (12m)
233. [FRED-MD	] Cross-Asset	1Y Treasury Yield
234. [FRED-MD	] Cross-Asset	Effective Fed Funds Rate
235. [FRED-MD	] Cross-Asset	10Y Treasury minus Fed Funds Spread
236. [FRED-MD	] Cross-Asset	5Y Treasury minus Fed Funds Spread
237. [FRED-MD	] Cross-Asset	Baa Corp minus Fed Funds Spread
238. [FRED-MD	] Cross-Asset	Aaa Corp minus Fed Funds Spread
239. [FRED-MD	] Cross-Asset	Moody Aaa Corporate Bond Yield
240. [FRED-MD	] Cross-Asset	Moody Baa Corporate Bond Yield
241. [Computed	] Cross-Asset	Term Spread 10Y-1Y
242. [Computed	] Cross-Asset	Credit Spread Baa-Aaa
243. [Computed	] Cross-Asset	Risk-Free Rate (1m T-bill)
244. [Computed	] Cross-Asset	FF Market Excess Return
245. [Computed	] Cross-Asset	FF SMB (Small Minus Big)
246. [Computed	] Cross-Asset	FF HML (High Minus Low)
247. [FRED-MD	] Commodities	WTI Crude Oil (spliced)
248. [FRED API	] Commodities	Regular Gasoline Price
249. [FRED API	] Commodities	No.2 Heating Oil
250. [FRED API	] Commodities	Copper (Global Price)
251. [WRDS	] Commodities	iShares GSCI Commodity ETF (GSG)
252. [WRDS	] Commodities	Invesco DB Commodity ETF (DBC)
253. [FRED-MD	] FX / Global Macro	Trade-Weighted Dollar Index
254. [FRED-MD	] FX / Global Macro	USD/JPY
255. [FRED-MD	] FX / Global Macro	GBP/USD
256. [FRED-MD	] FX / Global Macro	CAD/USD

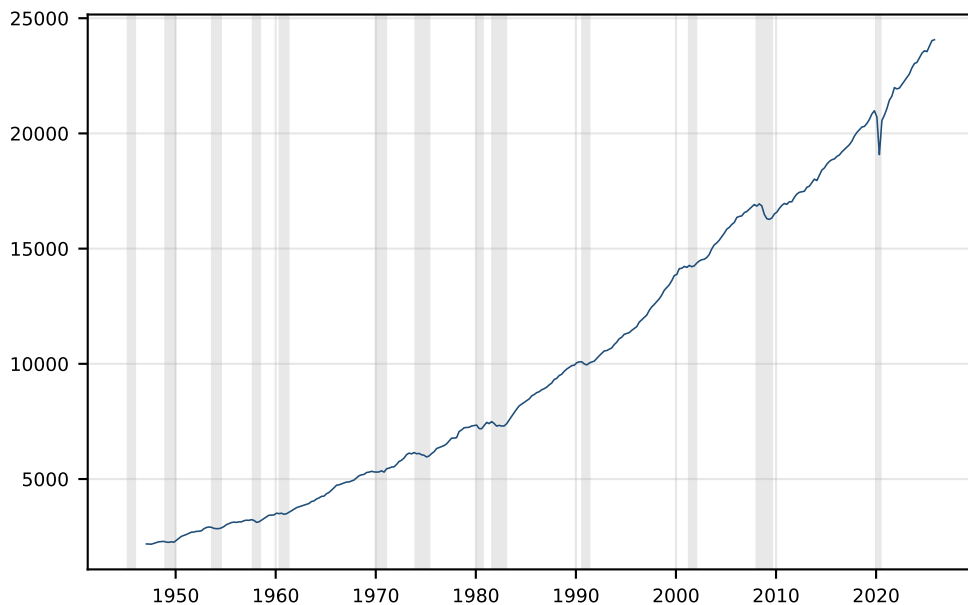
Table of Contents (cont.)

257. [FRED-MD	] FX / Global Macro	CHF/USD
258. [FRED API	] FX / Global Macro	EUR/USD
259. [FRED API	] FX / Global Macro	USD/CNY
260. [FRED API	] FX / Global Macro	Real Effective Exchange Rate (BIS)
261. [Computed	] FX / Global Macro	Dollar Realized Vol (12m)
262. [FRED API	] FX / Global Macro	OECD CLI USA
263. [FRED API	] FX / Global Macro	TED Spread
264. [FRED API	] FX / Global Macro	SOFR Rate
265. [FRED API	] Debt	Federal Debt: Total Public Debt
266. [FRED API	] Debt	Total Public Debt as Percent of GDP
267. [FRED API	] Debt	All Sectors; Debt Securities and Loans; Liability, Level
268. [FRED API	] Debt	Market Value of Gross Federal Debt
269. [FRED API	] Debt	Federal Debt Held by the Public as Percent of GDP
270. [FRED API	] Debt	Nonfinancial Corporate Business; Debt Securities; Liability, Level
271. [FRED API	] Debt	State and Local Governments; Debt Securities and Loans; Liability, Level
272. [FRED API	] Debt	Federal Government; Debt Securities and Loans; Liability, Level
273. [FRED API	] Debt	Security Brokers and Dealers; Debt Securities; Asset, Level
274. [FRED API	] Debt	Nonfinancial Business; Debt Securities and Loans; Liability, Level
275. [FRED API	] Debt	3-Month Treasury Bill Secondary Market Rate, Discount Basis
276. [FRED-MD	] Volatility / Stress	10Y Treasury Yield
277. [FRED API	] Debt	Market Yield on U.S. Treasury Securities at 3-Year Constant Maturity, Quoted on an Investment Basis
278. [FRED API	] Debt	Market Yield on U.S. Treasury Securities at 7-Year Constant Maturity, Quoted on an Investment Basis
279. [FRED API	] Debt	Market Yield on U.S. Treasury Securities at 6-Month Constant Maturity, Quoted on an Investment Basis
280. [FRED API	] Debt	Market Yield on U.S. Treasury Securities at 6-Month Constant Maturity, Quoted on an Investment Basis
281. [FRED API	] Sovereign credibility	Federal government current expenditures: Interest payments
282. [FRED API	] Sovereign credibility	Federal government current expenditures: Interest payments: to the rest of the world
283. [FRED API	] Sovereign credibility	Federal Government: Current Expenditures
284. [FRED API	] Demographics	Population Growth for the United States
285. [FRED API	] Demographics	Working-Age Population Total: From 15 to 64 Years for United States
286. [FRED API	] Demographics	Age Dependency Ratio: Older Dependents to Working-Age Population for the United States
287. [FRED API	] Demographics	Labor Force Participation Rate
288. [FRED API	] Demographics	Population ages 65 and above for the United States
289. [FRED API	] Demographics	Population Level - 55 Yrs. & over
290. [FRED API	] Demographics	Population Level - 25-54 Yrs.
291. [FRED API	] Demographics	Population Level - 16-19 Yrs.
292. [FRED API	] Demographics	Population Level - 20-24 Yrs.
293. [FRED API	] Demographics	Personal Saving Rate
294. [FRED API	] Productivity	Real Potential Gross Domestic Product
295. [FRED API	] Productivity	Nonfarm Business Sector: Labor Productivity (Output per Hour) for All Workers
296. [FRED API	] Productivity	Total Factor Productivity at Constant National Prices for United States
297. [FRED API	] Productivity	Net domestic investment: Private: Domestic business
298. [FRED API	] Productivity	Nonfinancial Corporate Business; Gross Fixed Investment, Transactions
299. [FRED API	] Productivity	Noncyclical Rate of Unemployment
300. [FRED API	] Productivity	Median usual weekly real earnings: Wage and salary workers: 16 years and over
301. [FRED-MD	] Cross-Asset	Effective Fed Funds Rate
302. [FRED-MD	] Cross-Asset	Effective Fed Funds Rate
303. [FRED API	] Structural macro	Composite Leading Indicators: Composite Leading Indicator (CLI) Trend Restored for United States
304. [FRED API	] Structural macro	Sticky Price Consumer Price Index less Food and Energy
305. [FRED API	] Structural macro	Sticky Price Consumer Price Index less Food and Energy
306. [FRED API	] Equities	Personal Income Receipts on Assets: Personal Dividend Income
307. [FRED API	] Equities	Net corporate dividend payments

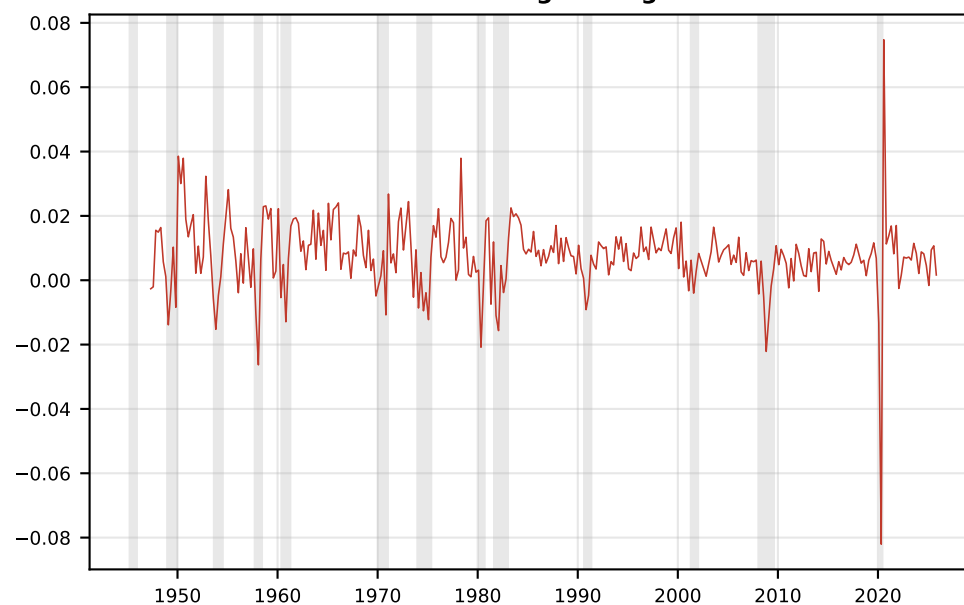
Real Gross Domestic Product

Block: Output / Production / Output / Production | Source: FRED API | Col: GDPC1 | Transform: 8: Log diff lag 3

Raw Series



After: 8: Log diff lag 3



SUMMARY STATISTICS

N : 1287
Valid : 316
NaN : 971
Start : 1947-01
End : 2025-10
Mean : 10323.5990
Std : 6366.1573
Min : 2172.4320
Max : 24065.9560
Skew : 0.481
Kurt : -1.034

STATIONARITY TESTS

ADF stat : -10.582
ADF p : 0.0000
ADF lags : 1
ADF : Stationary
KPSS stat : 0.456
KPSS p : 0.0528
KPSS : Stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

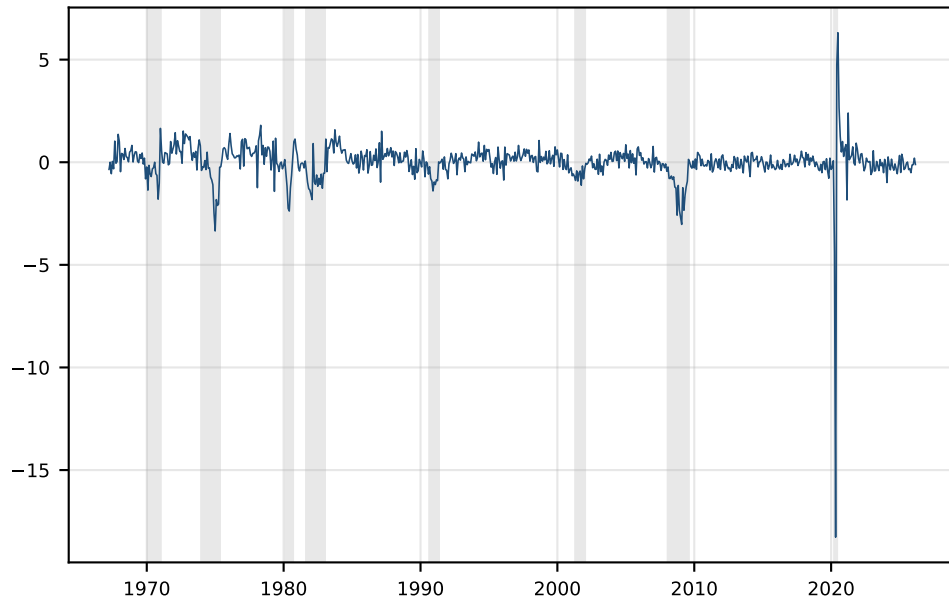
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	79	41.8%	58.2%	-0.0044
Q2	79	7.6%	92.4%	0.0057
Q3	78	2.6%	97.4%	0.0096
Q4 (highest)	79	0.0%	100.0%	0.0196

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

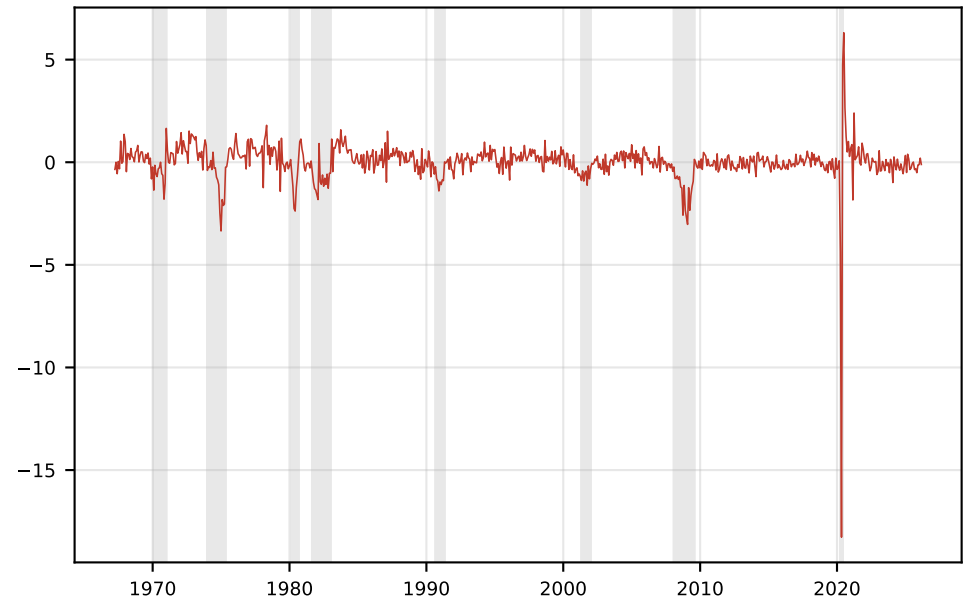
Chicago Fed National Activity Index

Block: Output / Production / Output / Production | Source: FRED API | Col: CFNAI | Transform: 1: No transform

Raw Series



After: 1: No transform



SUMMARY STATISTICS

N : 1287
Valid : 708
NaN : 579
Start : 1967-03
End : 2026-02
Mean : -0.0001
Std : 1.0001
Min : -18.2700
Max : 6.3100
Skew : -8.464
Kurt : 160.531

STATIONARITY TESTS

ADF stat : -7.979
ADF p : 0.0000
ADF lags : 5
ADF : Stationary
KPSS stat : 0.336
KPSS p : 0.1000
KPSS : Stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

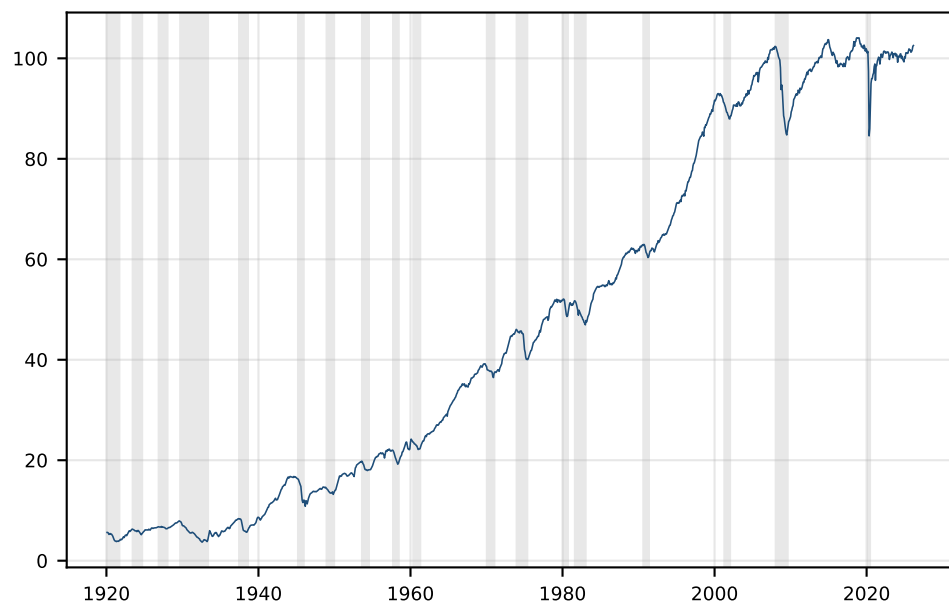
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	182	40.1%	59.9%	-0.8715
Q2	174	5.2%	94.8%	-0.0953
Q3	181	0.6%	99.4%	0.2079
Q4 (highest)	171	1.2%	98.8%	0.8043

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

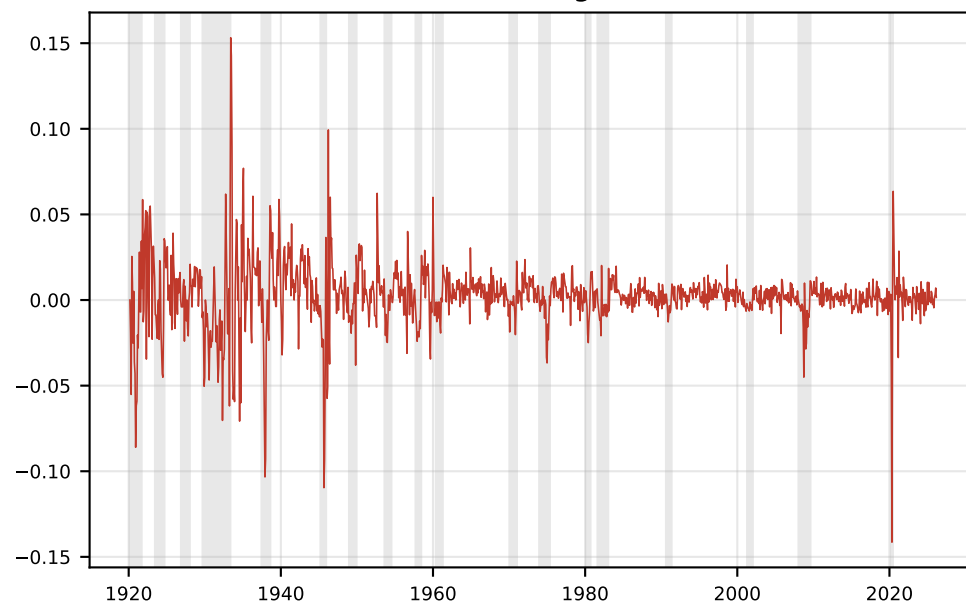
Industrial Production: Total Index

Block: Output / Production / Output / Production | Source: FRED API | Col: INDPRO | Transform: 5: Log diff

Raw Series



After: 5: Log diff



SUMMARY STATISTICS

N : 1287
Valid : 1274
NaN : 13
Start : 1920-01
End : 2026-02
Mean : 47.1915
Std : 35.1061
Min : 3.6891
Max : 104.1004
Skew : 0.337
Kurt : -1.387

STATIONARITY TESTS

ADF stat : -10.826
ADF p : 0.0000
ADF lags : 12
ADF : Stationary
KPSS stat : 0.128
KPSS p : 0.1000
KPSS : Stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

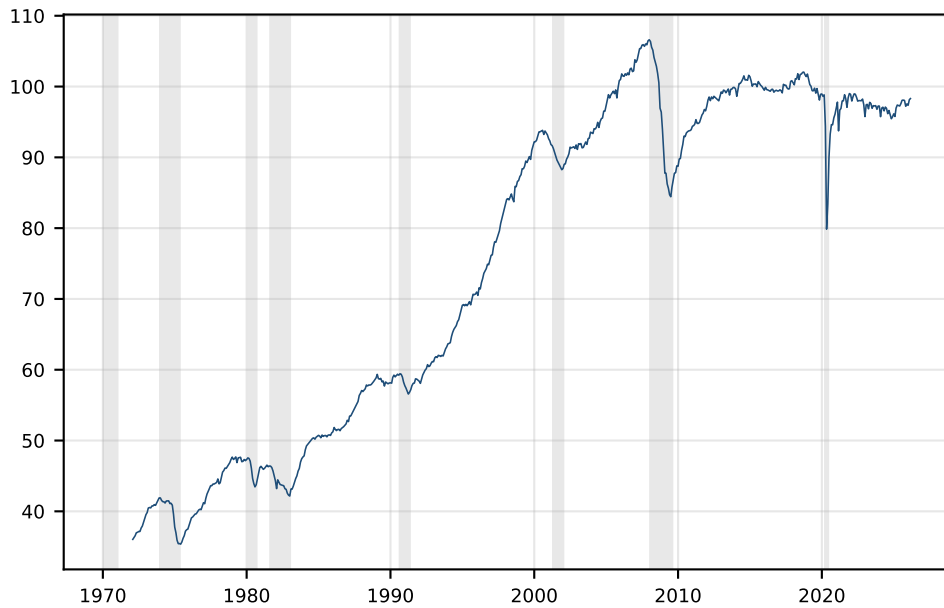
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	319	51.4%	48.6%	-0.0170
Q2	318	13.5%	86.5%	-0.0002
Q3	318	3.1%	96.9%	0.0052
Q4 (highest)	318	5.0%	95.0%	0.0212

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

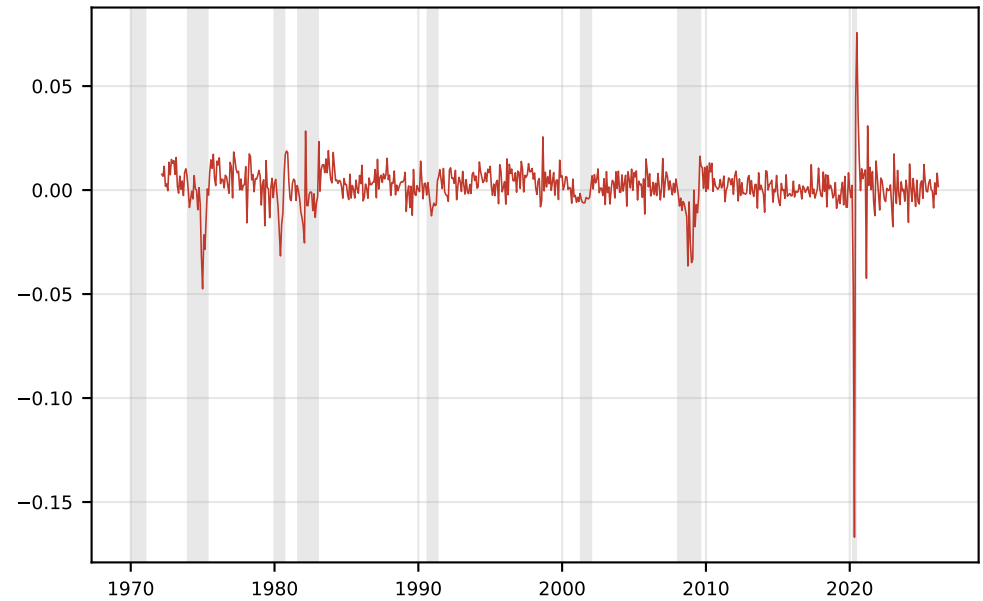
Industrial Production: Manufacturing

Block: Output / Production / Output / Production | Source: FRED API | Col: IPMAN | Transform: 5: Log diff

Raw Series



After: 5: Log diff



SUMMARY STATISTICS

N : 1287
Valid : 650
NaN : 637
Start : 1972-01
End : 2026-02
Mean : 75.2862
Std : 23.5671
Min : 35.3608
Max : 106.6125
Skew : -0.288
Kurt : -1.590

STATIONARITY TESTS

ADF stat : -19.993
ADF p : 0.0000
ADF lags : 0
ADF : Stationary
KPSS stat : 0.347
KPSS p : 0.1000
KPSS : Stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

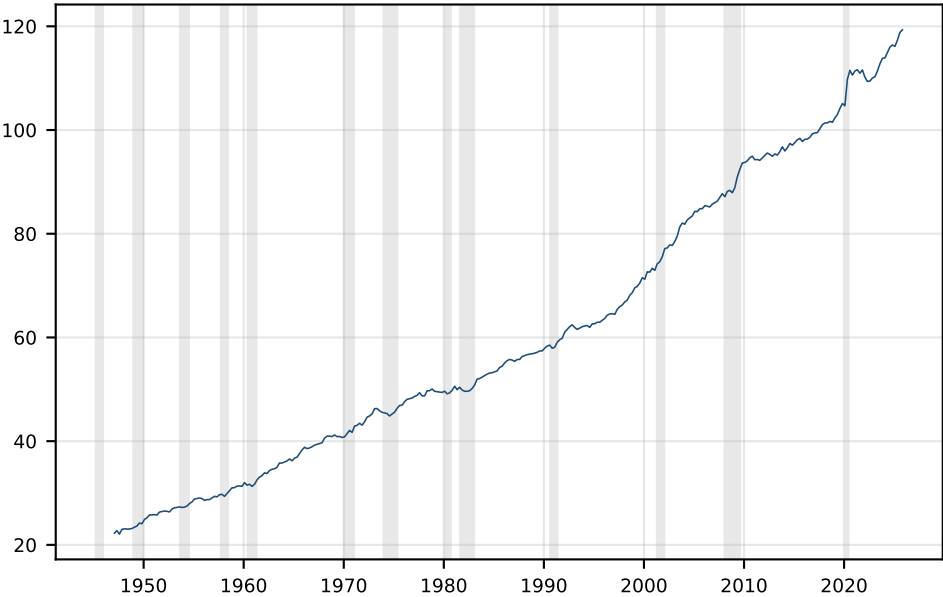
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	163	35.6%	64.4%	-0.0095
Q2	162	8.0%	92.0%	-0.0002
Q3	162	0.6%	99.4%	0.0042
Q4 (highest)	162	1.2%	98.8%	0.0117

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

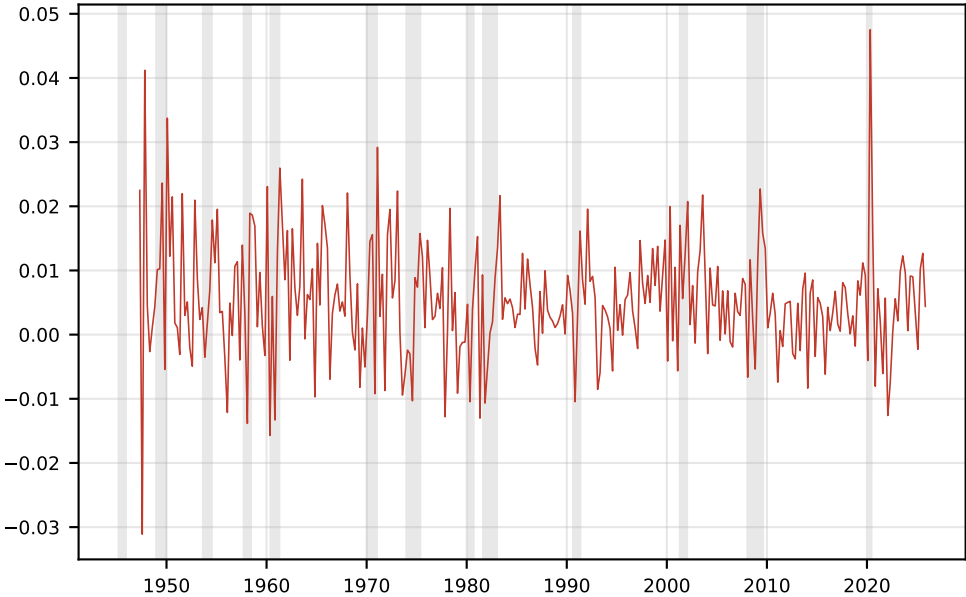
Nonfarm Business Sector: Labor Productivity (Output per Hour)

Block: Output / Production / Output / Production | Source: FRED API | Col: OPHNFB | Transform: 8: Log diff lag 3

Raw Series



After: 8: Log diff lag 3



SUMMARY STATISTICS

N : 1287
Valid : 316
NaN : 971
Start : 1947-01
End : 2025-10
Mean : 61.0877
Std : 27.3175
Min : 22.0650
Max : 119.3380
Skew : 0.462
Kurt : -0.985

STATIONARITY TESTS

ADF stat : -19.030
ADF p : 0.0000
ADF lags : 0
ADF : Stationary
KPSS stat : 0.259
KPSS p : 0.1000
KPSS : Stationary

QUARTILE-BUCKET ANALYSIS

(vs NBER recession/expansion)

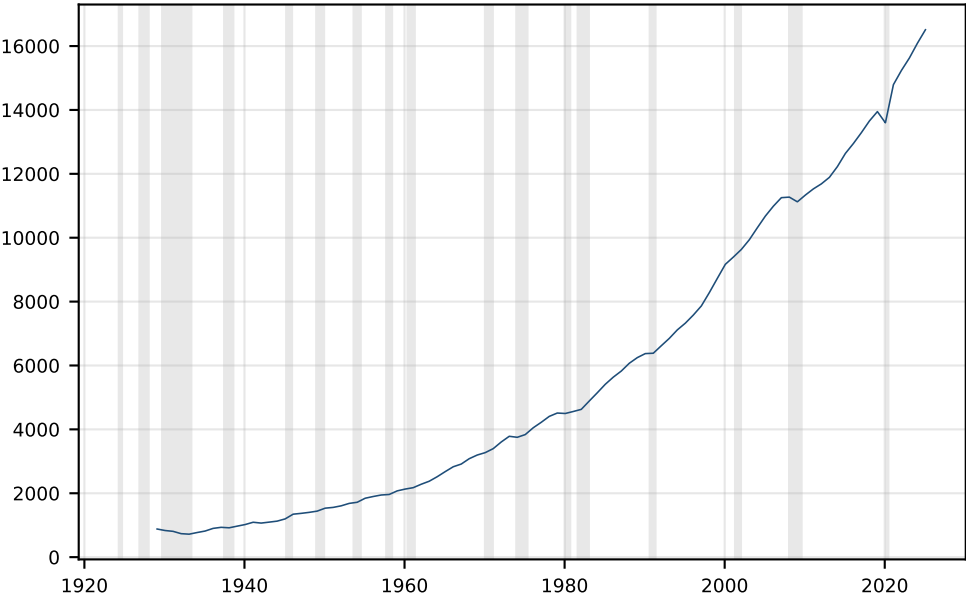
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	79	19.0%	81.0%	-0.0053
Q2	79	8.9%	91.1%	0.0028
Q3	78	7.7%	92.3%	0.0072
Q4 (highest)	79	16.5%	83.5%	0.0167

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

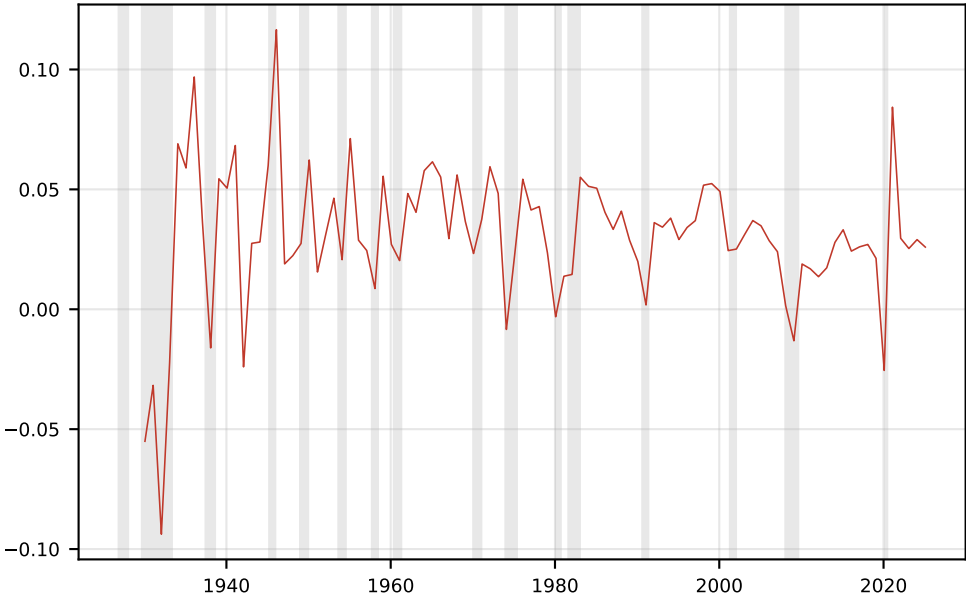
Real Personal Consumption Expenditures

Block: Consumption / Consumption | Source: FRED API | Col: PCECCA | Transform: 9: Log diff lag 12

Raw Series



After: 9: Log diff lag 12



SUMMARY STATISTICS

N	: 1287
Valid	: 97
NaN	: 1190
Start	: 1929-01
End	: 2025-01
Mean	: 5680.9872
Std	: 4607.9080
Min	: 718.9260
Max	: 16511.0960
Skew	: 0.754
Kurt	: -0.693

STATIONARITY TESTS

ADF stat	: -3.115
ADF p	: 0.0254
ADF lags	: 8
ADF	: Stationary
KPSS stat	: 0.131
KPSS p	: 0.1000
KPSS	: Stationary

QUARTILE-BUCKET ANALYSIS
(vs NBER recession/expansion)

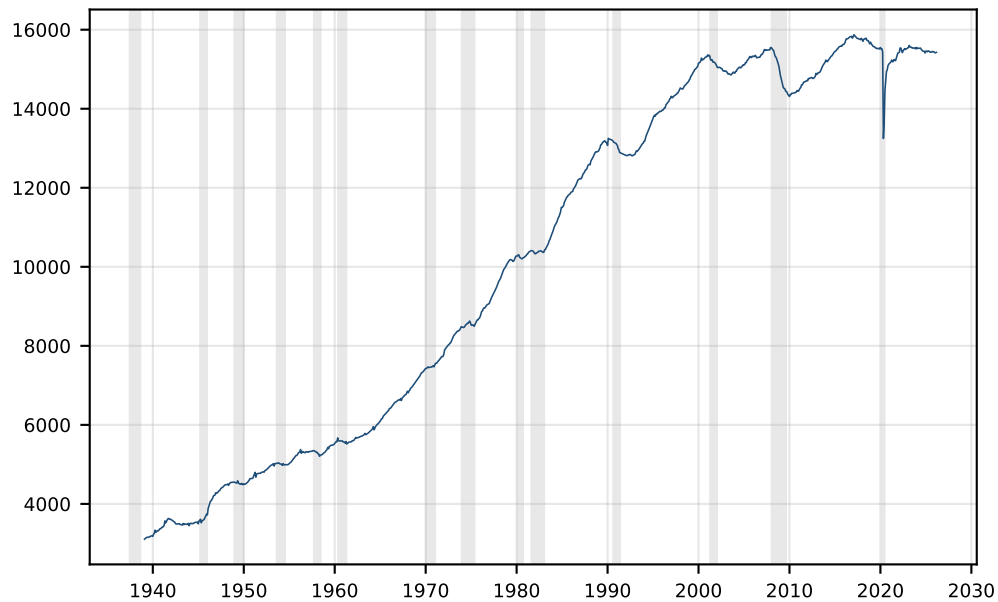
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	24	54.2%	45.8%	-0.0038
Q2	24	12.5%	87.5%	0.0259
Q3	24	0.0%	100.0%	0.0374
Q4 (highest)	24	0.0%	100.0%	0.0626

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

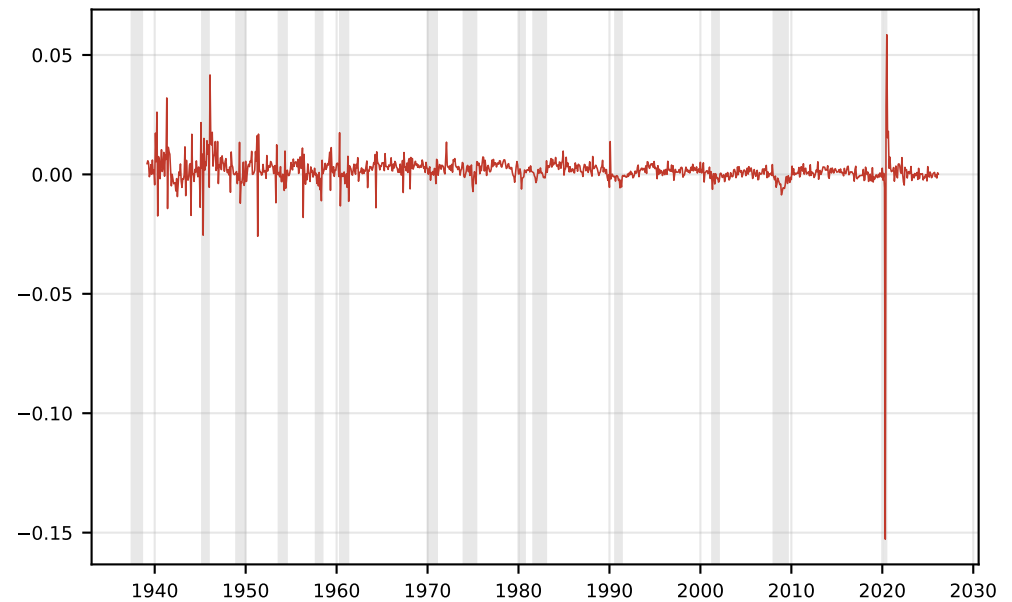
Retail Trade

Block: Consumption / Consumption | Source: FRED API | Col: USTRADE | Transform: 5: Log diff

Raw Series



After: 5: Log diff



SUMMARY STATISTICS

N : 1287
Valid : 1046
NaN : 241
Start : 1939-01
End : 2026-02
Mean : 10241.6950
Std : 4457.6338
Min : 3107.8000
Max : 15872.6000
Skew : -0.168
Kurt : -1.584

STATIONARITY TESTS

ADF stat : -8.706
ADF p : 0.0000
ADF lags : 7
ADF : Stationary
KPSS stat : 1.787
KPSS p : 0.0100
KPSS : Non-stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

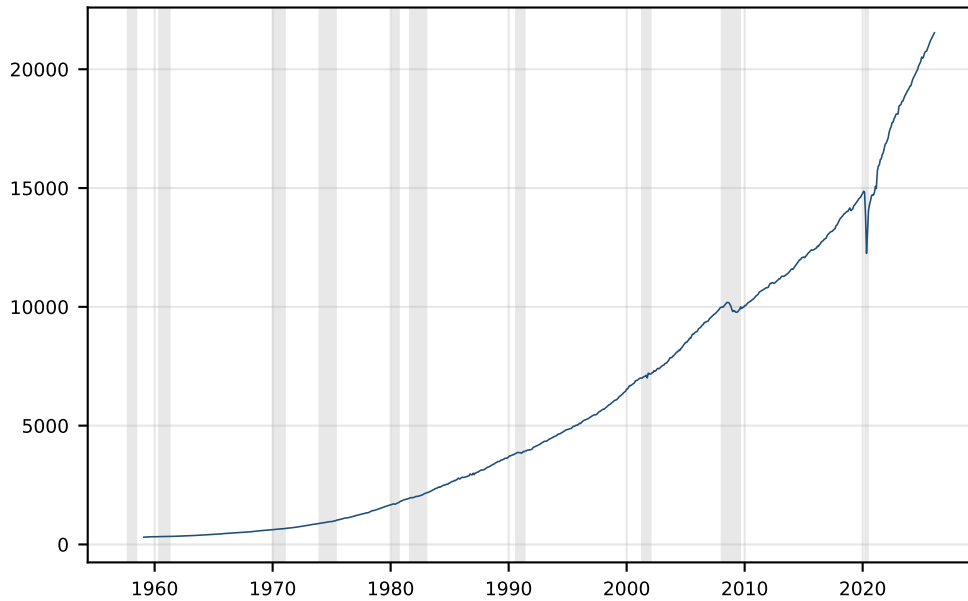
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	262	30.5%	69.5%	-0.0035
Q2	261	11.1%	88.9%	0.0006
Q3	261	3.8%	96.2%	0.0024
Q4 (highest)	261	5.0%	95.0%	0.0067

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

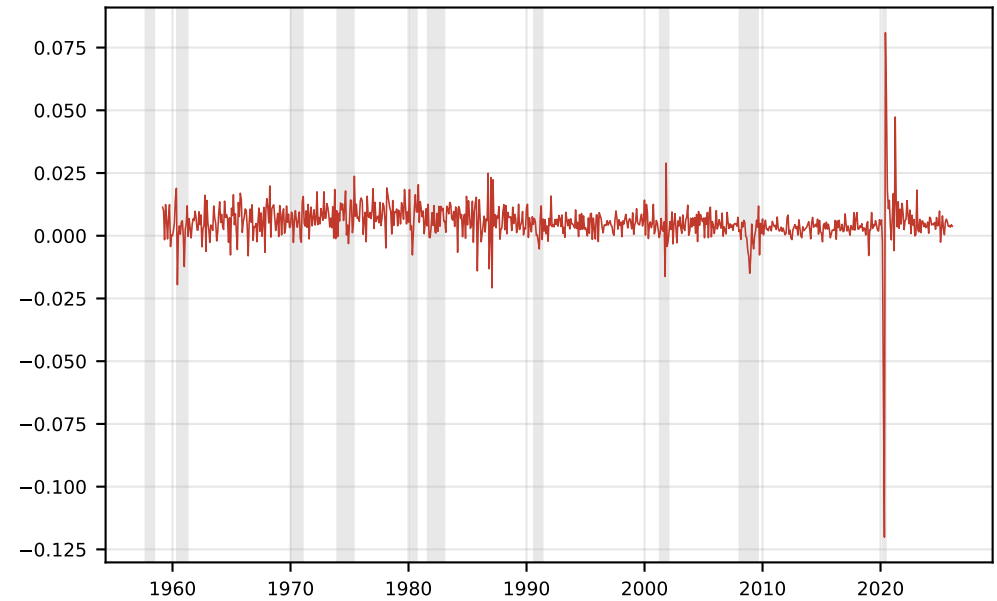
Personal Consumption Expenditures

Block: Consumption / PCE Inflation | Source: FRED API | Col: PCE | Transform: 5: Log diff

Raw Series



After: 5: Log diff



SUMMARY STATISTICS

N : 1287
Valid : 805
NaN : 482
Start : 1959-01
End : 2026-01
Mean : 6061.1675
Std : 5614.7377
Min : 306.1000
Max : 21536.6000
Skew : 0.902
Kurt : -0.142

STATIONARITY TESTS

ADF stat : -6.322
ADF p : 0.0000
ADF lags : 9
ADF : Stationary
KPSS stat : 1.864
KPSS p : 0.0100
KPSS : Non-stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

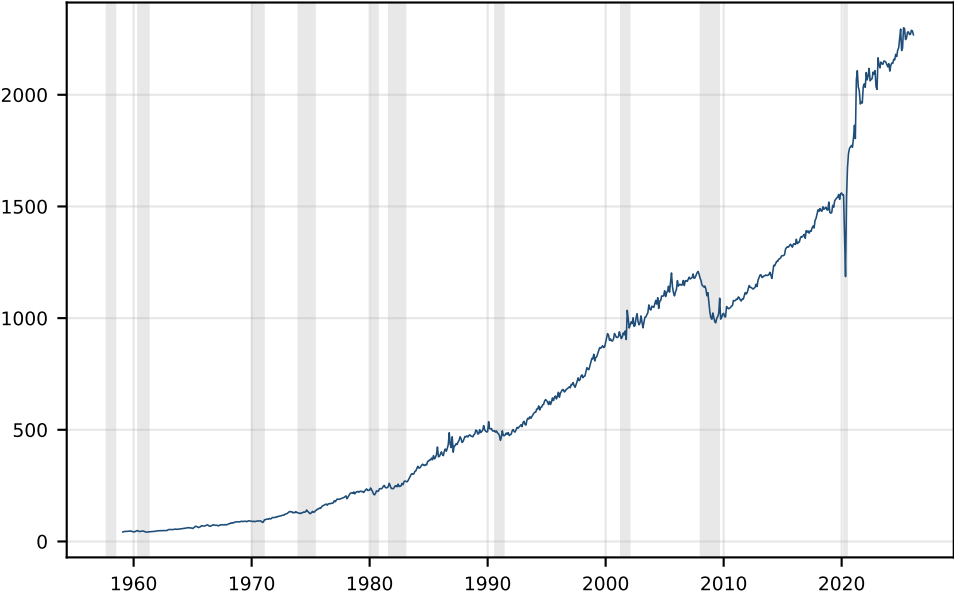
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	201	20.4%	79.6%	-0.0019
Q2	201	6.5%	93.5%	0.0037
Q3	201	9.0%	91.0%	0.0065
Q4 (highest)	201	11.4%	88.6%	0.0129

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

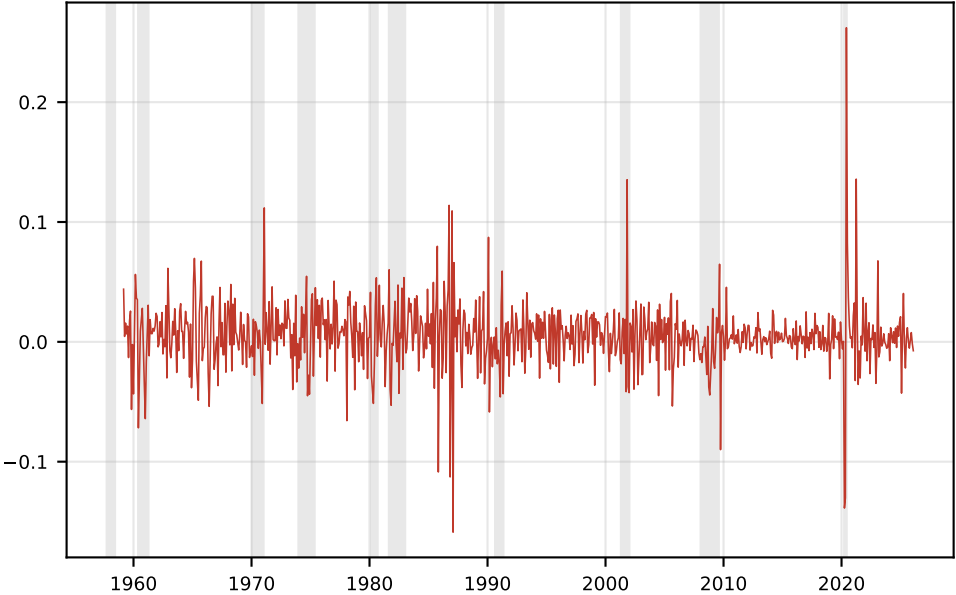
Personal Consumption Expenditures: Durable Goods

Block: Consumption / PCE Inflation | Source: FRED API | Col: PCEDG | Transform: 5: Log diff

Raw Series



After: 5: Log diff



SUMMARY STATISTICS

N : 1287
Valid : 805
NaN : 482
Start : 1959-01
End : 2026-01
Mean : 713.8826
Std : 614.5883
Min : 41.6000
Max : 2300.2000
Skew : 0.831
Kurt : -0.154

STATIONARITY TESTS

ADF stat : -10.264
ADF p : 0.0000
ADF lags : 7
ADF : Stationary
KPSS stat : 0.472
KPSS p : 0.0480
KPSS : Non-stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

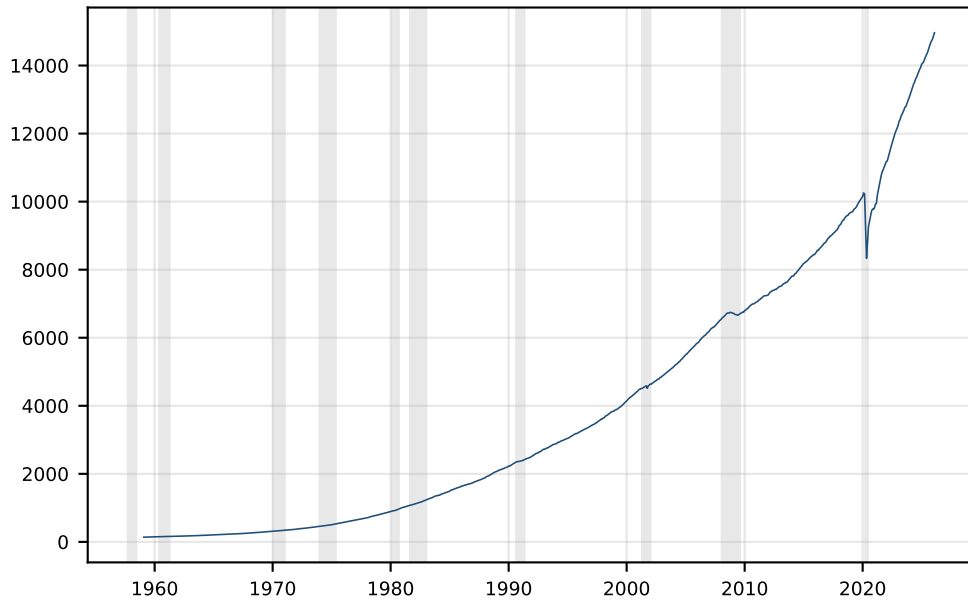
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	201	23.4%	76.6%	-0.0258
Q2	201	6.5%	93.5%	-0.0005
Q3	201	5.5%	94.5%	0.0110
Q4 (highest)	201	11.9%	88.1%	0.0352

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

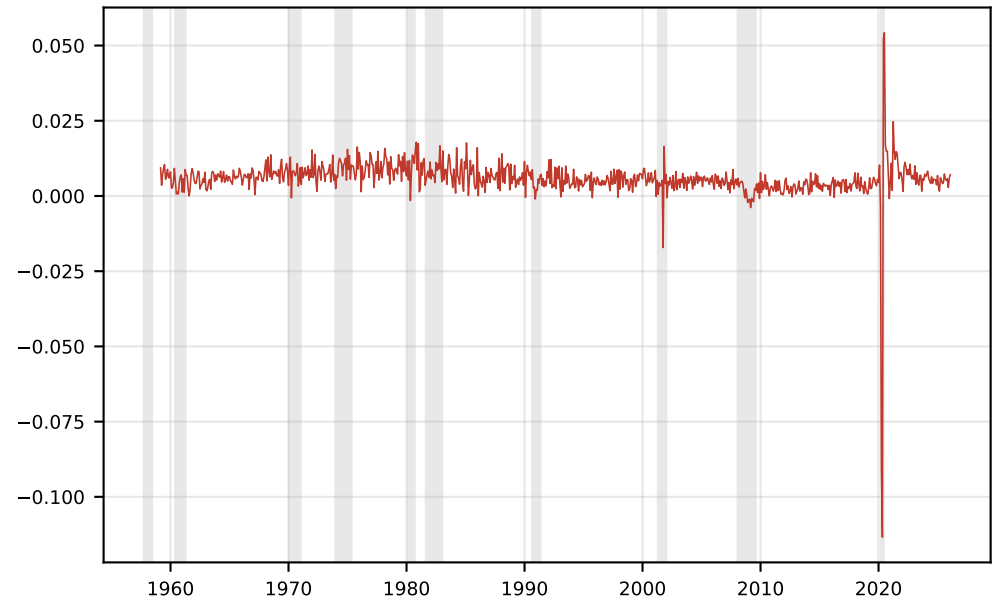
Personal Consumption Expenditures: Services

Block: Consumption / PCE Inflation | Source: FRED API | Col: PCES | Transform: 5: Log diff

Raw Series



After: 5: Log diff



SUMMARY STATISTICS

N : 1287
Valid : 805
NaN : 482
Start : 1959-01
End : 2026-01
Mean : 3963.8460
Std : 3860.0097
Min : 138.7000
Max : 14963.6000
Skew : 0.945
Kurt : -0.066

STATIONARITY TESTS

ADF stat : -6.172
ADF p : 0.0000
ADF lags : 9
ADF : Stationary
KPSS stat : 3.631
KPSS p : 0.0100
KPSS : Non-stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

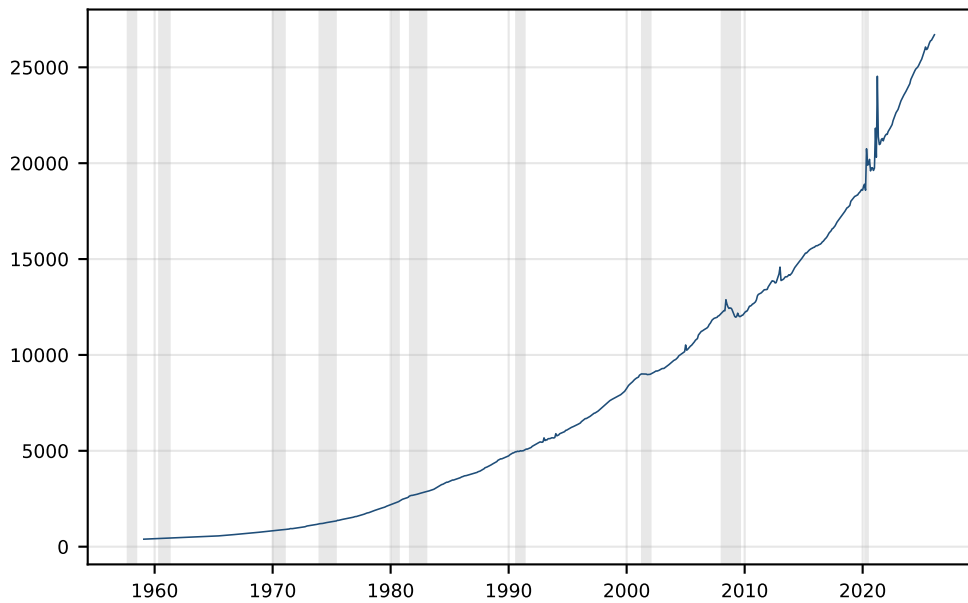
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	201	16.4%	83.6%	0.0008
Q2	201	7.5%	92.5%	0.0047
Q3	201	8.5%	91.5%	0.0067
Q4 (highest)	201	14.9%	85.1%	0.0112

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

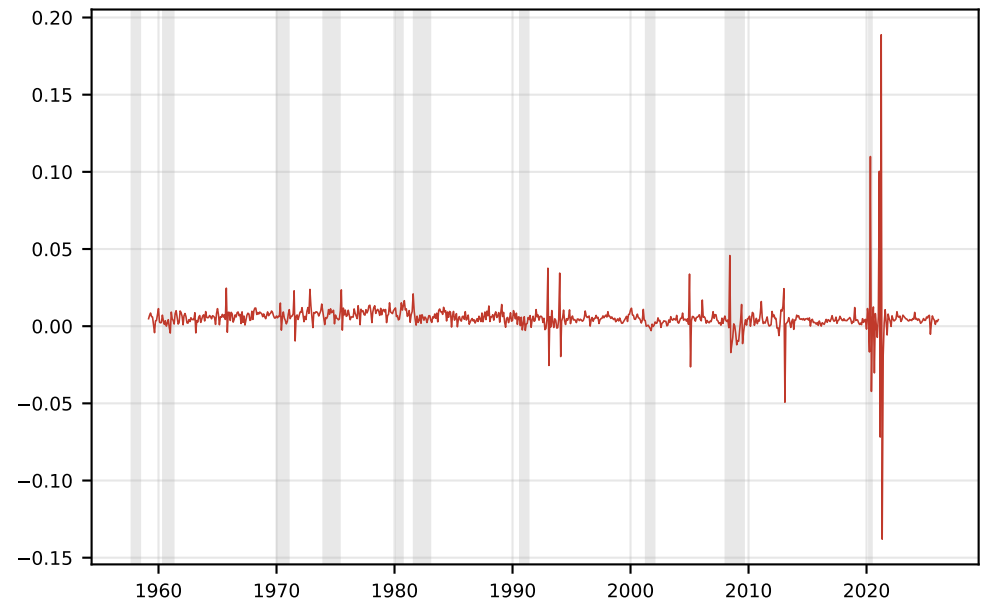
Personal Income

Block: Income / Income | Source: FRED API | Col: PI | Transform: 5: Log diff

Raw Series



After: 5: Log diff



SUMMARY STATISTICS

N : 1287
Valid : 805
NaN : 482
Start : 1959-01
End : 2026-01
Mean : 7658.1699
Std : 7061.8048
Min : 391.8000
Max : 26698.9000
Skew : 0.921
Kurt : -0.120

STATIONARITY TESTS

ADF stat : -4.741
ADF p : 0.0001
ADF lags : 11
ADF : Stationary
KPSS stat : 1.594
KPSS p : 0.0100
KPSS : Non-stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

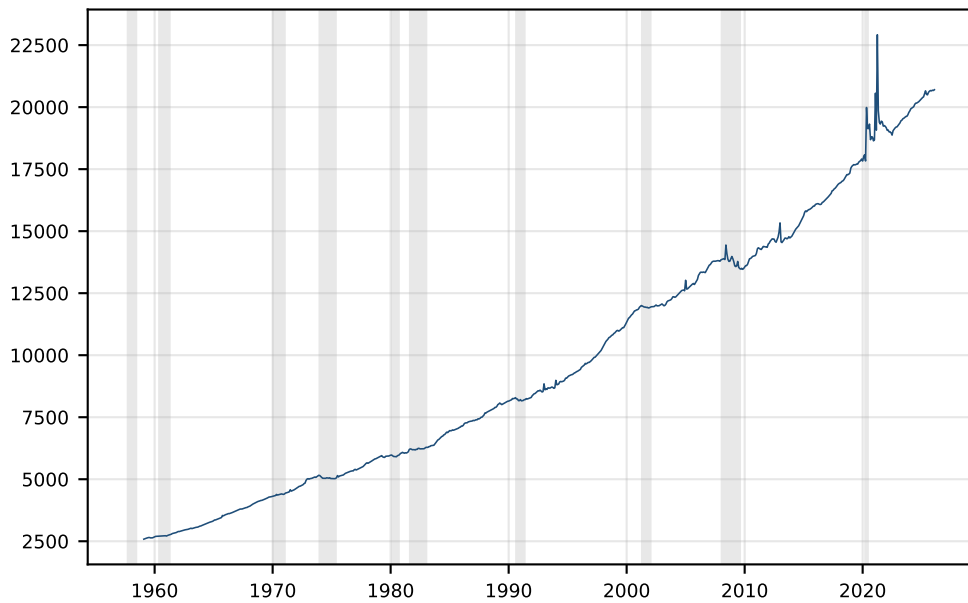
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	201	20.9%	79.1%	-0.0018
Q2	201	8.0%	92.0%	0.0041
Q3	201	10.4%	89.6%	0.0061
Q4 (highest)	201	8.0%	92.0%	0.0126

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

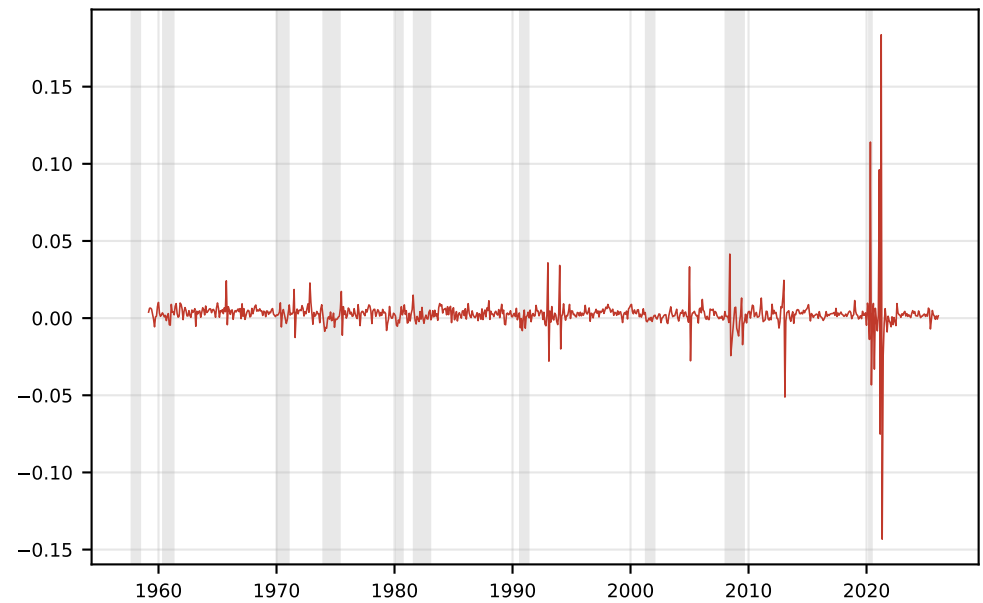
Real Personal Income

Block: Income / Income | Source: FRED API | Col: RPI | Transform: 5: Log diff

Raw Series



After: 5: Log diff



SUMMARY STATISTICS

N : 1287
Valid : 805
NaN : 482
Start : 1959-01
End : 2026-01
Mean : 9756.7897
Std : 5225.7651
Min : 2583.5600
Max : 22918.4220
Skew : 0.462
Kurt : -0.943

STATIONARITY TESTS

ADF stat : -6.162
ADF p : 0.0000
ADF lags : 12
ADF : Stationary
KPSS stat : 0.565
KPSS p : 0.0271
KPSS : Non-stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

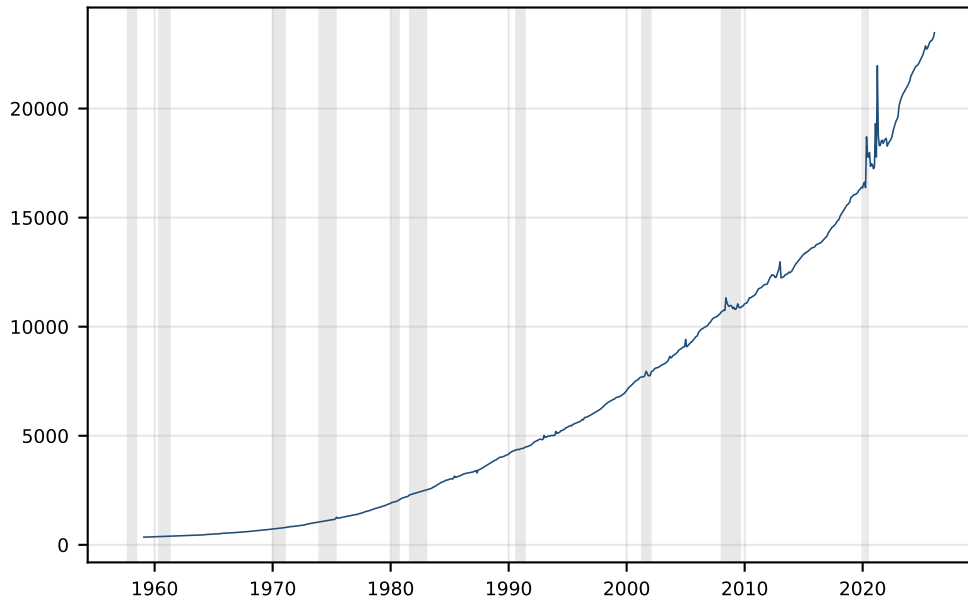
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	201	27.9%	72.1%	-0.0045
Q2	201	9.5%	90.5%	0.0017
Q3	201	4.5%	95.5%	0.0037
Q4 (highest)	201	5.5%	94.5%	0.0095

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

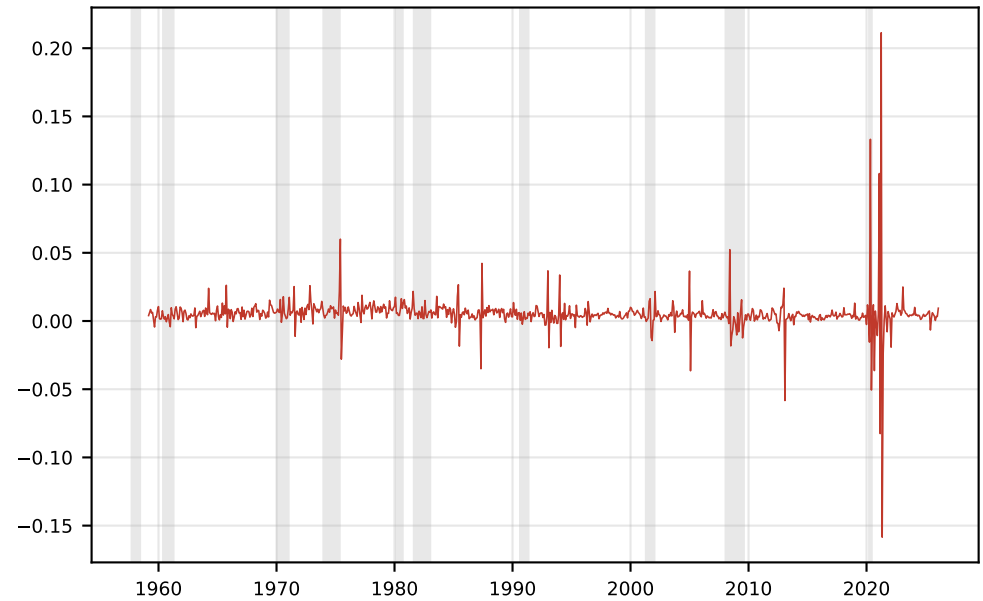
Disposable Personal Income

Block: Income / Income | Source: FRED API | Col: DSPI | Transform: 5: Log diff

Raw Series



After: 5: Log diff



SUMMARY STATISTICS

N : 1287
Valid : 805
NaN : 482
Start : 1959-01
End : 2026-01
Mean : 6734.4512
Std : 6199.1354
Min : 351.5000
Max : 23474.9000
Skew : 0.910
Kurt : -0.151

STATIONARITY TESTS

ADF stat : -5.559
ADF p : 0.0000
ADF lags : 12
ADF : Stationary
KPSS stat : 1.252
KPSS p : 0.0100
KPSS : Non-stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

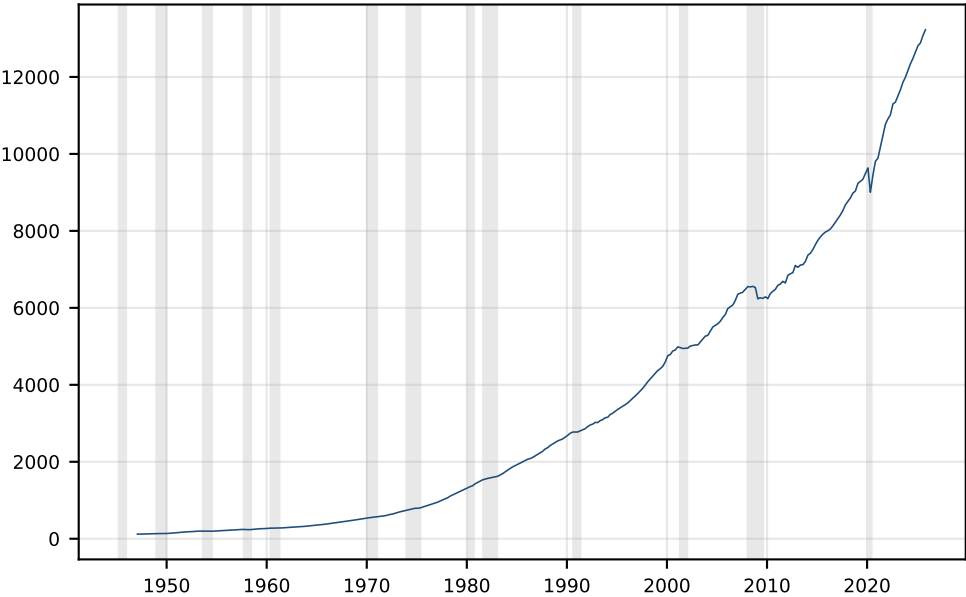
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	201	19.4%	80.6%	-0.0031
Q2	201	8.5%	91.5%	0.0038
Q3	201	9.0%	91.0%	0.0060
Q4 (highest)	201	10.4%	89.6%	0.0142

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

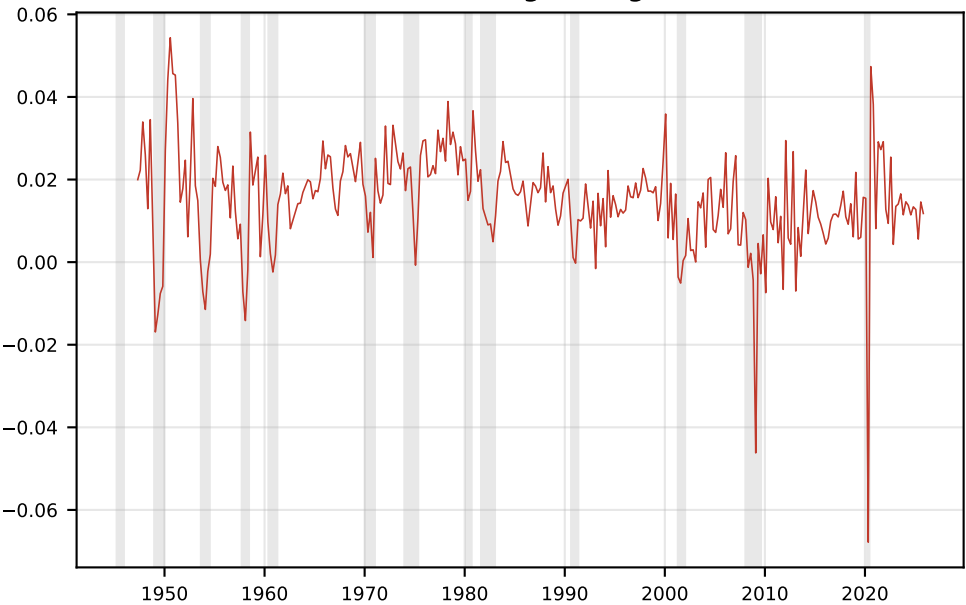
Gross Domestic Income: Wages and Salaries

Block: Income / Income | Source: FRED API | Col: A4102C1Q027SBEA | Transform: 8: Log diff lag 3

Raw Series



After: 8: Log diff lag 3



SUMMARY STATISTICS

N : 1287
Valid : 316
NaN : 971
Start : 1947-01
End : 2025-10
Mean : 3417.9794
Std : 3484.7907
Min : 118.7950
Max : 13227.1100
Skew : 1.003
Kurt : 0.019

STATIONARITY TESTS

ADF stat : -5.312
ADF p : 0.0000
ADF lags : 6
ADF : Stationary
KPSS stat : 0.825
KPSS p : 0.0100
KPSS : Non-stationary

QUARTILE-BUCKET ANALYSIS

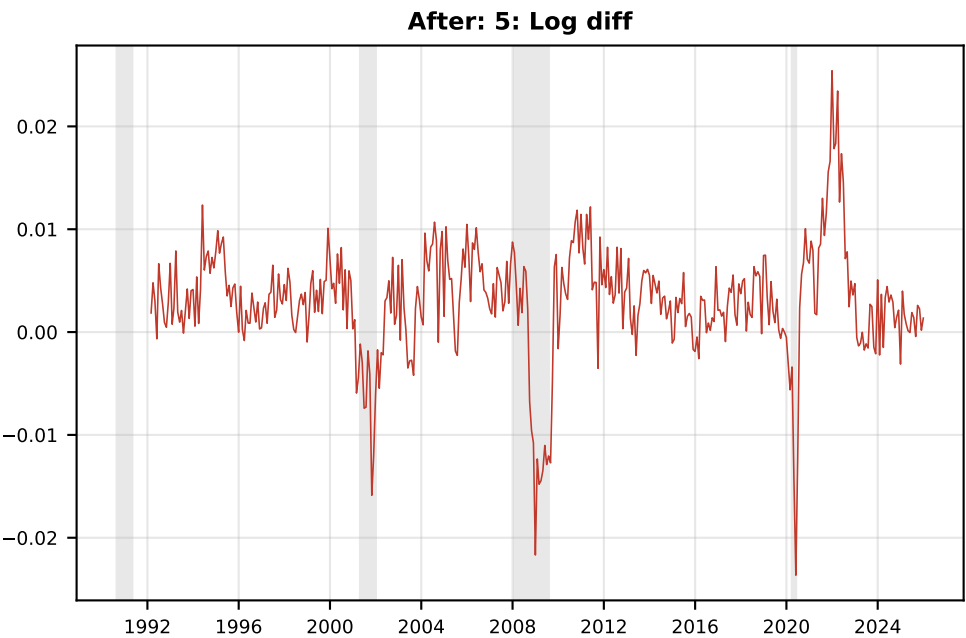
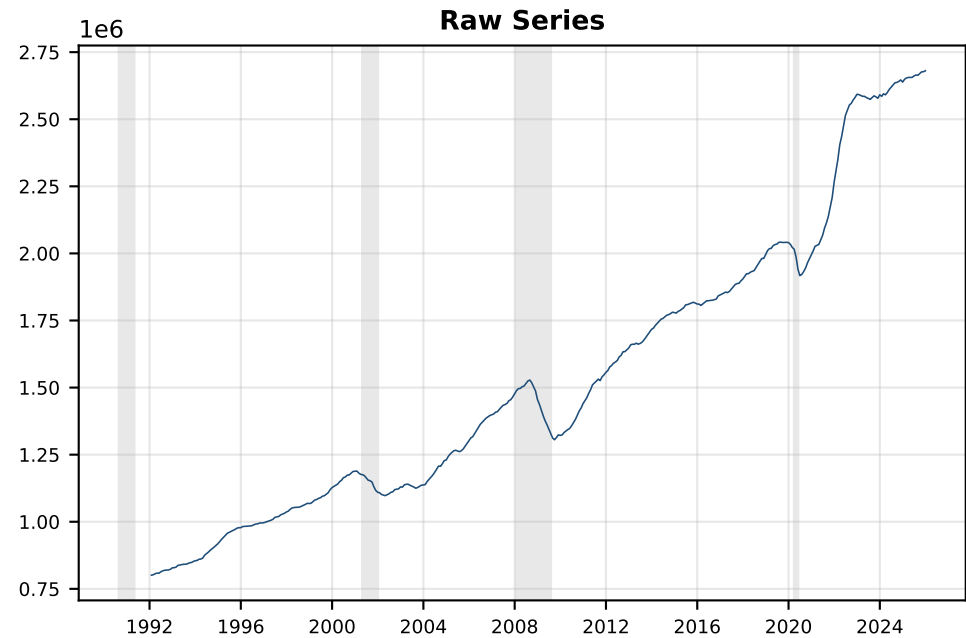
(vs NBER recession/expansion)

Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	79	36.7%	63.3%	0.0006
Q2	79	8.9%	91.1%	0.0125
Q3	78	3.8%	96.2%	0.0183
Q4 (highest)	79	2.5%	97.5%	0.0285

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

Total Business Inventories

Block: Orders / Inventories / Orders / Inventories | Source: FRED API | Col: BUSINV | Transform: 5: Log diff



SUMMARY STATISTICS

N : 1287
Valid : 408
NaN : 879
Start : 1992-01
End : 2025-12
Mean : 1534218.6005
Std : 527226.4067
Min : 800927.0000
Max : 2680732.0000
Skew : 0.677
Kurt : -0.488

STATIONARITY TESTS

ADF stat : -4.900
ADF p : 0.0000
ADF lags : 2
ADF : Stationary
KPSS stat : 0.039
KPSS p : 0.1000
KPSS : Stationary

QUARTILE-BUCKET ANALYSIS

(vs NBER recession/expansion)

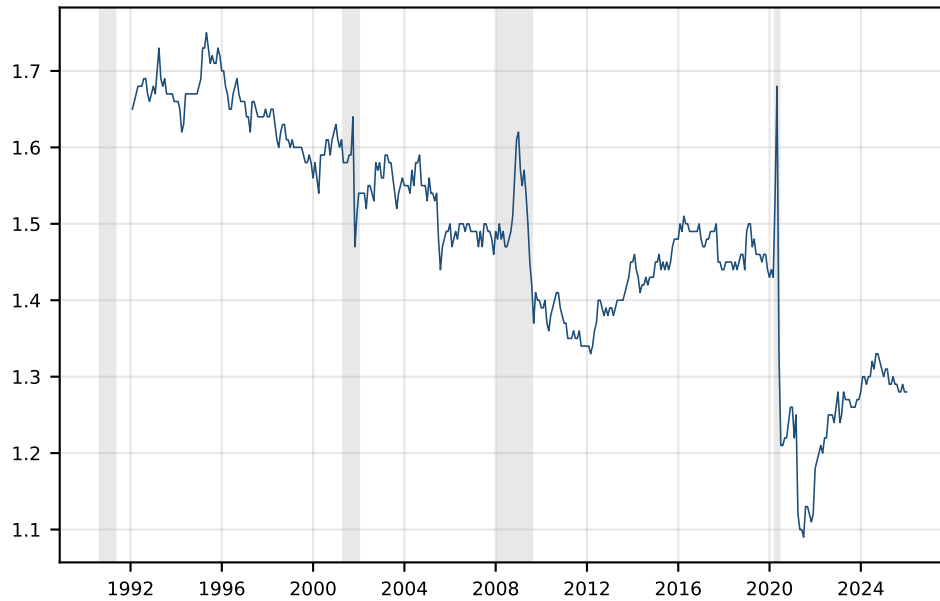
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	102	20.6%	79.4%	-0.0034
Q2	102	2.0%	98.0%	0.0020
Q3	101	2.0%	98.0%	0.0044
Q4 (highest)	102	2.9%	97.1%	0.0089

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

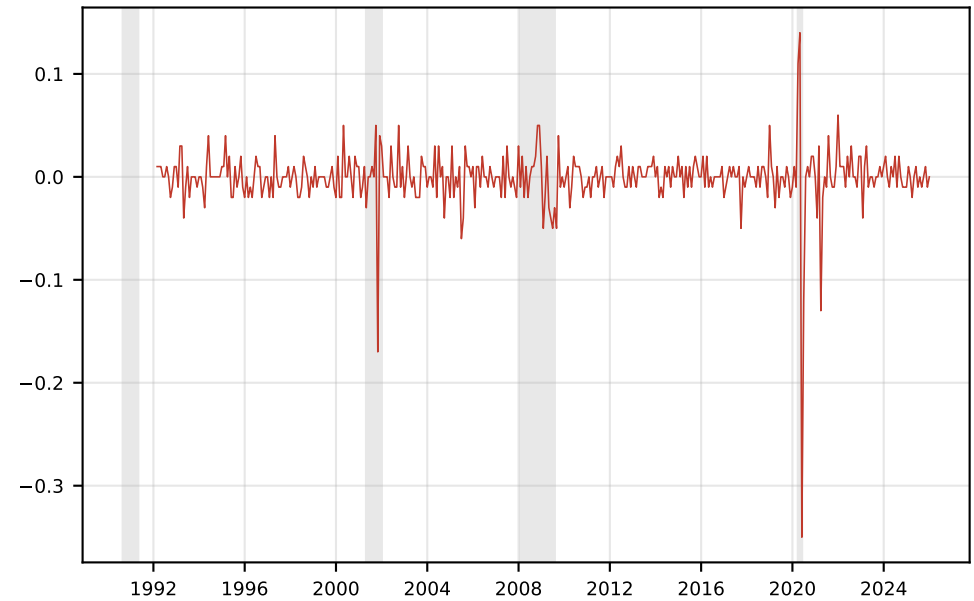
Retailers: Inventories to Sales Ratio

Block: Orders / Inventories / Orders / Inventories | Source: FRED API | Col: RETAILIRSA | Transform: 2: First diff

Raw Series



After: 2: First diff



SUMMARY STATISTICS

N : 1287
Valid : 408
NaN : 879
Start : 1992-01
End : 2025-12
Mean : 1.4828
Std : 0.1440
Min : 1.0900
Max : 1.7500
Skew : -0.431
Kurt : -0.380

STATIONARITY TESTS

ADF stat : -11.422
ADF p : 0.0000
ADF lags : 4
ADF : Stationary
KPSS stat : 0.032
KPSS p : 0.1000
KPSS : Stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

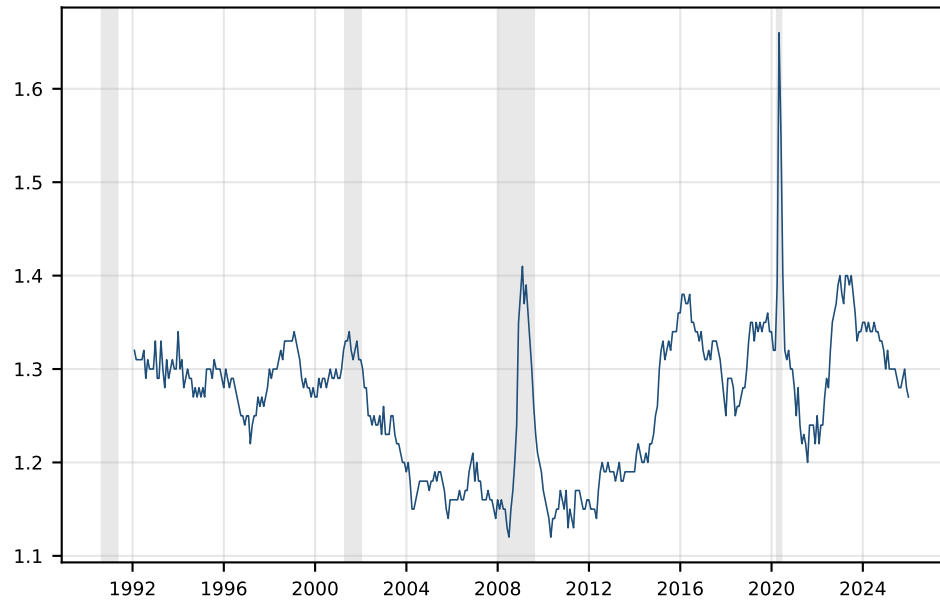
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	136	7.4%	92.6%	-0.0225
Q2	128	3.1%	96.9%	-0.0002
Q3	85	5.9%	94.1%	0.0100
Q4 (highest)	58	15.5%	84.5%	0.0321

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

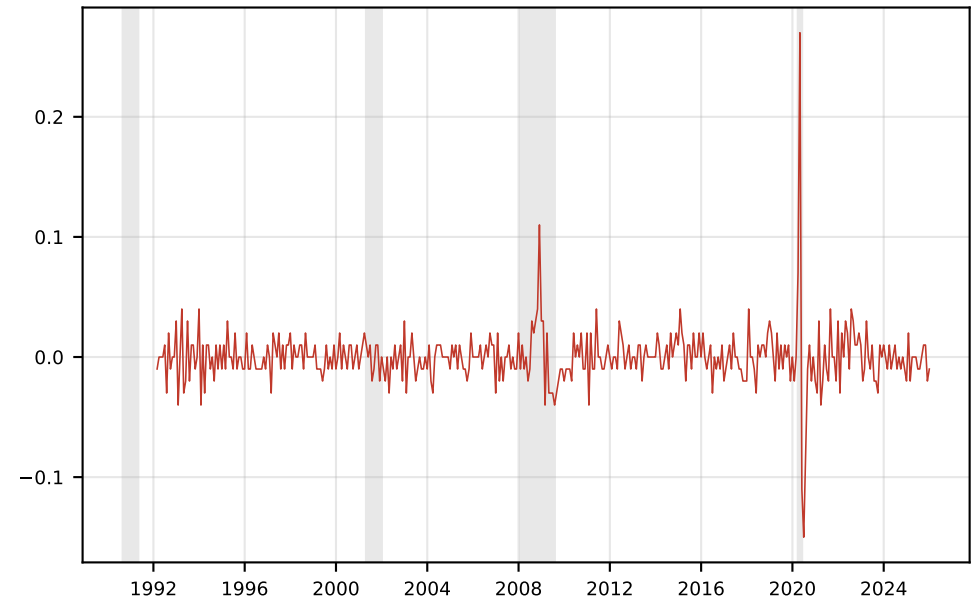
Wholesale Inventories to Sales Ratio

Block: Orders / Inventories / Orders / Inventories | Source: FRED API | Col: WHLSLRIRSA | Transform: 2: First diff

Raw Series



After: 2: First diff



SUMMARY STATISTICS

N : 1287
Valid : 408
NaN : 879
Start : 1992-01
End : 2025-12
Mean : 1.2649
Std : 0.0739
Min : 1.1200
Max : 1.6600
Skew : 0.275
Kurt : 1.196

STATIONARITY TESTS

ADF stat : -13.122
ADF p : 0.0000
ADF lags : 2
ADF : Stationary
KPSS stat : 0.035
KPSS p : 0.1000
KPSS : Stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

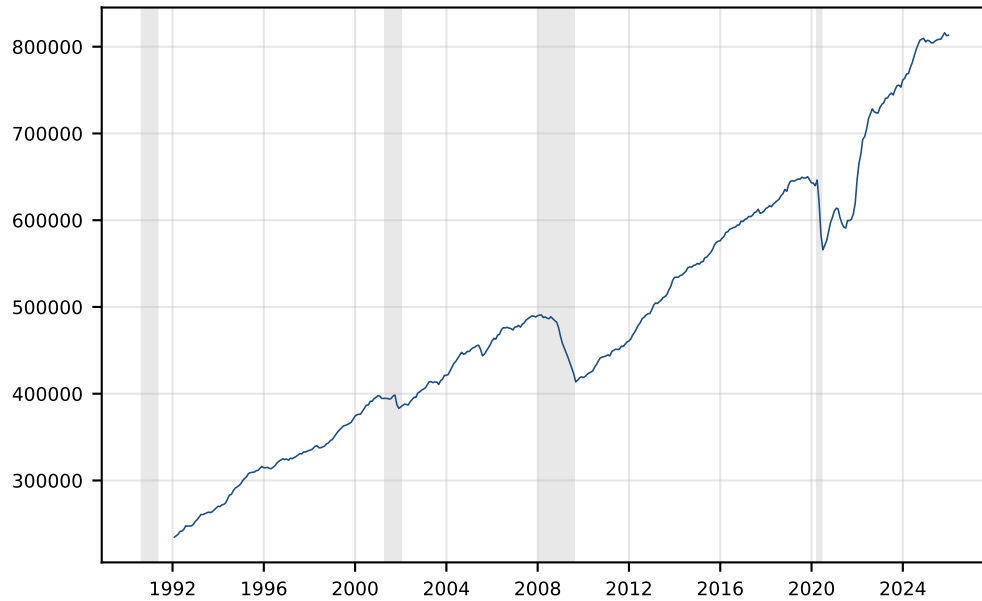
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	156	6.4%	93.6%	-0.0177
Q2	109	2.8%	97.2%	-0.0003
Q3	80	6.2%	93.8%	0.0100
Q4 (highest)	62	16.1%	83.9%	0.0313

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

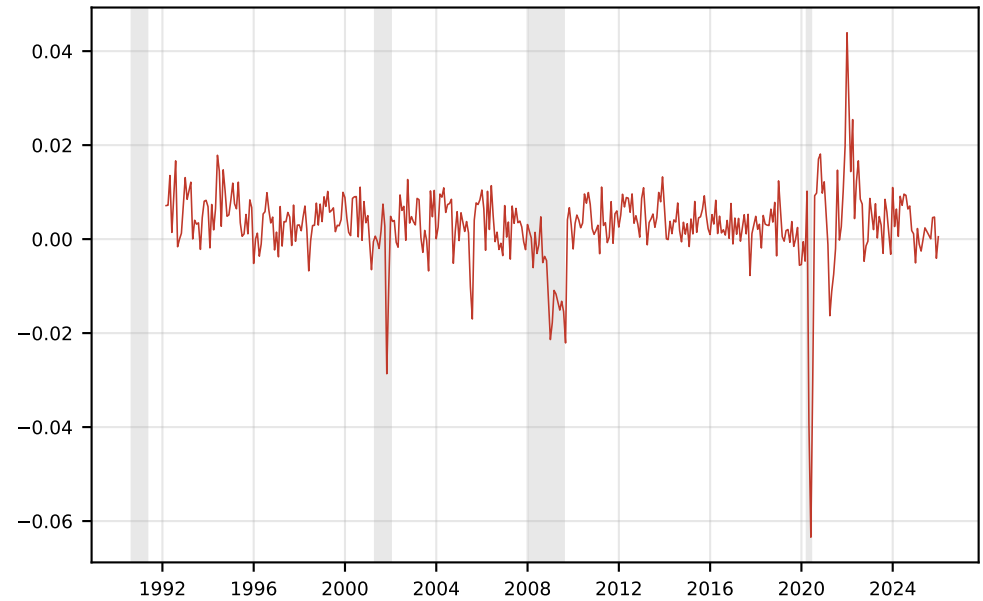
Retail Inventories

Block: Orders / Inventories / Orders / Inventories | Source: FRED API | Col: RETAILIMSA | Transform: 5: Log diff

Raw Series



After: 5: Log diff



SUMMARY STATISTICS

N : 1287
Valid : 408
NaN : 879
Start : 1992-01
End : 2025-12
Mean : 485983.1373
Std : 147611.5143
Min : 234641.0000
Max : 816106.0000
Skew : 0.453
Kurt : -0.516

STATIONARITY TESTS

ADF stat : -10.889
ADF p : 0.0000
ADF lags : 0
ADF : Stationary
KPSS stat : 0.158
KPSS p : 0.1000
KPSS : Stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	102	18.6%	81.4%	-0.0055
Q2	102	5.9%	94.1%	0.0020
Q3	101	1.0%	99.0%	0.0051
Q4 (highest)	102	2.0%	98.0%	0.0107

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

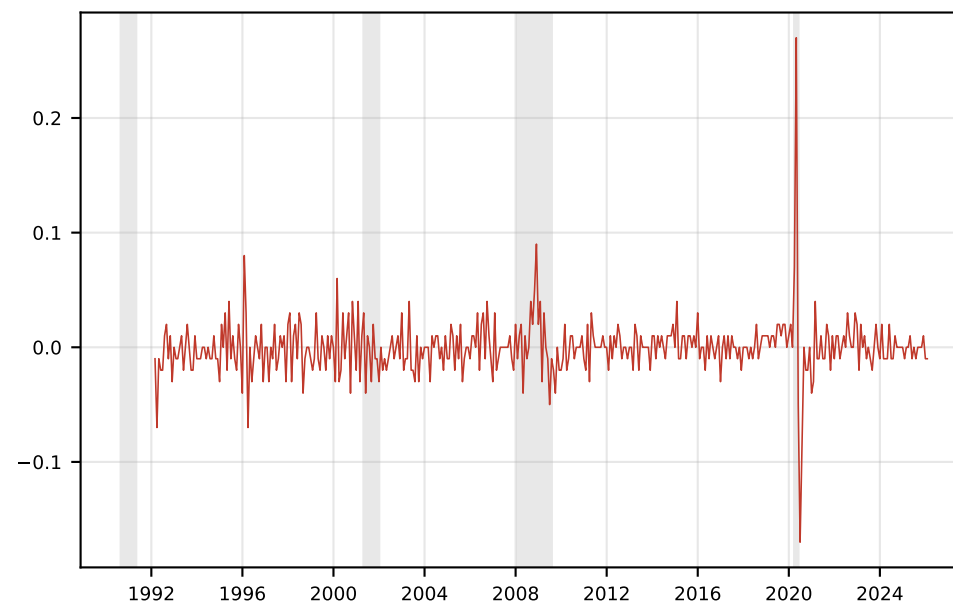
Manufacturers Total Inventories

Block: Orders / Inventories / Orders / Inventories | Source: FRED API | Col: AMTMIS | Transform: 2: First diff

Raw Series



After: 2: First diff



SUMMARY STATISTICS

N : 1287
Valid : 409
NaN : 878
Start : 1992-01
End : 2026-01
Mean : 1.3842
Std : 0.1257
Min : 1.1400
Max : 1.8800
Skew : 0.311
Kurt : -0.095

STATIONARITY TESTS

ADF stat : -16.346
ADF p : 0.0000
ADF lags : 1
ADF : Stationary
KPSS stat : 0.281
KPSS p : 0.1000
KPSS : Stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

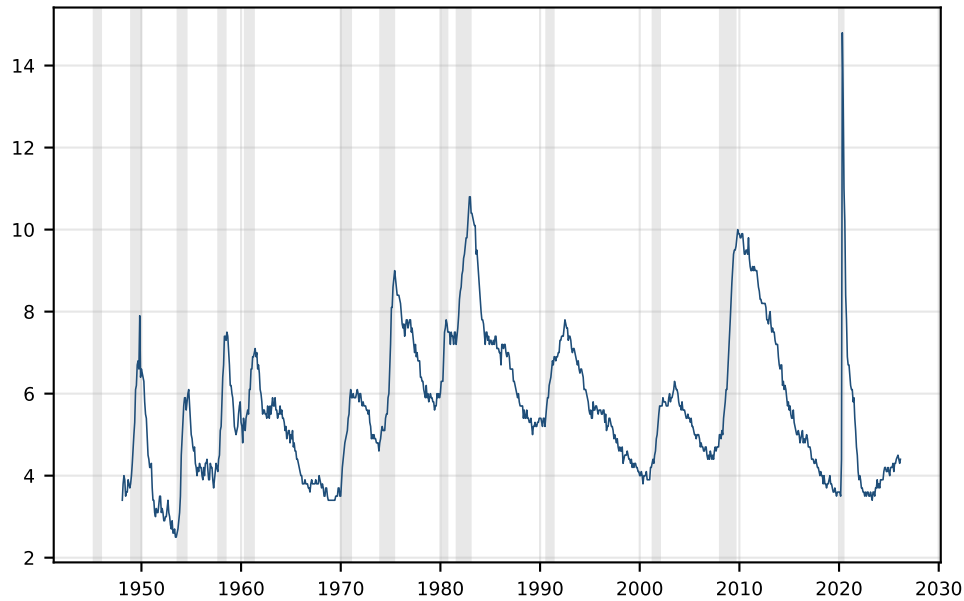
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	159	5.7%	94.3%	-0.0191
Q2	104	3.8%	96.2%	-0.0003
Q3	79	3.8%	96.2%	0.0100
Q4 (highest)	66	18.2%	81.8%	0.0329

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

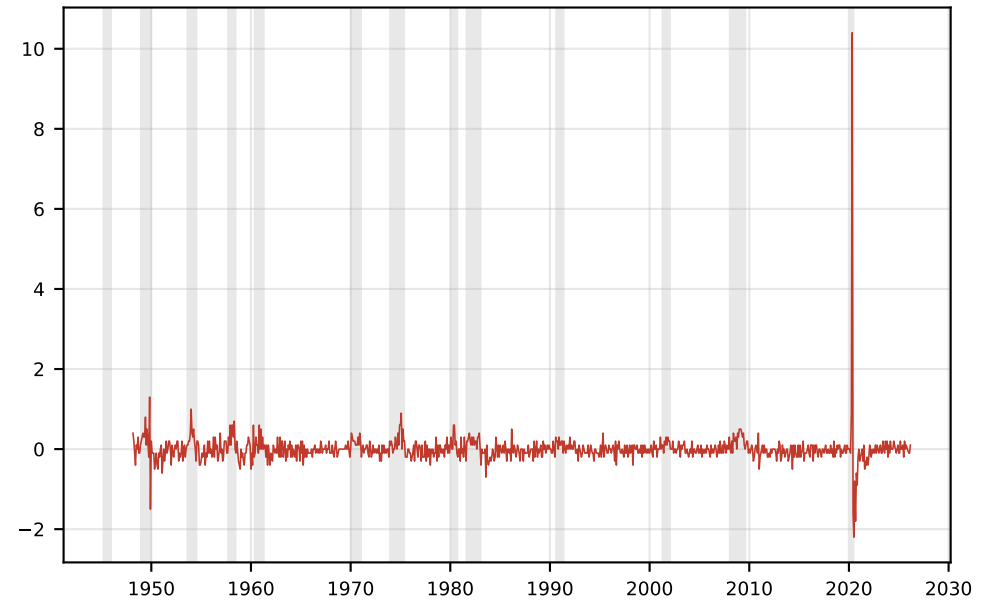
Unemployment Rate

Block: Labor market / Labor market | Source: FRED API | Col: UNRATE | Transform: 2: First diff

Raw Series



After: 2: First diff



SUMMARY STATISTICS

N : 1287
Valid : 937
NaN : 350
Start : 1948-01
End : 2026-02
Mean : 5.6637
Std : 1.7053
Min : 2.5000
Max : 14.8000
Skew : 0.883
Kurt : 1.099

STATIONARITY TESTS

ADF stat : -29.474
ADF p : 0.0000
ADF lags : 0
ADF : Stationary
KPSS stat : 0.039
KPSS p : 0.1000
KPSS : Stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

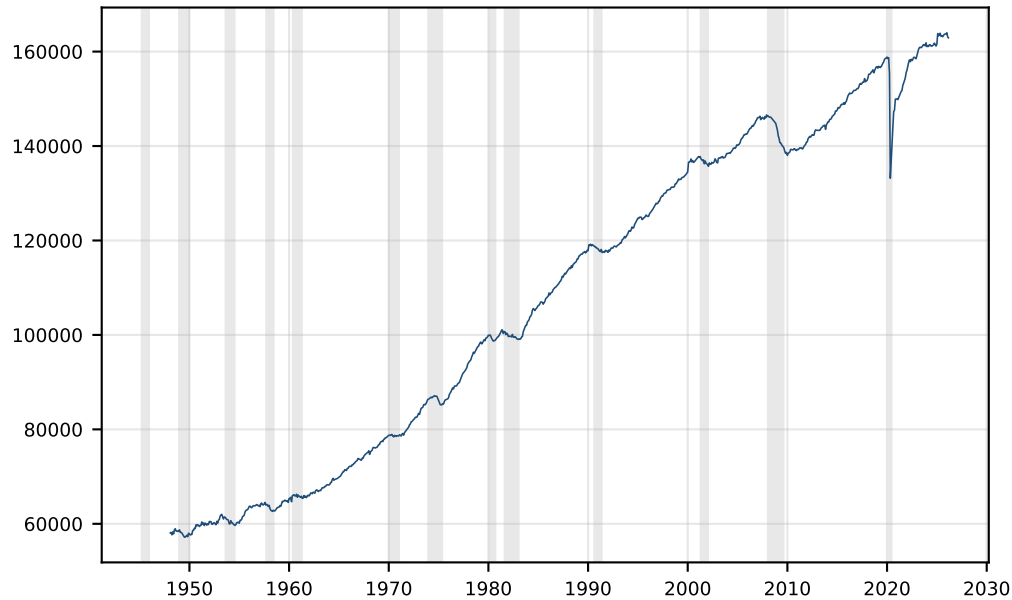
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	238	1.7%	98.3%	-0.2571
Q2	373	4.3%	95.7%	-0.0389
Q3	93	18.3%	81.7%	0.1000
Q4 (highest)	231	37.7%	62.3%	0.2913

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

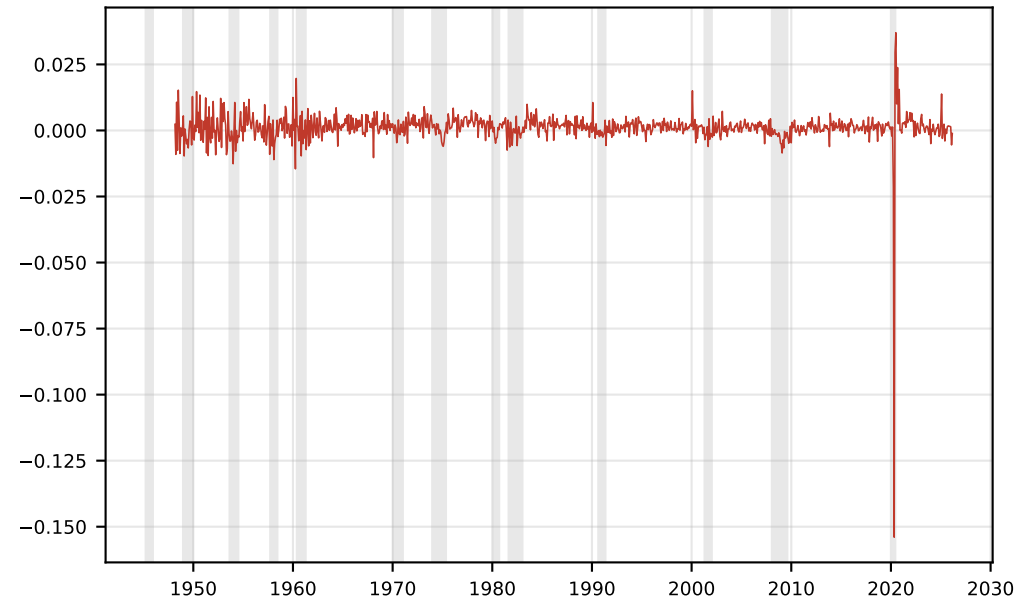
Employment Level

Block: Labor market / Labor market | Source: FRED API | Col: CE16OV | Transform: 5: Log diff

Raw Series



After: 5: Log diff



SUMMARY STATISTICS

N : 1287
Valid : 937
NaN : 350
Start : 1948-01
End : 2026-02
Mean : 108443.1153
Std : 34087.3391
Min : 57172.0000
Max : 163992.0000
Skew : -0.047
Kurt : -1.441

STATIONARITY TESTS

ADF stat : -23.445
ADF p : 0.0000
ADF lags : 1
ADF : Stationary
KPSS stat : 0.252
KPSS p : 0.1000
KPSS : Stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

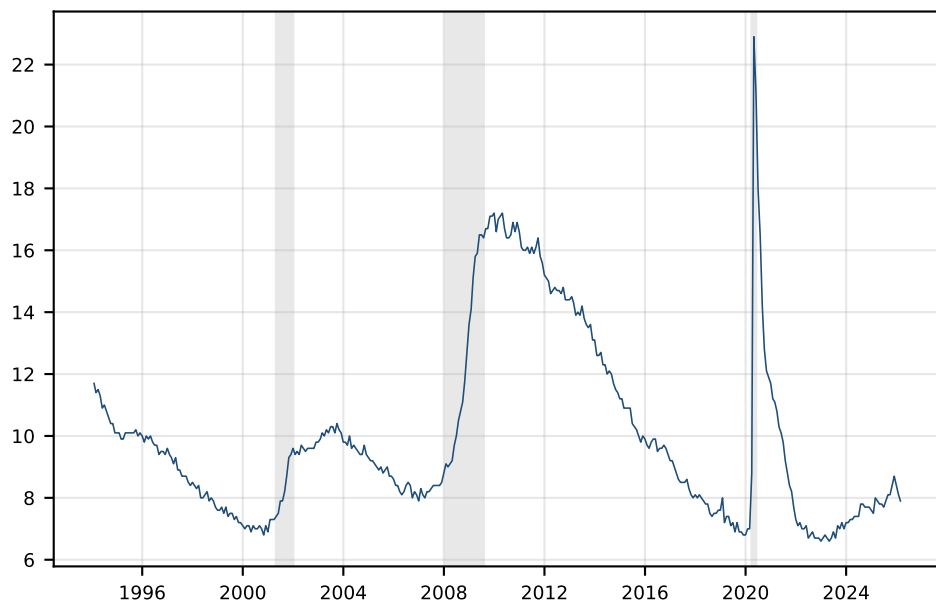
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	234	31.6%	68.4%	-0.0038
Q2	234	12.0%	88.0%	0.0004
Q3	233	4.7%	95.3%	0.0021
Q4 (highest)	234	4.7%	95.3%	0.0058

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

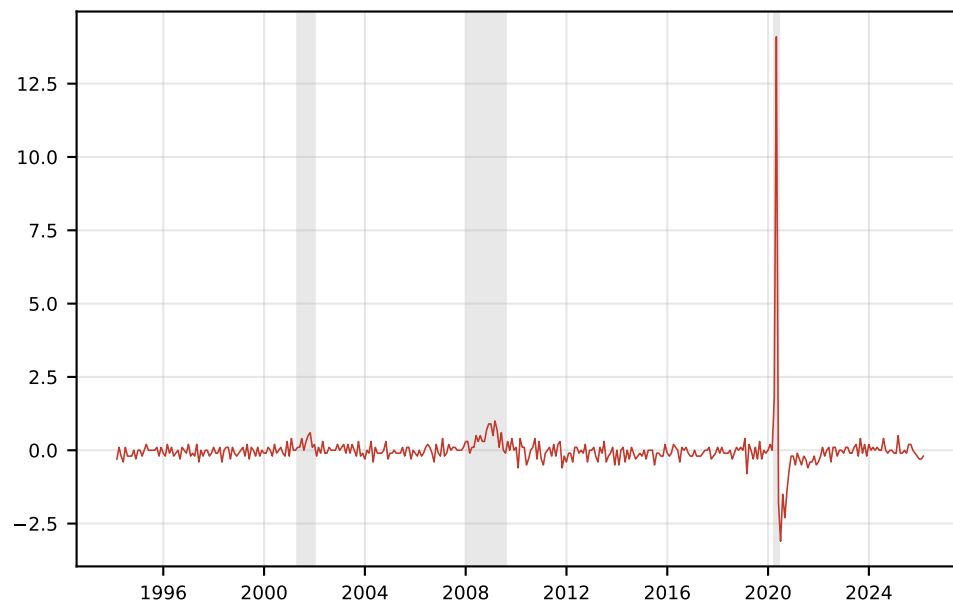
U-6 Unemployment Rate

Block: Labor market / Labor market | Source: FRED API | Col: U6RATE | Transform: 2: First diff

Raw Series



After: 2: First diff



SUMMARY STATISTICS

N : 1287
Valid : 385
NaN : 902
Start : 1994-01
End : 2026-02
Mean : 10.0166
Std : 2.9791
Min : 6.6000
Max : 22.9000
Skew : 1.304
Kurt : 1.161

STATIONARITY TESTS

ADF stat : -11.171
ADF p : 0.0000
ADF lags : 3
ADF : Stationary
KPSS stat : 0.045
KPSS p : 0.1000
KPSS : Stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

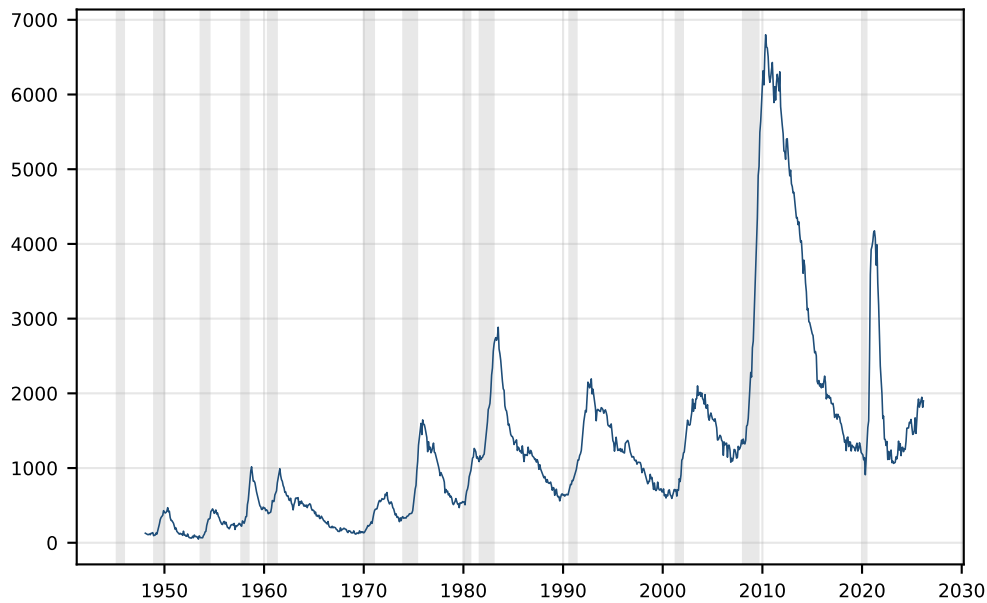
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	110	0.0%	100.0%	-0.3755
Q2	144	2.1%	97.9%	-0.0479
Q3	50	10.0%	90.0%	0.1000
Q4 (highest)	79	25.3%	74.7%	0.4911

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

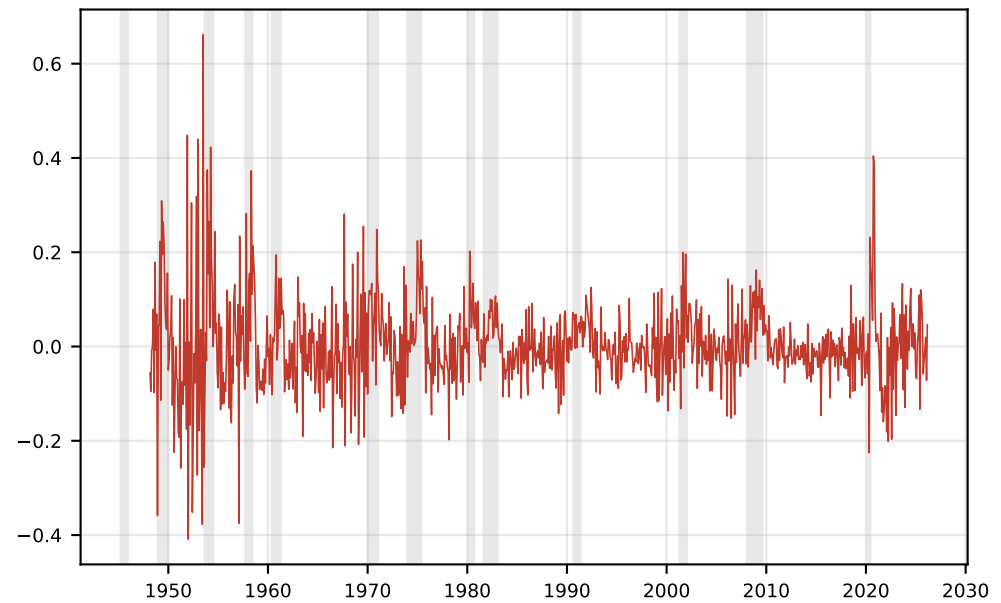
Long-term Unemployment

Block: Labor market / Labor market | Source: FRED API | Col: UEMP27OV | Transform: 5: Log diff

Raw Series



After: 5: Log diff



SUMMARY STATISTICS

N : 1287
Valid : 937
NaN : 350
Start : 1948-01
End : 2026-02
Mean : 1296.8036
Std : 1292.4053
Min : 48.0000
Max : 6800.0000
Skew : 2.215
Kurt : 5.230

STATIONARITY TESTS

ADF stat : -8.571
ADF p : 0.0000
ADF lags : 12
ADF : Stationary
KPSS stat : 0.036
KPSS p : 0.1000
KPSS : Stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

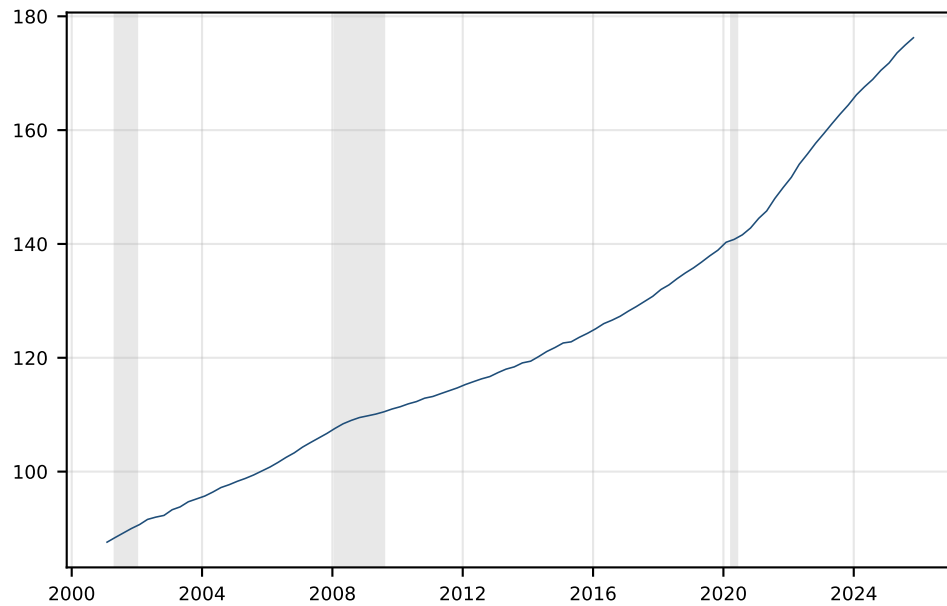
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	234	5.1%	94.9%	-0.1025
Q2	234	4.7%	95.3%	-0.0228
Q3	233	14.2%	85.8%	0.0200
Q4 (highest)	234	29.1%	70.9%	0.1167

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

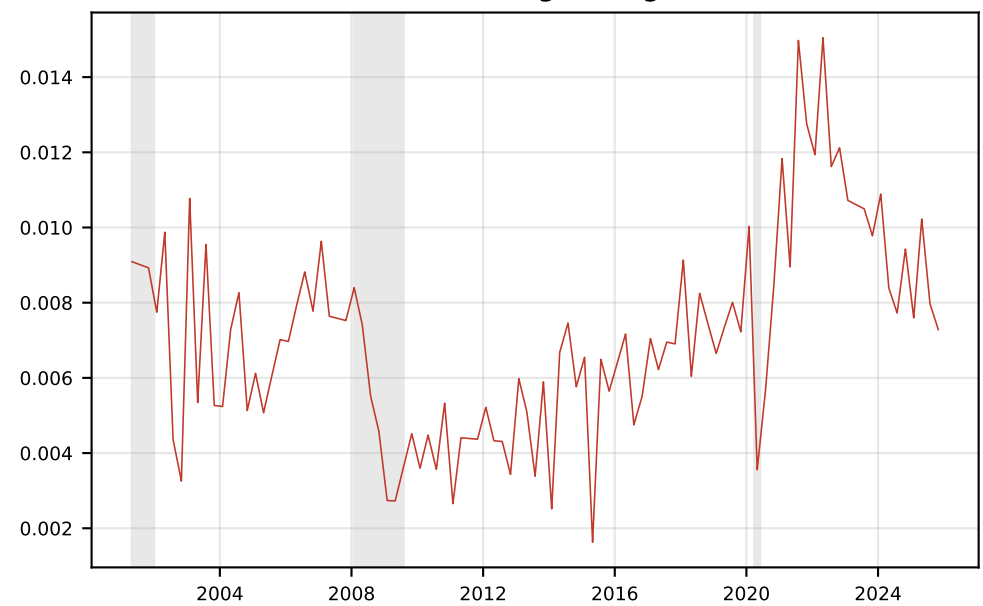
Employment Cost Index

Block: Wages / labor costs / Wages / labor costs | Source: FRED API | Col: ECIWAG | Transform: 8: Log diff lag 3

Raw Series



After: 8: Log diff lag 3



SUMMARY STATISTICS

N : 1287
Valid : 100
NaN : 1187
Start : 2001-01
End : 2025-10
Mean : 123.0185
Std : 23.8962
Min : 87.6000
Max : 176.2330
Skew : 0.589
Kurt : -0.562

STATIONARITY TESTS

ADF stat : -2.631
ADF p : 0.0867
ADF lags : 1
ADF : Non-stationary
KPSS stat : 0.589
KPSS p : 0.0236
KPSS : Non-stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

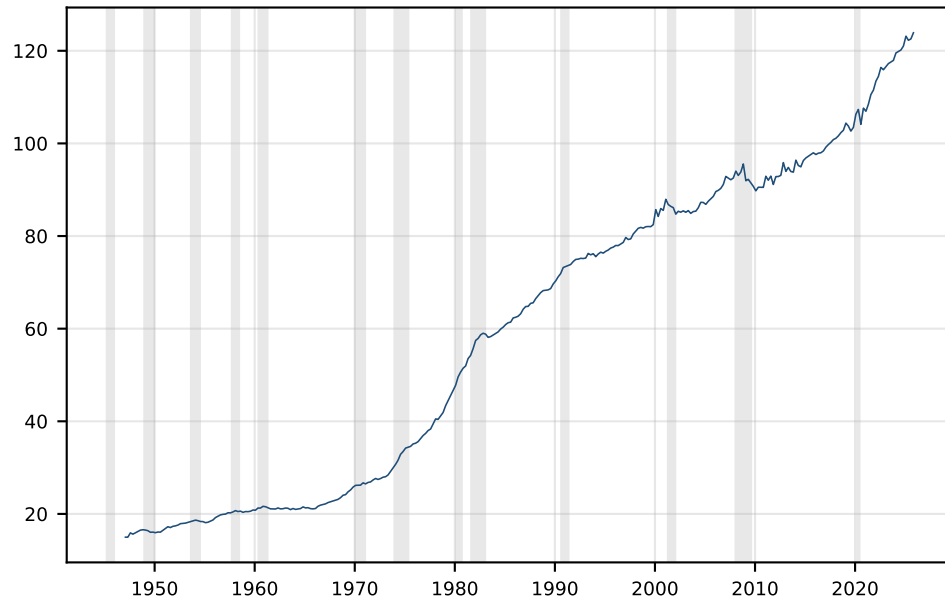
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	25	16.0%	84.0%	0.0039
Q2	25	4.0%	96.0%	0.0061
Q3	24	8.3%	91.7%	0.0077
Q4 (highest)	25	12.0%	88.0%	0.0106

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

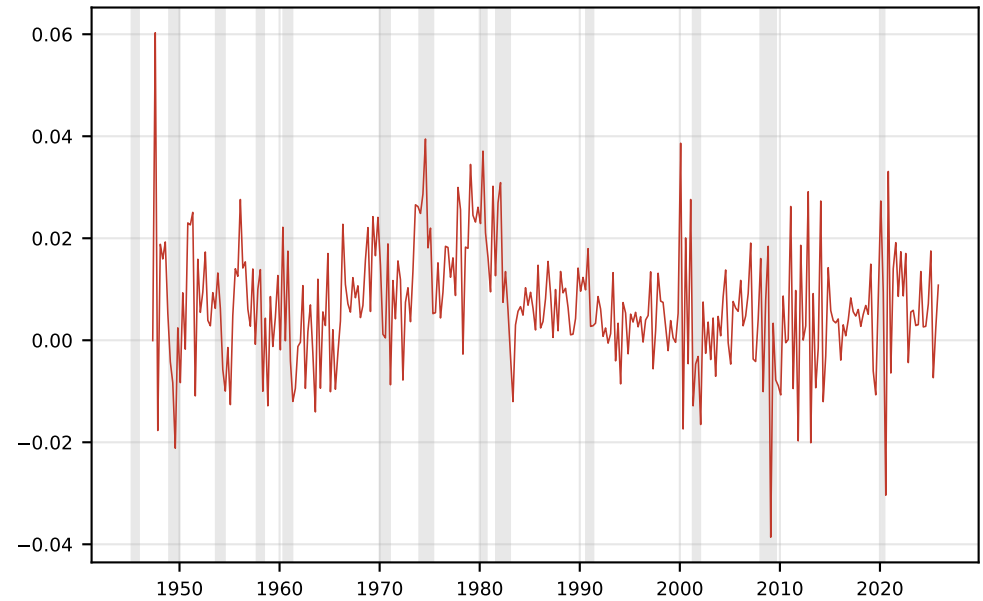
Unit Labor Costs

Block: Wages / labor costs / Wages / labor costs | Source: FRED API | Col: ULCNFB | Transform: 8: Log diff lag 3

Raw Series



After: 8: Log diff lag 3



SUMMARY STATISTICS

N : 1287
Valid : 316
NaN : 971
Start : 1947-01
End : 2025-10
Mean : 59.6507
Std : 33.5006
Min : 14.9800
Max : 123.8960
Skew : 0.075
Kurt : -1.433

STATIONARITY TESTS

ADF stat : -6.611
ADF p : 0.0000
ADF lags : 2
ADF : Stationary
KPSS stat : 0.463
KPSS p : 0.0500
KPSS : Non-stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

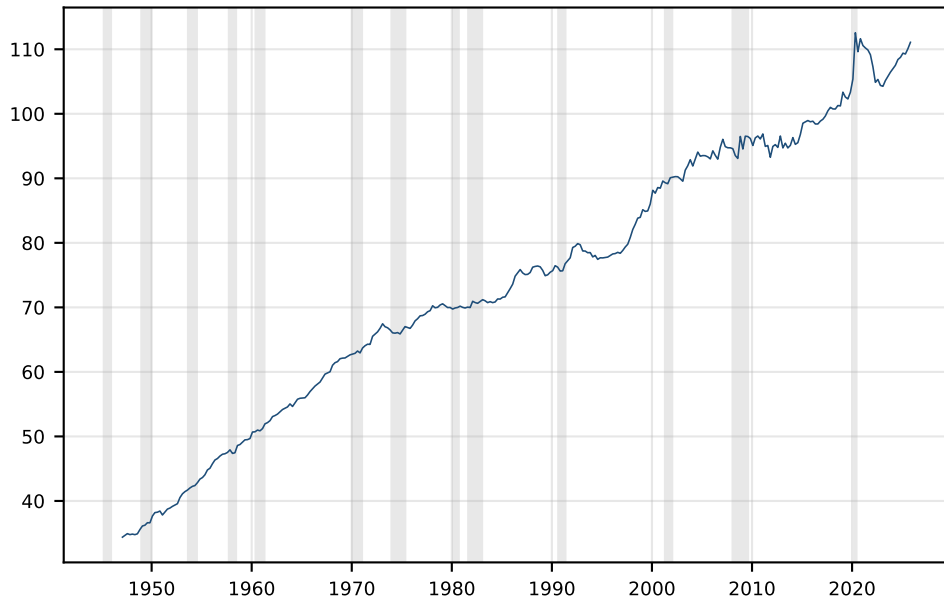
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	79	16.5%	83.5%	-0.0073
Q2	79	5.1%	94.9%	0.0035
Q3	78	10.3%	89.7%	0.0091
Q4 (highest)	79	20.3%	79.7%	0.0216

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

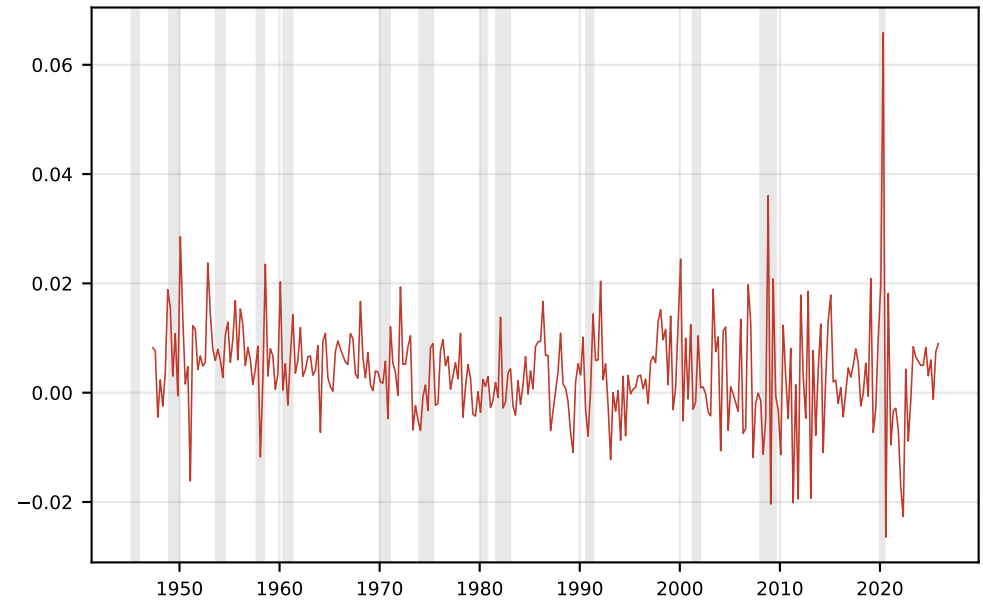
Real Hourly Compensation

Block: Wages / labor costs / Wages / labor costs | Source: FRED API | Col: COMPRNFB | Transform: 8: Log diff lag 3

Raw Series



After: 8: Log diff lag 3



SUMMARY STATISTICS

N : 1287
Valid : 316
NaN : 971
Start : 1947-01
End : 2025-10
Mean : 74.0326
Std : 20.8837
Min : 34.3860
Max : 112.5620
Skew : -0.119
Kurt : -0.951

STATIONARITY TESTS

ADF stat : -6.167
ADF p : 0.0000
ADF lags : 7
ADF : Stationary
KPSS stat : 0.846
KPSS p : 0.0100
KPSS : Non-stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

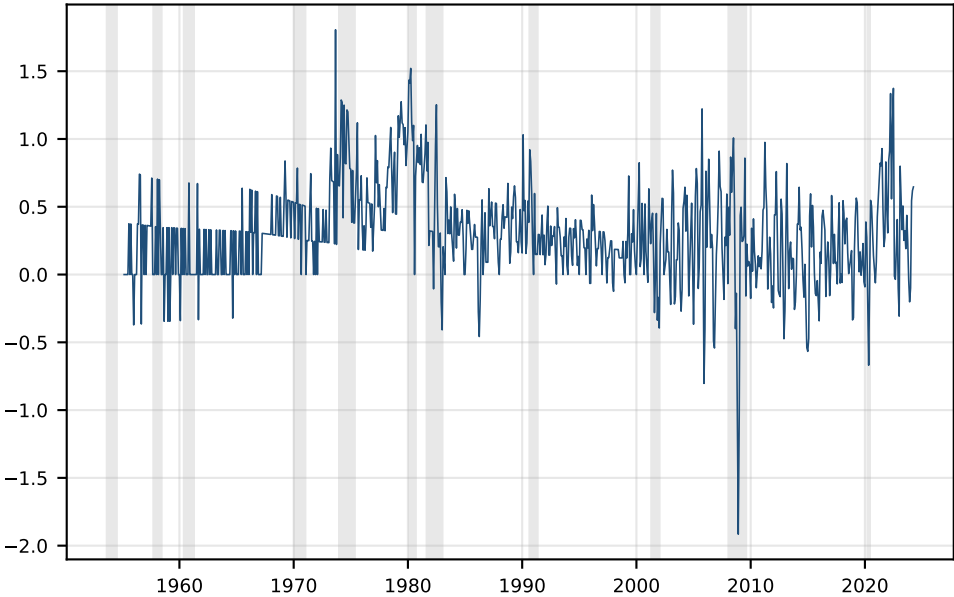
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	79	17.7%	82.3%	-0.0066
Q2	79	15.2%	84.8%	0.0014
Q3	78	7.7%	92.3%	0.0057
Q4 (highest)	79	11.4%	88.6%	0.0144

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

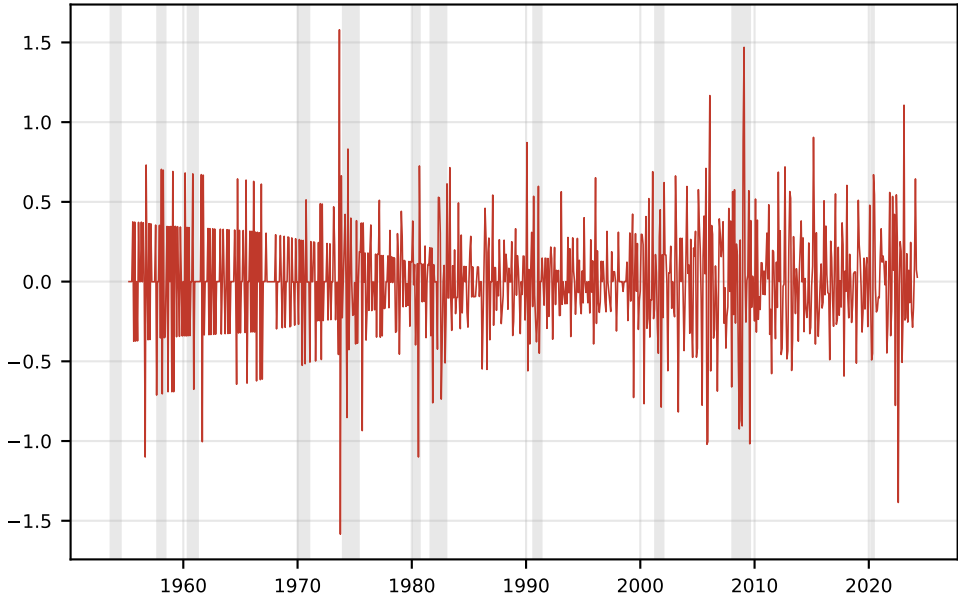
CPI: All Items: Total for United States

Block: Inflation / Prices / Consumer Inflation | Source: FRED API | Col: CPALTT01USM657N | Transform: 2: First diff

Raw Series



After: 2: First diff



SUMMARY STATISTICS

N	: 1287
Valid	: 830
NaN	: 457
Start	: 1955-02
End	: 2024-03
Mean	: 0.2974
Std	: 0.3559
Min	: -1.9153
Max	: 1.8059
Skew	: 0.103
Kurt	: 2.709

STATIONARITY TESTS

ADF stat	: -14.651
ADF p	: 0.0000
ADF lags	: 11
ADF	: Stationary
KPSS stat	: 0.084
KPSS p	: 0.1000
KPSS	: Stationary

QUARTILE-BUCKET ANALYSIS
(vs NBER recession/expansion)

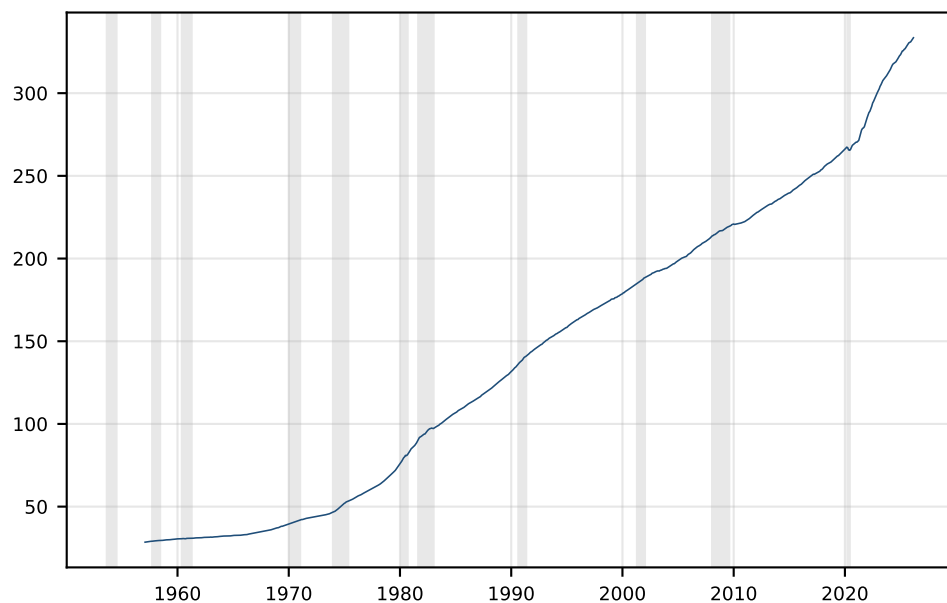
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	208	18.8%	81.2%	-0.4157
Q2	207	7.2%	92.8%	-0.0692
Q3	207	9.7%	90.3%	0.0739
Q4 (highest)	207	14.0%	86.0%	0.4161

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

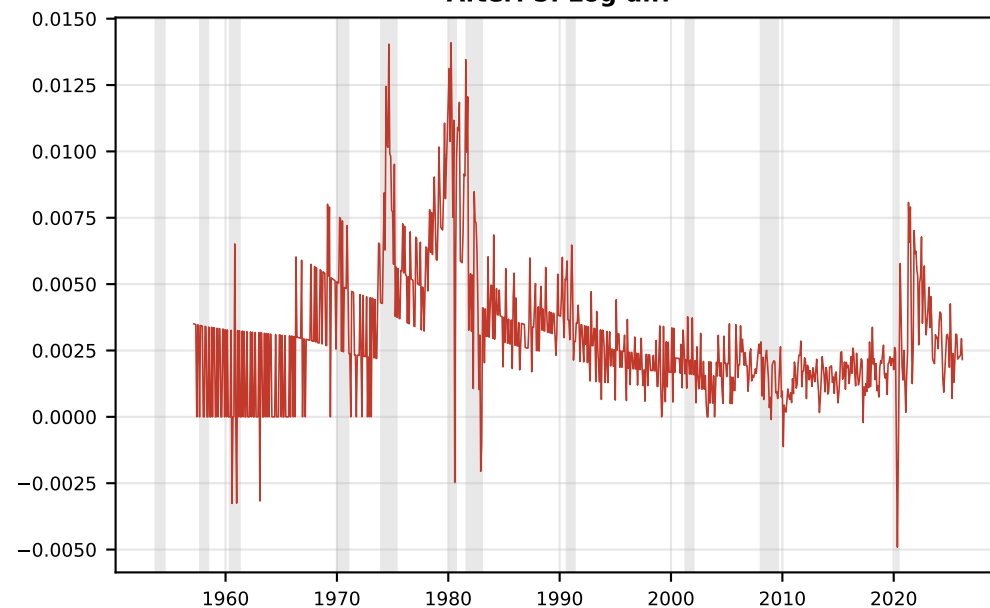
CPI Less Food and Energy

Block: Inflation / Prices / Consumer Inflation | Source: FRED API | Col: CPILFESL | Transform: 5: Log diff

Raw Series



After: 5: Log diff



SUMMARY STATISTICS

N : 1287
Valid : 829
NaN : 458
Start : 1957-01
End : 2026-02
Mean : 142.3109
Std : 88.8185
Min : 28.5000
Max : 333.5120
Skew : 0.254
Kurt : -1.138

STATIONARITY TESTS

ADF stat : -2.817
ADF p : 0.0559
ADF lags : 8
ADF : Non-stationary
KPSS stat : 0.870
KPSS p : 0.0100
KPSS : Non-stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

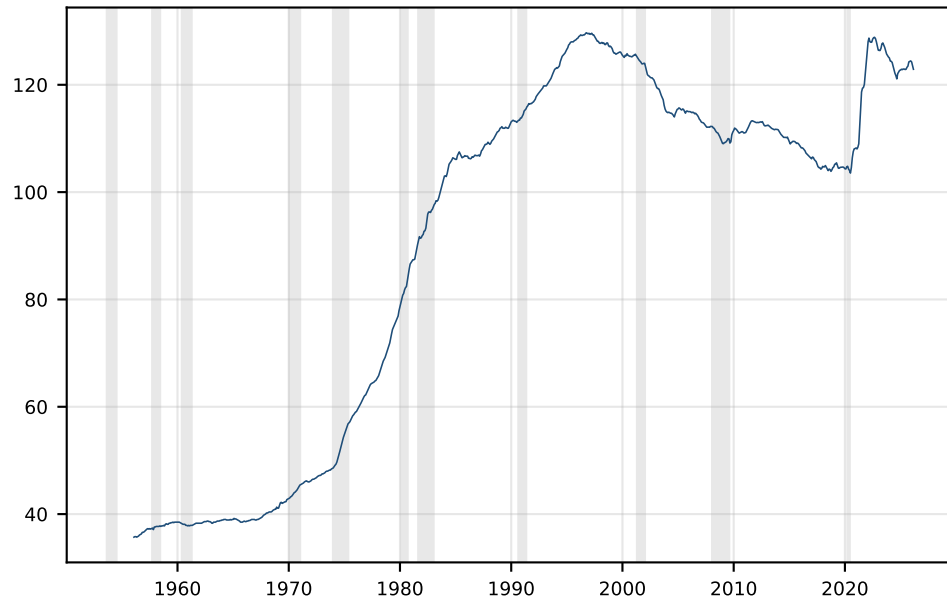
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	207	12.1%	87.9%	0.0005
Q2	207	8.2%	91.8%	0.0021
Q3	206	7.8%	92.2%	0.0031
Q4 (highest)	207	21.7%	78.3%	0.0062

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

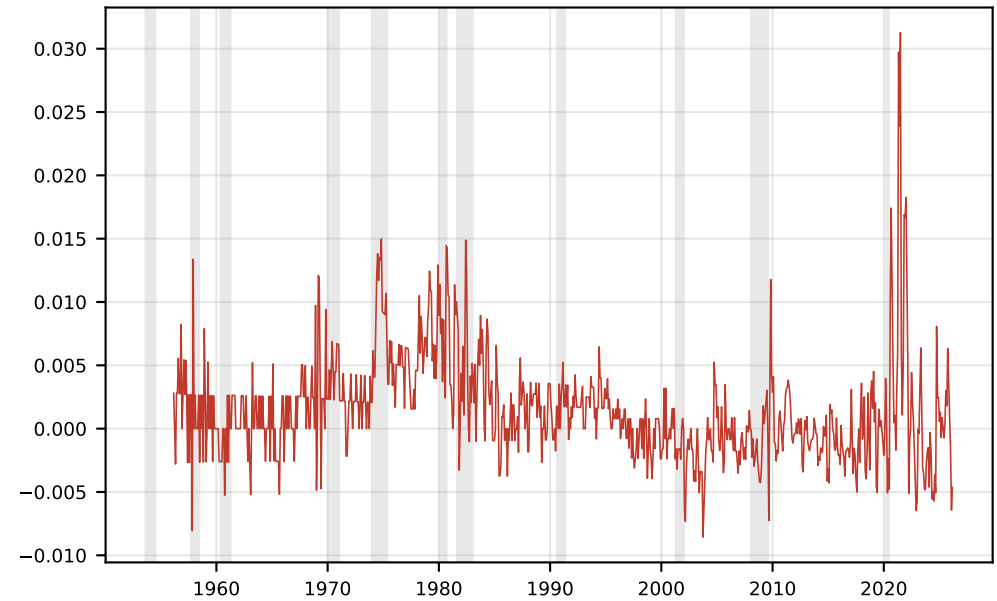
CPI Durables

Block: Inflation / Prices / Consumer Inflation | Source: FRED API | Col: CUSR0000SAD | Transform: 5: Log diff

Raw Series



After: 5: Log diff



SUMMARY STATISTICS

N : 1287
Valid : 841
NaN : 446
Start : 1956-01
End : 2026-02
Mean : 90.5371
Std : 33.8177
Min : 35.7000
Max : 129.7000
Skew : -0.583
Kurt : -1.359

STATIONARITY TESTS

ADF stat : -4.624
ADF p : 0.0001
ADF lags : 11
ADF : Stationary
KPSS stat : 1.016
KPSS p : 0.0100
KPSS : Non-stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

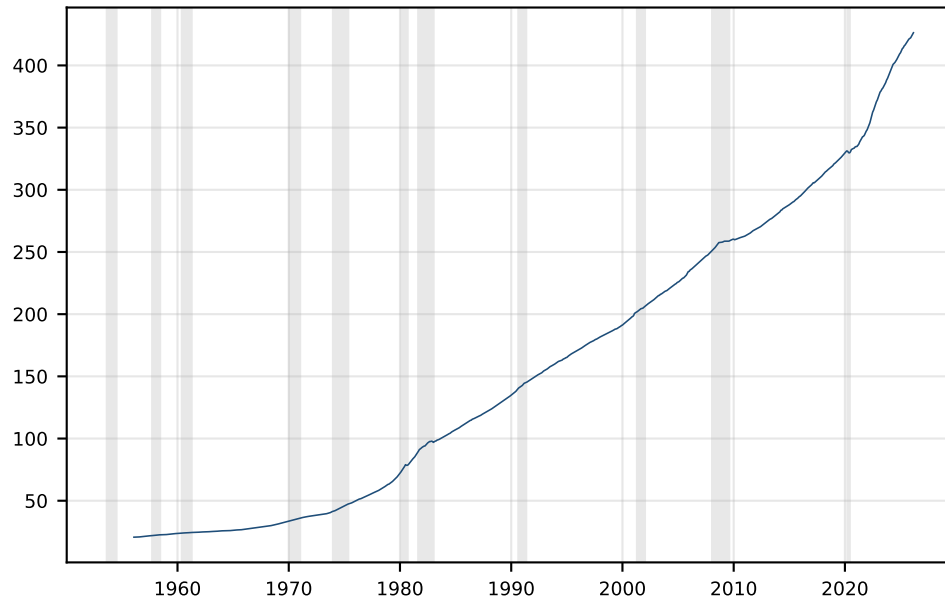
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	210	11.4%	88.6%	-0.0027
Q2	210	7.6%	92.4%	-0.0001
Q3	210	10.0%	90.0%	0.0020
Q4 (highest)	209	20.1%	79.9%	0.0067

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

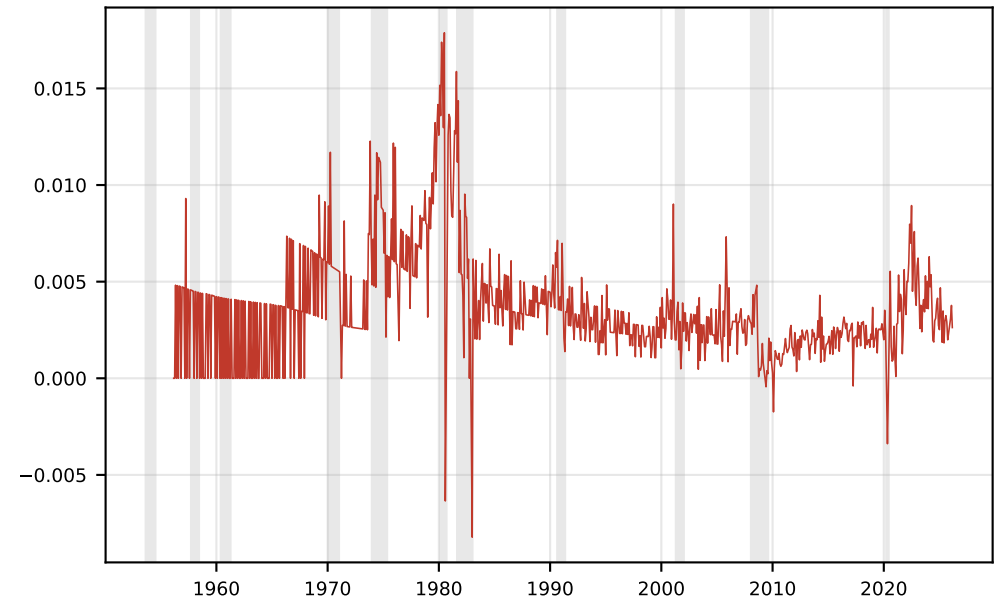
CPI Services

Block: Inflation / Prices / Consumer Inflation | Source: FRED API | Col: CUSR0000SAS | Transform: 5: Log diff

Raw Series



After: 5: Log diff



SUMMARY STATISTICS

N : 1287
Valid : 841
NaN : 446
Start : 1956-01
End : 2026-02
Mean : 157.2456
Std : 114.8316
Min : 20.7000
Max : 426.2770
Skew : 0.465
Kurt : -0.930

STATIONARITY TESTS

ADF stat : -3.401
ADF p : 0.0109
ADF lags : 8
ADF : Stationary
KPSS stat : 0.884
KPSS p : 0.0100
KPSS : Non-stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

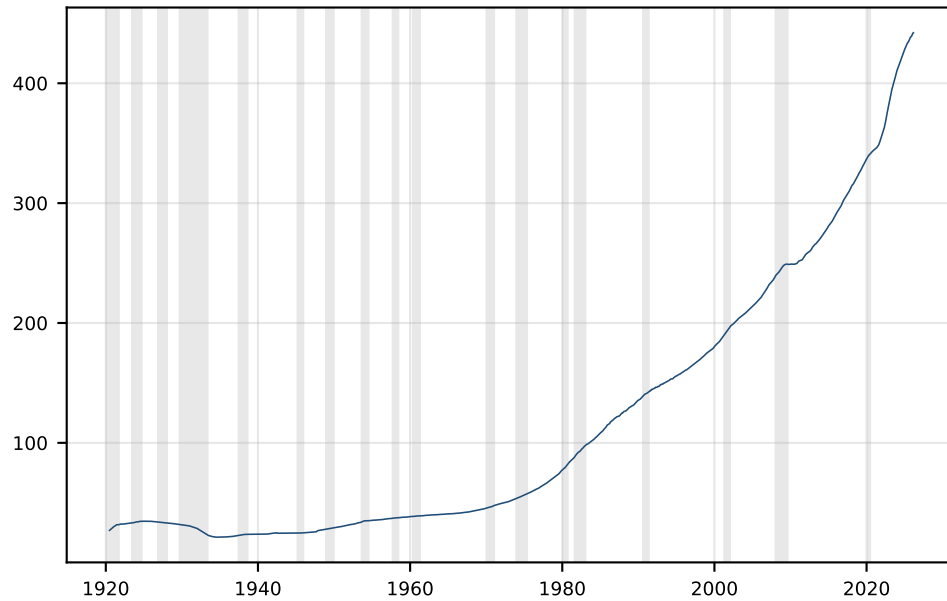
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	210	13.3%	86.7%	0.0008
Q2	210	3.3%	96.7%	0.0026
Q3	209	10.5%	89.5%	0.0038
Q4 (highest)	210	21.9%	78.1%	0.0073

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

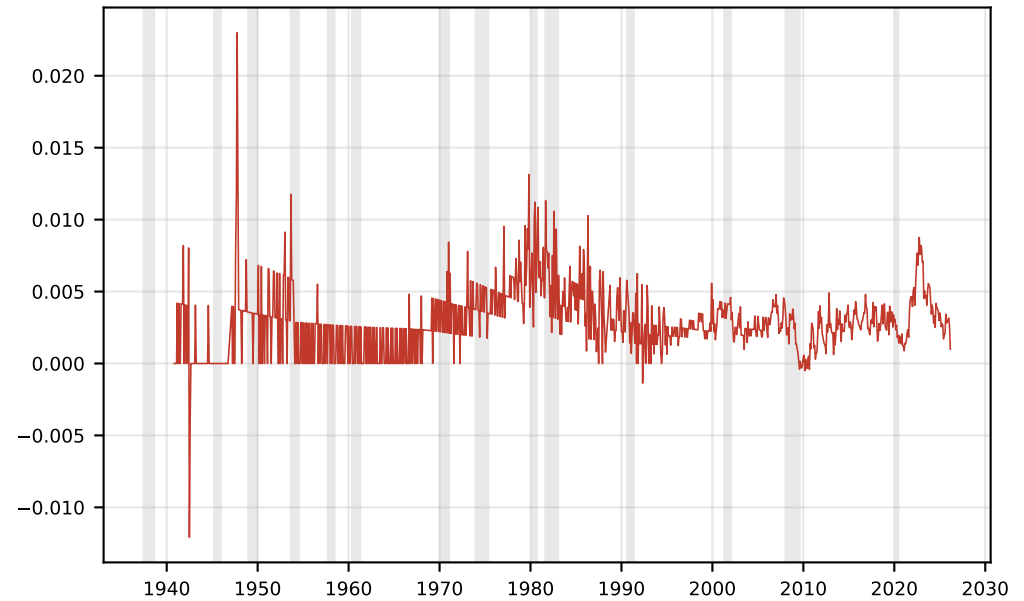
CPI Rent of Primary Residence

Block: Inflation / Prices / Consumer Inflation | Source: FRED API | Col: CUUR0000SEHA | Transform: 5: Log diff

Raw Series



After: 5: Log diff



SUMMARY STATISTICS

N : 1287
Valid : 1066
NaN : 221
Start : 1920-06
End : 2026-02
Mean : 131.6038
Std : 111.8236
Min : 21.3000
Max : 442.1570
Skew : 0.953
Kurt : -0.136

STATIONARITY TESTS

ADF stat : -3.491
ADF p : 0.0082
ADF lags : 11
ADF : Stationary
KPSS stat : 0.438
KPSS p : 0.0607
KPSS : Stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

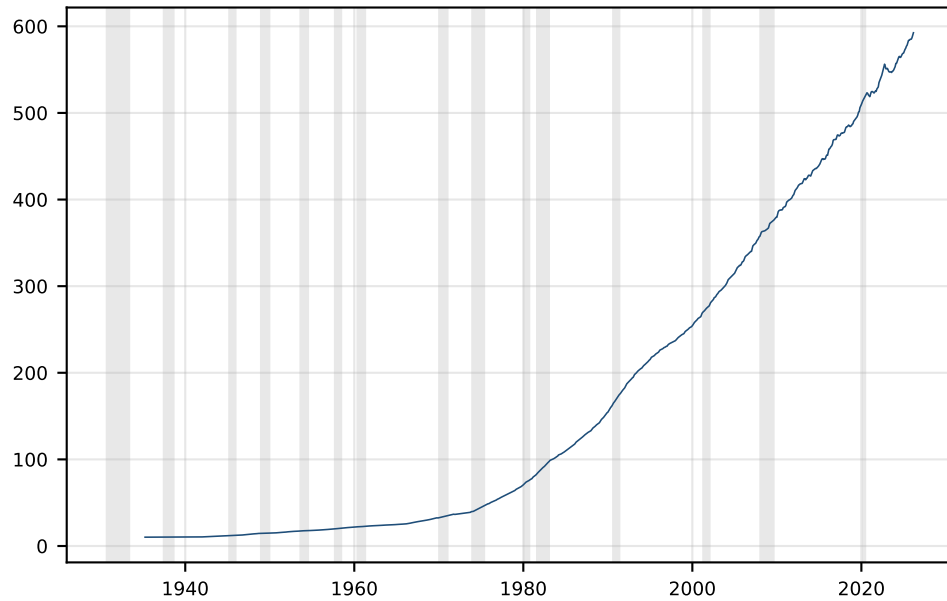
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	249	9.2%	90.8%	0.0003
Q2	249	9.6%	90.4%	0.0023
Q3	249	14.5%	85.5%	0.0033
Q4 (highest)	249	16.5%	83.5%	0.0057

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

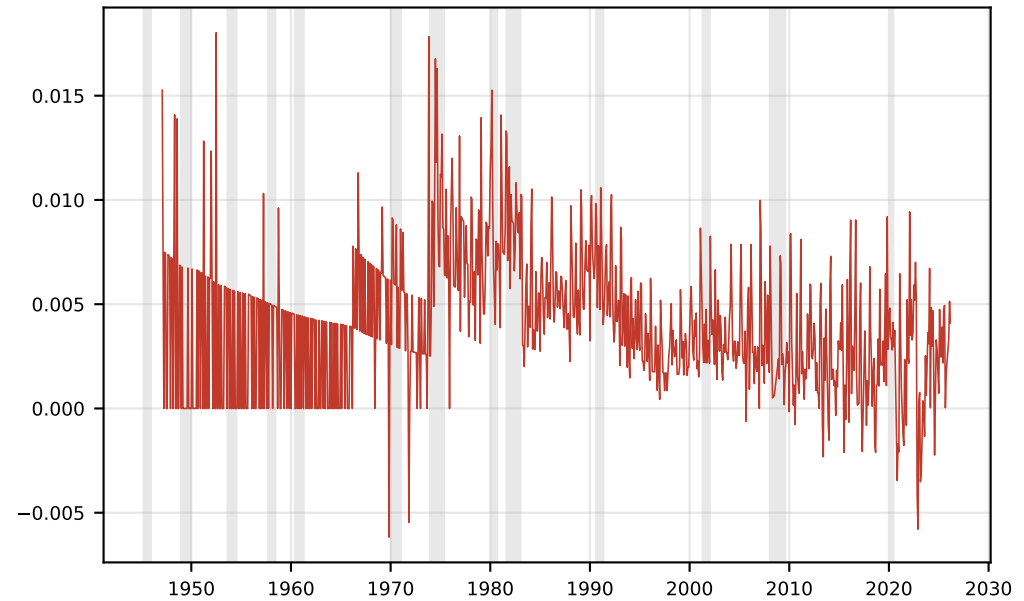
CPI Medical Care

Block: Inflation / Prices / Consumer Inflation | Source: FRED API | Col: CPIMEDNS | Transform: 5: Log diff

Raw Series



After: 5: Log diff



SUMMARY STATISTICS

N : 1287
Valid : 997
NaN : 290
Start : 1935-03
End : 2026-02
Mean : 182.6349
Std : 179.6668
Min : 10.2000
Max : 592.5930
Skew : 0.771
Kurt : -0.796

STATIONARITY TESTS

ADF stat : -3.314
ADF p : 0.0142
ADF lags : 11
ADF : Stationary
KPSS stat : 1.085
KPSS p : 0.0100
KPSS : Non-stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

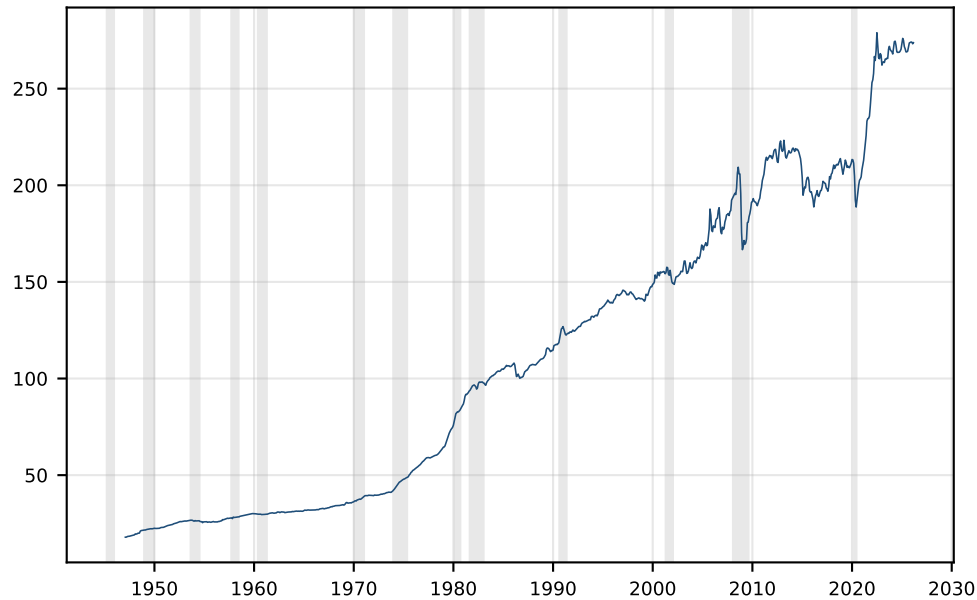
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	237	12.2%	87.8%	0.0001
Q2	237	8.0%	92.0%	0.0028
Q3	237	12.2%	87.8%	0.0049
Q4 (highest)	237	19.8%	80.2%	0.0083

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

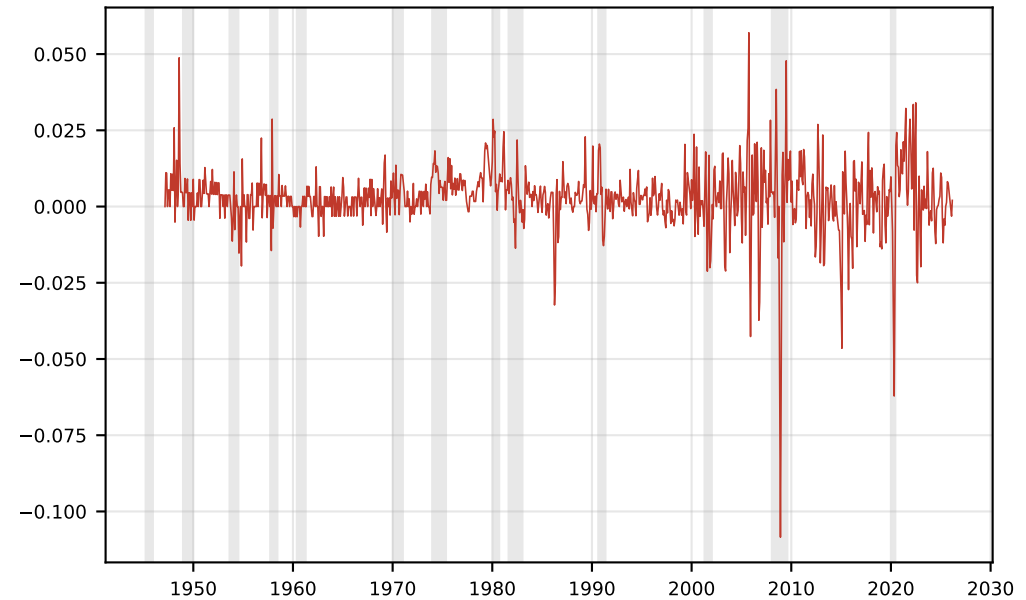
CPI Transportation

Block: Inflation / Prices / Consumer Inflation | Source: FRED API | Col: CPITRNSL | Transform: 5: Log diff

Raw Series



After: 5: Log diff



SUMMARY STATISTICS

N : 1287
Valid : 949
NaN : 338
Start : 1947-01
End : 2026-02
Mean : 111.3328
Std : 76.3289
Min : 17.9000
Max : 278.9320
Skew : 0.408
Kurt : -1.061

STATIONARITY TESTS

ADF stat : -8.203
ADF p : 0.0000
ADF lags : 11
ADF : Stationary
KPSS stat : 0.290
KPSS p : 0.1000
KPSS : Stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

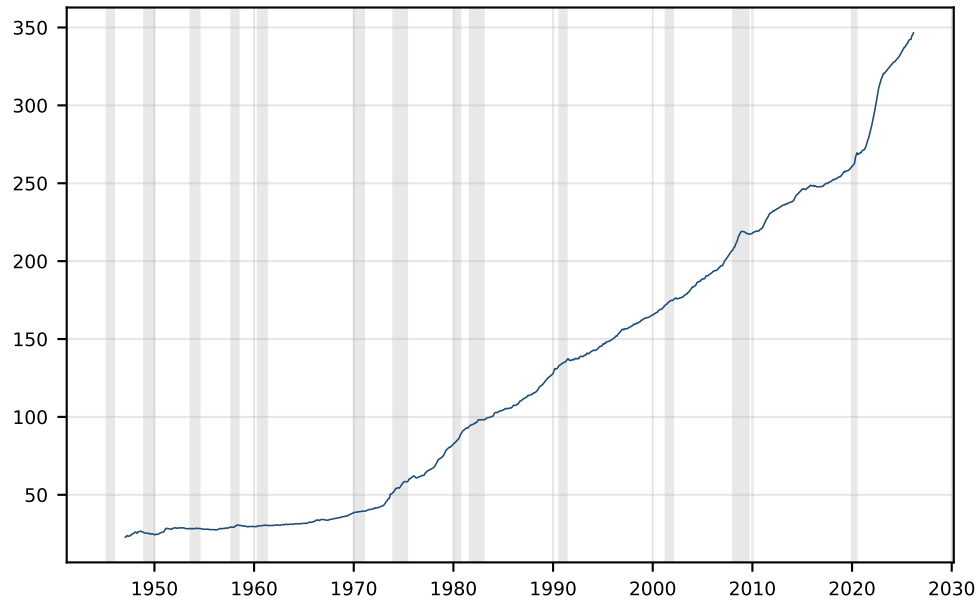
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	237	13.9%	86.1%	-0.0081
Q2	237	9.3%	90.7%	0.0011
Q3	236	13.1%	86.9%	0.0047
Q4 (highest)	237	16.0%	84.0%	0.0138

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

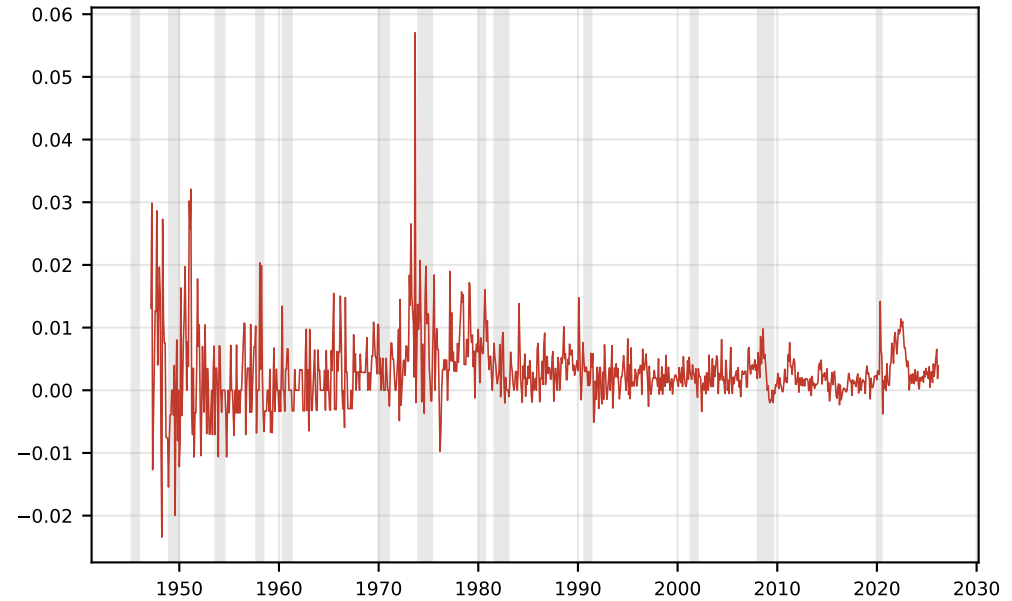
CPI Food

Block: Inflation / Prices / Consumer Inflation | Source: FRED API | Col: CPIUFDSL | Transform: 5: Log diff

Raw Series



After: 5: Log diff



SUMMARY STATISTICS

N : 1287
Valid : 949
NaN : 338
Start : 1947-01
End : 2026-02
Mean : 126.0787
Std : 91.4052
Min : 22.8000
Max : 346.6220
Skew : 0.561
Kurt : -0.829

STATIONARITY TESTS

ADF stat : -7.055
ADF p : 0.0000
ADF lags : 11
ADF : Stationary
KPSS stat : 0.282
KPSS p : 0.1000
KPSS : Stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

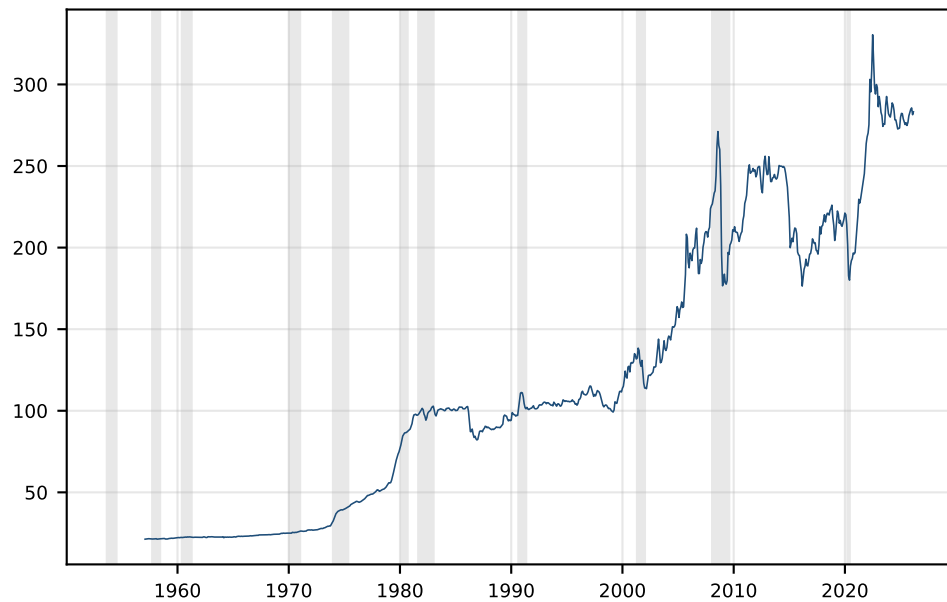
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	263	16.3%	83.7%	-0.0022
Q2	211	8.5%	91.5%	0.0014
Q3	236	11.4%	88.6%	0.0033
Q4 (highest)	237	15.2%	84.8%	0.0094

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

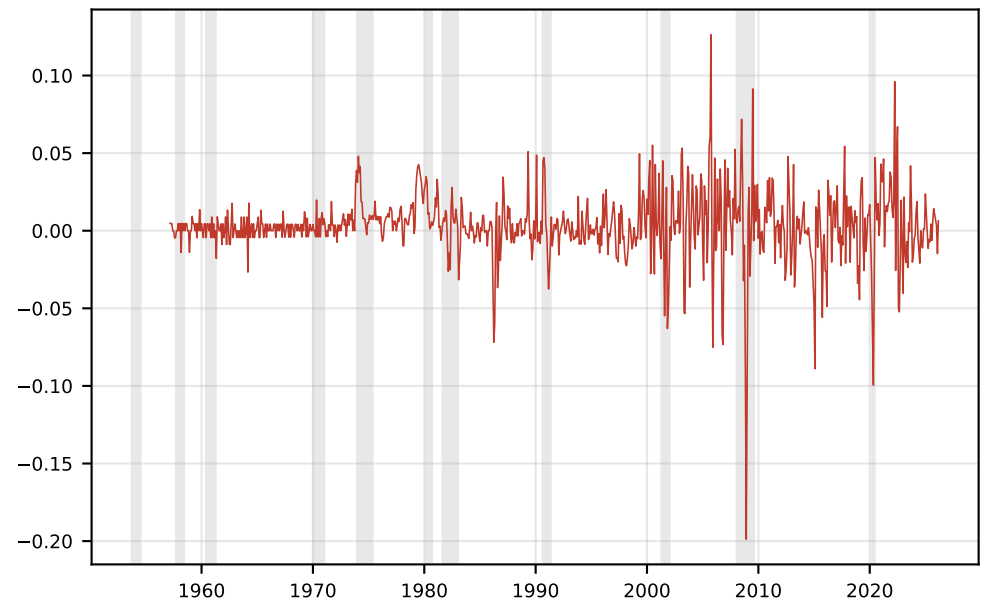
CPI Energy

Block: Inflation / Prices / Consumer Inflation | Source: FRED API | Col: CPIENGSL | Transform: 5: Log diff

Raw Series



After: 5: Log diff



SUMMARY STATISTICS

N : 1287
Valid : 829
NaN : 458
Start : 1957-01
End : 2026-02
Mean : 118.5845
Std : 83.1106
Min : 21.3000
Max : 330.4560
Skew : 0.529
Kurt : -0.906

STATIONARITY TESTS

ADF stat : -8.382
ADF p : 0.0000
ADF lags : 11
ADF : Stationary
KPSS stat : 0.113
KPSS p : 0.1000
KPSS : Stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

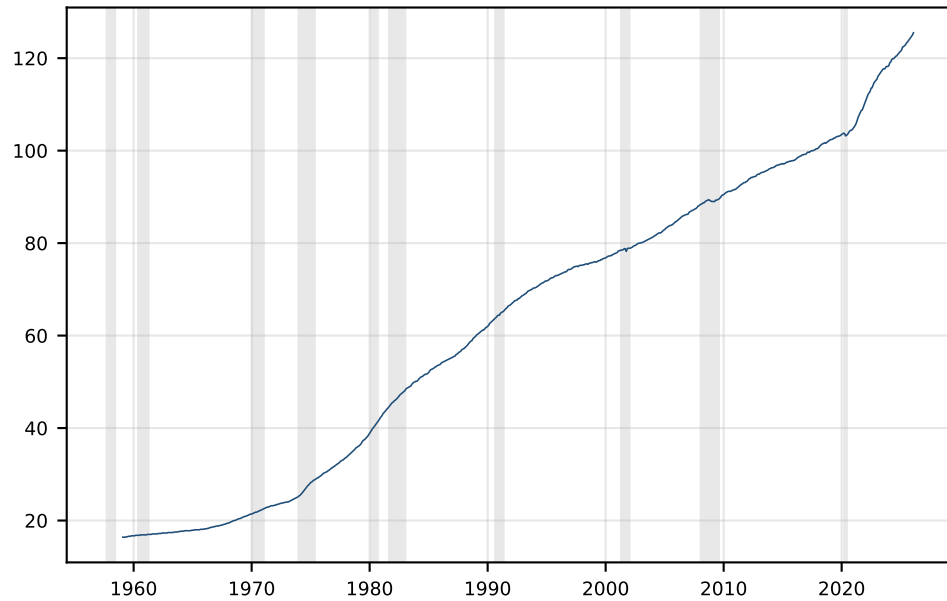
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	207	12.6%	87.4%	-0.0195
Q2	207	7.7%	92.3%	-0.0006
Q3	206	15.0%	85.0%	0.0061
Q4 (highest)	207	14.5%	85.5%	0.0265

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

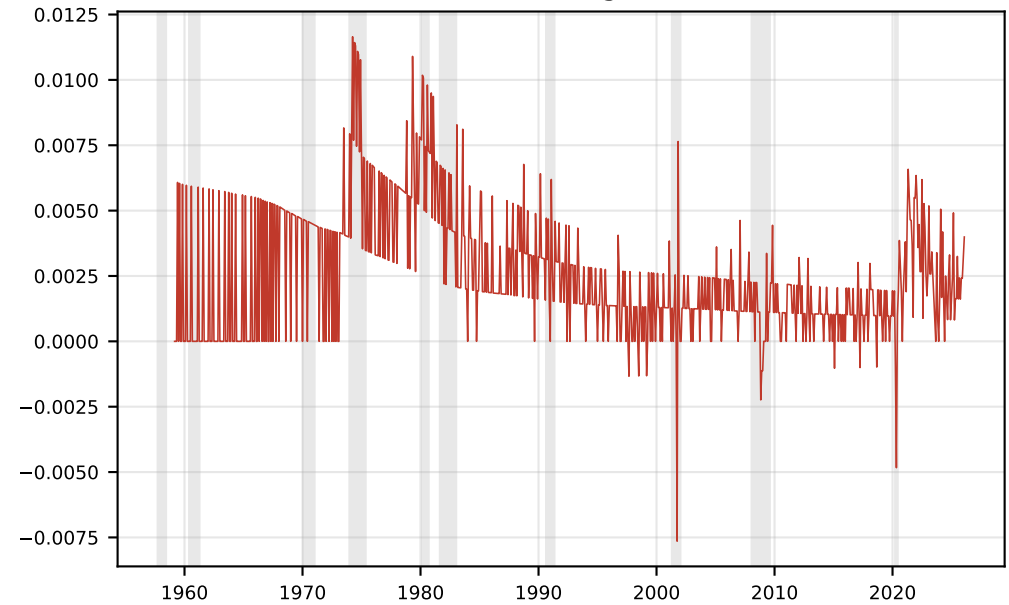
PCE Less Food, Energy, Housing

Block: Inflation / Prices / Consumer Inflation | Source: FRED API | Col: IA001176M | Transform: 5: Log diff

Raw Series



After: 5: Log diff



SUMMARY STATISTICS

N : 1287
Valid : 805
NaN : 482
Start : 1959-01
End : 2026-01
Mean : 62.8353
Std : 32.1819
Min : 16.4000
Max : 125.5000
Skew : -0.013
Kurt : -1.250

STATIONARITY TESTS

ADF stat : -3.117
ADF p : 0.0253
ADF lags : 6
ADF : Stationary
KPSS stat : 1.240
KPSS p : 0.0100
KPSS : Non-stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

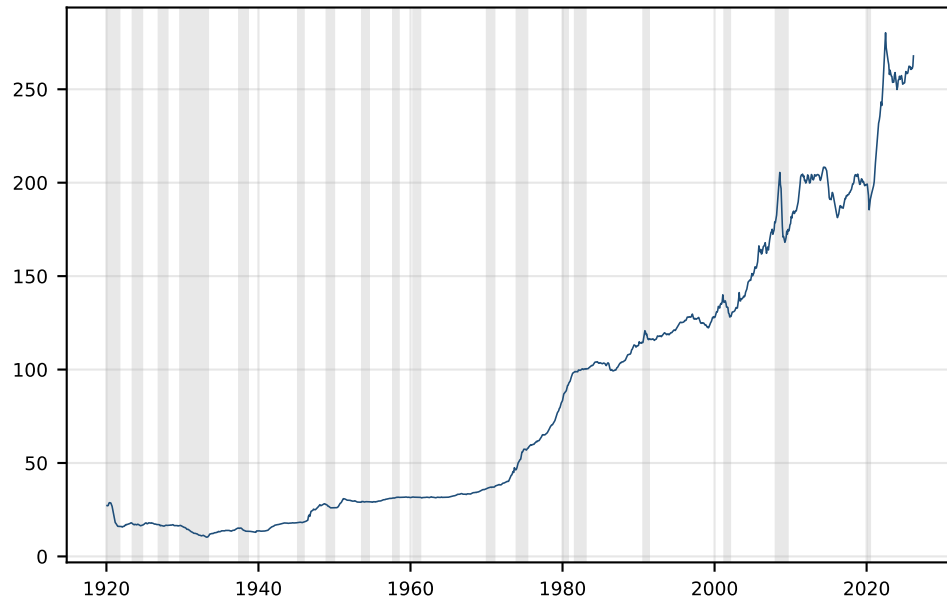
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	201	11.4%	88.6%	0.0000
Q2	202	5.4%	94.6%	0.0014
Q3	200	6.0%	94.0%	0.0029
Q4 (highest)	201	24.4%	75.6%	0.0059

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

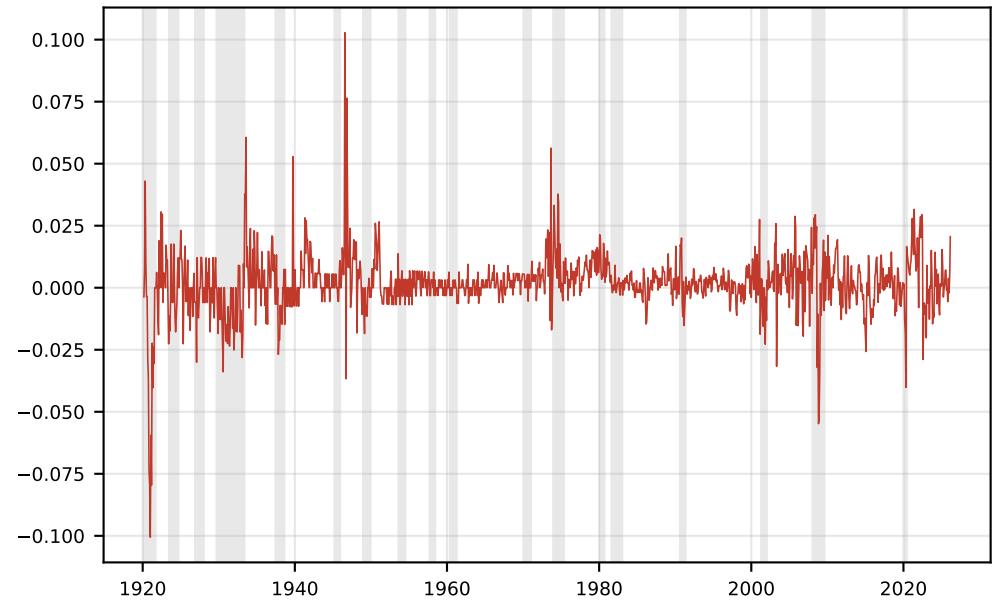
PPI: All Commodities

Block: Inflation / Prices / Producer inflation | Source: FRED API | Col: PPIACO | Transform: 5: Log diff

Raw Series



After: 5: Log diff



SUMMARY STATISTICS

N : 1287
Valid : 1274
NaN : 13
Start : 1920-01
End : 2026-02
Mean : 84.1194
Std : 72.6981
Min : 10.3000
Max : 280.2510
Skew : 0.825
Kurt : -0.512

STATIONARITY TESTS

ADF stat : -10.046
ADF p : 0.0000
ADF lags : 12
ADF : Stationary
KPSS stat : 0.424
KPSS p : 0.0669
KPSS : Stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

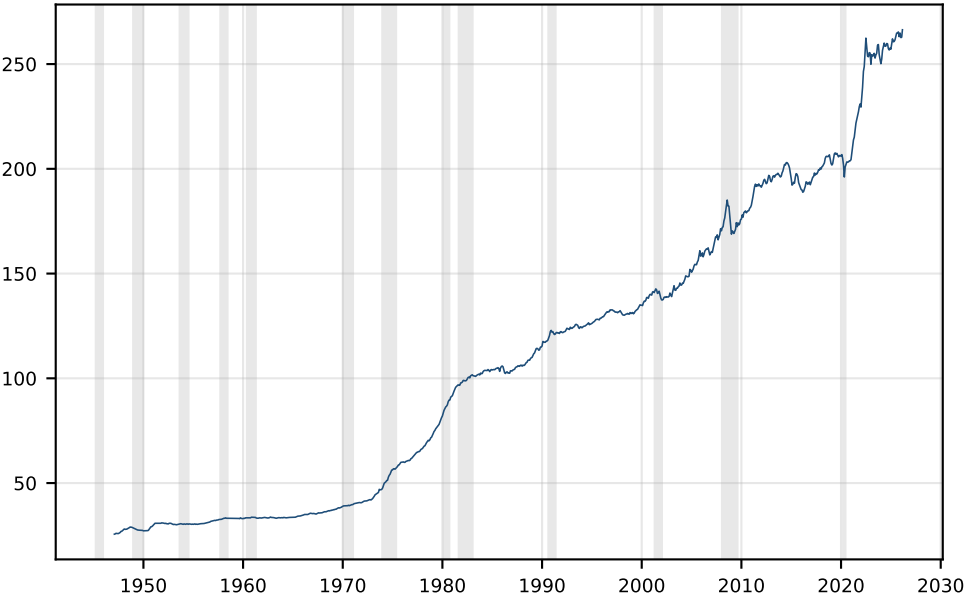
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	319	34.2%	65.8%	-0.0110
Q2	318	15.4%	84.6%	-0.0002
Q3	318	11.0%	89.0%	0.0040
Q4 (highest)	318	12.6%	87.4%	0.0145

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

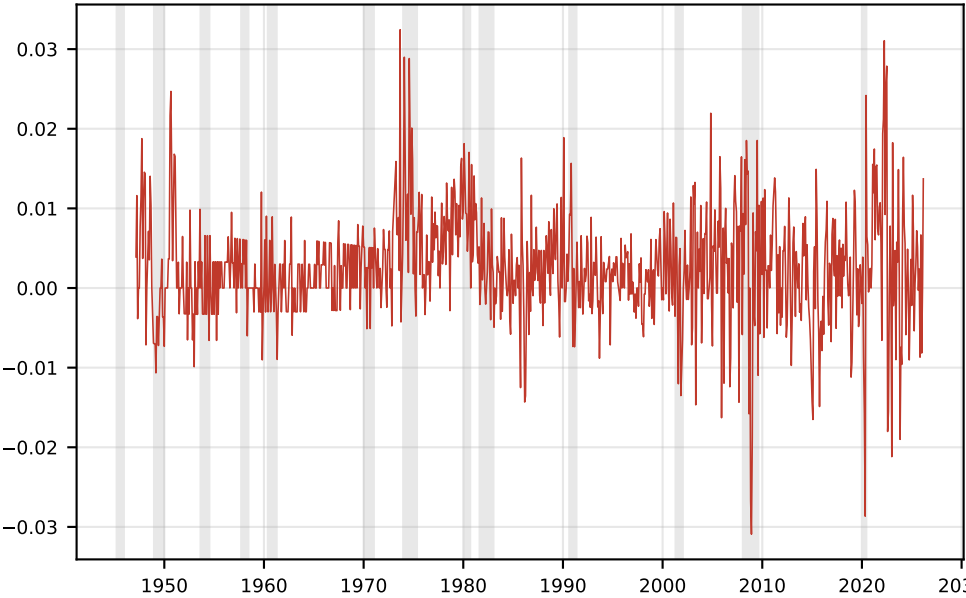
PPI Finished Goods

Block: Inflation / Prices / Producer inflation | Source: FRED API | Col: WPUFD49207 | Transform: 5: Log diff

Raw Series



After: 5: Log diff



SUMMARY STATISTICS

N : 1287
Valid : 950
NaN : 337
Start : 1947-01
End : 2026-02
Mean : 108.3277
Std : 69.4384
Min : 25.6000
Max : 266.3920
Skew : 0.460
Kurt : -0.899

STATIONARITY TESTS

ADF stat : -6.142
ADF p : 0.0000
ADF lags : 12
ADF : Stationary
KPSS stat : 0.215
KPSS p : 0.1000
KPSS : Stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

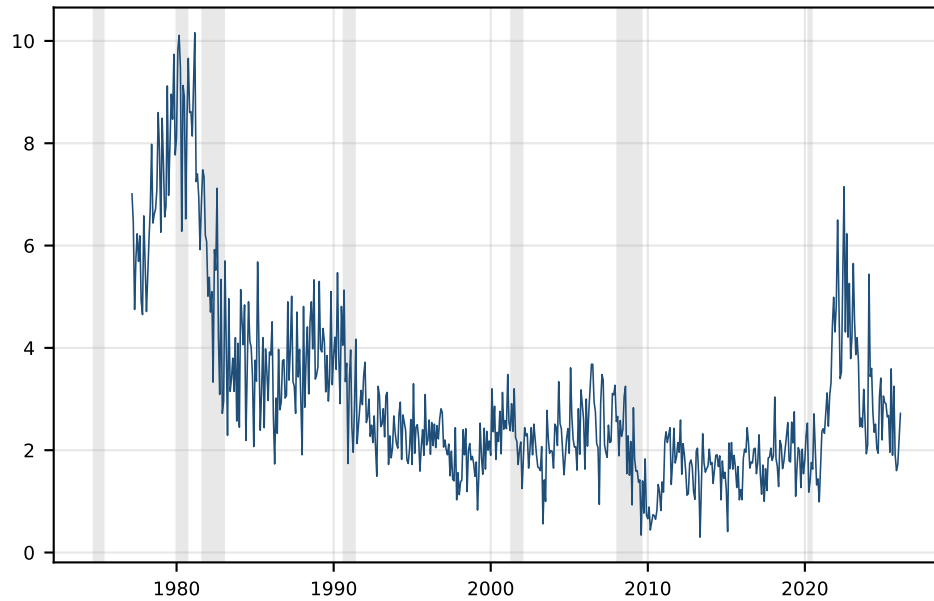
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	238	16.0%	84.0%	-0.0054
Q2	237	9.3%	90.7%	0.0005
Q3	237	10.1%	89.9%	0.0039
Q4 (highest)	237	16.9%	83.1%	0.0109

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

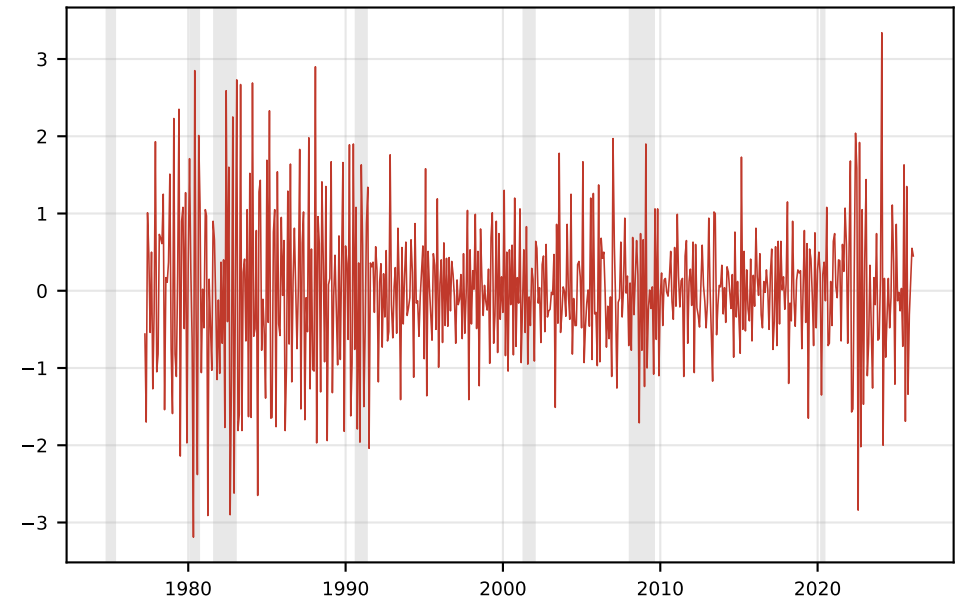
Trimmed Mean PCE Inflation Rate

Block: Inflation / Prices / Alternative inflation | Source: FRED API | Col: PCETRM1M158SFRBDAL | Transform: 2: First diff

Raw Series



After: 2: First diff



SUMMARY STATISTICS

N : 1287
Valid : 588
NaN : 699
Start : 1977-02
End : 2026-01
Mean : 3.0311
Std : 1.8389
Min : 0.3000
Max : 10.1600
Skew : 1.641
Kurt : 2.592

STATIONARITY TESTS

ADF stat : -16.625
ADF p : 0.0000
ADF lags : 4
ADF : Stationary
KPSS stat : 0.334
KPSS p : 0.1000
KPSS : Stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

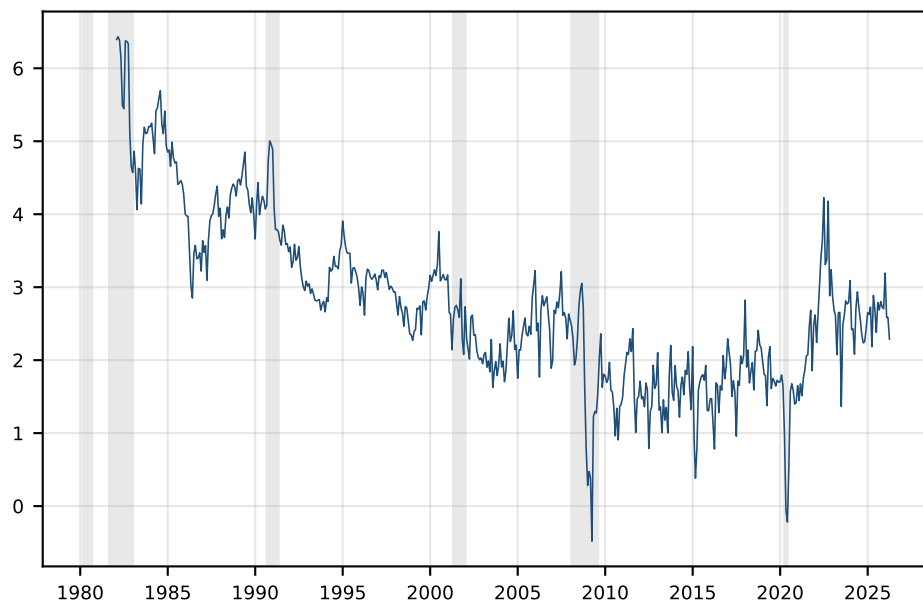
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	148	13.5%	86.5%	-1.1798
Q2	147	8.2%	91.8%	-0.2696
Q3	145	8.3%	91.7%	0.2605
Q4 (highest)	147	9.5%	90.5%	1.1713

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

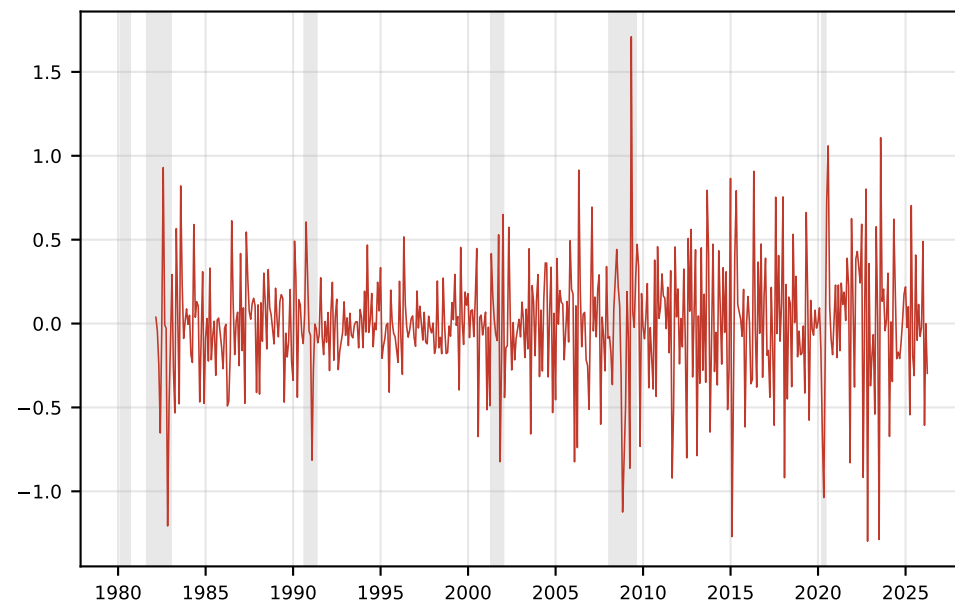
1-Year Expected Inflation

Block: Inflation / Prices / Inflation Expectations | Source: FRED API | Col: EXPINF1YR | Transform: 2: First diff

Raw Series



After: 2: First diff



SUMMARY STATISTICS

N : 1287
Valid : 531
NaN : 756
Start : 1982-01
End : 2026-03
Mean : 2.7780
Std : 1.1619
Min : -0.4807
Max : 6.4321
Skew : 0.599
Kurt : 0.422

STATIONARITY TESTS

ADF stat : -13.306
ADF p : 0.0000
ADF lags : 5
ADF : Stationary
KPSS stat : 0.189
KPSS p : 0.1000
KPSS : Stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

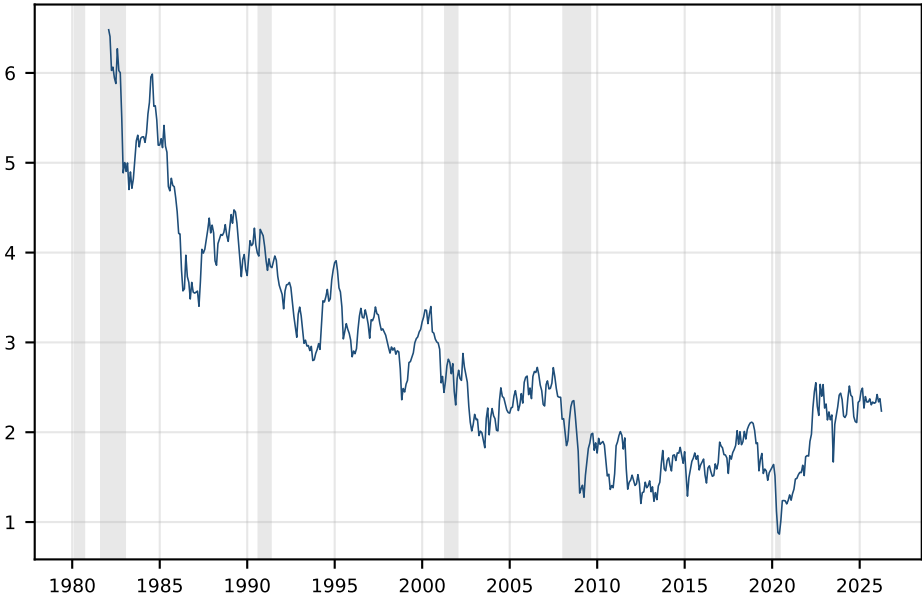
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	133	12.8%	87.2%	-0.4364
Q2	132	9.1%	90.9%	-0.0762
Q3	132	5.3%	94.7%	0.0744
Q4 (highest)	132	7.6%	92.4%	0.4127

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

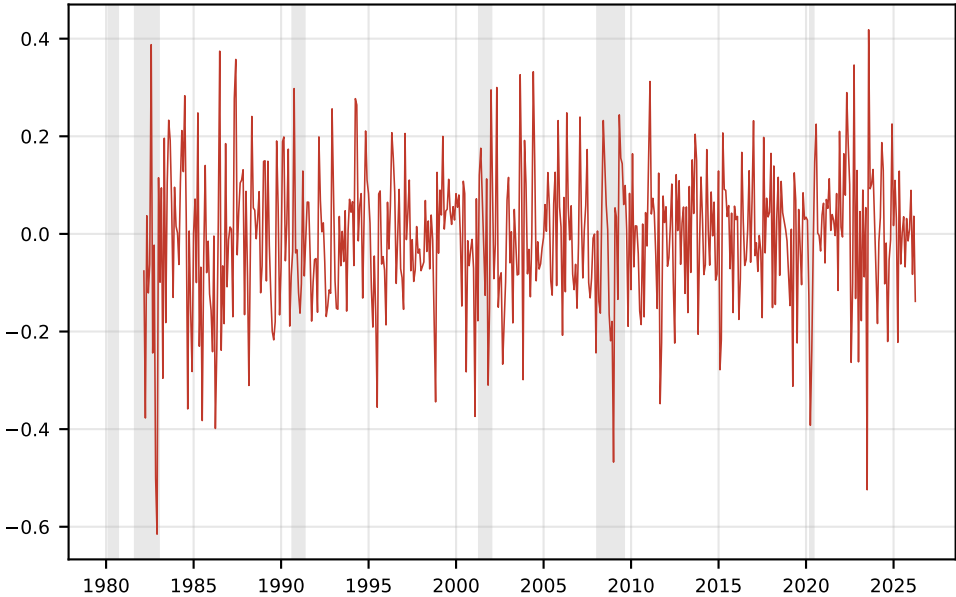
5-Year Expected Inflation

Block: Inflation / Prices / Inflation Expectations | Source: FRED API | Col: EXPINF5YR | Transform: 2: First diff

Raw Series



After: 2: First diff



SUMMARY STATISTICS

N : 1287
Valid : 531
NaN : 756
Start : 1982-01
End : 2026-03
Mean : 2.7824
Std : 1.1641
Min : 0.8649
Max : 6.4809
Skew : 0.894
Kurt : 0.247

STATIONARITY TESTS

ADF stat : -21.969
ADF p : 0.0000
ADF lags : 0
ADF : Stationary
KPSS stat : 0.282
KPSS p : 0.1000
KPSS : Stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

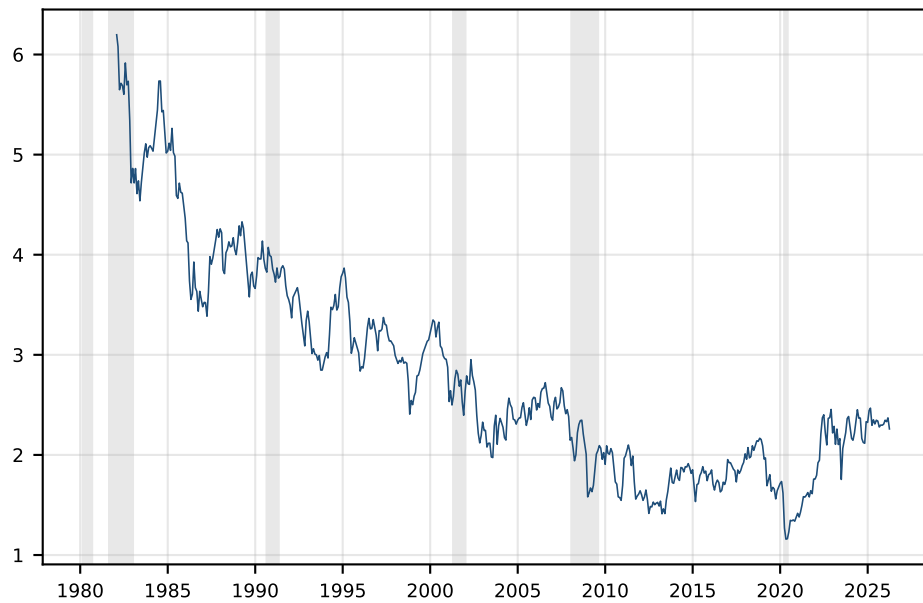
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	133	15.0%	85.0%	-0.1907
Q2	132	5.3%	94.7%	-0.0448
Q3	132	6.1%	93.9%	0.0404
Q4 (highest)	132	8.3%	91.7%	0.1654

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

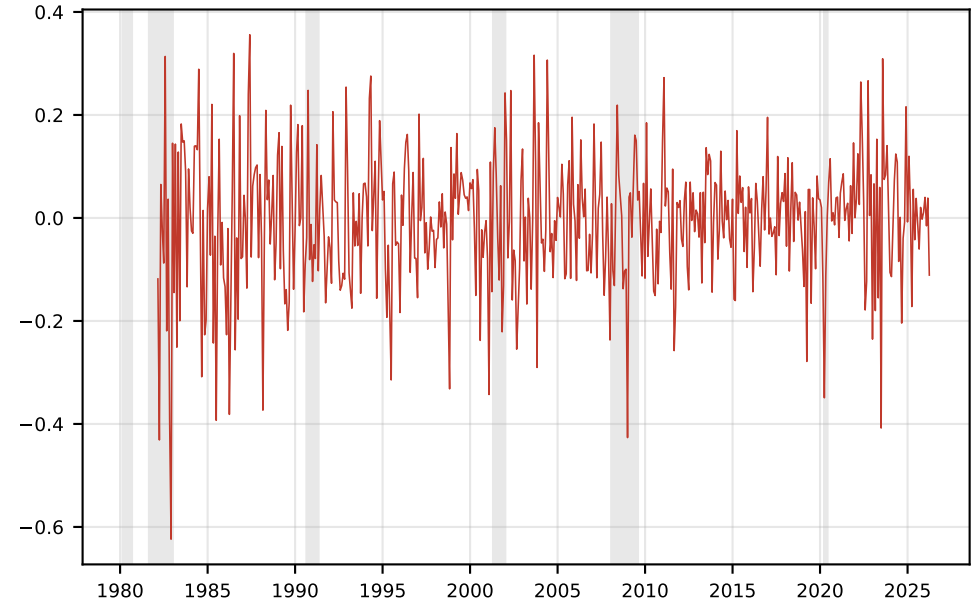
10-Year Expected Inflation

Block: Inflation / Prices / Inflation Expectations | Source: FRED API | Col: EXPINF10YR | Transform: 2: First diff

Raw Series



After: 2: First diff



SUMMARY STATISTICS

N : 1287
Valid : 531
NaN : 756
Start : 1982-01
End : 2026-03
Mean : 2.7973
Std : 1.0644
Min : 1.1583
Max : 6.1976
Skew : 0.901
Kurt : 0.211

STATIONARITY TESTS

ADF stat : -11.949
ADF p : 0.0000
ADF lags : 4
ADF : Stationary
KPSS stat : 0.282
KPSS p : 0.1000
KPSS : Stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

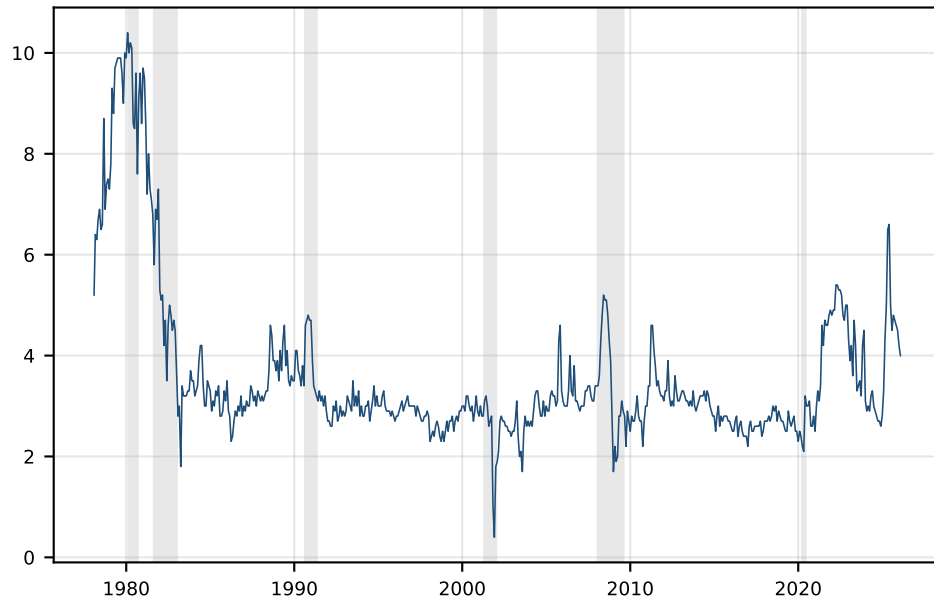
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	133	14.3%	85.7%	-0.1693
Q2	132	5.3%	94.7%	-0.0354
Q3	132	7.6%	92.4%	0.0344
Q4 (highest)	132	7.6%	92.4%	0.1425

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

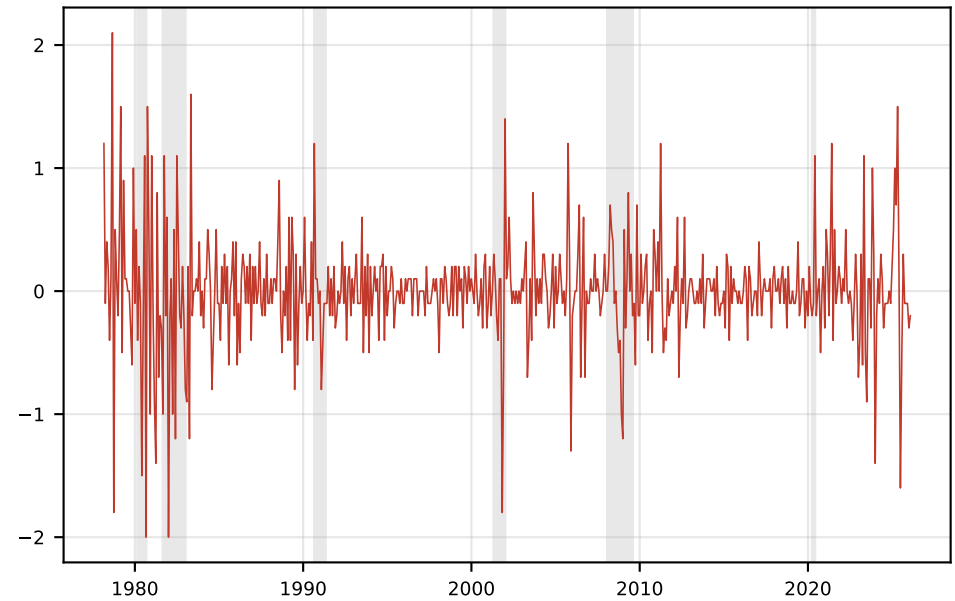
University of Michigan: Inflation Expectation

Block: Inflation / Prices / Inflation Expectations | Source: FRED API | Col: MICH | Transform: 2: First diff

Raw Series



After: 2: First diff



SUMMARY STATISTICS

```
N          : 1287
Valid       : 577
NaN         : MICH    710
MICH       709
dtype: int64
Start      : 1978-01
End        : 2026-01
Mean       : 3.6069
Std        : 1.6145
Min        : 0.4000
Max        : 10.4000
Skew       : 2.460
Kurt       : 6.095
```

STATIONARITY TESTS

```
ADF stat    : -6.202
ADF p       : 0.0000
ADF lags    : 11
ADF         : Stationary
KPSS stat   : 0.055
KPSS p      : 0.1000
KPSS        : Stationary
```

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	148	16.2%	83.8%	-0.4730
Q2	193	5.2%	94.8%	-0.0539
Q3	98	8.2%	91.8%	0.1153
Q4 (highest)	137	11.7%	88.3%	0.4956

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

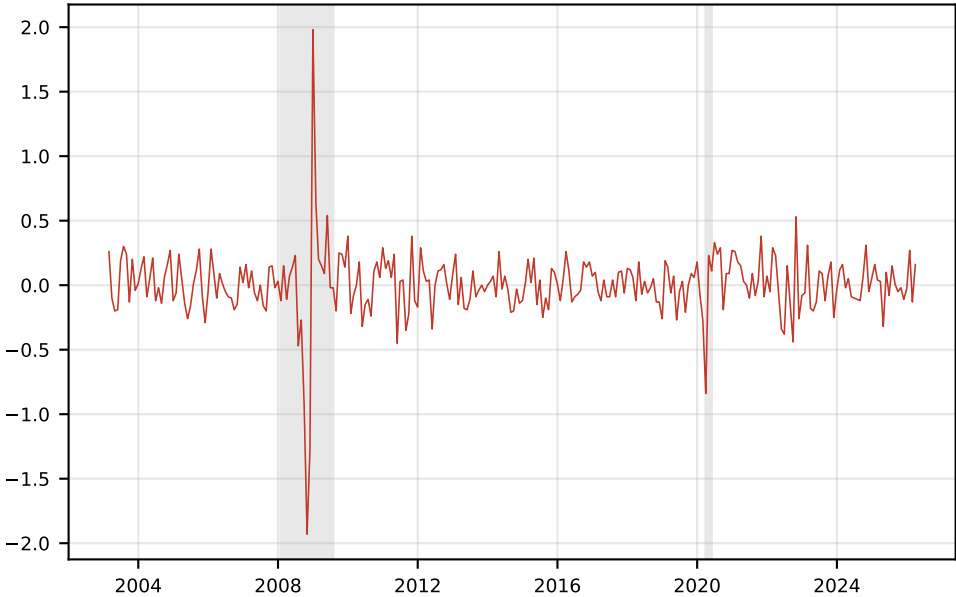
5-Year Breakeven Inflation Rate

Block: Inflation / Prices / Inflation Expectations | Source: FRED API | Col: T5YIE | Transform: 2: First diff

Raw Series



After: 2: First diff



SUMMARY STATISTICS

N : 1287
Valid : 279
NaN : T5YIE 1008
T5YIE 1008
dtype: int64
Start : 2003-01
End : 2026-03
Mean : 1.9565
Std : 0.5978
Min : -2.2400
Max : 3.3400
Skew : -1.924
Kurt : 10.163

STATIONARITY TESTS

ADF stat : -6.472
ADF p : 0.0000
ADF lags : 12
ADF : Stationary
KPSS stat : 0.028
KPSS p : 0.1000
KPSS : Stationary

QUARTILE-BUCKET ANALYSIS

(vs NBER recession/expansion)

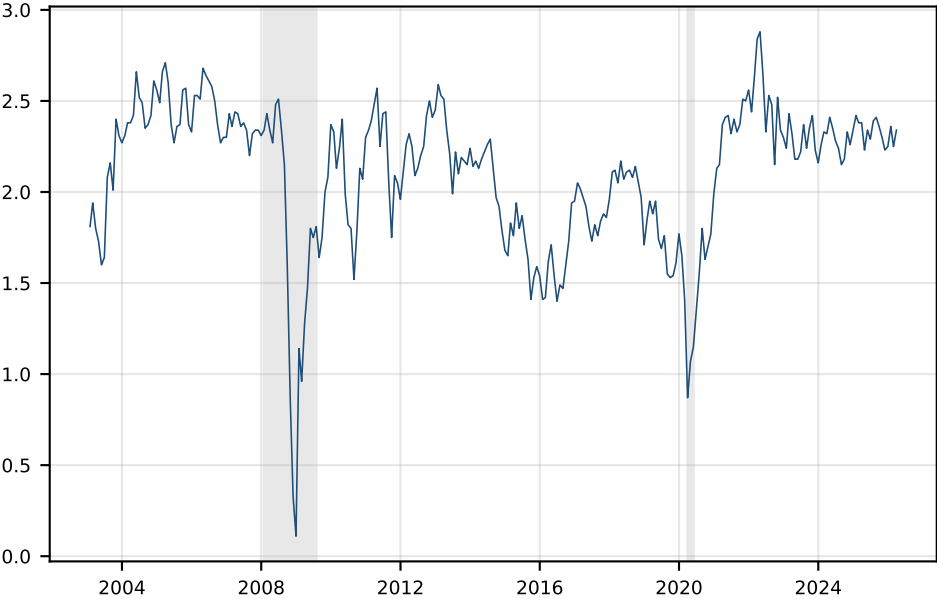
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	71	9.9%	90.1%	-0.2587
Q2	70	2.9%	97.1%	-0.0501
Q3	67	3.0%	97.0%	0.0722
Q4 (highest)	69	13.0%	87.0%	0.2601

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

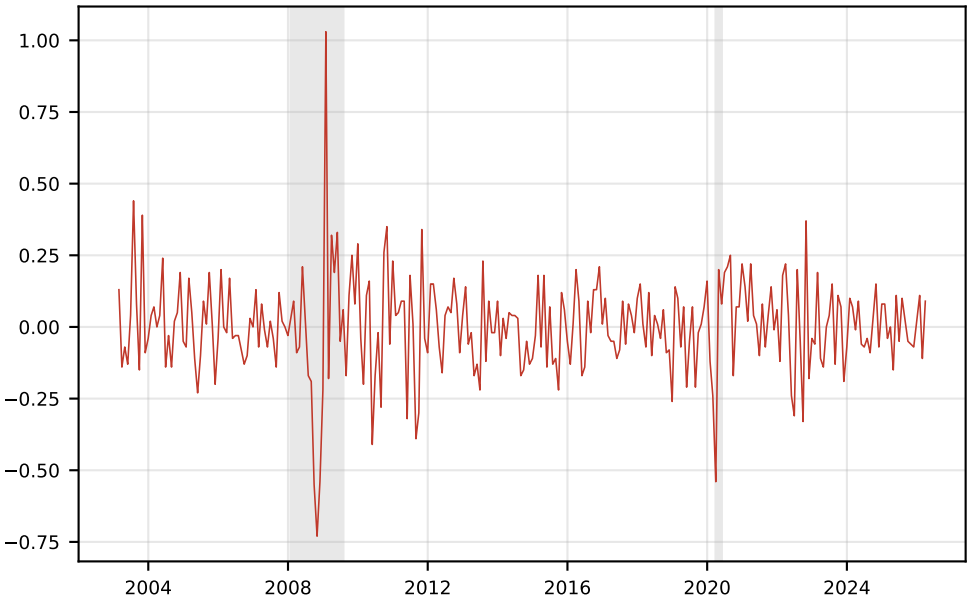
10-Year Breakeven Inflation Rate

Block: Inflation / Prices / Inflation Expectations | Source: FRED API | Col: T10YIE | Transform: 2: First diff

Raw Series



After: 2: First diff



SUMMARY STATISTICS

N : 1287
Valid : 279
NaN : T10YIE 1008
T10YIE 1008
dtype: int64
Start : 2003-01
End : 2026-03
Mean : 2.1057
Std : 0.3961
Min : 0.1100
Max : 2.8800
Skew : -1.373
Kurt : 3.270

STATIONARITY TESTS

ADF stat : -9.179
ADF p : 0.0000
ADF lags : 4
ADF : Stationary
KPSS stat : 0.024
KPSS p : 0.1000
KPSS : Stationary

QUARTILE-BUCKET ANALYSIS

(vs NBER recession/expansion)

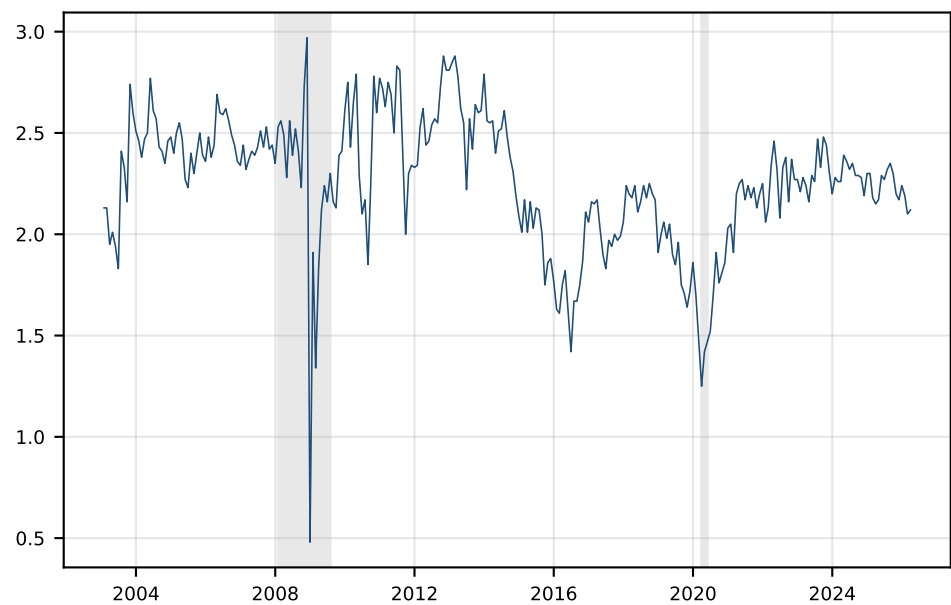
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	71	12.7%	87.3%	-0.1921
Q2	70	2.9%	97.1%	-0.0409
Q3	68	2.9%	97.1%	0.0518
Q4 (highest)	68	10.3%	89.7%	0.1974

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

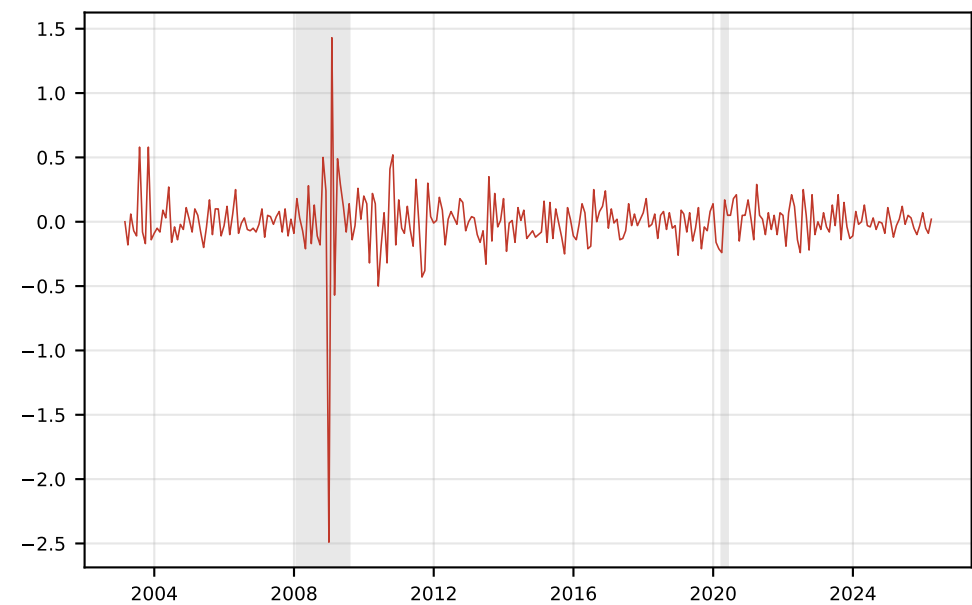
5-Year, 5-Year Forward Inflation Expectation Rate

Block: Inflation / Prices / Inflation Expectations | Source: FRED API | Col: T5YIFR | Transform: 2: First diff

Raw Series



After: 2: First diff



SUMMARY STATISTICS

N : 1287
Valid : 279
NaN : T5YIFR 1008
T5YIFR 1008
dtype: int64
Start : 2003-01
End : 2026-03
Mean : 2.2553
Std : 0.3269
Min : 0.4800
Max : 2.9700
Skew : -0.958
Kurt : 2.828

STATIONARITY TESTS

ADF stat : -8.861
ADF p : 0.0000
ADF lags : 6
ADF : Stationary
KPSS stat : 0.033
KPSS p : 0.1000
KPSS : Stationary

QUARTILE-BUCKET ANALYSIS

(vs NBER recession/expansion)

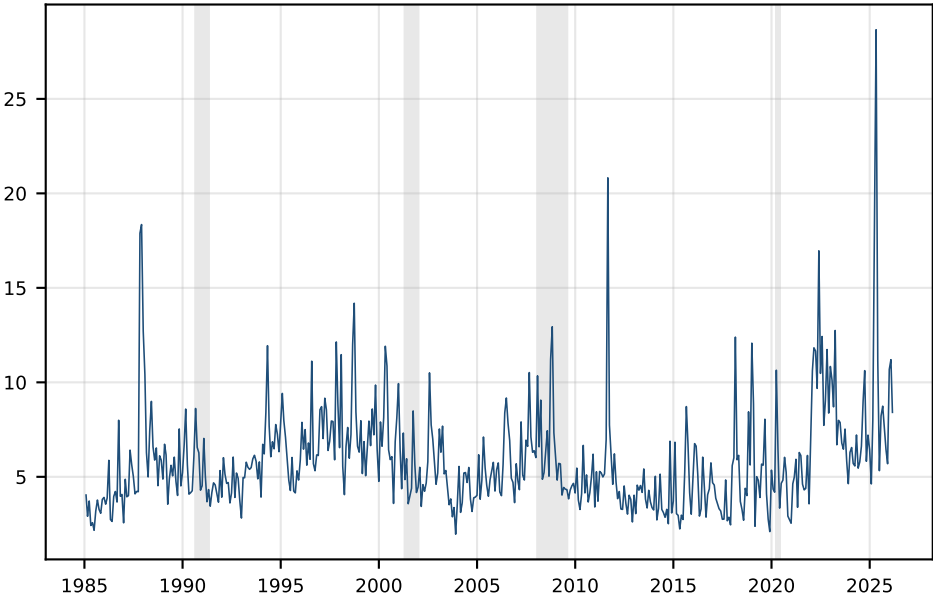
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	71	9.9%	90.1%	-0.2096
Q2	69	2.9%	97.1%	-0.0488
Q3	69	1.4%	98.6%	0.0442
Q4 (highest)	68	14.7%	85.3%	0.2231

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

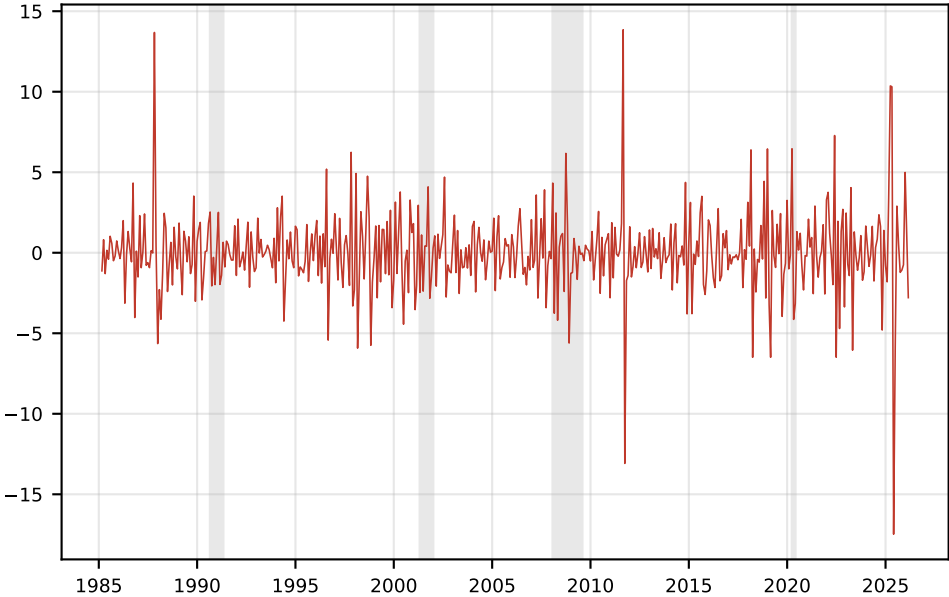
Equity Volatility: Inflation

Block: Inflation / Prices / Inflation Uncertainty | Source: FRED API | Col: EMVMACROINFLATION | Transform: 2: First diff

Raw Series



After: 2: First diff



SUMMARY STATISTICS

N : 1287
Valid : 494
NaN : 793
Start : 1985-01
End : 2026-02
Mean : 5.8710
Std : 2.7556
Min : 1.9653
Max : 28.6591
Skew : 2.601
Kurt : 12.998

STATIONARITY TESTS

ADF stat : -12.654
ADF p : 0.0000
ADF lags : 7
ADF : Stationary
KPSS stat : 0.080
KPSS p : 0.1000
KPSS : Stationary

QUARTILE-BUCKET ANALYSIS
(vs NBER recession/expansion)

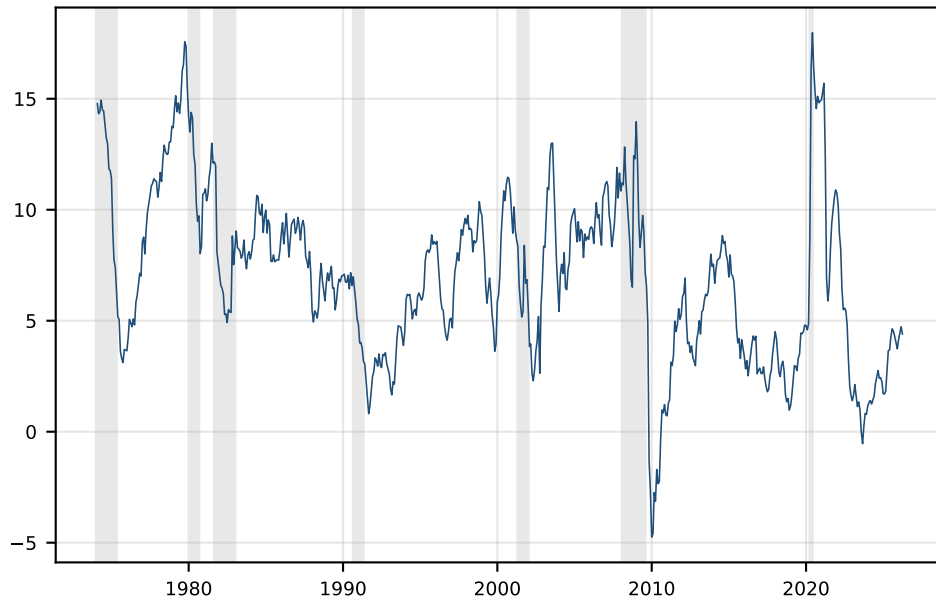
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	124	12.9%	87.1%	-2.7663
Q2	123	0.8%	99.2%	-0.5937
Q3	123	8.9%	91.1%	0.5237
Q4 (highest)	123	6.5%	93.5%	2.8943

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

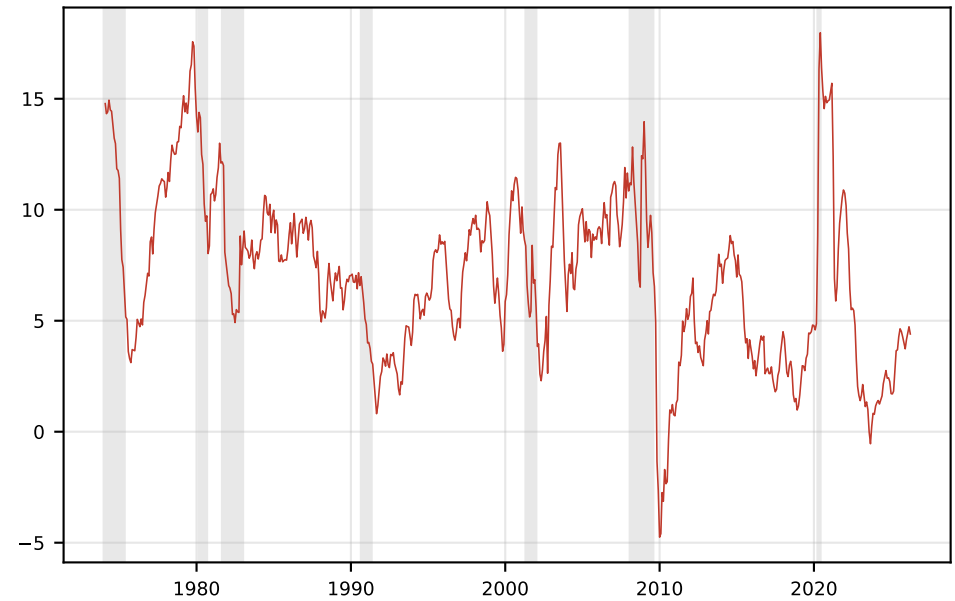
Asset Growth YoY

Block: Selected export / Credit creation / bank channel | Source: FRED API | Col: derived_asset_growth_yoy | Transform: nan

Raw Series



After: nan



SUMMARY STATISTICS

N : 1287
Valid : 627
NaN : 660
Start : 1974-01
End : 2026-03
Mean : 6.9691
Std : 3.7590
Min : -4.7509
Max : 17.9745
Skew : 0.150
Kurt : 0.050

STATIONARITY TESTS

ADF stat : -2.888
ADF p : 0.0468
ADF lags : 12
ADF : Stationary
KPSS stat : 0.849
KPSS p : 0.0100
KPSS : Non-stationary

QUARTILE-BUCKET ANALYSIS

(vs NBER recession/expansion)

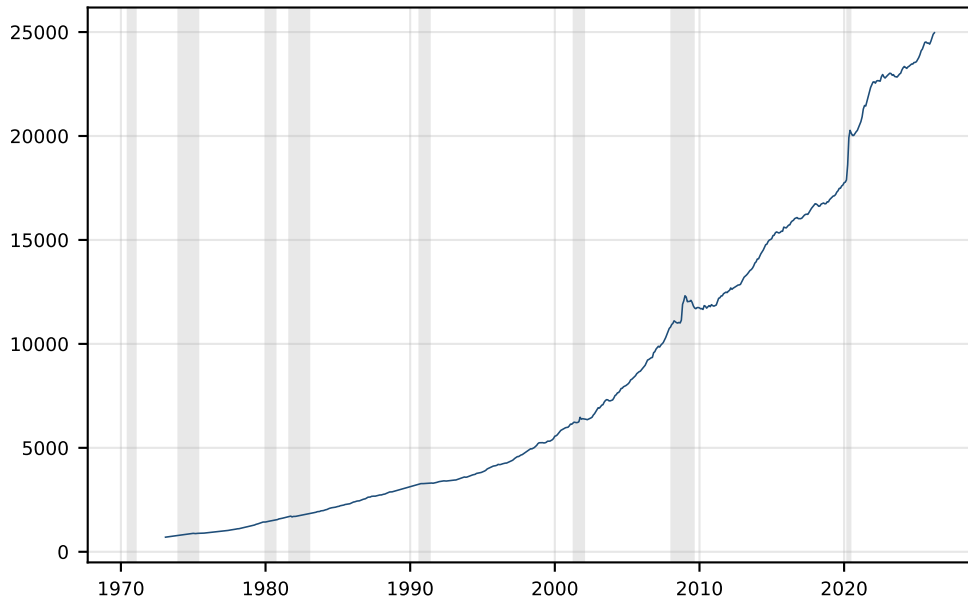
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	157	1.9%	98.1%	2.3047
Q2	156	14.1%	85.9%	5.6278
Q3	156	9.6%	90.4%	8.1962
Q4 (highest)	157	21.0%	79.0%	11.7634

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

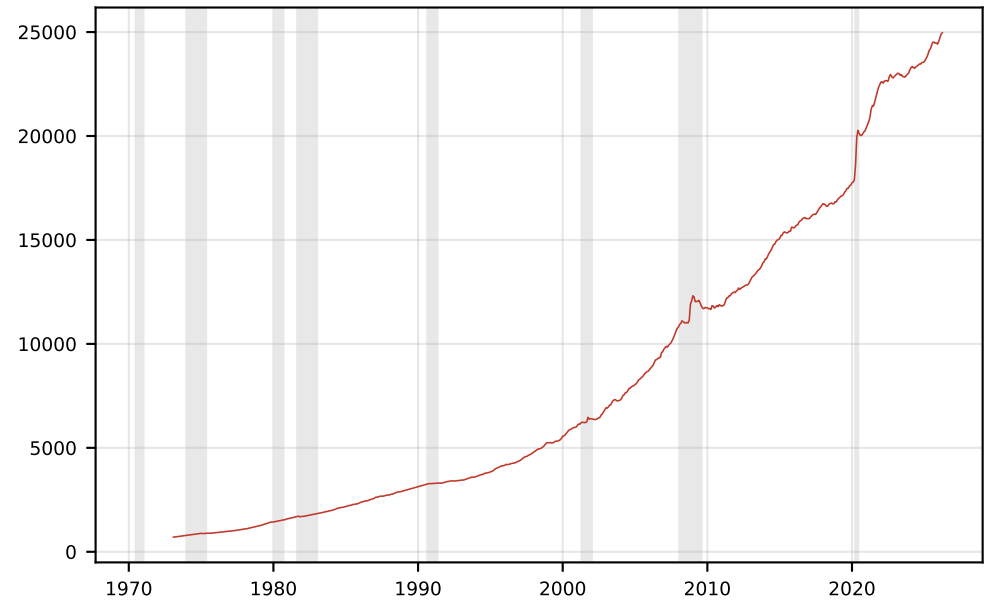
Assets, All Commercial Banks

Block: Selected export / Credit creation / bank channel | Source: FRED API | Col: derived_assets_all_commercial_banks | Transform: nan

Raw Series



After: nan



SUMMARY STATISTICS

N : 1287
Valid : 639
NaN : 648
Start : 1973-01
End : 2026-03
Mean : 8304.9049
Std : 7122.0113
Min : 702.8384
Max : 24967.0010
Skew : 0.842
Kurt : -0.523

STATIONARITY TESTS

ADF stat : 4.876
ADF p : 1.0000
ADF lags : 2
ADF : Non-stationary
KPSS stat : 3.533
KPSS p : 0.0100
KPSS : Non-stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

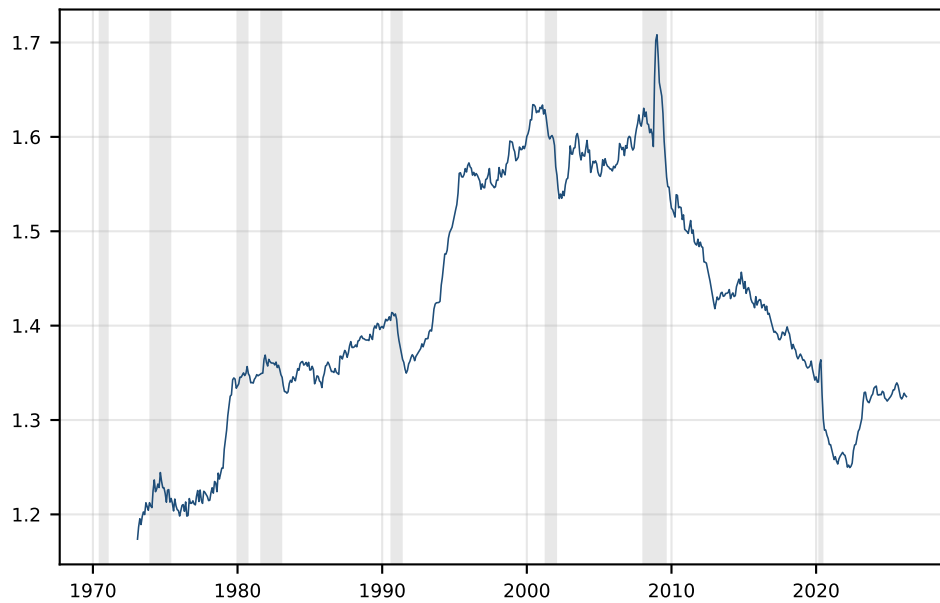
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	160	23.8%	76.2%	1429.4184
Q2	159	5.0%	95.0%	3660.5615
Q3	159	16.4%	83.6%	9182.1927
Q4 (highest)	160	1.2%	98.8%	18819.7646

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

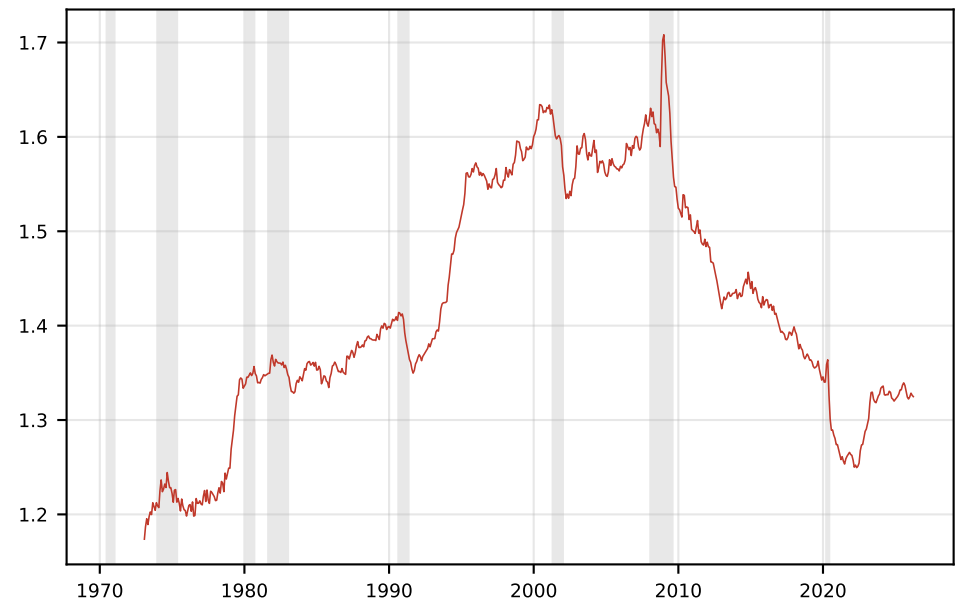
Assets-to-Deposits Ratio

Block: Selected export / Credit creation / bank channel | Source: FRED API | Col: derived_assets_deposits_ratio | Transform: nan

Raw Series



After: nan



SUMMARY STATISTICS

N : 1287
Valid : 639
NaN : 648
Start : 1973-01
End : 2026-03
Mean : 1.4172
Std : 0.1258
Min : 1.1737
Max : 1.7082
Skew : 0.134
Kurt : -1.014

STATIONARITY TESTS

ADF stat : -1.702
ADF p : 0.4303
ADF lags : 6
ADF : Non-stationary
KPSS stat : 1.192
KPSS p : 0.0100
KPSS : Non-stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

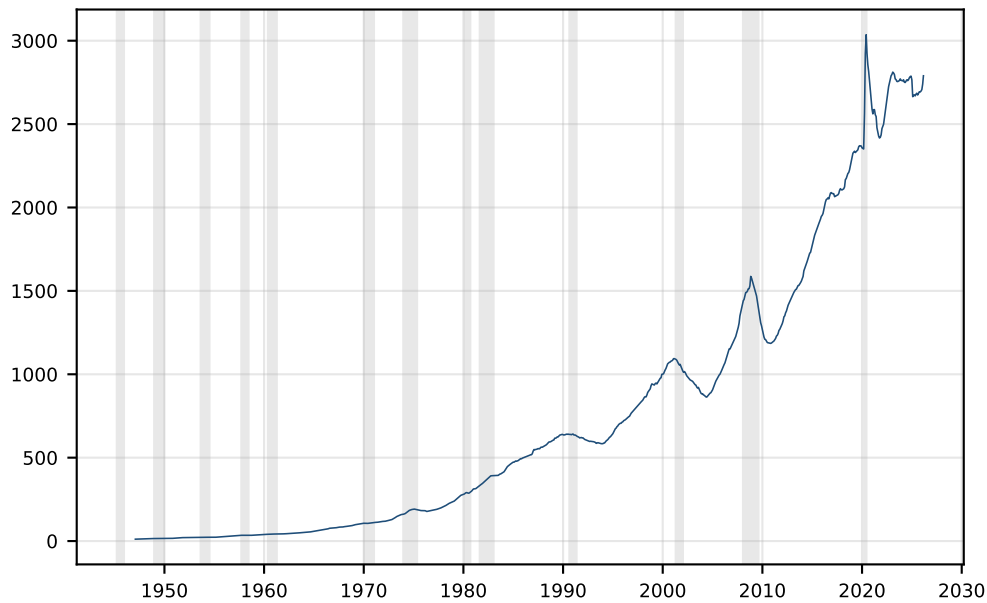
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	160	10.0%	90.0%	1.2651
Q2	159	16.4%	83.6%	1.3615
Q3	159	3.8%	96.2%	1.4539
Q4 (highest)	160	16.2%	83.8%	1.5888

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

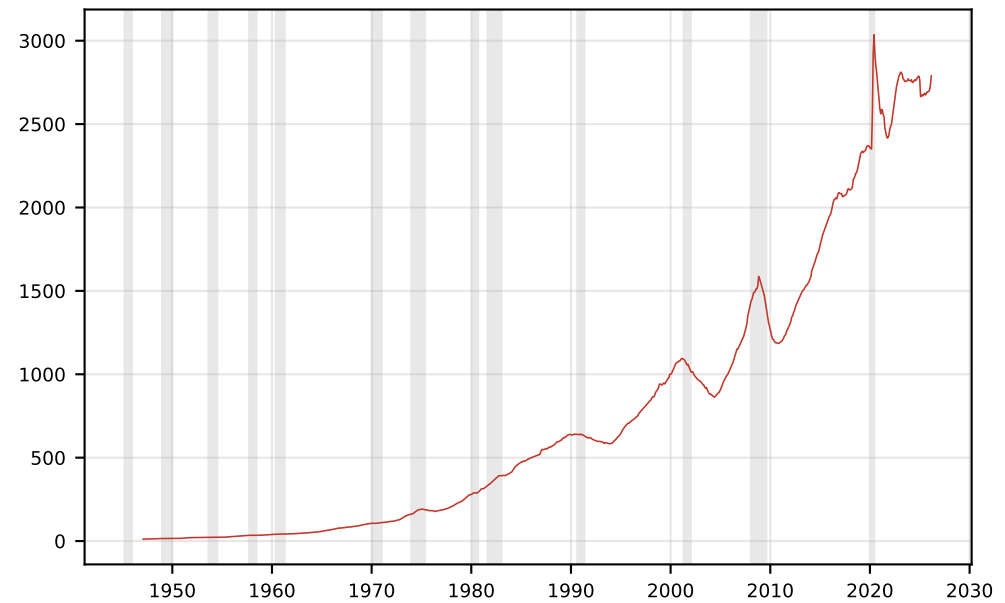
Commercial and Industrial Loans Monthly

Block: Selected export / Credit creation / bank channel | Source: FRED API | Col: derived_busloans_monthly | Transform: nan

Raw Series



After: nan



SUMMARY STATISTICS

N : 1287
Valid : 950
NaN : 337
Start : 1947-01
End : 2026-02
Mean : 755.9113
Std : 816.6150
Min : 11.2894
Max : 3035.7977
Skew : 1.157
Kurt : 0.288

STATIONARITY TESTS

ADF stat : 2.180
ADF p : 0.9989
ADF lags : 4
ADF : Non-stationary
KPSS stat : 4.114
KPSS p : 0.0100
KPSS : Non-stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

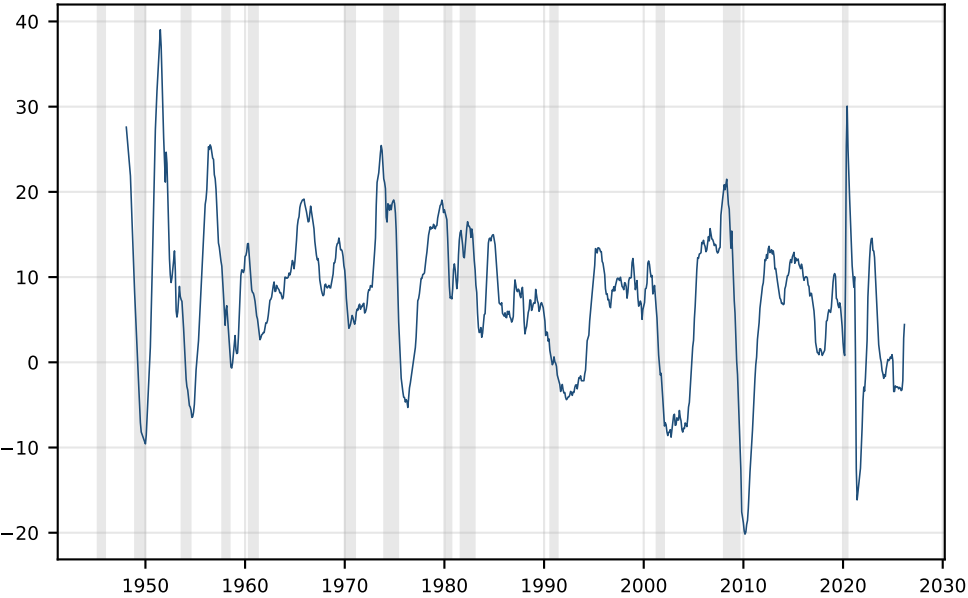
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	238	16.4%	83.6%	33.5245
Q2	237	20.7%	79.3%	235.2383
Q3	237	6.8%	93.2%	779.8272
Q4 (highest)	238	8.4%	91.6%	1972.9680

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

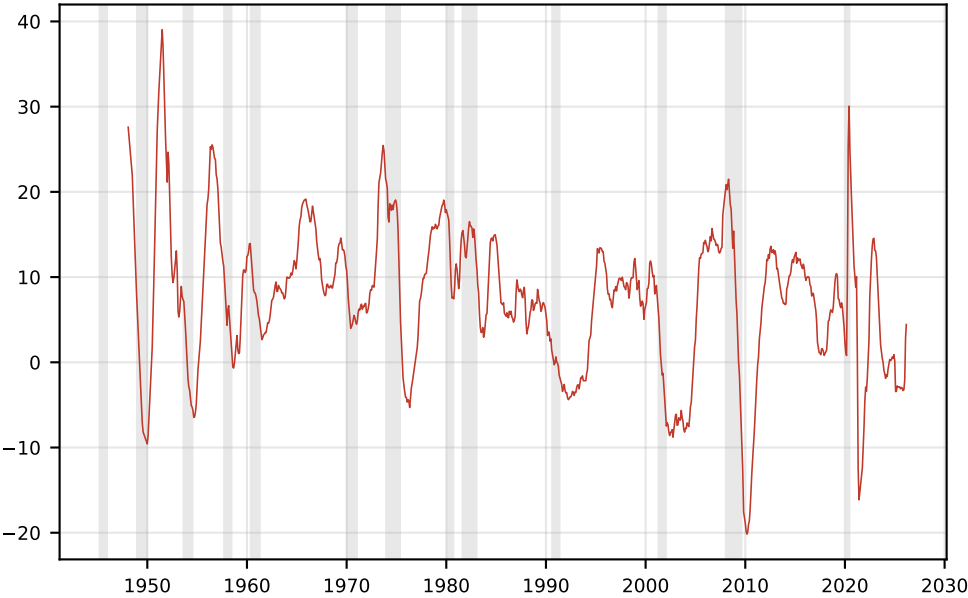
Commercial and Industrial Loans YoY

Block: Selected export / Credit creation / bank channel | Source: FRED API | Col: derived_busloans_yoy_2 | Transform: nan

Raw Series



After: nan



SUMMARY STATISTICS

N : 1287
Valid : 938
NaN : 349
Start : 1948-01
End : 2026-02
Mean : 7.4519
Std : 8.6161
Min : -20.1688
Max : 39.0263
Skew : -0.130
Kurt : 0.818

STATIONARITY TESTS

ADF stat : -4.489
ADF p : 0.0002
ADF lags : 12
ADF : Stationary
KPSS stat : 0.645
KPSS p : 0.0186
KPSS : Non-stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

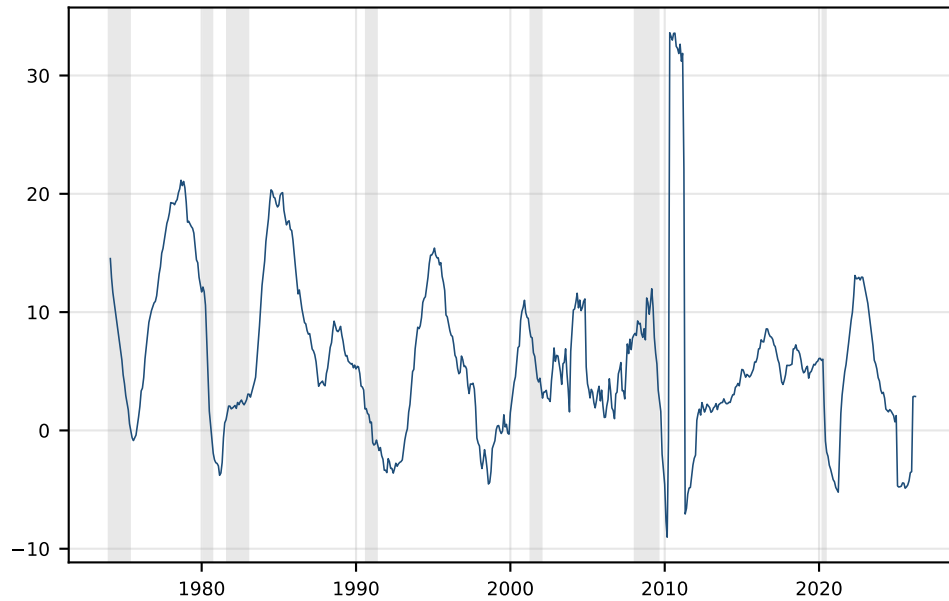
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	235	14.0%	86.0%	-3.9000
Q2	234	14.1%	85.9%	5.8545
Q3	234	6.0%	94.0%	10.1774
Q4 (highest)	235	18.7%	81.3%	17.6805

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

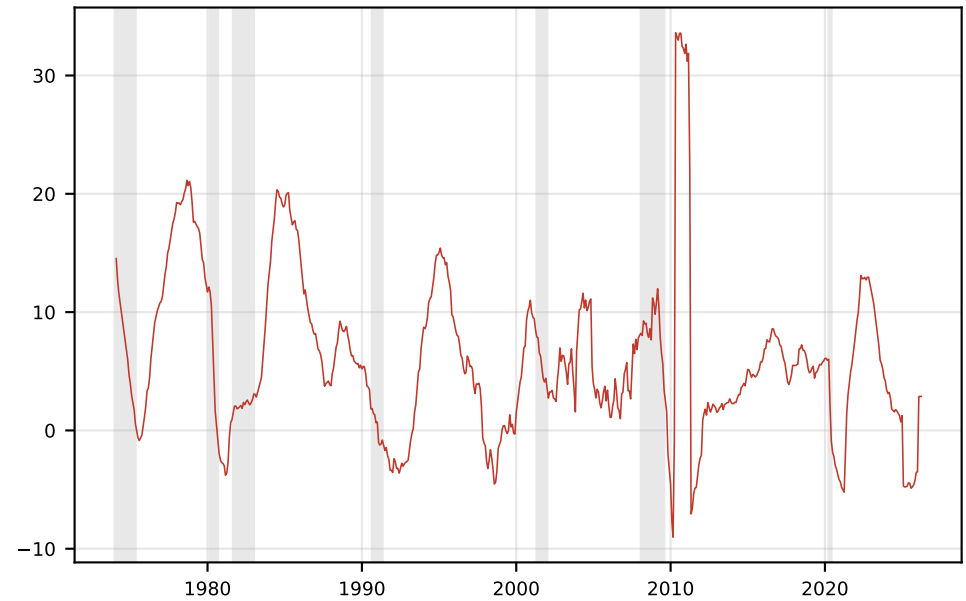
Consumer Loans YoY

Block: Selected export / Credit creation / bank channel | Source: FRED API | Col: derived_consumer_loans_yoy | Transform: nan

Raw Series



After: nan



SUMMARY STATISTICS

N : 1287
Valid : 627
NaN : 660
Start : 1974-01
End : 2026-03
Mean : 6.0973
Std : 6.9313
Min : -9.0225
Max : 33.6151
Skew : 1.149
Kurt : 2.448

STATIONARITY TESTS

ADF stat : -3.060
ADF p : 0.0296
ADF lags : 12
ADF : Stationary
KPSS stat : 0.328
KPSS p : 0.1000
KPSS : Stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

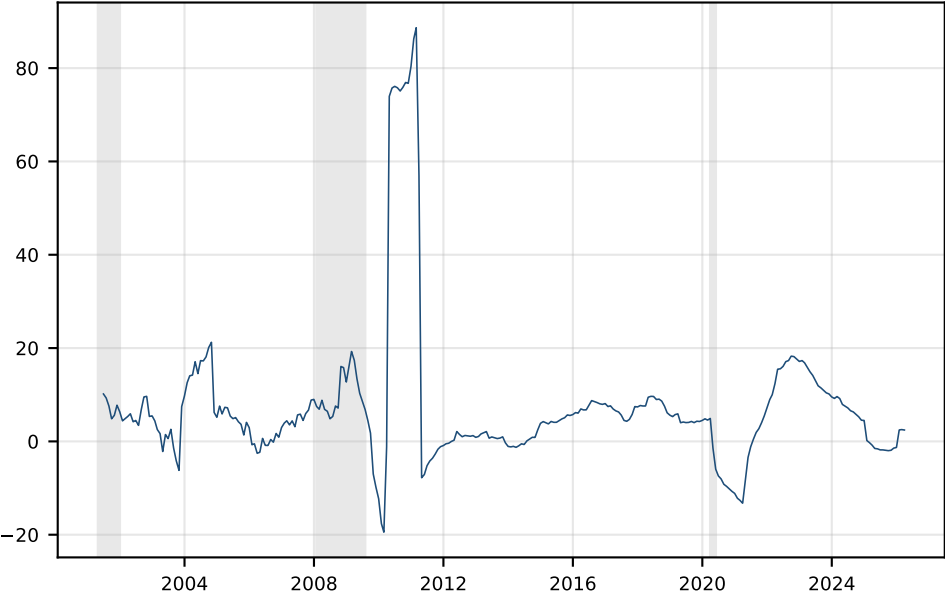
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	157	9.6%	90.4%	-1.3525
Q2	156	13.5%	86.5%	3.4925
Q3	156	14.1%	85.9%	6.9122
Q4 (highest)	157	9.6%	90.4%	15.3460

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

Credit Cards YoY

Block: Selected export / Credit creation / bank channel | Source: FRED API | Col: derived_credit_cards_yoy | Transform: nan

Raw Series



After: nan



SUMMARY STATISTICS

N : 1287
Valid : 298
NaN : 989
Start : 2001-06
End : 2026-03
Mean : 7.3250
Std : 15.6842
Min : -19.4545
Max : 88.6519
Skew : 3.488
Kurt : 13.772

STATIONARITY TESTS

ADF stat : -2.780
ADF p : 0.0612
ADF lags : 12
ADF : Non-stationary
KPSS stat : 0.112
KPSS p : 0.1000
KPSS : Stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

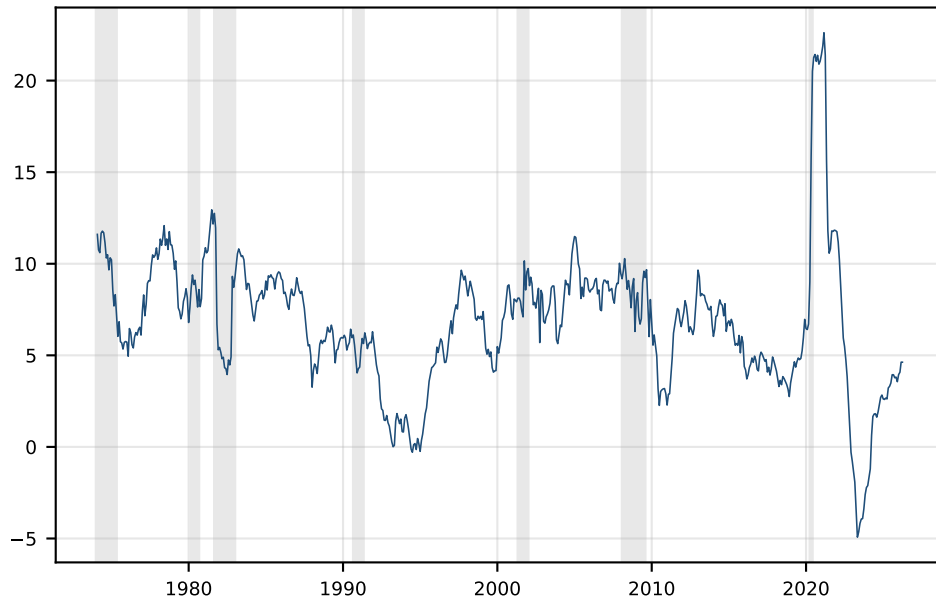
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	75	1.3%	98.7%	-3.5578
Q2	74	1.4%	98.6%	3.1721
Q3	74	16.2%	83.8%	6.5052
Q4 (highest)	74	16.2%	83.8%	23.3935

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

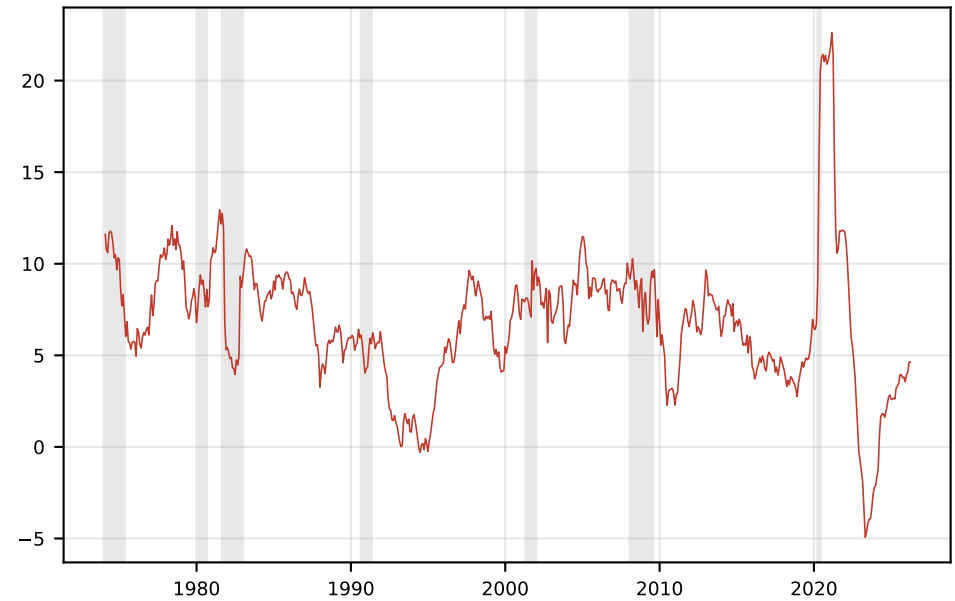
Deposit Growth YoY

Block: Selected export / Credit creation / bank channel | Source: FRED API | Col: derived_deposit_growth_yoy | Transform: nan

Raw Series



After: nan



SUMMARY STATISTICS

N : 1287
Valid : 627
NaN : 660
Start : 1974-01
End : 2026-03
Mean : 6.7544
Std : 3.6653
Min : -4.9283
Max : 22.6128
Skew : 0.550
Kurt : 3.679

STATIONARITY TESTS

ADF stat : -3.200
ADF p : 0.0200
ADF lags : 12
ADF : Stationary
KPSS stat : 0.224
KPSS p : 0.1000
KPSS : Stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

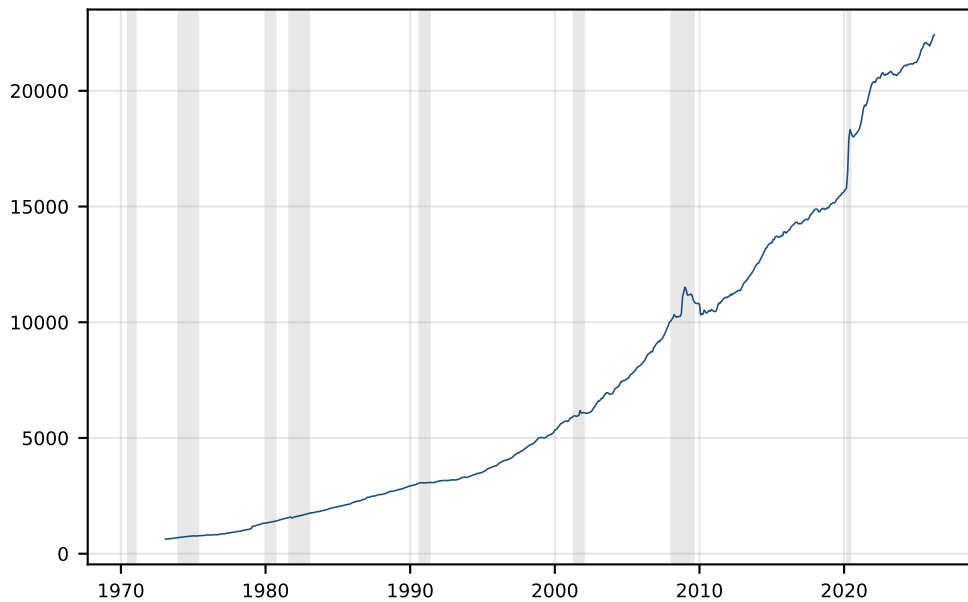
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	157	5.7%	94.3%	2.3931
Q2	156	8.3%	91.7%	5.8452
Q3	156	12.8%	87.2%	7.9152
Q4 (highest)	157	19.7%	80.3%	10.8794

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

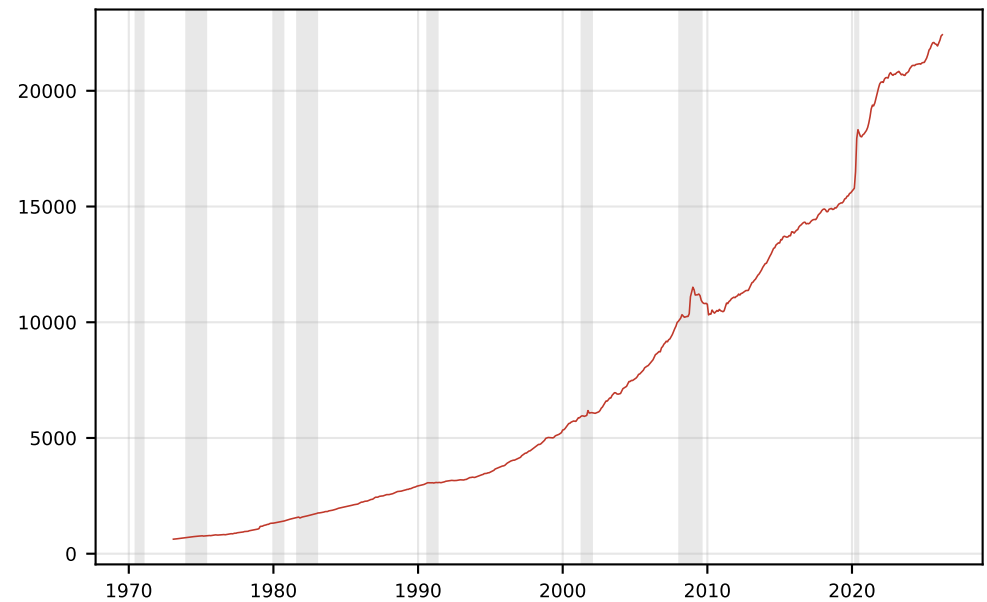
Liabilities, All Commercial Banks

Block: Selected export / Credit creation / bank channel | Source: FRED API | Col: derived_liabilities_all_commercial_banks | Transform: nan

Raw Series



After: nan



SUMMARY STATISTICS

N : 1287
Valid : 639
NaN : 648
Start : 1973-01
End : 2026-03
Mean : 7549.5568
Std : 6370.9918
Min : 627.8978
Max : 22421.5056
Skew : 0.830
Kurt : -0.492

STATIONARITY TESTS

ADF stat : 2.570
ADF p : 0.9991
ADF lags : 11
ADF : Non-stationary
KPSS stat : 3.546
KPSS p : 0.0100
KPSS : Non-stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

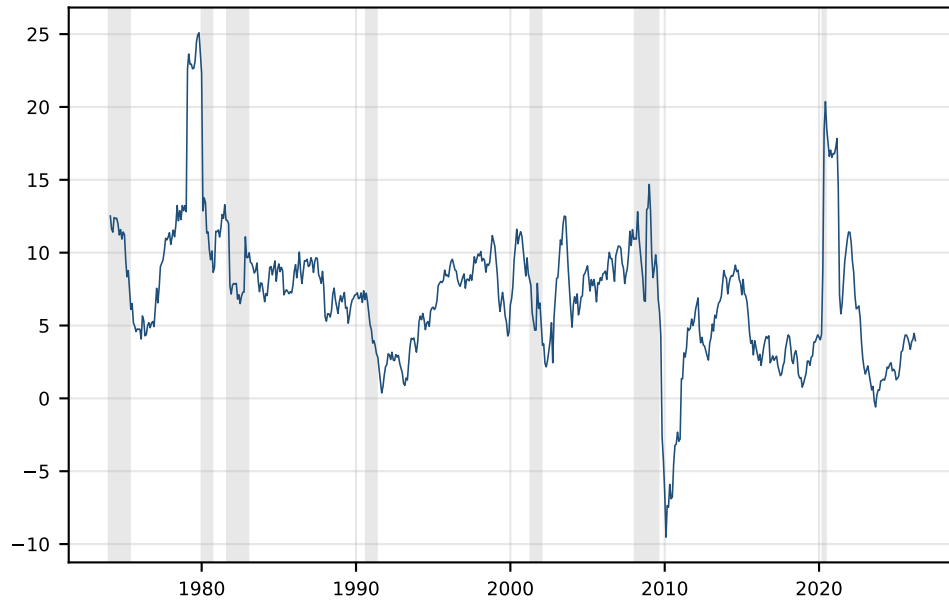
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	160	23.8%	76.2%	1305.3113
Q2	159	5.0%	95.0%	3422.5985
Q3	159	15.7%	84.3%	8489.1022
Q4 (highest)	160	1.9%	98.1%	16868.3444

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

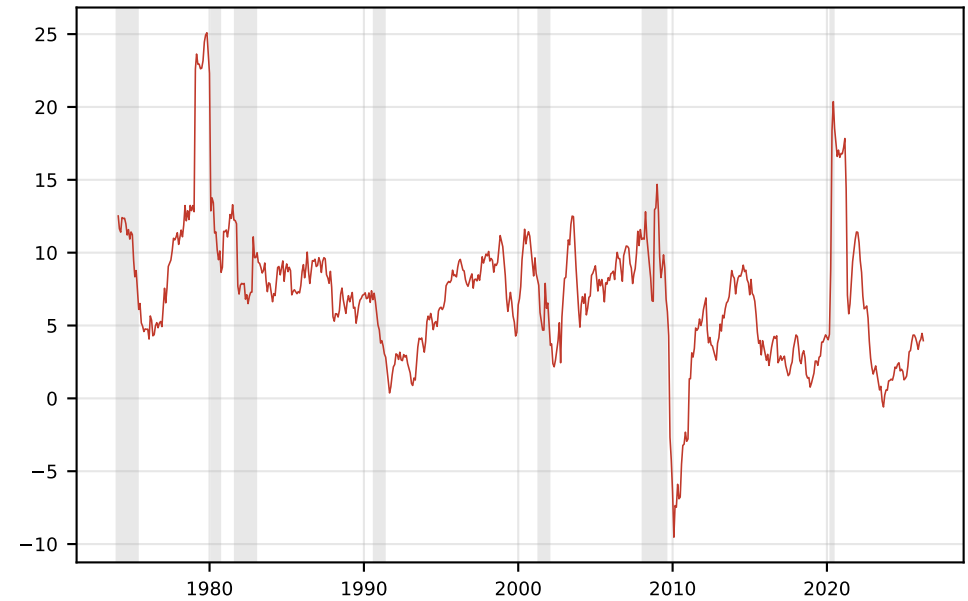
Liability Growth YoY

Block: Selected export / Credit creation / bank channel | Source: FRED API | Col: derived_liability_growth_yoy | Transform: nan

Raw Series



After: nan



SUMMARY STATISTICS

N : 1287
Valid : 627
NaN : 660
Start : 1974-01
End : 2026-03
Mean : 7.0400
Std : 4.5081
Min : -9.5269
Max : 25.0945
Skew : 0.561
Kurt : 3.047

STATIONARITY TESTS

ADF stat : -3.032
ADF p : 0.0320
ADF lags : 12
ADF : Stationary
KPSS stat : 0.779
KPSS p : 0.0100
KPSS : Non-stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

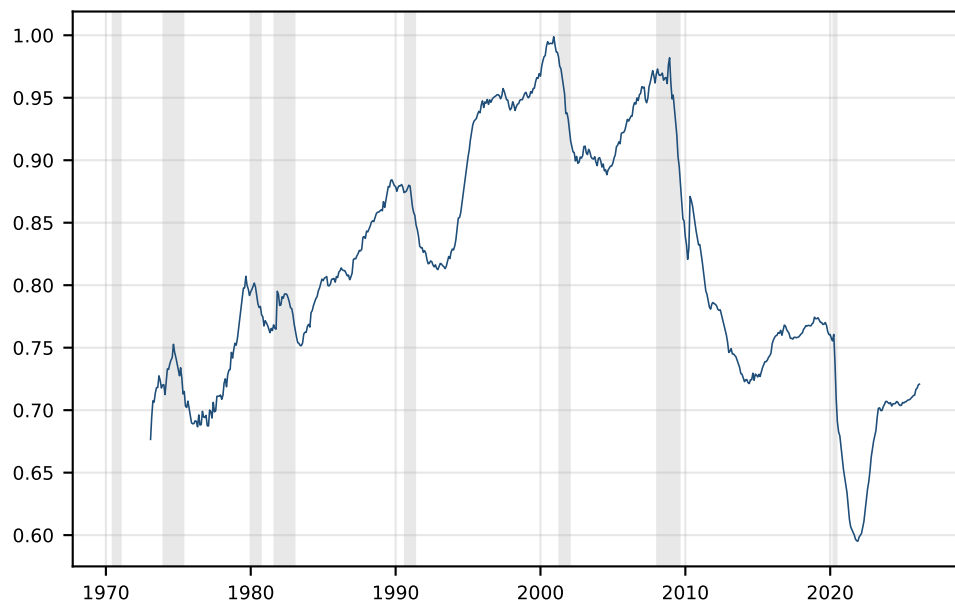
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	157	1.9%	98.1%	1.7883
Q2	156	10.9%	89.1%	5.7878
Q3	156	11.5%	88.5%	8.2575
Q4 (highest)	157	22.3%	77.7%	12.3459

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

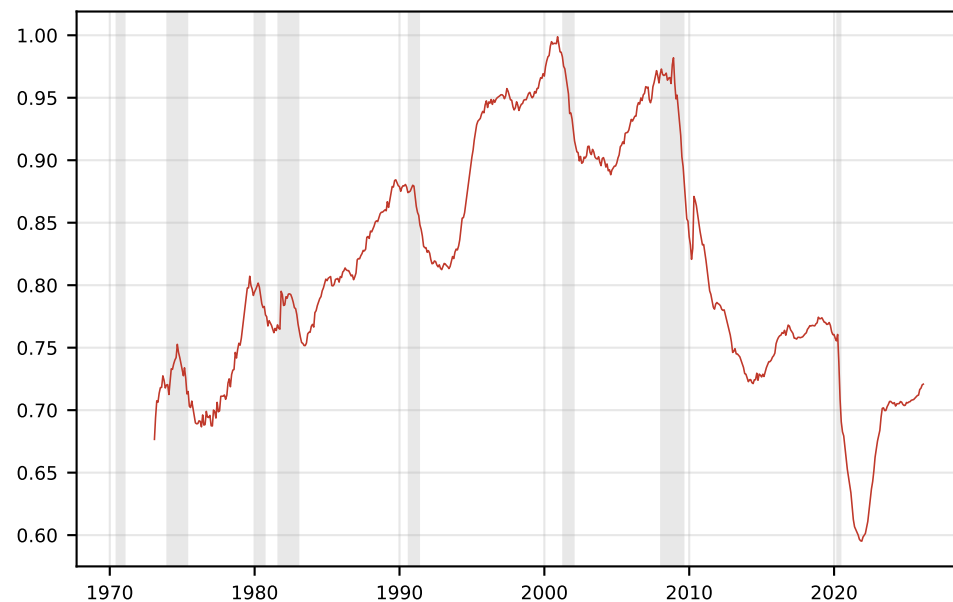
Loan-to-Deposit Ratio

Block: Selected export / Credit creation / bank channel | Source: FRED API | Col: derived_loan_deposit_ratio | Transform: nan

Raw Series



After: nan



SUMMARY STATISTICS

N : 1287
Valid : 638
NaN : 649
Start : 1973-01
End : 2026-02
Mean : 0.8142
Std : 0.0961
Min : 0.5952
Max : 0.9988
Skew : 0.092
Kurt : -0.894

STATIONARITY TESTS

ADF stat : -1.992
ADF p : 0.2900
ADF lags : 12
ADF : Non-stationary
KPSS stat : 0.838
KPSS p : 0.0100
KPSS : Non-stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

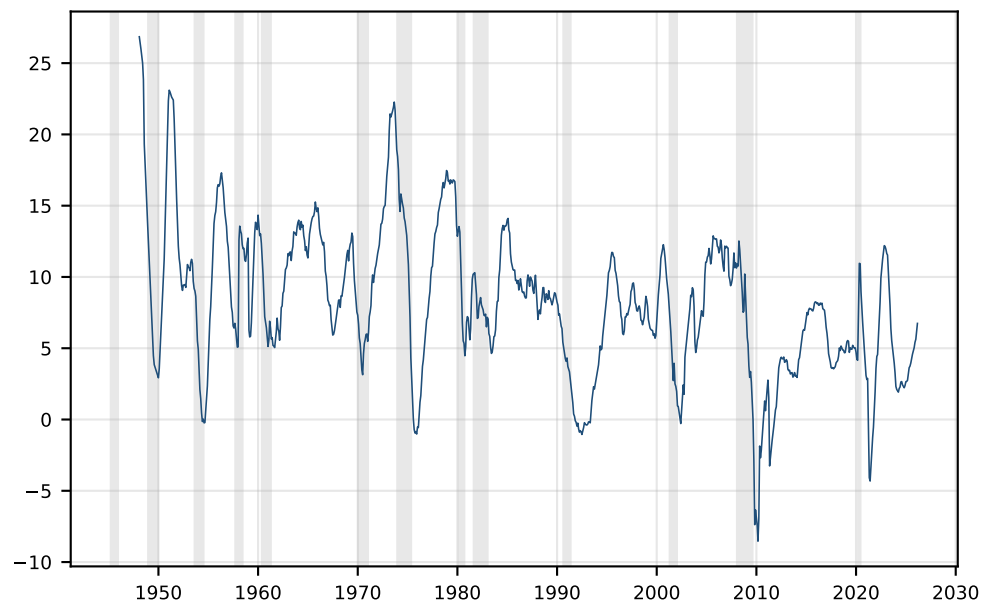
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	160	8.1%	91.9%	0.6971
Q2	159	16.4%	83.6%	0.7700
Q3	159	5.7%	94.3%	0.8444
Q4 (highest)	160	16.2%	83.8%	0.9450

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

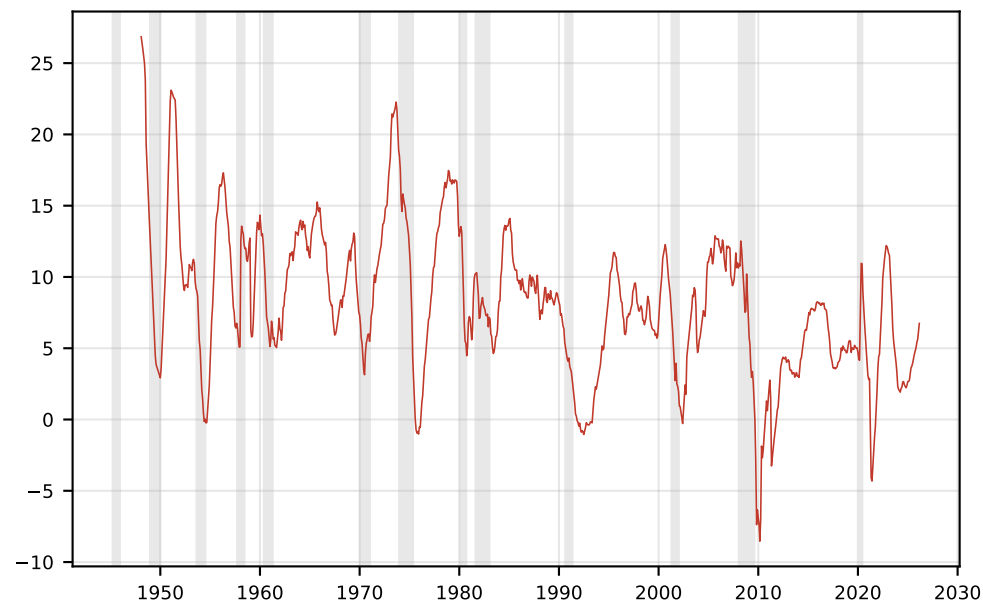
Loan Growth YoY

Block: Selected export / Credit creation / bank channel | Source: FRED API | Col: derived_loan_growth_yoy | Transform: nan

Raw Series



After: nan



SUMMARY STATISTICS

N : 1287
Valid : 938
NaN : 349
Start : 1948-01
End : 2026-02
Mean : 8.2045
Std : 5.1511
Min : -8.5348
Max : 26.8518
Skew : 0.278
Kurt : 0.813

STATIONARITY TESTS

ADF stat : -3.746
ADF p : 0.0035
ADF lags : 12
ADF : Stationary
KPSS stat : 1.572
KPSS p : 0.0100
KPSS : Non-stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

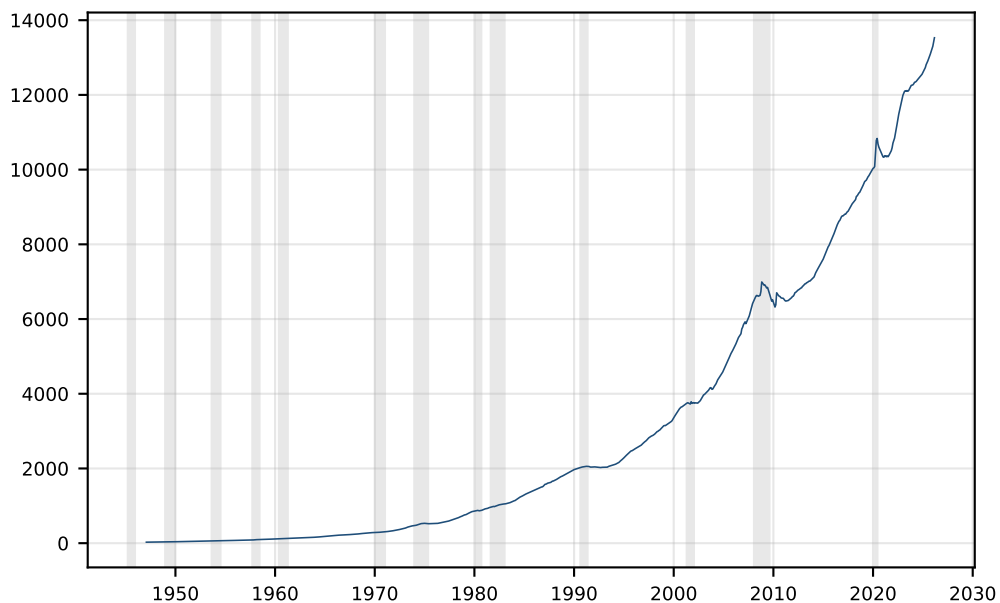
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	235	13.6%	86.4%	1.9459
Q2	234	18.4%	81.6%	6.5379
Q3	234	11.5%	88.5%	9.6046
Q4 (highest)	235	9.4%	90.6%	14.7285

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

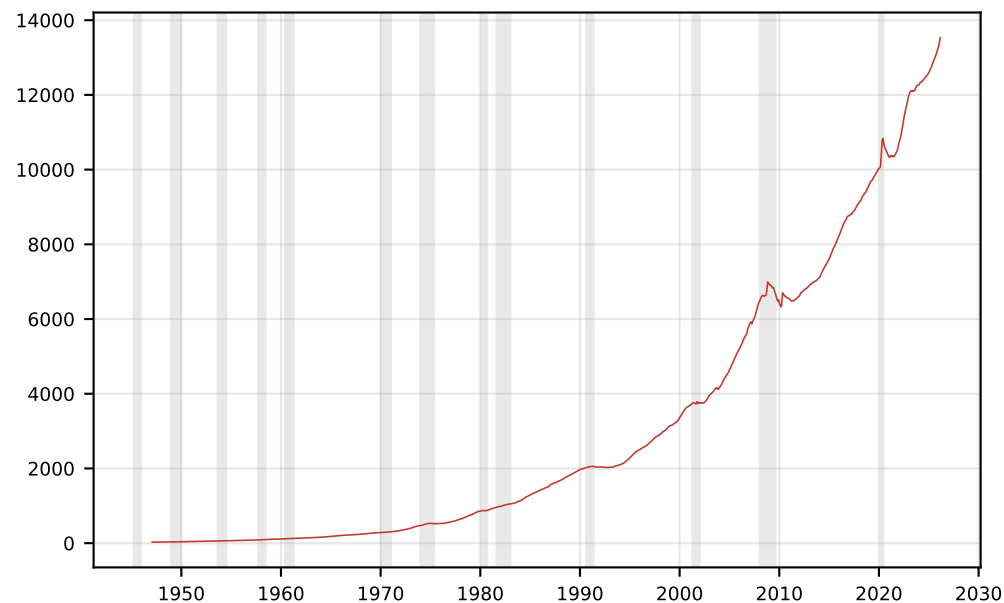
Loans and Leases in Bank Credit Monthly

Block: Selected export / Credit creation / bank channel | Source: FRED API | Col: derived_loans_monthly | Transform: nan

Raw Series



After: nan



SUMMARY STATISTICS

N : 1287
Valid : 950
NaN : 337
Start : 1947-01
End : 2026-02
Mean : 3108.2948
Std : 3666.2299
Min : 26.7375
Max : 13530.3897
Skew : 1.191
Kurt : 0.252

STATIONARITY TESTS

ADF stat : 6.456
ADF p : 1.0000
ADF lags : 4
ADF : Non-stationary
KPSS stat : 4.067
KPSS p : 0.0100
KPSS : Non-stationary

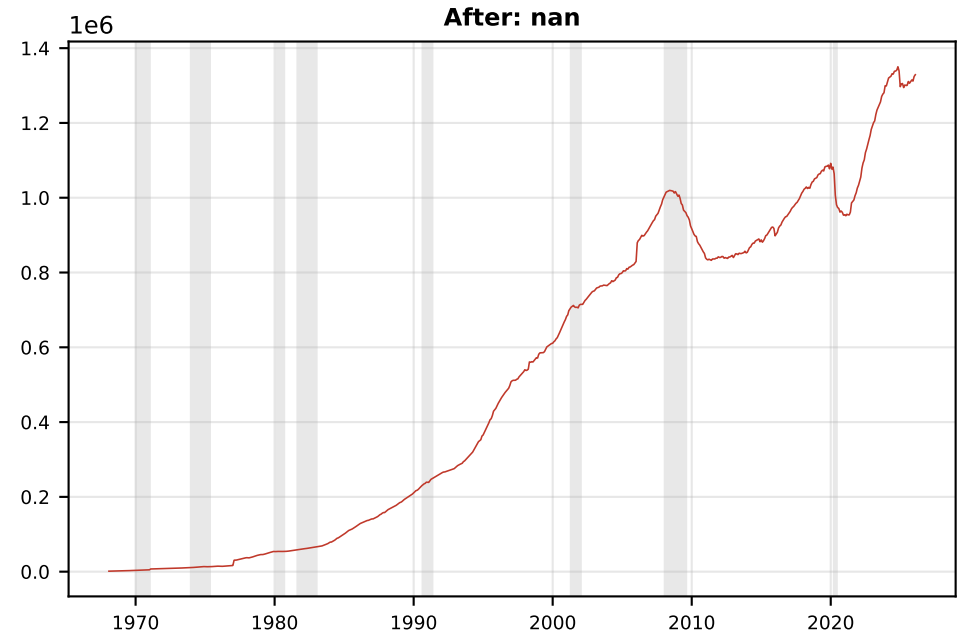
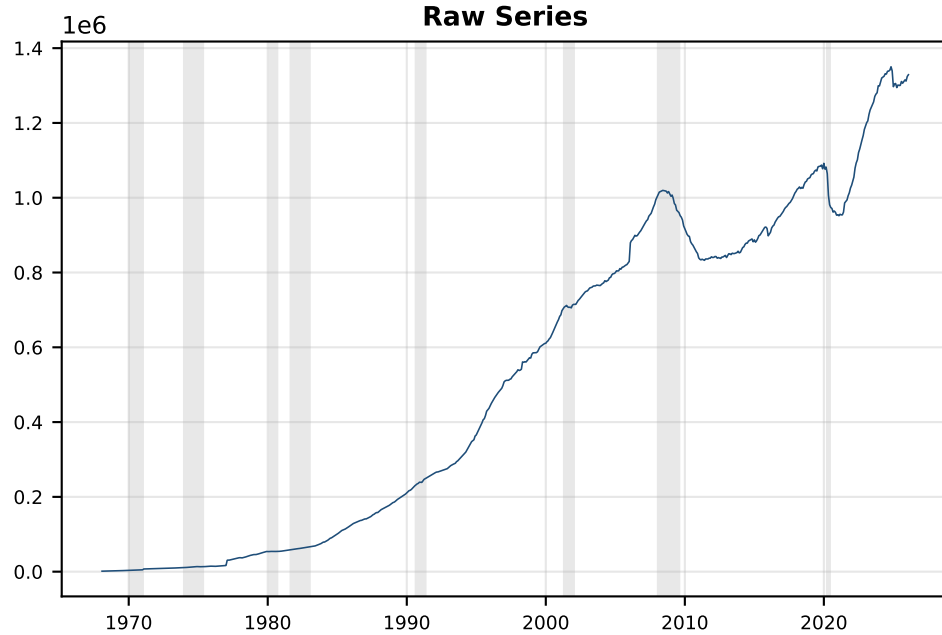
QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	238	16.4%	83.6%	94.7013
Q2	237	20.7%	79.3%	673.1611
Q3	237	6.8%	93.2%	2869.8358
Q4 (highest)	238	8.4%	91.6%	8784.2473

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

Revolving Consumer Credit Monthly

Block: Selected export / Credit creation / bank channel | Source: FRED API | Col: derived_revolsl_monthly | Transform: nan



SUMMARY STATISTICS

N : 1287
Valid : 697
NaN : 590
Start : 1968-01
End : 2026-01
Mean : 514218.9772
Std : 425856.2477
Min : 1316.7700
Max : 1350299.0200
Skew : 0.204
Kurt : -1.424

STATIONARITY TESTS

ADF stat : 0.336
ADF p : 0.9789
ADF lags : 11
ADF : Non-stationary
KPSS stat : 3.873
KPSS p : 0.0100
KPSS : Non-stationary

QUARTILE-BUCKET ANALYSIS

(vs NBER recession/expansion)

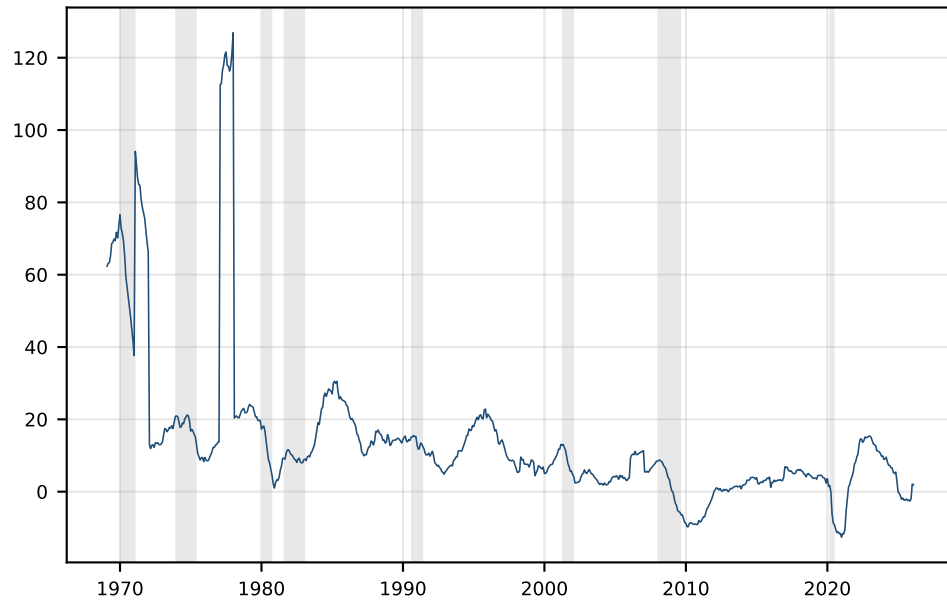
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	175	25.7%	74.3%	24089.0011
Q2	174	6.9%	93.1%	225516.0998
Q3	174	4.6%	95.4%	750505.4033
Q4 (highest)	174	11.5%	88.5%	1059582.2433

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

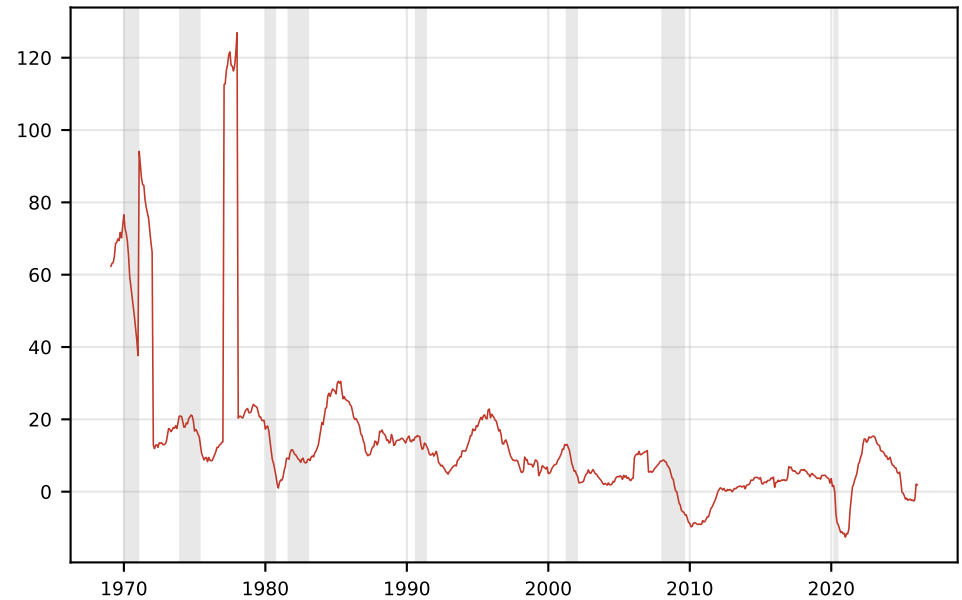
Revolving Consumer Credit YoY

Block: Selected export / Credit creation / bank channel | Source: FRED API | Col: derived_revolsl_yoy_2 | Transform: nan

Raw Series



After: nan



SUMMARY STATISTICS

N : 1287
Valid : 685
NaN : 602
Start : 1969-01
End : 2026-01
Mean : 13.8748
Std : 21.0082
Min : -12.5812
Max : 126.9071
Skew : 3.124
Kurt : 11.175

STATIONARITY TESTS

ADF stat : -3.761
ADF p : 0.0033
ADF lags : 12
ADF : Stationary
KPSS stat : 1.906
KPSS p : 0.0100
KPSS : Non-stationary

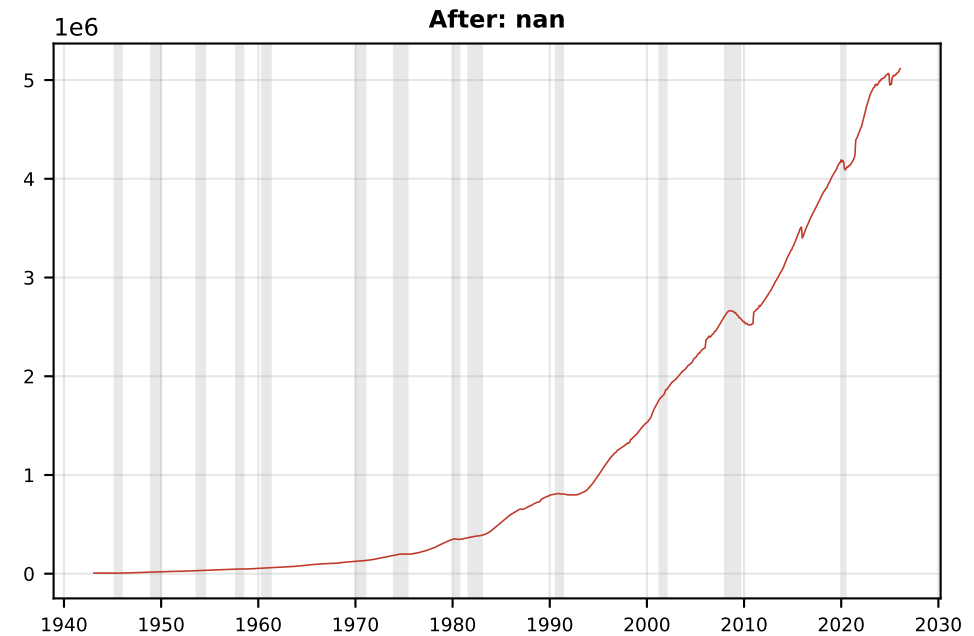
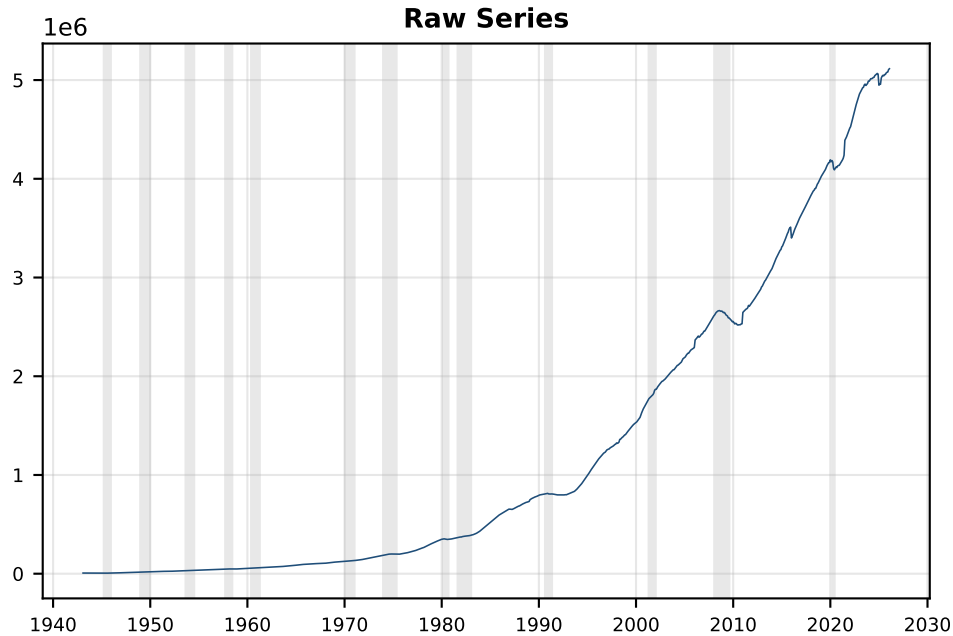
QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	172	7.0%	93.0%	-0.6707
Q2	171	12.9%	87.1%	6.4852
Q3	171	12.3%	87.7%	12.2268
Q4 (highest)	171	17.5%	82.5%	37.5429

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

Total Consumer Credit Monthly

Block: Selected export / Credit creation / bank channel | Source: FRED API | Col: derived_totalsl_monthly | Transform: nan



SUMMARY STATISTICS

N : 1287
Valid : 997
NaN : 290
Start : 1943-01
End : 2026-01
Mean : 1238585.9430
Std : 1490418.3307
Min : 5354.3600
Max : 5114679.4400
Skew : 1.152
Kurt : 0.096

STATIONARITY TESTS

ADF stat : 4.098
ADF p : 1.0000
ADF lags : 9
ADF : Non-stationary
KPSS stat : 4.263
KPSS p : 0.0100
KPSS : Non-stationary

QUARTILE-BUCKET ANALYSIS

(vs NBER recession/expansion)

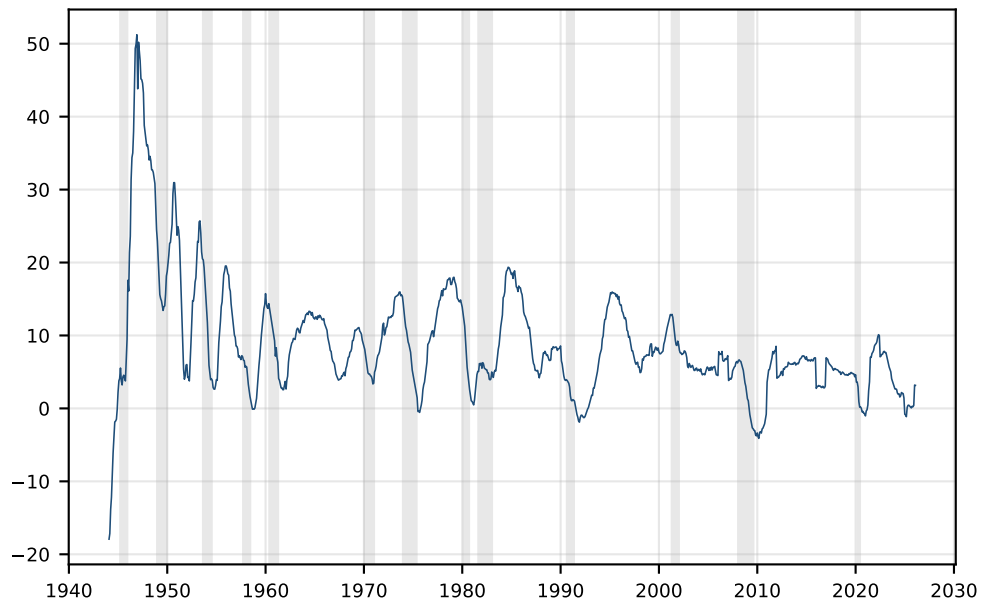
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	250	18.8%	81.2%	32700.4258
Q2	249	19.7%	80.3%	215772.0669
Q3	249	6.4%	93.6%	1176316.4532
Q4 (highest)	249	8.0%	92.0%	3534397.7398

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

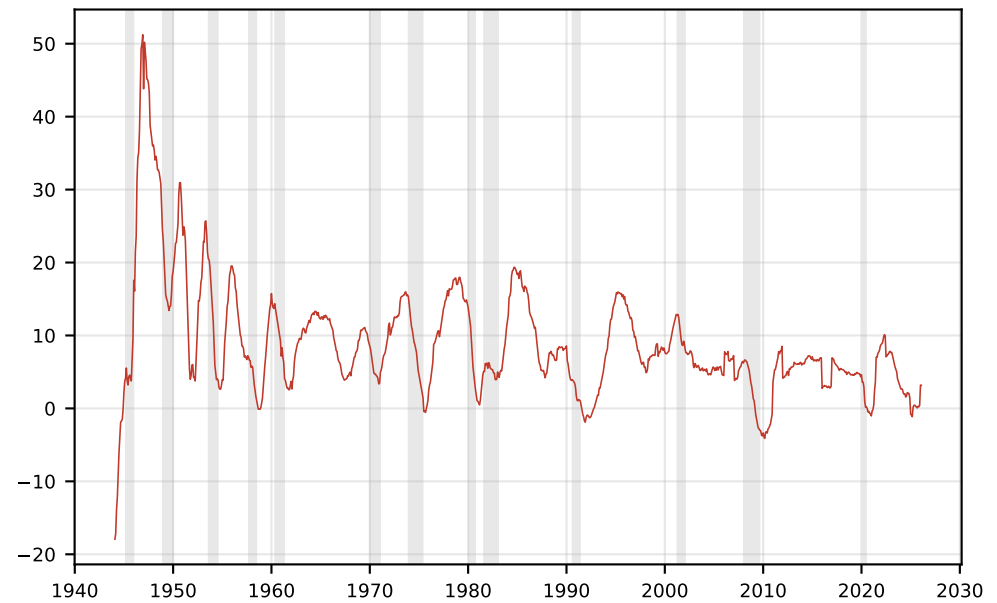
Total Consumer Credit YoY

Block: Selected export / Credit creation / bank channel | Source: FRED API | Col: derived_totalsl_yoy_2 | Transform: nan

Raw Series



After: nan



SUMMARY STATISTICS

N : 1287
Valid : 985
NaN : 302
Start : 1944-01
End : 2026-01
Mean : 8.8544
Std : 8.0826
Min : -17.9278
Max : 51.2297
Skew : 1.899
Kurt : 6.512

STATIONARITY TESTS

ADF stat : -3.270
ADF p : 0.0163
ADF lags : 12
ADF : Stationary
KPSS stat : 1.287
KPSS p : 0.0100
KPSS : Non-stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

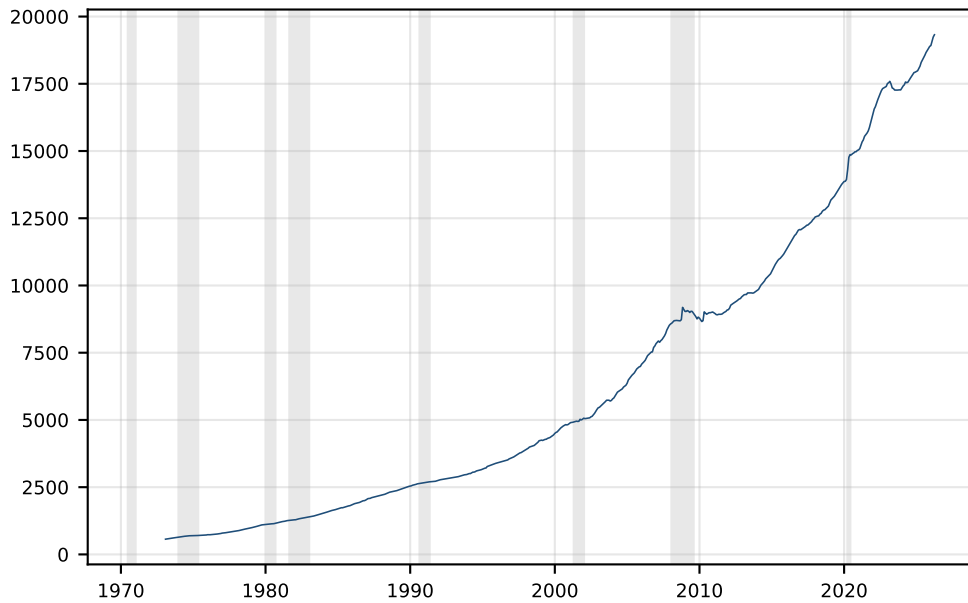
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	247	16.6%	83.4%	1.4220
Q2	246	16.3%	83.7%	5.7505
Q3	246	11.4%	88.6%	9.0709
Q4 (highest)	246	9.3%	90.7%	19.2043

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

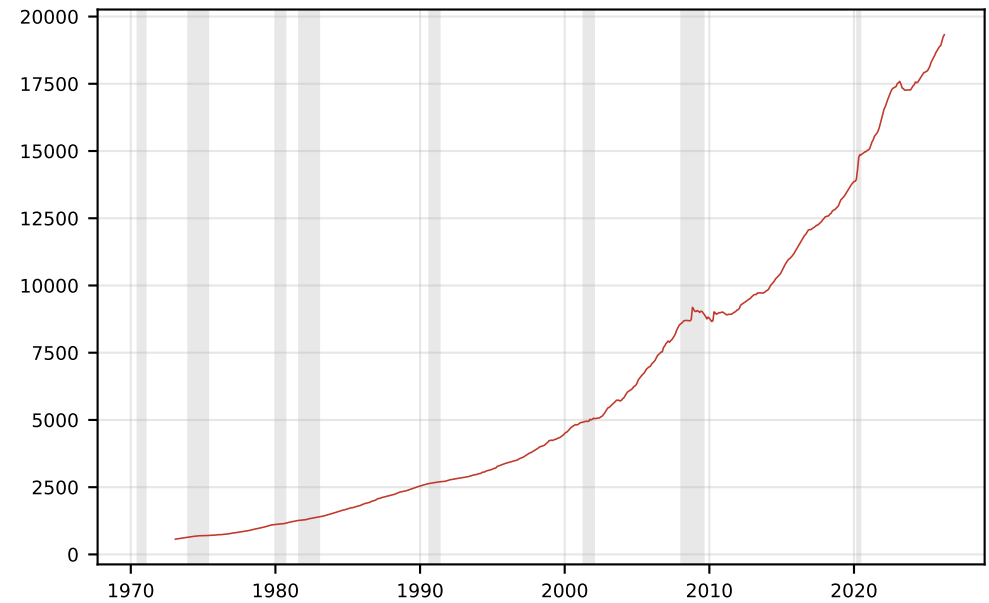
Total Bank Credit Monthly

Block: Selected export / Credit creation / bank channel | Source: FRED API | Col: derived_totbkcr_monthly | Transform: nan

Raw Series



After: nan



SUMMARY STATISTICS

N : 1287
Valid : 639
NaN : 648
Start : 1973-01
End : 2026-03
Mean : 6334.0742
Std : 5292.3038
Min : 566.5982
Max : 19323.7638
Skew : 0.843
Kurt : -0.440

STATIONARITY TESTS

ADF stat : 4.287
ADF p : 1.0000
ADF lags : 7
ADF : Non-stationary
KPSS stat : 3.546
KPSS p : 0.0100
KPSS : Non-stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

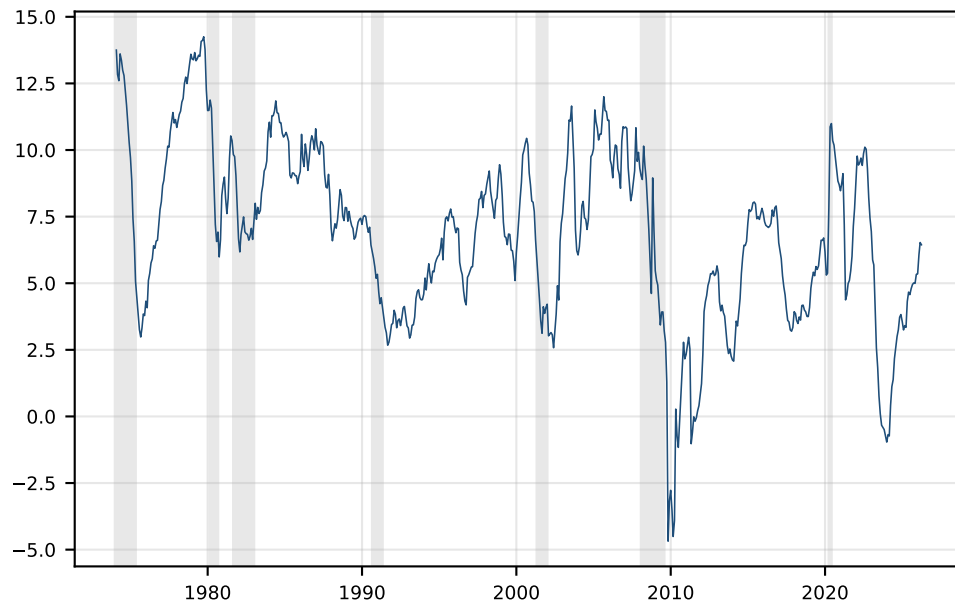
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	160	23.8%	76.2%	1111.2438
Q2	159	5.0%	95.0%	2979.6639
Q3	159	16.4%	83.6%	7117.1525
Q4 (highest)	160	1.2%	98.8%	14030.9801

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

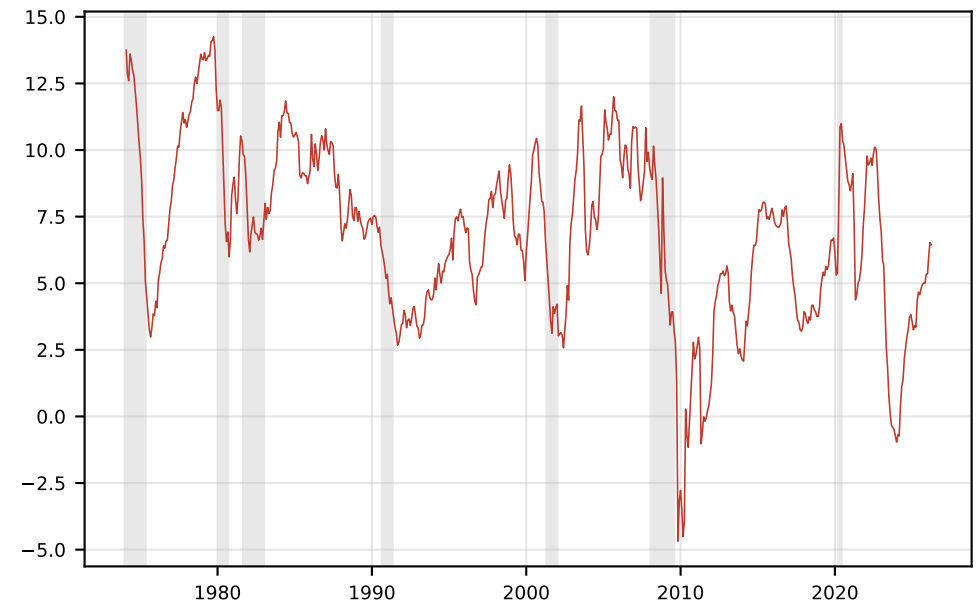
Total Bank Credit YoY

Block: Selected export / Credit creation / bank channel | Source: FRED API | Col: derived_totbkcr_yoy_2 | Transform: nan

Raw Series



After: nan



SUMMARY STATISTICS

N : 1287
Valid : 627
NaN : 660
Start : 1974-01
End : 2026-03
Mean : 6.8484
Std : 3.2704
Min : -4.6822
Max : 14.2519
Skew : -0.354
Kurt : 0.284

STATIONARITY TESTS

ADF stat : -2.601
ADF p : 0.0927
ADF lags : 12
ADF : Non-stationary
KPSS stat : 1.016
KPSS p : 0.0100
KPSS : Non-stationary

QUARTILE-BUCKET ANALYSIS

(vs NBER recession/expansion)

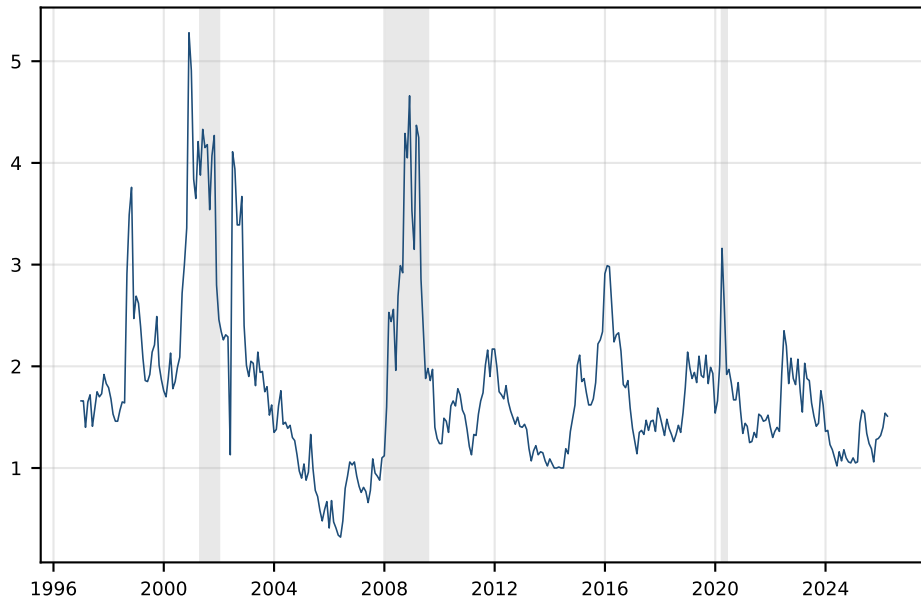
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	157	7.6%	92.4%	2.6197
Q2	156	17.3%	82.7%	5.8973
Q3	156	9.0%	91.0%	8.0198
Q4 (highest)	157	12.7%	87.3%	10.8608

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

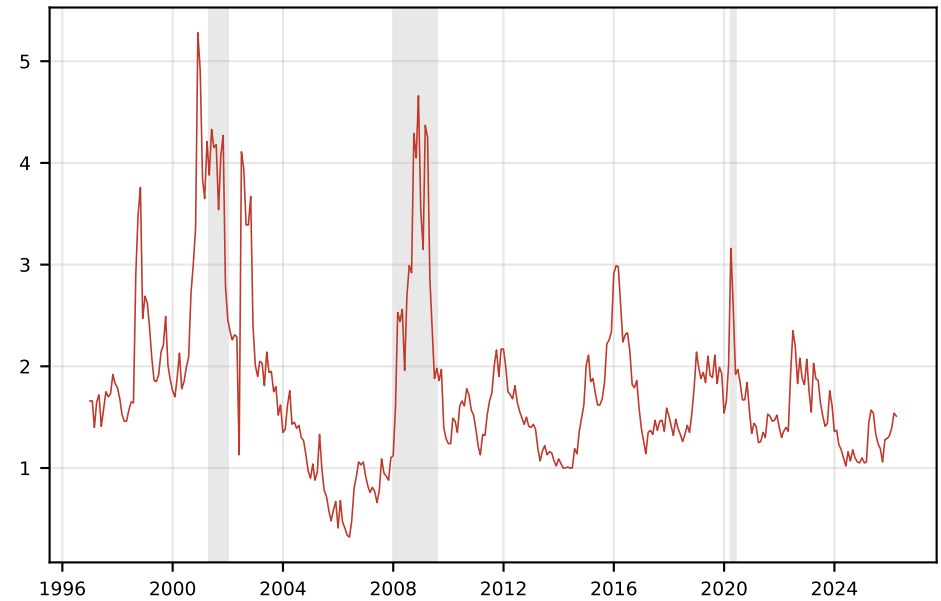
B Minus BB OAS

Block: Selected export / Credit-derived variables | Source: Derived | Col: derived_b_minus_bb_oas | Transform: nan

Raw Series



After: nan



SUMMARY STATISTICS

N : 1287
Valid : 352
NaN : 935
Start : 1996-12
End : 2026-03
Mean : 1.7767
Std : 0.8224
Min : 0.3200
Max : 5.2800
Skew : 1.563
Kurt : 2.930

STATIONARITY TESTS

ADF stat : -4.220
ADF p : 0.0006
ADF lags : 0
ADF : Stationary
KPSS stat : 0.380
KPSS p : 0.0859
KPSS : Stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

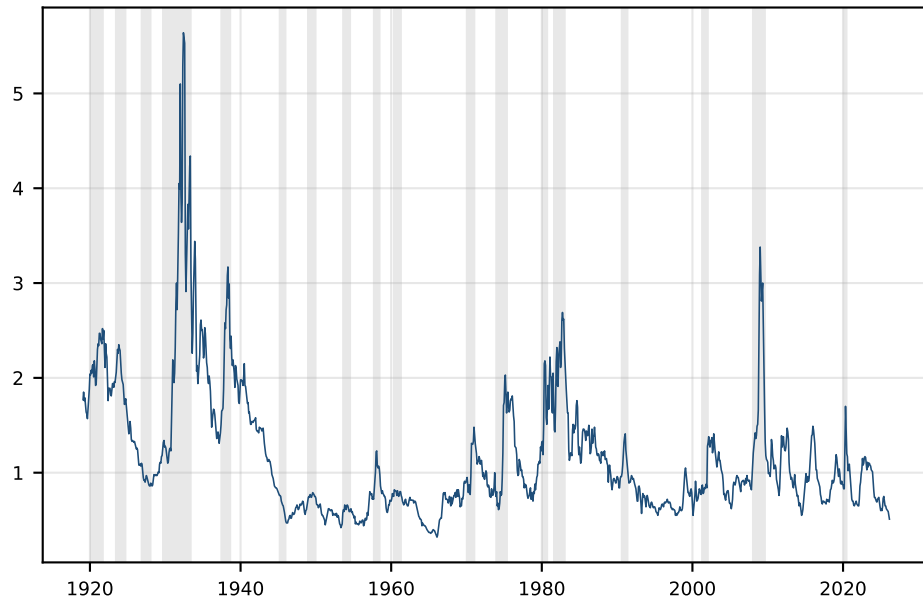
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	89	0.0%	100.0%	0.9988
Q2	88	1.1%	98.9%	1.4453
Q3	86	2.3%	97.7%	1.8014
Q4 (highest)	88	28.4%	71.6%	2.8738

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

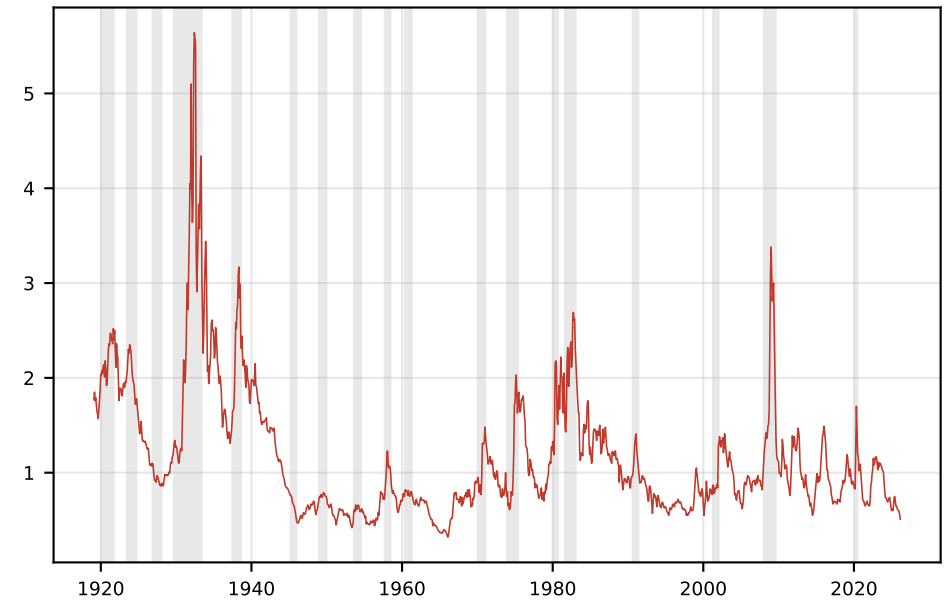
BAA Minus AAA

Block: Selected export / Credit-derived variables | Source: Derived | Col: derived_baa_aaa | Transform: nan

Raw Series



After: nan



SUMMARY STATISTICS

N : 1287
Valid : 1286
NaN : 1
Start : 1919-01
End : 2026-02
Mean : 1.1601
Std : 0.6820
Min : 0.3200
Max : 5.6400
Skew : 2.214
Kurt : 7.567

STATIONARITY TESTS

ADF stat : -4.186
ADF p : 0.0007
ADF lags : 9
ADF : Stationary
KPSS stat : 1.156
KPSS p : 0.0100
KPSS : Non-stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	324	6.5%	93.5%	0.5903
Q2	317	15.8%	84.2%	0.8138
Q3	315	14.3%	85.7%	1.1312
Q4 (highest)	318	36.8%	63.2%	2.0921

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

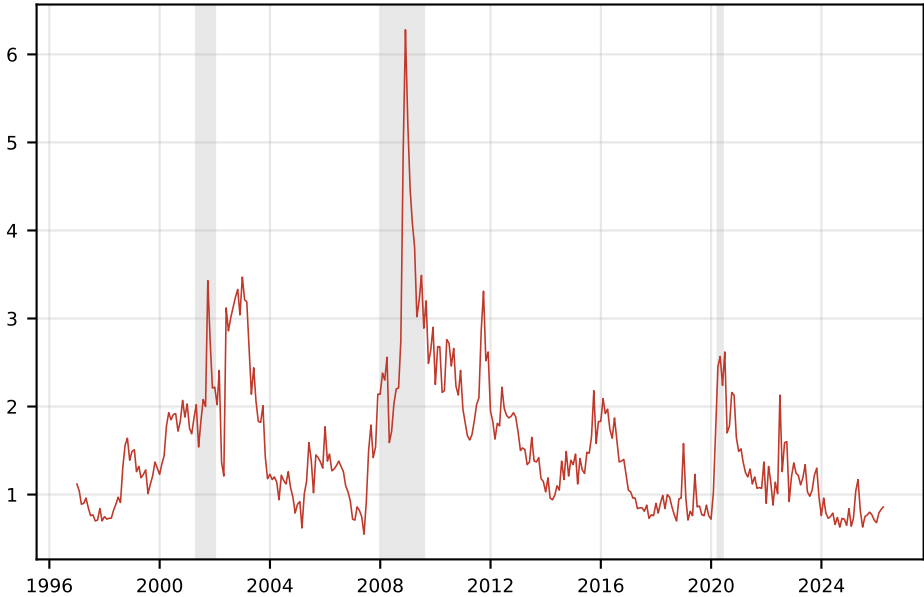
BB Minus BBB OAS

Block: Selected export / Credit-derived variables | Source: Derived | Col: derived_bb_minus_bbb_oas | Transform: nan

Raw Series



After: nan



SUMMARY STATISTICS

N : 1287
Valid : 352
NaN : 935
Start : 1996-12
End : 2026-03
Mean : 1.5382
Std : 0.7918
Min : 0.5500
Max : 6.2800
Skew : 1.851
Kurt : 5.552

STATIONARITY TESTS

ADF stat : -2.914
ADF p : 0.0438
ADF lags : 6
ADF : Stationary
KPSS stat : 0.493
KPSS p : 0.0432
KPSS : Non-stationary

QUARTILE-BUCKET ANALYSIS

(vs NBER recession/expansion)

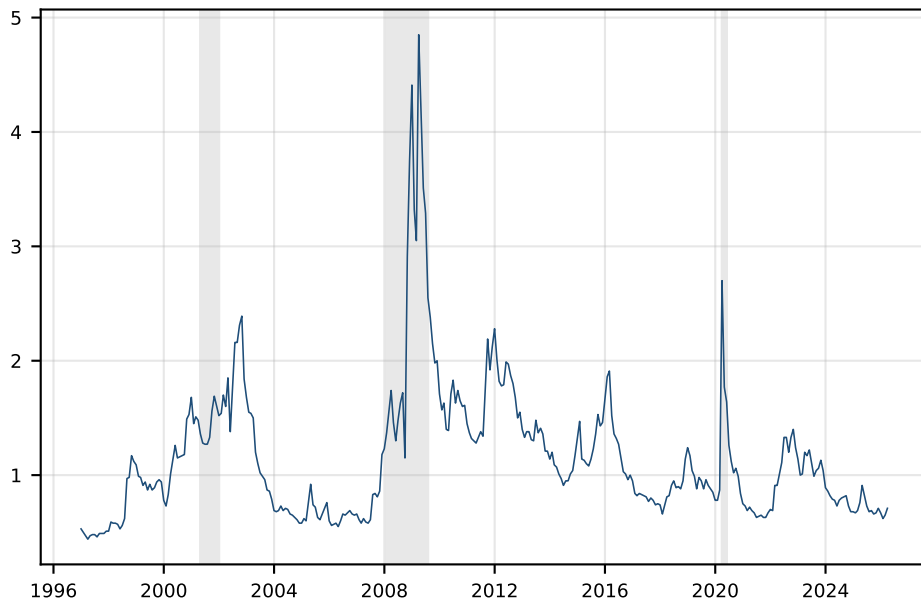
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	88	0.0%	100.0%	0.7991
Q2	88	0.0%	100.0%	1.1552
Q3	87	4.6%	95.4%	1.5897
Q4 (highest)	88	27.3%	72.7%	2.6169

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

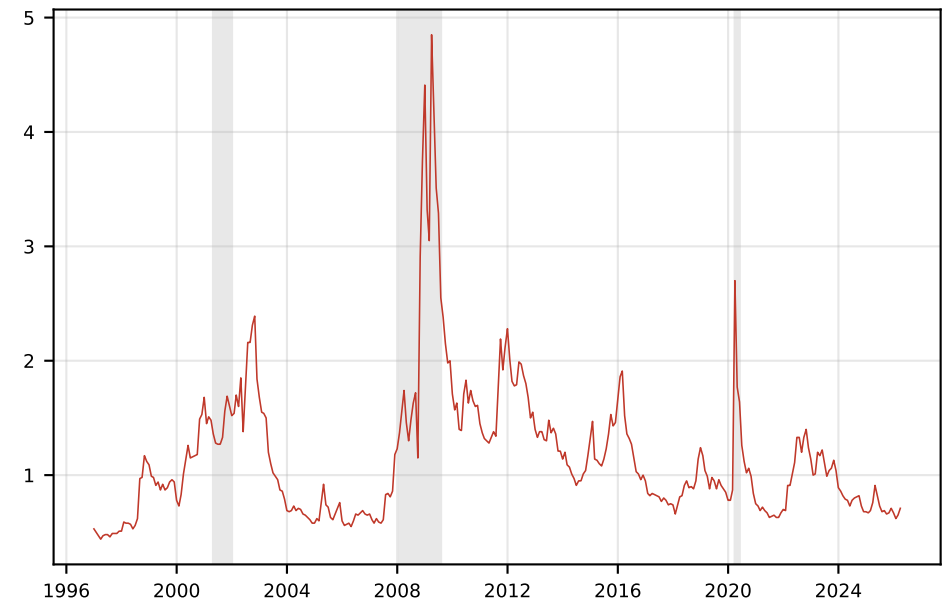
BBB Minus AAA OAS

Block: Selected export / Credit-derived variables | Source: Derived | Col: derived_bbb_minus_aaa_oas | Transform: nan

Raw Series



After: nan



SUMMARY STATISTICS

N : 1287
Valid : 352
NaN : 935
Start : 1996-12
End : 2026-03
Mean : 1.1559
Std : 0.6076
Min : 0.4400
Max : 4.8500
Skew : 2.395
Kurt : 8.895

STATIONARITY TESTS

ADF stat : -3.983
ADF p : 0.0015
ADF lags : 5
ADF : Stationary
KPSS stat : 0.249
KPSS p : 0.1000
KPSS : Stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	90	0.0%	100.0%	0.6257
Q2	86	0.0%	100.0%	0.8726
Q3	87	9.2%	90.8%	1.1970
Q4 (highest)	88	22.7%	77.3%	1.9394

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

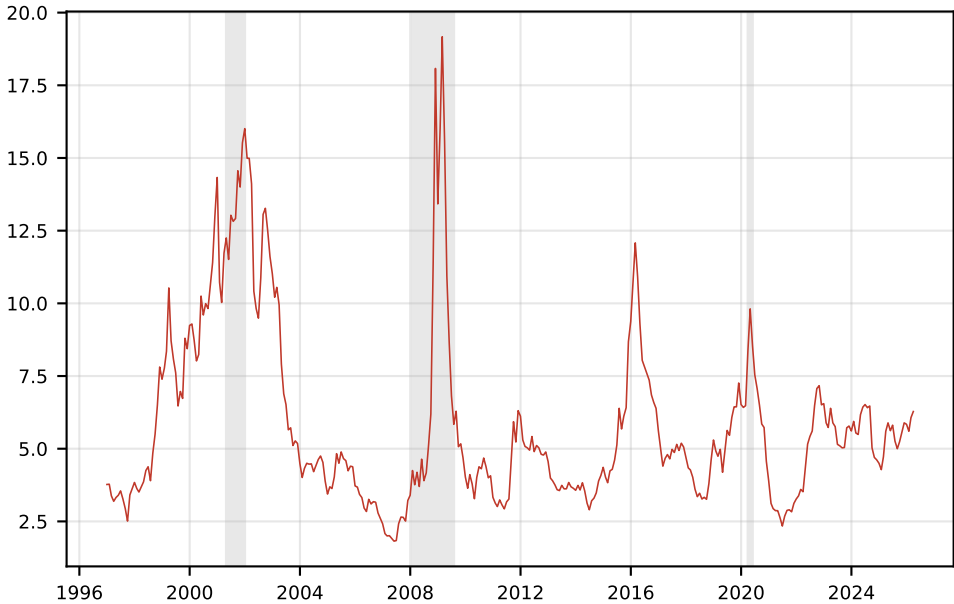
CCC Minus B OAS

Block: Selected export / Credit-derived variables | Source: Derived | Col: derived_ccc_minus_b_oas | Transform: nan

Raw Series



After: nan



SUMMARY STATISTICS

N : 1287
Valid : 352
NaN : 935
Start : 1996-12
End : 2026-03
Mean : 5.8302
Std : 3.0712
Min : 1.8200
Max : 19.1700
Skew : 1.640
Kurt : 2.633

STATIONARITY TESTS

ADF stat : -4.100
ADF p : 0.0010
ADF lags : 3
ADF : Stationary
KPSS stat : 0.347
KPSS p : 0.1000
KPSS : Stationary

QUARTILE-BUCKET ANALYSIS
(vs NBER recession/expansion)

Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	89	1.1%	98.9%	3.1471
Q2	87	6.9%	93.1%	4.3341
Q3	88	2.3%	97.7%	5.6588
Q4 (highest)	87	21.8%	78.2%	10.2392

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

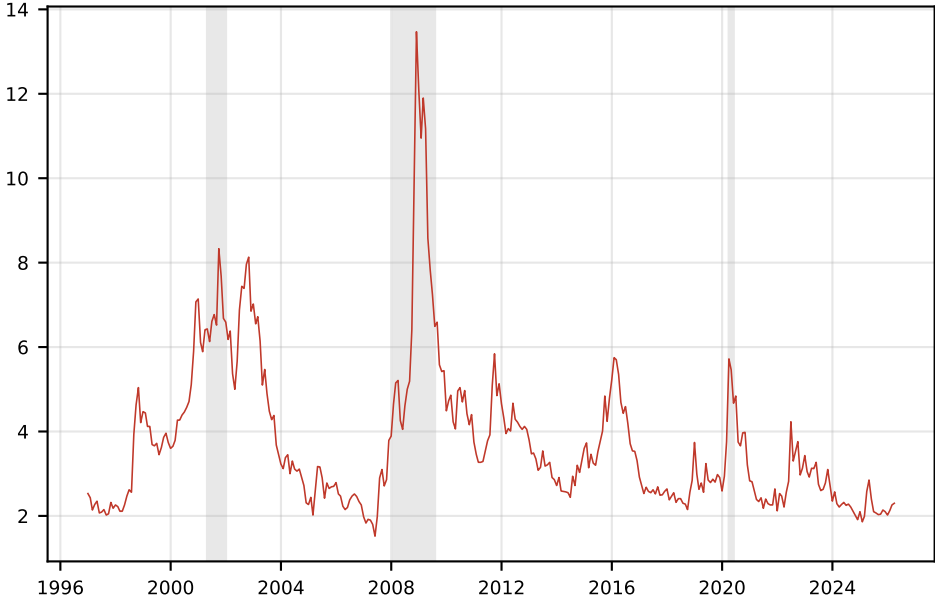
High Yield Minus Investment Grade

Block: Selected export / Credit-derived variables | Source: Derived | Col: derived_hy_minus_ig | Transform: nan

Raw Series



After: nan



SUMMARY STATISTICS

N : 1287
Valid : 352
NaN : 935
Start : 1996-12
End : 2026-03
Mean : 3.7475
Std : 1.7709
Min : 1.5200
Max : 13.4700
Skew : 2.049
Kurt : 5.989

STATIONARITY TESTS

ADF stat : -3.709
ADF p : 0.0040
ADF lags : 3
ADF : Stationary
KPSS stat : 0.524
KPSS p : 0.0363
KPSS : Non-stationary

QUARTILE-BUCKET ANALYSIS

(vs NBER recession/expansion)

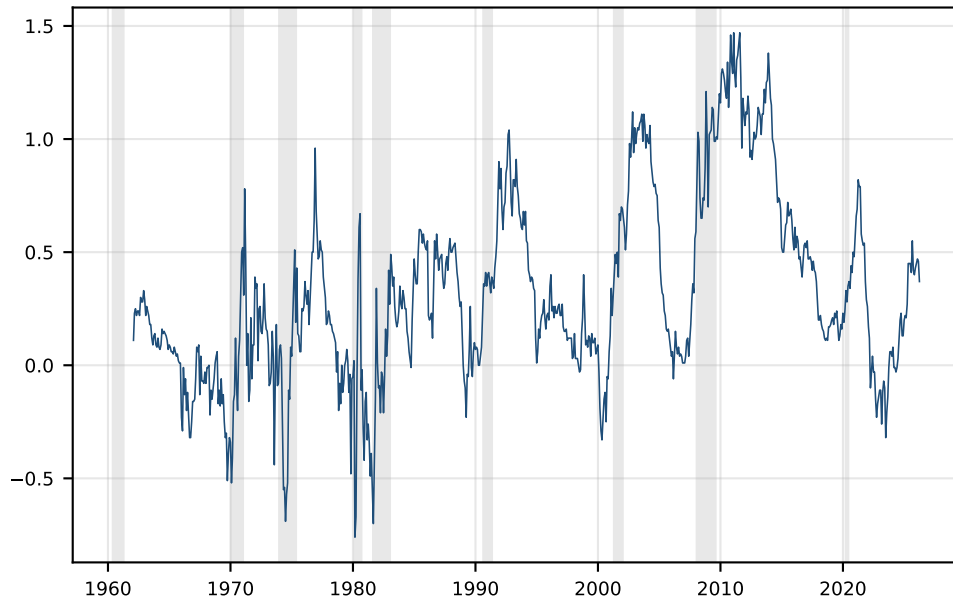
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	89	0.0%	100.0%	2.2294
Q2	87	0.0%	100.0%	2.8310
Q3	87	2.3%	97.7%	3.7946
Q4 (highest)	88	29.5%	70.5%	6.1589

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

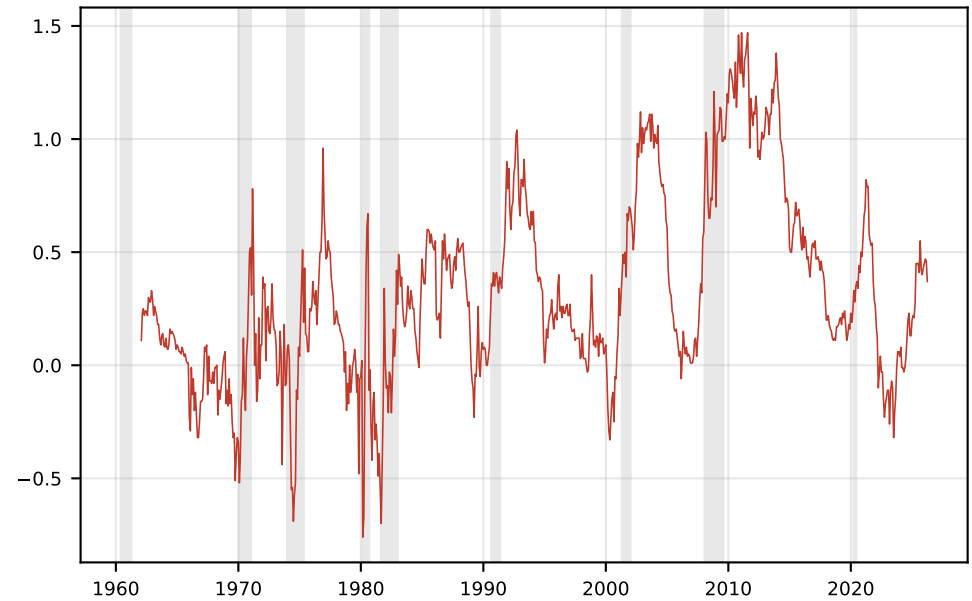
Belly Slope

Block: Selected export / Curve-derived variables | Source: Derived | Col: derived_belly_slope | Transform: nan

Raw Series



After: nan



SUMMARY STATISTICS

N : 1287
Valid : 771
NaN : 516
Start : 1962-01
End : 2026-03
Mean : 0.3234
Std : 0.4096
Min : -0.7600
Max : 1.4700
Skew : 0.554
Kurt : 0.044

STATIONARITY TESTS

ADF stat : -3.402
ADF p : 0.0109
ADF lags : 12
ADF : Stationary
KPSS stat : 1.249
KPSS p : 0.0100
KPSS : Non-stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	194	16.5%	83.5%	-0.1289
Q2	191	4.2%	95.8%	0.1426
Q3	193	11.4%	88.6%	0.3860
Q4 (highest)	192	12.0%	88.0%	0.8970

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

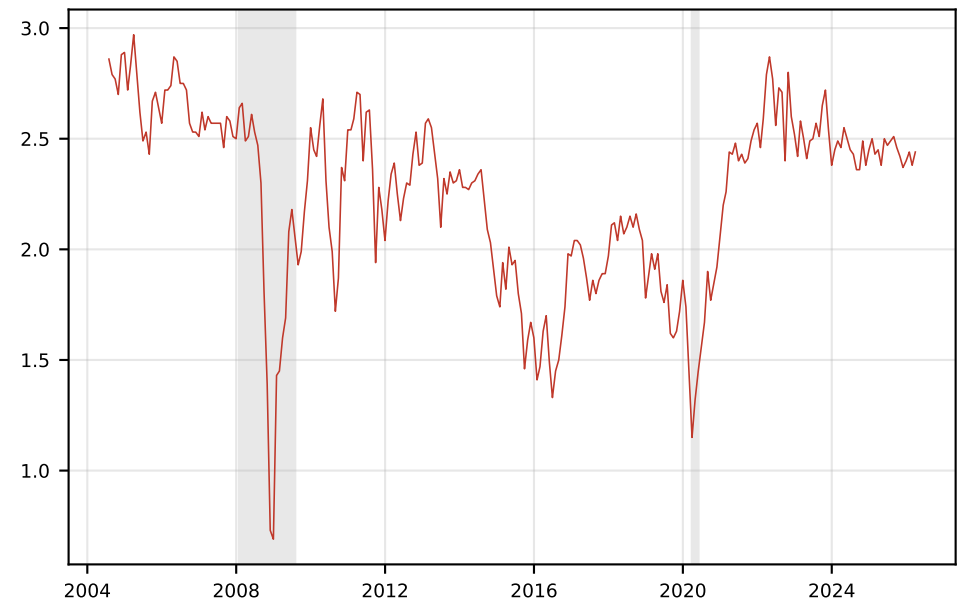
20Y Breakeven Inflation

Block: Selected export / Curve-derived variables | Source: Derived | Col: derived_breakeven_20y | Transform: nan

Raw Series



After: nan



SUMMARY STATISTICS

N : 1287
Valid : 261
NaN : 1026
Start : 2004-07
End : 2026-03
Mean : 2.2447
Std : 0.4049
Min : 0.6900
Max : 2.9700
Skew : -0.862
Kurt : 0.599

STATIONARITY TESTS

ADF stat : -3.501
ADF p : 0.0080
ADF lags : 1
ADF : Stationary
KPSS stat : 0.383
KPSS p : 0.0844
KPSS : Stationary

QUARTILE-BUCKET ANALYSIS

(vs NBER recession/expansion)

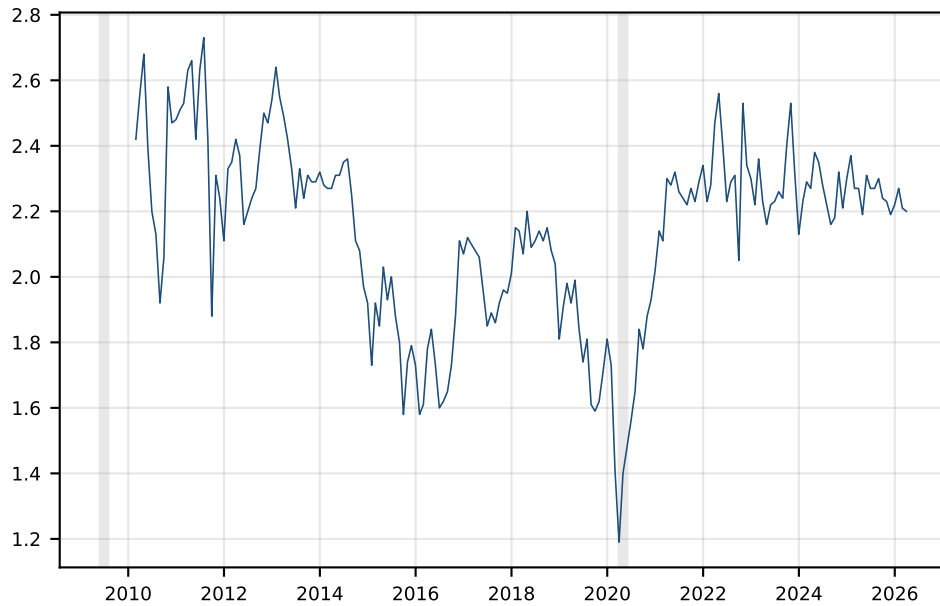
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	65	15.4%	84.6%	1.6708
Q2	65	4.6%	95.4%	2.1780
Q3	65	6.2%	93.8%	2.4552
Q4 (highest)	65	4.6%	95.4%	2.6717

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

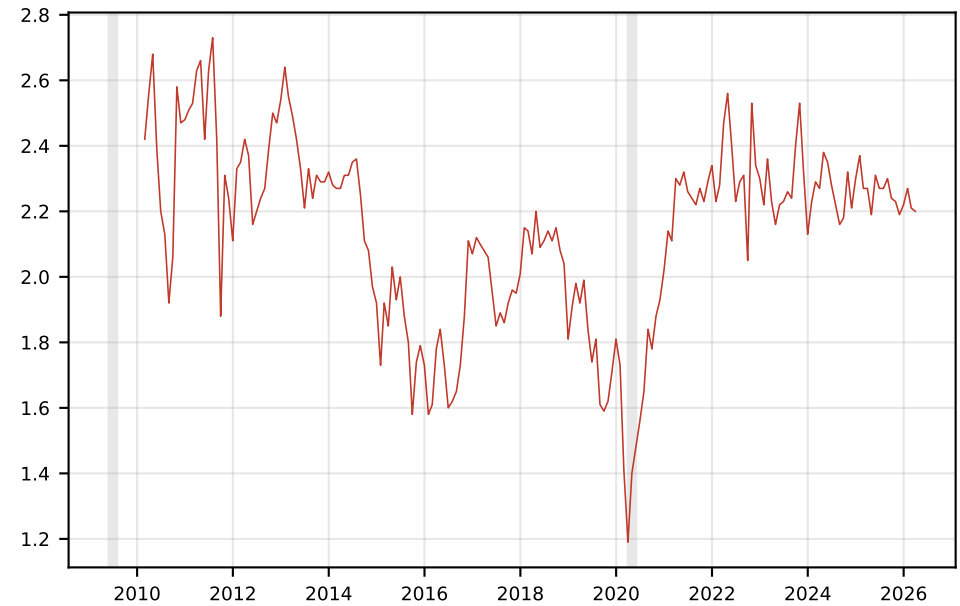
30Y Breakeven Inflation

Block: Selected export / Curve-derived variables | Source: Derived | Col: derived_breakeven_30y | Transform: nan

Raw Series



After: nan



SUMMARY STATISTICS

N : 1287
Valid : 194
NaN : 1093
Start : 2010-02
End : 2026-03
Mean : 2.1382
Std : 0.2815
Min : 1.1900
Max : 2.7300
Skew : -0.612
Kurt : 0.158

STATIONARITY TESTS

ADF stat : -2.915
ADF p : 0.0436
ADF lags : 2
ADF : Stationary
KPSS stat : 0.344
KPSS p : 0.1000
KPSS : Stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

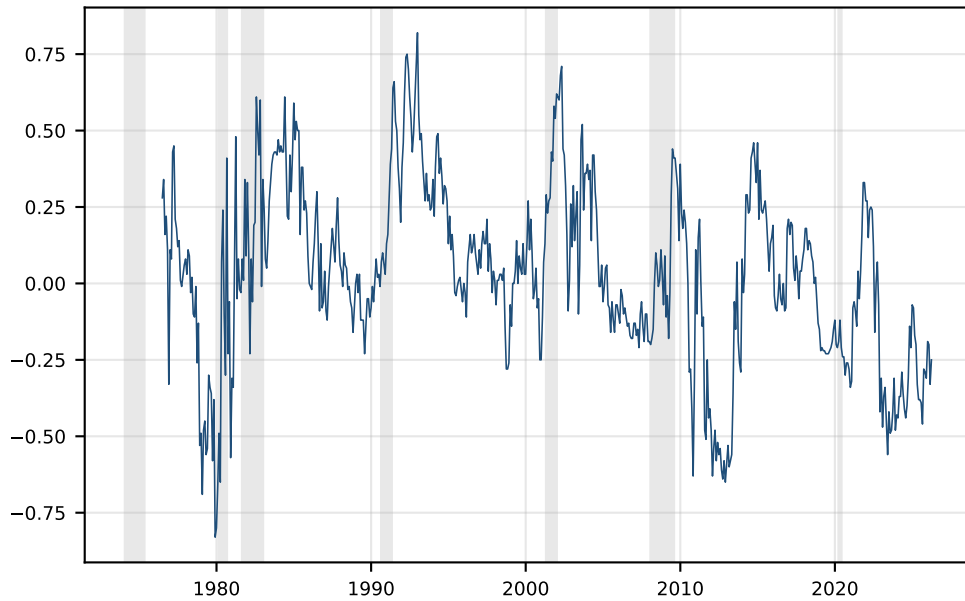
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	49	4.1%	95.9%	1.7435
Q2	51	0.0%	100.0%	2.1088
Q3	45	0.0%	100.0%	2.2680
Q4 (highest)	48	0.0%	100.0%	2.4494

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

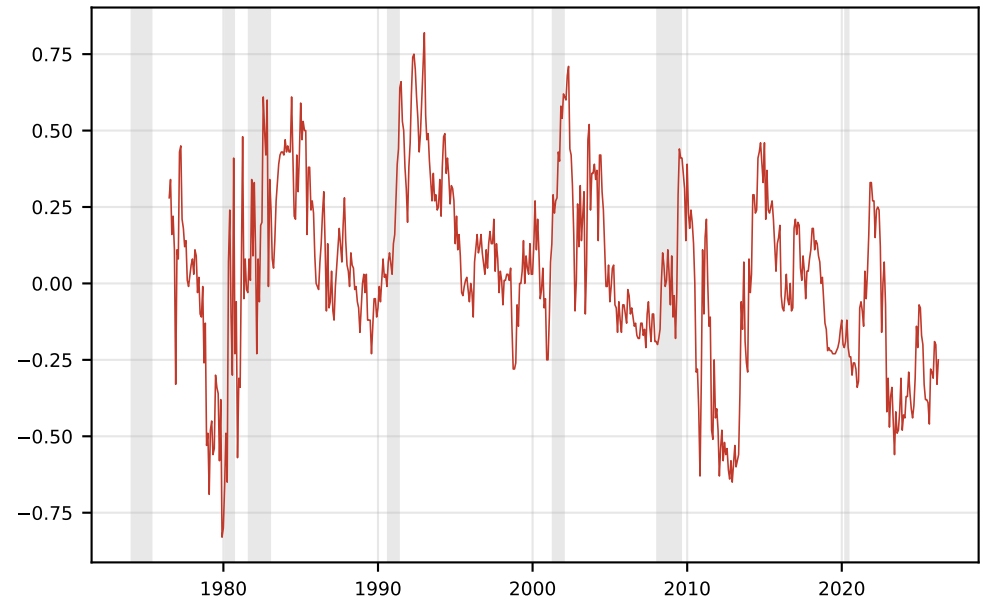
Curve Curvature

Block: Selected export / Curve-derived variables | Source: Derived | Col: derived_curve_curvature | Transform: nan

Raw Series



After: nan



SUMMARY STATISTICS

N : 1287
Valid : 598
NaN : 689
Start : 1976-06
End : 2026-03
Mean : 0.0330
Std : 0.2917
Min : -0.8300
Max : 0.8200
Skew : -0.137
Kurt : -0.073

STATIONARITY TESTS

ADF stat : -4.117
ADF p : 0.0009
ADF lags : 12
ADF : Stationary
KPSS stat : 0.662
KPSS p : 0.0170
KPSS : Non-stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

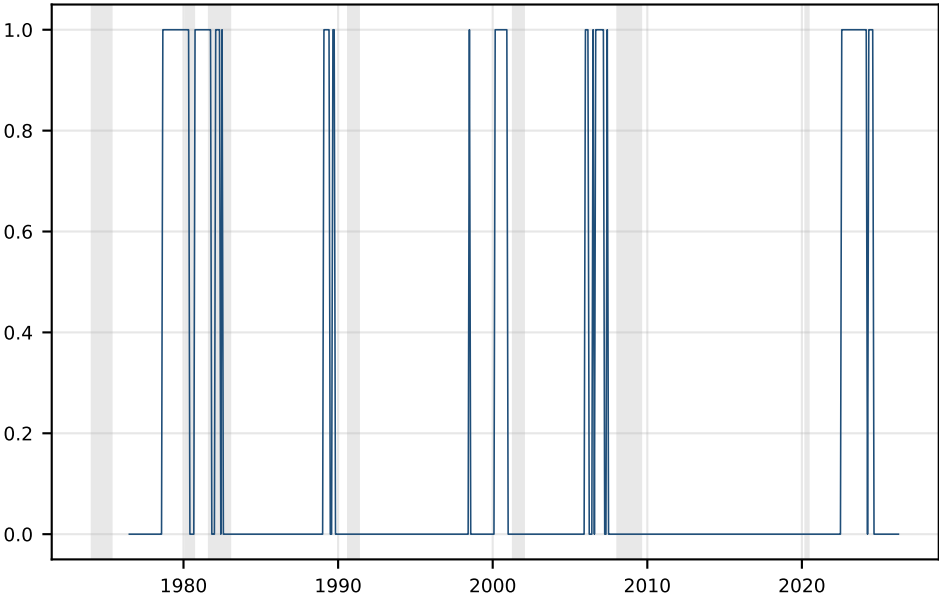
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	151	6.0%	94.0%	-0.3414
Q2	149	9.4%	90.6%	-0.0419
Q3	148	10.8%	89.2%	0.1197
Q4 (highest)	149	12.8%	87.2%	0.4032

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

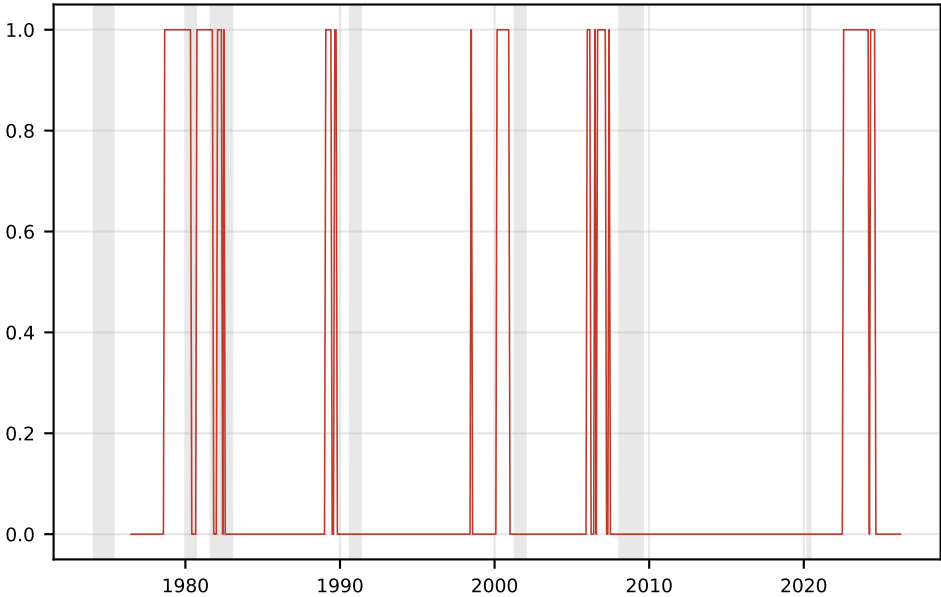
Inversion Dummy 10Y Minus 2Y

Block: Selected export / Curve-derived variables | Source: Derived | Col: derived_inv_10y_2y | Transform: nan

Raw Series



After: nan



SUMMARY STATISTICS

N : 1287
Valid : 598
NaN : 689
Start : 1976-06
End : 2026-03
Mean : 0.1555
Std : 0.3627
Min : 0.0000
Max : 1.0000
Skew : 1.906
Kurt : 1.638

STATIONARITY TESTS

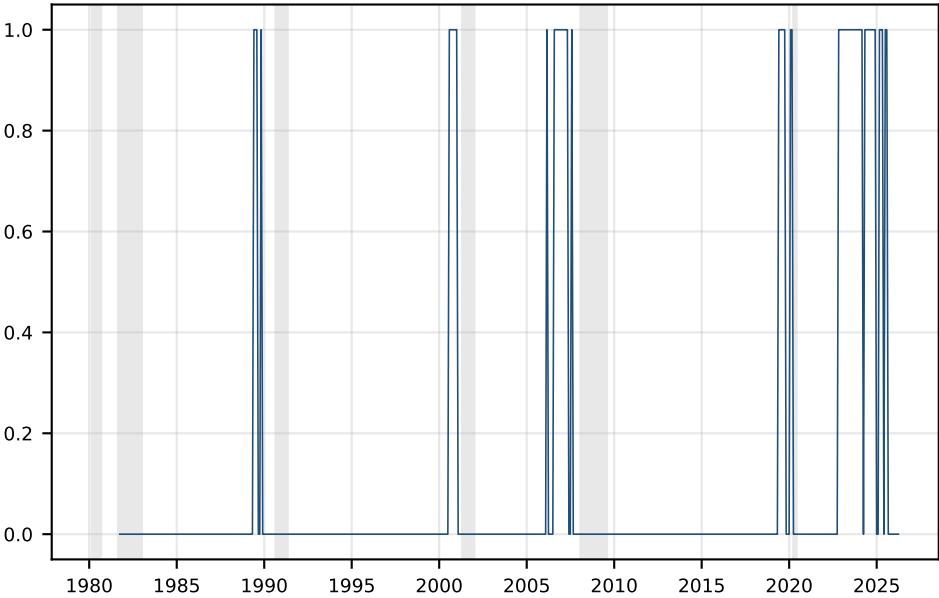
ADF stat : -4.696
ADF p : 0.0001
ADF lags : 4
ADF : Stationary
KPSS stat : 0.211
KPSS p : 0.1000
KPSS : Stationary

Quartile analysis not available

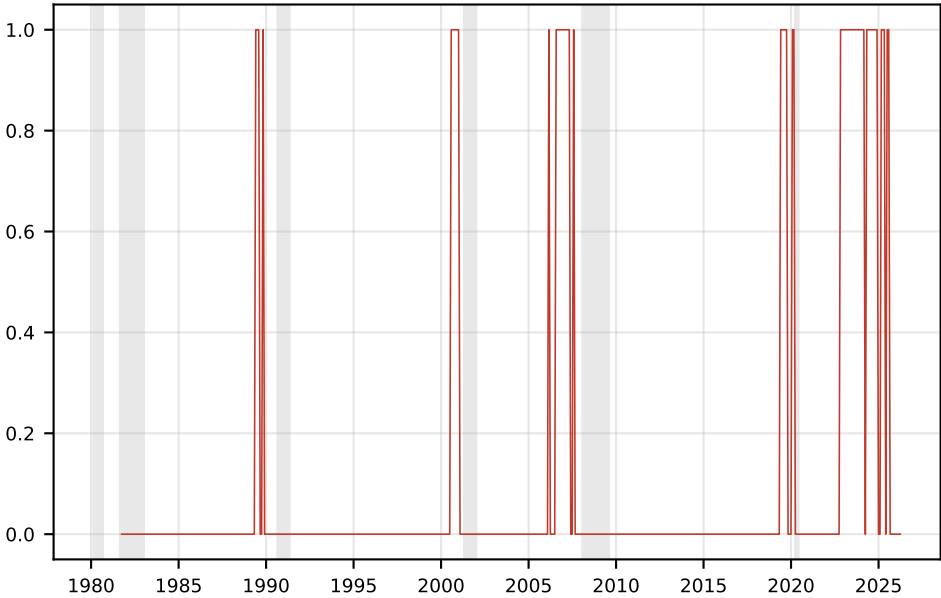
Inversion Dummy 10Y Minus 3M

Block: Selected export / Curve-derived variables | Source: Derived | Col: derived_inv_10y_3m | Transform: nan

Raw Series



After: nan



SUMMARY STATISTICS

N : 1287
Valid : 535
NaN : 752
Start : 1981-09
End : 2026-03
Mean : 0.1103
Std : 0.3135
Min : 0.0000
Max : 1.0000
Skew : 2.495
Kurt : 4.242

STATIONARITY TESTS

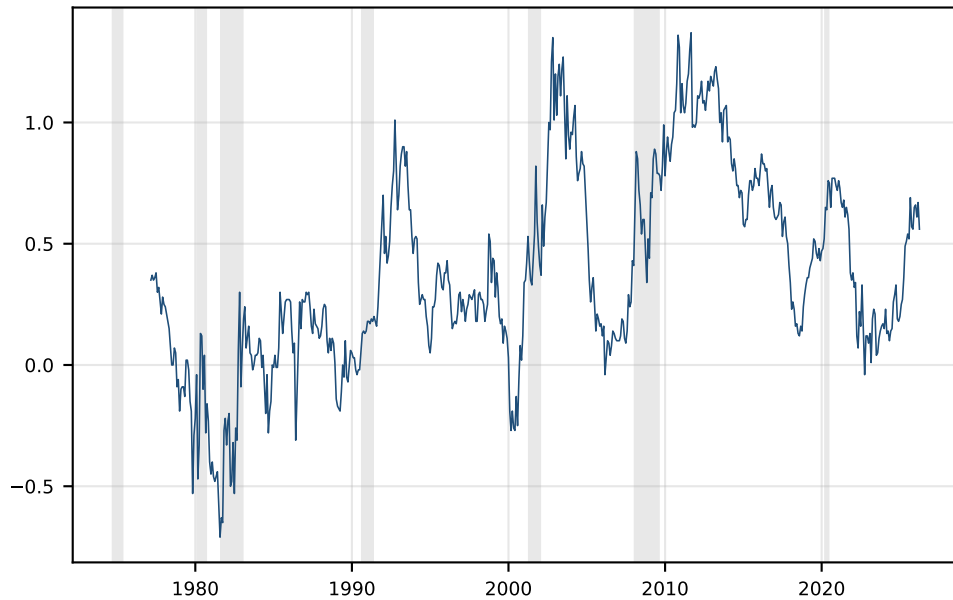
ADF stat : -4.444
ADF p : 0.0002
ADF lags : 6
ADF : Stationary
KPSS stat : 0.656
KPSS p : 0.0176
KPSS : Non-stationary

Quartile analysis not available

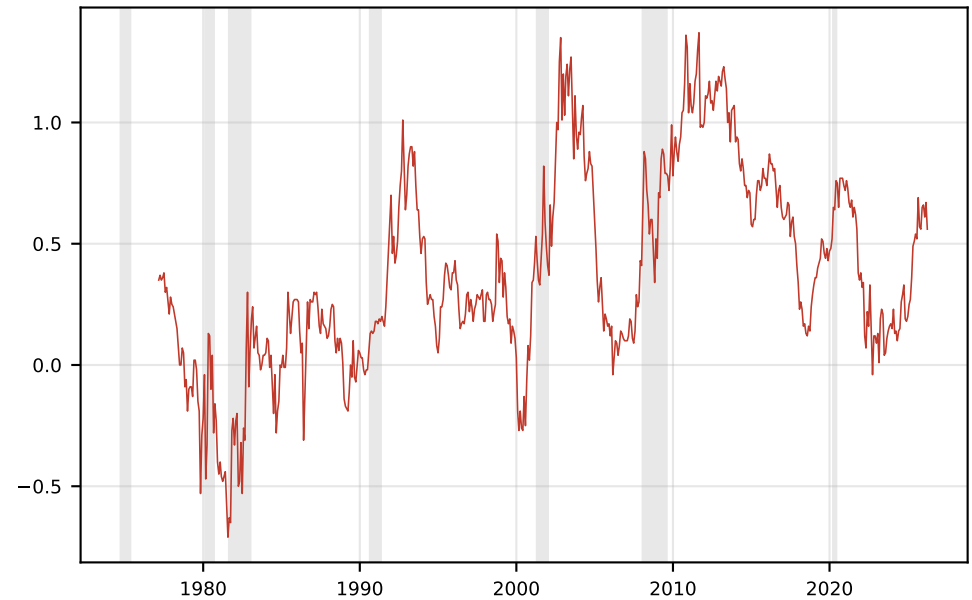
Long-End Slope

Block: Selected export / Curve-derived variables | Source: Derived | Col: derived_long_end_slope | Transform: nan

Raw Series



After: nan



SUMMARY STATISTICS

N : 1287
Valid : 590
NaN : 697
Start : 1977-02
End : 2026-03
Mean : 0.3782
Std : 0.4020
Min : -0.7100
Max : 1.3700
Skew : 0.227
Kurt : -0.339

STATIONARITY TESTS

ADF stat : -2.547
ADF p : 0.1045
ADF lags : 2
ADF : Non-stationary
KPSS stat : 1.383
KPSS p : 0.0100
KPSS : Non-stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	149	13.4%	86.6%	-0.0850
Q2	149	6.0%	94.0%	0.2030
Q3	145	13.1%	86.9%	0.4765
Q4 (highest)	146	6.8%	93.2%	0.9311

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

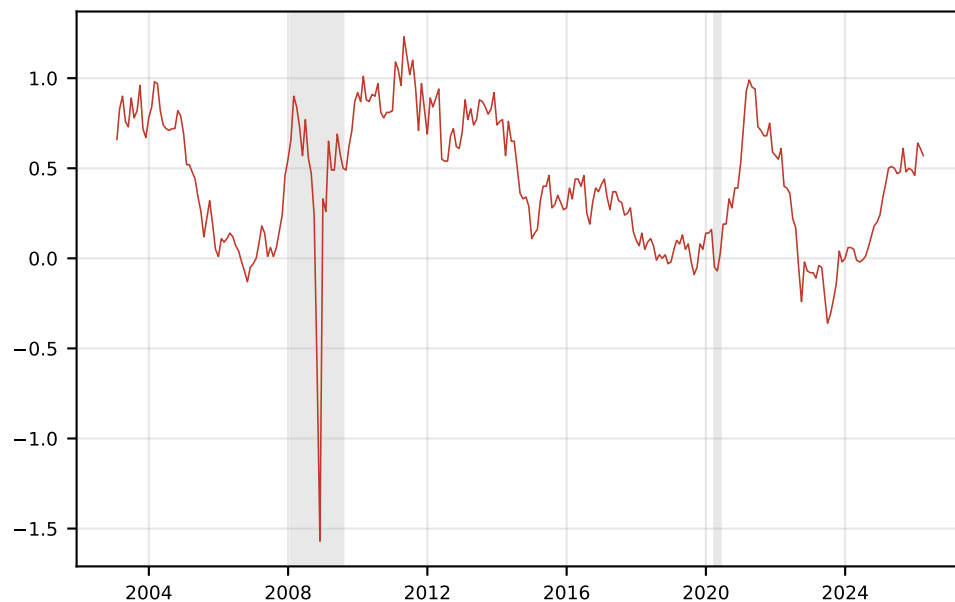
Real Slope 10Y Minus 5Y

Block: Selected export / Curve-derived variables | Source: Derived | Col: derived_real_slope_10y_5y | Transform: nan

Raw Series



After: nan



SUMMARY STATISTICS

N : 1287
Valid : 279
NaN : 1008
Start : 2003-01
End : 2026-03
Mean : 0.4165
Std : 0.3683
Min : -1.5700
Max : 1.2300
Skew : -0.536
Kurt : 1.802

STATIONARITY TESTS

ADF stat : -4.039
ADF p : 0.0012
ADF lags : 0
ADF : Stationary
KPSS stat : 0.500
KPSS p : 0.0416
KPSS : Non-stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	70	5.7%	94.3%	-0.0407
Q2	70	4.3%	95.7%	0.2646
Q3	68	13.2%	86.8%	0.5738
Q4 (highest)	70	5.7%	94.3%	0.8706

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

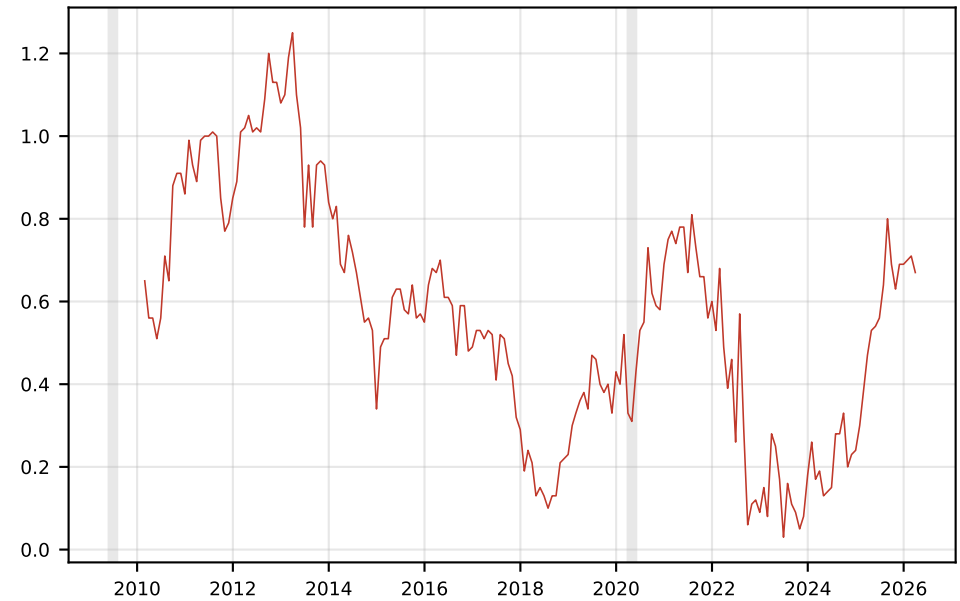
Real Slope 30Y Minus 10Y

Block: Selected export / Curve-derived variables | Source: Derived | Col: derived_real_slope_30y_10y | Transform: nan

Raw Series



After: nan



SUMMARY STATISTICS

N : 1287
Valid : 194
NaN : 1093
Start : 2010-02
End : 2026-03
Mean : 0.5630
Std : 0.2876
Min : 0.0300
Max : 1.2500
Skew : 0.155
Kurt : -0.673

STATIONARITY TESTS

ADF stat : -1.672
ADF p : 0.4456
ADF lags : 1
ADF : Non-stationary
KPSS stat : 1.002
KPSS p : 0.0100
KPSS : Non-stationary

QUARTILE-BUCKET ANALYSIS

(vs NBER recession/expansion)

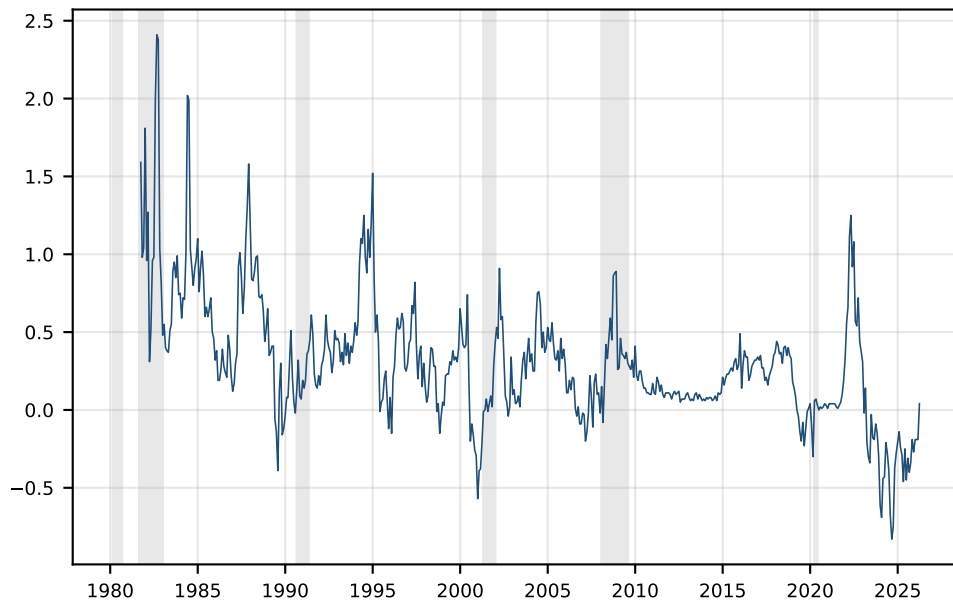
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	50	4.0%	96.0%	0.1948
Q2	50	0.0%	100.0%	0.4824
Q3	45	0.0%	100.0%	0.6511
Q4 (highest)	48	0.0%	100.0%	0.9456

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

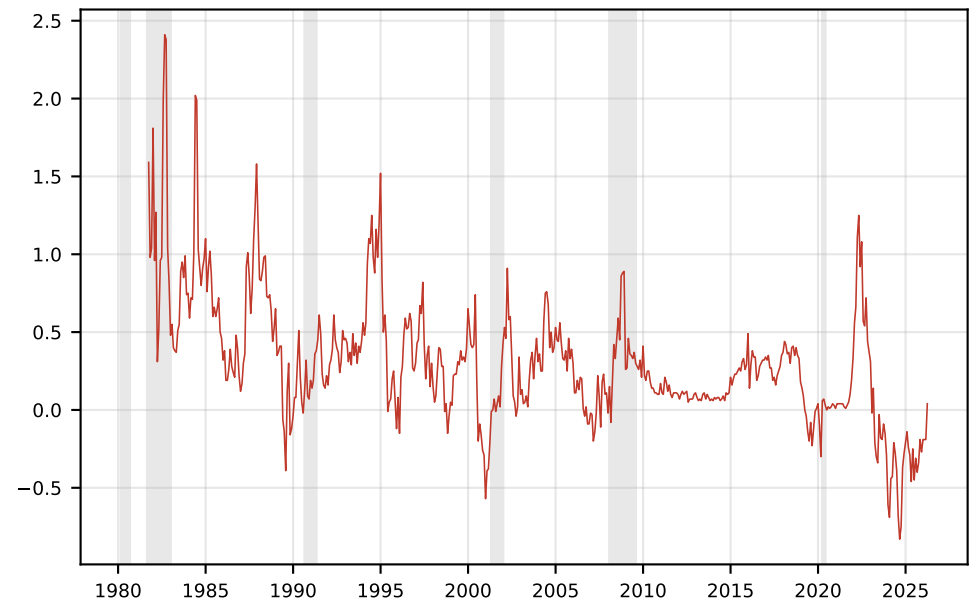
Short-End Slope

Block: Selected export / Curve-derived variables | Source: Derived | Col: derived_short_end_slope | Transform: nan

Raw Series



After: nan



SUMMARY STATISTICS

N : 1287
Valid : 535
NaN : 752
Start : 1981-09
End : 2026-03
Mean : 0.3053
Std : 0.4080
Min : -0.8300
Max : 2.4100
Skew : 1.230
Kurt : 4.017

STATIONARITY TESTS

ADF stat : -4.474
ADF p : 0.0002
ADF lags : 4
ADF : Stationary
KPSS stat : 1.662
KPSS p : 0.0100
KPSS : Non-stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

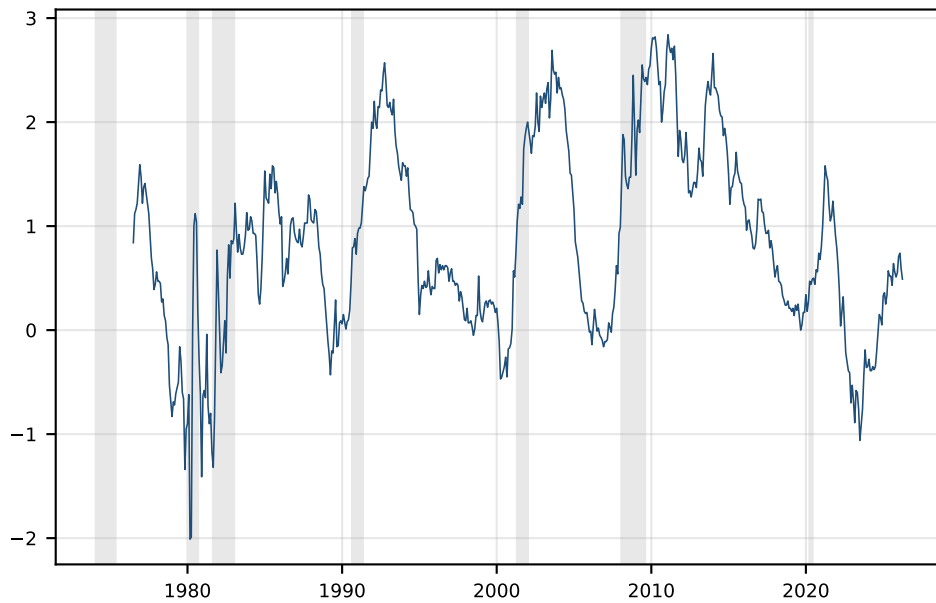
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	138	6.5%	93.5%	-0.1130
Q2	131	7.6%	92.4%	0.1598
Q3	132	9.1%	90.9%	0.3559
Q4 (highest)	133	15.0%	85.0%	0.8343

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

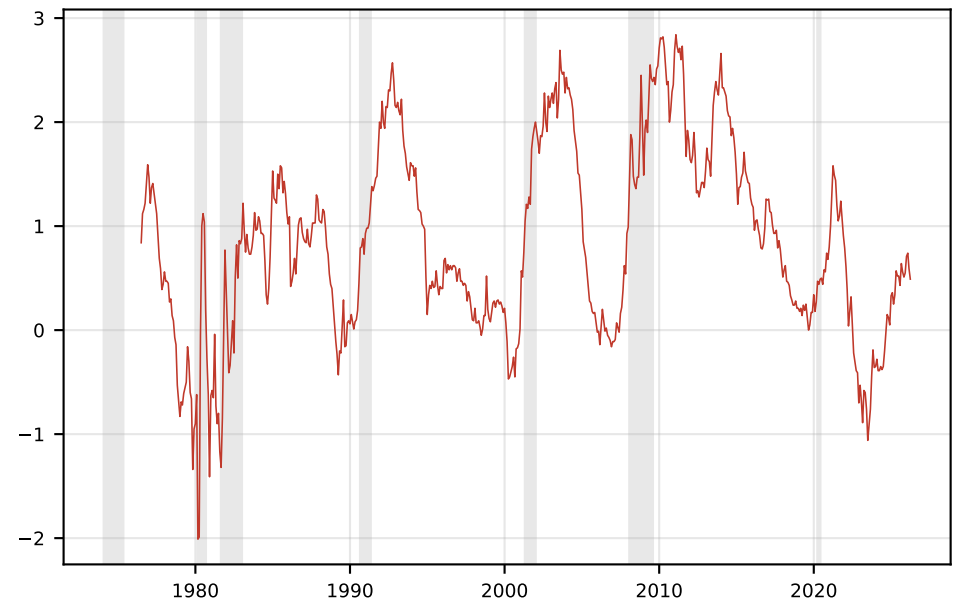
Slope 10Y Minus 2Y

Block: Selected export / Curve-derived variables | Source: Derived | Col: derived_slope_10y_2y | Transform: nan

Raw Series



After: nan



SUMMARY STATISTICS

N : 1287
Valid : 598
NaN : 689
Start : 1976-06
End : 2026-03
Mean : 0.8475
Std : 0.9113
Min : -2.0100
Max : 2.8400
Skew : 0.088
Kurt : -0.434

STATIONARITY TESTS

ADF stat : -3.367
ADF p : 0.0121
ADF lags : 9
ADF : Stationary
KPSS stat : 0.320
KPSS p : 0.1000
KPSS : Stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

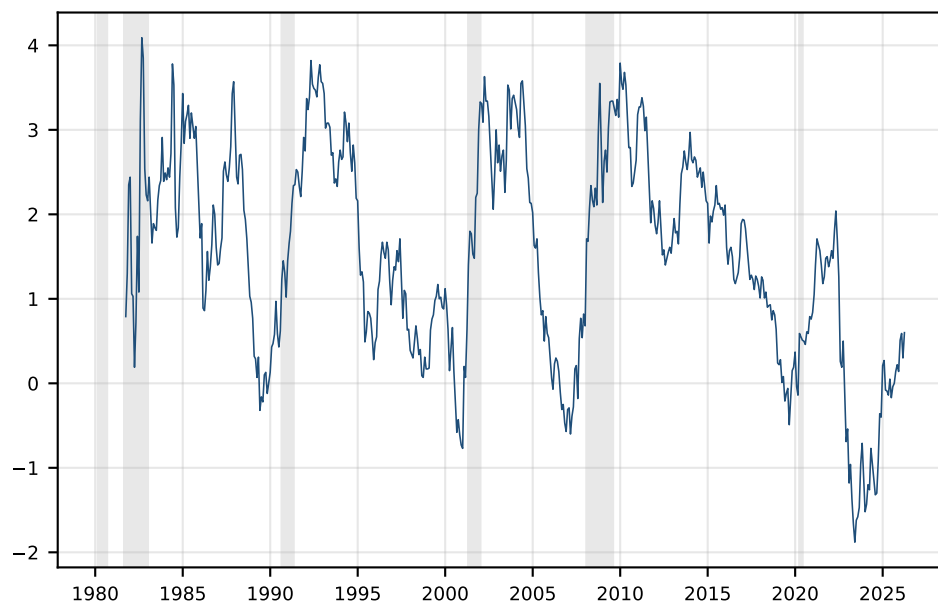
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	150	8.0%	92.0%	-0.2577
Q2	149	5.4%	94.6%	0.4662
Q3	150	14.7%	85.3%	1.1225
Q4 (highest)	148	10.8%	89.2%	2.0753

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

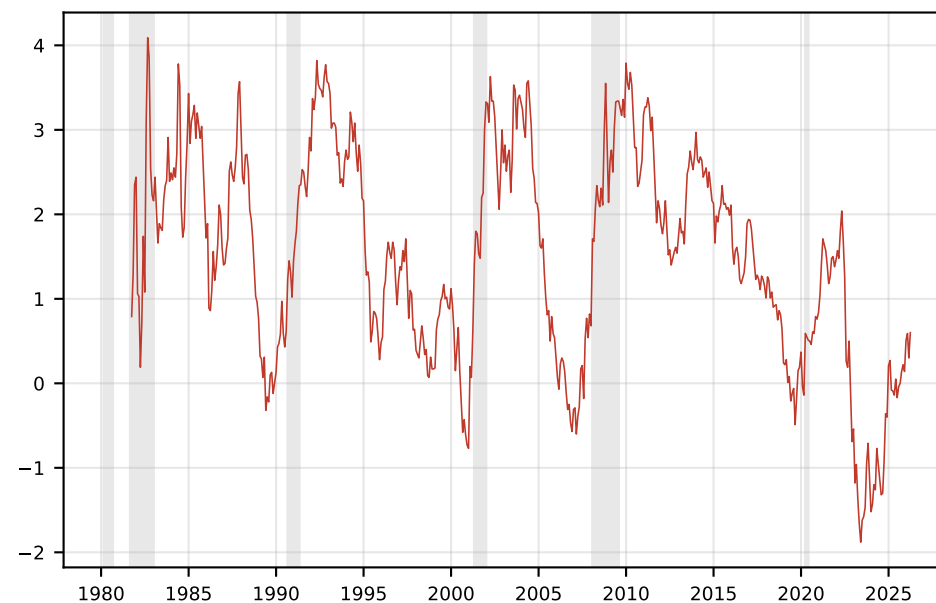
Slope 10Y Minus 3M

Block: Selected export / Curve-derived variables | Source: Derived | Col: derived_slope_10y_3m | Transform: nan

Raw Series



After: nan



SUMMARY STATISTICS

N : 1287
Valid : 535
NaN : 752
Start : 1981-09
End : 2026-03
Mean : 1.5331
Std : 1.2500
Min : -1.8800
Max : 4.0900
Skew : -0.270
Kurt : -0.608

STATIONARITY TESTS

ADF stat : -3.605
ADF p : 0.0057
ADF lags : 8
ADF : Stationary
KPSS stat : 0.581
KPSS p : 0.0243
KPSS : Non-stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

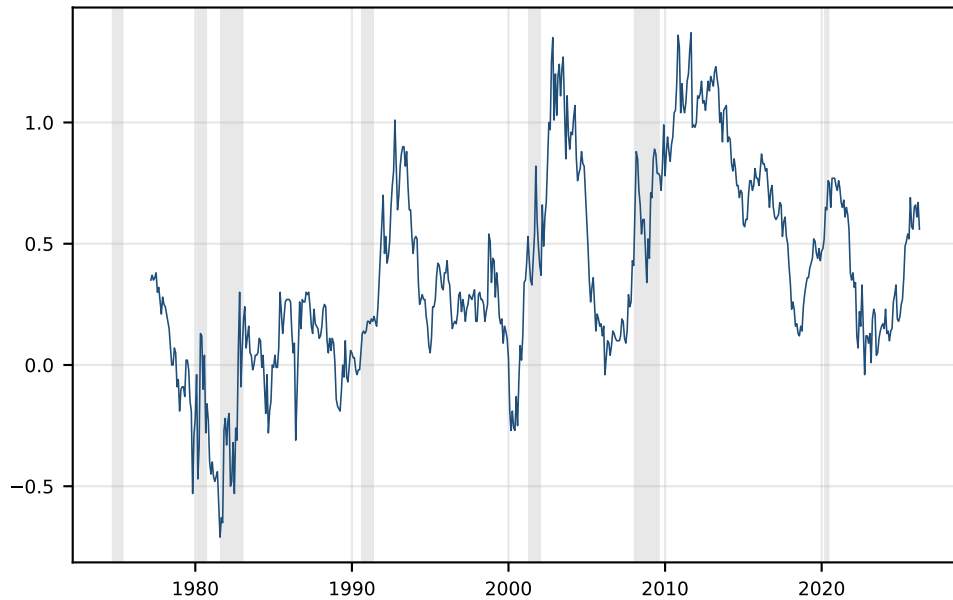
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	134	1.5%	98.5%	-0.1213
Q2	133	11.3%	88.7%	1.1195
Q3	134	15.7%	84.3%	2.0716
Q4 (highest)	133	9.8%	90.2%	3.0780

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

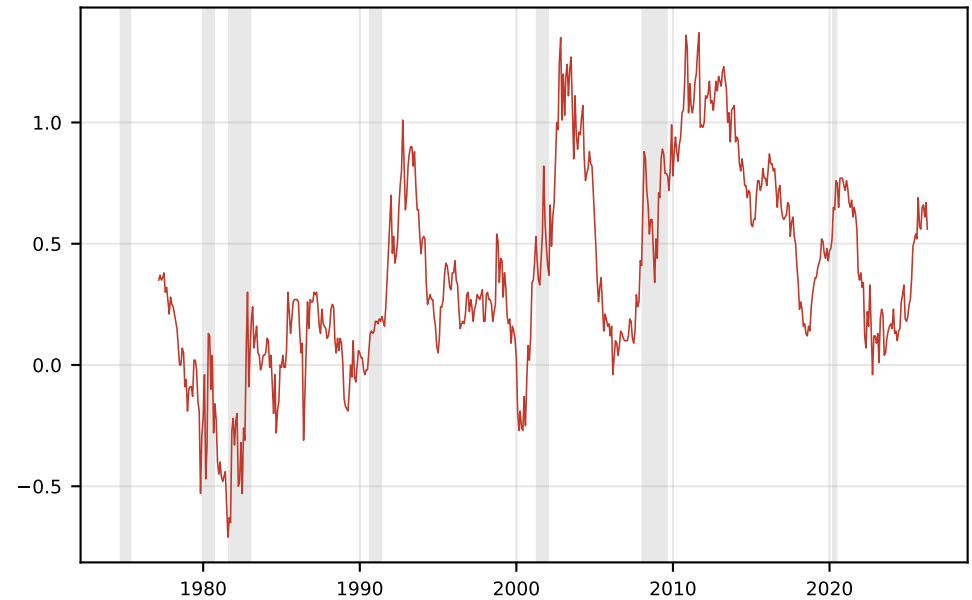
Slope 30Y Minus 10Y

Block: Selected export / Curve-derived variables | Source: Derived | Col: derived_slope_30y_10y | Transform: nan

Raw Series



After: nan



SUMMARY STATISTICS

N : 1287
Valid : 590
NaN : 697
Start : 1977-02
End : 2026-03
Mean : 0.3782
Std : 0.4020
Min : -0.7100
Max : 1.3700
Skew : 0.227
Kurt : -0.339

STATIONARITY TESTS

ADF stat : -2.547
ADF p : 0.1045
ADF lags : 2
ADF : Non-stationary
KPSS stat : 1.383
KPSS p : 0.0100
KPSS : Non-stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

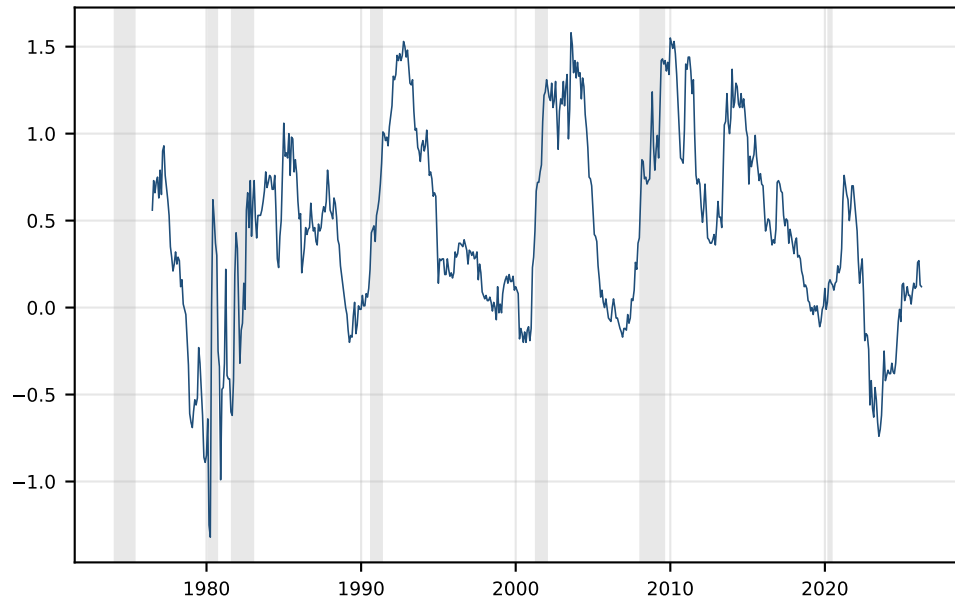
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	149	13.4%	86.6%	-0.0850
Q2	149	6.0%	94.0%	0.2030
Q3	145	13.1%	86.9%	0.4765
Q4 (highest)	146	6.8%	93.2%	0.9311

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

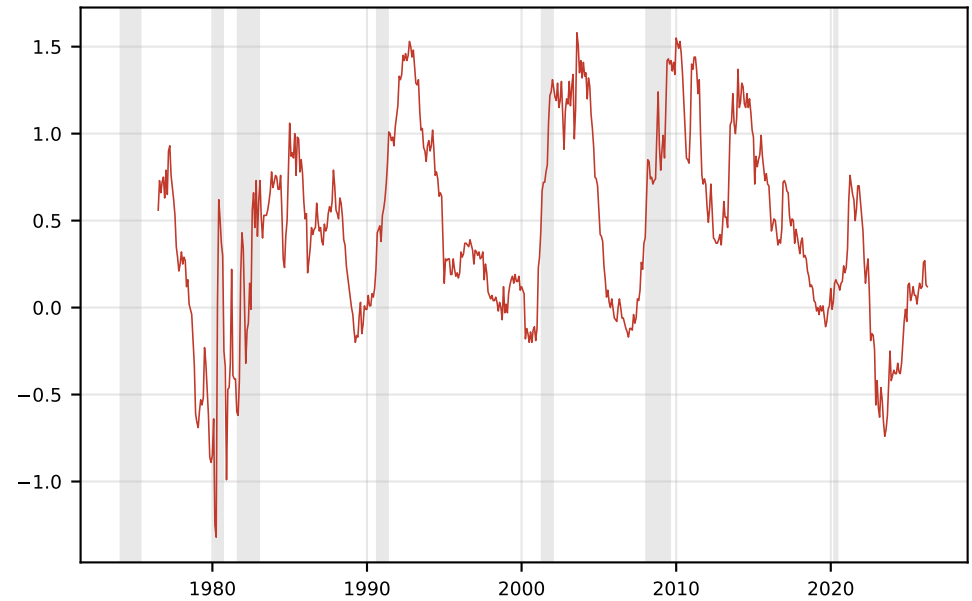
Slope 5Y Minus 2Y

Block: Selected export / Curve-derived variables | Source: Derived | Col: derived_slope_5y_2y | Transform: nan

Raw Series



After: nan



SUMMARY STATISTICS

N : 1287
Valid : 598
NaN : 689
Start : 1976-06
End : 2026-03
Mean : 0.4403
Std : 0.5370
Min : -1.3200
Max : 1.5800
Skew : -0.029
Kurt : -0.292

STATIONARITY TESTS

ADF stat : -3.614
ADF p : 0.0055
ADF lags : 9
ADF : Stationary
KPSS stat : 0.223
KPSS p : 0.1000
KPSS : Stationary

QUARTILE-BUCKET ANALYSIS

(vs NBER recession/expansion)

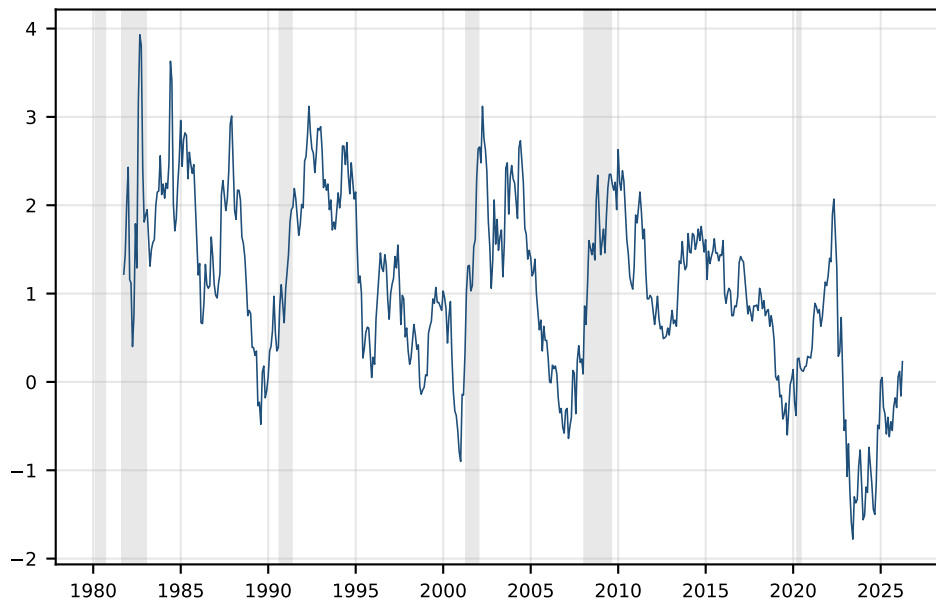
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	150	6.7%	93.3%	-0.2197
Q2	149	4.7%	95.3%	0.2344
Q3	149	16.8%	83.2%	0.6042
Q4 (highest)	149	10.7%	89.3%	1.1487

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

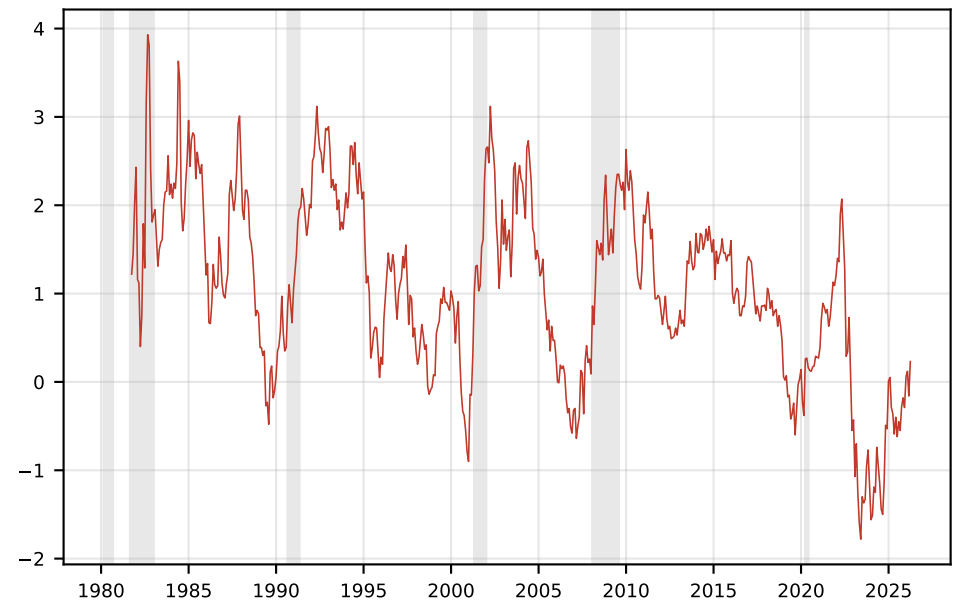
Slope 5Y Minus 3M

Block: Selected export / Curve-derived variables | Source: Derived | Col: derived_slope_5y_3m | Transform: nan

Raw Series



After: nan



SUMMARY STATISTICS

N : 1287
Valid : 535
NaN : 752
Start : 1981-09
End : 2026-03
Mean : 1.0863
Std : 1.0141
Min : -1.7800
Max : 3.9300
Skew : -0.205
Kurt : -0.201

STATIONARITY TESTS

ADF stat : -3.646
ADF p : 0.0049
ADF lags : 2
ADF : Stationary
KPSS stat : 1.073
KPSS p : 0.0100
KPSS : Non-stationary

QUARTILE-BUCKET ANALYSIS

(vs NBER recession/expansion)

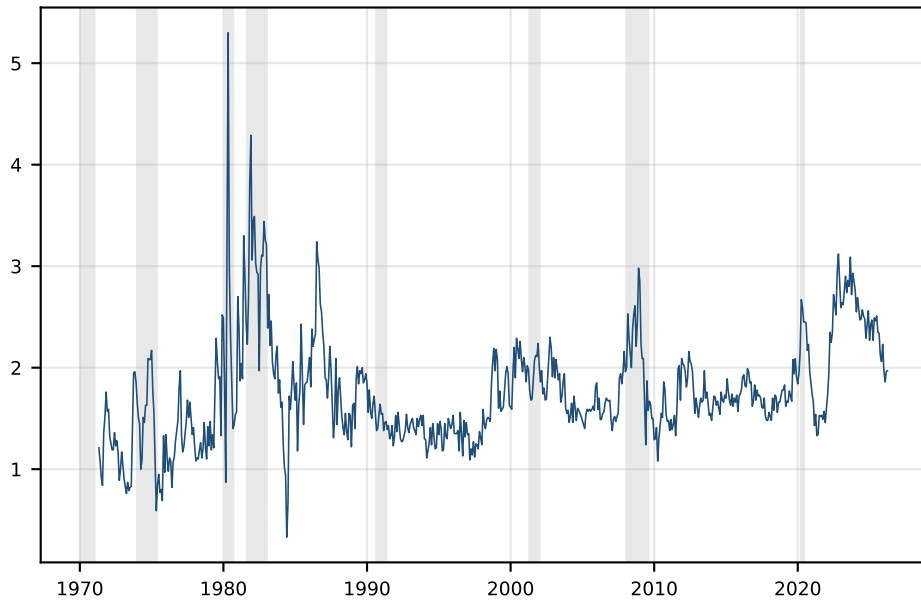
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	135	1.5%	98.5%	-0.2356
Q2	132	9.1%	90.9%	0.7900
Q3	134	17.9%	82.1%	1.4526
Q4 (highest)	133	9.8%	90.2%	2.3595

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

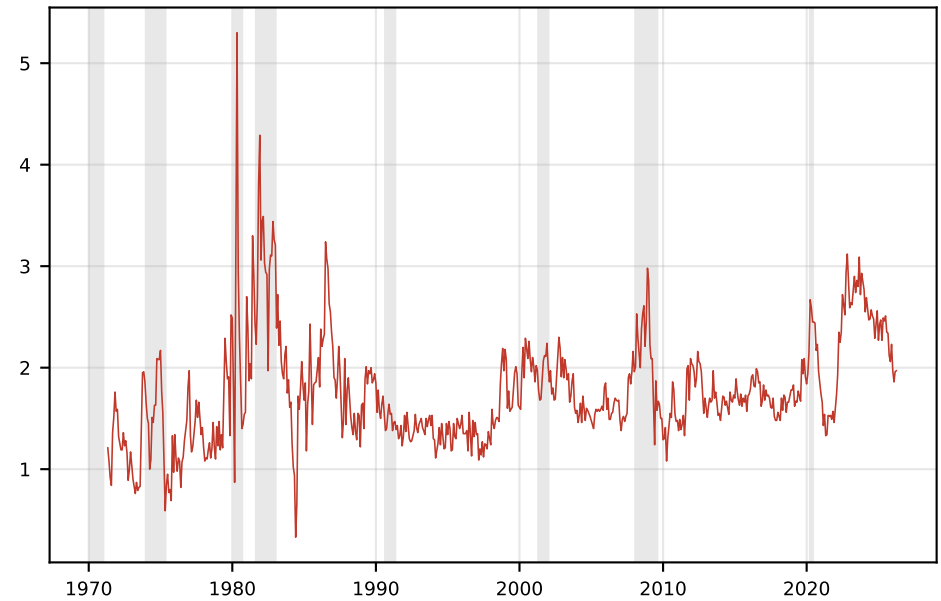
30Y Mortgage Rate Minus 10Y Treasury Yield

Block: Selected export / Housing-derived variables | Source: Derived | Col: derived_mortgage30_minus_10y | Transform: nan

Raw Series



After: nan



SUMMARY STATISTICS

N : 1287
Valid : 660
NaN : 627
Start : 1971-04
End : 2026-03
Mean : 1.7619
Std : 0.5301
Min : 0.3300
Max : 5.3000
Skew : 1.315
Kurt : 4.157

STATIONARITY TESTS

ADF stat : -3.890
ADF p : 0.0021
ADF lags : 7
ADF : Stationary
KPSS stat : 0.459
KPSS p : 0.0516
KPSS : Stationary

QUARTILE-BUCKET ANALYSIS

(vs NBER recession/expansion)

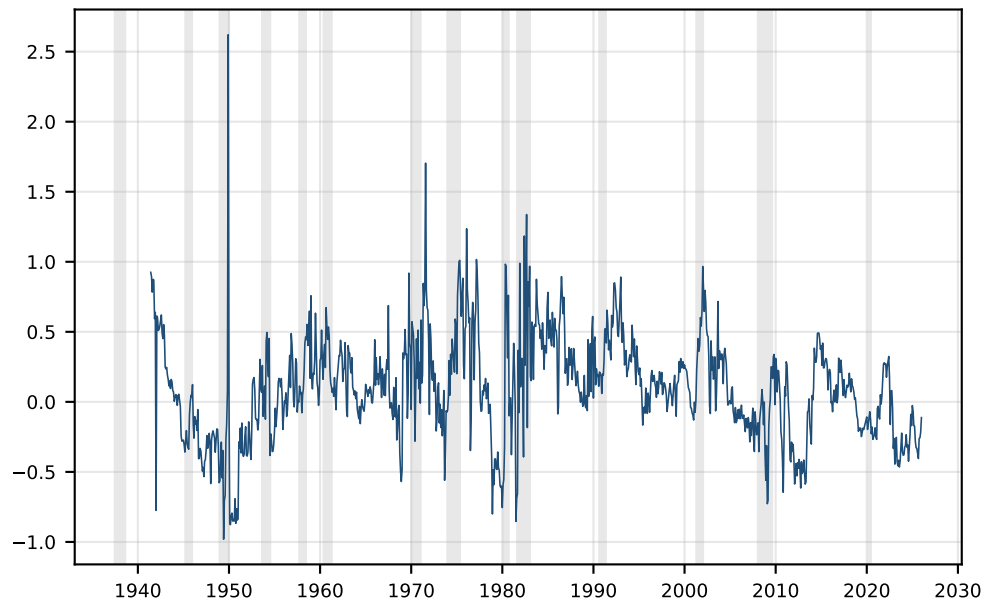
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	167	4.8%	95.2%	1.2164
Q2	166	8.4%	91.6%	1.5502
Q3	161	5.0%	95.0%	1.8180
Q4 (highest)	165	26.7%	73.3%	2.4712

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

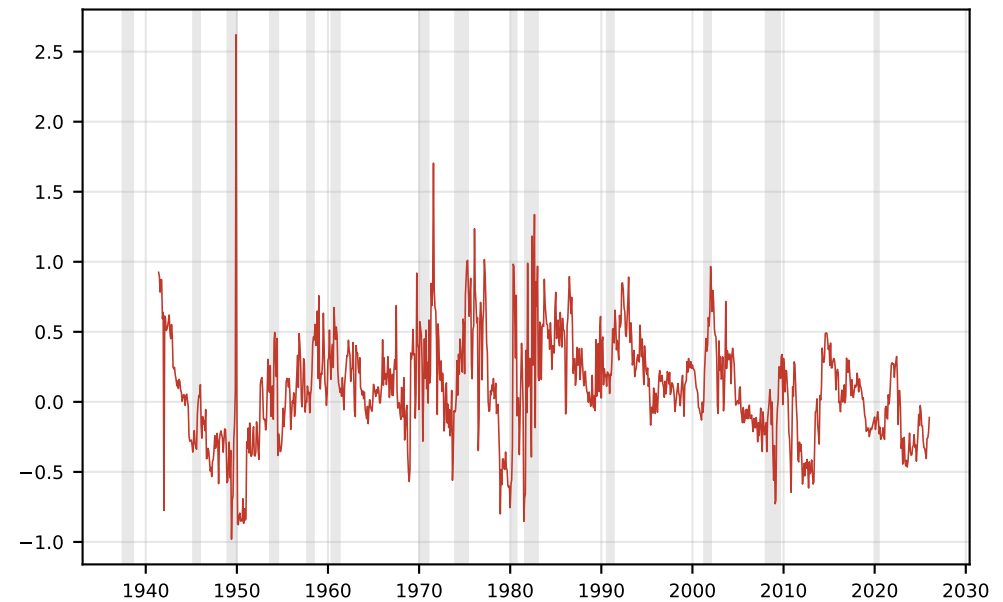
CRSP Curve Curvature

Block: Selected export / Treasury term structure | Source: WRDS | Col: derived_curve_curvature_wrds | Transform: nan

Raw Series



After: nan



SUMMARY STATISTICS

N : 1287
Valid : 1016
NaN : 271
Start : 1941-05
End : 2025-12
Mean : 0.0944
Std : 0.3688
Min : -0.9802
Max : 2.6208
Skew : 0.414
Kurt : 2.390

STATIONARITY TESTS

ADF stat : -4.875
ADF p : 0.0000
ADF lags : 8
ADF : Stationary
KPSS stat : 0.434
KPSS p : 0.0626
KPSS : Stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

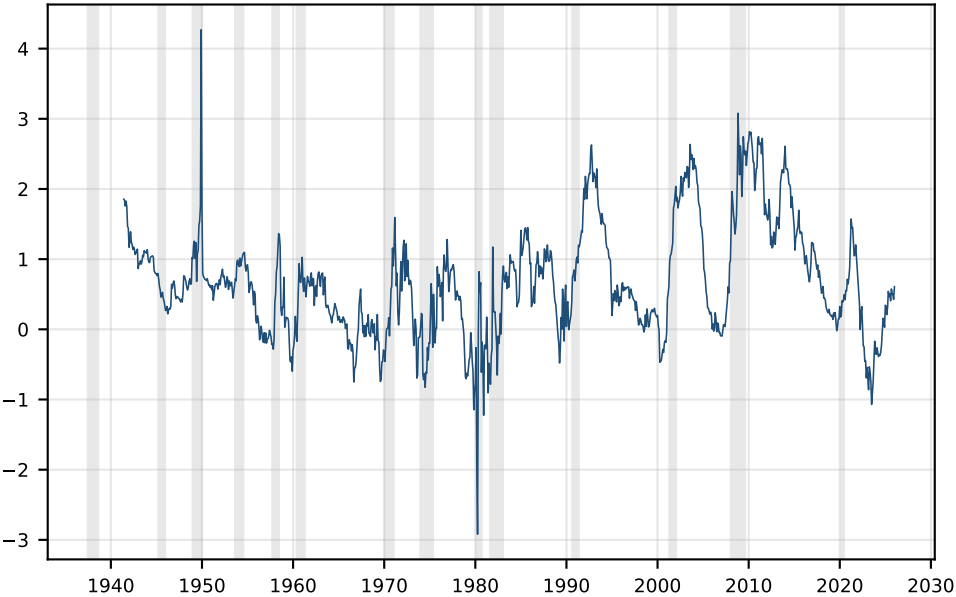
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	254	12.2%	87.8%	-0.3589
Q2	254	9.8%	90.2%	-0.0122
Q3	254	12.6%	87.4%	0.1897
Q4 (highest)	254	17.3%	82.7%	0.5590

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

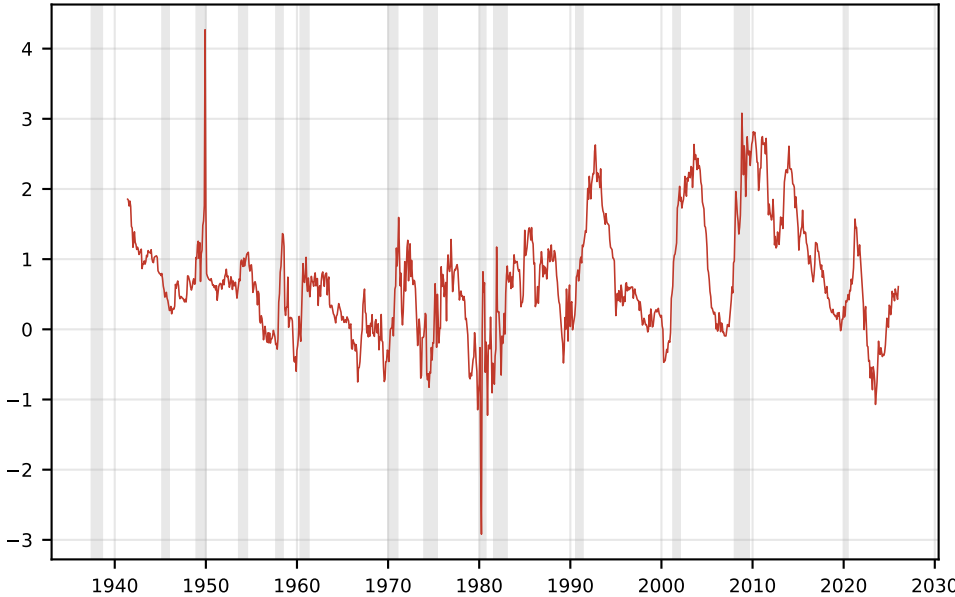
CRSP Slope 10Y Minus 2Y

Block: Selected export / Treasury term structure | Source: WRDS | Col: derived_slope_10y_2y_wrds | Transform: nan

Raw Series



After: nan



SUMMARY STATISTICS

N	: 1287
Valid	: 1016
NaN	: 271
Start	: 1941-05
End	: 2025-12
Mean	: 0.6799
Std	: 0.7929
Min	: -2.9191
Max	: 4.2692
Skew	: 0.517
Kurt	: 0.794

STATIONARITY TESTS

ADF stat	: -4.365
ADF p	: 0.0003
ADF lags	: 11
ADF	: Stationary
KPSS stat	: 0.577
KPSS p	: 0.0248
KPSS	: Non-stationary

QUARTILE-BUCKET ANALYSIS
(vs NBER recession/expansion)

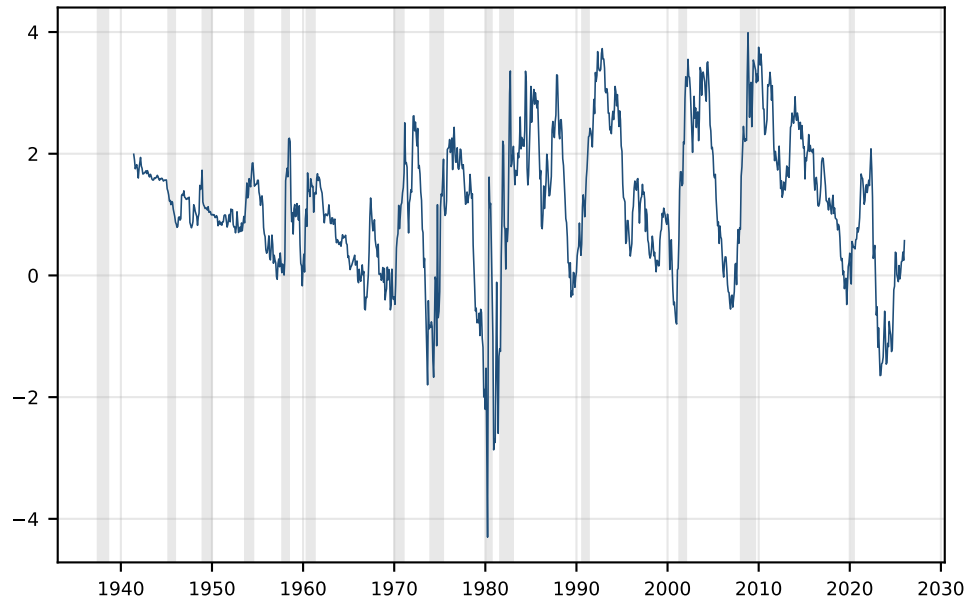
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	254	13.4%	86.6%	-0.2247
Q2	254	10.2%	89.8%	0.3805
Q3	254	15.4%	84.6%	0.8116
Q4 (highest)	254	13.0%	87.0%	1.7521

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

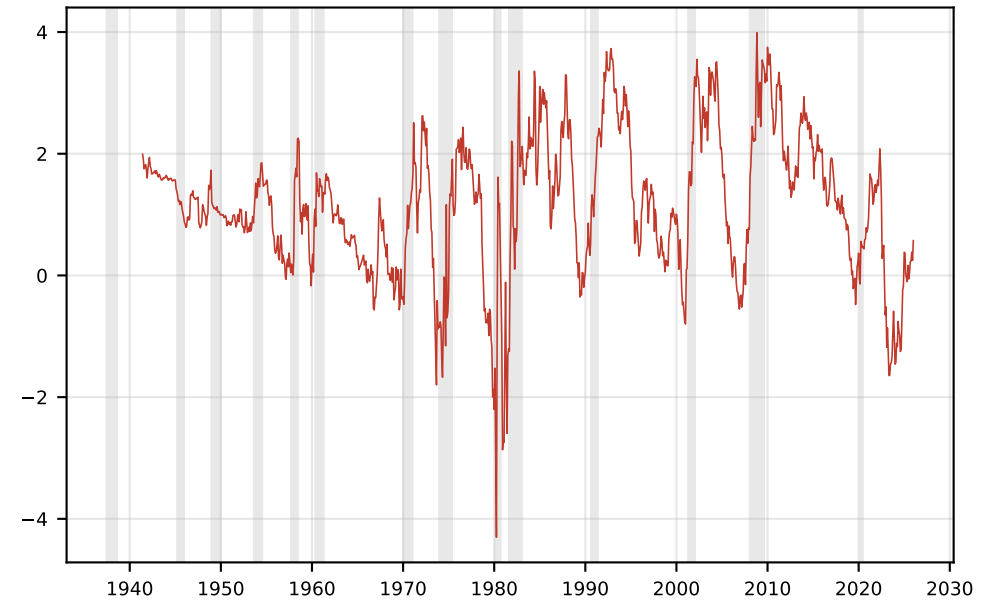
CRSP Slope 10Y Minus 3M

Block: Selected export / Treasury term structure | Source: WRDS | Col: derived_slope_10y_3m_wrds | Transform: nan

Raw Series



After: nan



SUMMARY STATISTICS

N : 1287
Valid : 1016
NaN : 271
Start : 1941-05
End : 2025-12
Mean : 1.1605
Std : 1.1457
Min : -4.3013
Max : 3.9884
Skew : -0.305
Kurt : 0.662

STATIONARITY TESTS

ADF stat : -5.329
ADF p : 0.0000
ADF lags : 10
ADF : Stationary
KPSS stat : 0.277
KPSS p : 0.1000
KPSS : Stationary

QUARTILE-BUCKET ANALYSIS

(vs NBER recession/expansion)

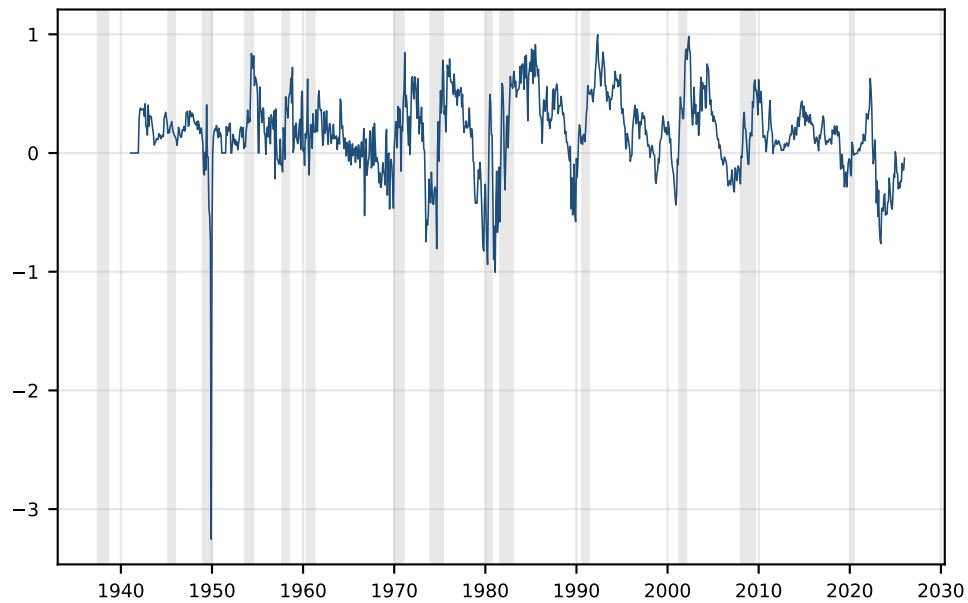
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	254	9.4%	90.6%	-0.2983
Q2	254	14.2%	85.8%	0.8493
Q3	254	18.1%	81.9%	1.4940
Q4 (highest)	254	10.2%	89.8%	2.5969

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

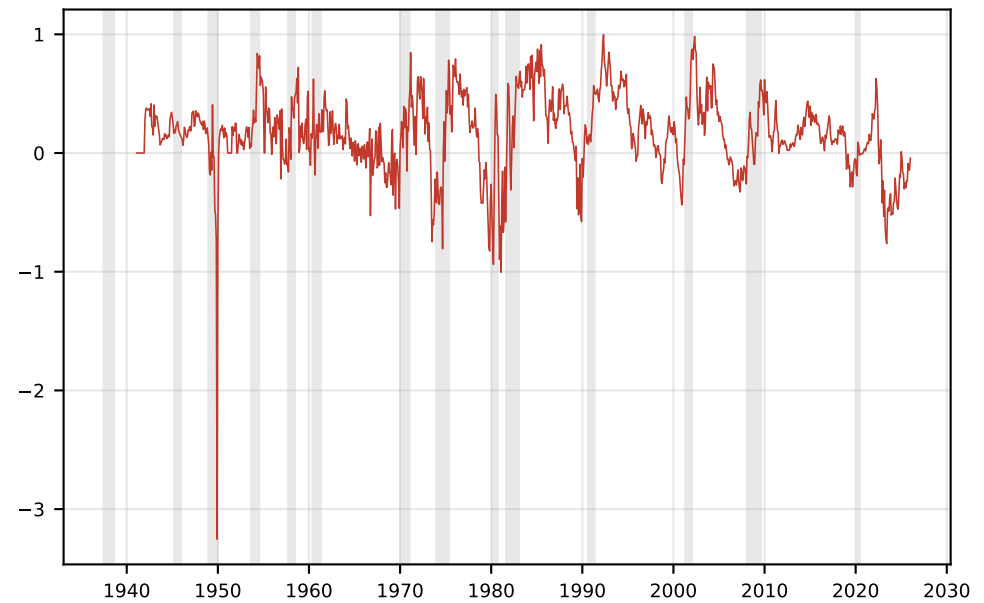
CRSP Slope 2Y Minus 1Y

Block: Selected export / Treasury term structure | Source: WRDS | Col: derived_slope_2y_1y_wrds | Transform: nan

Raw Series



After: nan



SUMMARY STATISTICS

N : 1287
Valid : 1020
NaN : 267
Start : 1941-01
End : 2025-12
Mean : 0.1641
Std : 0.3284
Min : -3.2532
Max : 0.9966
Skew : -1.385
Kurt : 11.517

STATIONARITY TESTS

ADF stat : -6.435
ADF p : 0.0000
ADF lags : 4
ADF : Stationary
KPSS stat : 0.177
KPSS p : 0.1000
KPSS : Stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

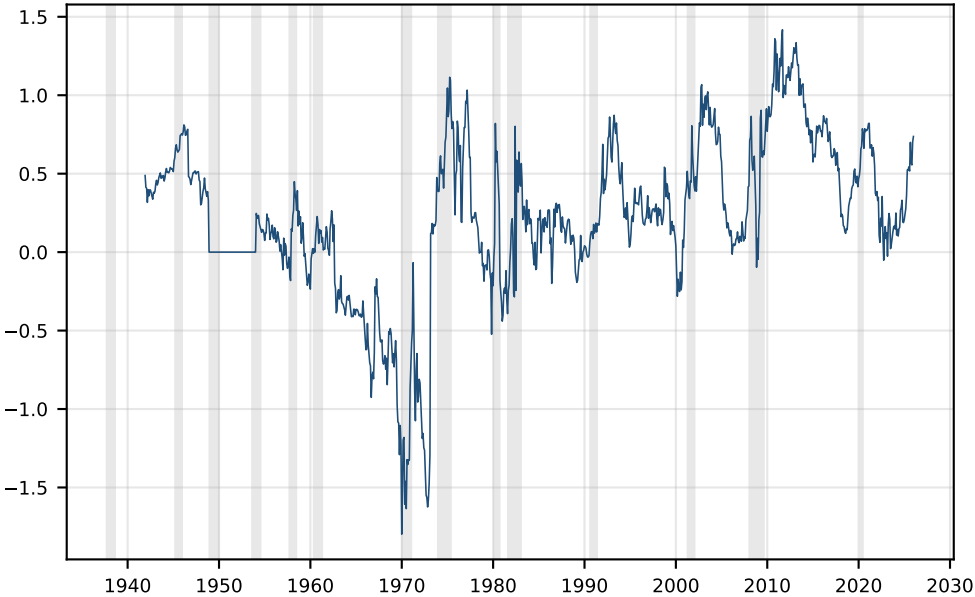
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	264	12.9%	87.1%	-0.2318
Q2	246	10.2%	89.8%	0.0979
Q3	255	16.1%	83.9%	0.2525
Q4 (highest)	255	12.5%	87.5%	0.5494

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

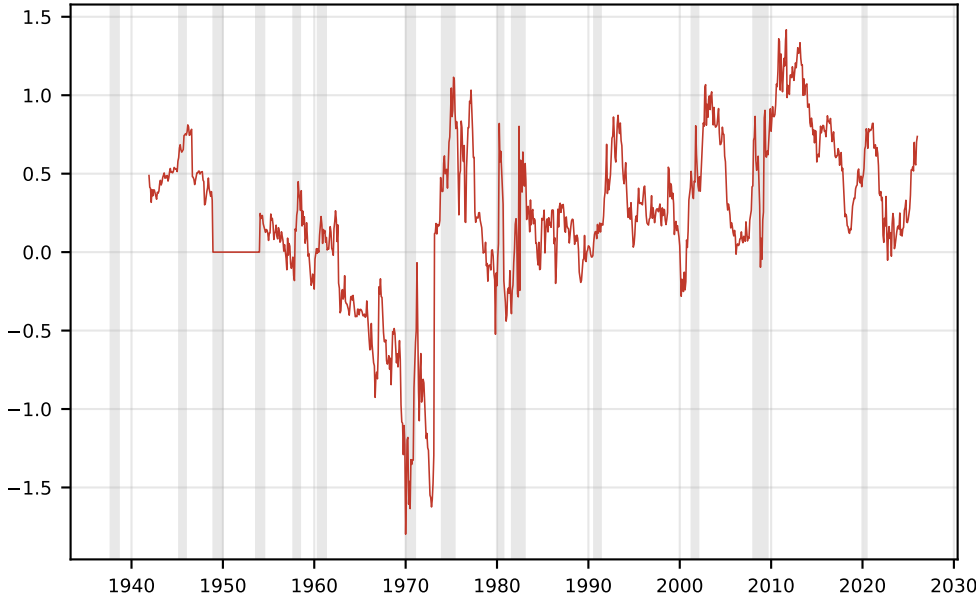
CRSP Slope 30Y Minus 10Y

Block: Selected export / Treasury term structure | Source: WRDS | Col: derived_slope_30y_10y_wrds | Transform: nan

Raw Series



After: nan



SUMMARY STATISTICS

N : 1287
Valid : 1010
NaN : 277
Start : 1941-11
End : 2025-12
Mean : 0.2204
Std : 0.4998
Min : -1.7977
Max : 1.4174
Skew : -0.780
Kurt : 1.690

STATIONARITY TESTS

ADF stat : -3.204
ADF p : 0.0198
ADF lags : 12
ADF : Stationary
KPSS stat : 1.350
KPSS p : 0.0100
KPSS : Non-stationary

QUARTILE-BUCKET ANALYSIS
(vs NBER recession/expansion)

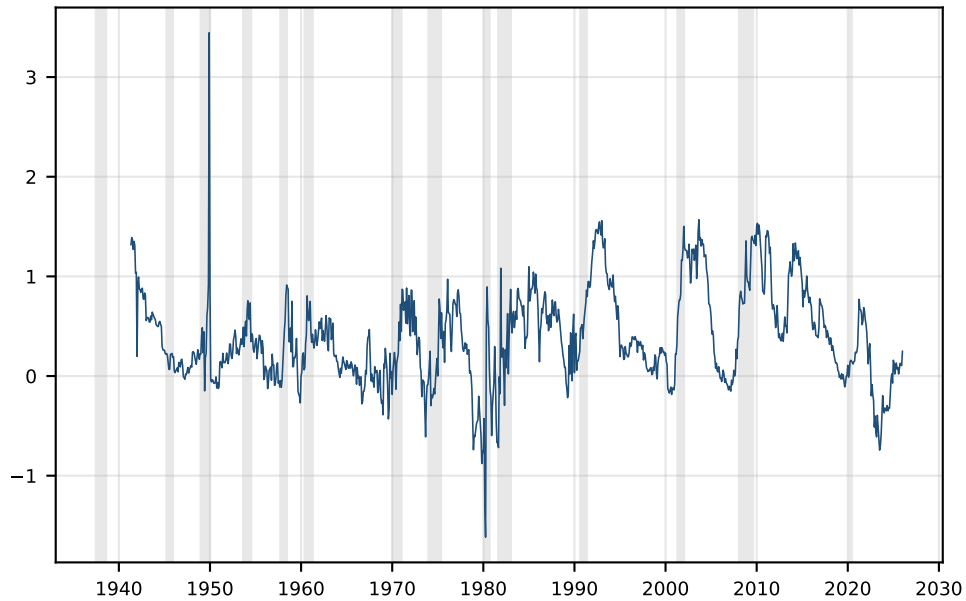
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	277	13.7%	86.3%	-0.3640
Q2	228	11.0%	89.0%	0.1218
Q3	252	11.5%	88.5%	0.3615
Q4 (highest)	253	15.8%	84.2%	0.8086

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

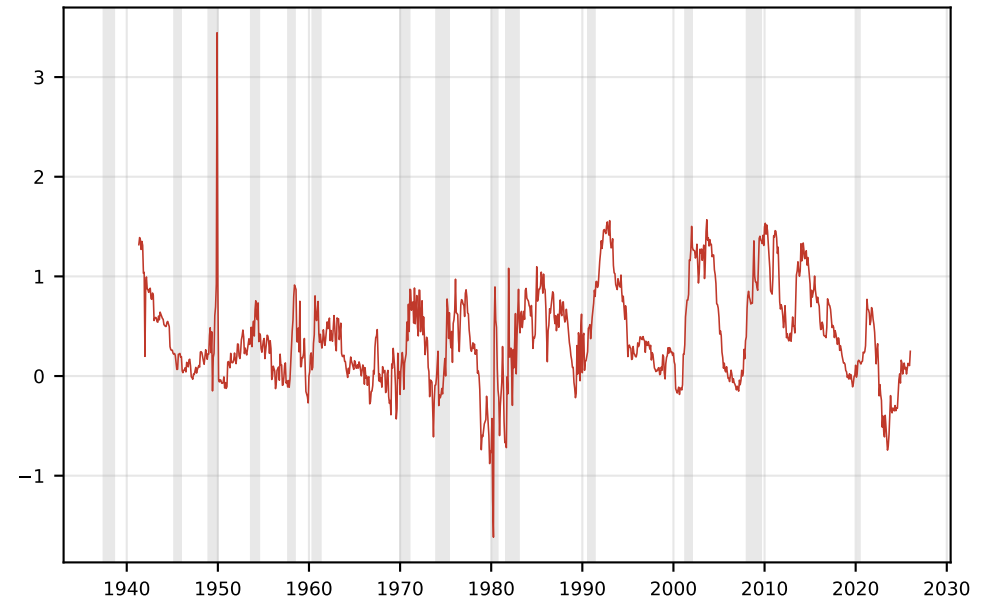
CRSP Slope 5Y Minus 2Y

Block: Selected export / Treasury term structure | Source: WRDS | Col: derived_slope_5y_2y_wrds | Transform: nan

Raw Series



After: nan



SUMMARY STATISTICS

N : 1287
Valid : 1017
NaN : 270
Start : 1941-04
End : 2025-12
Mean : 0.3881
Std : 0.4719
Min : -1.6170
Max : 3.4450
Skew : 0.500
Kurt : 1.762

STATIONARITY TESTS

ADF stat : -4.853
ADF p : 0.0000
ADF lags : 11
ADF : Stationary
KPSS stat : 0.398
KPSS p : 0.0782
KPSS : Stationary

QUARTILE-BUCKET ANALYSIS

(vs NBER recession/expansion)

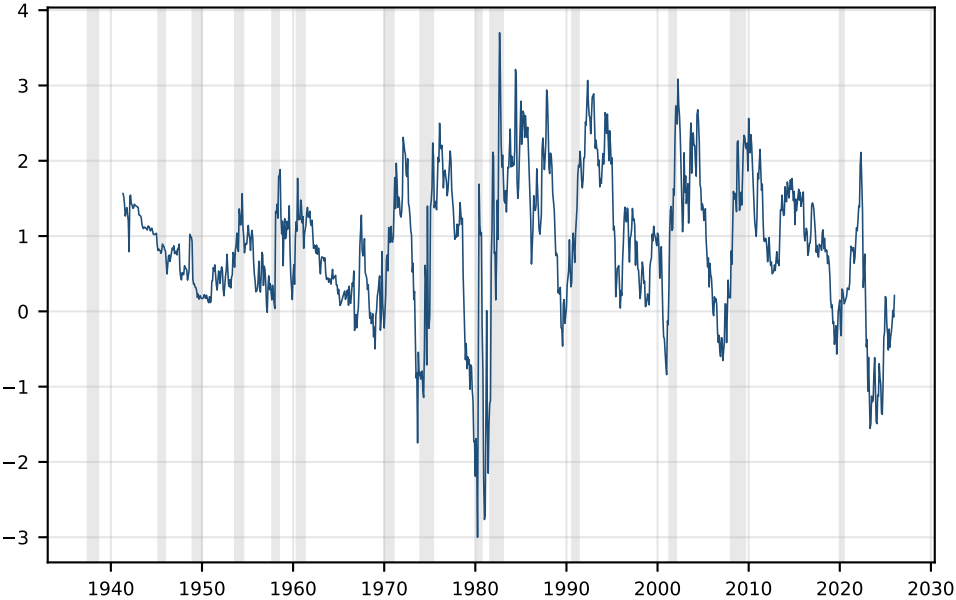
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	255	9.8%	90.2%	-0.1431
Q2	254	12.2%	87.8%	0.1898
Q3	254	15.4%	84.6%	0.4853
Q4 (highest)	254	14.6%	85.4%	1.0223

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

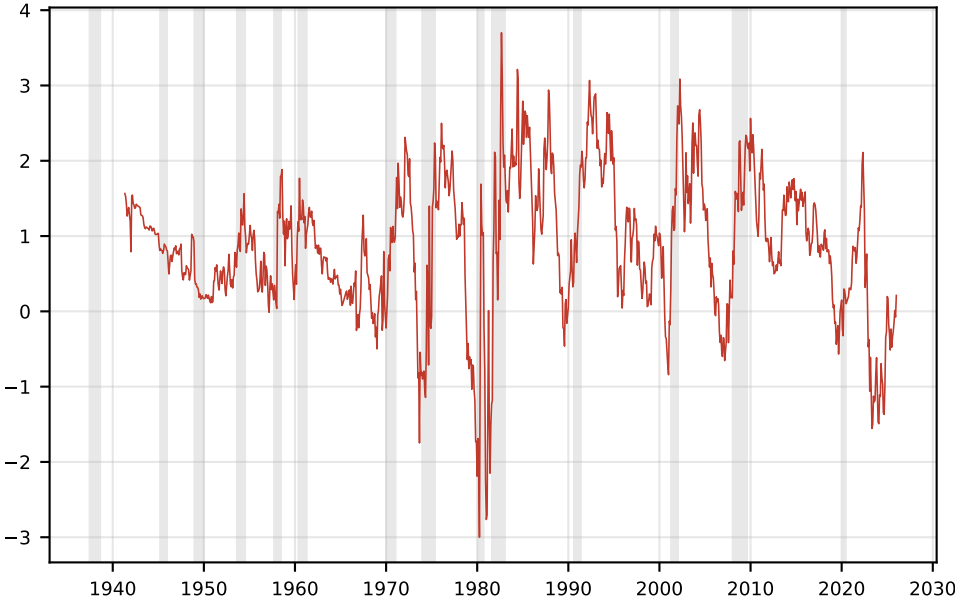
CRSP Slope 5Y Minus 3M

Block: Selected export / Treasury term structure | Source: WRDS | Col: derived_slope_5y_3m_wrds | Transform: nan

Raw Series



After: nan



SUMMARY STATISTICS

N : 1287
Valid : 1017
NaN : 270
Start : 1941-04
End : 2025-12
Mean : 0.8684
Std : 0.9209
Min : -2.9992
Max : 3.7005
Skew : -0.374
Kurt : 0.863

STATIONARITY TESTS

ADF stat : -5.530
ADF p : 0.0000
ADF lags : 10
ADF : Stationary
KPSS stat : 0.200
KPSS p : 0.1000
KPSS : Stationary

QUARTILE-BUCKET ANALYSIS

(vs NBER recession/expansion)

Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	255	11.0%	89.0%	-0.2831
Q2	254	12.2%	87.8%	0.6039
Q3	254	16.9%	83.1%	1.1563
Q4 (highest)	254	11.8%	88.2%	2.0011

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

CRSP 10Y Treasury Yield

Block: Selected export / Treasury term structure | Source: WRDS | Col: raw_y_10Y_wrds | Transform: nan

Raw Series



After: nan



SUMMARY STATISTICS

N : 1287
Valid : 1016
NaN : 271
Start : 1941-05
End : 2025-12
Mean : 4.9144
Std : 2.8178
Min : 0.5273
Max : 15.2141
Skew : 0.968
Kurt : 0.622

STATIONARITY TESTS

ADF stat : -1.740
ADF p : 0.4108
ADF lags : 0
ADF : Non-stationary
KPSS stat : 1.092
KPSS p : 0.0100
KPSS : Non-stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

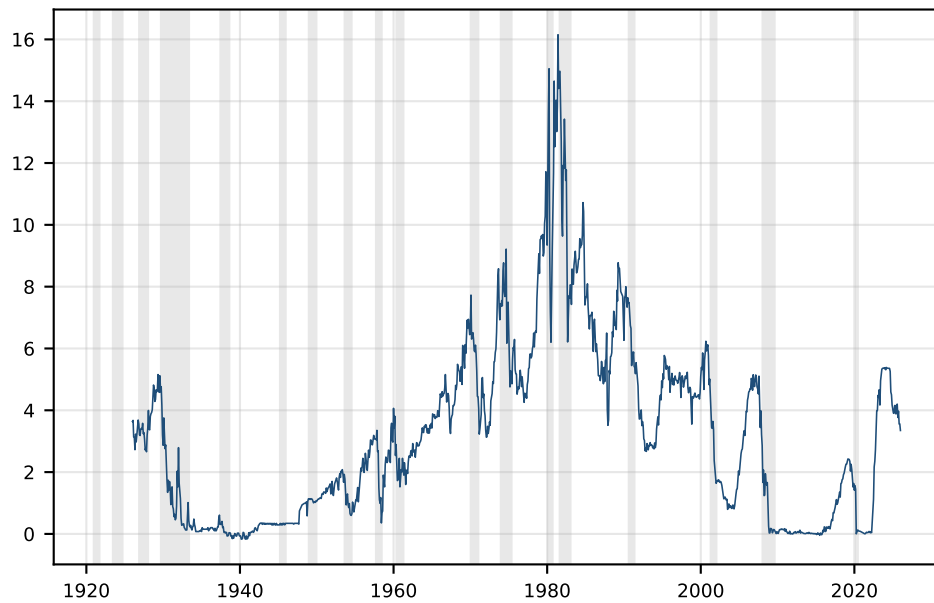
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	254	10.2%	89.8%	1.9352
Q2	254	15.7%	84.3%	3.4916
Q3	254	3.9%	96.1%	5.3744
Q4 (highest)	254	22.0%	78.0%	8.8565

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

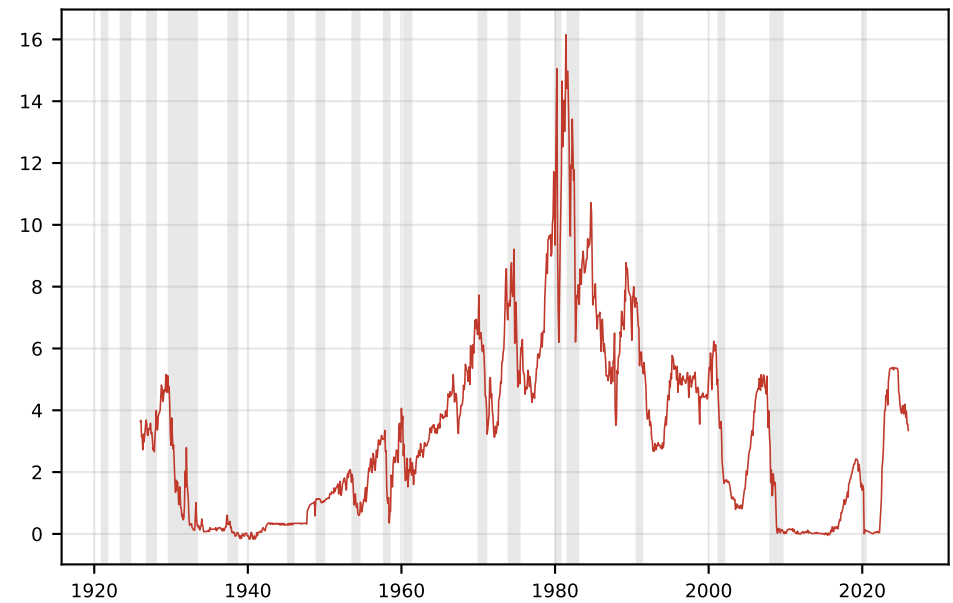
CRSP 1M Treasury Yield

Block: Selected export / Treasury term structure | Source: WRDS | Col: raw_y_1M_wrds | Transform: nan

Raw Series



After: nan



SUMMARY STATISTICS

N : 1287
Valid : 1201
NaN : 86
Start : 1925-12
End : 2025-12
Mean : 3.1835
Std : 2.9244
Min : -0.1699
Max : 16.1505
Skew : 1.106
Kurt : 1.478

STATIONARITY TESTS

ADF stat : -2.889
ADF p : 0.0466
ADF lags : 12
ADF : Stationary
KPSS stat : 1.077
KPSS p : 0.0100
KPSS : Non-stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

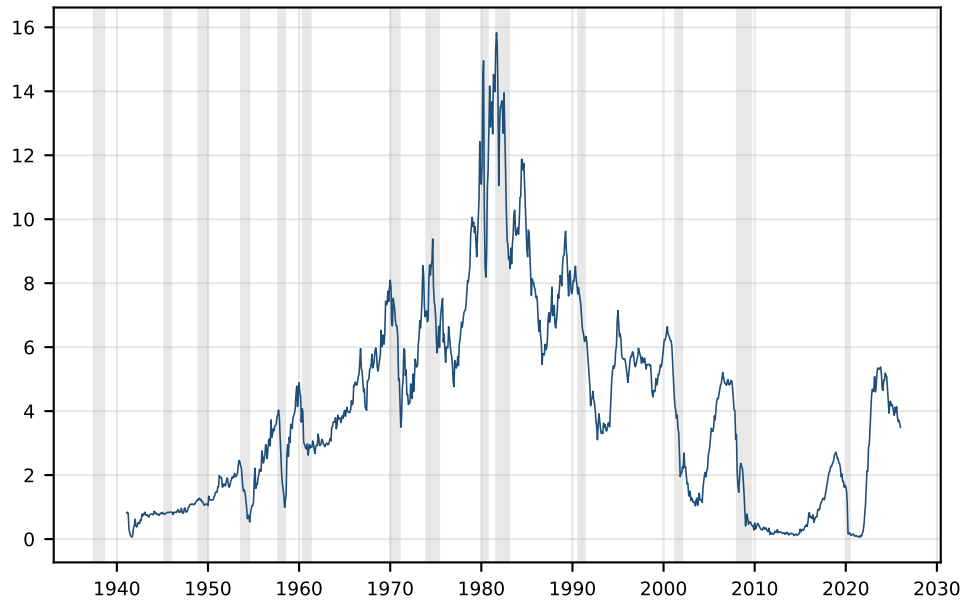
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	301	13.3%	86.7%	0.1317
Q2	300	26.0%	74.0%	1.5707
Q3	300	9.3%	90.7%	3.9100
Q4 (highest)	300	18.3%	81.7%	7.1316

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

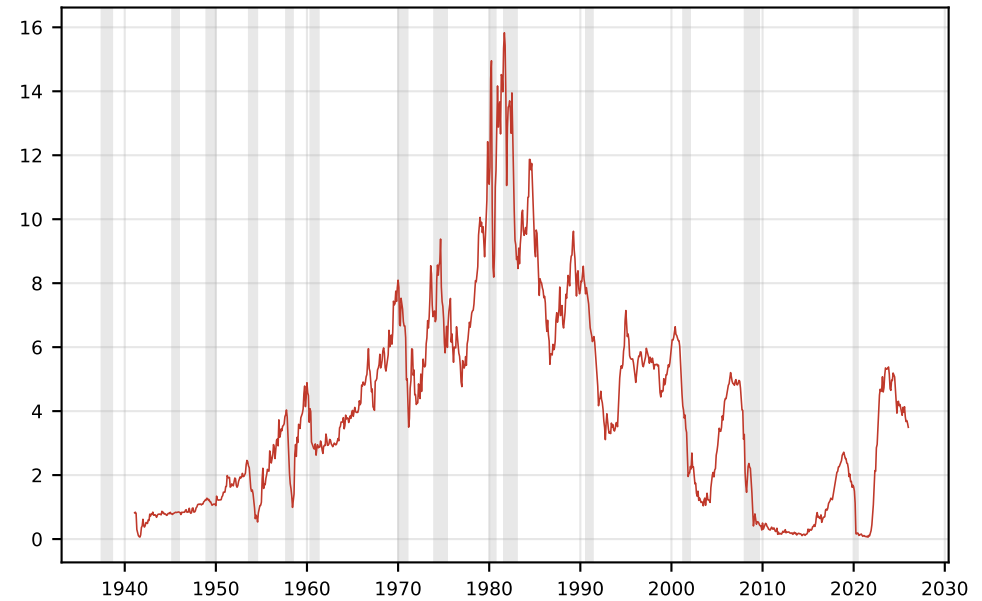
CRSP 1Y Treasury Yield

Block: Selected export / Treasury term structure | Source: WRDS | Col: raw_y_1Y_wrds | Transform: nan

Raw Series



After: nan



SUMMARY STATISTICS

N : 1287
Valid : 1020
NaN : 267
Start : 1941-01
End : 2025-12
Mean : 4.0565
Std : 3.1586
Min : 0.0592
Max : 15.8300
Skew : 0.899
Kurt : 0.675

STATIONARITY TESTS

ADF stat : -2.534
ADF p : 0.1075
ADF lags : 11
ADF : Non-stationary
KPSS stat : 0.962
KPSS p : 0.0100
KPSS : Non-stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	255	11.4%	88.6%	0.6024
Q2	255	15.7%	84.3%	2.4337
Q3	255	3.1%	96.9%	4.8428
Q4 (highest)	255	21.6%	78.4%	8.3473

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

CRSP 20Y Treasury Yield

Block: Selected export / Treasury term structure | Source: WRDS | Col: raw_y_20Y_wrds | Transform: nan

Raw Series



After: nan



SUMMARY STATISTICS

N : 1287
Valid : 1008
NaN : 279
Start : 1942-01
End : 2025-12
Mean : 5.1765
Std : 2.7323
Min : 0.9371
Max : 15.0035
Skew : 1.019
Kurt : 0.634

STATIONARITY TESTS

ADF stat : -1.923
ADF p : 0.3214
ADF lags : 11
ADF : Non-stationary
KPSS stat : 1.093
KPSS p : 0.0100
KPSS : Non-stationary

QUARTILE-BUCKET ANALYSIS

(vs NBER recession/expansion)

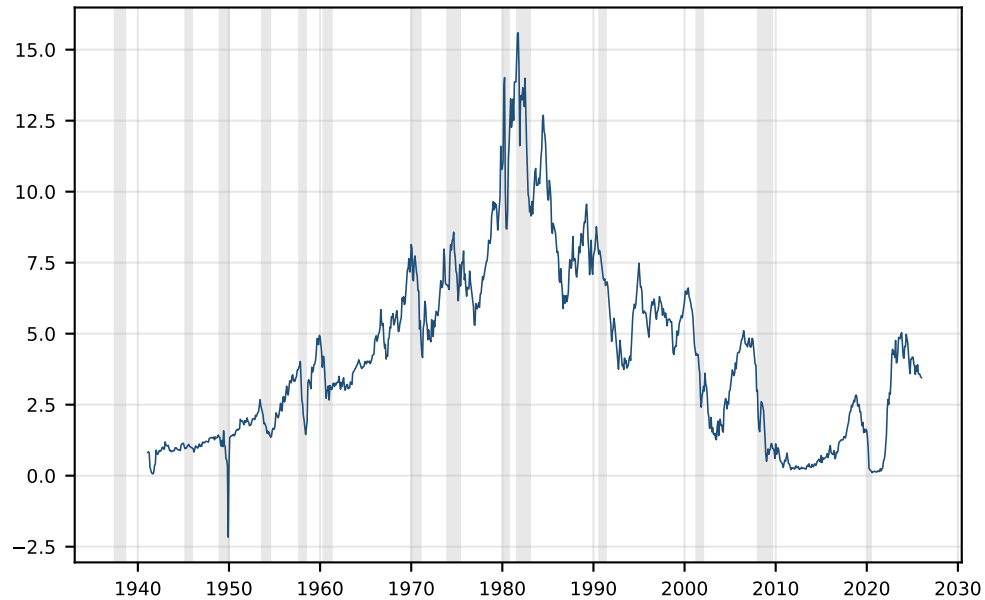
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	252	11.9%	88.1%	2.3412
Q2	252	13.1%	86.9%	3.7931
Q3	252	8.7%	91.3%	5.5019
Q4 (highest)	252	18.7%	81.3%	9.0699

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

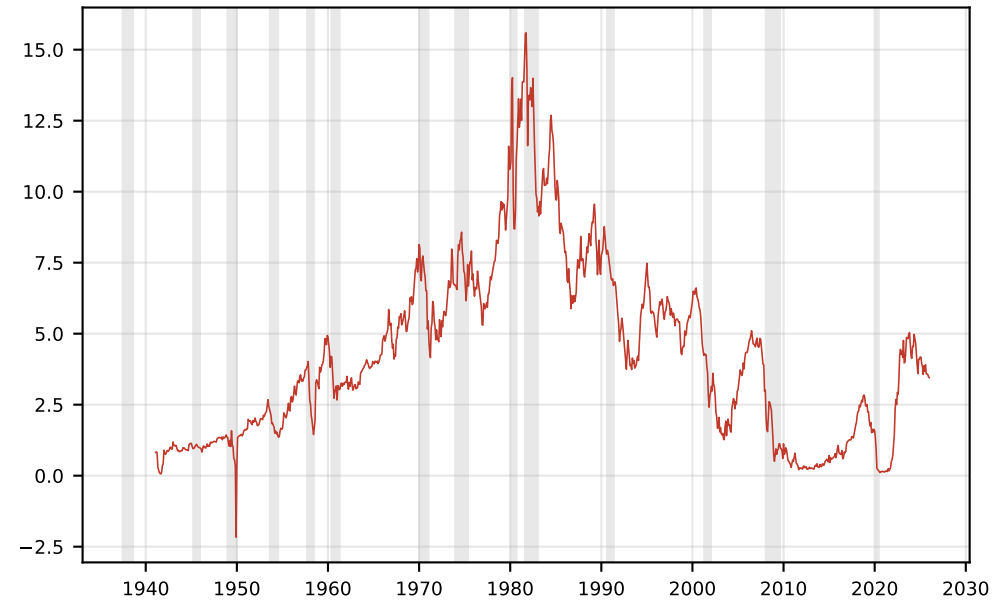
CRSP 2Y Treasury Yield

Block: Selected export / Treasury term structure | Source: WRDS | Col: raw_y_2Y_wrds | Transform: nan

Raw Series



After: nan



SUMMARY STATISTICS

N : 1287
Valid : 1020
NaN : 267
Start : 1941-01
End : 2025-12
Mean : 4.2206
Std : 3.1295
Min : -2.1651
Max : 15.5963
Skew : 0.857
Kurt : 0.503

STATIONARITY TESTS

ADF stat : -2.322
ADF p : 0.1649
ADF lags : 11
ADF : Non-stationary
KPSS stat : 1.011
KPSS p : 0.0100
KPSS : Non-stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	255	11.4%	88.6%	0.7788
Q2	255	16.1%	83.9%	2.6564
Q3	255	2.4%	97.6%	4.9413
Q4 (highest)	255	22.0%	78.0%	8.5060

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

CRSP 30Y Treasury Yield

Block: Selected export / Treasury term structure | Source: WRDS | Col: raw_y_30Y_wrds | Transform: nan

Raw Series



After: nan



SUMMARY STATISTICS

N : 1287
Valid : 1010
NaN : 277
Start : 1941-11
End : 2025-12
Mean : 5.1527
Std : 2.7043
Min : 1.1892
Max : 15.0035
Skew : 1.083
Kurt : 0.743

STATIONARITY TESTS

ADF stat : -1.871
ADF p : 0.3458
ADF lags : 11
ADF : Non-stationary
KPSS stat : 1.103
KPSS p : 0.0100
KPSS : Non-stationary

QUARTILE-BUCKET ANALYSIS

(vs NBER recession/expansion)

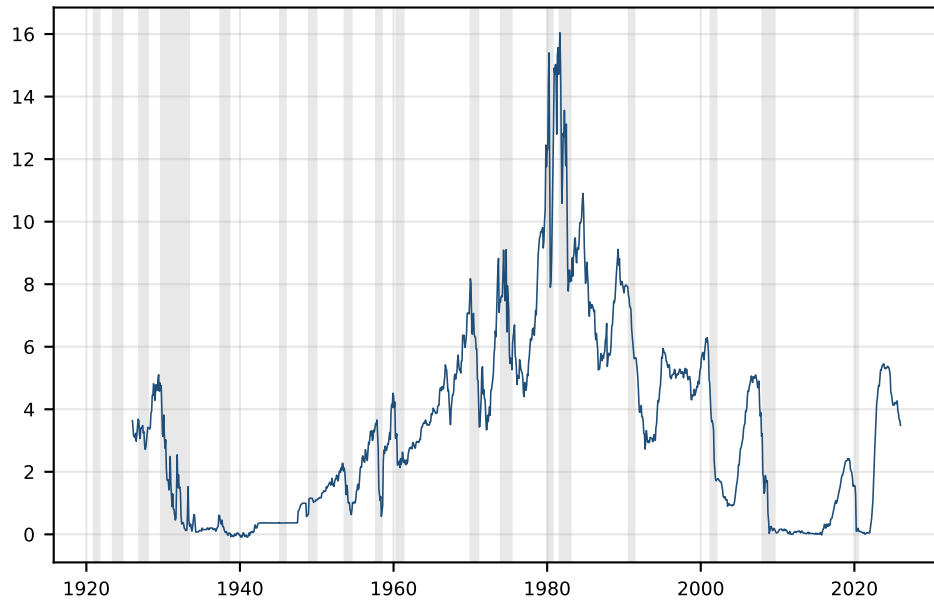
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	253	12.3%	87.7%	2.4221
Q2	252	13.1%	86.9%	3.7386
Q3	252	8.7%	91.3%	5.3803
Q4 (highest)	253	18.2%	81.8%	9.0650

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

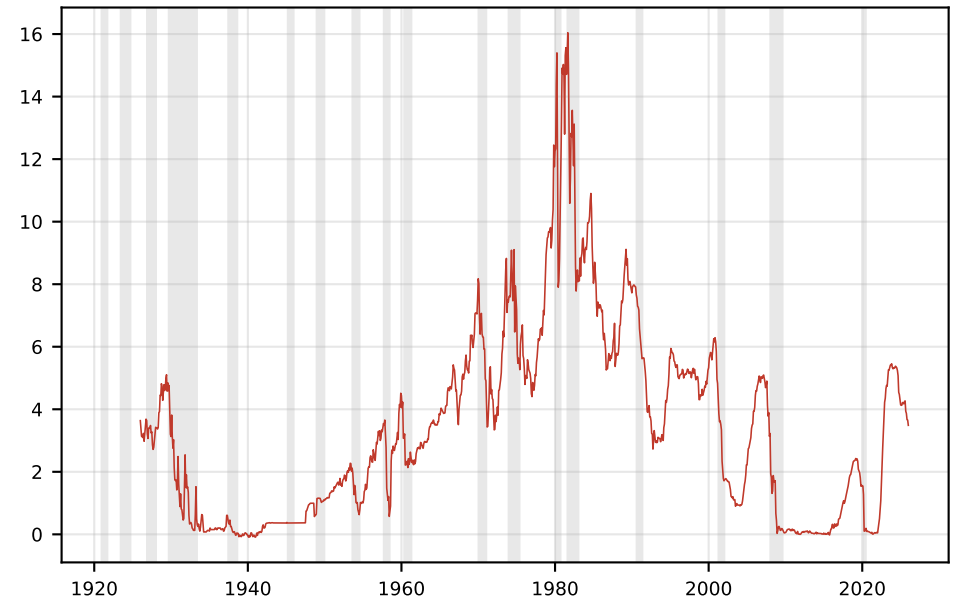
CRSP 3M Treasury Yield

Block: Selected export / Treasury term structure | Source: WRDS | Col: raw_y_3M_wrds | Transform: nan

Raw Series



After: nan



SUMMARY STATISTICS

N : 1287
Valid : 1201
NaN : 86
Start : 1925-12
End : 2025-12
Mean : 3.3802
Std : 3.0757
Min : -0.0905
Max : 16.0421
Skew : 1.102
Kurt : 1.416

STATIONARITY TESTS

ADF stat : -2.563
ADF p : 0.1009
ADF lags : 12
ADF : Non-stationary
KPSS stat : 1.092
KPSS p : 0.0100
KPSS : Non-stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	301	14.0%	86.0%	0.1682
Q2	300	25.3%	74.7%	1.7038
Q3	300	9.0%	91.0%	4.1251
Q4 (highest)	300	18.7%	81.3%	7.5345

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

CRSP 5Y Treasury Yield

Block: Selected export / Treasury term structure | Source: WRDS | Col: raw_y_5Y_wrds | Transform: nan

Raw Series



After: nan



SUMMARY STATISTICS

N : 1287
Valid : 1017
NaN : 270
Start : 1941-04
End : 2025-12
Mean : 4.6187
Std : 3.0058
Min : 0.2270
Max : 15.5891
Skew : 0.887
Kurt : 0.440

STATIONARITY TESTS

ADF stat : -1.865
ADF p : 0.3489
ADF lags : 2
ADF : Non-stationary
KPSS stat : 1.067
KPSS p : 0.0100
KPSS : Non-stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	255	10.6%	89.4%	1.3575
Q2	254	16.1%	83.9%	3.1105
Q3	254	3.1%	96.9%	5.2246
Q4 (highest)	254	22.0%	78.0%	8.7951

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

CRSP 7Y Treasury Yield

Block: Selected export / Treasury term structure | Source: WRDS | Col: raw_y_7Y_wrds | Transform: nan

Raw Series



After: nan



SUMMARY STATISTICS

N : 1287
Valid : 1017
NaN : 270
Start : 1941-04
End : 2025-12
Mean : 4.7779
Std : 2.9329
Min : 0.3779
Max : 15.1039
Skew : 0.890
Kurt : 0.407

STATIONARITY TESTS

ADF stat : -1.734
ADF p : 0.4137
ADF lags : 3
ADF : Non-stationary
KPSS stat : 1.092
KPSS p : 0.0100
KPSS : Non-stationary

QUARTILE-BUCKET ANALYSIS

(vs NBER recession/expansion)

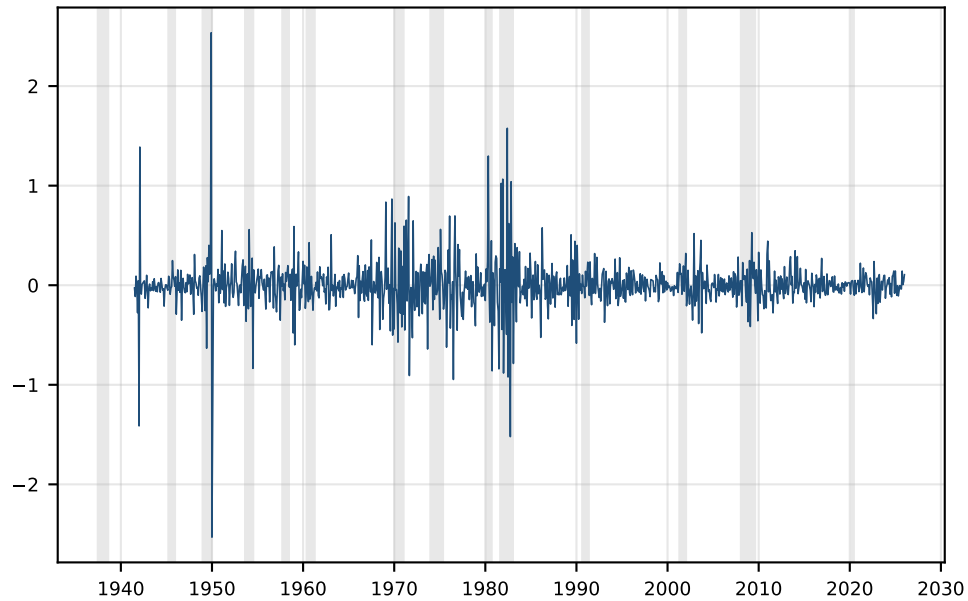
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	255	10.6%	89.4%	1.5965
Q2	254	16.1%	83.9%	3.3265
Q3	254	4.3%	95.7%	5.3325
Q4 (highest)	254	20.9%	79.1%	8.8687

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

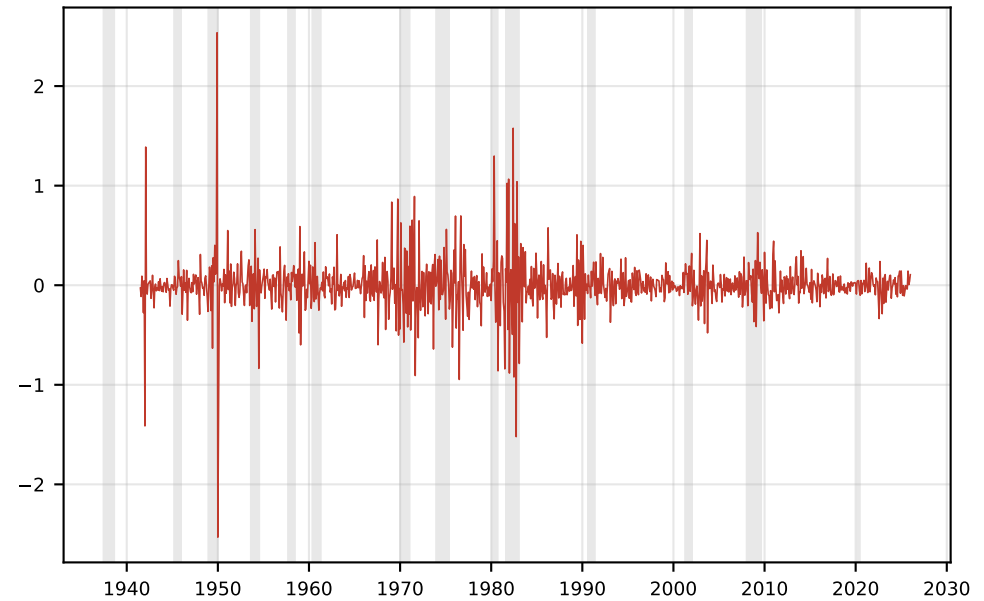
CRSP Curve Curvature First Difference

Block: Selected export / Treasury term structure | Source: WRDS | Col: derived_curve_curvature_wrds_diff_1_wrds | Transform: nan

Raw Series



After: nan



SUMMARY STATISTICS

N : 1287
Valid : 1015
NaN : 272
Start : 1941-06
End : 2025-12
Mean : -0.0010
Std : 0.2574
Min : -2.5305
Max : 2.5358
Skew : 0.161
Kurt : 24.220

STATIONARITY TESTS

ADF stat : -16.603
ADF p : 0.0000
ADF lags : 7
ADF : Stationary
KPSS stat : 0.057
KPSS p : 0.1000
KPSS : Stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

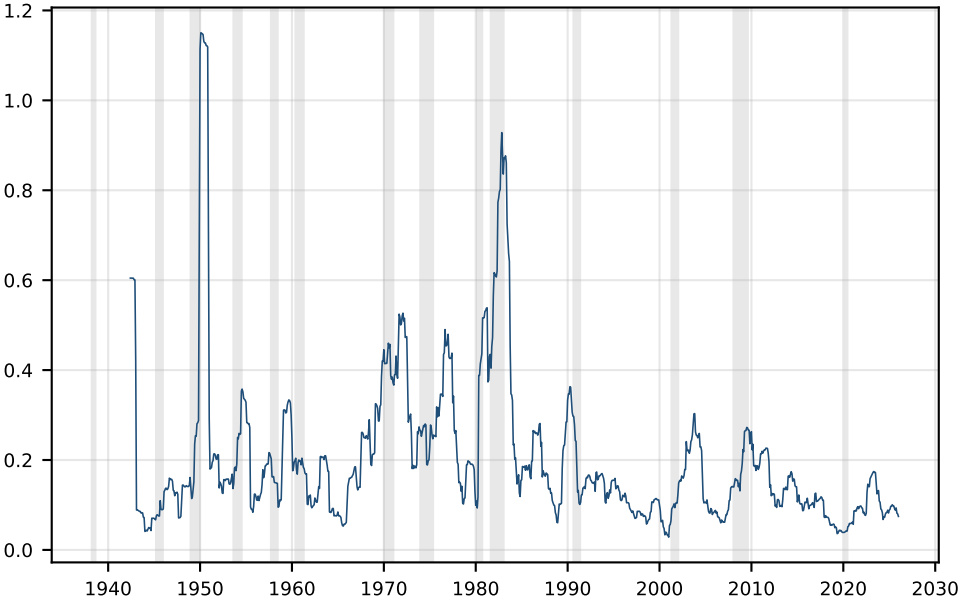
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	254	13.8%	86.2%	-0.2532
Q2	254	6.7%	93.3%	-0.0464
Q3	253	9.1%	90.9%	0.0391
Q4 (highest)	254	22.4%	77.6%	0.2566

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

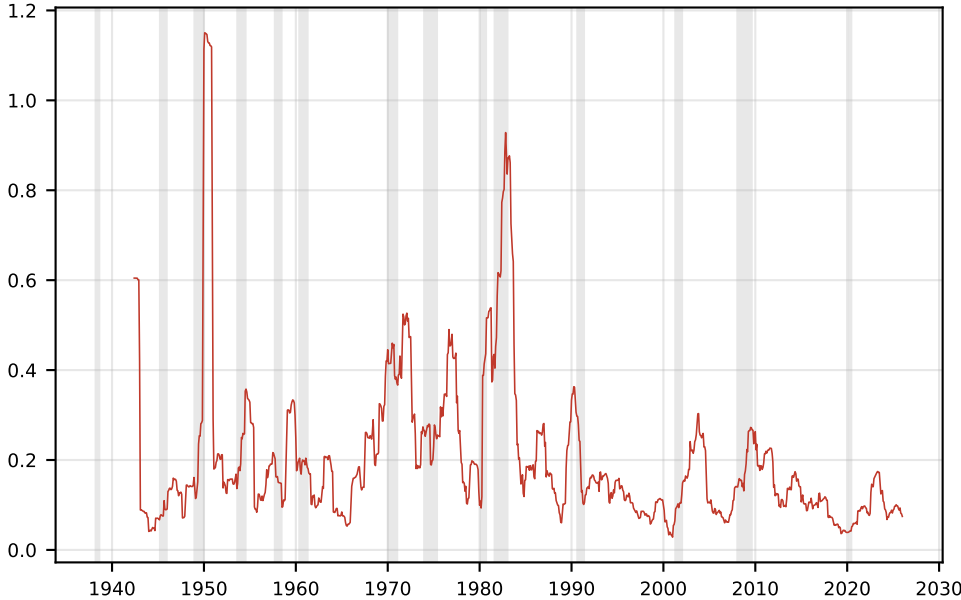
CRSP Curve Curvature 12M Rolling Volatility

Block: Selected export / Treasury term structure | Source: WRDS | Col: derived_curve_curvature_wrds_vol_12m_wrds | Transform: nan

Raw Series



After: nan



SUMMARY STATISTICS

N : 1287
Valid : 1004
NaN : 283
Start : 1942-05
End : 2025-12
Mean : 0.2016
Std : 0.1714
Min : 0.0283
Max : 1.1504
Skew : 2.850
Kurt : 10.446

STATIONARITY TESTS

ADF stat : -3.831
ADF p : 0.0026
ADF lags : 12
ADF : Stationary
KPSS stat : 0.808
KPSS p : 0.0100
KPSS : Non-stationary

QUARTILE-BUCKET ANALYSIS

(vs NBER recession/expansion)

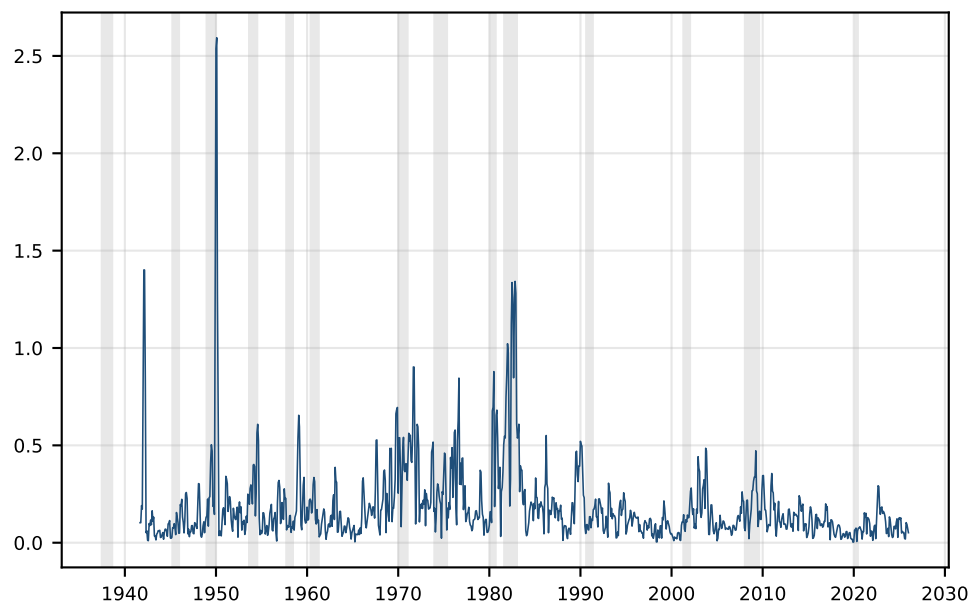
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	251	5.6%	94.4%	0.0746
Q2	251	9.6%	90.4%	0.1252
Q3	251	14.3%	85.7%	0.1864
Q4 (highest)	251	23.1%	76.9%	0.4202

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

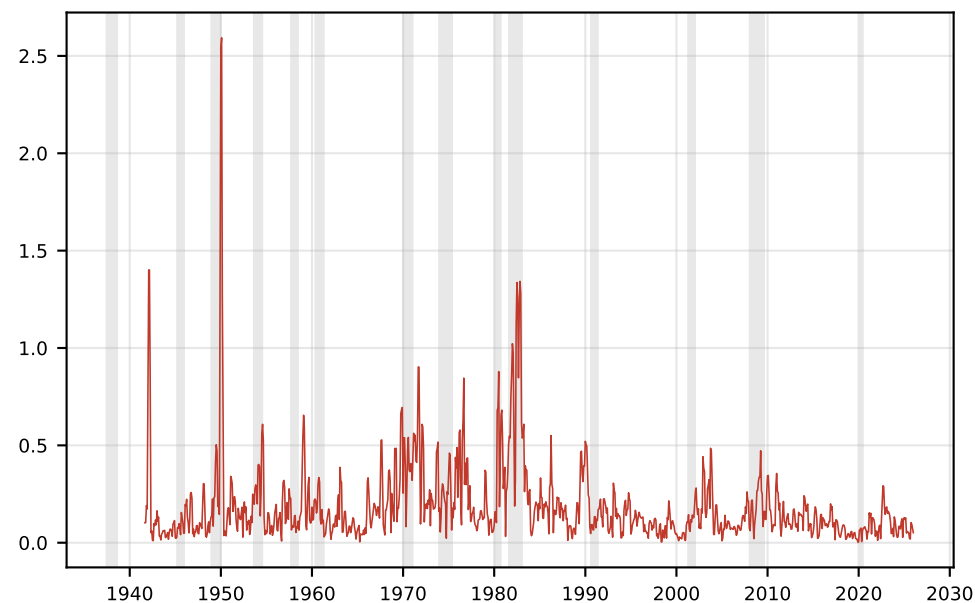
CRSP Curve Curvature 3M Rolling Volatility

Block: Selected export / Treasury term structure | Source: WRDS | Col: derived_curve_curvature_wrds_vol_3m_wrds | Transform: nan

Raw Series



After: nan



SUMMARY STATISTICS

N : 1287
Valid : 1013
NaN : 274
Start : 1941-08
End : 2025-12
Mean : 0.1870
Std : 0.2163
Min : 0.0028
Max : 2.5933
Skew : 4.614
Kurt : 35.081

STATIONARITY TESTS

ADF stat : -6.209
ADF p : 0.0000
ADF lags : 9
ADF : Stationary
KPSS stat : 0.753
KPSS p : 0.0100
KPSS : Non-stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

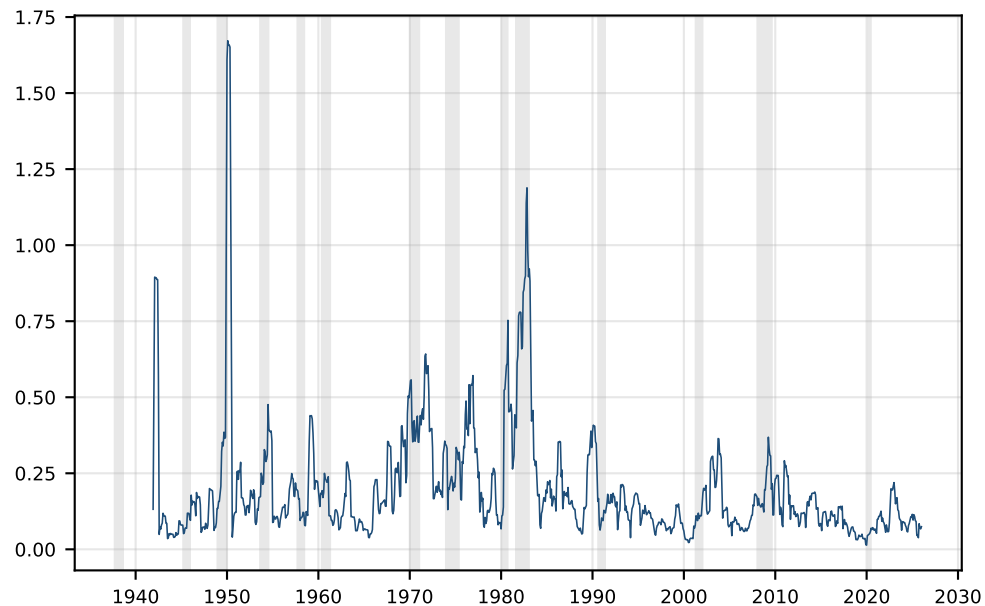
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	254	4.7%	95.3%	0.0450
Q2	253	10.3%	89.7%	0.0989
Q3	253	11.5%	88.5%	0.1642
Q4 (highest)	253	25.7%	74.3%	0.4405

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

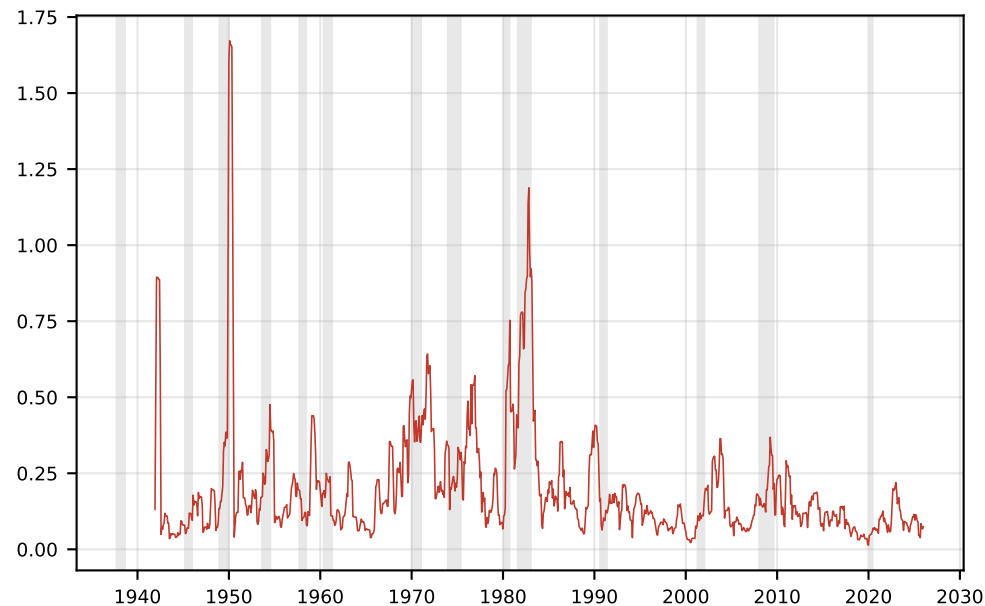
CRSP Curve Curvature 6M Rolling Volatility

Block: Selected export / Treasury term structure | Source: WRDS | Col: derived_curve_curvature_wrds_vol_6m_wrds | Transform: nan

Raw Series



After: nan



SUMMARY STATISTICS

N : 1287
Valid : 1010
NaN : 277
Start : 1941-11
End : 2025-12
Mean : 0.1969
Std : 0.1908
Min : 0.0136
Max : 1.6721
Skew : 3.611
Kurt : 19.178

STATIONARITY TESTS

ADF stat : -5.006
ADF p : 0.0000
ADF lags : 12
ADF : Stationary
KPSS stat : 0.782
KPSS p : 0.0100
KPSS : Non-stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

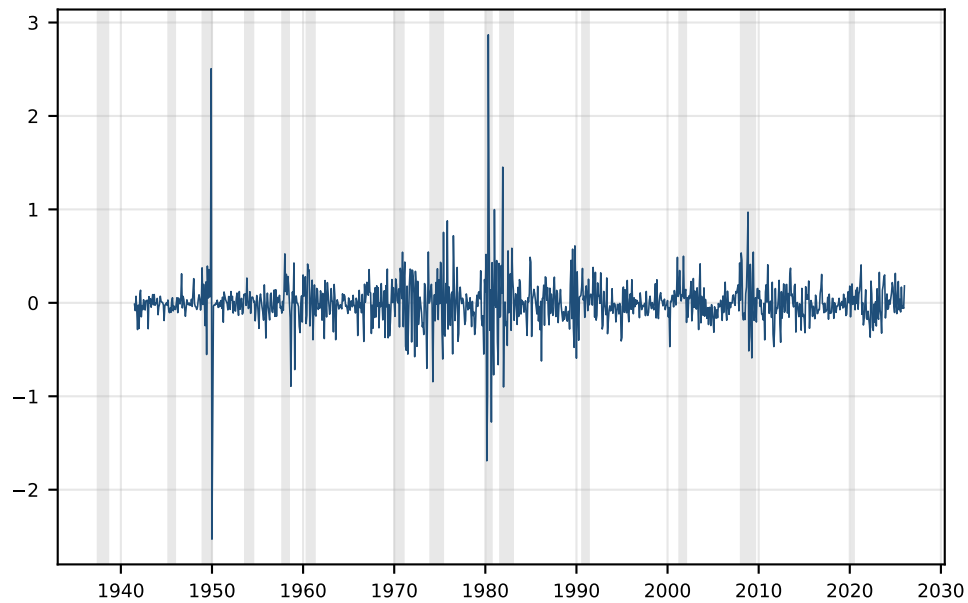
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	253	4.0%	96.0%	0.0649
Q2	252	11.1%	88.9%	0.1135
Q3	252	12.3%	87.7%	0.1769
Q4 (highest)	253	24.9%	75.1%	0.4318

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

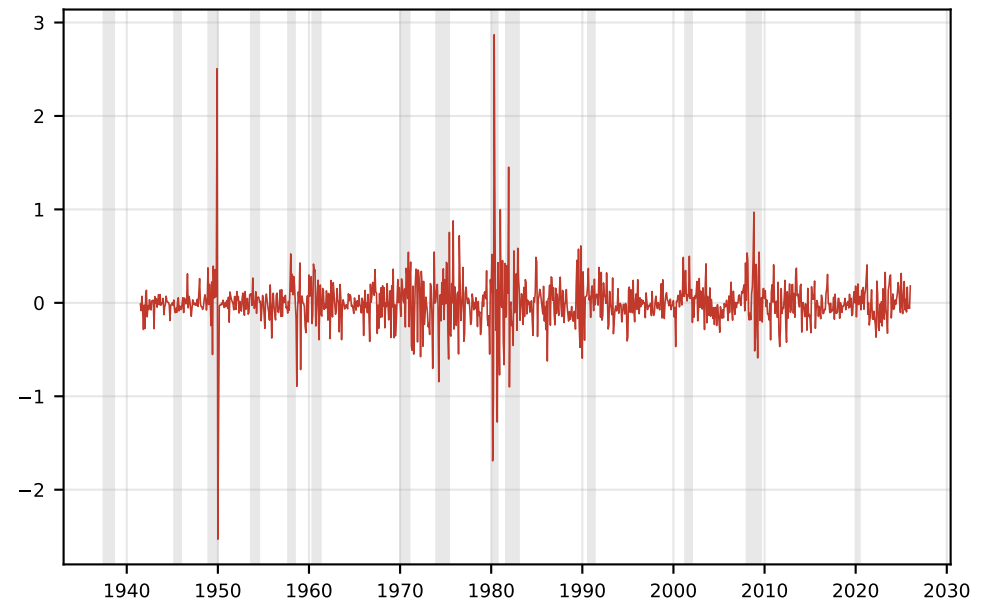
CRSP Slope 10Y Minus 2Y First Difference

Block: Selected export / Treasury term structure | Source: WRDS | Col: derived_slope_10y_2y_wrds_diff_1_wrds | Transform: nan

Raw Series



After: nan



SUMMARY STATISTICS

N : 1287
Valid : 1015
NaN : 272
Start : 1941-06
End : 2025-12
Mean : -0.0012
Std : 0.2672
Min : -2.5294
Max : 2.8693
Skew : 0.965
Kurt : 31.323

STATIONARITY TESTS

ADF stat : -15.412
ADF p : 0.0000
ADF lags : 5
ADF : Stationary
KPSS stat : 0.030
KPSS p : 0.1000
KPSS : Stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

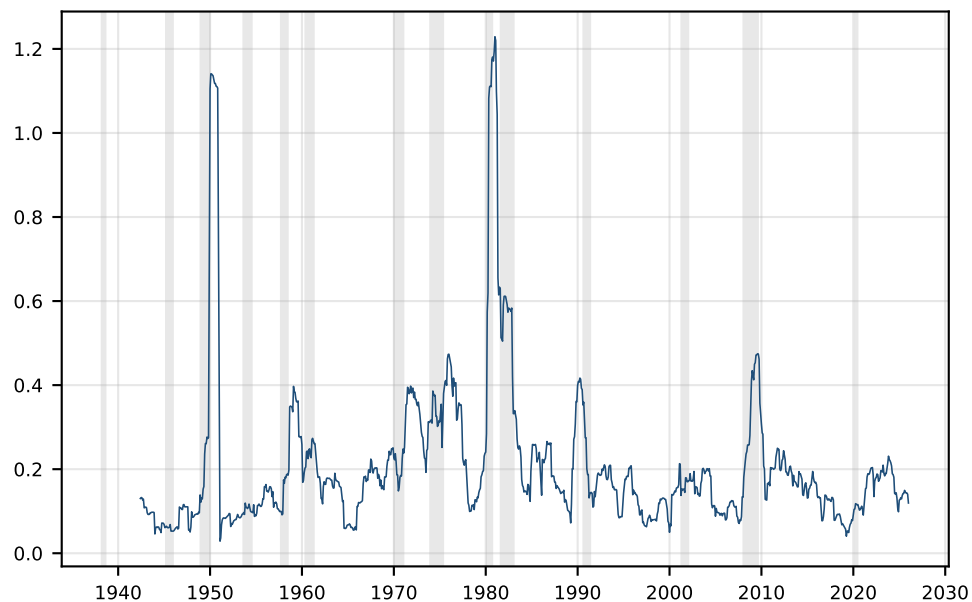
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	254	13.0%	87.0%	-0.2690
Q2	254	7.5%	92.5%	-0.0483
Q3	253	9.5%	90.5%	0.0417
Q4 (highest)	254	22.0%	78.0%	0.2708

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

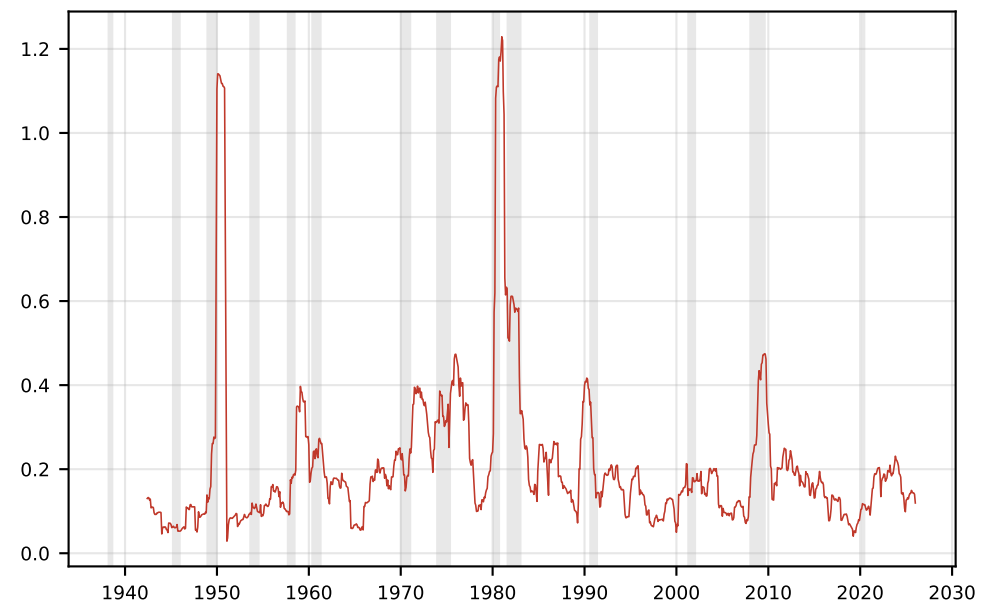
CRSP Slope 10Y Minus 2Y 12M Rolling Volatility

Block: Selected export / Treasury term structure | Source: WRDS | Col: derived_slope_10y_2y_wrds_vol_12m_wrds | Transform: nan

Raw Series



After: nan



SUMMARY STATISTICS

N : 1287
Valid : 1004
NaN : 283
Start : 1942-05
End : 2025-12
Mean : 0.2063
Std : 0.1799
Min : 0.0287
Max : 1.2289
Skew : 3.433
Kurt : 14.295

STATIONARITY TESTS

ADF stat : -4.011
ADF p : 0.0014
ADF lags : 12
ADF : Stationary
KPSS stat : 0.282
KPSS p : 0.1000
KPSS : Stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

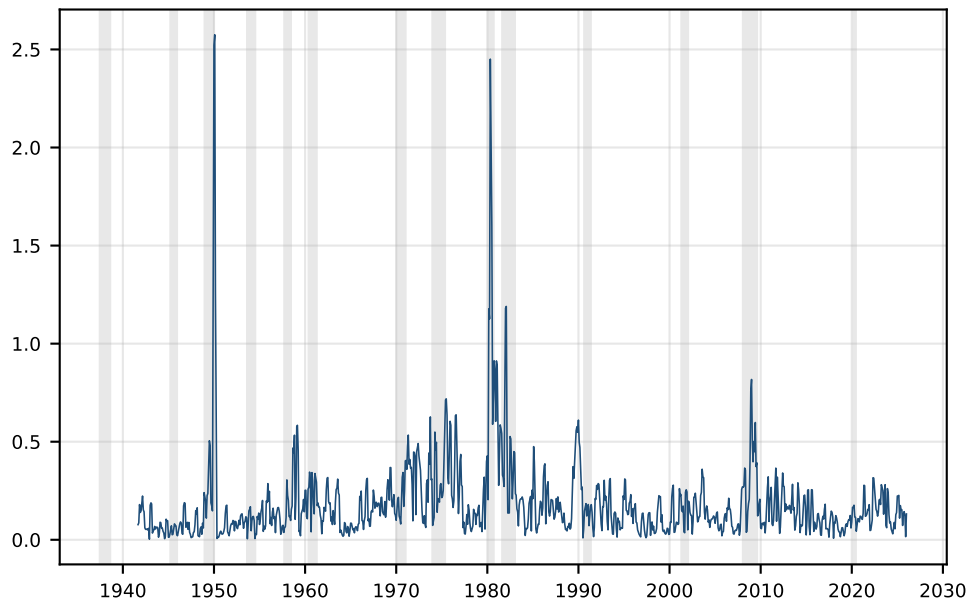
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	251	7.6%	92.4%	0.0821
Q2	251	6.8%	93.2%	0.1353
Q3	251	8.8%	91.2%	0.1892
Q4 (highest)	251	29.5%	70.5%	0.4187

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

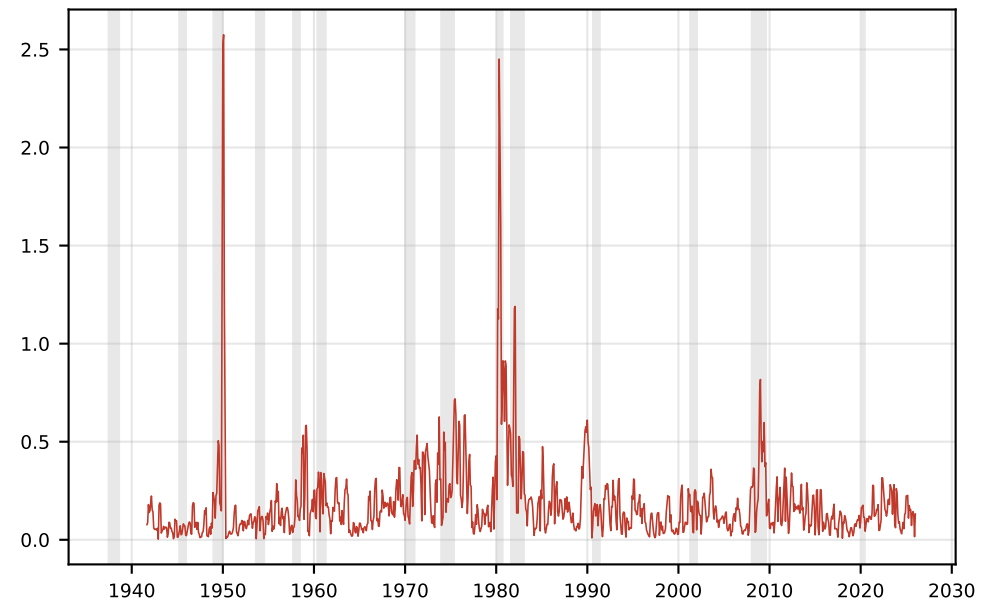
CRSP Slope 10Y Minus 2Y 3M Rolling Volatility

Block: Selected export / Treasury term structure | Source: WRDS | Col: derived_slope_10y_2y_wrds_vol_3m_wrds | Transform: nan

Raw Series



After: nan



SUMMARY STATISTICS

N : 1287
Valid : 1013
NaN : 274
Start : 1941-08
End : 2025-12
Mean : 0.1835
Std : 0.2173
Min : 0.0031
Max : 2.5750
Skew : 5.520
Kurt : 46.883

STATIONARITY TESTS

ADF stat : -6.584
ADF p : 0.0000
ADF lags : 9
ADF : Stationary
KPSS stat : 0.312
KPSS p : 0.1000
KPSS : Stationary

QUARTILE-BUCKET ANALYSIS

(vs NBER recession/expansion)

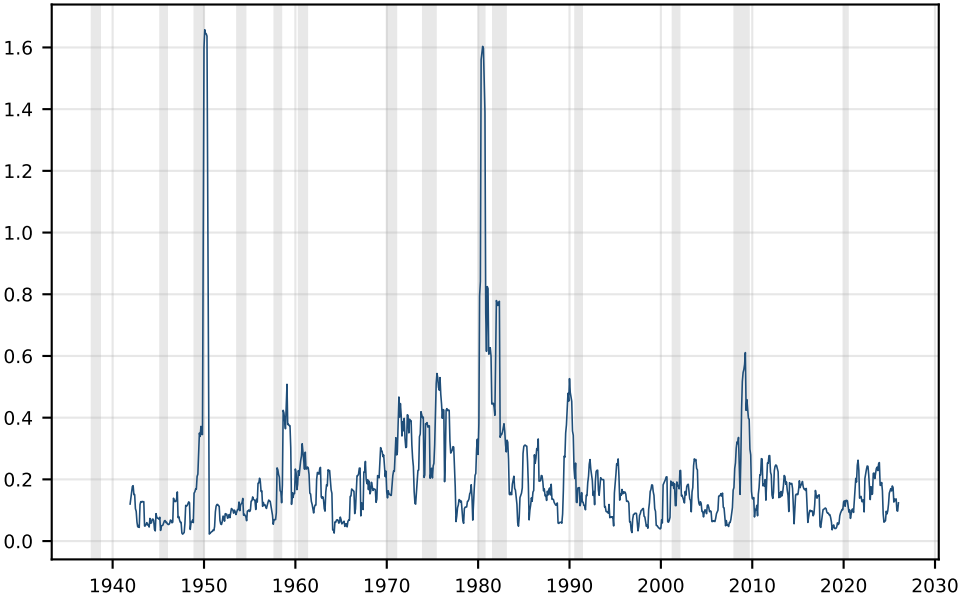
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	254	5.5%	94.5%	0.0431
Q2	253	9.1%	90.9%	0.0997
Q3	253	13.8%	86.2%	0.1712
Q4 (highest)	253	23.7%	76.3%	0.4205

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

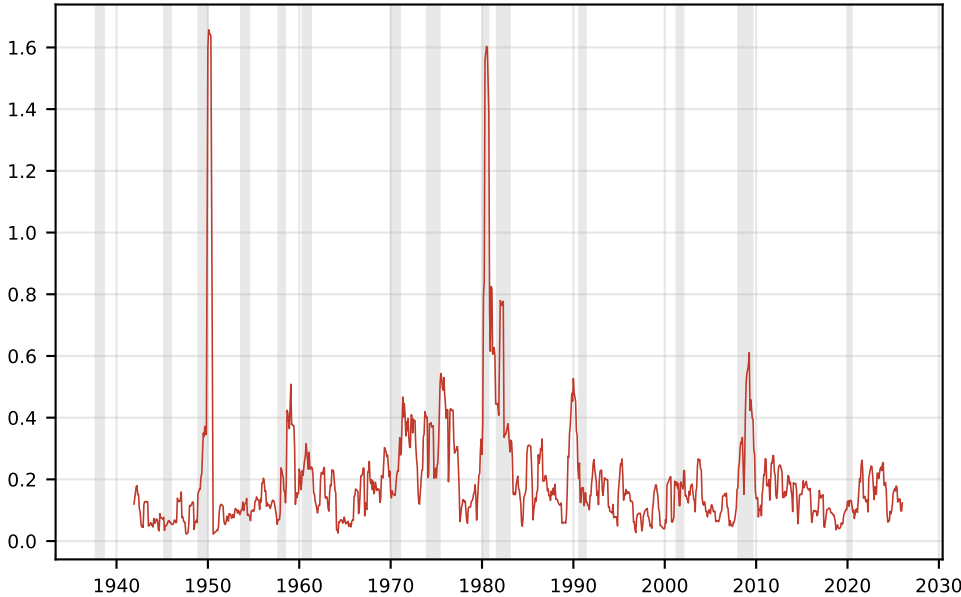
CRSP Slope 10Y Minus 2Y 6M Rolling Volatility

Block: Selected export / Treasury term structure | Source: WRDS | Col: derived_slope_10y_2y_wrds_vol_6m_wrds | Transform: nan

Raw Series



After: nan



SUMMARY STATISTICS

N : 1287
Valid : 1010
NaN : 277
Start : 1941-11
End : 2025-12
Mean : 0.1967
Std : 0.1959
Min : 0.0224
Max : 1.6574
Skew : 4.455
Kurt : 26.829

STATIONARITY TESTS

ADF stat : -5.326
ADF p : 0.0000
ADF lags : 12
ADF : Stationary
KPSS stat : 0.278
KPSS p : 0.1000
KPSS : Stationary

QUARTILE-BUCKET ANALYSIS

(vs NBER recession/expansion)

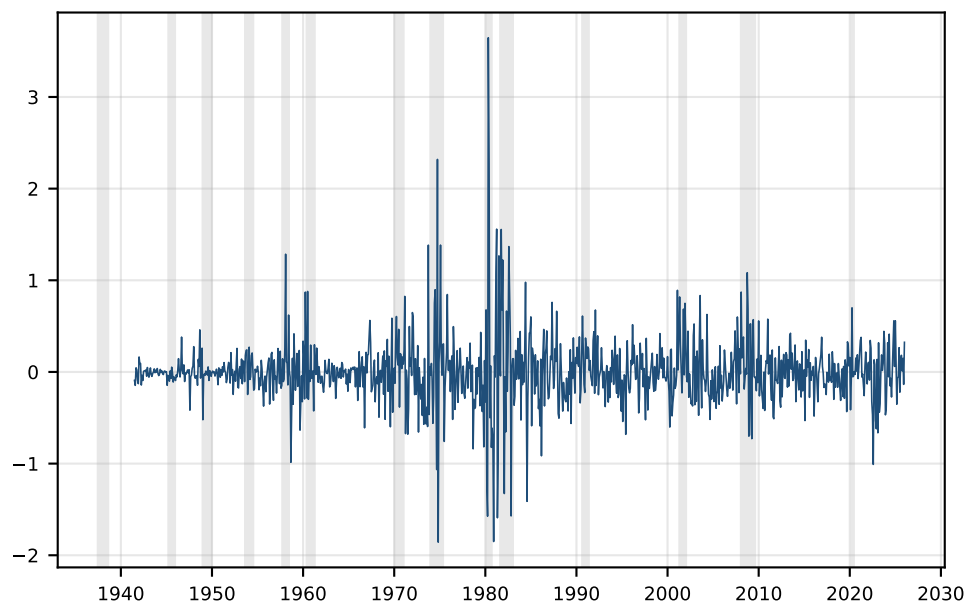
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	253	5.1%	94.9%	0.0660
Q2	252	7.5%	92.5%	0.1232
Q3	252	12.3%	87.7%	0.1821
Q4 (highest)	253	27.3%	72.7%	0.4151

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

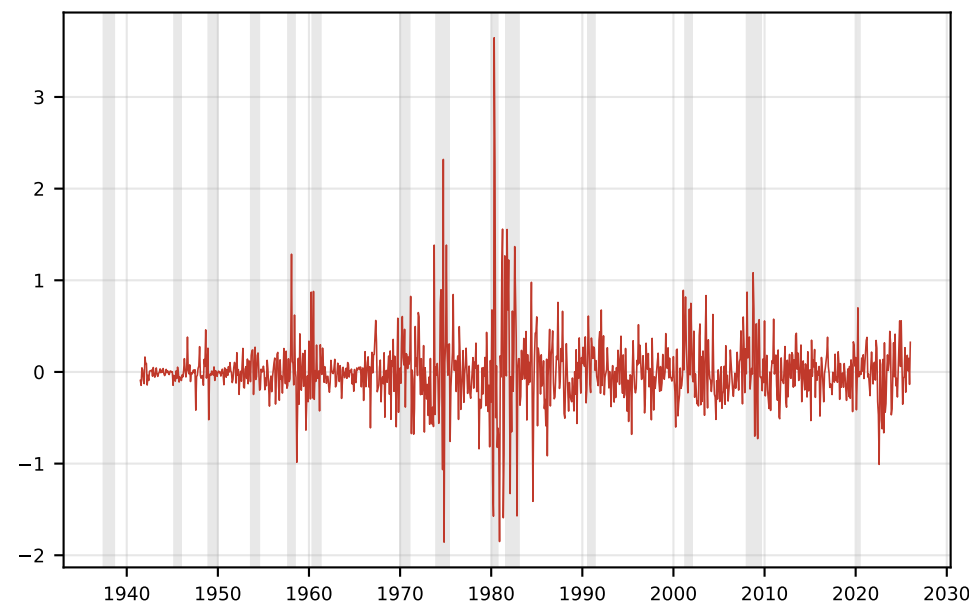
CRSP Slope 10Y Minus 3M First Difference

Block: Selected export / Treasury term structure | Source: WRDS | Col: derived_slope_10y_3m_wrds_diff_1_wrds | Transform: nan

Raw Series



After: nan



SUMMARY STATISTICS

N : 1287
Valid : 1015
NaN : 272
Start : 1941-06
End : 2025-12
Mean : -0.0014
Std : 0.3662
Min : -1.8572
Max : 3.6463
Skew : 1.304
Kurt : 15.911

STATIONARITY TESTS

ADF stat : -9.673
ADF p : 0.0000
ADF lags : 11
ADF : Stationary
KPSS stat : 0.015
KPSS p : 0.1000
KPSS : Stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

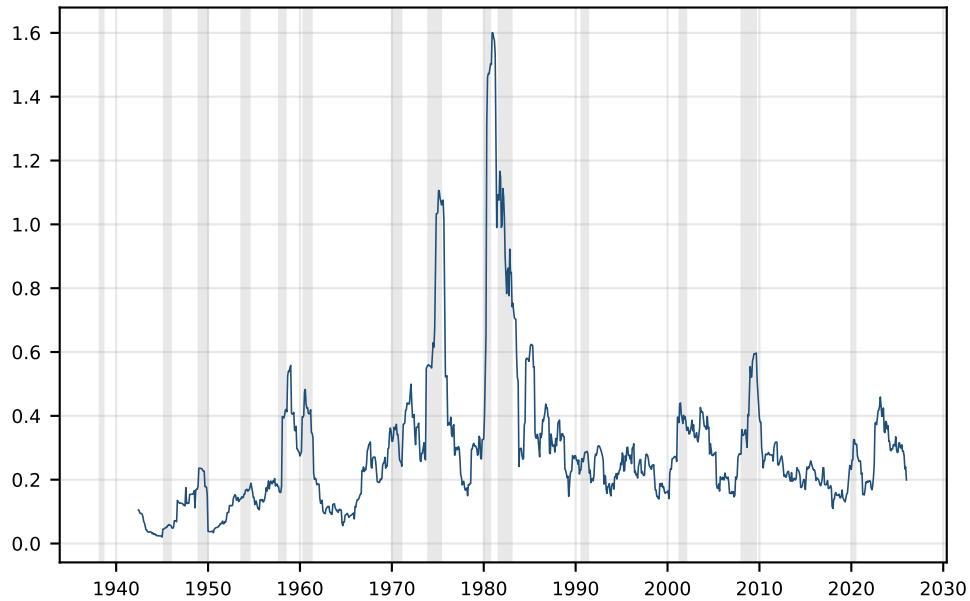
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	254	9.1%	90.9%	-0.3835
Q2	254	13.0%	87.0%	-0.0699
Q3	253	8.7%	91.3%	0.0504
Q4 (highest)	254	21.3%	78.7%	0.3975

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

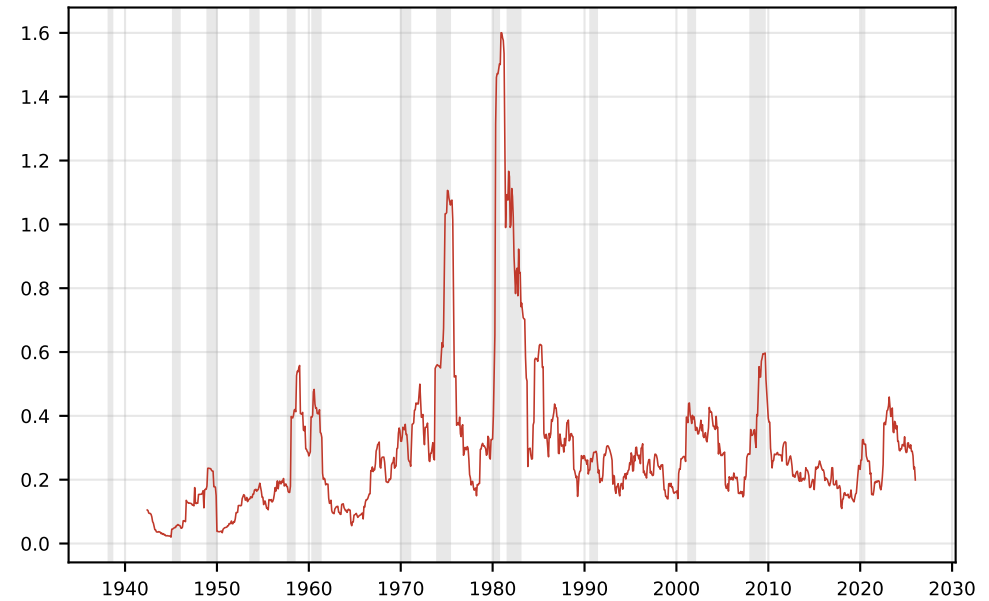
CRSP Slope 10Y Minus 3M 12M Rolling Volatility

Block: Selected export / Treasury term structure | Source: WRDS | Col: derived_slope_10y_3m_wrds_vol_12m_wrds | Transform: nan

Raw Series



After: nan



SUMMARY STATISTICS

N : 1287
Valid : 1004
NaN : 283
Start : 1942-05
End : 2025-12
Mean : 0.2912
Std : 0.2301
Min : 0.0199
Max : 1.6004
Skew : 2.845
Kurt : 10.558

STATIONARITY TESTS

ADF stat : -3.400
ADF p : 0.0109
ADF lags : 12
ADF : Stationary
KPSS stat : 0.506
KPSS p : 0.0402
KPSS : Non-stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

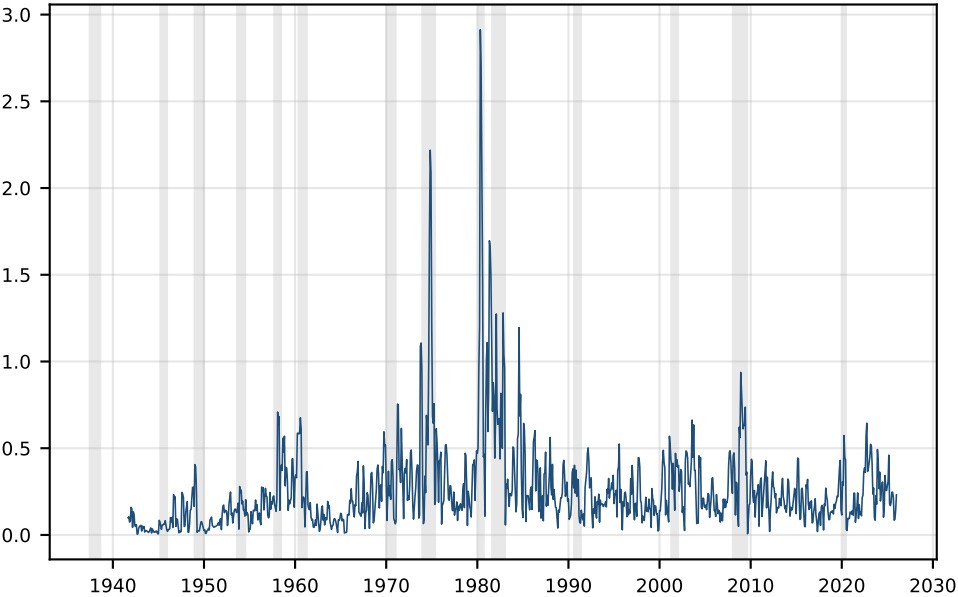
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	251	6.8%	93.2%	0.1063
Q2	251	6.4%	93.6%	0.2036
Q3	251	8.0%	92.0%	0.2877
Q4 (highest)	251	31.5%	68.5%	0.5672

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

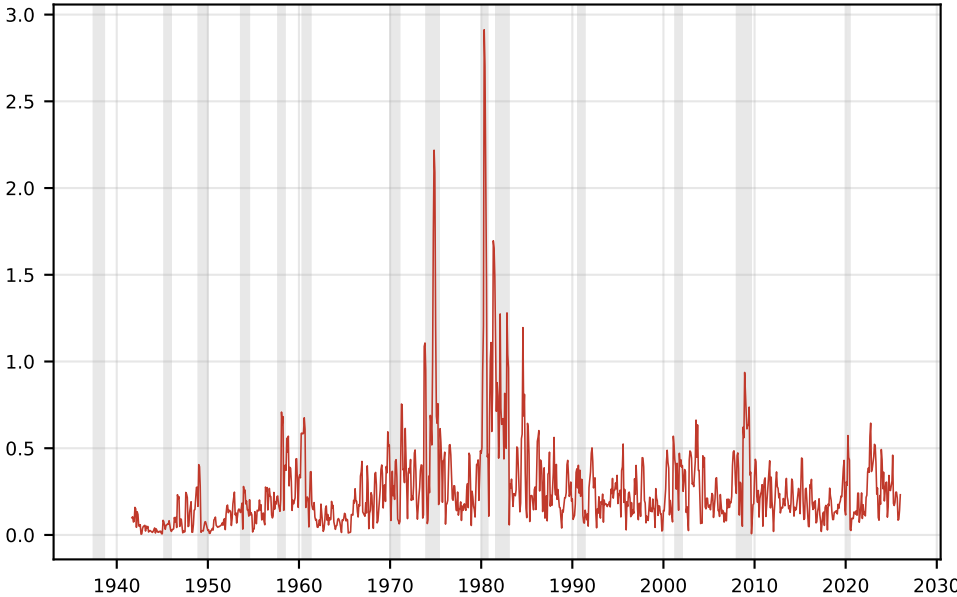
CRSP Slope 10Y Minus 3M 3M Rolling Volatility

Block: Selected export / Treasury term structure | Source: WRDS | Col: derived_slope_10y_3m_wrds_vol_3m_wrds | Transform: nan

Raw Series



After: nan



SUMMARY STATISTICS

N : 1287
Valid : 1013
NaN : 274
Start : 1941-08
End : 2025-12
Mean : 0.2584
Std : 0.2658
Min : 0.0045
Max : 2.9127
Skew : 4.110
Kurt : 27.141

STATIONARITY TESTS

ADF stat : -4.439
ADF p : 0.0003
ADF lags : 12
ADF : Stationary
KPSS stat : 0.581
KPSS p : 0.0244
KPSS : Non-stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

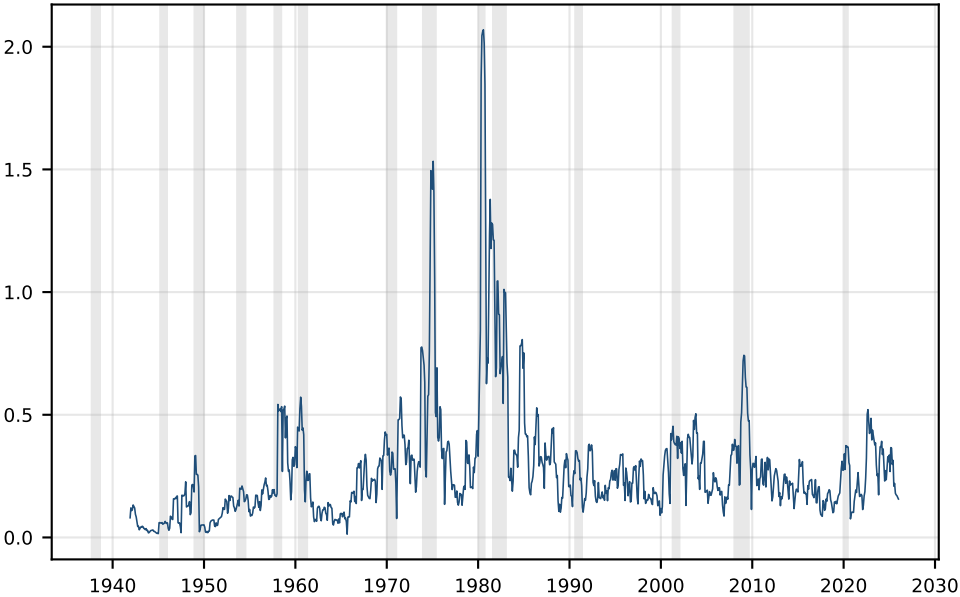
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	254	10.2%	89.8%	0.0613
Q2	253	5.1%	94.9%	0.1509
Q3	253	9.5%	90.5%	0.2476
Q4 (highest)	253	27.3%	72.7%	0.5747

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

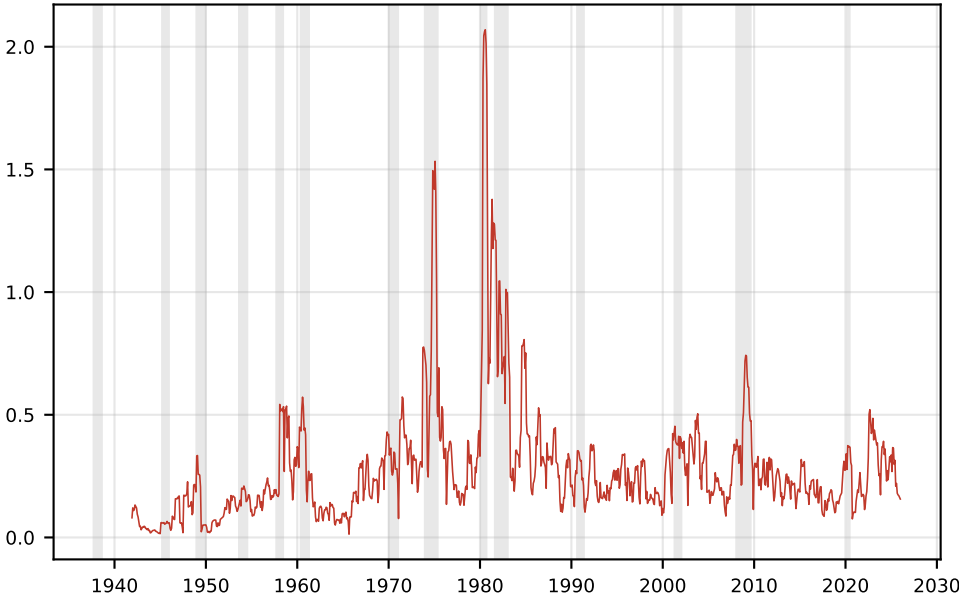
CRSP Slope 10Y Minus 3M 6M Rolling Volatility

Block: Selected export / Treasury term structure | Source: WRDS | Col: derived_slope_10y_3m_wrds_vol_6m_wrds | Transform: nan

Raw Series



After: nan



SUMMARY STATISTICS

N : 1287
Valid : 1010
NaN : 277
Start : 1941-11
End : 2025-12
Mean : 0.2790
Std : 0.2455
Min : 0.0133
Max : 2.0693
Skew : 3.469
Kurt : 17.280

STATIONARITY TESTS

ADF stat : -4.468
ADF p : 0.0002
ADF lags : 6
ADF : Stationary
KPSS stat : 0.528
KPSS p : 0.0353
KPSS : Non-stationary

QUARTILE-BUCKET ANALYSIS
(vs NBER recession/expansion)

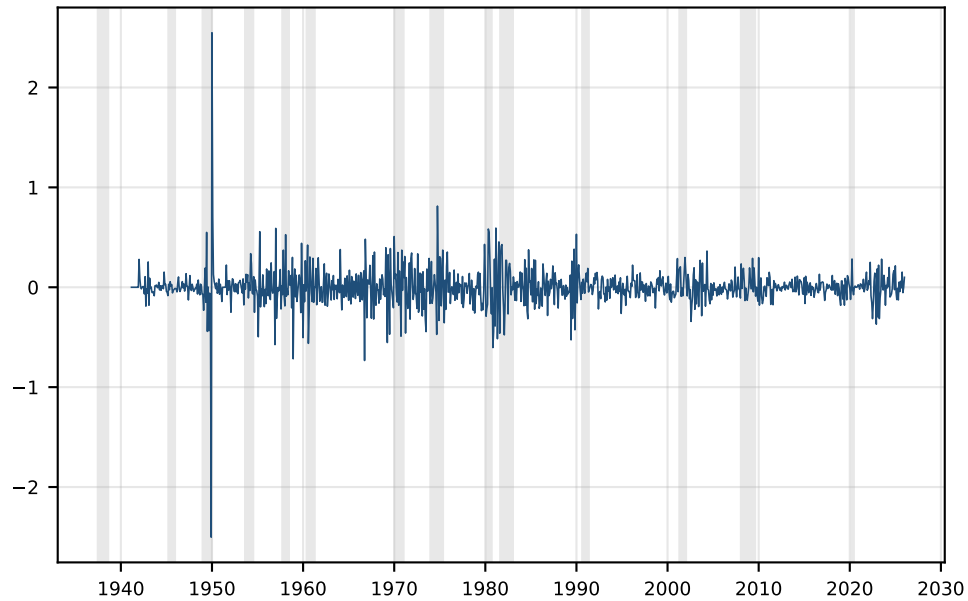
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	253	6.7%	93.3%	0.0913
Q2	252	6.0%	94.0%	0.1826
Q3	252	7.9%	92.1%	0.2769
Q4 (highest)	253	31.6%	68.4%	0.5647

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

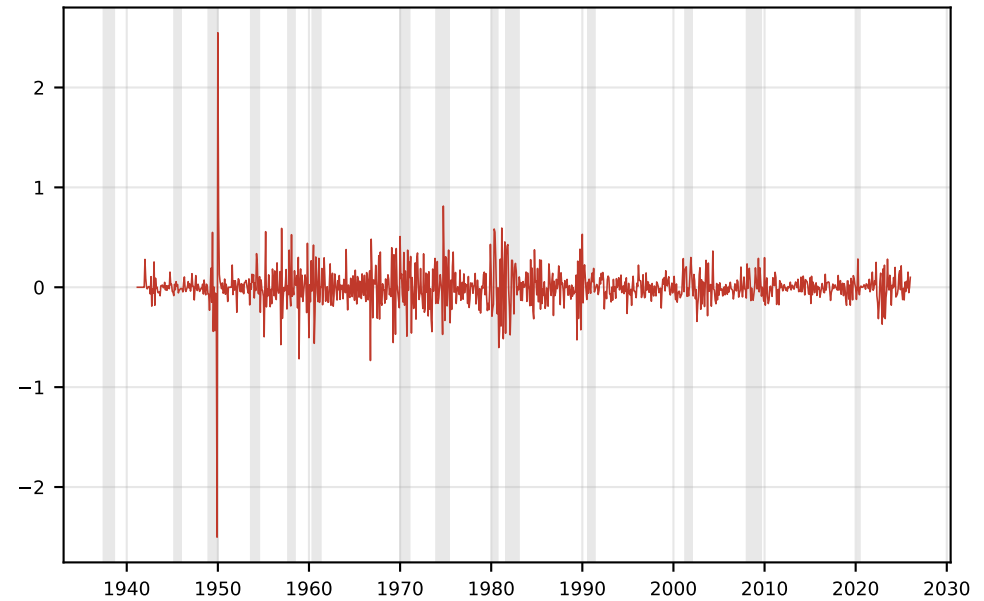
CRSP Slope 2Y Minus 1Y First Difference

Block: Selected export / Treasury term structure | Source: WRDS | Col: derived_slope_2y_1y_wrds_diff_1_wrds | Transform: nan

Raw Series



After: nan



SUMMARY STATISTICS

N : 1287
Valid : 1019
NaN : 268
Start : 1941-02
End : 2025-12
Mean : -0.0000
Std : 0.1973
Min : -2.5021
Max : 2.5482
Skew : 0.183
Kurt : 53.053

STATIONARITY TESTS

ADF stat : -14.933
ADF p : 0.0000
ADF lags : 7
ADF : Stationary
KPSS stat : 0.021
KPSS p : 0.1000
KPSS : Stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

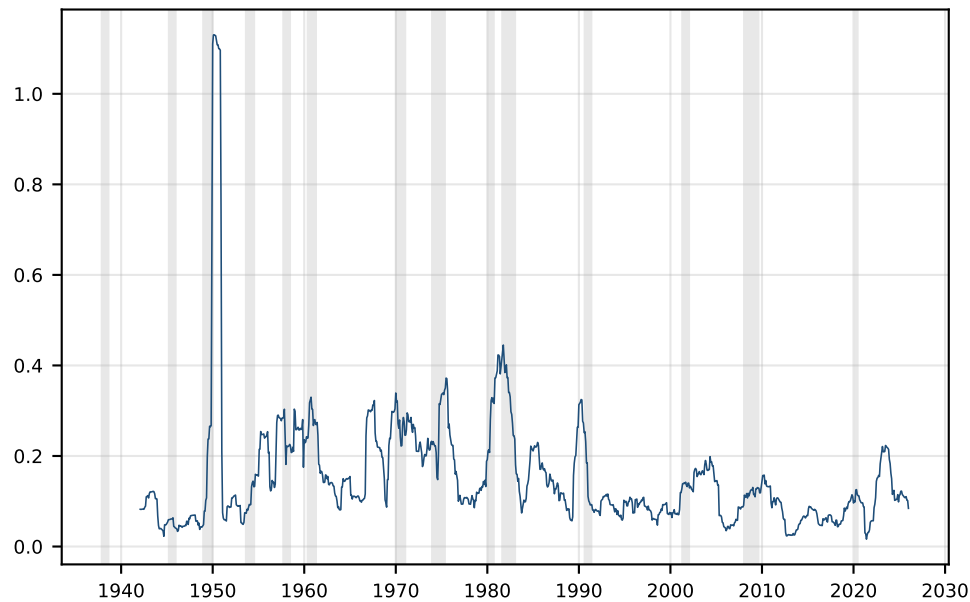
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	255	14.1%	85.9%	-0.1943
Q2	255	11.4%	88.6%	-0.0347
Q3	254	5.5%	94.5%	0.0261
Q4 (highest)	255	20.8%	79.2%	0.2029

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

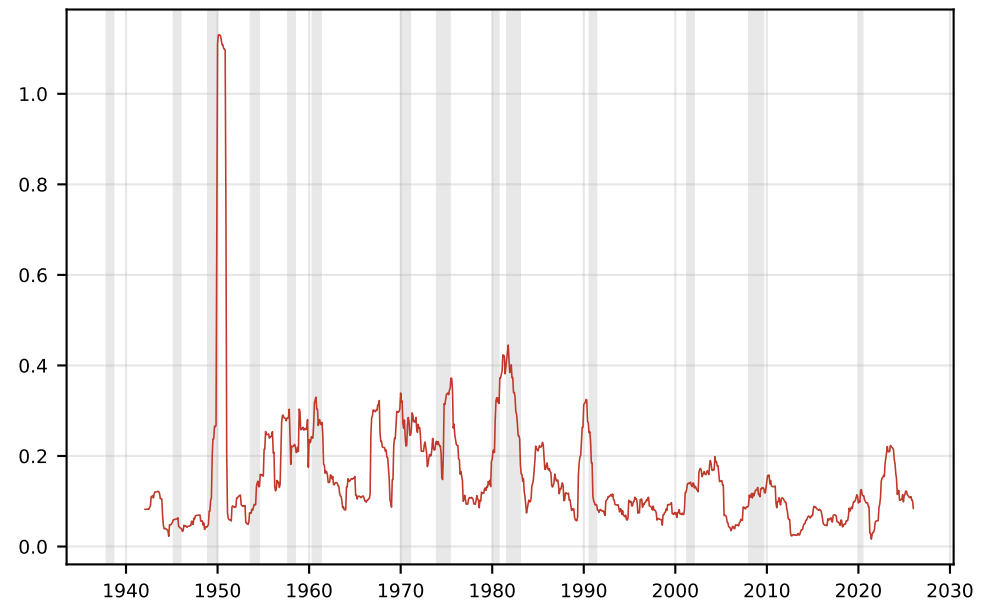
CRSP Slope 2Y Minus 1Y 12M Rolling Volatility

Block: Selected export / Treasury term structure | Source: WRDS | Col: derived_slope_2y_1y_wrds_vol_12m_wrds | Transform: nan

Raw Series



After: nan



SUMMARY STATISTICS

N : 1287
Valid : 1008
NaN : 279
Start : 1942-01
End : 2025-12
Mean : 0.1537
Std : 0.1346
Min : 0.0162
Max : 1.1305
Skew : 4.302
Kurt : 26.770

STATIONARITY TESTS

ADF stat : -4.329
ADF p : 0.0004
ADF lags : 12
ADF : Stationary
KPSS stat : 0.875
KPSS p : 0.0100
KPSS : Non-stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

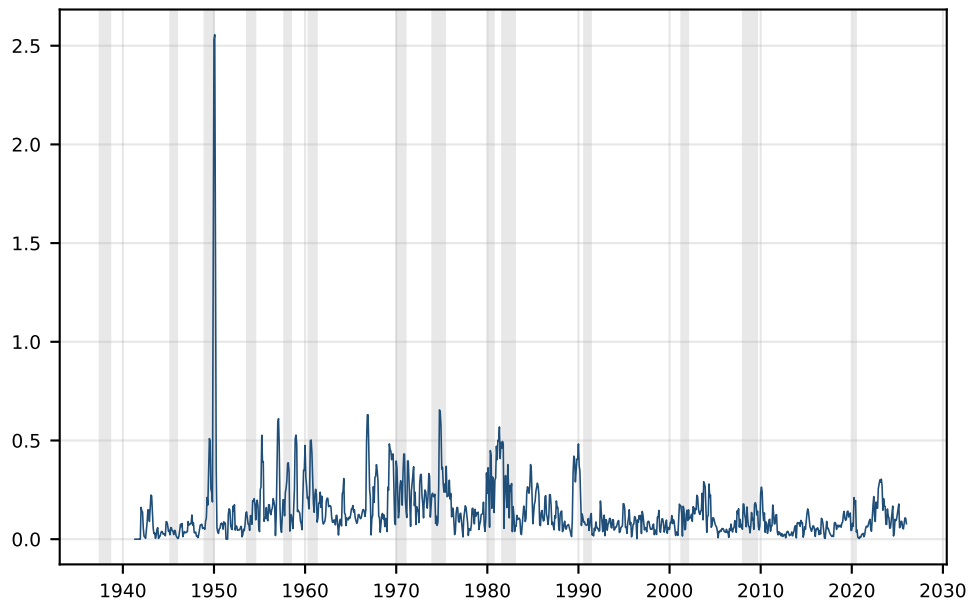
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	252	5.2%	94.8%	0.0564
Q2	252	6.7%	93.3%	0.0980
Q3	252	12.7%	87.3%	0.1488
Q4 (highest)	252	27.8%	72.2%	0.3118

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

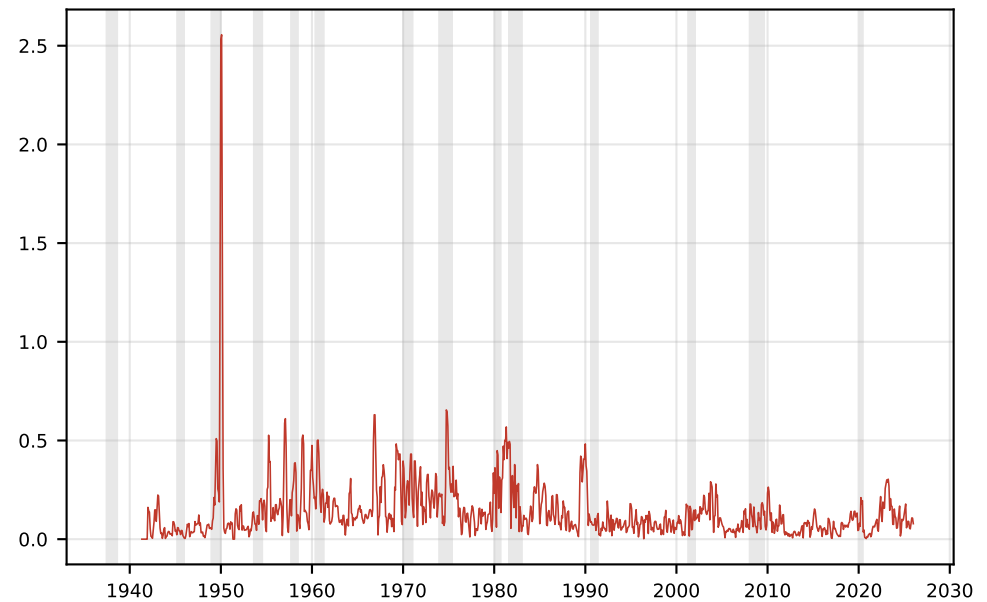
CRSP Slope 2Y Minus 1Y 3M Rolling Volatility

Block: Selected export / Treasury term structure | Source: WRDS | Col: derived_slope_2y_1y_wrds_vol_3m_wrds | Transform: nan

Raw Series



After: nan



SUMMARY STATISTICS

N : 1287
Valid : 1017
NaN : 270
Start : 1941-04
End : 2025-12
Mean : 0.1393
Std : 0.1647
Min : 0.0000
Max : 2.5557
Skew : 7.366
Kurt : 93.368

STATIONARITY TESTS

ADF stat : -9.283
ADF p : 0.0000
ADF lags : 3
ADF : Stationary
KPSS stat : 0.945
KPSS p : 0.0100
KPSS : Non-stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

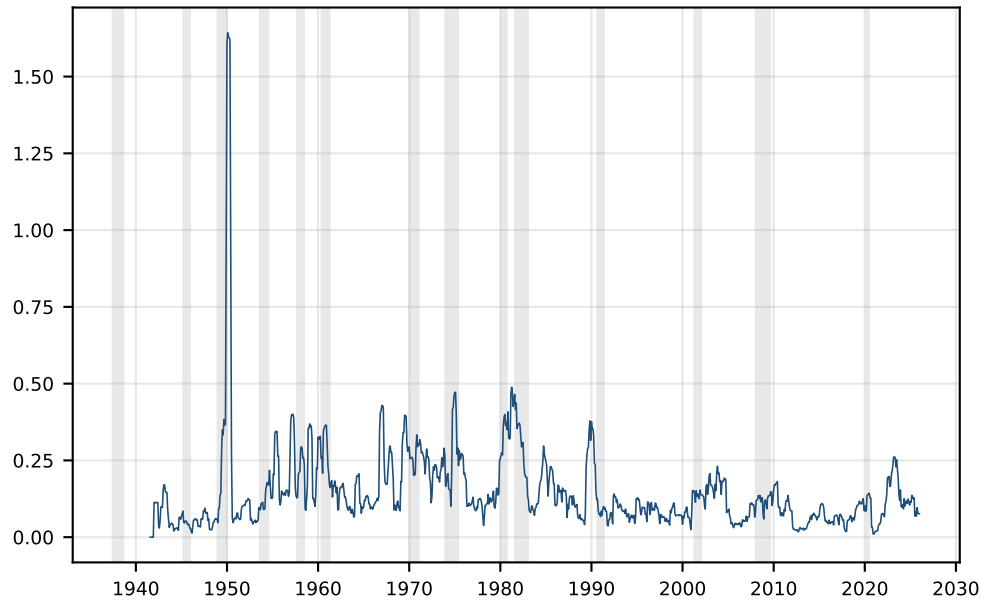
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	255	6.3%	93.7%	0.0320
Q2	254	8.7%	91.3%	0.0762
Q3	254	10.6%	89.4%	0.1297
Q4 (highest)	254	26.4%	73.6%	0.3196

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

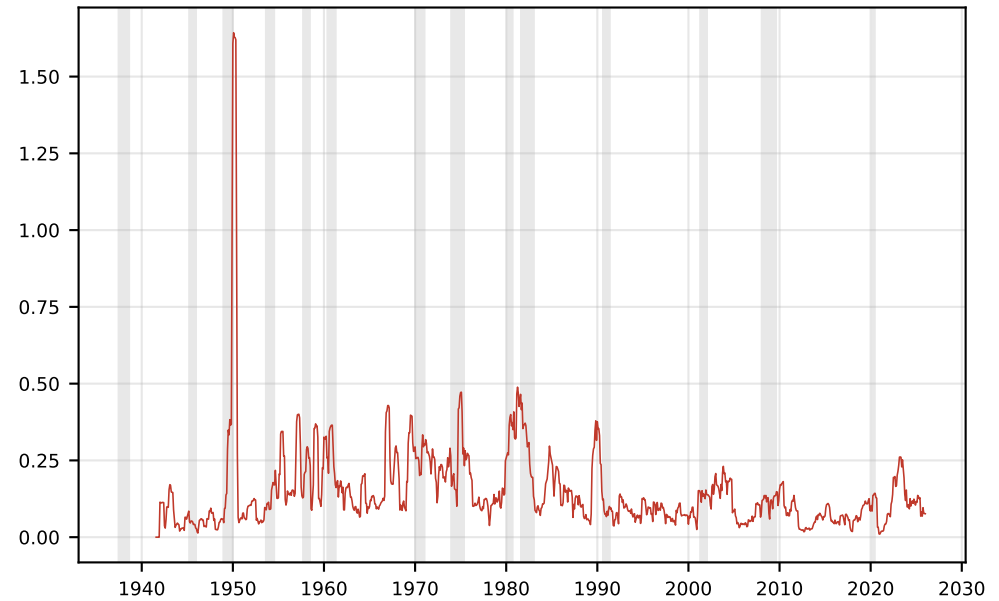
CRSP Slope 2Y Minus 1Y 6M Rolling Volatility

Block: Selected export / Treasury term structure | Source: WRDS | Col: derived_slope_2y_1y_wrds_vol_6m_wrds | Transform: nan

Raw Series



After: nan



SUMMARY STATISTICS

N : 1287
Valid : 1014
NaN : 273
Start : 1941-07
End : 2025-12
Mean : 0.1478
Std : 0.1466
Min : 0.0000
Max : 1.6428
Skew : 5.706
Kurt : 50.960

STATIONARITY TESTS

ADF stat : -5.645
ADF p : 0.0000
ADF lags : 12
ADF : Stationary
KPSS stat : 0.843
KPSS p : 0.0100
KPSS : Non-stationary

QUARTILE-BUCKET ANALYSIS

(vs NBER recession/expansion)

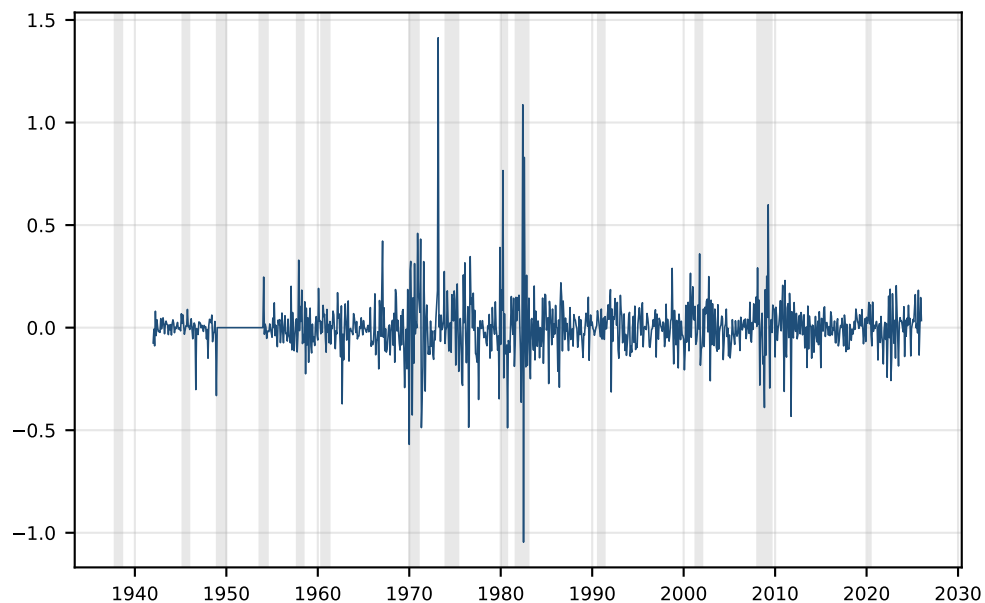
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	254	4.3%	95.7%	0.0474
Q2	253	6.7%	93.3%	0.0912
Q3	253	14.6%	85.4%	0.1383
Q4 (highest)	254	26.4%	73.6%	0.3140

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

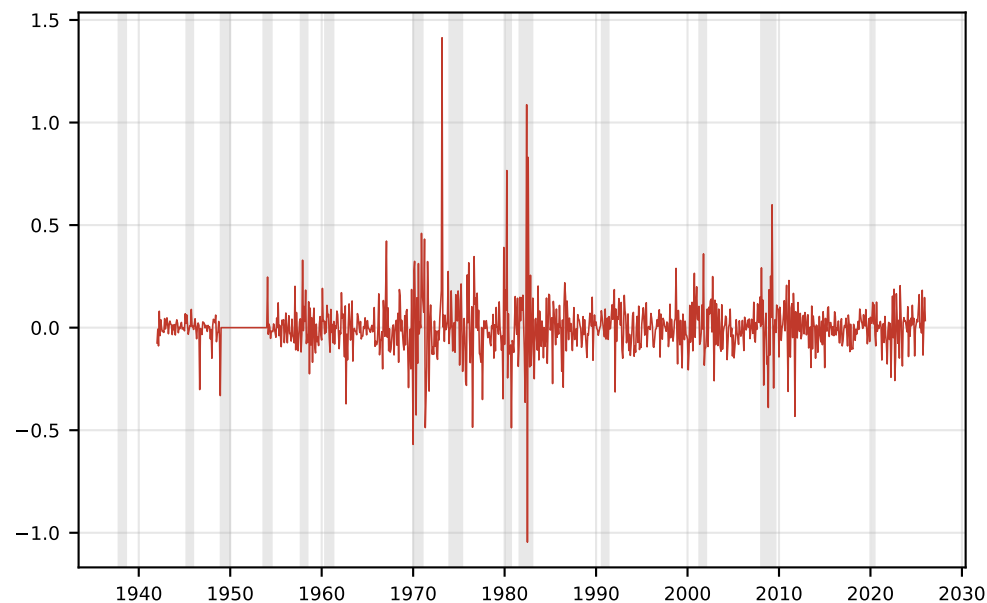
CRSP Slope 30Y Minus 10Y First Difference

Block: Selected export / Treasury term structure | Source: WRDS | Col: derived_slope_30y_10y_wrds_diff_1_wrds | Transform: nan

Raw Series



After: nan



SUMMARY STATISTICS

N : 1287
Valid : 1009
NaN : 278
Start : 1941-12
End : 2025-12
Mean : 0.0002
Std : 0.1318
Min : -1.0458
Max : 1.4134
Skew : 1.555
Kurt : 25.031

STATIONARITY TESTS

ADF stat : -10.893
ADF p : 0.0000
ADF lags : 10
ADF : Stationary
KPSS stat : 0.027
KPSS p : 0.1000
KPSS : Stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

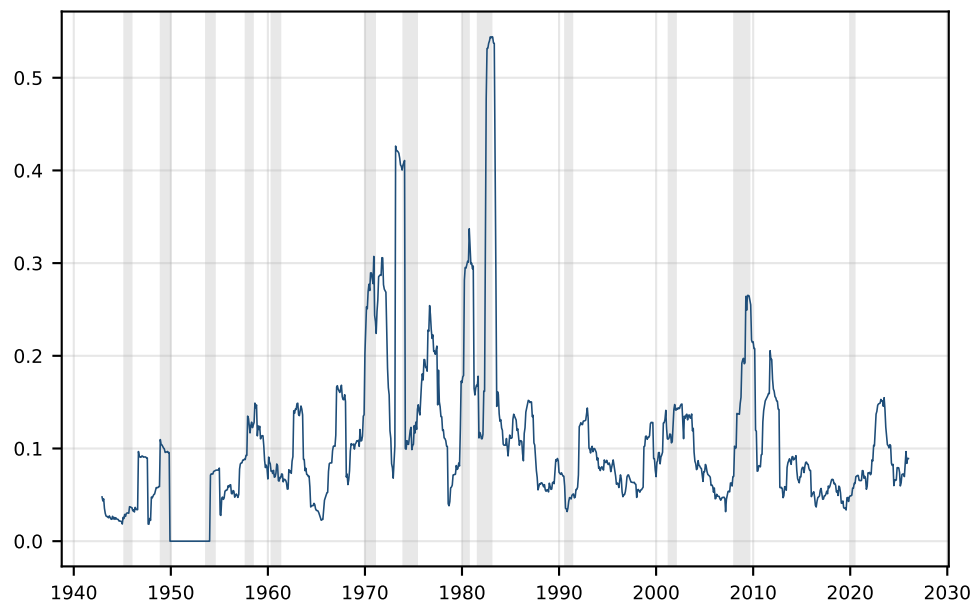
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	253	11.9%	88.1%	-0.1349
Q2	277	11.6%	88.4%	-0.0163
Q3	227	10.6%	89.4%	0.0205
Q4 (highest)	252	18.3%	81.7%	0.1359

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

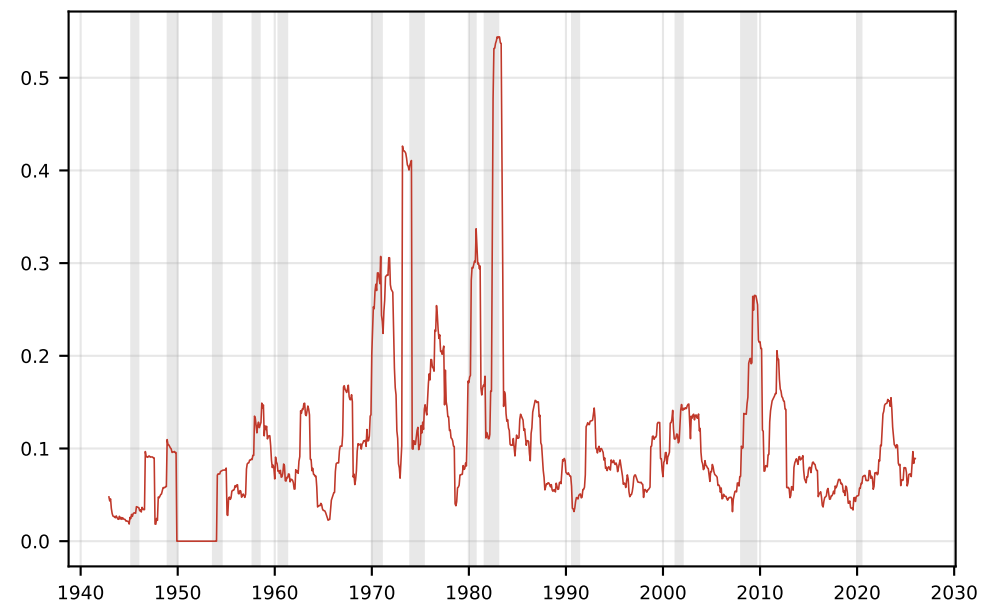
CRSP Slope 30Y Minus 10Y 12M Rolling Volatility

Block: Selected export / Treasury term structure | Source: WRDS | Col: derived_slope_30y_10y_wrds_vol_12m_wrds | Transform: nan

Raw Series



After: nan



SUMMARY STATISTICS

N : 1287
Valid : 998
NaN : 289
Start : 1942-11
End : 2025-12
Mean : 0.1045
Std : 0.0844
Min : 0.0000
Max : 0.5443
Skew : 2.440
Kurt : 8.148

STATIONARITY TESTS

ADF stat : -3.750
ADF p : 0.0035
ADF lags : 12
ADF : Stationary
KPSS stat : 0.487
KPSS p : 0.0446
KPSS : Non-stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

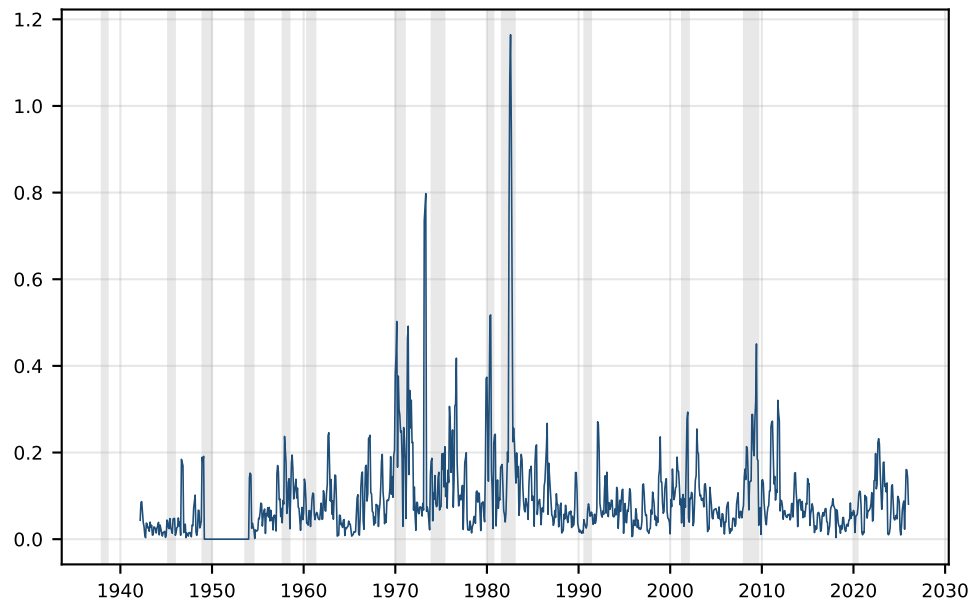
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	250	8.4%	91.6%	0.0337
Q2	249	6.8%	93.2%	0.0698
Q3	249	18.5%	81.5%	0.1029
Q4 (highest)	250	19.2%	80.8%	0.2112

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

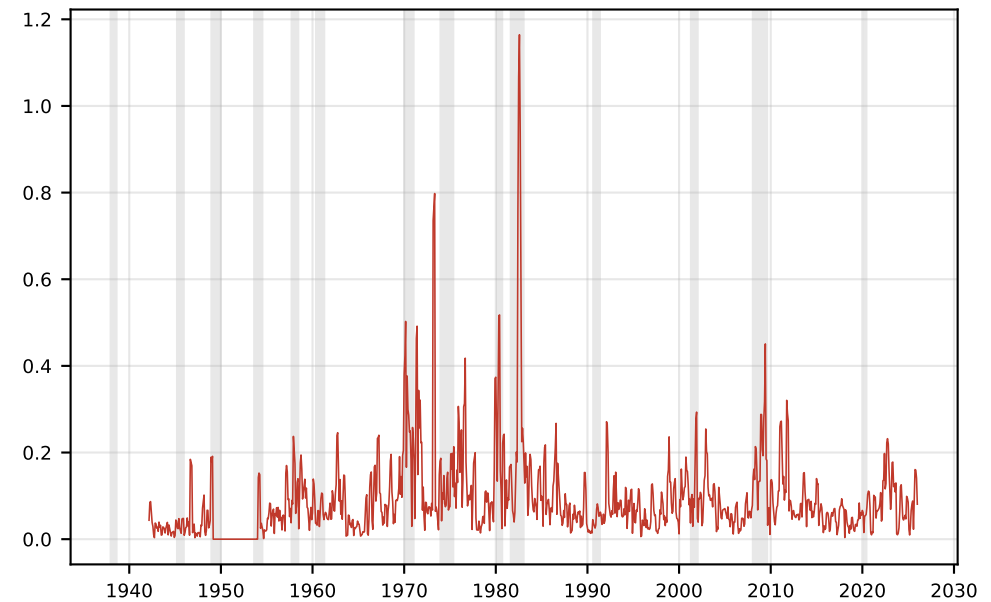
CRSP Slope 30Y Minus 10Y 3M Rolling Volatility

Block: Selected export / Treasury term structure | Source: WRDS | Col: derived_slope_30y_10y_wrds_vol_3m_wrds | Transform: nan

Raw Series



After: nan



SUMMARY STATISTICS

N : 1287
Valid : 1007
NaN : 280
Start : 1942-02
End : 2025-12
Mean : 0.0914
Std : 0.1033
Min : 0.0000
Max : 1.1644
Skew : 4.408
Kurt : 31.265

STATIONARITY TESTS

ADF stat : -9.147
ADF p : 0.0000
ADF lags : 4
ADF : Stationary
KPSS stat : 0.511
KPSS p : 0.0391
KPSS : Non-stationary

QUARTILE-BUCKET ANALYSIS

(vs NBER recession/expansion)

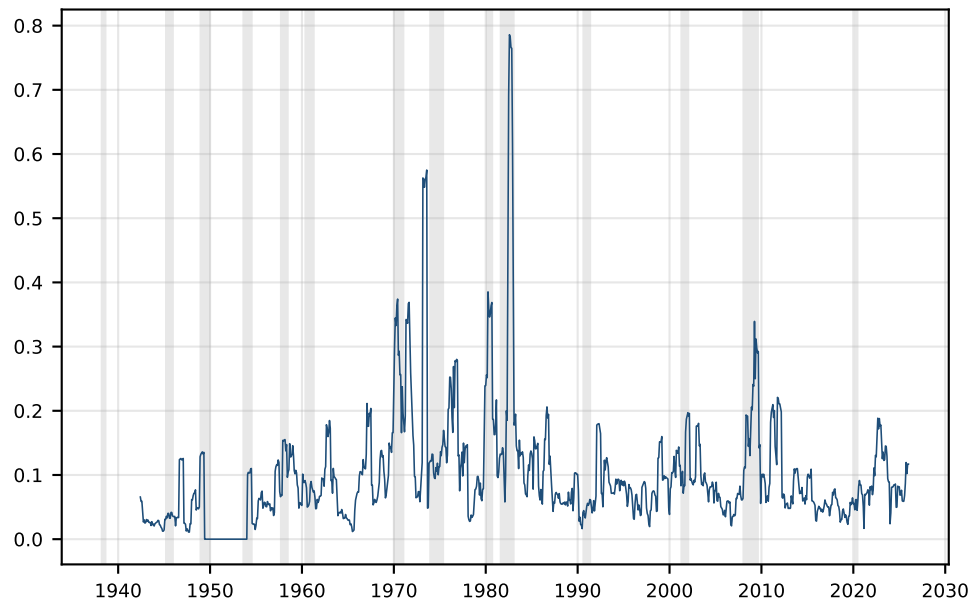
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	252	10.7%	89.3%	0.0160
Q2	252	7.5%	92.5%	0.0510
Q3	251	9.2%	90.8%	0.0862
Q4 (highest)	252	25.0%	75.0%	0.2124

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

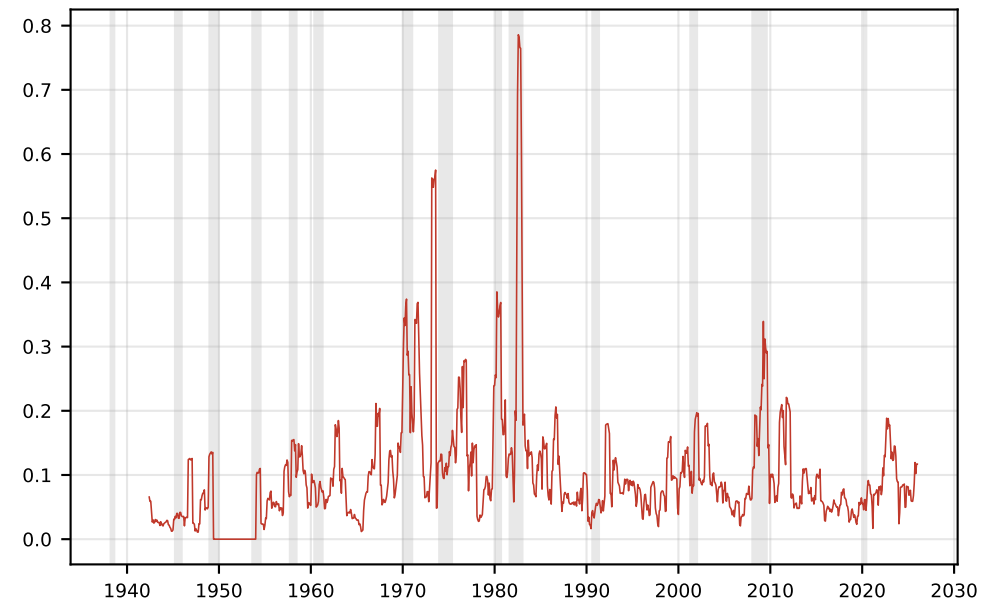
CRSP Slope 30Y Minus 10Y 6M Rolling Volatility

Block: Selected export / Treasury term structure | Source: WRDS | Col: derived_slope_30y_10y_wrds_vol_6m_wrds | Transform: nan

Raw Series



After: nan



SUMMARY STATISTICS

N : 1287
Valid : 1004
NaN : 283
Start : 1942-05
End : 2025-12
Mean : 0.0989
Std : 0.0918
Min : 0.0000
Max : 0.7858
Skew : 3.386
Kurt : 17.450

STATIONARITY TESTS

ADF stat : -5.050
ADF p : 0.0000
ADF lags : 12
ADF : Stationary
KPSS stat : 0.489
KPSS p : 0.0442
KPSS : Non-stationary

QUARTILE-BUCKET ANALYSIS

(vs NBER recession/expansion)

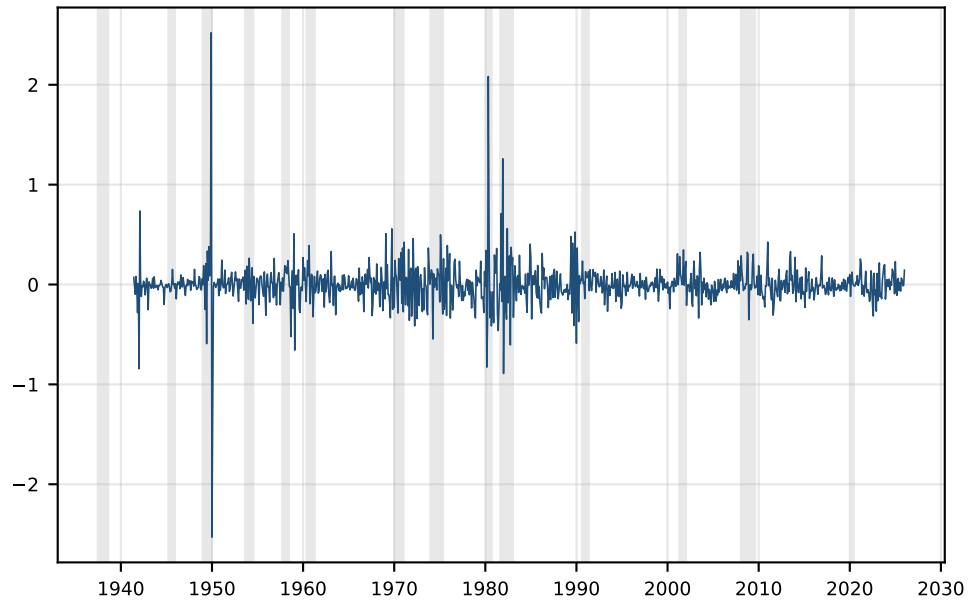
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	251	9.2%	90.8%	0.0264
Q2	251	5.6%	94.4%	0.0629
Q3	251	12.4%	87.6%	0.0966
Q4 (highest)	251	25.5%	74.5%	0.2096

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

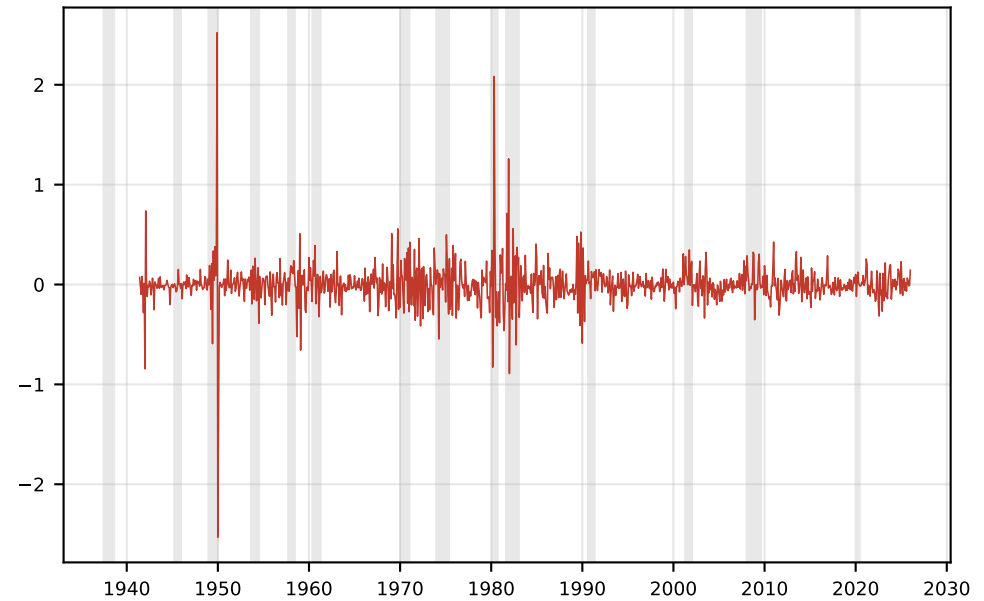
CRSP Slope 5Y Minus 2Y First Difference

Block: Selected export / Treasury term structure | Source: WRDS | Col: derived_slope_5y_2y_wrds_diff_1_wrds | Transform: nan

Raw Series



After: nan



SUMMARY STATISTICS

N : 1287
Valid : 1016
NaN : 271
Start : 1941-05
End : 2025-12
Mean : -0.0011
Std : 0.2111
Min : -2.5299
Max : 2.5210
Skew : 0.923
Kurt : 51.106

STATIONARITY TESTS

ADF stat : -16.696
ADF p : 0.0000
ADF lags : 5
ADF : Stationary
KPSS stat : 0.035
KPSS p : 0.1000
KPSS : Stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

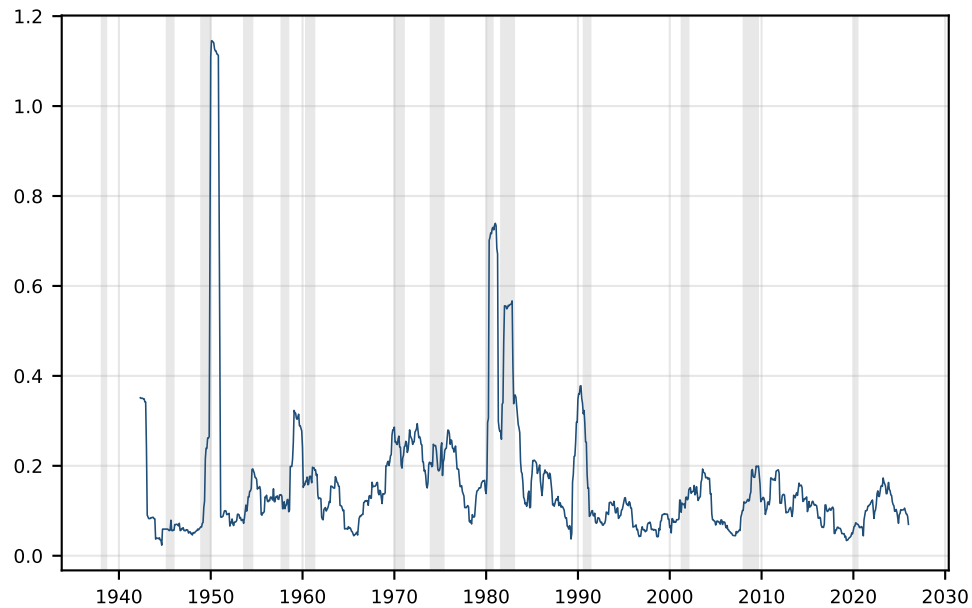
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	254	10.6%	89.4%	-0.2009
Q2	254	5.9%	94.1%	-0.0341
Q3	254	9.8%	90.2%	0.0298
Q4 (highest)	254	25.6%	74.4%	0.2010

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

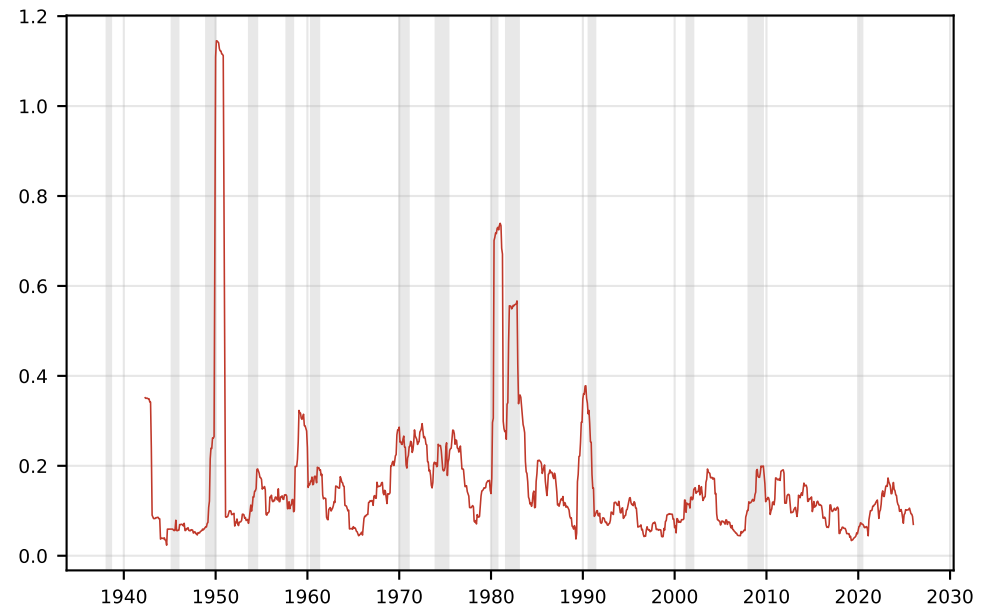
CRSP Slope 5Y Minus 2Y 12M Rolling Volatility

Block: Selected export / Treasury term structure | Source: WRDS | Col: derived_slope_5y_2y_wrds_vol_12m_wrds | Transform: nan

Raw Series



After: nan



SUMMARY STATISTICS

N : 1287
Valid : 1005
NaN : 282
Start : 1942-04
End : 2025-12
Mean : 0.1575
Std : 0.1490
Min : 0.0235
Max : 1.1453
Skew : 4.025
Kurt : 20.476

STATIONARITY TESTS

ADF stat : -4.132
ADF p : 0.0009
ADF lags : 12
ADF : Stationary
KPSS stat : 0.489
KPSS p : 0.0441
KPSS : Non-stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

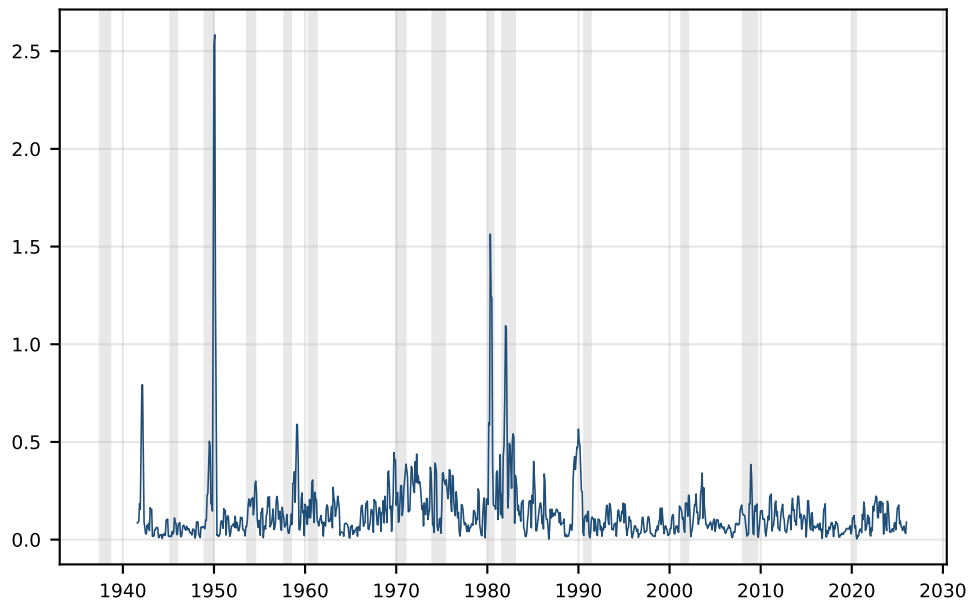
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	252	5.2%	94.8%	0.0607
Q2	251	8.0%	92.0%	0.1006
Q3	251	12.7%	87.3%	0.1440
Q4 (highest)	251	26.7%	73.3%	0.3250

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

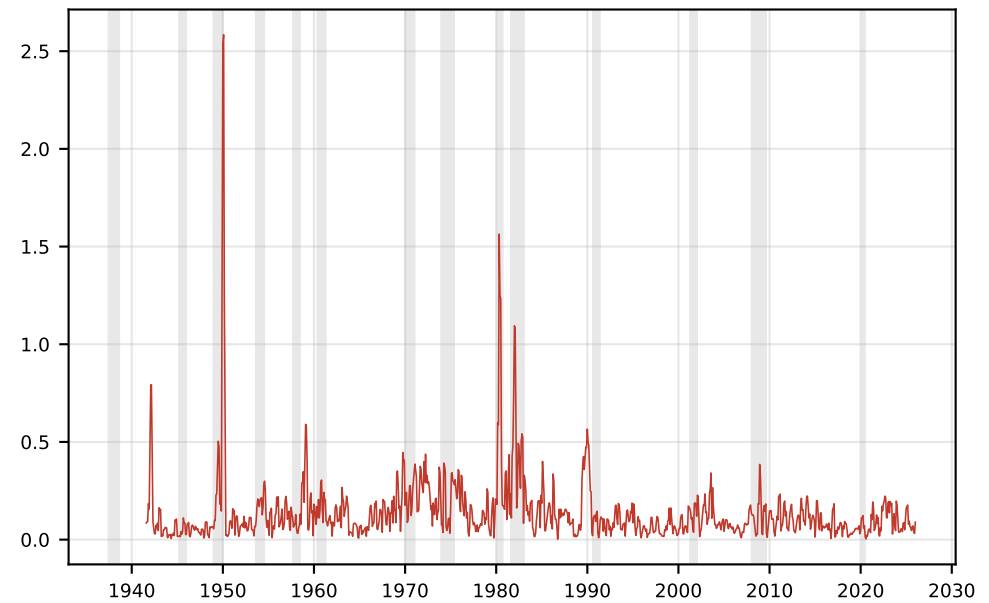
CRSP Slope 5Y Minus 2Y 3M Rolling Volatility

Block: Selected export / Treasury term structure | Source: WRDS | Col: derived_slope_5y_2y_wrds_vol_3m_wrds | Transform: nan

Raw Series



After: nan



SUMMARY STATISTICS

N : 1287
Valid : 1014
NaN : 273
Start : 1941-07
End : 2025-12
Mean : 0.1395
Std : 0.1804
Min : 0.0023
Max : 2.5842
Skew : 6.926
Kurt : 73.955

STATIONARITY TESTS

ADF stat : -8.175
ADF p : 0.0000
ADF lags : 6
ADF : Stationary
KPSS stat : 0.548
KPSS p : 0.0310
KPSS : Non-stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

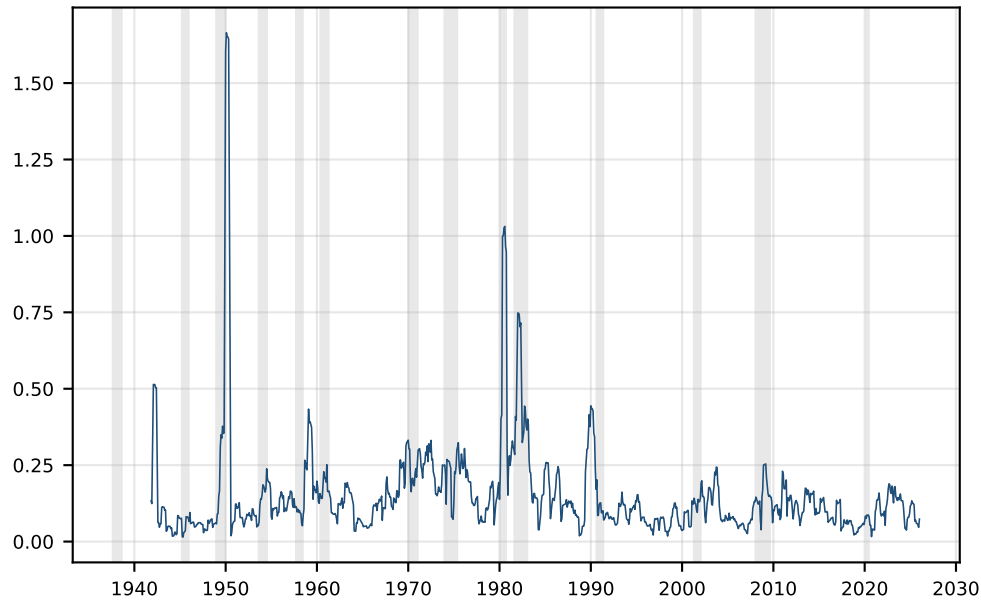
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	254	7.9%	92.1%	0.0332
Q2	253	6.7%	93.3%	0.0734
Q3	253	13.4%	86.6%	0.1286
Q4 (highest)	254	24.0%	76.0%	0.3226

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

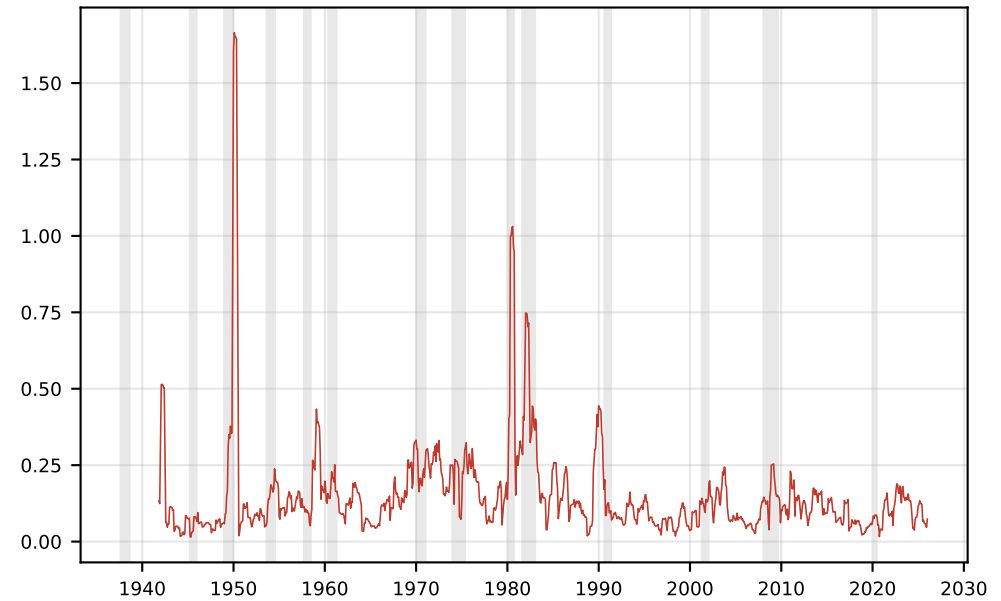
CRSP Slope 5Y Minus 2Y 6M Rolling Volatility

Block: Selected export / Treasury term structure | Source: WRDS | Col: derived_slope_5y_2y_wrds_vol_6m_wrds | Transform: nan

Raw Series



After: nan



SUMMARY STATISTICS

N : 1287
Valid : 1011
NaN : 276
Start : 1941-10
End : 2025-12
Mean : 0.1506
Std : 0.1626
Min : 0.0144
Max : 1.6647
Skew : 5.336
Kurt : 39.421

STATIONARITY TESTS

ADF stat : -5.585
ADF p : 0.0000
ADF lags : 12
ADF : Stationary
KPSS stat : 0.511
KPSS p : 0.0393
KPSS : Non-stationary

QUARTILE-BUCKET ANALYSIS

(vs NBER recession/expansion)

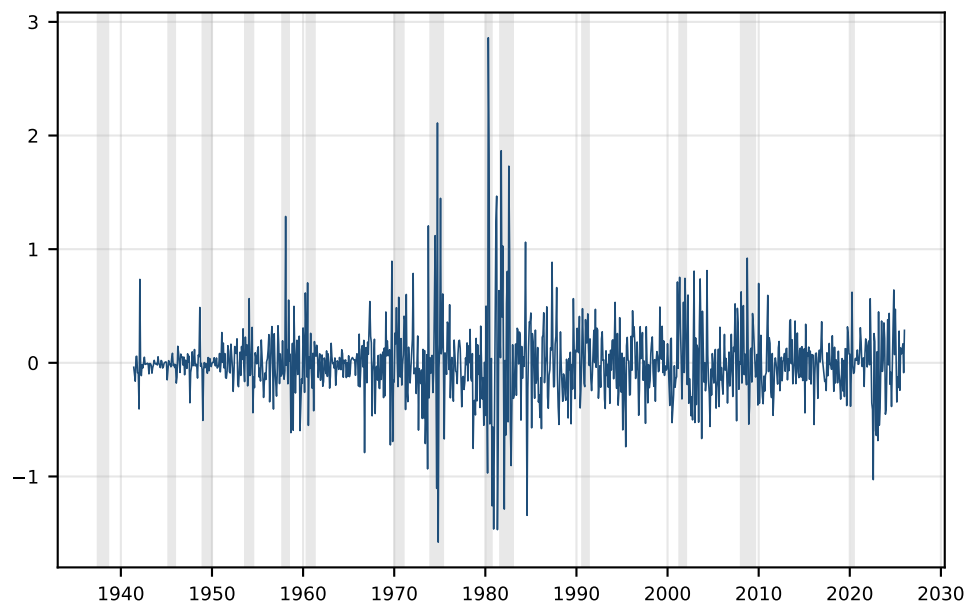
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	253	4.0%	96.0%	0.0508
Q2	253	9.1%	90.9%	0.0908
Q3	252	12.7%	87.3%	0.1380
Q4 (highest)	253	26.5%	73.5%	0.3228

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

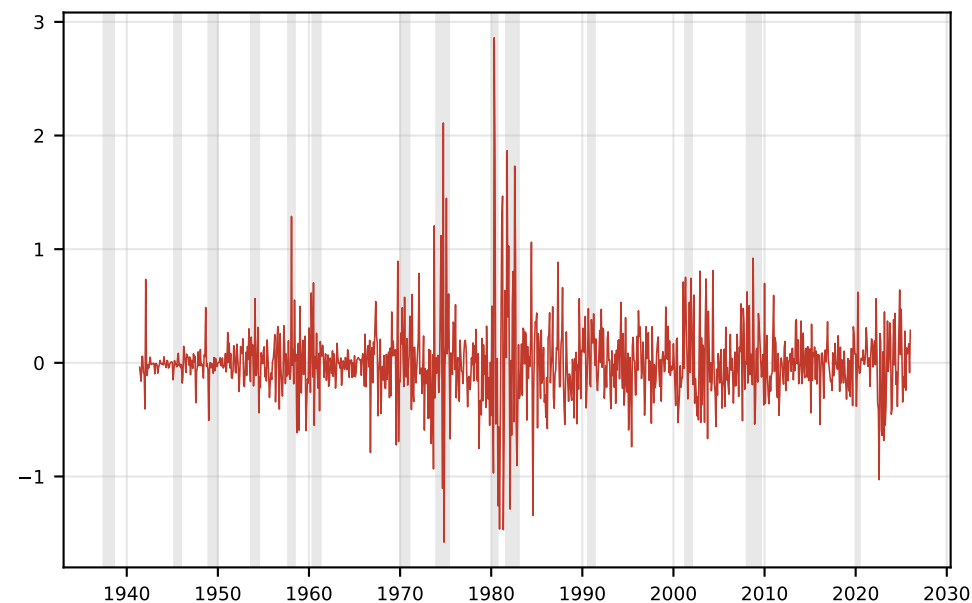
CRSP Slope 5Y Minus 3M First Difference

Block: Selected export / Treasury term structure | Source: WRDS | Col: derived_slope_5y_3m_wrds_diff_1_wrds | Transform: nan

Raw Series



After: nan



SUMMARY STATISTICS

N : 1287
Valid : 1016
NaN : 271
Start : 1941-05
End : 2025-12
Mean : -0.0013
Std : 0.3418
Min : -1.5782
Max : 2.8605
Skew : 1.185
Kurt : 10.591

STATIONARITY TESTS

ADF stat : -16.009
ADF p : 0.0000
ADF lags : 5
ADF : Stationary
KPSS stat : 0.013
KPSS p : 0.1000
KPSS : Stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

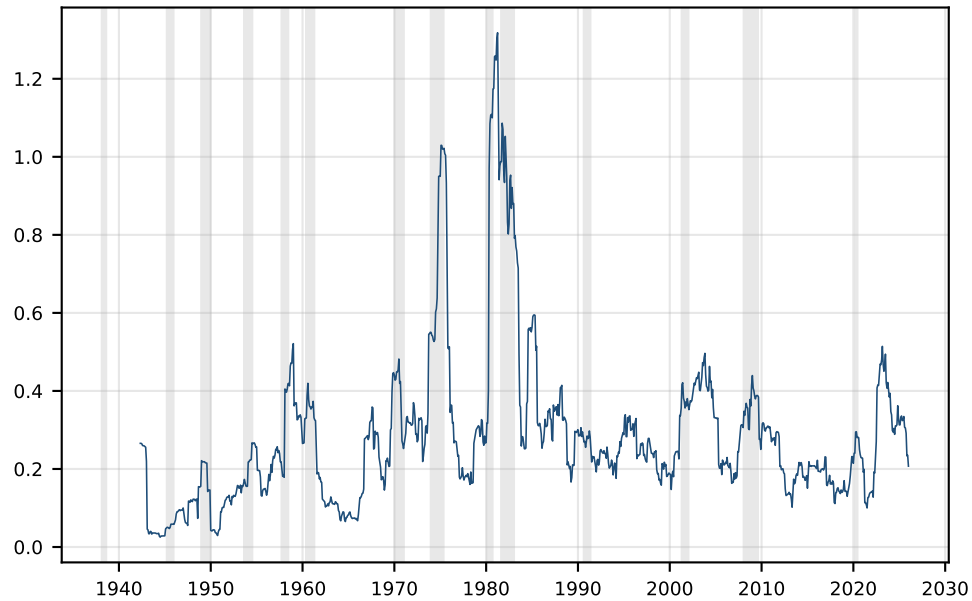
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	254	11.4%	88.6%	-0.3667
Q2	254	12.2%	87.8%	-0.0716
Q3	254	9.8%	90.2%	0.0510
Q4 (highest)	254	18.5%	81.5%	0.3821

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

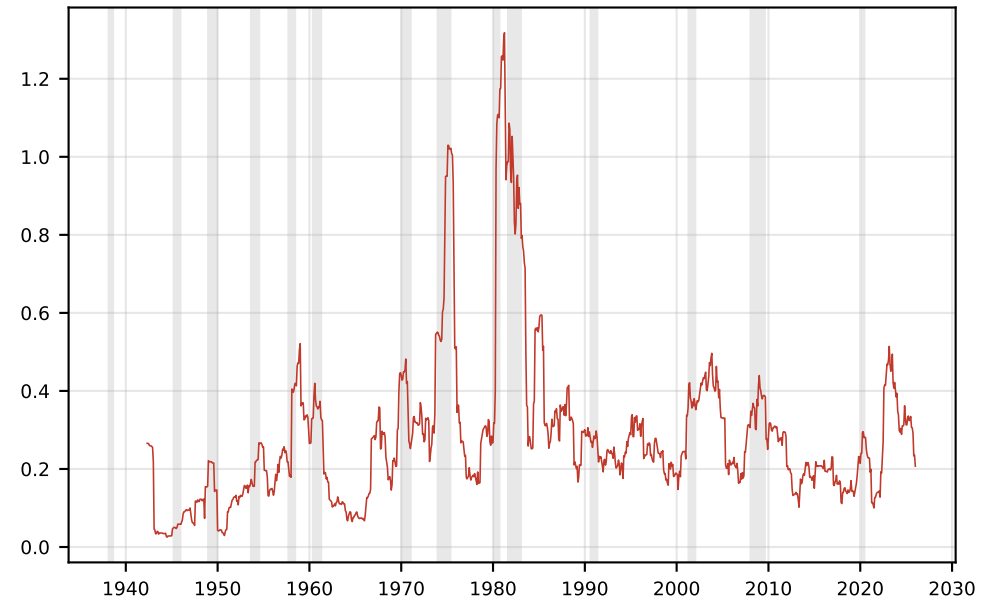
CRSP Slope 5Y Minus 3M 12M Rolling Volatility

Block: Selected export / Treasury term structure | Source: WRDS | Col: derived_slope_5y_3m_wrds_vol_12m_wrds | Transform: nan

Raw Series



After: nan



SUMMARY STATISTICS

N : 1287
Valid : 1005
NaN : 282
Start : 1942-04
End : 2025-12
Mean : 0.2829
Std : 0.2023
Min : 0.0252
Max : 1.3180
Skew : 2.339
Kurt : 6.913

STATIONARITY TESTS

ADF stat : -3.466
ADF p : 0.0089
ADF lags : 12
ADF : Stationary
KPSS stat : 0.509
KPSS p : 0.0396
KPSS : Non-stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

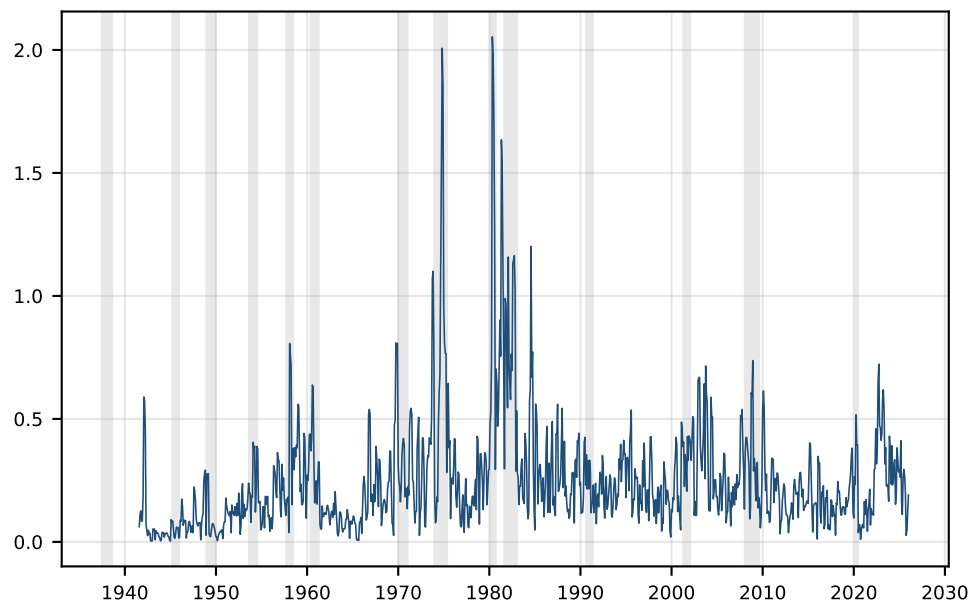
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	252	6.0%	94.0%	0.1063
Q2	251	7.2%	92.8%	0.2060
Q3	251	6.4%	93.6%	0.2883
Q4 (highest)	251	33.1%	66.9%	0.5318

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

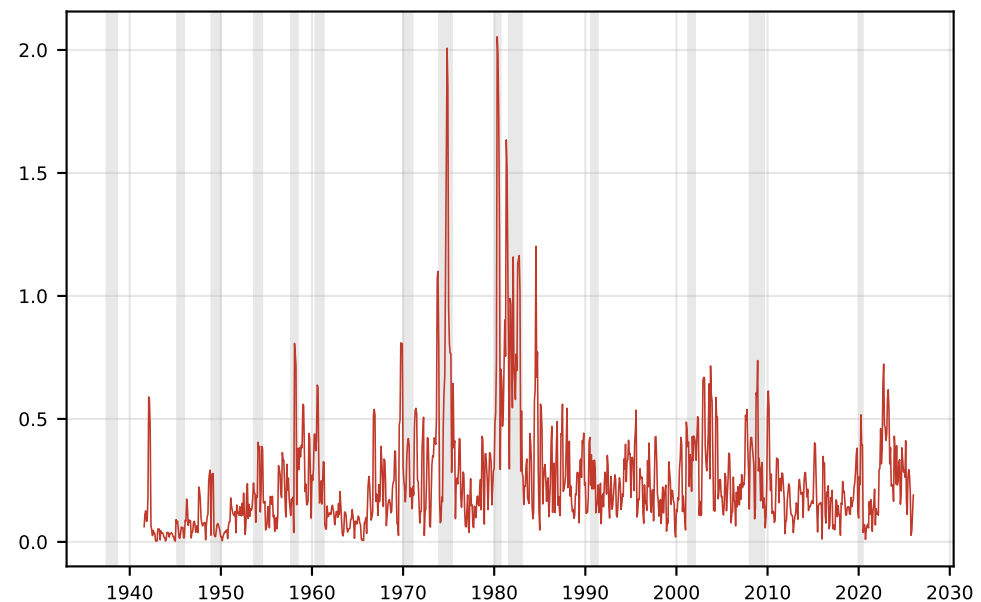
CRSP Slope 5Y Minus 3M 3M Rolling Volatility

Block: Selected export / Treasury term structure | Source: WRDS | Col: derived_slope_5y_3m_wrds_vol_3m_wrds | Transform: nan

Raw Series



After: nan



SUMMARY STATISTICS

N : 1287
Valid : 1014
NaN : 273
Start : 1941-07
End : 2025-12
Mean : 0.2525
Std : 0.2386
Min : 0.0029
Max : 2.0538
Skew : 3.172
Kurt : 15.708

STATIONARITY TESTS

ADF stat : -4.730
ADF p : 0.0001
ADF lags : 12
ADF : Stationary
KPSS stat : 0.567
KPSS p : 0.0265
KPSS : Non-stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

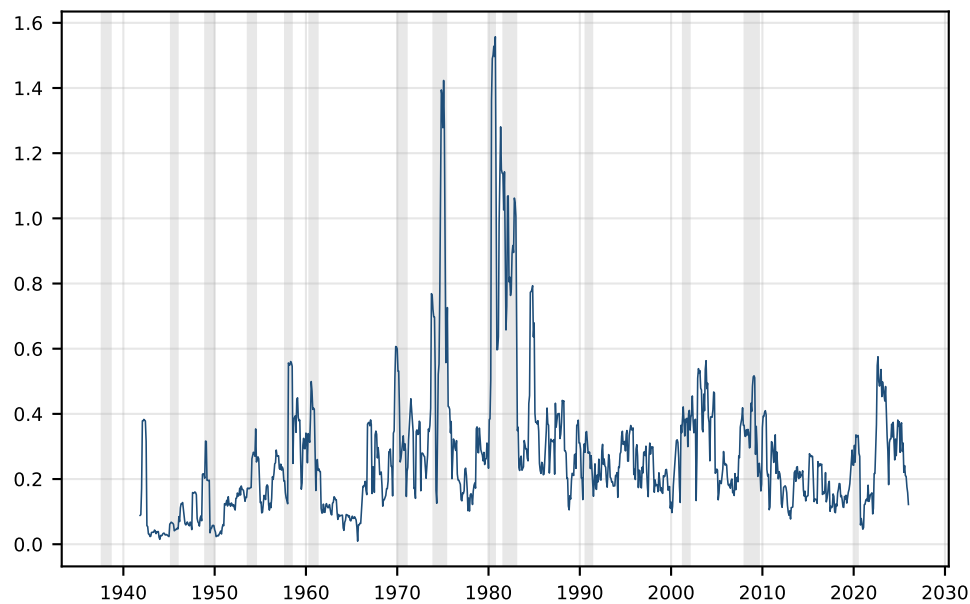
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	254	8.3%	91.7%	0.0631
Q2	253	7.1%	92.9%	0.1475
Q3	253	11.5%	88.5%	0.2494
Q4 (highest)	254	25.2%	74.8%	0.5497

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

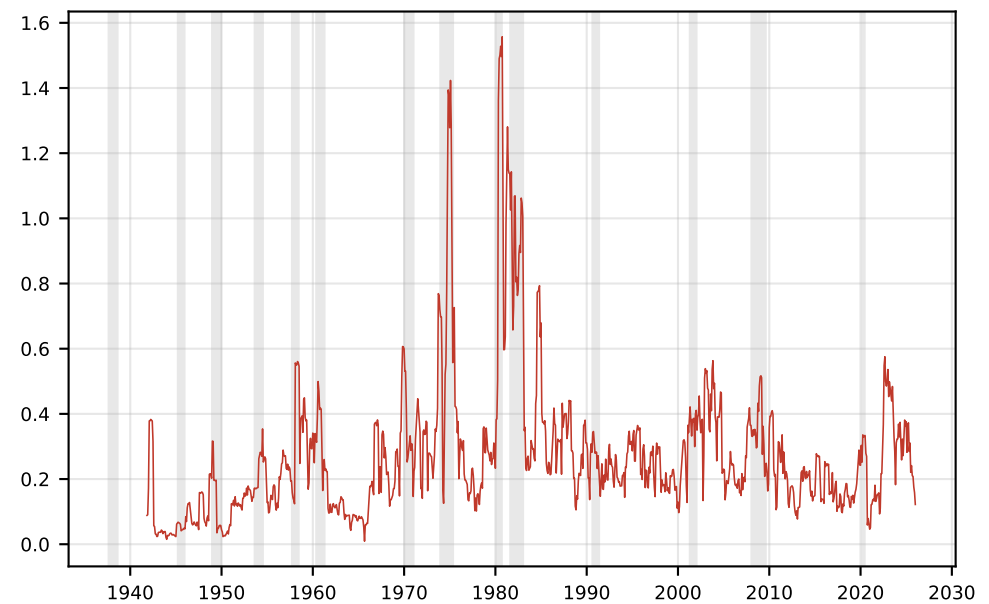
CRSP Slope 5Y Minus 3M 6M Rolling Volatility

Block: Selected export / Treasury term structure | Source: WRDS | Col: derived_slope_5y_3m_wrds_vol_6m_wrds | Transform: nan

Raw Series



After: nan



SUMMARY STATISTICS

N : 1287
Valid : 1011
NaN : 276
Start : 1941-10
End : 2025-12
Mean : 0.2723
Std : 0.2172
Min : 0.0094
Max : 1.5573
Skew : 2.747
Kurt : 10.380

STATIONARITY TESTS

ADF stat : -4.466
ADF p : 0.0002
ADF lags : 6
ADF : Stationary
KPSS stat : 0.525
KPSS p : 0.0360
KPSS : Non-stationary

QUARTILE-BUCKET ANALYSIS

(vs NBER recession/expansion)

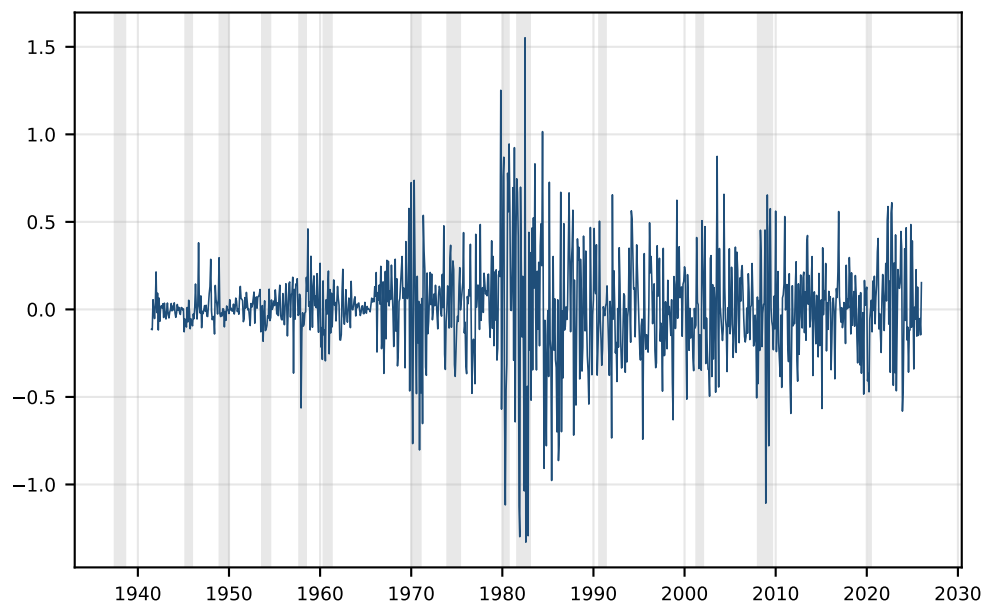
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	253	6.7%	93.3%	0.0905
Q2	253	5.1%	94.9%	0.1857
Q3	252	14.3%	85.7%	0.2773
Q4 (highest)	253	26.1%	73.9%	0.5358

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

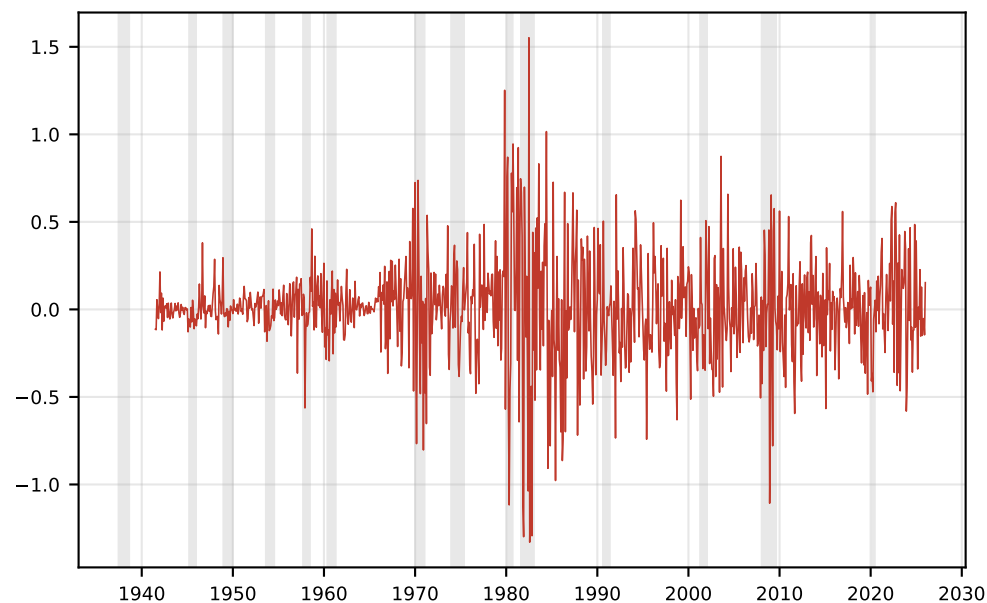
CRSP 10Y Treasury Yield First Difference

Block: Selected export / Treasury term structure | Source: WRDS | Col: derived_y_10Y_wrds_diff_1_wrds | Transform: nan

Raw Series



After: nan



SUMMARY STATISTICS

N : 1287
Valid : 1015
NaN : 272
Start : 1941-06
End : 2025-12
Mean : 0.0020
Std : 0.2818
Min : -1.3296
Max : 1.5519
Skew : -0.200
Kurt : 3.976

STATIONARITY TESTS

ADF stat : -8.480
ADF p : 0.0000
ADF lags : 12
ADF : Stationary
KPSS stat : 0.141
KPSS p : 0.1000
KPSS : Stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

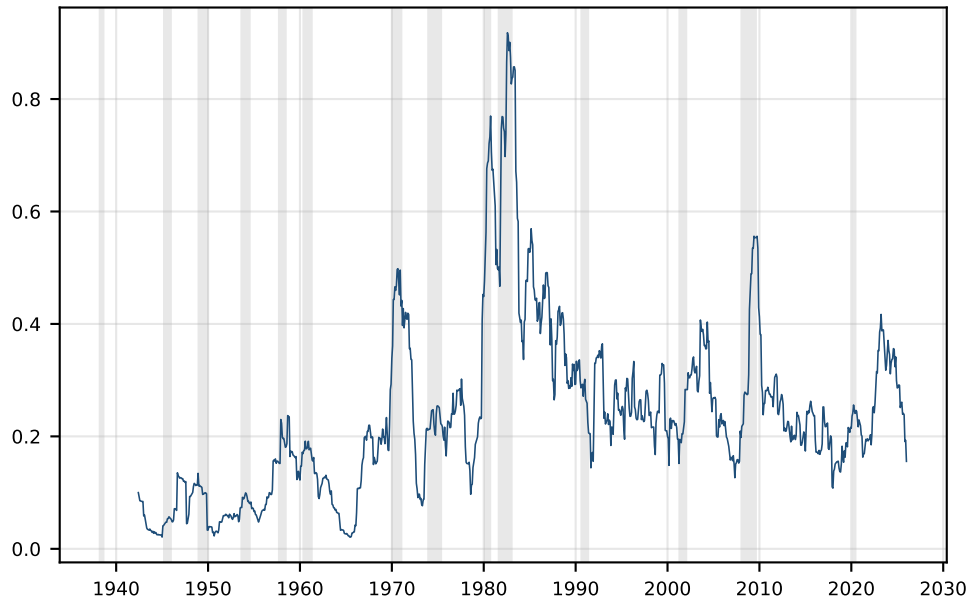
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	254	17.7%	82.3%	-0.3306
Q2	254	14.6%	85.4%	-0.0449
Q3	253	8.3%	91.7%	0.0570
Q4 (highest)	254	11.4%	88.6%	0.3266

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

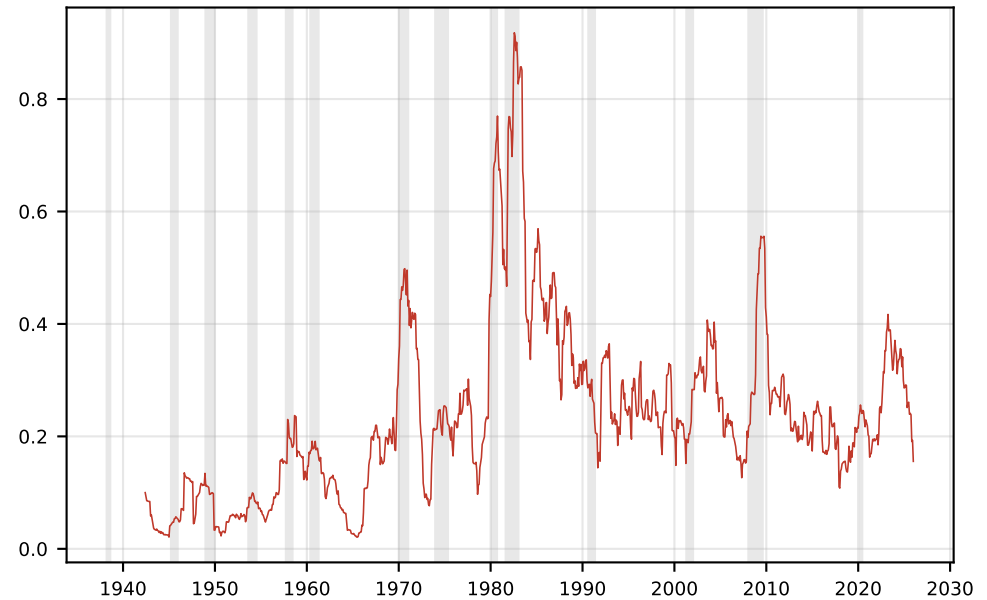
CRSP 10Y Treasury Yield 12M Rolling Volatility

Block: Selected export / Treasury term structure | Source: WRDS | Col: derived_y_10Y_wrds_vol_12m_wrds | Transform: nan

Raw Series



After: nan



SUMMARY STATISTICS

N : 1287
Valid : 1004
NaN : 283
Start : 1942-05
End : 2025-12
Mean : 0.2360
Std : 0.1558
Min : 0.0207
Max : 0.9179
Skew : 1.452
Kurt : 3.163

STATIONARITY TESTS

ADF stat : -2.643
ADF p : 0.0844
ADF lags : 12
ADF : Non-stationary
KPSS stat : 1.307
KPSS p : 0.0100
KPSS : Non-stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

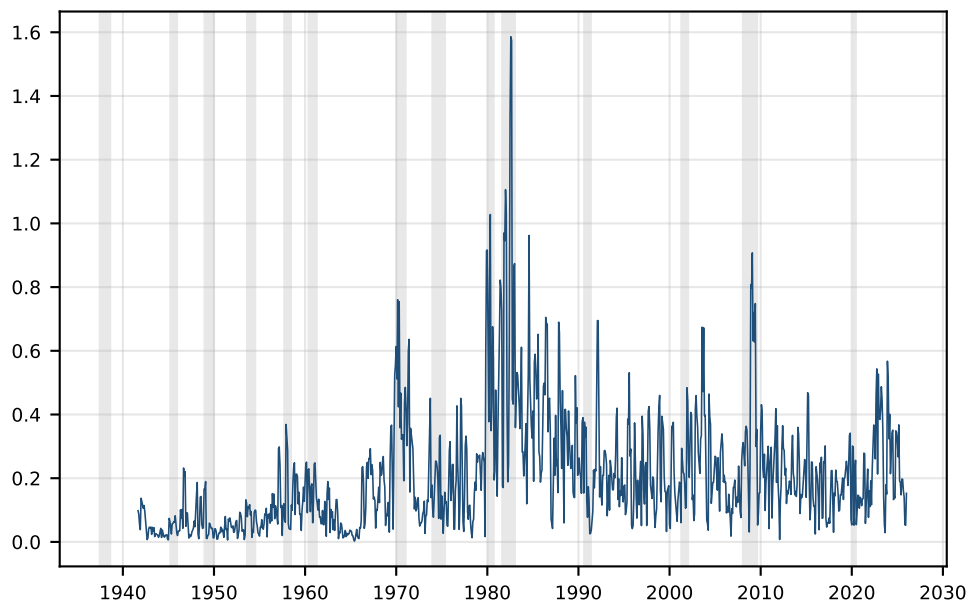
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	251	11.6%	88.4%	0.0714
Q2	251	10.8%	89.2%	0.1801
Q3	251	13.1%	86.9%	0.2493
Q4 (highest)	251	17.1%	82.9%	0.4431

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

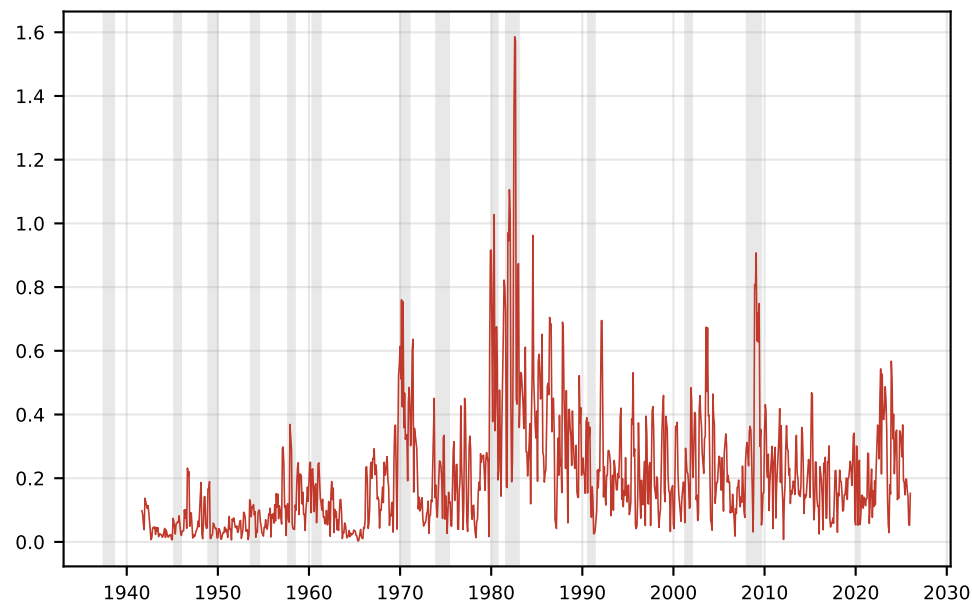
CRSP 10Y Treasury Yield 3M Rolling Volatility

Block: Selected export / Treasury term structure | Source: WRDS | Col: derived_y_10Y_wrds_vol_3m_wrds | Transform: nan

Raw Series



After: nan



SUMMARY STATISTICS

N : 1287
Valid : 1013
NaN : 274
Start : 1941-08
End : 2025-12
Mean : 0.2077
Std : 0.1884
Min : 0.0023
Max : 1.5860
Skew : 2.271
Kurt : 8.780

STATIONARITY TESTS

ADF stat : -3.819
ADF p : 0.0027
ADF lags : 12
ADF : Stationary
KPSS stat : 1.356
KPSS p : 0.0100
KPSS : Non-stationary

QUARTILE-BUCKET ANALYSIS

(vs NBER recession/expansion)

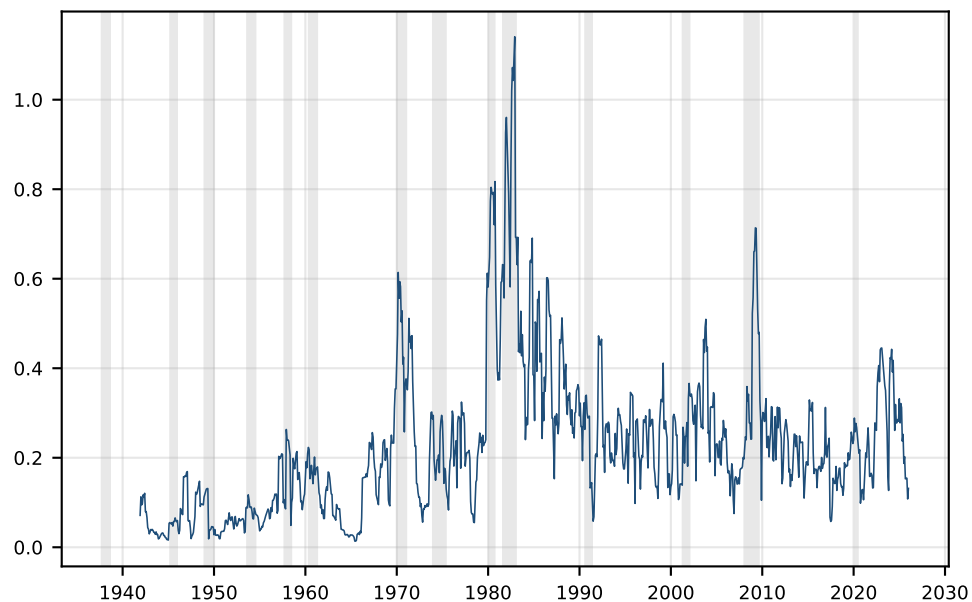
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	254	12.2%	87.8%	0.0394
Q2	253	8.7%	91.3%	0.1175
Q3	253	11.1%	88.9%	0.2158
Q4 (highest)	253	20.2%	79.8%	0.4586

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

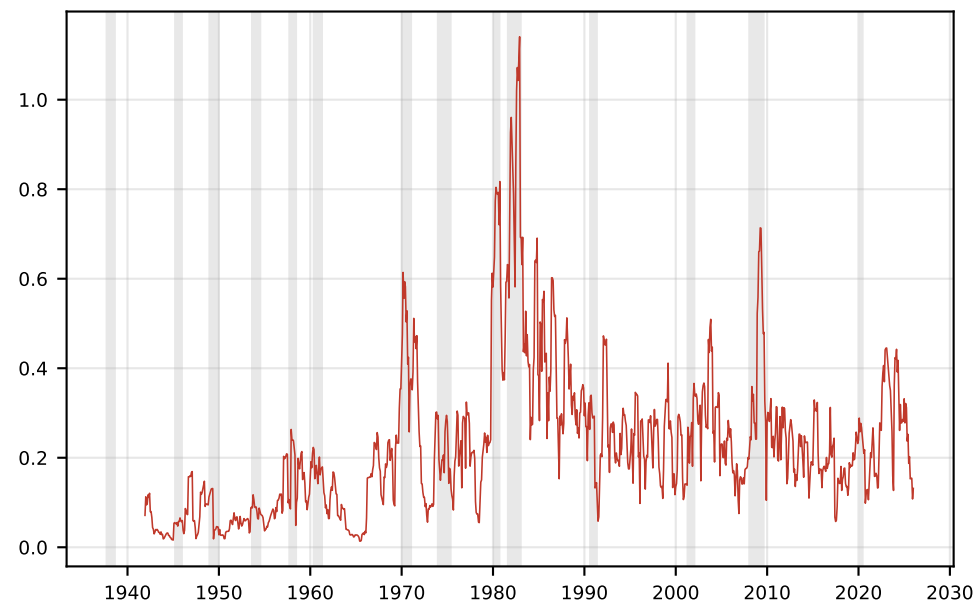
CRSP 10Y Treasury Yield 6M Rolling Volatility

Block: Selected export / Treasury term structure | Source: WRDS | Col: derived_y_10Y_wrds_vol_6m_wrds | Transform: nan

Raw Series



After: nan



SUMMARY STATISTICS

N : 1287
Valid : 1010
NaN : 277
Start : 1941-11
End : 2025-12
Mean : 0.2275
Std : 0.1674
Min : 0.0136
Max : 1.1407
Skew : 1.710
Kurt : 4.494

STATIONARITY TESTS

ADF stat : -3.463
ADF p : 0.0090
ADF lags : 12
ADF : Stationary
KPSS stat : 1.321
KPSS p : 0.0100
KPSS : Non-stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

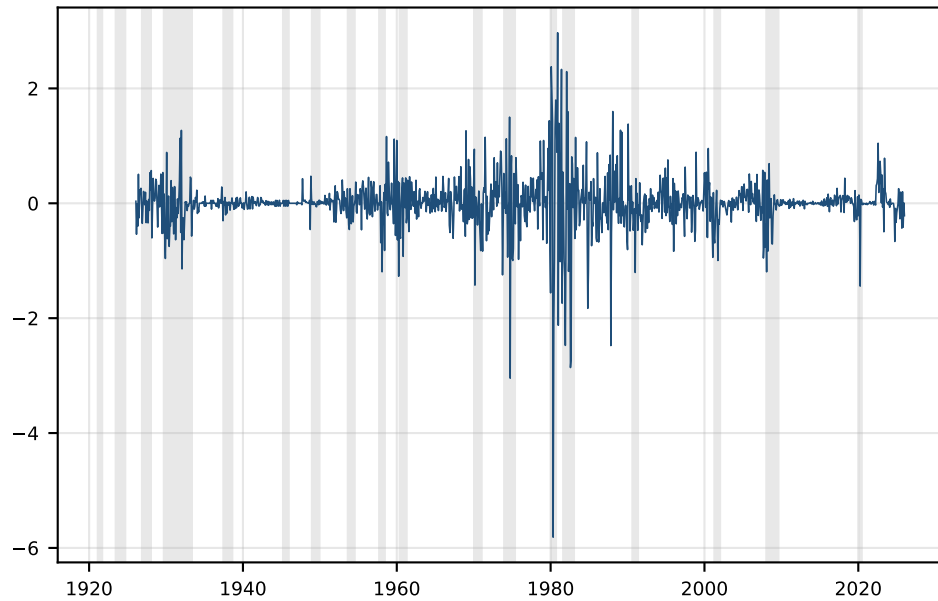
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	253	9.9%	90.1%	0.0603
Q2	252	10.7%	89.3%	0.1579
Q3	252	11.5%	88.5%	0.2446
Q4 (highest)	253	20.2%	79.8%	0.4469

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

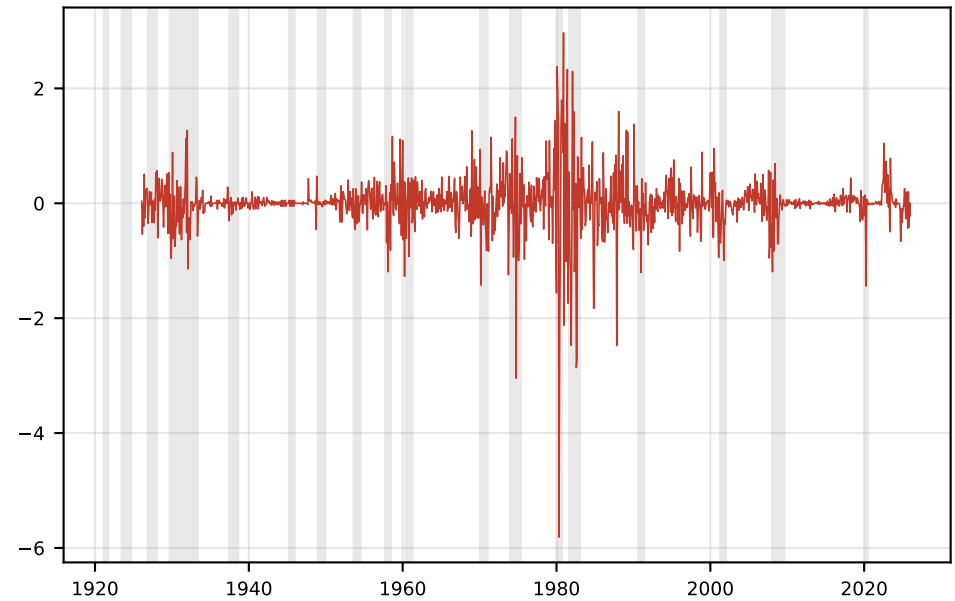
CRSP 1M Treasury Yield First Difference

Block: Selected export / Treasury term structure | Source: WRDS | Col: derived_y_1M_wrds_diff_1_wrds | Transform: nan

Raw Series



After: nan



SUMMARY STATISTICS

N : 1287
Valid : 1200
NaN : 87
Start : 1926-01
End : 2025-12
Mean : -0.0002
Std : 0.4775
Min : -5.8127
Max : 2.9692
Skew : -1.784
Kurt : 26.257

STATIONARITY TESTS

ADF stat : -8.928
ADF p : 0.0000
ADF lags : 12
ADF : Stationary
KPSS stat : 0.039
KPSS p : 0.1000
KPSS : Stationary

QUARTILE-BUCKET ANALYSIS

(vs NBER recession/expansion)

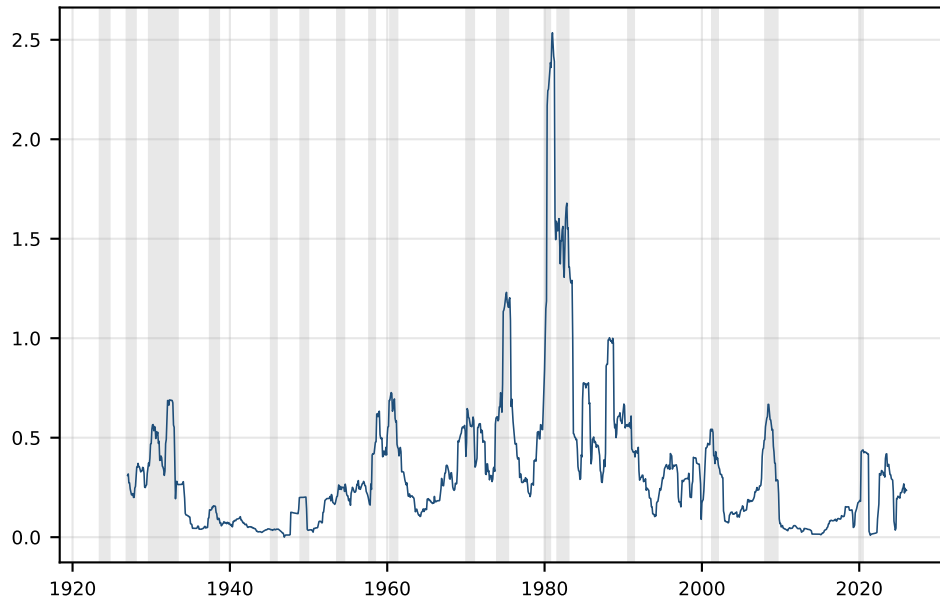
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	300	28.3%	71.7%	-0.4778
Q2	300	11.3%	88.7%	-0.0329
Q3	300	11.7%	88.3%	0.0492
Q4 (highest)	300	15.7%	84.3%	0.4606

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

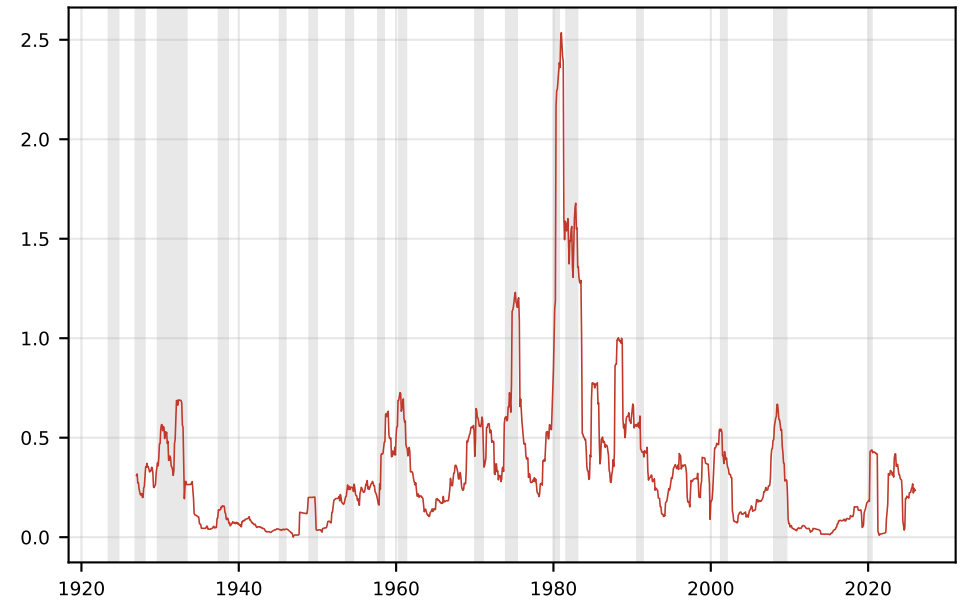
CRSP 1M Treasury Yield 12M Rolling Volatility

Block: Selected export / Treasury term structure | Source: WRDS | Col: derived_y_1M_wrds_vol_12m_wrds | Transform: nan

Raw Series



After: nan



SUMMARY STATISTICS

N : 1287
Valid : 1189
NaN : 98
Start : 1926-12
End : 2025-12
Mean : 0.3311
Std : 0.3520
Min : 0.0000
Max : 2.5353
Skew : 2.939
Kurt : 11.930

STATIONARITY TESTS

ADF stat : -2.947
ADF p : 0.0401
ADF lags : 12
ADF : Stationary
KPSS stat : 0.556
KPSS p : 0.0291
KPSS : Non-stationary

QUARTILE-BUCKET ANALYSIS

(vs NBER recession/expansion)

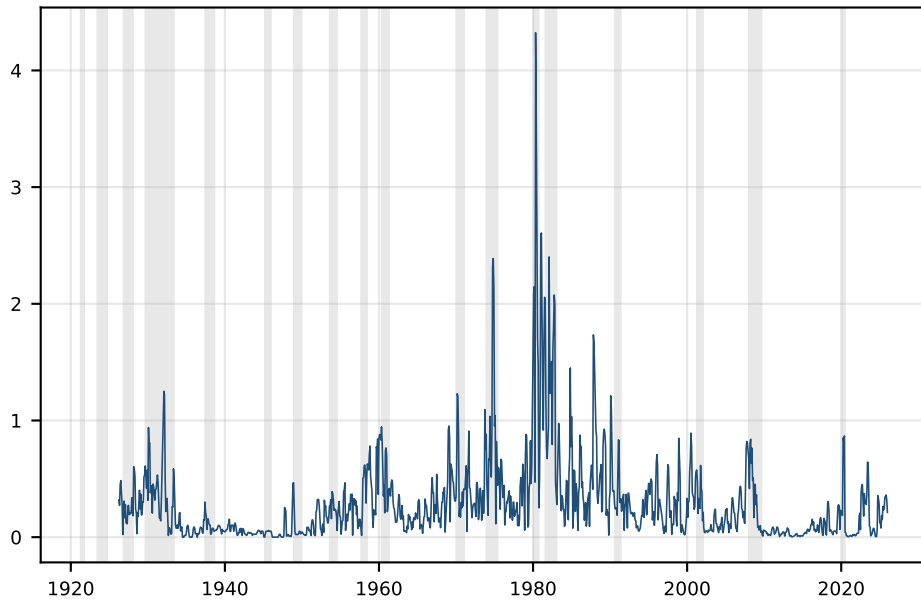
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	298	3.7%	96.3%	0.0507
Q2	297	13.1%	86.9%	0.1787
Q3	297	13.1%	86.9%	0.3306
Q4 (highest)	297	37.4%	62.6%	0.7653

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

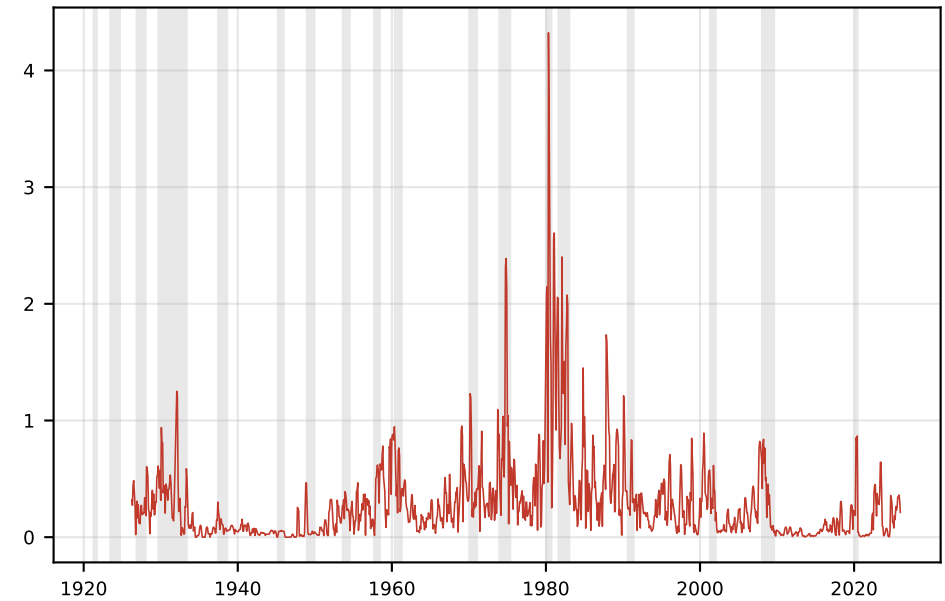
CRSP 1M Treasury Yield 3M Rolling Volatility

Block: Selected export / Treasury term structure | Source: WRDS | Col: derived_y_1M_wrds_vol_3m_wrds | Transform: nan

Raw Series



After: nan



SUMMARY STATISTICS

N : 1287
Valid : 1198
NaN : 89
Start : 1926-03
End : 2025-12
Mean : 0.2942
Std : 0.3829
Min : 0.0000
Max : 4.3231
Skew : 3.763
Kurt : 22.549

STATIONARITY TESTS

ADF stat : -3.999
ADF p : 0.0014
ADF lags : 12
ADF : Stationary
KPSS stat : 0.590
KPSS p : 0.0235
KPSS : Non-stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

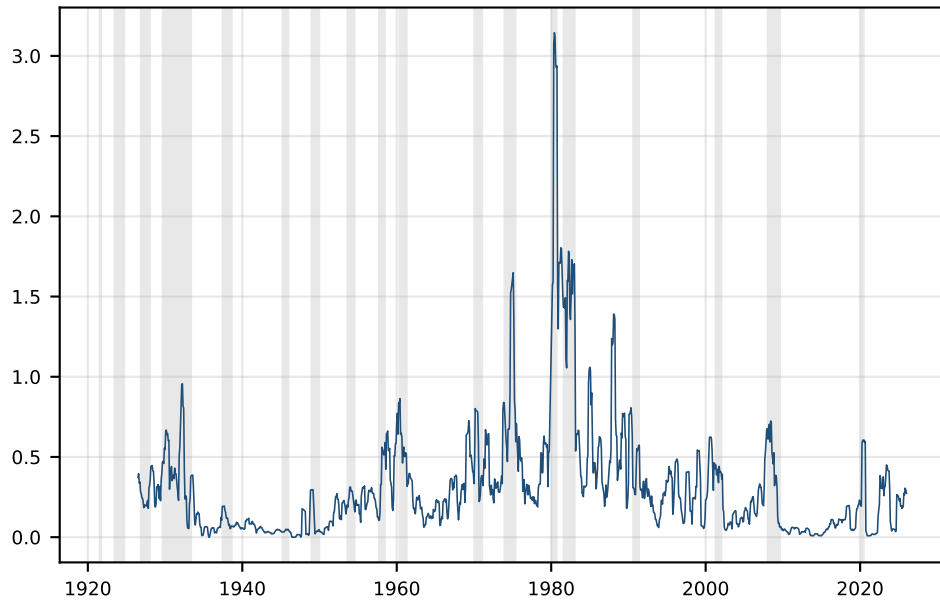
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	300	8.0%	92.0%	0.0291
Q2	299	11.0%	89.0%	0.1149
Q3	299	17.1%	82.9%	0.2726
Q4 (highest)	300	31.0%	69.0%	0.7596

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

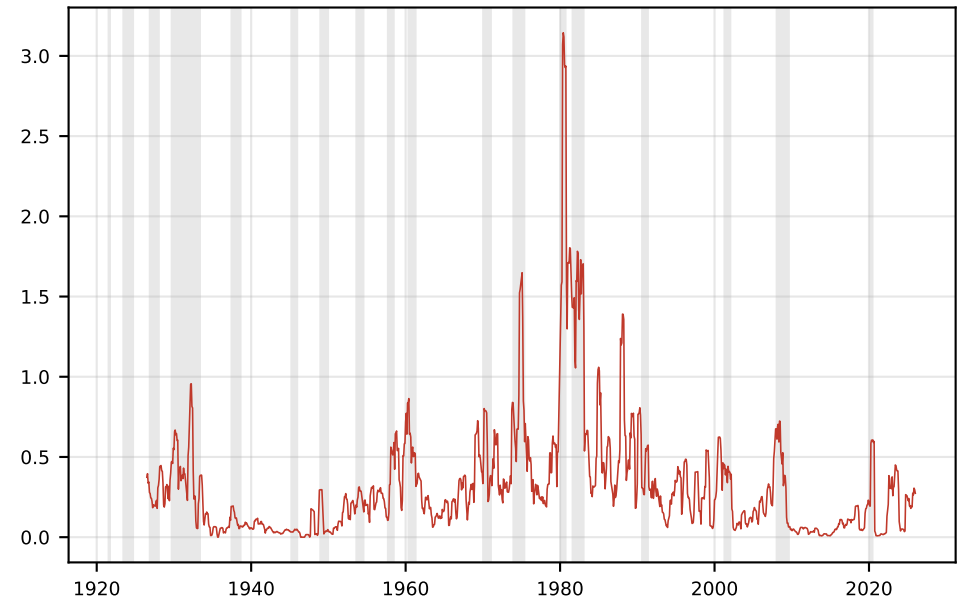
CRSP 1M Treasury Yield 6M Rolling Volatility

Block: Selected export / Treasury term structure | Source: WRDS | Col: derived_y_1M_wrds_vol_6m_wrds | Transform: nan

Raw Series



After: nan



SUMMARY STATISTICS

N : 1287
Valid : 1195
NaN : 92
Start : 1926-06
End : 2025-12
Mean : 0.3163
Std : 0.3695
Min : 0.0000
Max : 3.1444
Skew : 3.326
Kurt : 16.481

STATIONARITY TESTS

ADF stat : -3.828
ADF p : 0.0026
ADF lags : 12
ADF : Stationary
KPSS stat : 0.541
KPSS p : 0.0324
KPSS : Non-stationary

QUARTILE-BUCKET ANALYSIS

(vs NBER recession/expansion)

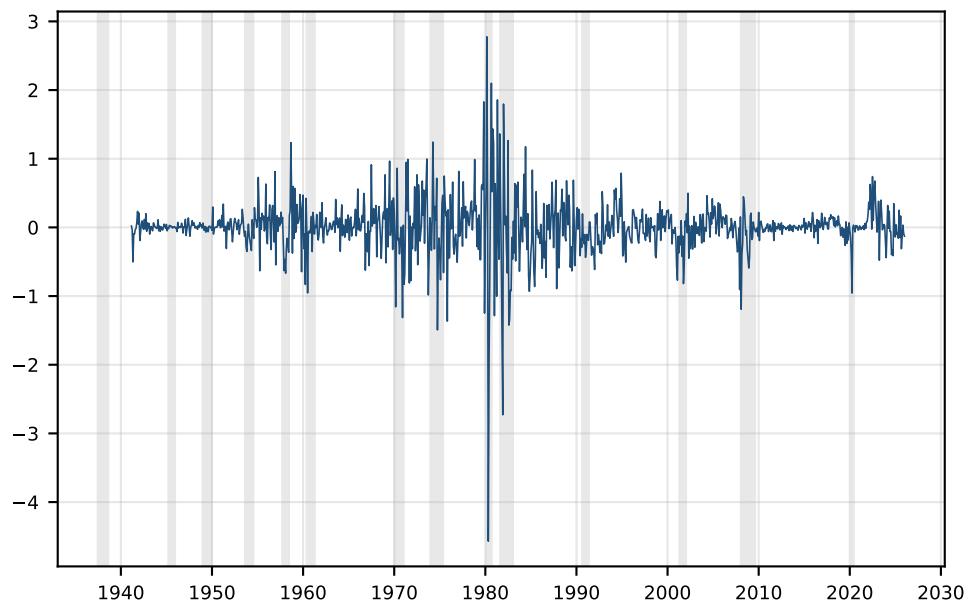
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	299	7.0%	93.0%	0.0399
Q2	299	9.7%	90.3%	0.1512
Q3	298	16.1%	83.9%	0.3045
Q4 (highest)	299	34.4%	65.6%	0.7694

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

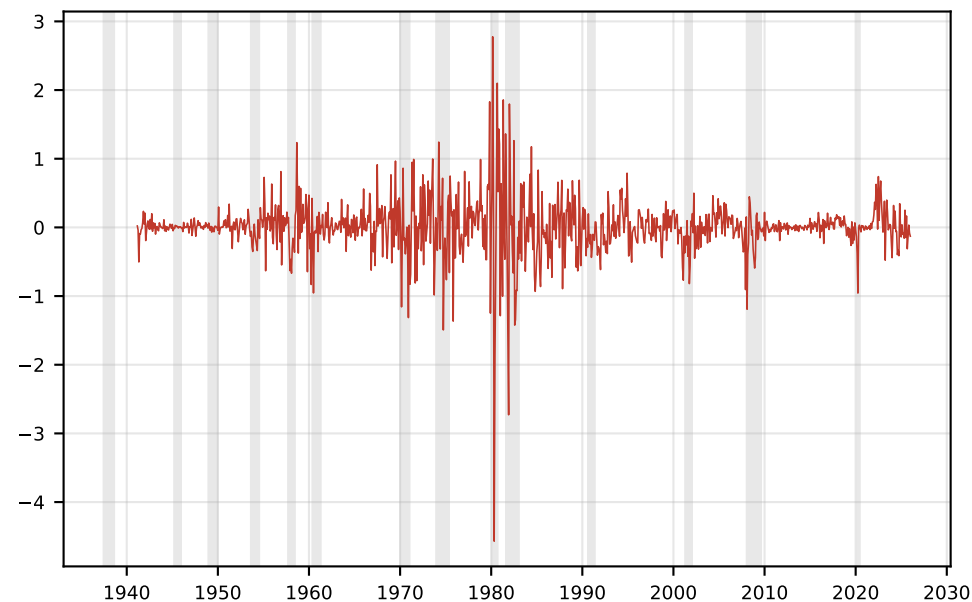
CRSP 1Y Treasury Yield First Difference

Block: Selected export / Treasury term structure | Source: WRDS | Col: derived_y_1Y_wrds_diff_1_wrds | Transform: nan

Raw Series



After: nan



SUMMARY STATISTICS

N : 1287
Valid : 1019
NaN : 268
Start : 1941-02
End : 2025-12
Mean : 0.0026
Std : 0.4035
Min : -4.5693
Max : 2.7765
Skew : -1.278
Kurt : 23.263

STATIONARITY TESTS

ADF stat : -8.319
ADF p : 0.0000
ADF lags : 10
ADF : Stationary
KPSS stat : 0.081
KPSS p : 0.1000
KPSS : Stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

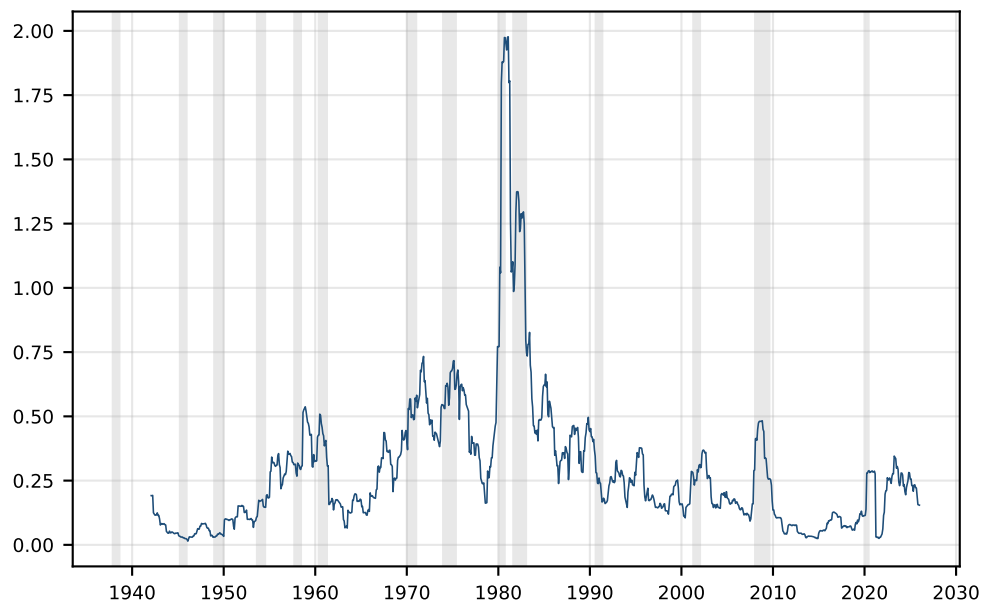
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	255	25.5%	74.5%	-0.4175
Q2	255	9.8%	90.2%	-0.0414
Q3	254	7.9%	92.1%	0.0595
Q4 (highest)	255	8.6%	91.4%	0.4102

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

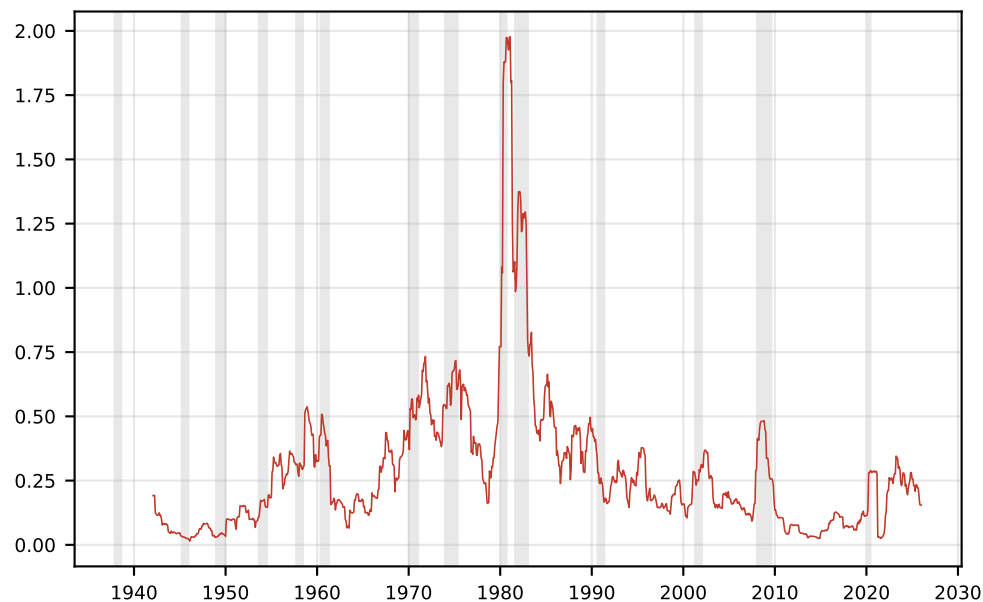
CRSP 1Y Treasury Yield 12M Rolling Volatility

Block: Selected export / Treasury term structure | Source: WRDS | Col: derived_y_1Y_wrds_vol_12m_wrds | Transform: nan

Raw Series



After: nan



SUMMARY STATISTICS

N : 1287
Valid : 1008
NaN : 279
Start : 1942-01
End : 2025-12
Mean : 0.2895
Std : 0.2819
Min : 0.0143
Max : 1.9772
Skew : 3.039
Kurt : 12.909

STATIONARITY TESTS

ADF stat : -2.706
ADF p : 0.0731
ADF lags : 12
ADF : Non-stationary
KPSS stat : 0.624
KPSS p : 0.0204
KPSS : Non-stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

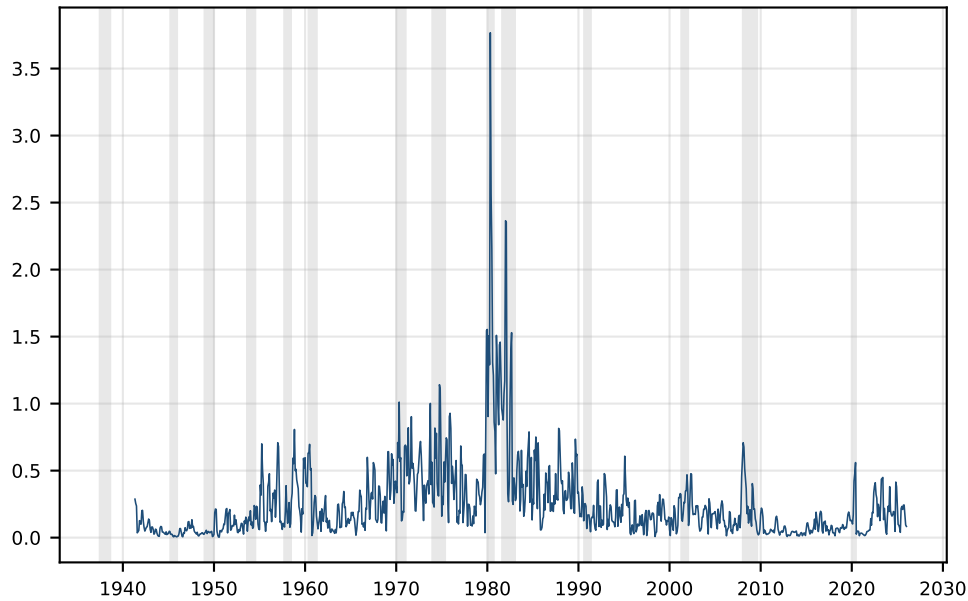
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	252	7.9%	92.1%	0.0657
Q2	252	4.0%	96.0%	0.1627
Q3	252	12.3%	87.7%	0.2941
Q4 (highest)	252	28.2%	71.8%	0.6353

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

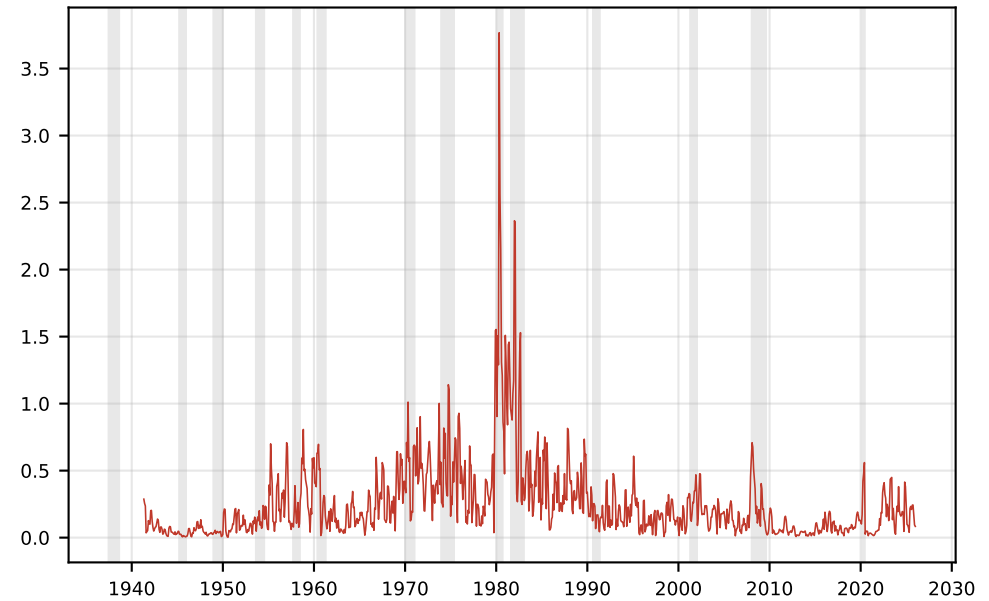
CRSP 1Y Treasury Yield 3M Rolling Volatility

Block: Selected export / Treasury term structure | Source: WRDS | Col: derived_y_1Y_wrds_vol_3m_wrds | Transform: nan

Raw Series



After: nan



SUMMARY STATISTICS

N : 1287
Valid : 1017
NaN : 270
Start : 1941-04
End : 2025-12
Mean : 0.2482
Std : 0.2999
Min : 0.0034
Max : 3.7673
Skew : 4.132
Kurt : 29.871

STATIONARITY TESTS

ADF stat : -3.682
ADF p : 0.0044
ADF lags : 11
ADF : Stationary
KPSS stat : 0.694
KPSS p : 0.0141
KPSS : Non-stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

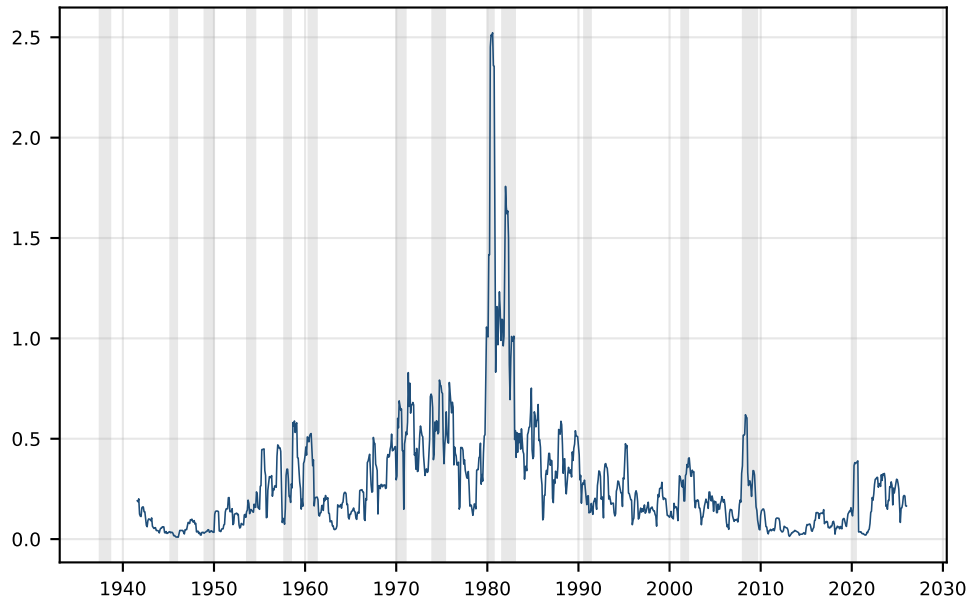
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	255	9.0%	91.0%	0.0386
Q2	254	9.1%	90.9%	0.1106
Q3	254	13.8%	86.2%	0.2272
Q4 (highest)	254	20.1%	79.9%	0.6171

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

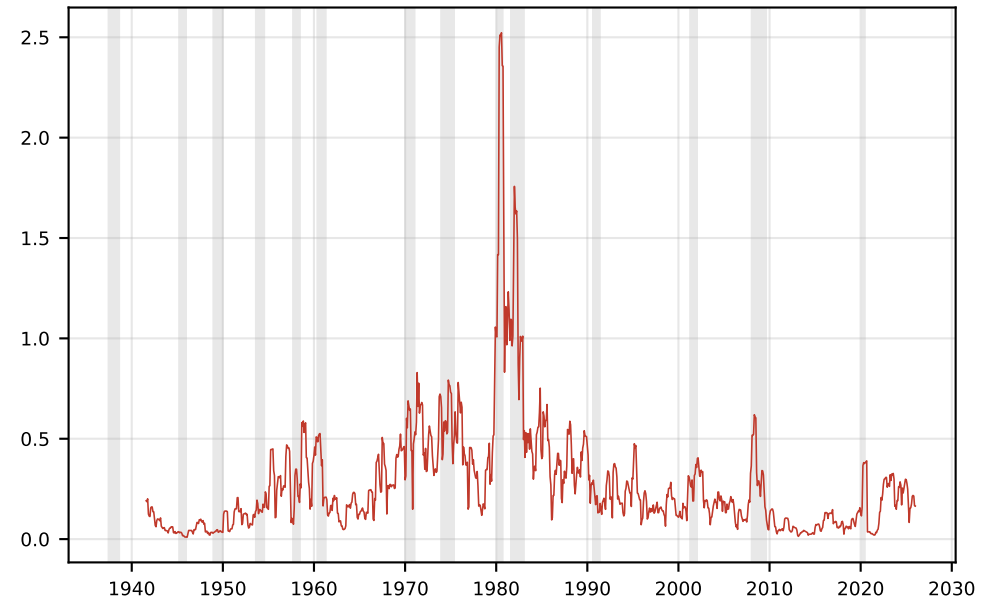
CRSP 1Y Treasury Yield 6M Rolling Volatility

Block: Selected export / Treasury term structure | Source: WRDS | Col: derived_y_1Y_wrds_vol_6m_wrds | Transform: nan

Raw Series



After: nan



SUMMARY STATISTICS

N : 1287
Valid : 1014
NaN : 273
Start : 1941-07
End : 2025-12
Mean : 0.2746
Std : 0.2920
Min : 0.0097
Max : 2.5226
Skew : 3.633
Kurt : 20.180

STATIONARITY TESTS

ADF stat : -3.447
ADF p : 0.0095
ADF lags : 12
ADF : Stationary
KPSS stat : 0.648
KPSS p : 0.0183
KPSS : Non-stationary

QUARTILE-BUCKET ANALYSIS

(vs NBER recession/expansion)

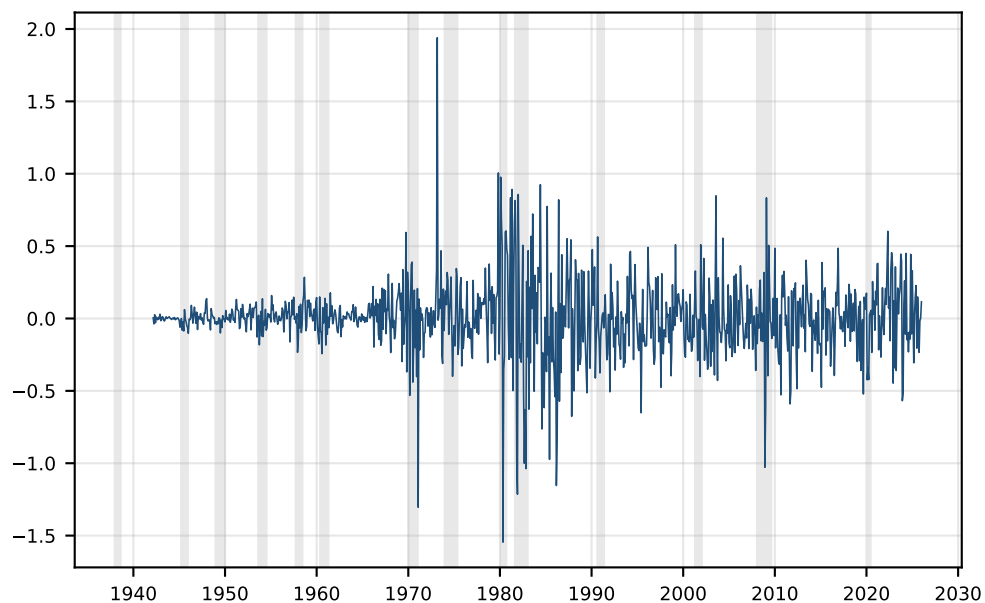
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	254	7.9%	92.1%	0.0549
Q2	253	7.1%	92.9%	0.1456
Q3	253	12.3%	87.7%	0.2688
Q4 (highest)	254	24.8%	75.2%	0.6285

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

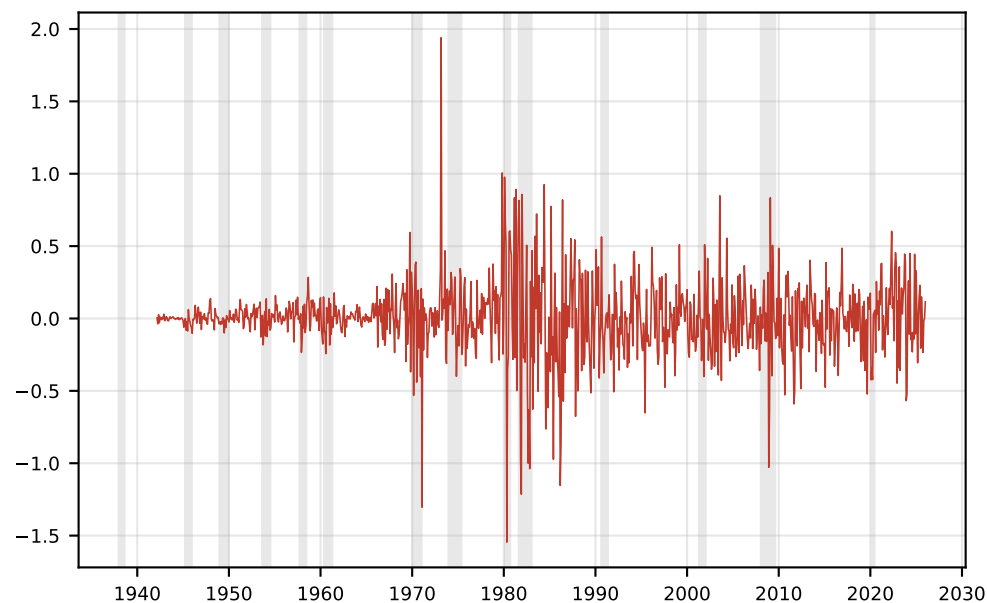
CRSP 20Y Treasury Yield First Difference

Block: Selected export / Treasury term structure | Source: WRDS | Col: derived_y_20Y_wrds_diff_1_wrds | Transform: nan

Raw Series



After: nan



SUMMARY STATISTICS

N : 1287
Valid : 1007
NaN : 280
Start : 1942-02
End : 2025-12
Mean : 0.0023
Std : 0.2573
Min : -1.5454
Max : 1.9396
Skew : -0.050
Kurt : 7.567

STATIONARITY TESTS

ADF stat : -8.188
ADF p : 0.0000
ADF lags : 10
ADF : Stationary
KPSS stat : 0.146
KPSS p : 0.1000
KPSS : Stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

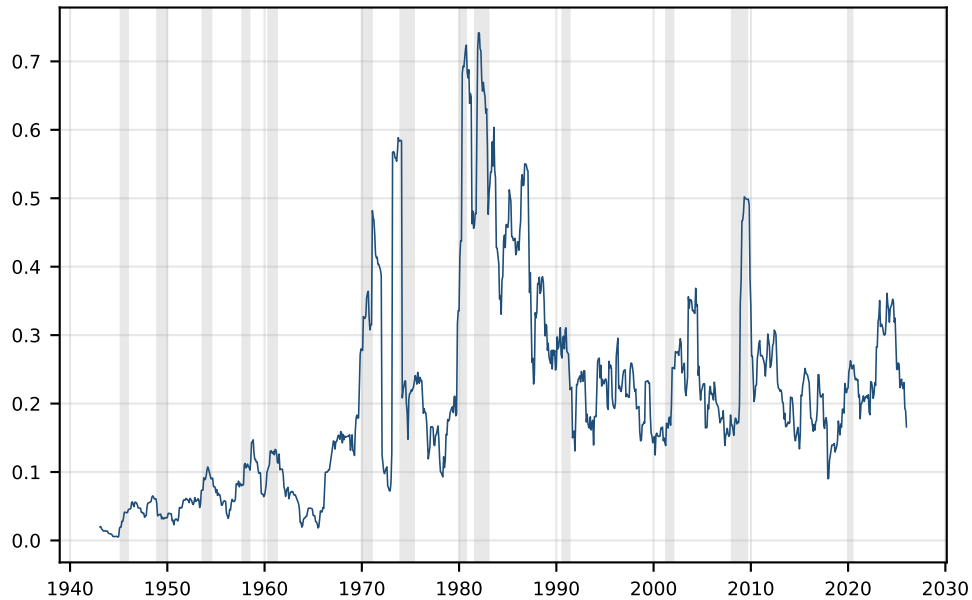
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	252	16.7%	83.3%	-0.2908
Q2	252	14.7%	85.3%	-0.0388
Q3	251	8.8%	91.2%	0.0487
Q4 (highest)	252	12.3%	87.7%	0.2902

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

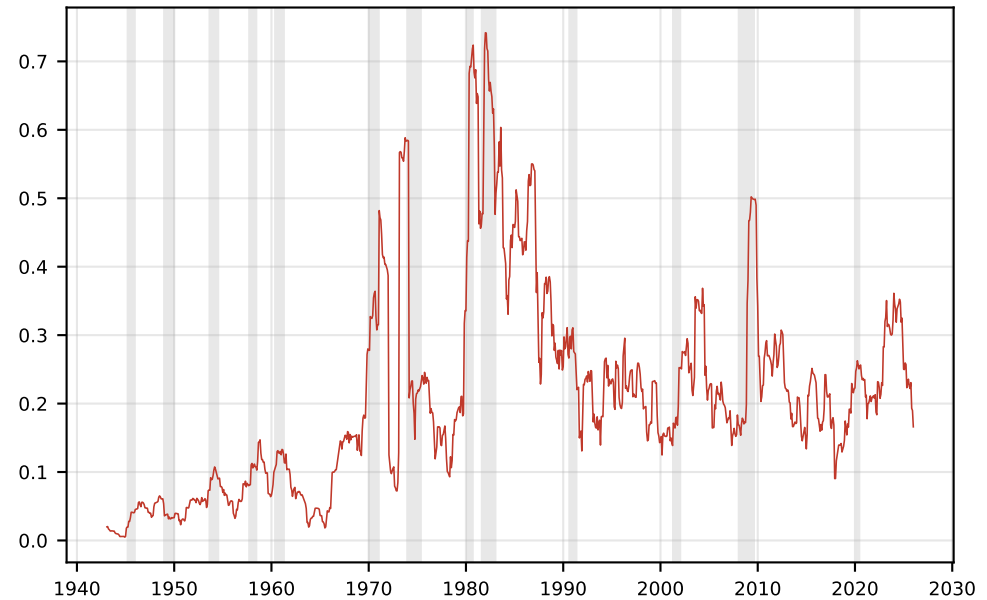
CRSP 20Y Treasury Yield 12M Rolling Volatility

Block: Selected export / Treasury term structure | Source: WRDS | Col: derived_y_20Y_wrds_vol_12m_wrds | Transform: nan

Raw Series



After: nan



SUMMARY STATISTICS

N : 1287
Valid : 996
NaN : 291
Start : 1943-01
End : 2025-12
Mean : 0.2101
Std : 0.1489
Min : 0.0048
Max : 0.7419
Skew : 1.179
Kurt : 1.407

STATIONARITY TESTS

ADF stat : -2.627
ADF p : 0.0876
ADF lags : 12
ADF : Non-stationary
KPSS stat : 1.436
KPSS p : 0.0100
KPSS : Non-stationary

QUARTILE-BUCKET ANALYSIS

(vs NBER recession/expansion)

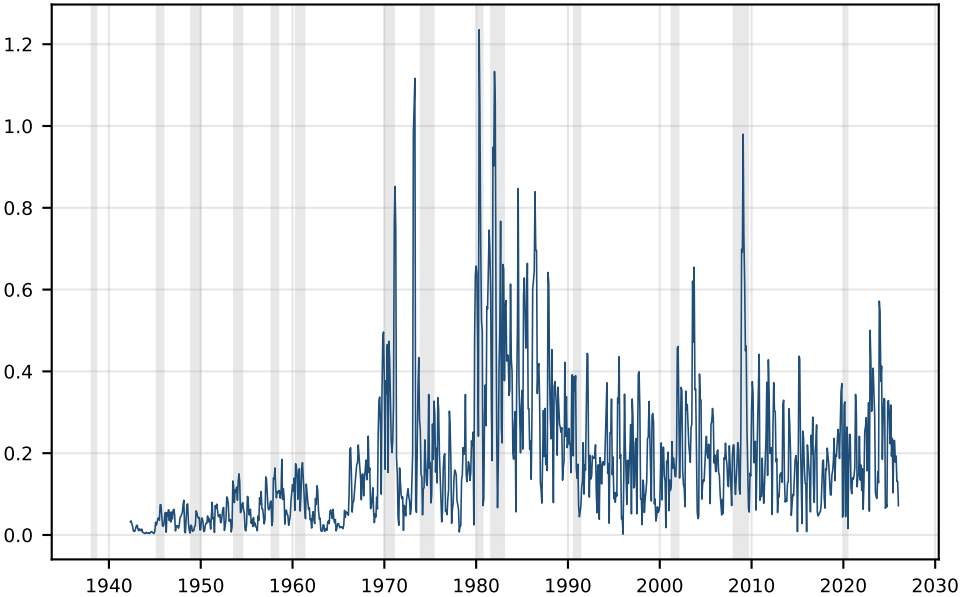
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	249	11.2%	88.8%	0.0512
Q2	249	14.5%	85.5%	0.1488
Q3	249	6.8%	93.2%	0.2263
Q4 (highest)	249	20.5%	79.5%	0.4140

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

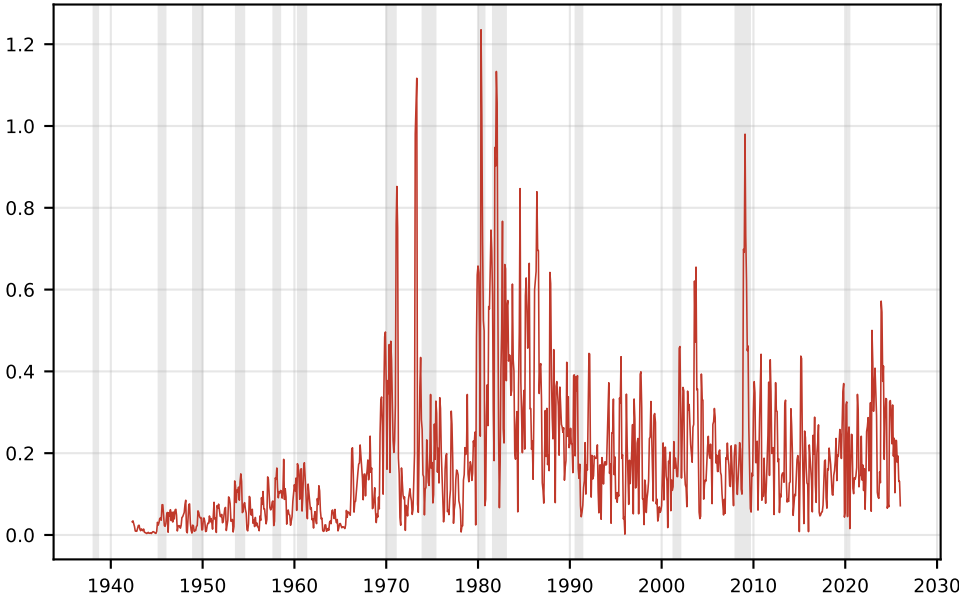
CRSP 20Y Treasury Yield 3M Rolling Volatility

Block: Selected export / Treasury term structure | Source: WRDS | Col: derived_y_20Y_wrds_vol_3m_wrds | Transform: nan

Raw Series



After: nan



SUMMARY STATISTICS

N : 1287
Valid : 1005
NaN : 282
Start : 1942-04
End : 2025-12
Mean : 0.1826
Std : 0.1754
Min : 0.0020
Max : 1.2354
Skew : 2.136
Kurt : 6.317

STATIONARITY TESTS

ADF stat : -4.364
ADF p : 0.0003
ADF lags : 11
ADF : Stationary
KPSS stat : 1.646
KPSS p : 0.0100
KPSS : Non-stationary

QUARTILE-BUCKET ANALYSIS

(vs NBER recession/expansion)

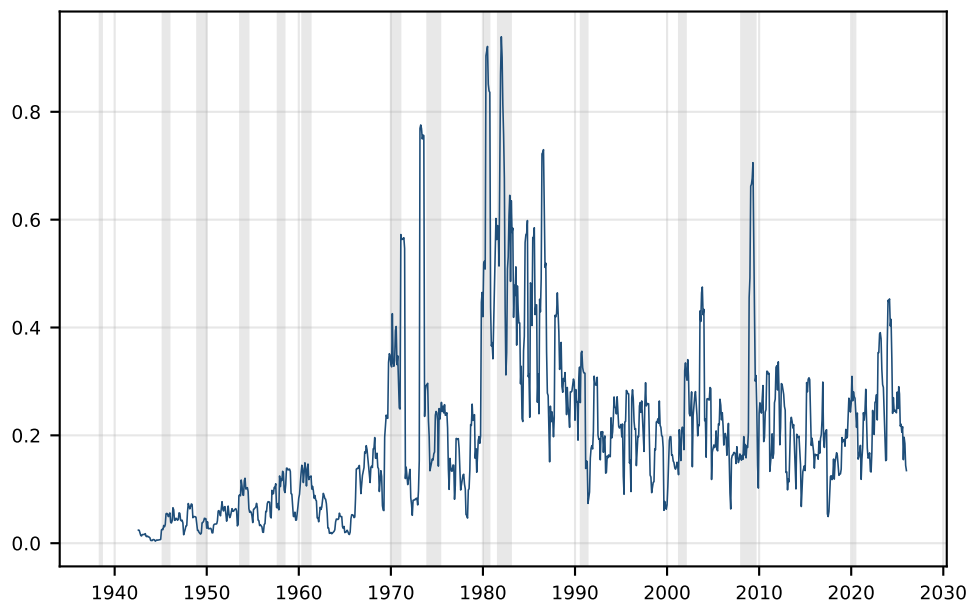
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	252	9.9%	90.1%	0.0316
Q2	251	11.6%	88.4%	0.0966
Q3	251	15.5%	84.5%	0.1836
Q4 (highest)	251	15.5%	84.5%	0.4190

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

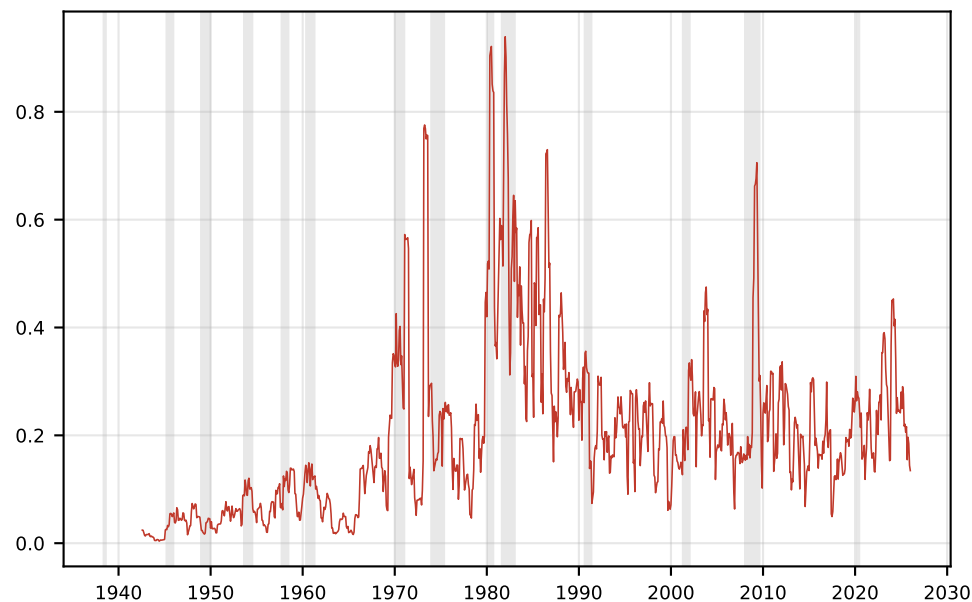
CRSP 20Y Treasury Yield 6M Rolling Volatility

Block: Selected export / Treasury term structure | Source: WRDS | Col: derived_y_20Y_wrds_vol_6m_wrds | Transform: nan

Raw Series



After: nan



SUMMARY STATISTICS

N : 1287
Valid : 1002
NaN : 285
Start : 1942-07
End : 2025-12
Mean : 0.2011
Std : 0.1602
Min : 0.0038
Max : 0.9392
Skew : 1.672
Kurt : 3.674

STATIONARITY TESTS

ADF stat : -3.558
ADF p : 0.0066
ADF lags : 12
ADF : Stationary
KPSS stat : 1.496
KPSS p : 0.0100
KPSS : Non-stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

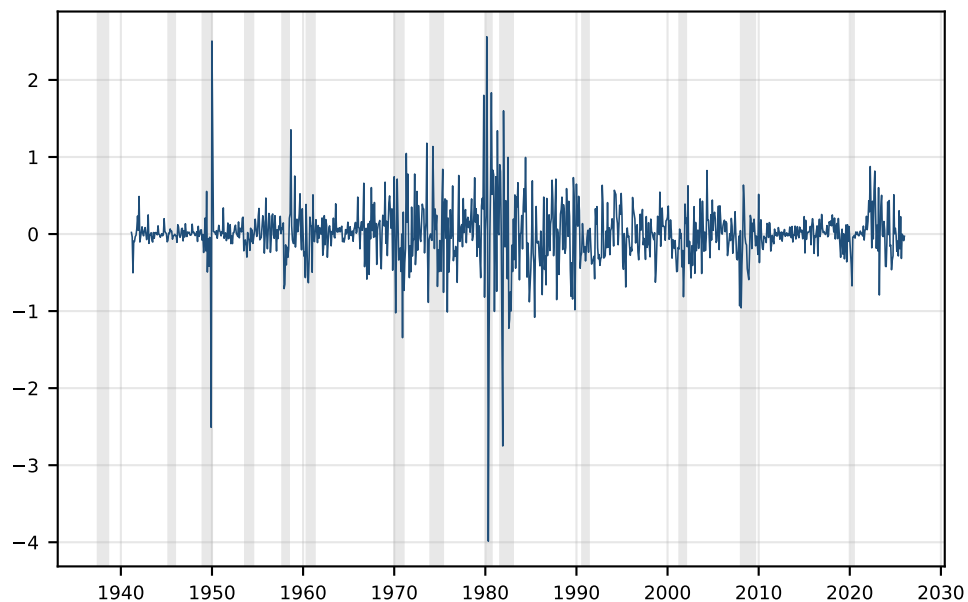
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	251	8.4%	91.6%	0.0444
Q2	250	16.4%	83.6%	0.1319
Q3	250	8.0%	92.0%	0.2142
Q4 (highest)	251	19.9%	80.1%	0.4138

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

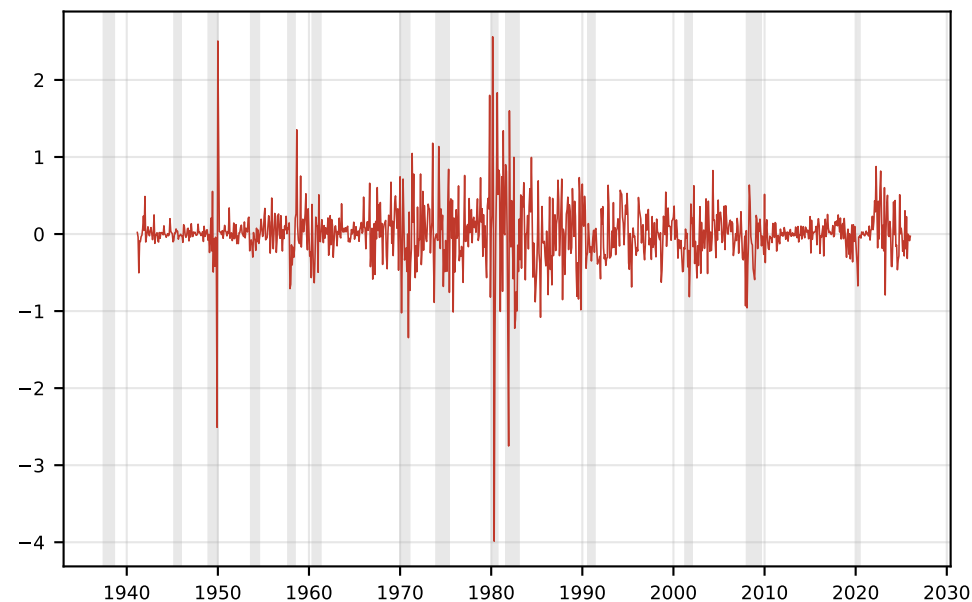
CRSP 2Y Treasury Yield First Difference

Block: Selected export / Treasury term structure | Source: WRDS | Col: derived_y_2Y_wrds_diff_1_wrds | Transform: nan

Raw Series



After: nan



SUMMARY STATISTICS

N : 1287
Valid : 1019
NaN : 268
Start : 1941-02
End : 2025-12
Mean : 0.0026
Std : 0.3894
Min : -3.9861
Max : 2.5607
Skew : -0.931
Kurt : 19.333

STATIONARITY TESTS

ADF stat : -8.368
ADF p : 0.0000
ADF lags : 10
ADF : Stationary
KPSS stat : 0.103
KPSS p : 0.1000
KPSS : Stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

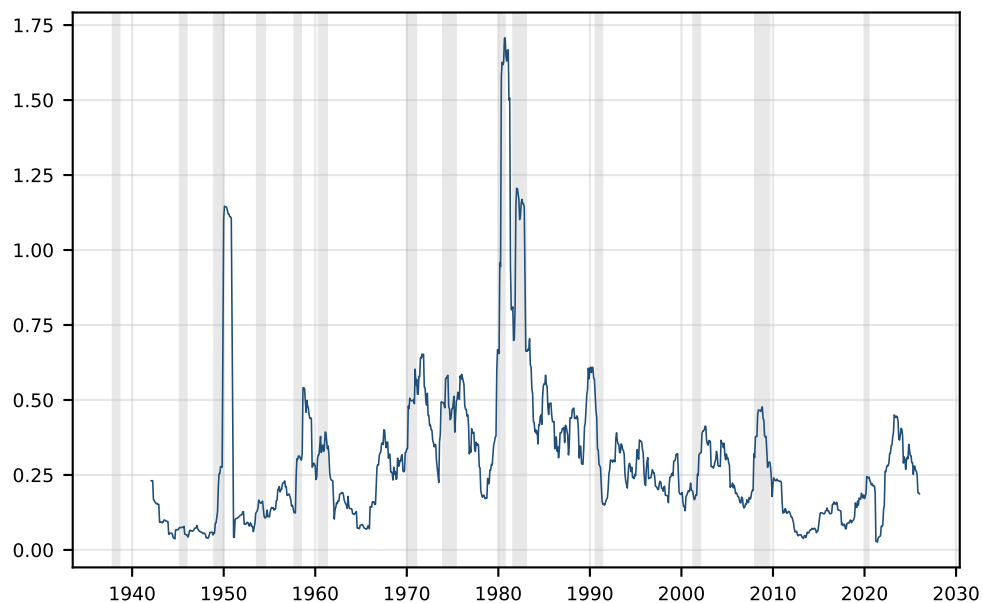
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	255	22.7%	77.3%	-0.4120
Q2	255	11.4%	88.6%	-0.0523
Q3	254	7.9%	92.1%	0.0617
Q4 (highest)	255	9.8%	90.2%	0.4132

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

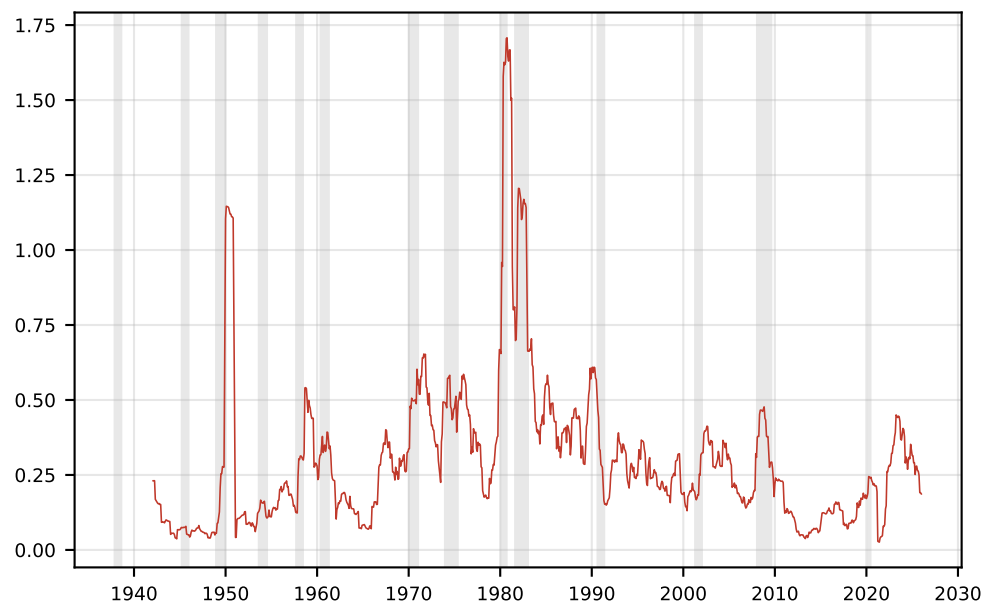
CRSP 2Y Treasury Yield 12M Rolling Volatility

Block: Selected export / Treasury term structure | Source: WRDS | Col: derived_y_2Y_wrds_vol_12m_wrds | Transform: nan

Raw Series



After: nan



SUMMARY STATISTICS

N : 1287
Valid : 1008
NaN : 279
Start : 1942-01
End : 2025-12
Mean : 0.2994
Std : 0.2528
Min : 0.0259
Max : 1.7078
Skew : 2.631
Kurt : 9.529

STATIONARITY TESTS

ADF stat : -3.307
ADF p : 0.0146
ADF lags : 12
ADF : Stationary
KPSS stat : 0.476
KPSS p : 0.0470
KPSS : Non-stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

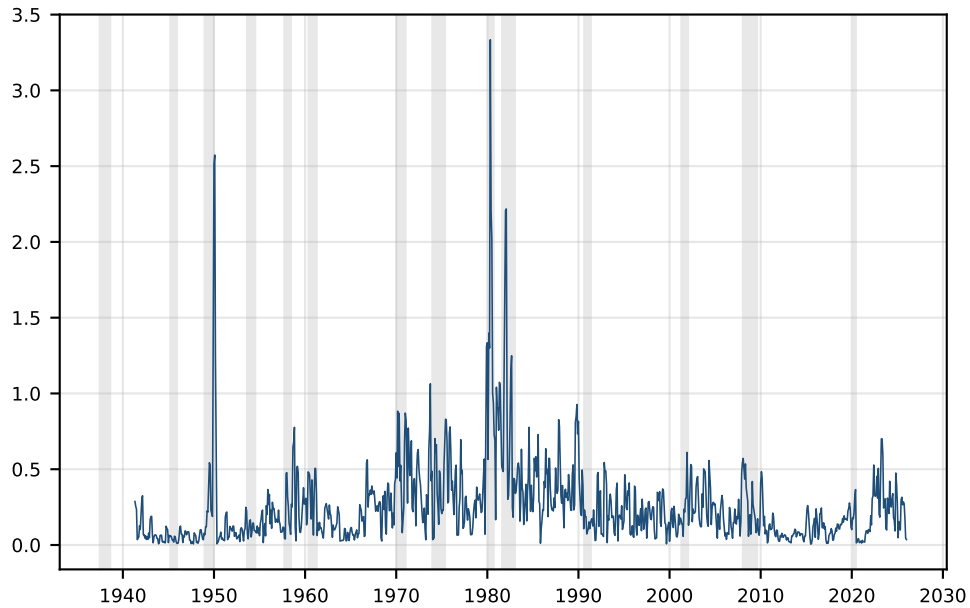
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	252	6.7%	93.3%	0.0857
Q2	252	7.1%	92.9%	0.1891
Q3	252	12.3%	87.7%	0.3092
Q4 (highest)	252	26.2%	73.8%	0.6137

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

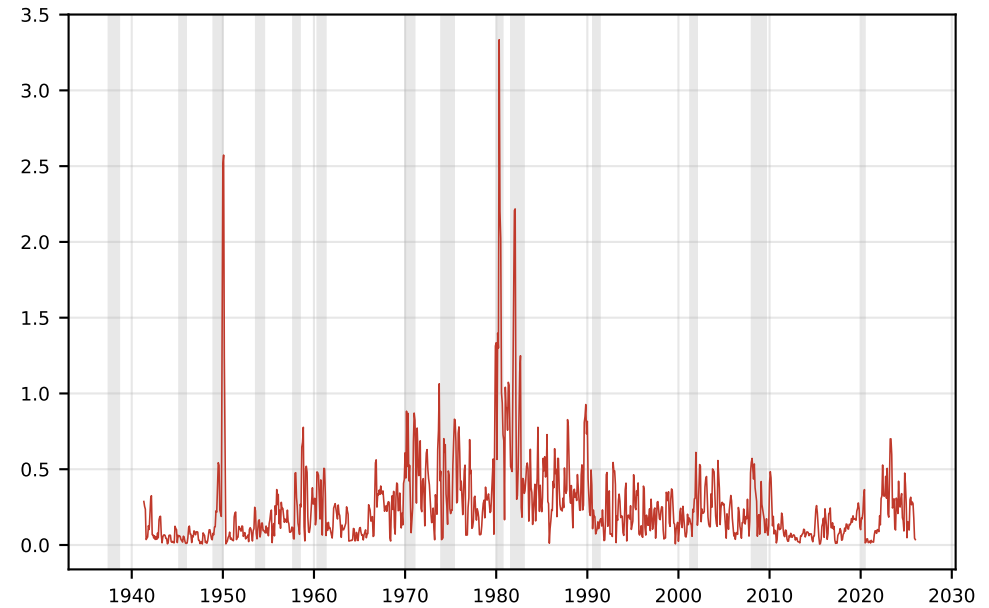
CRSP 2Y Treasury Yield 3M Rolling Volatility

Block: Selected export / Treasury term structure | Source: WRDS | Col: derived_y_2Y_wrds_vol_3m_wrds | Transform: nan

Raw Series



After: nan



SUMMARY STATISTICS

N : 1287
Valid : 1017
NaN : 270
Start : 1941-04
End : 2025-12
Mean : 0.2544
Std : 0.2872
Min : 0.0054
Max : 3.3342
Skew : 4.168
Kurt : 28.522

STATIONARITY TESTS

ADF stat : -5.529
ADF p : 0.0000
ADF lags : 9
ADF : Stationary
KPSS stat : 0.550
KPSS p : 0.0304
KPSS : Non-stationary

QUARTILE-BUCKET ANALYSIS

(vs NBER recession/expansion)

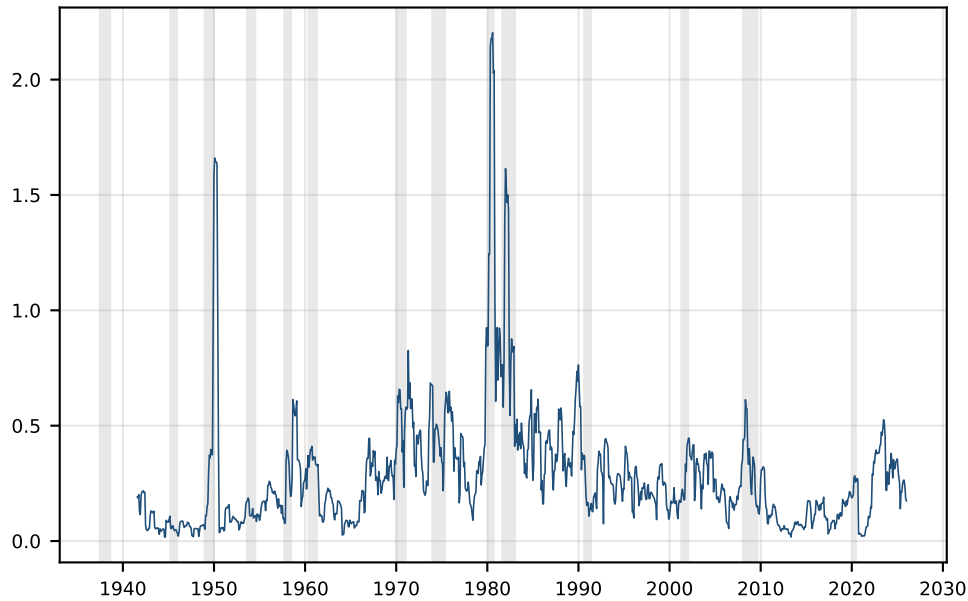
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	255	9.4%	90.6%	0.0470
Q2	254	8.3%	91.7%	0.1269
Q3	254	13.8%	86.2%	0.2470
Q4 (highest)	254	20.5%	79.5%	0.5975

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

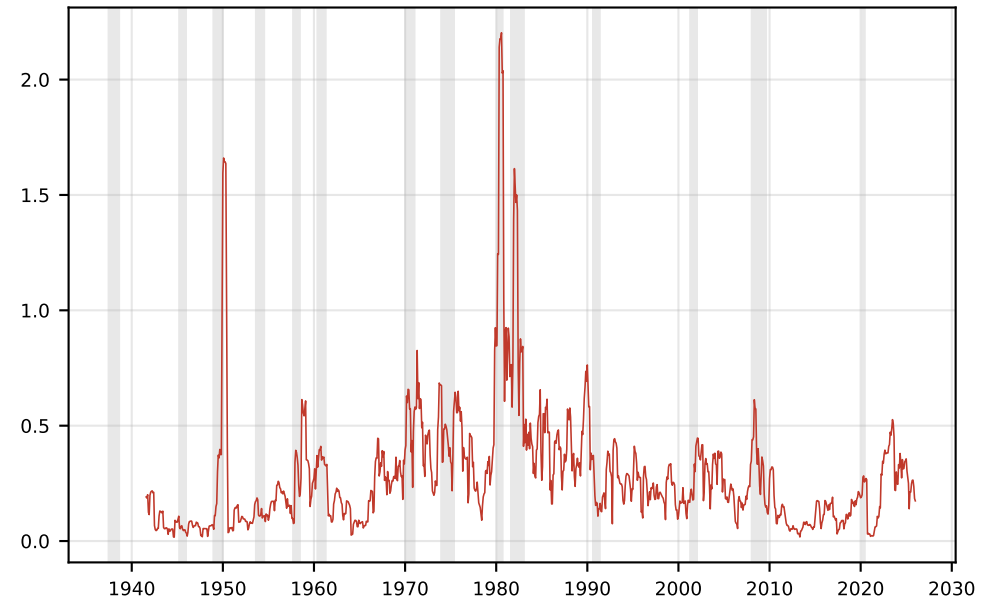
CRSP 2Y Treasury Yield 6M Rolling Volatility

Block: Selected export / Treasury term structure | Source: WRDS | Col: derived_y_2Y_wrds_vol_6m_wrds | Transform: nan

Raw Series



After: nan



SUMMARY STATISTICS

N : 1287
Valid : 1014
NaN : 273
Start : 1941-07
End : 2025-12
Mean : 0.2836
Std : 0.2688
Min : 0.0168
Max : 2.2031
Skew : 3.408
Kurt : 17.085

STATIONARITY TESTS

ADF stat : -4.361
ADF p : 0.0003
ADF lags : 12
ADF : Stationary
KPSS stat : 0.499
KPSS p : 0.0418
KPSS : Non-stationary

QUARTILE-BUCKET ANALYSIS

(vs NBER recession/expansion)

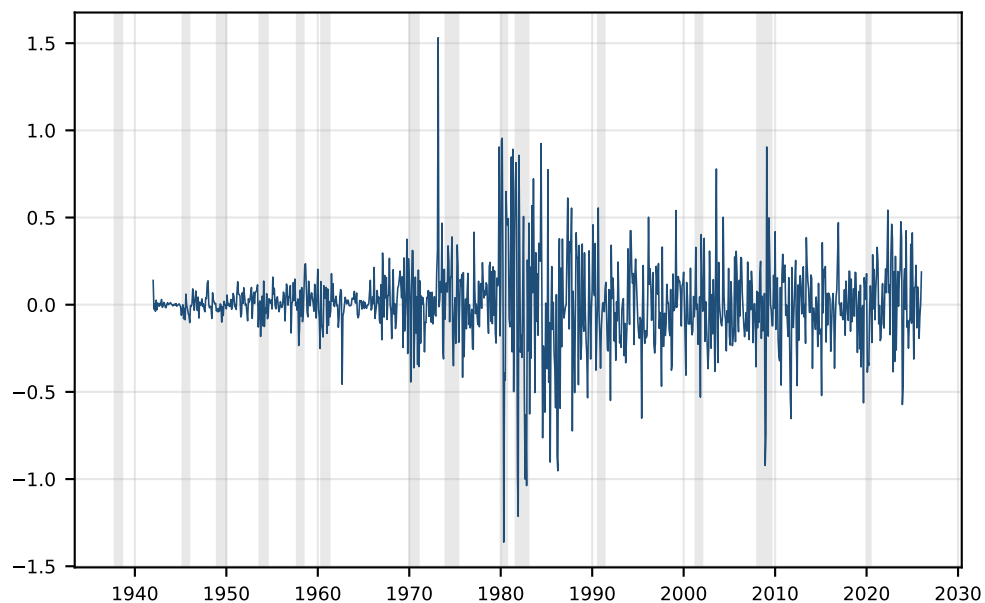
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	254	7.9%	92.1%	0.0706
Q2	253	7.5%	92.5%	0.1705
Q3	253	10.7%	89.3%	0.2908
Q4 (highest)	254	26.0%	74.0%	0.6021

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

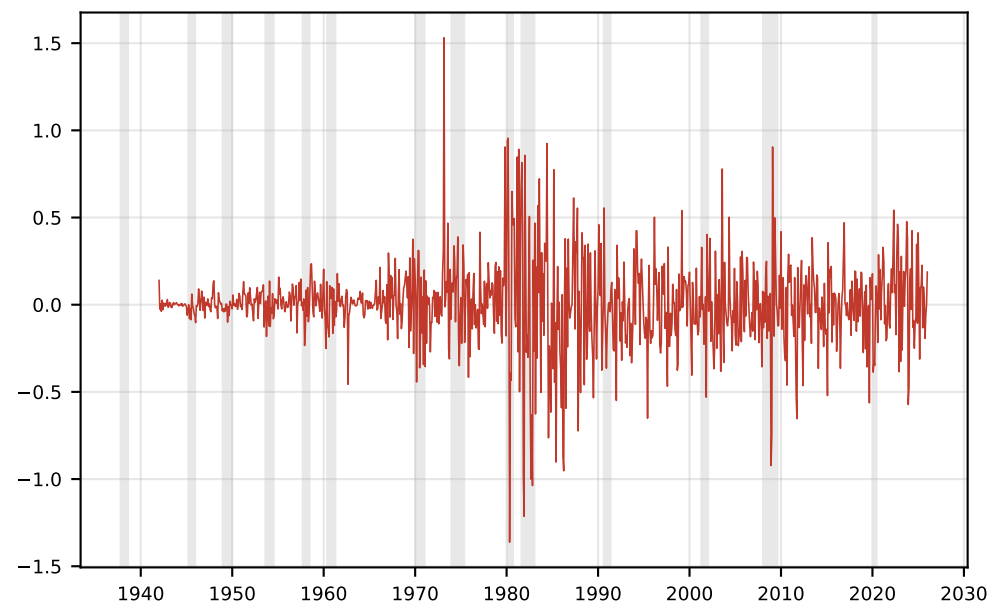
CRSP 30Y Treasury Yield First Difference

Block: Selected export / Treasury term structure | Source: WRDS | Col: derived_y_30Y_wrds_diff_1_wrds | Transform: nan

Raw Series



After: nan



SUMMARY STATISTICS

N : 1287
Valid : 1009
NaN : 278
Start : 1941-12
End : 2025-12
Mean : 0.0024
Std : 0.2428
Min : -1.3619
Max : 1.5312
Skew : -0.090
Kurt : 5.773

STATIONARITY TESTS

ADF stat : -8.229
ADF p : 0.0000
ADF lags : 10
ADF : Stationary
KPSS stat : 0.156
KPSS p : 0.1000
KPSS : Stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

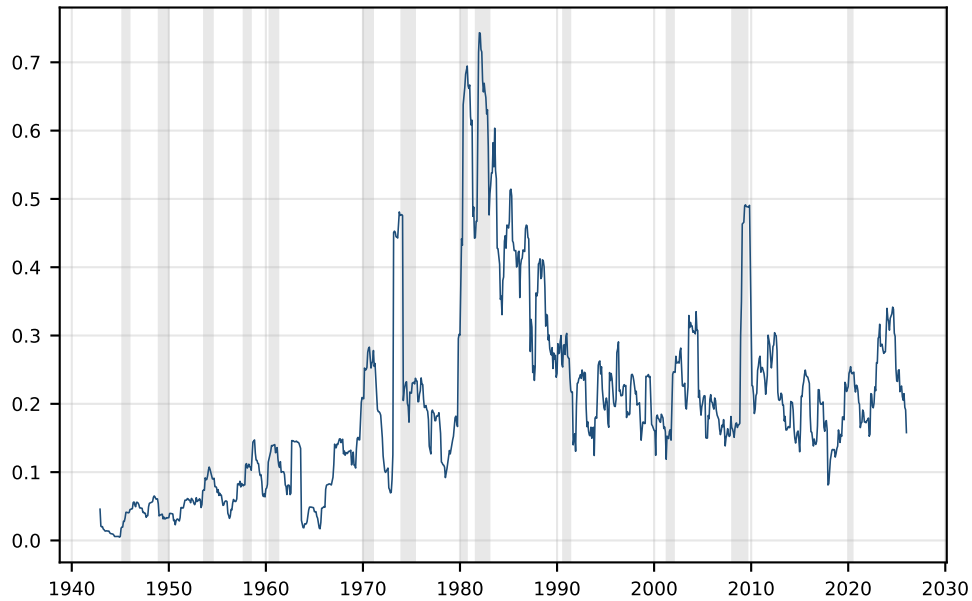
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	253	18.2%	81.8%	-0.2743
Q2	252	12.3%	87.7%	-0.0373
Q3	252	9.9%	90.1%	0.0462
Q4 (highest)	252	11.9%	88.1%	0.2762

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

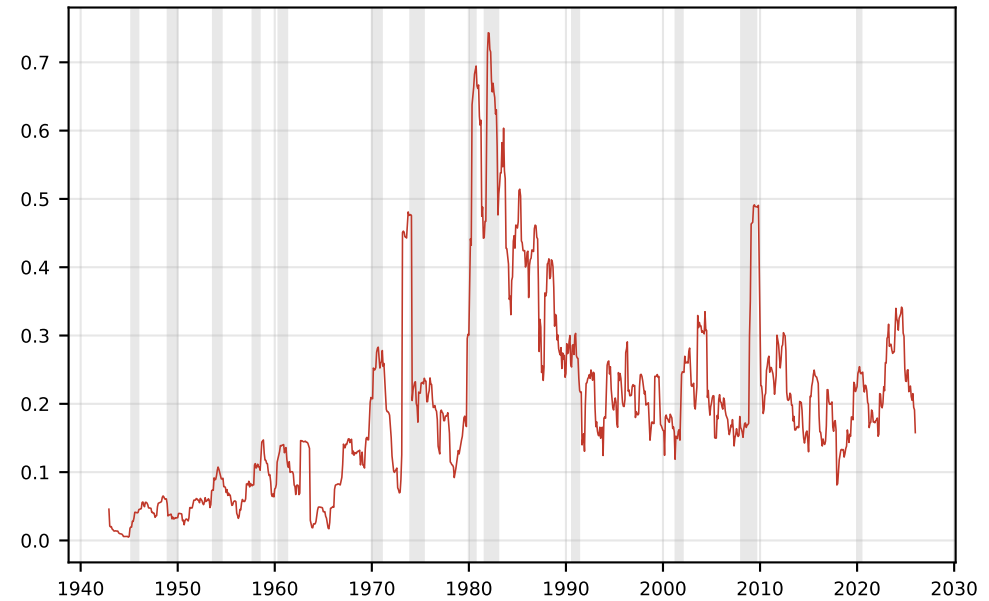
CRSP 30Y Treasury Yield 12M Rolling Volatility

Block: Selected export / Treasury term structure | Source: WRDS | Col: derived_y_30Y_wrds_vol_12m_wrds | Transform: nan

Raw Series



After: nan



SUMMARY STATISTICS

N : 1287
Valid : 998
NaN : 289
Start : 1942-11
End : 2025-12
Mean : 0.1988
Std : 0.1387
Min : 0.0048
Max : 0.7433
Skew : 1.304
Kurt : 2.056

STATIONARITY TESTS

ADF stat : -2.438
ADF p : 0.1314
ADF lags : 12
ADF : Non-stationary
KPSS stat : 1.493
KPSS p : 0.0100
KPSS : Non-stationary

QUARTILE-BUCKET ANALYSIS

(vs NBER recession/expansion)

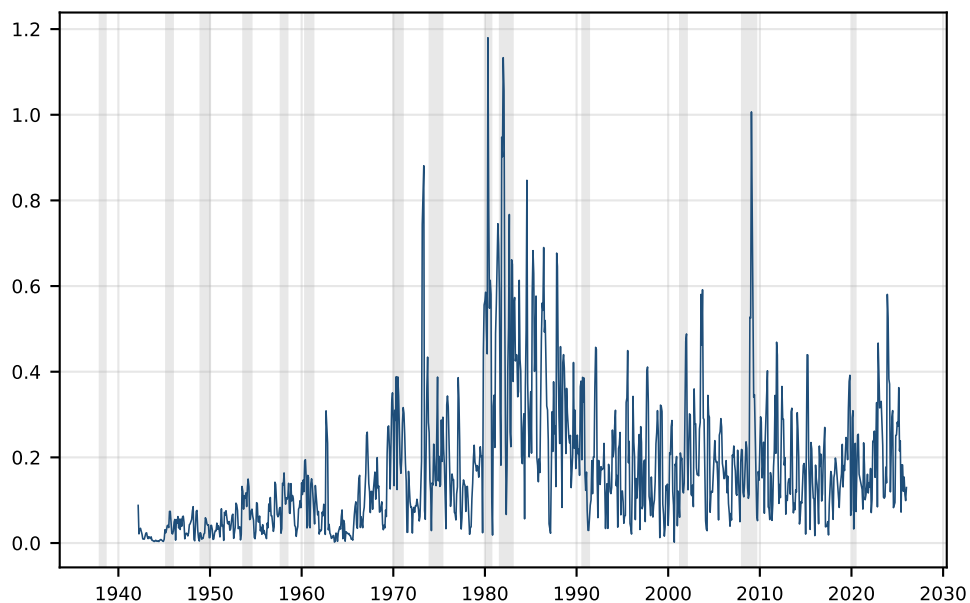
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	250	11.2%	88.8%	0.0522
Q2	249	14.5%	85.5%	0.1449
Q3	249	6.8%	93.2%	0.2133
Q4 (highest)	250	20.4%	79.6%	0.3847

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

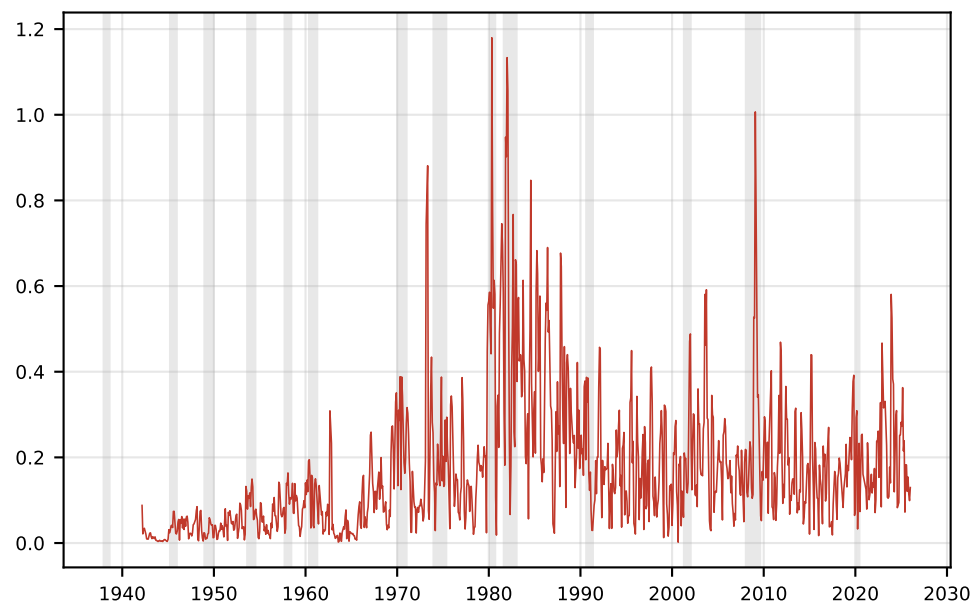
CRSP 30Y Treasury Yield 3M Rolling Volatility

Block: Selected export / Treasury term structure | Source: WRDS | Col: derived_y_30Y_wrds_vol_3m_wrds | Transform: nan

Raw Series



After: nan



SUMMARY STATISTICS

N : 1287
Valid : 1007
NaN : 280
Start : 1942-02
End : 2025-12
Mean : 0.1726
Std : 0.1626
Min : 0.0020
Max : 1.1798
Skew : 2.073
Kurt : 6.236

STATIONARITY TESTS

ADF stat : -3.829
ADF p : 0.0026
ADF lags : 12
ADF : Stationary
KPSS stat : 1.618
KPSS p : 0.0100
KPSS : Non-stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

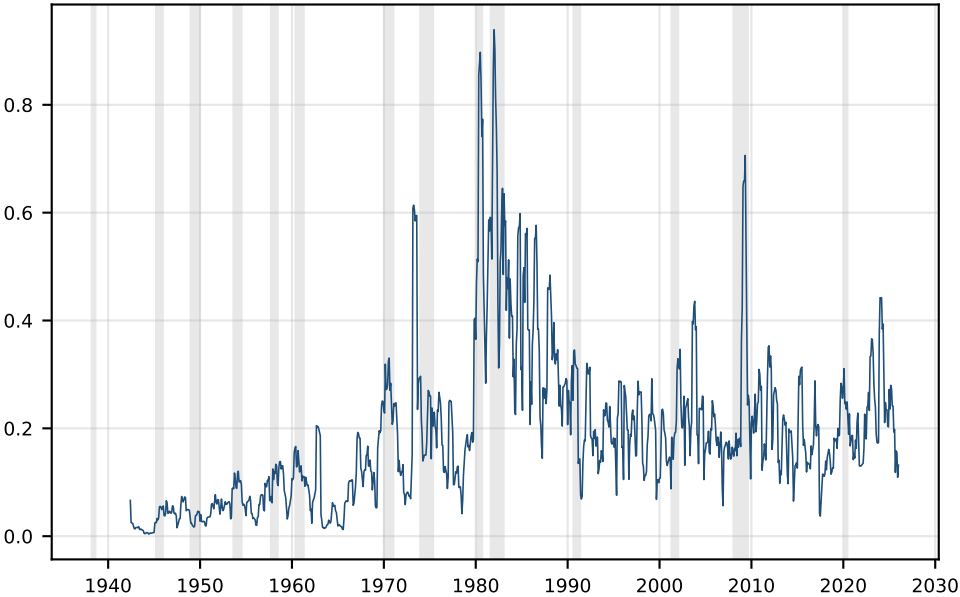
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	252	9.9%	90.1%	0.0290
Q2	252	10.3%	89.7%	0.0936
Q3	251	15.9%	84.1%	0.1756
Q4 (highest)	252	16.3%	83.7%	0.3920

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

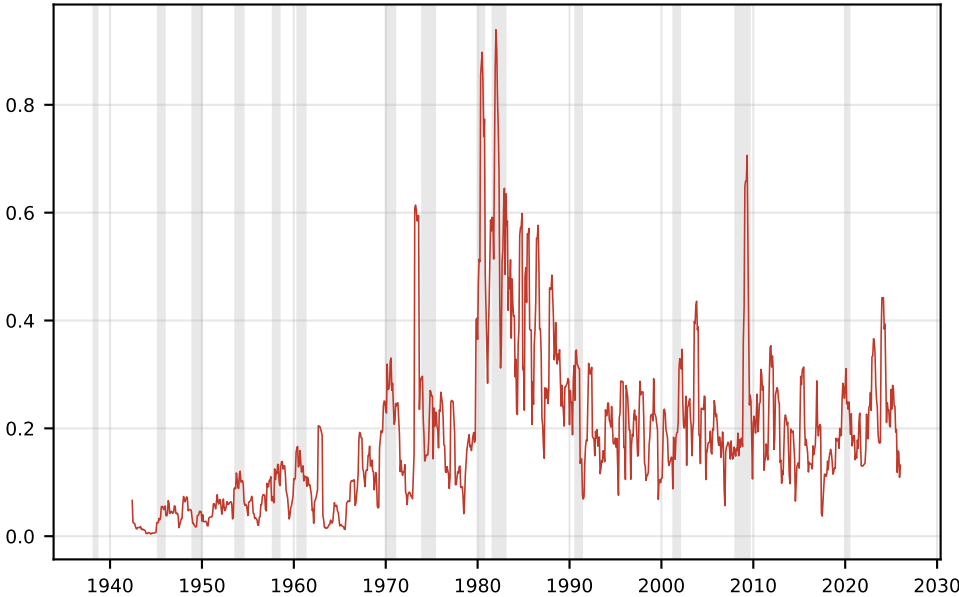
CRSP 30Y Treasury Yield 6M Rolling Volatility

Block: Selected export / Treasury term structure | Source: WRDS | Col: derived_y_30Y_wrds_vol_6m_wrds | Transform: nan

Raw Series



After: nan



SUMMARY STATISTICS

N : 1287
Valid : 1004
NaN : 283
Start : 1942-05
End : 2025-12
Mean : 0.1902
Std : 0.1488
Min : 0.0038
Max : 0.9392
Skew : 1.718
Kurt : 4.144

STATIONARITY TESTS

ADF stat : -3.304
ADF p : 0.0147
ADF lags : 12
ADF : Stationary
KPSS stat : 1.541
KPSS p : 0.0100
KPSS : Non-stationary

QUARTILE-BUCKET ANALYSIS

(vs NBER recession/expansion)

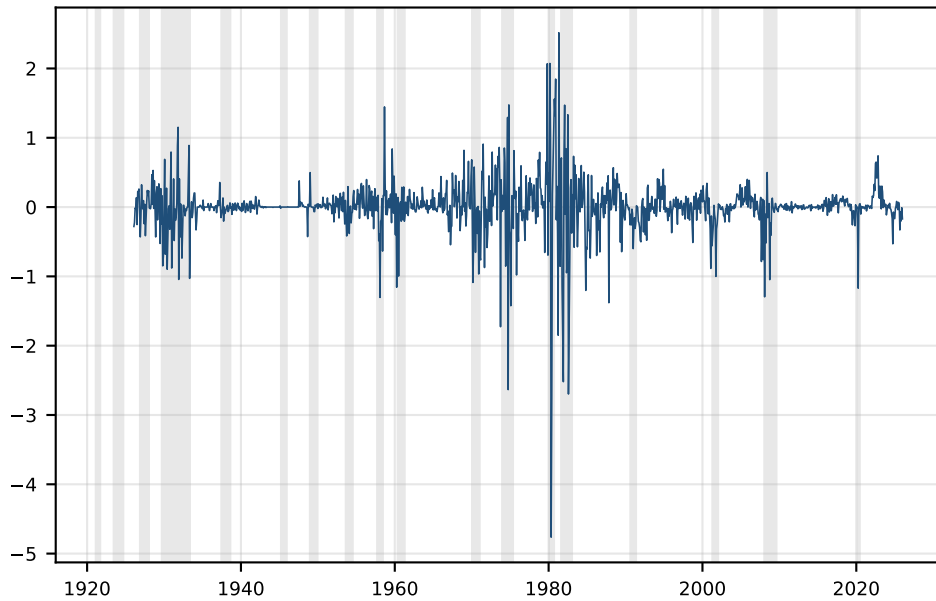
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	251	8.4%	91.6%	0.0442
Q2	251	15.9%	84.1%	0.1272
Q3	251	8.0%	92.0%	0.2024
Q4 (highest)	251	20.3%	79.7%	0.3871

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

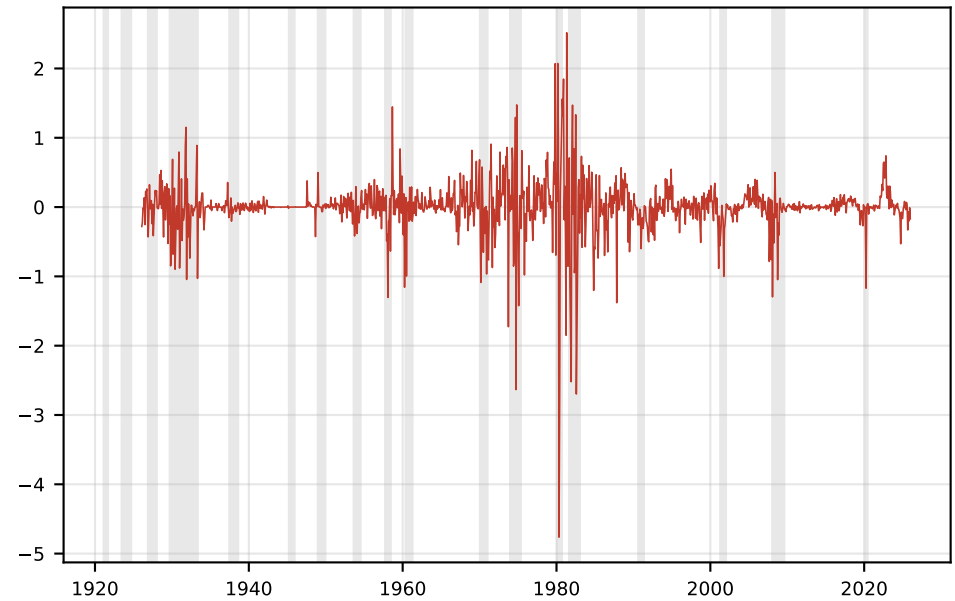
CRSP 3M Treasury Yield First Difference

Block: Selected export / Treasury term structure | Source: WRDS | Col: derived_y_3M_wrds_diff_1_wrds | Transform: nan

Raw Series



After: nan



SUMMARY STATISTICS

N : 1287
Valid : 1200
NaN : 87
Start : 1926-01
End : 2025-12
Mean : -0.0001
Std : 0.3993
Min : -4.7631
Max : 2.5156
Skew : -2.151
Kurt : 27.745

STATIONARITY TESTS

ADF stat : -9.320
ADF p : 0.0000
ADF lags : 11
ADF : Stationary
KPSS stat : 0.044
KPSS p : 0.1000
KPSS : Stationary

QUARTILE-BUCKET ANALYSIS

(vs NBER recession/expansion)

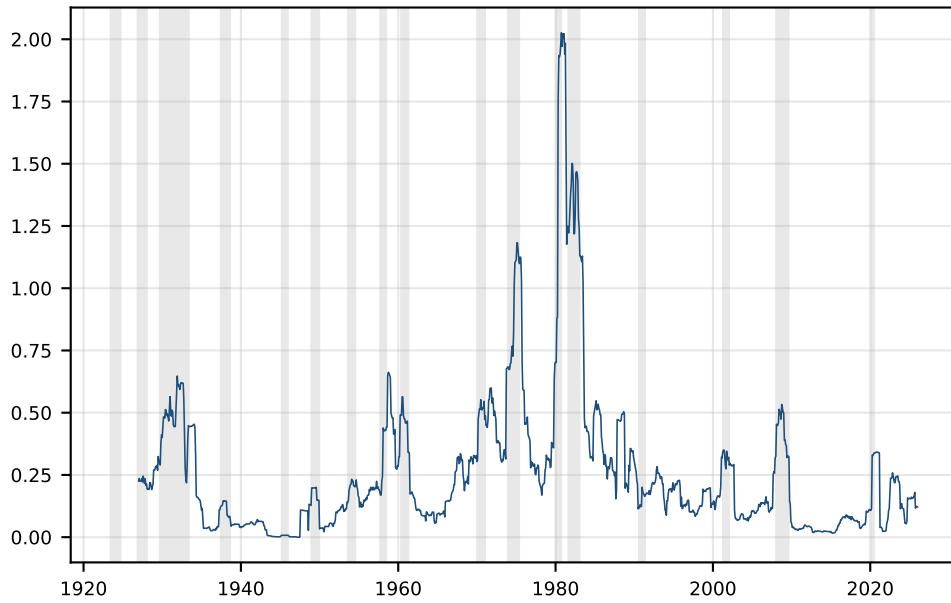
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	300	31.3%	68.7%	-0.3833
Q2	300	10.7%	89.3%	-0.0191
Q3	300	10.7%	89.3%	0.0401
Q4 (highest)	300	14.3%	85.7%	0.3618

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

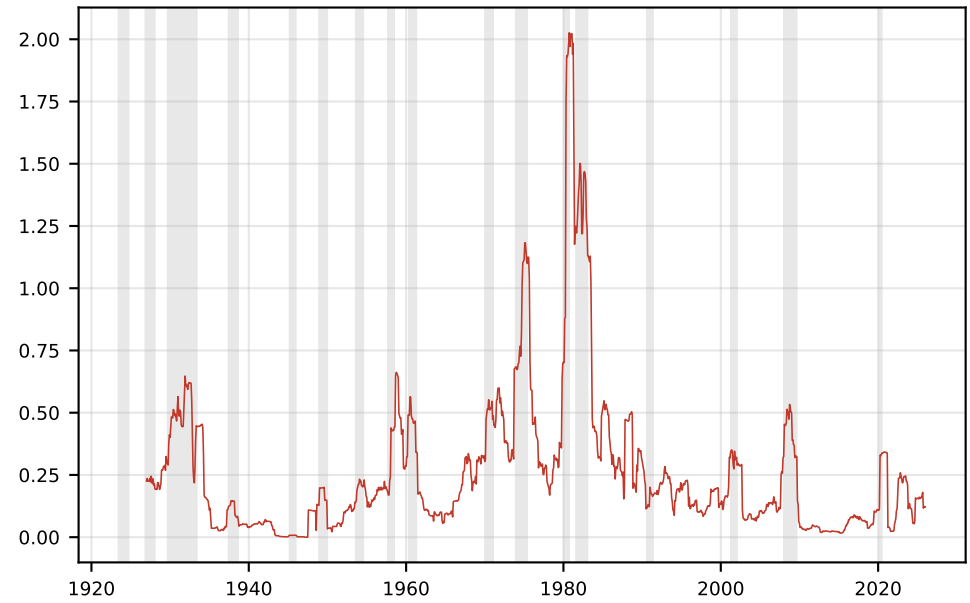
CRSP 3M Treasury Yield 12M Rolling Volatility

Block: Selected export / Treasury term structure | Source: WRDS | Col: derived_y_3m_wrds_vol_12m_wrds | Transform: nan

Raw Series



After: nan



SUMMARY STATISTICS

N : 1287
Valid : 1189
NaN : 98
Start : 1926-12
End : 2025-12
Mean : 0.2588
Std : 0.3030
Min : 0.0000
Max : 2.0262
Skew : 3.008
Kurt : 11.582

STATIONARITY TESTS

ADF stat : -3.368
ADF p : 0.0121
ADF lags : 12
ADF : Stationary
KPSS stat : 0.449
KPSS p : 0.0561
KPSS : Stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

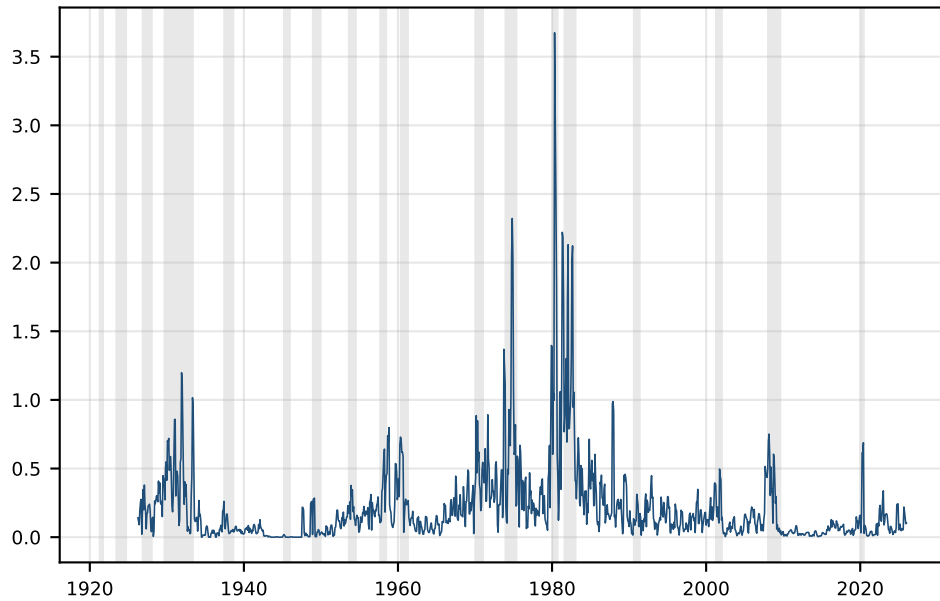
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	298	2.7%	97.3%	0.0377
Q2	297	7.7%	92.3%	0.1205
Q3	297	15.5%	84.5%	0.2412
Q4 (highest)	297	41.4%	58.6%	0.6366

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

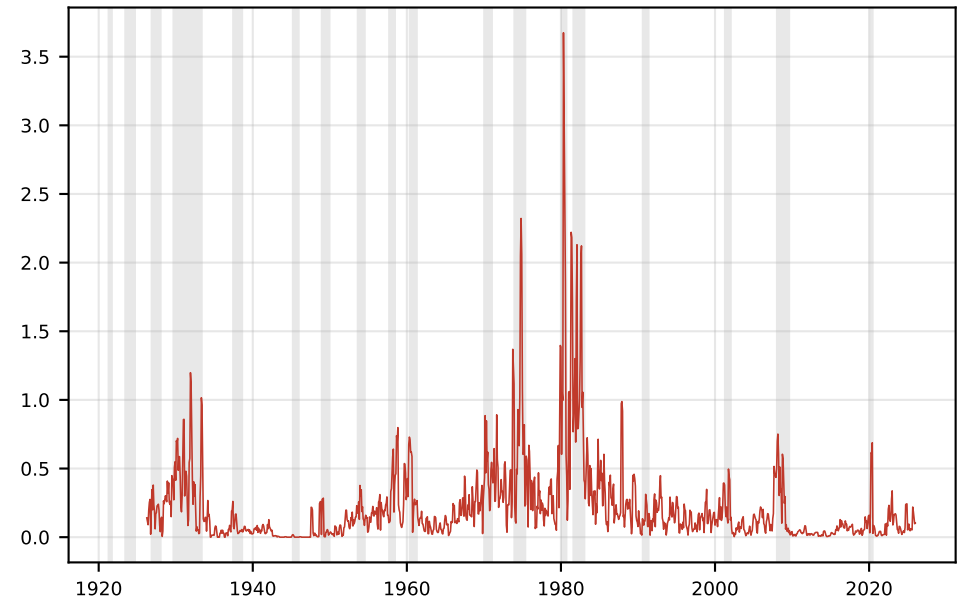
CRSP 3M Treasury Yield 3M Rolling Volatility

Block: Selected export / Treasury term structure | Source: WRDS | Col: derived_y_3m_wrds_vol_3m_wrds | Transform: nan

Raw Series



After: nan



SUMMARY STATISTICS

N : 1287
Valid : 1198
NaN : 89
Start : 1926-03
End : 2025-12
Mean : 0.2171
Std : 0.3178
Min : 0.0000
Max : 3.6745
Skew : 4.326
Kurt : 27.796

STATIONARITY TESTS

ADF stat : -4.240
ADF p : 0.0006
ADF lags : 12
ADF : Stationary
KPSS stat : 0.492
KPSS p : 0.0434
KPSS : Non-stationary

QUARTILE-BUCKET ANALYSIS

(vs NBER recession/expansion)

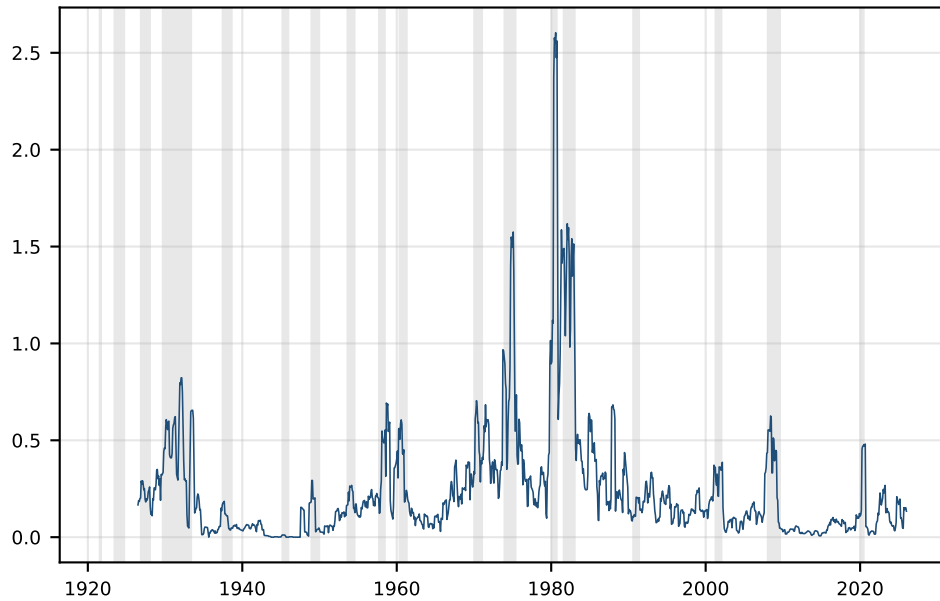
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	300	7.7%	92.3%	0.0200
Q2	299	9.7%	90.3%	0.0790
Q3	299	13.0%	87.0%	0.1810
Q4 (highest)	300	36.7%	63.3%	0.5880

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

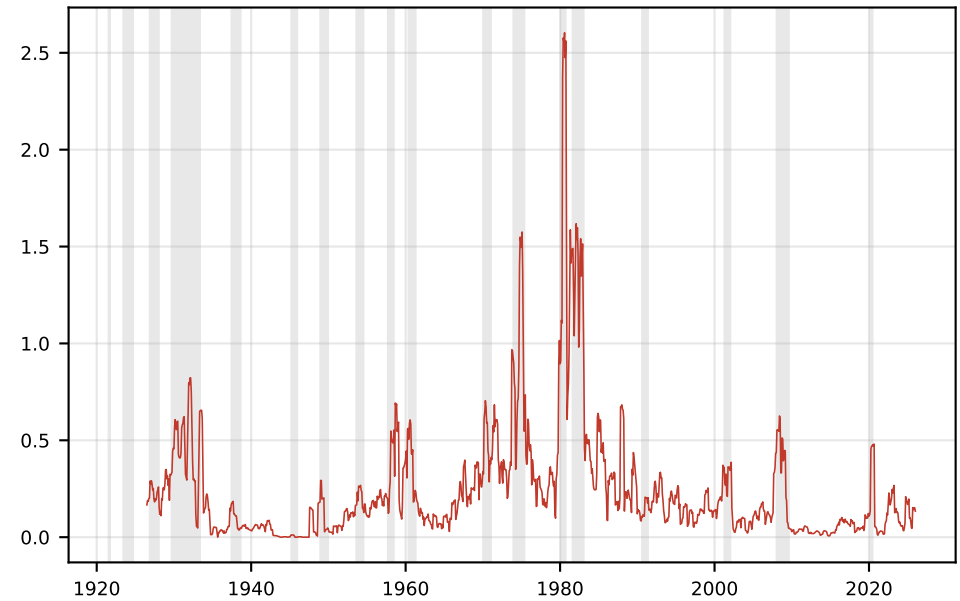
CRSP 3M Treasury Yield 6M Rolling Volatility

Block: Selected export / Treasury term structure | Source: WRDS | Col: derived_y_3M_wrds_vol_6m_wrds | Transform: nan

Raw Series



After: nan



SUMMARY STATISTICS

N : 1287
Valid : 1195
NaN : 92
Start : 1926-06
End : 2025-12
Mean : 0.2413
Std : 0.3122
Min : 0.0000
Max : 2.6036
Skew : 3.555
Kurt : 17.602

STATIONARITY TESTS

ADF stat : -4.011
ADF p : 0.0014
ADF lags : 12
ADF : Stationary
KPSS stat : 0.465
KPSS p : 0.0496
KPSS : Non-stationary

QUARTILE-BUCKET ANALYSIS

(vs NBER recession/expansion)

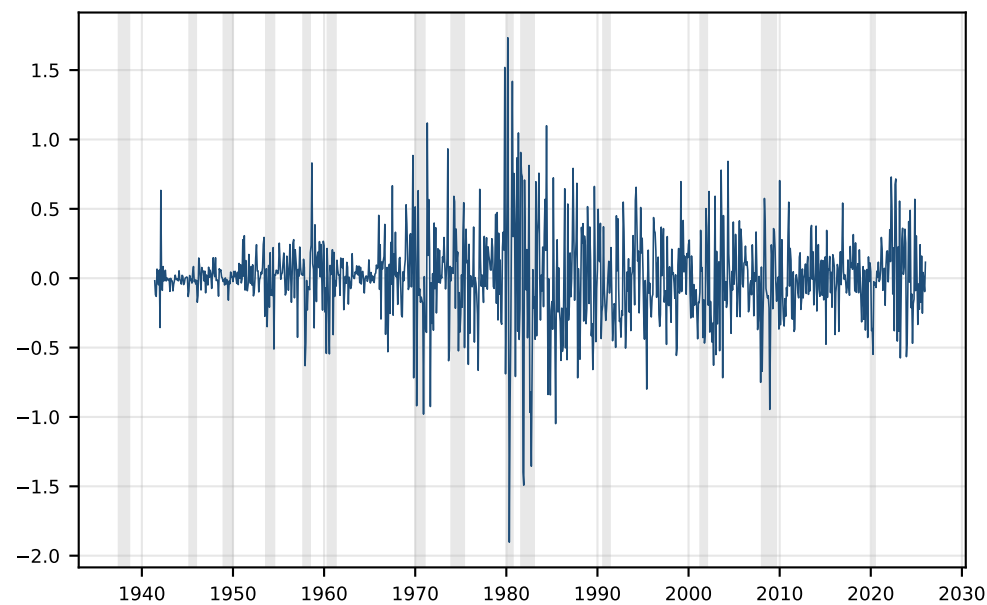
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	299	6.7%	93.3%	0.0296
Q2	299	5.0%	95.0%	0.1027
Q3	298	17.8%	82.2%	0.2114
Q4 (highest)	299	37.8%	62.2%	0.6214

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

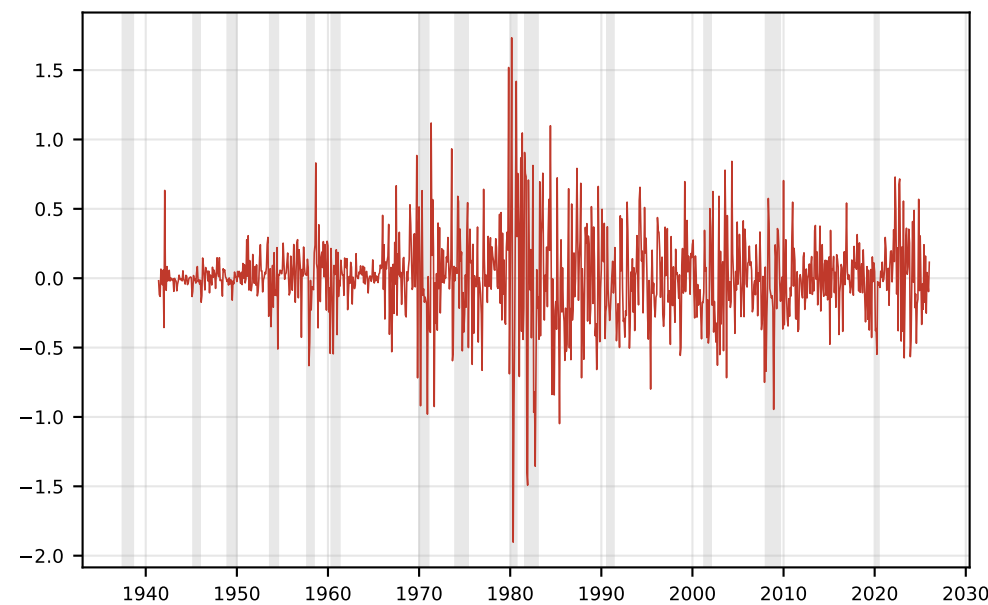
CRSP 5Y Treasury Yield First Difference

Block: Selected export / Treasury term structure | Source: WRDS | Col: derived_y_5Y_wrds_diff_1_wrds | Transform: nan

Raw Series



After: nan



SUMMARY STATISTICS

N : 1287
Valid : 1016
NaN : 271
Start : 1941-05
End : 2025-12
Mean : 0.0020
Std : 0.3141
Min : -1.9026
Max : 1.7332
Skew : -0.049
Kurt : 4.722

STATIONARITY TESTS

ADF stat : -22.853
ADF p : 0.0000
ADF lags : 1
ADF : Stationary
KPSS stat : 0.134
KPSS p : 0.1000
KPSS : Stationary

QUARTILE-BUCKET ANALYSIS
(vs NBER recession/expansion)

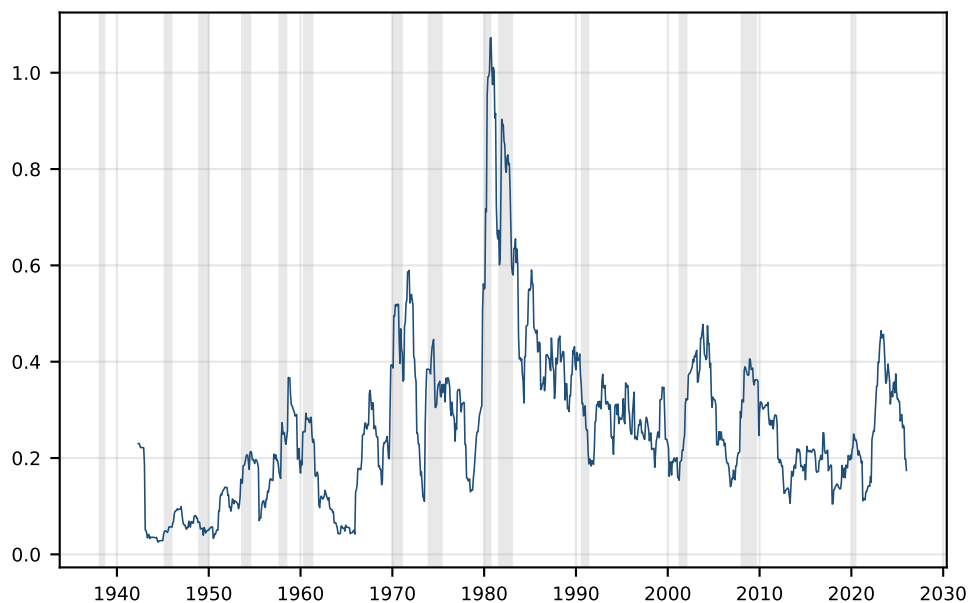
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	254	17.7%	82.3%	-0.3665
Q2	254	13.4%	86.6%	-0.0541
Q3	254	9.4%	90.6%	0.0615
Q4 (highest)	254	11.4%	88.6%	0.3673

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

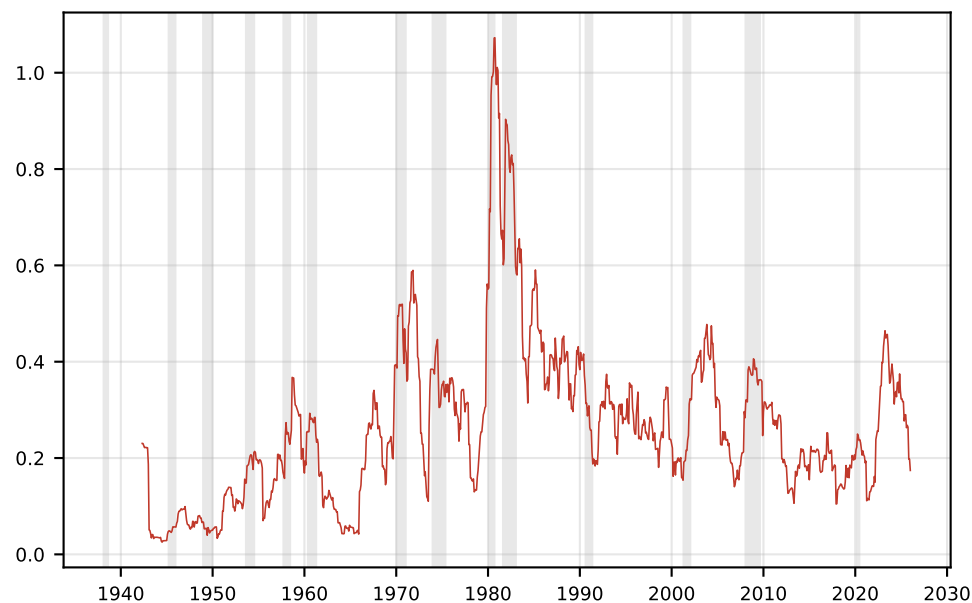
CRSP 5Y Treasury Yield 12M Rolling Volatility

Block: Selected export / Treasury term structure | Source: WRDS | Col: derived_y_5Y_wrds_vol_12m_wrds | Transform: nan

Raw Series



After: nan



SUMMARY STATISTICS

N : 1287
Valid : 1005
NaN : 282
Start : 1942-04
End : 2025-12
Mean : 0.2664
Std : 0.1683
Min : 0.0253
Max : 1.0726
Skew : 1.561
Kurt : 4.172

STATIONARITY TESTS

ADF stat : -2.860
ADF p : 0.0502
ADF lags : 12
ADF : Non-stationary
KPSS stat : 0.900
KPSS p : 0.0100
KPSS : Non-stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

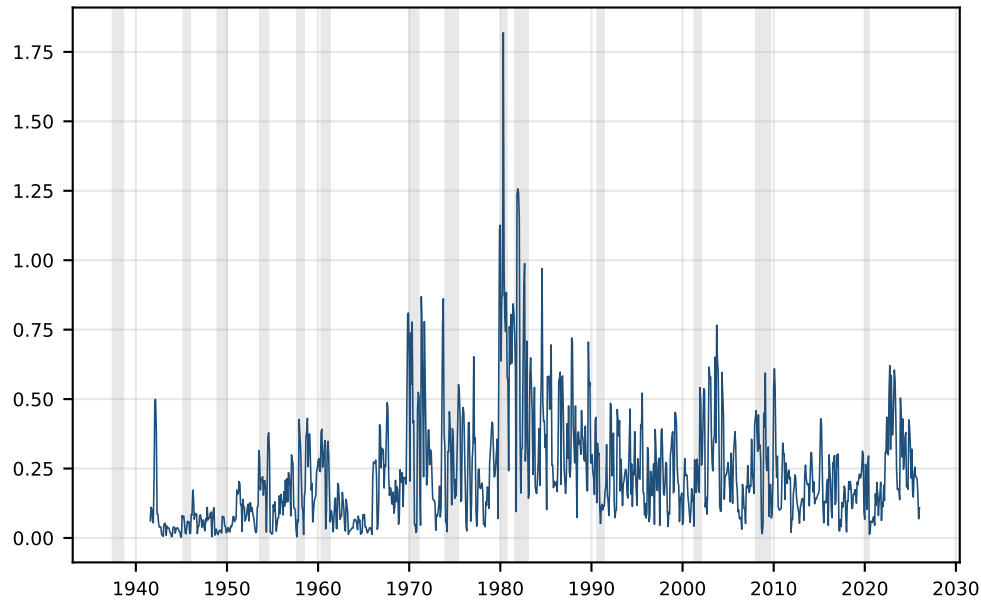
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	252	7.9%	92.1%	0.0901
Q2	251	7.2%	92.8%	0.2006
Q3	251	13.9%	86.1%	0.2927
Q4 (highest)	251	23.5%	76.5%	0.4828

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

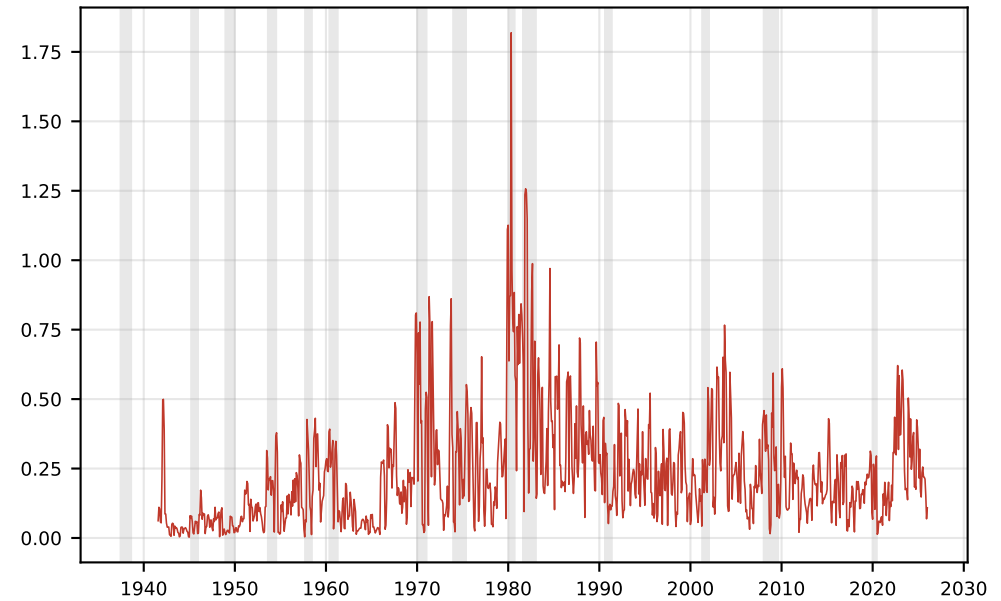
CRSP 5Y Treasury Yield 3M Rolling Volatility

Block: Selected export / Treasury term structure | Source: WRDS | Col: derived_y_5Y_wrds_vol_3m_wrds | Transform: nan

Raw Series



After: nan



SUMMARY STATISTICS

N : 1287
Valid : 1014
NaN : 273
Start : 1941-07
End : 2025-12
Mean : 0.2325
Std : 0.2008
Min : 0.0029
Max : 1.8189
Skew : 2.020
Kurt : 7.048

STATIONARITY TESTS

ADF stat : -3.993
ADF p : 0.0014
ADF lags : 11
ADF : Stationary
KPSS stat : 0.952
KPSS p : 0.0100
KPSS : Non-stationary

QUARTILE-BUCKET ANALYSIS

(vs NBER recession/expansion)

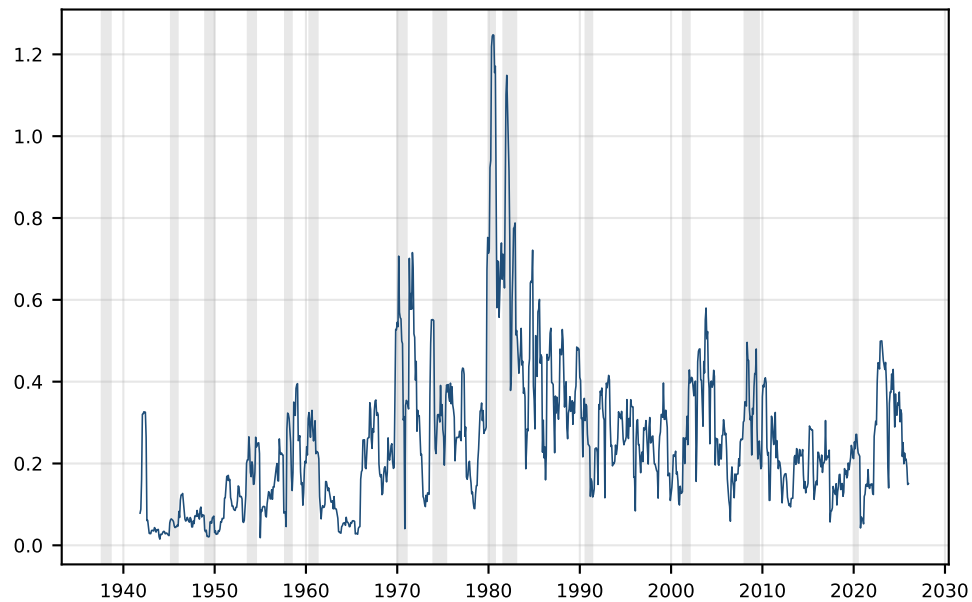
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	254	13.8%	86.2%	0.0478
Q2	253	7.5%	92.5%	0.1366
Q3	253	10.7%	89.3%	0.2416
Q4 (highest)	254	20.1%	79.9%	0.5039

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

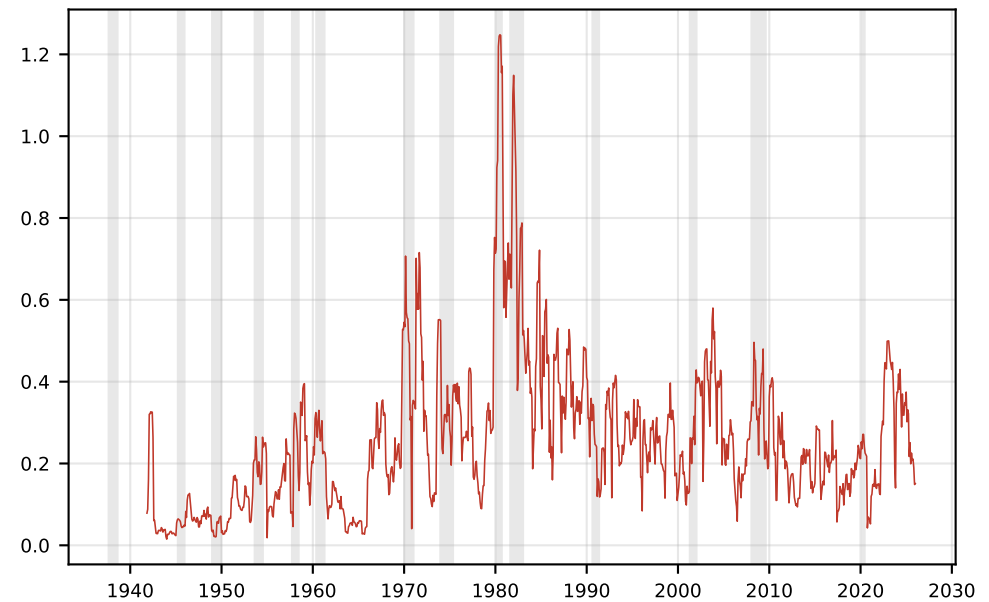
CRSP 5Y Treasury Yield 6M Rolling Volatility

Block: Selected export / Treasury term structure | Source: WRDS | Col: derived_y_5Y_wrds_vol_6m_wrds | Transform: nan

Raw Series



After: nan



SUMMARY STATISTICS

N : 1287
Valid : 1011
NaN : 276
Start : 1941-10
End : 2025-12
Mean : 0.2562
Std : 0.1815
Min : 0.0150
Max : 1.2480
Skew : 1.834
Kurt : 5.892

STATIONARITY TESTS

ADF stat : -3.654
ADF p : 0.0048
ADF lags : 12
ADF : Stationary
KPSS stat : 0.917
KPSS p : 0.0100
KPSS : Non-stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

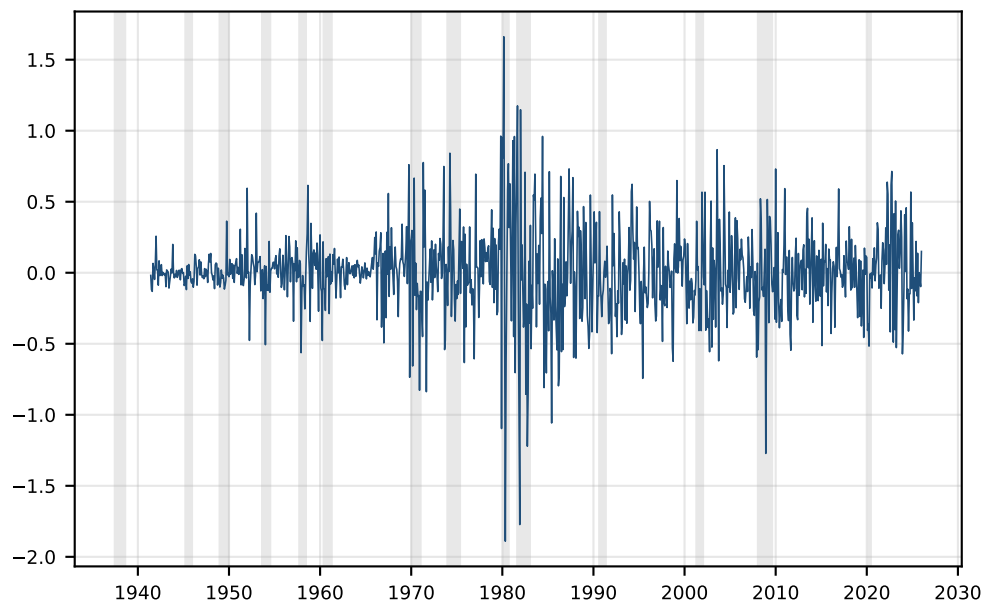
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	253	9.1%	90.9%	0.0741
Q2	253	5.9%	94.1%	0.1804
Q3	252	18.7%	81.3%	0.2789
Q4 (highest)	253	18.6%	81.4%	0.4915

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

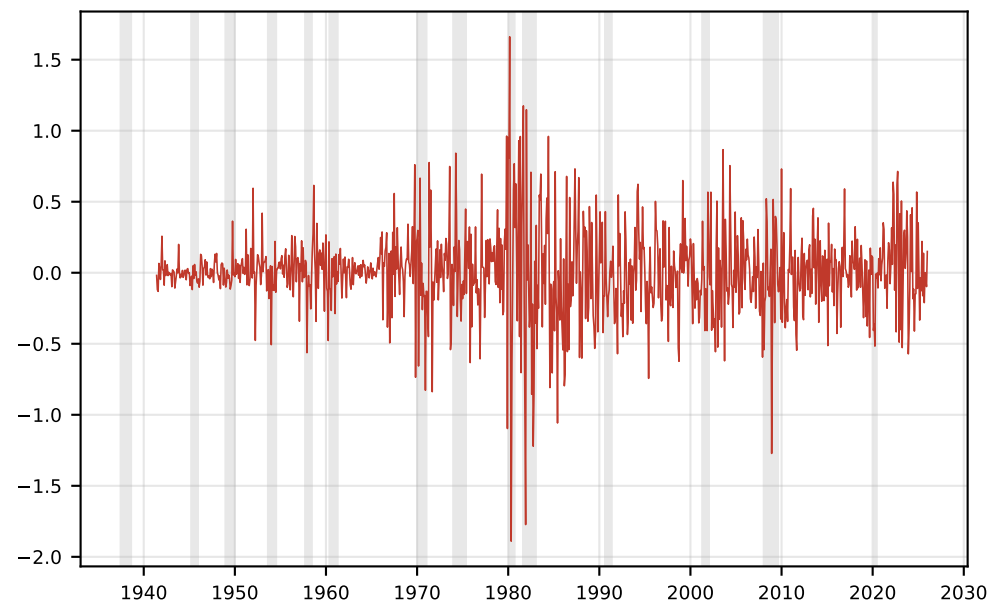
CRSP 7Y Treasury Yield First Difference

Block: Selected export / Treasury term structure | Source: WRDS | Col: derived_y_7Y_wrds_diff_1_wrds | Transform: nan

Raw Series



After: nan



SUMMARY STATISTICS

N : 1287
Valid : 1016
NaN : 271
Start : 1941-05
End : 2025-12
Mean : 0.0022
Std : 0.2989
Min : -1.8898
Max : 1.6616
Skew : -0.239
Kurt : 5.145

STATIONARITY TESTS

ADF stat : -20.173
ADF p : 0.0000
ADF lags : 2
ADF : Stationary
KPSS stat : 0.151
KPSS p : 0.1000
KPSS : Stationary

QUARTILE-BUCKET ANALYSIS

(vs NBER recession/expansion)

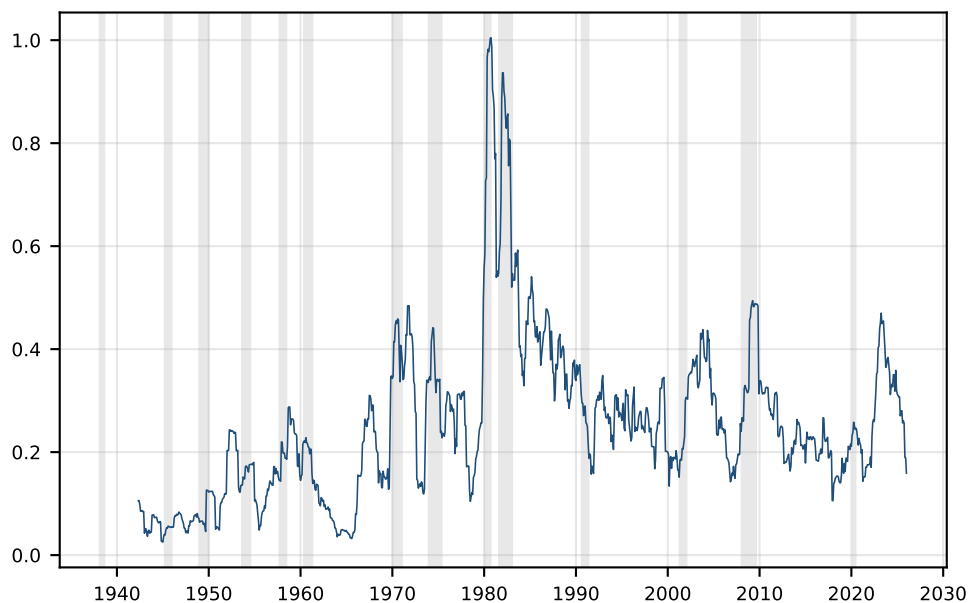
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	254	18.1%	81.9%	-0.3446
Q2	254	14.6%	85.4%	-0.0555
Q3	254	8.3%	91.7%	0.0598
Q4 (highest)	254	11.0%	89.0%	0.3492

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

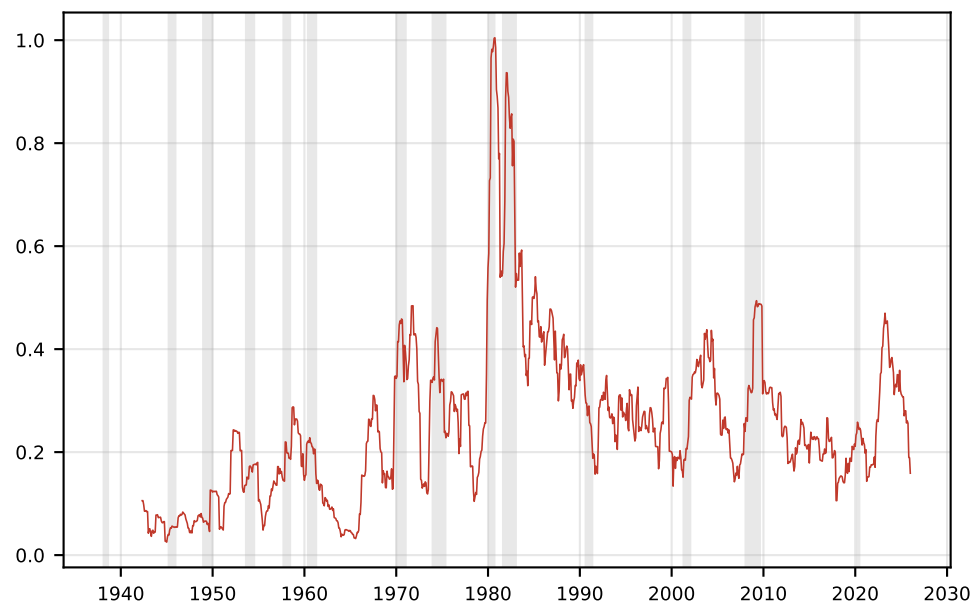
CRSP 7Y Treasury Yield 12M Rolling Volatility

Block: Selected export / Treasury term structure | Source: WRDS | Col: derived_y_7Y_wrds_vol_12m_wrds | Transform: nan

Raw Series



After: nan



SUMMARY STATISTICS

N : 1287
Valid : 1005
NaN : 282
Start : 1942-04
End : 2025-12
Mean : 0.2544
Std : 0.1601
Min : 0.0251
Max : 1.0047
Skew : 1.663
Kurt : 4.705

STATIONARITY TESTS

ADF stat : -2.757
ADF p : 0.0646
ADF lags : 12
ADF : Non-stationary
KPSS stat : 1.088
KPSS p : 0.0100
KPSS : Non-stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

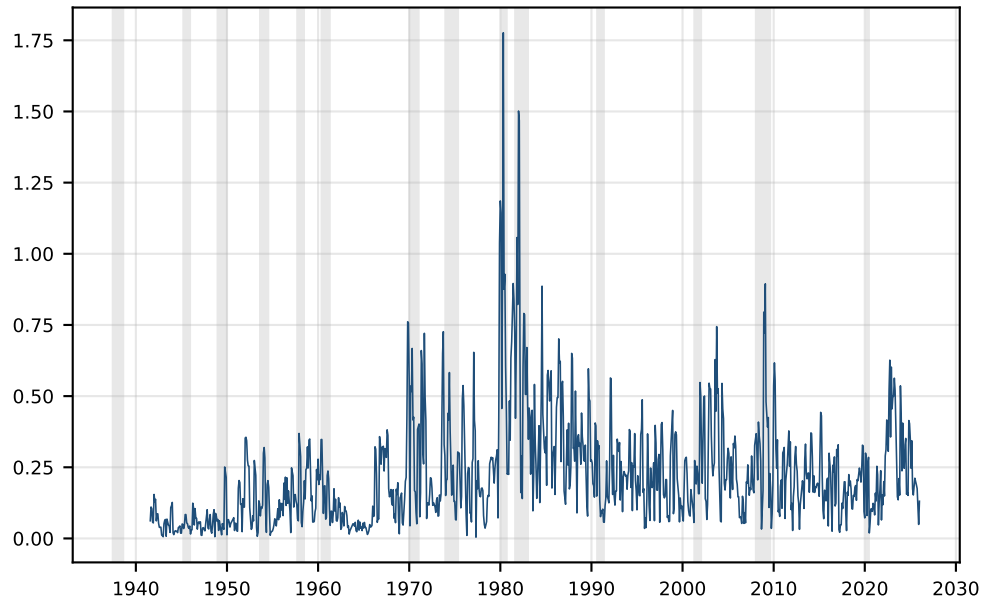
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	252	9.5%	90.5%	0.0889
Q2	251	12.0%	88.0%	0.1926
Q3	251	8.4%	91.6%	0.2769
Q4 (highest)	251	22.7%	77.3%	0.4599

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

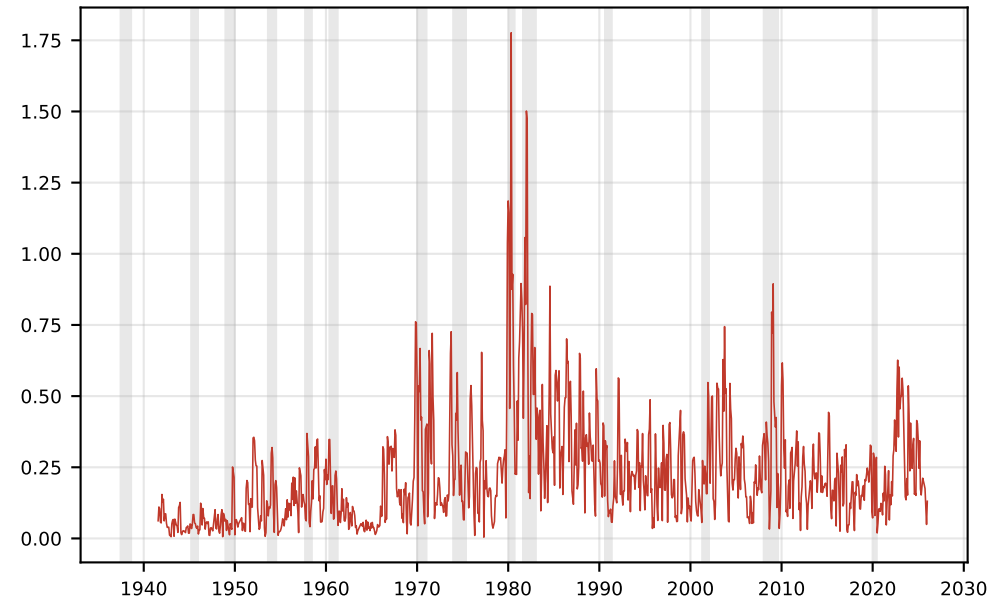
CRSP 7Y Treasury Yield 3M Rolling Volatility

Block: Selected export / Treasury term structure | Source: WRDS | Col: derived_y_7Y_wrds_vol_3m_wrds | Transform: nan

Raw Series



After: nan



SUMMARY STATISTICS

N : 1287
Valid : 1014
NaN : 273
Start : 1941-07
End : 2025-12
Mean : 0.2224
Std : 0.1927
Min : 0.0044
Max : 1.7766
Skew : 2.291
Kurt : 9.534

STATIONARITY TESTS

ADF stat : -4.439
ADF p : 0.0003
ADF lags : 11
ADF : Stationary
KPSS stat : 1.185
KPSS p : 0.0100
KPSS : Non-stationary

QUARTILE-BUCKET ANALYSIS

(vs NBER recession/expansion)

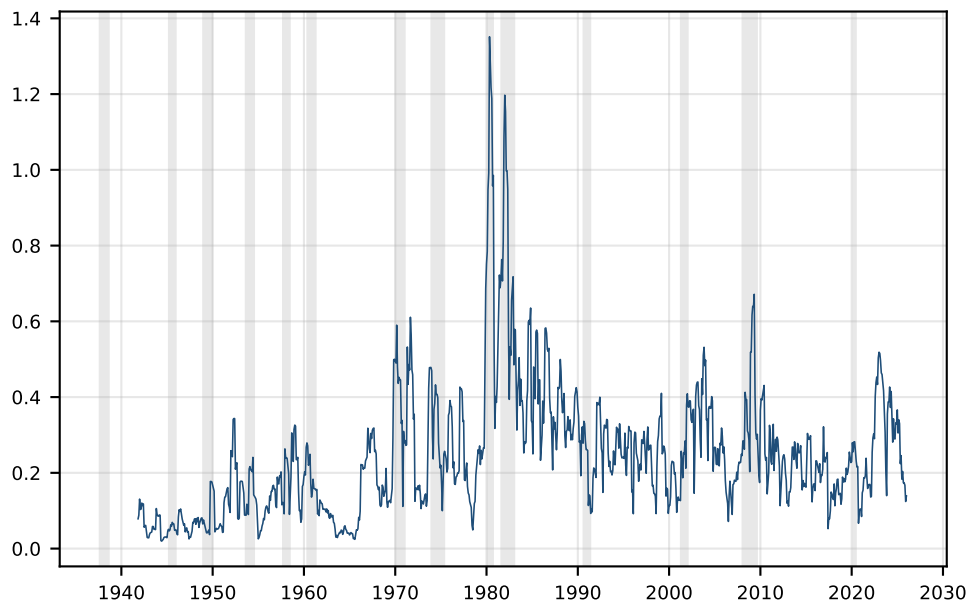
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	254	12.6%	87.4%	0.0484
Q2	253	9.5%	90.5%	0.1307
Q3	253	10.7%	89.3%	0.2324
Q4 (highest)	254	19.3%	80.7%	0.4778

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

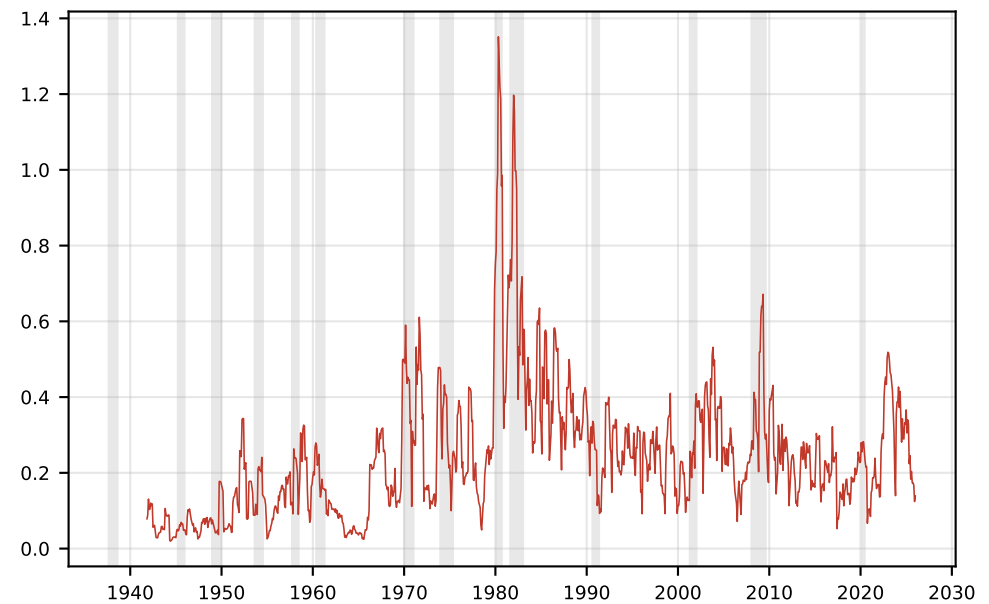
CRSP 7Y Treasury Yield 6M Rolling Volatility

Block: Selected export / Treasury term structure | Source: WRDS | Col: derived_y_7Y_wrds_vol_6m_wrds | Transform: nan

Raw Series



After: nan



SUMMARY STATISTICS

N : 1287
Valid : 1011
NaN : 276
Start : 1941-10
End : 2025-12
Mean : 0.2445
Std : 0.1747
Min : 0.0196
Max : 1.3515
Skew : 2.070
Kurt : 7.562

STATIONARITY TESTS

ADF stat : -3.802
ADF p : 0.0029
ADF lags : 12
ADF : Stationary
KPSS stat : 1.126
KPSS p : 0.0100
KPSS : Non-stationary

QUARTILE-BUCKET ANALYSIS

(vs NBER recession/expansion)

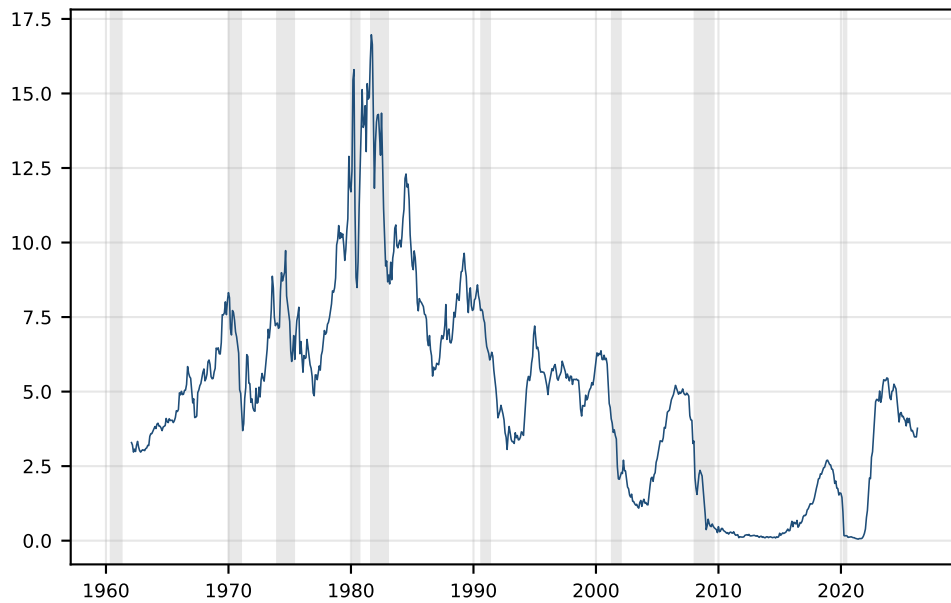
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	253	10.3%	89.7%	0.0740
Q2	253	8.7%	91.3%	0.1717
Q3	252	13.1%	86.9%	0.2624
Q4 (highest)	253	20.2%	79.8%	0.4701

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

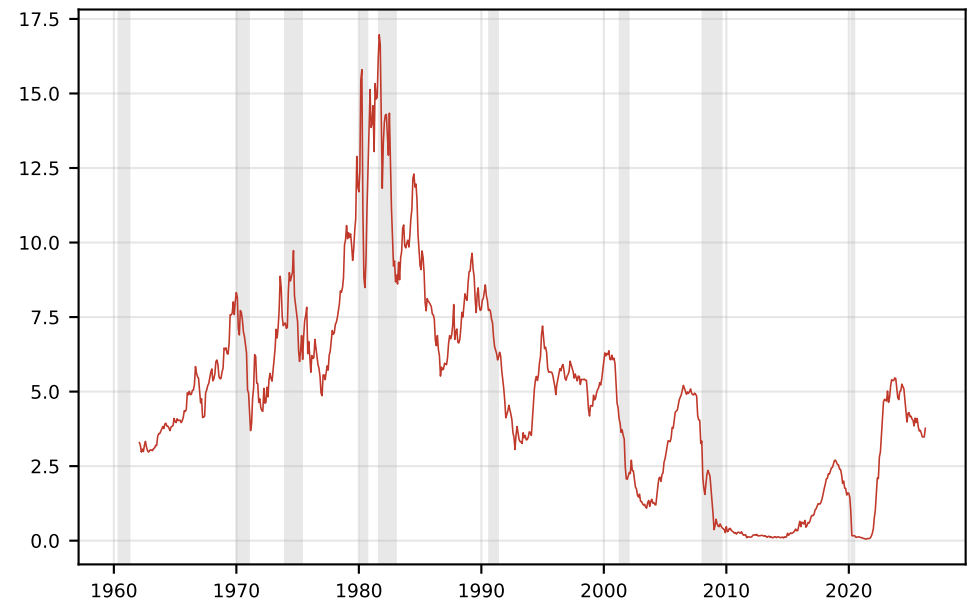
1Y Treasury Yield

Block: Selected export / Treasury term structure | Source: FRED API | Col: raw_DGS1 | Transform: nan

Raw Series



After: nan



SUMMARY STATISTICS

N : 1287
Valid : 771
NaN : 516
Start : 1962-01
End : 2026-03
Mean : 4.8864
Std : 3.3591
Min : 0.0500
Max : 16.9700
Skew : 0.674
Kurt : 0.612

STATIONARITY TESTS

ADF stat : -2.272
ADF p : 0.1813
ADF lags : 12
ADF : Non-stationary
KPSS stat : 2.042
KPSS p : 0.0100
KPSS : Non-stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

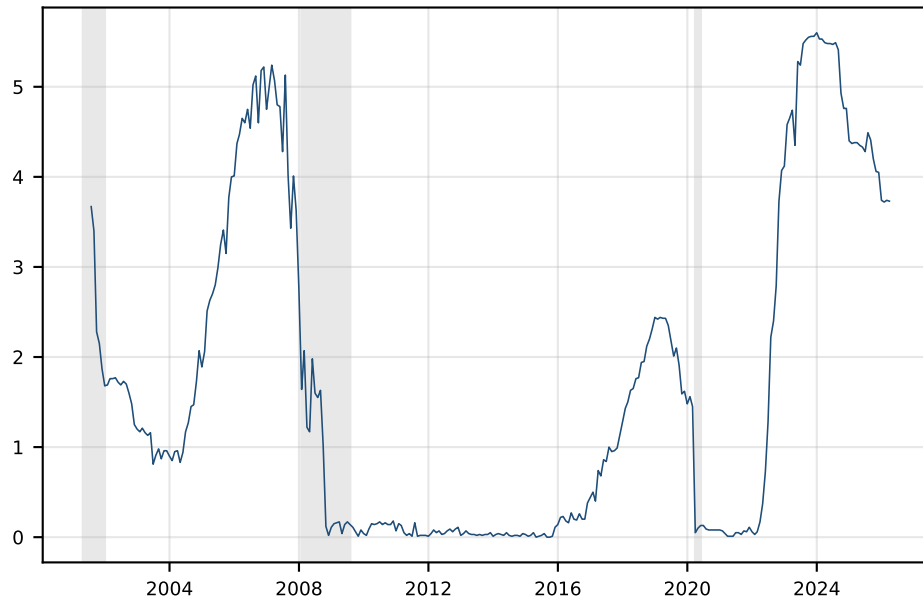
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	193	10.9%	89.1%	0.7741
Q2	192	3.6%	96.4%	3.8188
Q3	193	4.7%	95.3%	5.6771
Q4 (highest)	192	25.0%	75.0%	9.2986

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

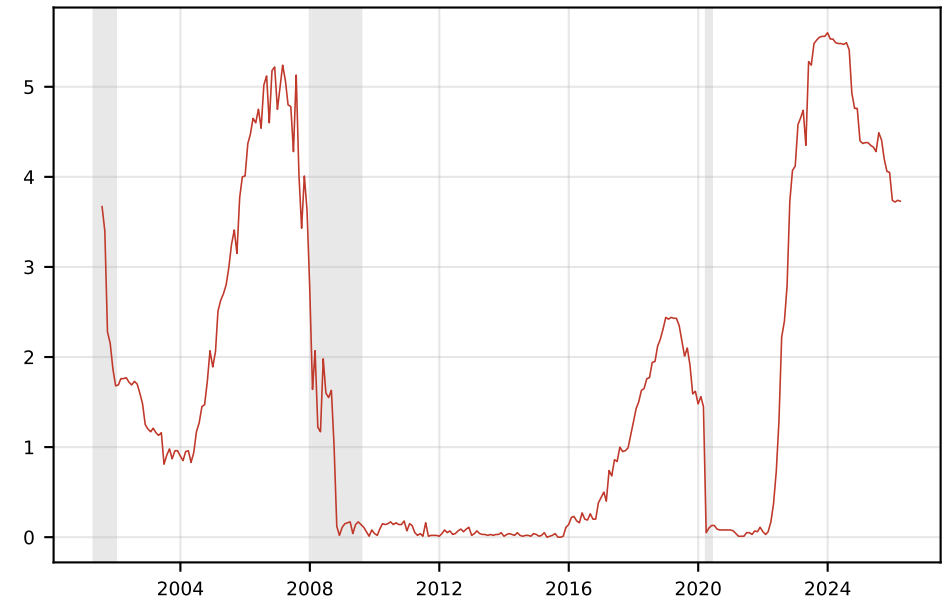
1M Treasury Yield

Block: Selected export / Treasury term structure | Source: FRED API | Col: raw_DGS1MO | Transform: nan

Raw Series



After: nan



SUMMARY STATISTICS

N : 1287
Valid : 297
NaN : 990
Start : 2001-07
End : 2026-03
Mean : 1.6930
Std : 1.8386
Min : 0.0000
Max : 5.6000
Skew : 0.825
Kurt : -0.745

STATIONARITY TESTS

ADF stat : -3.128
ADF p : 0.0245
ADF lags : 9
ADF : Stationary
KPSS stat : 0.408
KPSS p : 0.0735
KPSS : Stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	79	3.8%	96.2%	0.0365
Q2	69	11.6%	88.4%	0.3958
Q3	74	16.2%	83.8%	1.7909
Q4 (highest)	74	2.7%	97.3%	4.5455

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

2Y Treasury Yield

Block: Selected export / Treasury term structure | Source: FRED API | Col: raw_DGS2 | Transform: nan

Raw Series



After: nan



SUMMARY STATISTICS

N : 1287
Valid : 598
NaN : 689
Start : 1976-06
End : 2026-03
Mean : 4.9553
Std : 3.6777
Min : 0.1100
Max : 16.7300
Skew : 0.693
Kurt : 0.016

STATIONARITY TESTS

ADF stat : -1.880
ADF p : 0.3417
ADF lags : 11
ADF : Non-stationary
KPSS stat : 2.647
KPSS p : 0.0100
KPSS : Non-stationary

QUARTILE-BUCKET ANALYSIS

(vs NBER recession/expansion)

Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	150	8.7%	91.3%	0.6966
Q2	150	10.0%	90.0%	3.3500
Q3	148	2.0%	98.0%	5.8045
Q4 (highest)	149	18.1%	81.9%	10.0224

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

20Y Treasury Yield

Block: Selected export / Treasury term structure | Source: FRED API | Col: raw_DGS20 | Transform: nan

Raw Series



After: nan



SUMMARY STATISTICS

N : 1287
Valid : 690
NaN : 597
Start : 1962-01
End : 2026-03
Mean : 5.8515
Std : 2.8536
Min : 0.9800
Max : 15.7800
Skew : 1.006
Kurt : 0.820

STATIONARITY TESTS

ADF stat : -1.463
ADF p : 0.5516
ADF lags : 2
ADF : Non-stationary
KPSS stat : 1.588
KPSS p : 0.0100
KPSS : Non-stationary

QUARTILE-BUCKET ANALYSIS

(vs NBER recession/expansion)

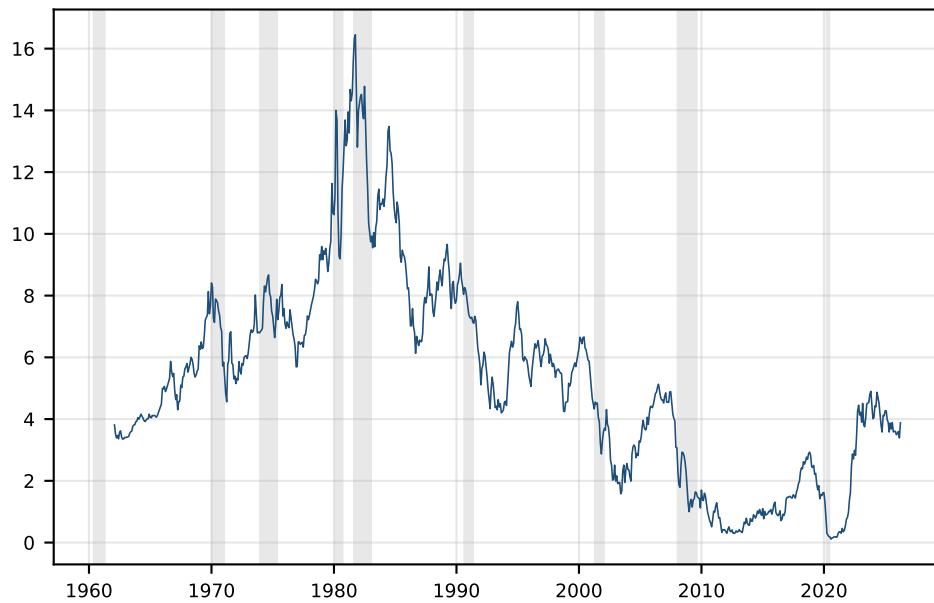
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	173	4.0%	96.0%	2.8234
Q2	173	8.1%	91.9%	4.6023
Q3	171	9.9%	90.1%	6.1981
Q4 (highest)	172	22.7%	77.3%	9.8147

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

3Y Treasury Yield

Block: Selected export / Treasury term structure | Source: FRED API | Col: raw_DGS3 | Transform: nan

Raw Series



After: nan



SUMMARY STATISTICS

N : 1287
Valid : 771
NaN : 516
Start : 1962-01
End : 2026-03
Mean : 5.2474
Std : 3.2669
Min : 0.1100
Max : 16.4500
Skew : 0.622
Kurt : 0.344

STATIONARITY TESTS

ADF stat : -2.027
ADF p : 0.2749
ADF lags : 11
ADF : Non-stationary
KPSS stat : 2.175
KPSS p : 0.0100
KPSS : Non-stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	193	9.3%	90.7%	1.2749
Q2	192	5.2%	94.8%	4.0824
Q3	192	5.2%	94.8%	6.0896
Q4 (highest)	193	24.4%	75.6%	9.5480

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

30Y Treasury Yield

Block: Selected export / Treasury term structure | Source: FRED API | Col: raw_DGS30 | Transform: nan

Raw Series



After: nan



SUMMARY STATISTICS

N : 1287
Valid : 590
NaN : 697
Start : 1977-02
End : 2026-03
Mean : 6.1587
Std : 3.0028
Min : 1.2000
Max : 15.1900
Skew : 0.728
Kurt : -0.095

STATIONARITY TESTS

ADF stat : -1.312
ADF p : 0.6237
ADF lags : 2
ADF : Non-stationary
KPSS stat : 2.995
KPSS p : 0.0100
KPSS : Non-stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

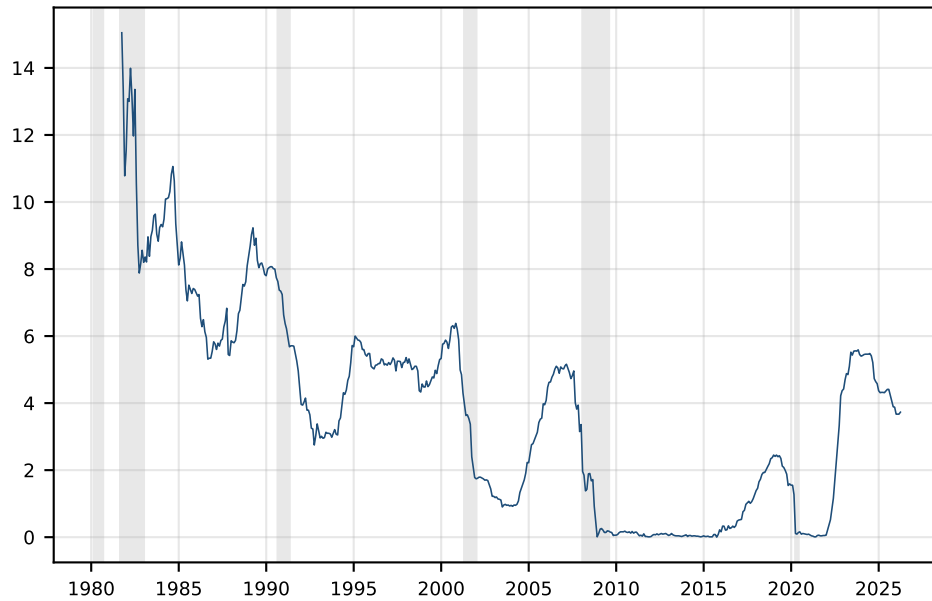
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	148	4.7%	95.3%	2.8174
Q2	148	11.5%	88.5%	4.7088
Q3	146	2.7%	97.3%	6.7949
Q4 (highest)	147	20.4%	79.6%	10.3594

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

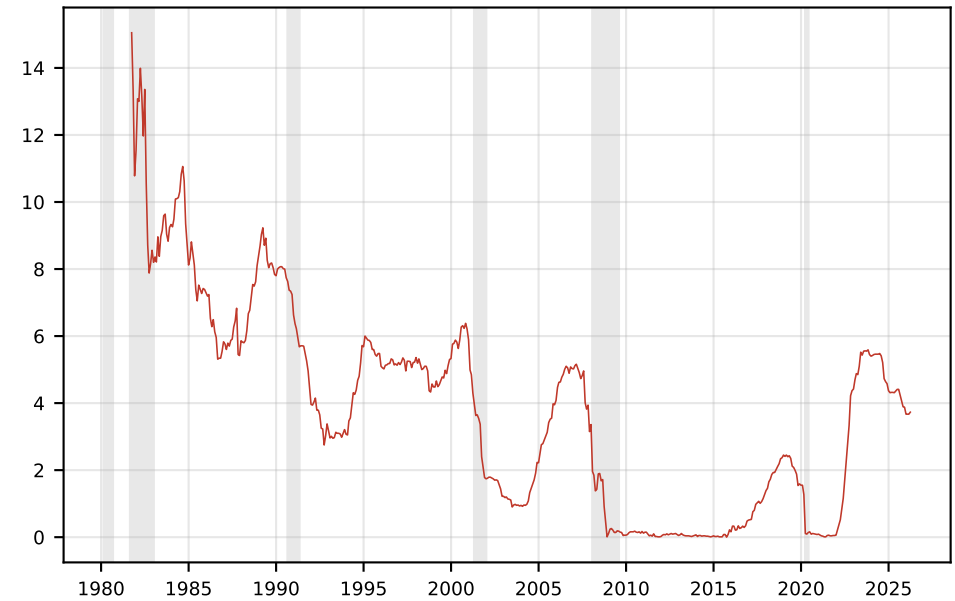
3M Treasury Yield

Block: Selected export / Treasury term structure | Source: FRED API | Col: raw_DGS3MO | Transform: nan

Raw Series



After: nan



SUMMARY STATISTICS

N : 1287
Valid : 535
NaN : 752
Start : 1981-09
End : 2026-03
Mean : 3.8181
Std : 3.1018
Min : 0.0000
Max : 15.0500
Skew : 0.599
Kurt : 0.004

STATIONARITY TESTS

ADF stat : -2.359
ADF p : 0.1536
ADF lags : 10
ADF : Non-stationary
KPSS stat : 2.157
KPSS p : 0.0100
KPSS : Non-stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	136	8.8%	91.2%	0.1774
Q2	131	12.2%	87.8%	2.3273
Q3	133	0.0%	100.0%	4.9384
Q4 (highest)	134	17.2%	82.8%	7.8593

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

5Y Treasury Yield

Block: Selected export / Treasury term structure | Source: FRED API | Col: raw_DGS5 | Transform: nan

Raw Series



After: nan



SUMMARY STATISTICS

N : 1287
Valid : 771
NaN : 516
Start : 1962-01
End : 2026-03
Mean : 5.4894
Std : 3.1451
Min : 0.2100
Max : 16.2700
Skew : 0.668
Kurt : 0.358

STATIONARITY TESTS

ADF stat : -1.726
ADF p : 0.4179
ADF lags : 2
ADF : Non-stationary
KPSS stat : 2.185
KPSS p : 0.0100
KPSS : Non-stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

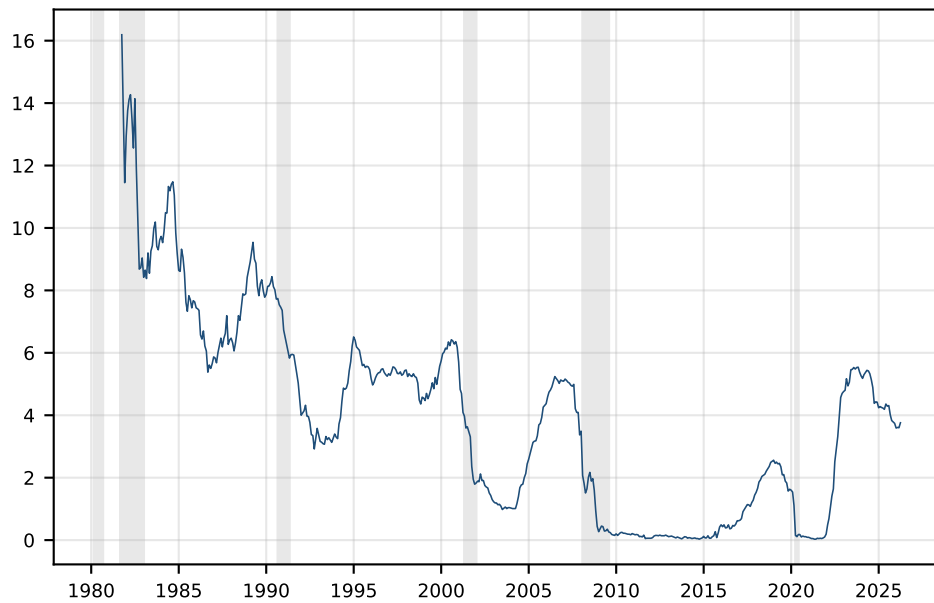
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	193	9.3%	90.7%	1.7642
Q2	193	5.2%	94.8%	4.2498
Q3	192	5.7%	94.3%	6.3021
Q4 (highest)	192	24.0%	76.0%	9.6754

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

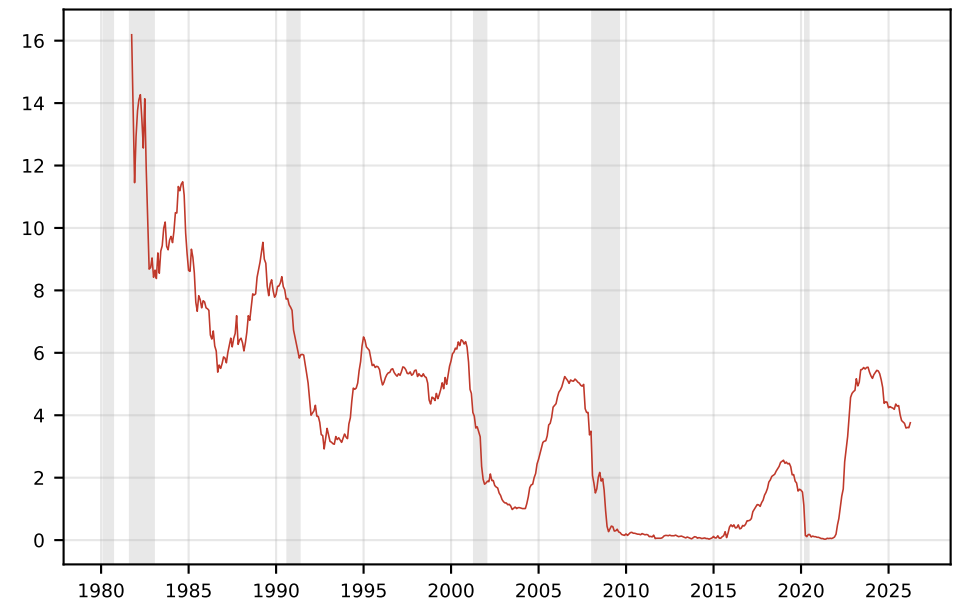
6M Treasury Yield

Block: Selected export / Treasury term structure | Source: FRED API | Col: raw_DGS6MO | Transform: nan

Raw Series



After: nan



SUMMARY STATISTICS

N : 1287
Valid : 535
NaN : 752
Start : 1981-09
End : 2026-03
Mean : 3.9823
Std : 3.2048
Min : 0.0300
Max : 16.1900
Skew : 0.679
Kurt : 0.248

STATIONARITY TESTS

ADF stat : -2.569
ADF p : 0.0995
ADF lags : 10
ADF : Non-stationary
KPSS stat : 2.205
KPSS p : 0.0100
KPSS : Non-stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	134	8.2%	91.8%	0.2499
Q2	135	12.6%	87.4%	2.4703
Q3	131	0.0%	100.0%	5.0515
Q4 (highest)	134	17.2%	82.8%	8.1943

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

7Y Treasury Yield

Block: Selected export / Treasury term structure | Source: FRED API | Col: raw_DGS7 | Transform: nan

Raw Series



After: nan



SUMMARY STATISTICS

N : 1287
Valid : 681
NaN : 606
Start : 1969-07
End : 2026-03
Mean : 5.8081
Std : 3.2087
Min : 0.3900
Max : 16.0500
Skew : 0.576
Kurt : -0.006

STATIONARITY TESTS

ADF stat : -1.522
ADF p : 0.5224
ADF lags : 2
ADF : Non-stationary
KPSS stat : 2.792
KPSS p : 0.0100
KPSS : Non-stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

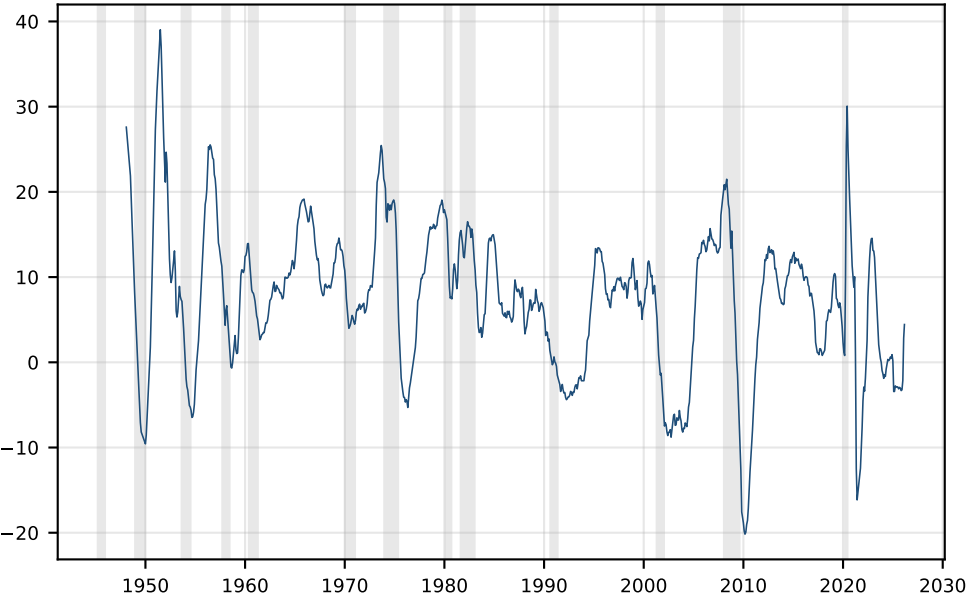
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	171	7.6%	92.4%	1.9833
Q2	169	8.9%	91.1%	4.4495
Q3	171	10.5%	89.5%	6.7906
Q4 (highest)	169	23.1%	76.9%	10.0526

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

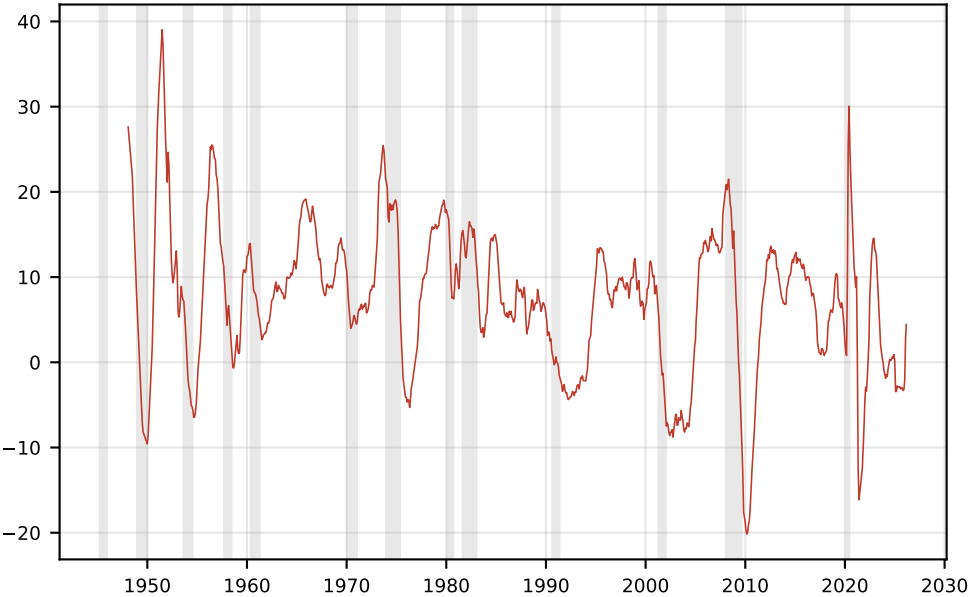
Commercial and Industrial Loans YoY

Block: Selected export / YoY growth variables | Source: Derived | Col: derived_busloans_yoy | Transform: nan

Raw Series



After: nan



SUMMARY STATISTICS

N : 1287
Valid : 938
NaN : 349
Start : 1948-01
End : 2026-02
Mean : 7.4519
Std : 8.6161
Min : -20.1688
Max : 39.0263
Skew : -0.130
Kurt : 0.818

STATIONARITY TESTS

ADF stat : -4.489
ADF p : 0.0002
ADF lags : 12
ADF : Stationary
KPSS stat : 0.645
KPSS p : 0.0186
KPSS : Non-stationary

QUARTILE-BUCKET ANALYSIS
(vs NBER recession/expansion)

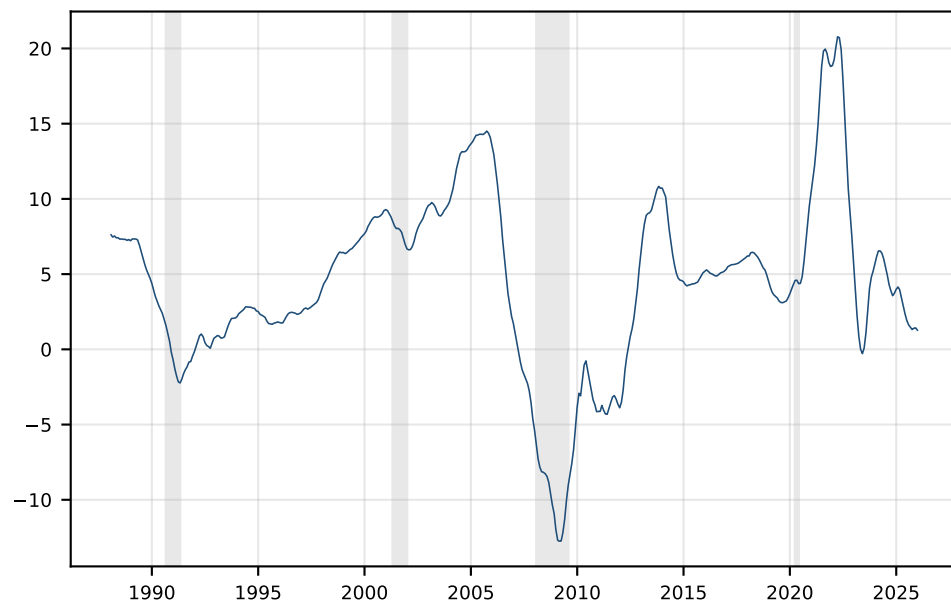
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	235	14.0%	86.0%	-3.9000
Q2	234	14.1%	85.9%	5.8545
Q3	234	6.0%	94.0%	10.1774
Q4 (highest)	235	18.7%	81.3%	17.6805

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

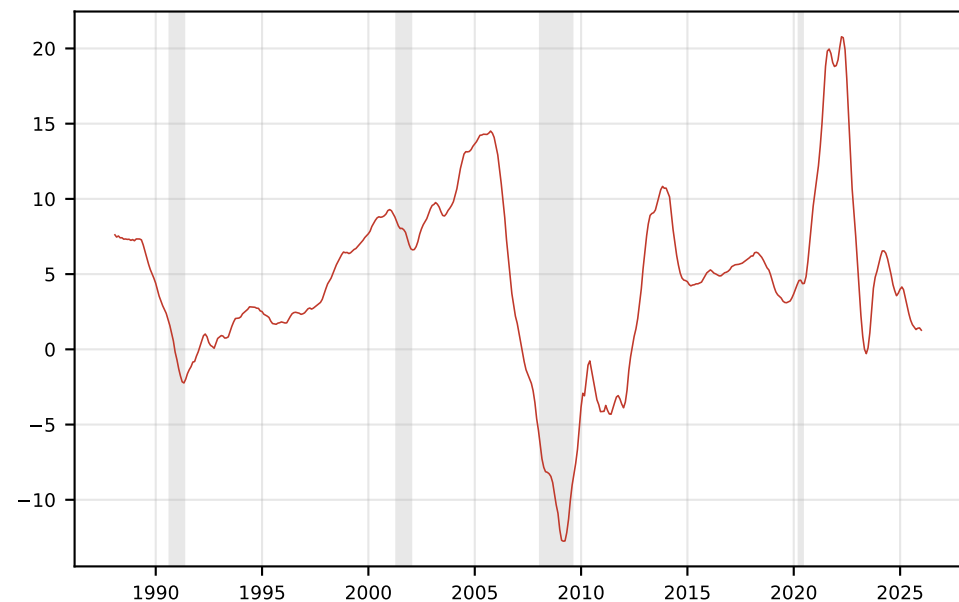
Case-Shiller National Home Price Index YoY

Block: Selected export / YoY growth variables | Source: Derived | Col: derived_csushpinsa_yoy | Transform: nan

Raw Series



After: nan



SUMMARY STATISTICS

N : 1287
Valid : 456
NaN : 831
Start : 1988-01
End : 2025-12
Mean : 4.4687
Std : 5.8272
Min : -12.7508
Max : 20.7783
Skew : -0.103
Kurt : 0.982

STATIONARITY TESTS

ADF stat : -2.663
ADF p : 0.0807
ADF lags : 10
ADF : Non-stationary
KPSS stat : 0.194
KPSS p : 0.1000
KPSS : Stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

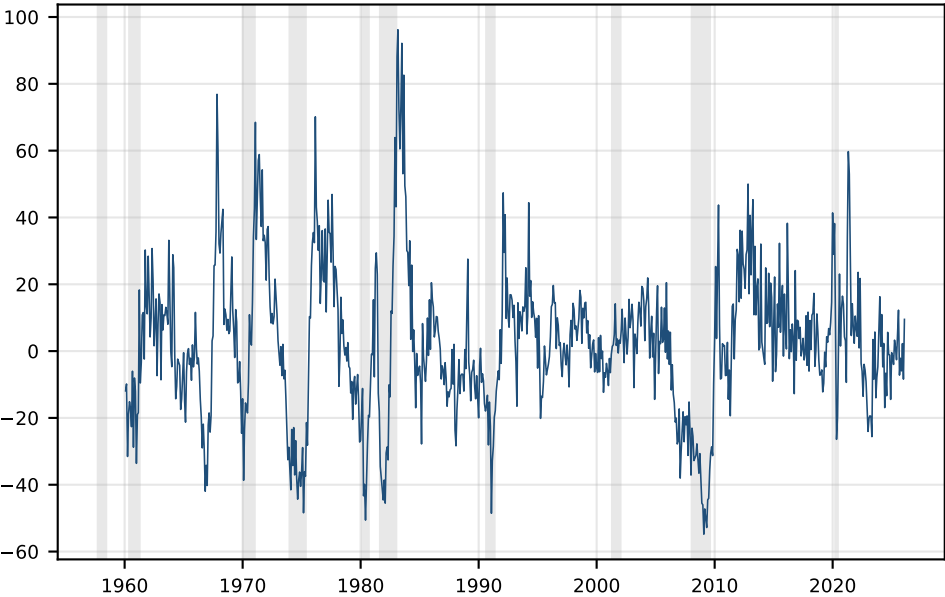
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	114	22.8%	77.2%	-2.7504
Q2	114	1.8%	98.2%	3.0586
Q3	114	1.8%	98.2%	6.0716
Q4 (highest)	114	5.3%	94.7%	11.4951

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

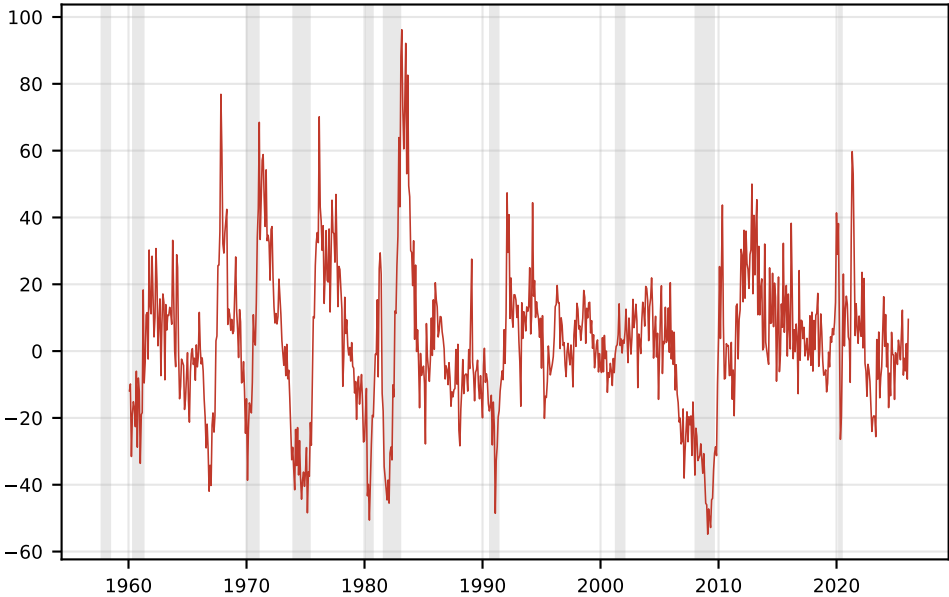
Housing Starts YoY

Block: Selected export / YoY growth variables | Source: Derived | Col: derived_houst_yoy | Transform: nan

Raw Series



After: nan



SUMMARY STATISTICS

N : 1287
Valid : 793
NaN : 494
Start : 1960-01
End : 2026-01
Mean : 2.1411
Std : 21.6544
Min : -54.7970
Max : 96.1894
Skew : 0.498
Kurt : 1.560

STATIONARITY TESTS

ADF stat : -5.589
ADF p : 0.0000
ADF lags : 12
ADF : Stationary
KPSS stat : 0.035
KPSS p : 0.1000
KPSS : Stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

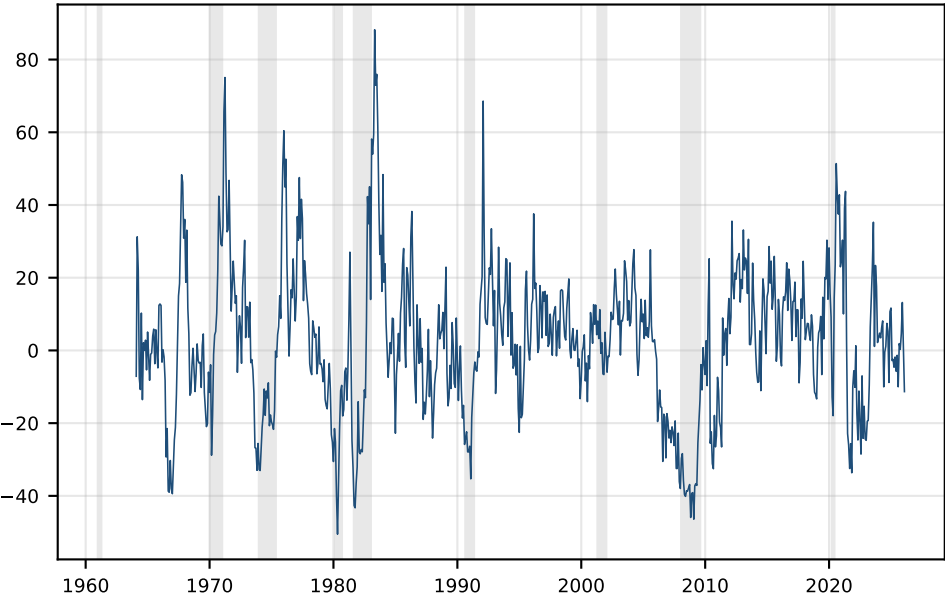
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	199	36.7%	63.3%	-23.7810
Q2	198	2.5%	97.5%	-3.7860
Q3	198	5.6%	94.4%	7.0584
Q4 (highest)	198	3.0%	97.0%	29.2039

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

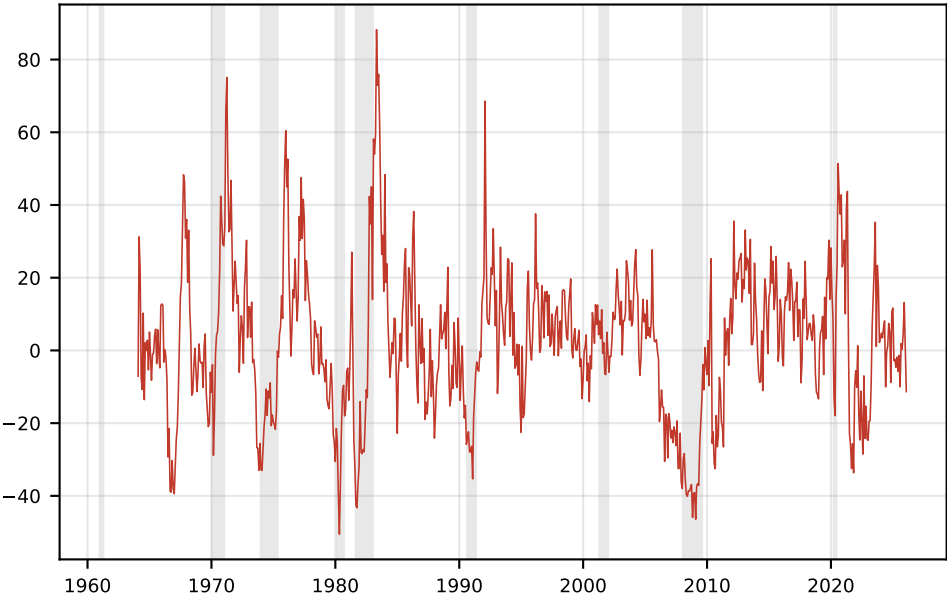
New One Family Houses Sold YoY

Block: Selected export / YoY growth variables | Source: Derived | Col: derived_hsn1f_yoy | Transform: nan

Raw Series



After: nan



SUMMARY STATISTICS

N : 1287
Valid : 745
NaN : 542
Start : 1964-01
End : 2026-01
Mean : 2.3100
Std : 20.1670
Min : -50.5348
Max : 88.2006
Skew : 0.370
Kurt : 0.984

STATIONARITY TESTS

ADF stat : -5.385
ADF p : 0.0000
ADF lags : 12
ADF : Stationary
KPSS stat : 0.050
KPSS p : 0.1000
KPSS : Stationary

QUARTILE-BUCKET ANALYSIS

(vs NBER recession/expansion)

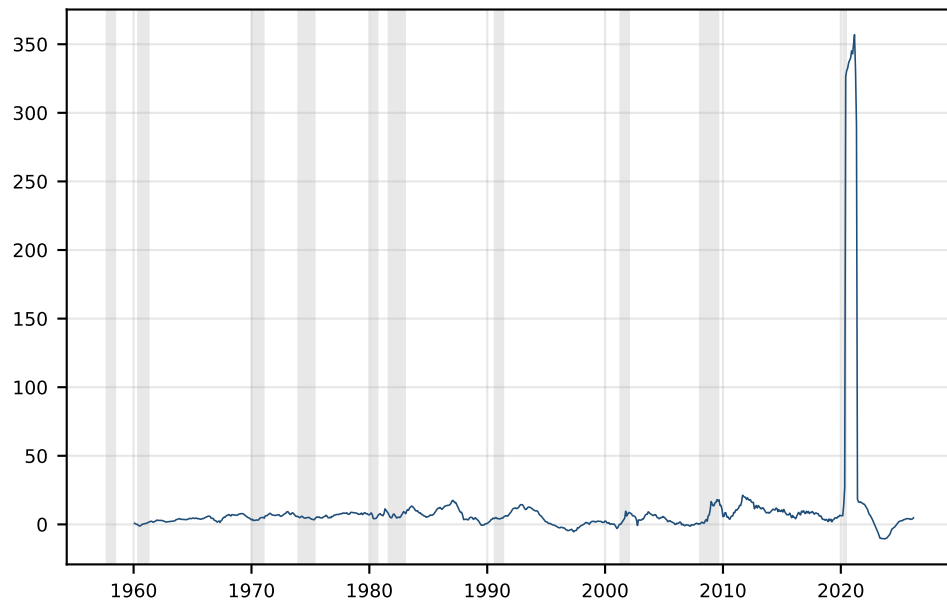
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	187	33.7%	66.3%	-22.7724
Q2	186	3.8%	96.2%	-3.1444
Q3	186	4.3%	95.7%	7.9091
Q4 (highest)	186	3.8%	96.2%	27.3824

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

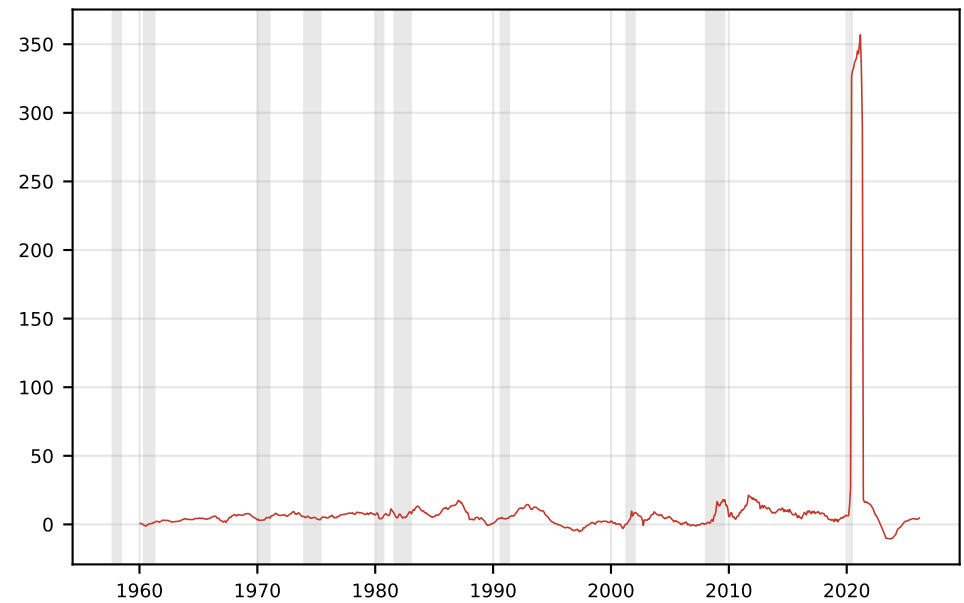
M1 YoY

Block: Selected export / YoY growth variables | Source: Derived | Col: derived_m1sl_yoy | Transform: nan

Raw Series



After: nan



SUMMARY STATISTICS

N : 1287
Valid : 794
NaN : 493
Start : 1960-01
End : 2026-02
Mean : 10.5106
Std : 40.5962
Min : -10.7720
Max : 356.9182
Skew : 7.804
Kurt : 60.178

STATIONARITY TESTS

ADF stat : -4.326
ADF p : 0.0004
ADF lags : 12
ADF : Stationary
KPSS stat : 0.326
KPSS p : 0.1000
KPSS : Stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

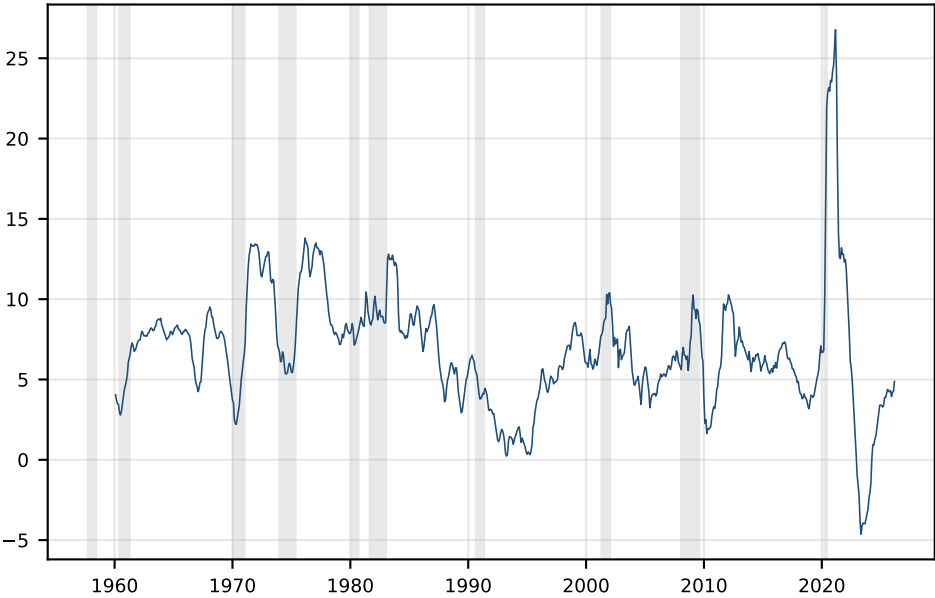
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	199	9.5%	90.5%	-0.4962
Q2	198	23.2%	76.8%	4.1030
Q3	198	8.1%	91.9%	6.8788
Q4 (highest)	199	7.0%	93.0%	31.5062

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

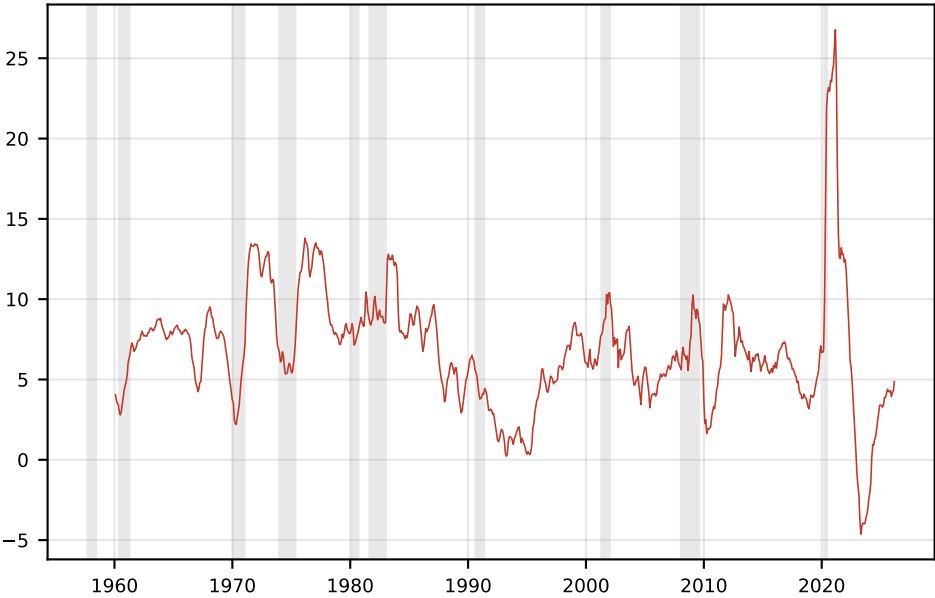
M2 YoY

Block: Selected export / YoY growth variables | Source: Derived | Col: derived_m2sl_yoy | Transform: nan

Raw Series



After: nan



SUMMARY STATISTICS

N	: 1287
Valid	: 794
NaN	: 493
Start	: 1960-01
End	: 2026-02
Mean	: 6.8187
Std	: 3.7987
Min	: -4.6304
Max	: 26.7753
Skew	: 1.067
Kurt	: 5.012

STATIONARITY TESTS

ADF stat	: -3.658
ADF p	: 0.0047
ADF lags	: 12
ADF	: Stationary
KPSS stat	: 0.381
KPSS p	: 0.0855
KPSS	: Stationary

QUARTILE-BUCKET ANALYSIS
(vs NBER recession/expansion)

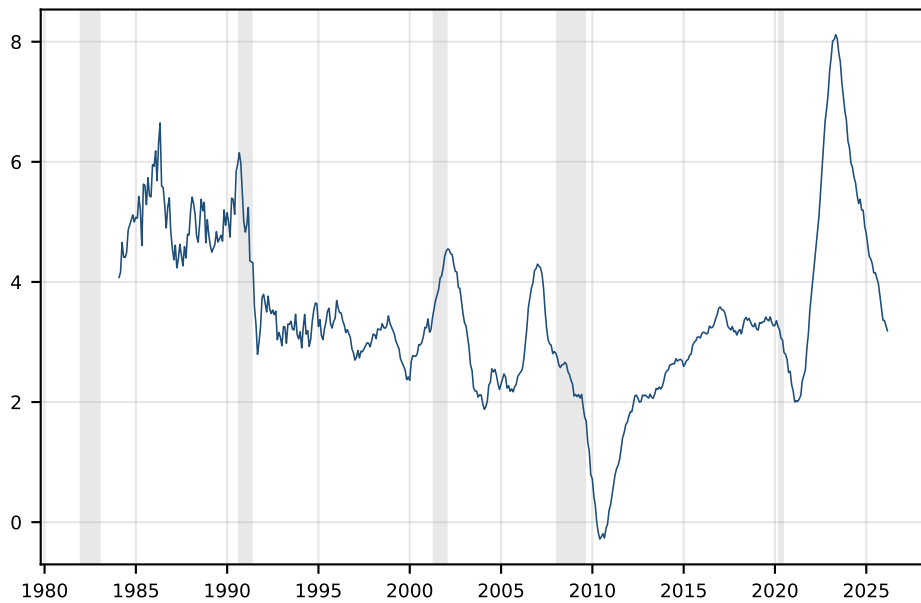
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	199	10.6%	89.4%	2.5962
Q2	198	15.2%	84.8%	5.7457
Q3	198	6.1%	93.9%	7.5874
Q4 (highest)	199	16.1%	83.9%	11.3439

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

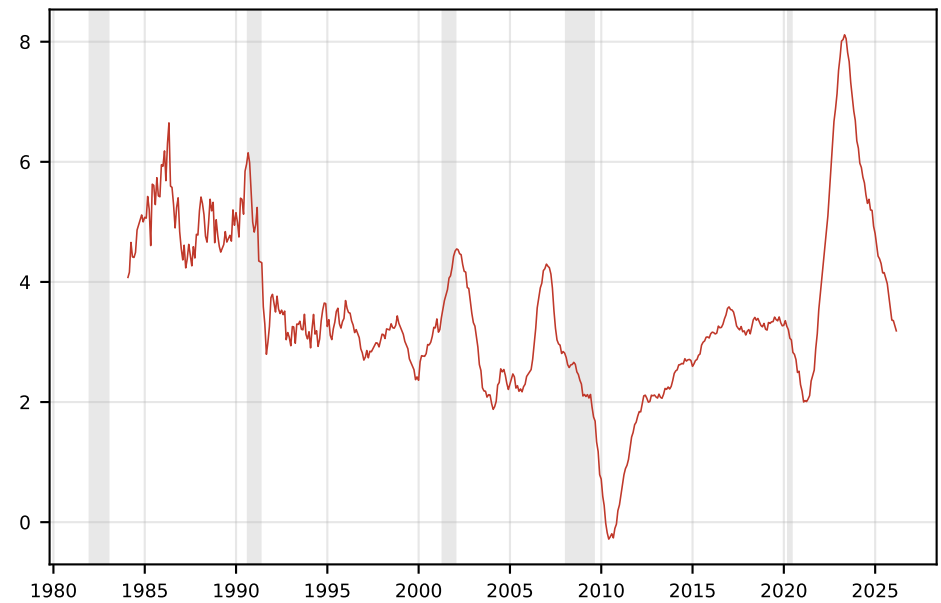
Owners' Equivalent Rent YoY

Block: Selected export / YoY growth variables | Source: Derived | Col: derived_oer_yoy | Transform: nan

Raw Series



After: nan



SUMMARY STATISTICS

N : 1287
Valid : 505
NaN : 782
Start : 1984-01
End : 2026-02
Mean : 3.4702
Std : 1.4138
Min : -0.2832
Max : 8.1162
Skew : 0.497
Kurt : 1.084

STATIONARITY TESTS

ADF stat : -2.141
ADF p : 0.2283
ADF lags : 12
ADF : Non-stationary
KPSS stat : 0.683
KPSS p : 0.0151
KPSS : Non-stationary

QUARTILE-BUCKET ANALYSIS

(vs NBER recession/expansion)

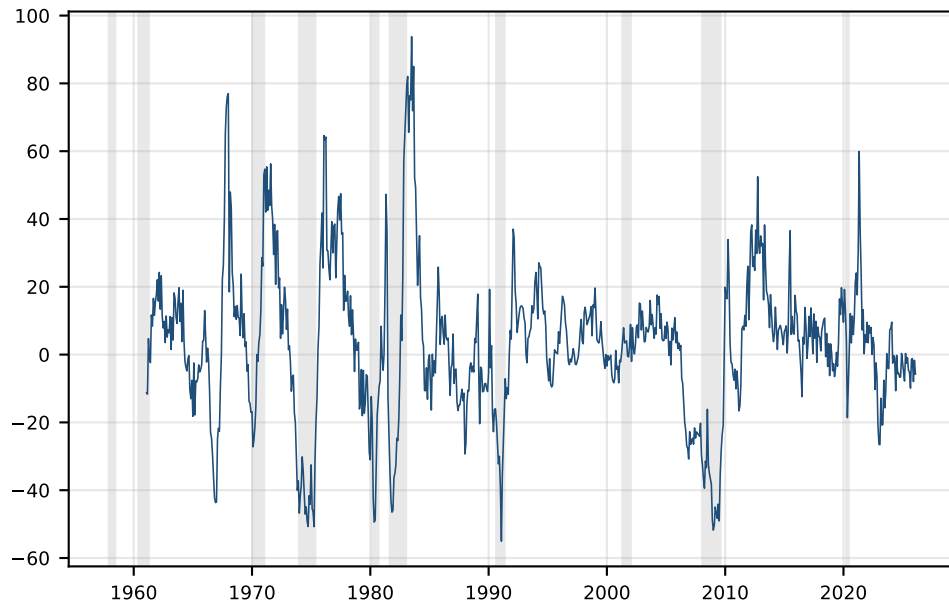
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	127	12.6%	87.4%	1.9031
Q2	126	3.2%	96.8%	2.9964
Q3	126	5.6%	94.4%	3.6047
Q4 (highest)	126	7.1%	92.9%	5.3891

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

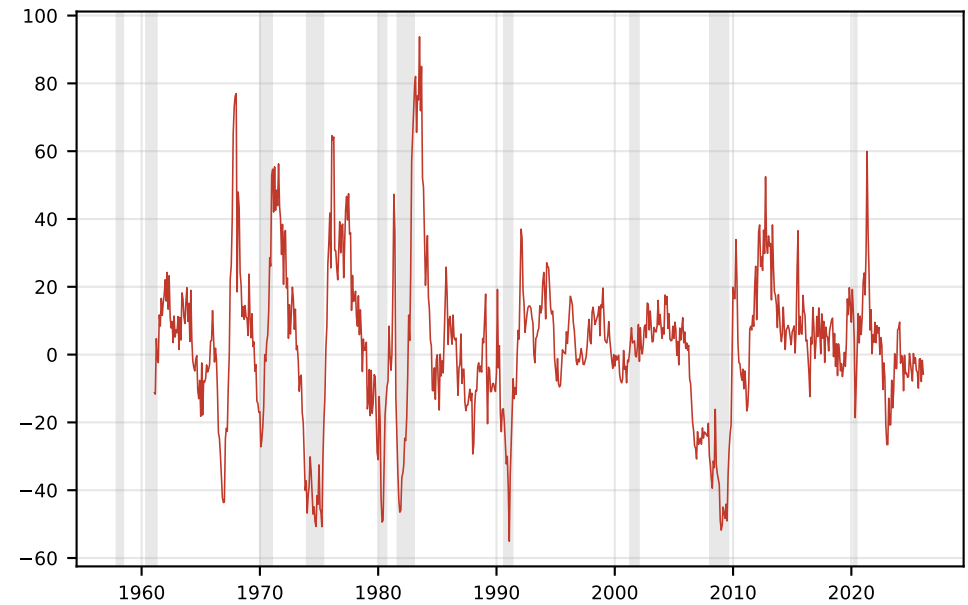
Building Permits YoY

Block: Selected export / YoY growth variables | Source: Derived | Col: derived_permit_yoy | Transform: nan

Raw Series



After: nan



SUMMARY STATISTICS

N : 1287
Valid : 781
NaN : 506
Start : 1961-01
End : 2026-01
Mean : 2.9982
Std : 22.2433
Min : -55.0343
Max : 93.7568
Skew : 0.407
Kurt : 1.679

STATIONARITY TESTS

ADF stat : -4.790
ADF p : 0.0001
ADF lags : 12
ADF : Stationary
KPSS stat : 0.066
KPSS p : 0.1000
KPSS : Stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

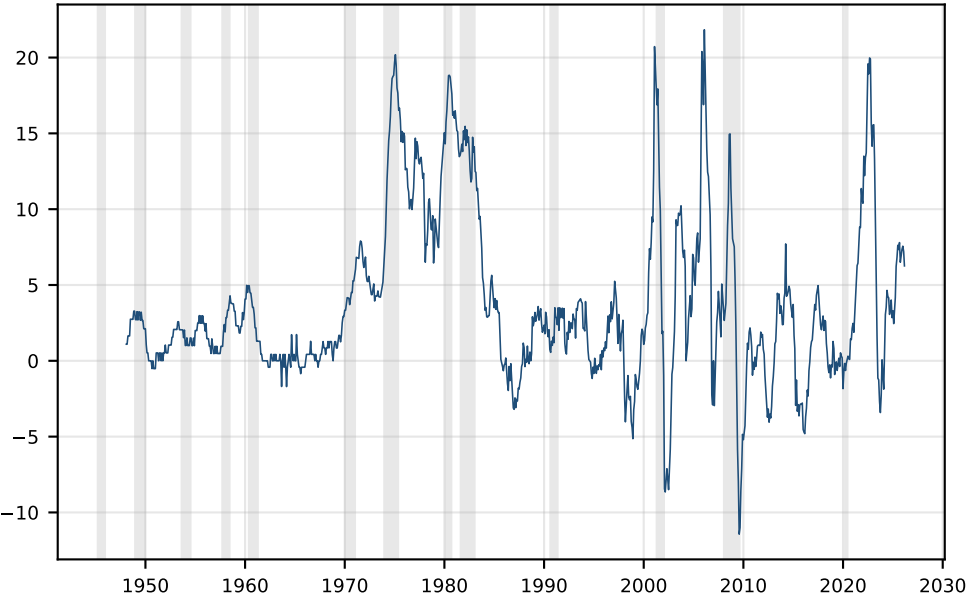
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	196	33.2%	66.8%	-23.8651
Q2	195	4.1%	95.9%	-1.6049
Q3	195	4.6%	95.4%	7.7688
Q4 (highest)	195	2.6%	97.4%	29.8318

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

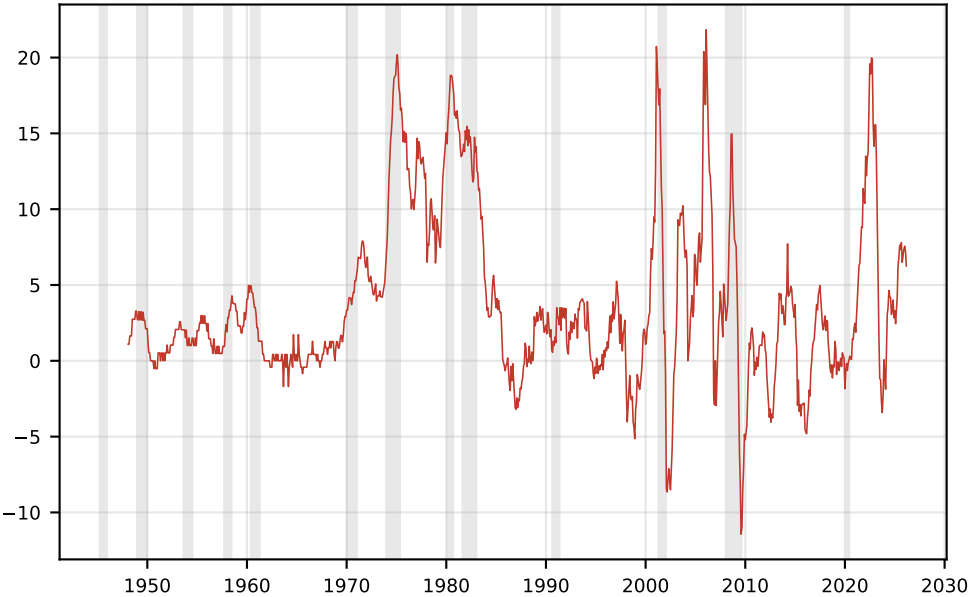
Rent of Primary Residence YoY

Block: Selected export / YoY growth variables | Source: Derived | Col: derived_rent_yoy | Transform: nan

Raw Series



After: nan



SUMMARY STATISTICS

N	: 1287
Valid	: 937
NaN	: 350
Start	: 1948-01
End	: 2026-02
Mean	: 3.7234
Std	: 5.4665
Min	: -11.4338
Max	: 21.8346
Skew	: 1.045
Kurt	: 0.959

STATIONARITY TESTS

ADF stat	: -3.164
ADF p	: 0.0221
ADF lags	: 12
ADF	: Stationary
KPSS stat	: 0.257
KPSS p	: 0.1000
KPSS	: Stationary

QUARTILE-BUCKET ANALYSIS
(vs NBER recession/expansion)

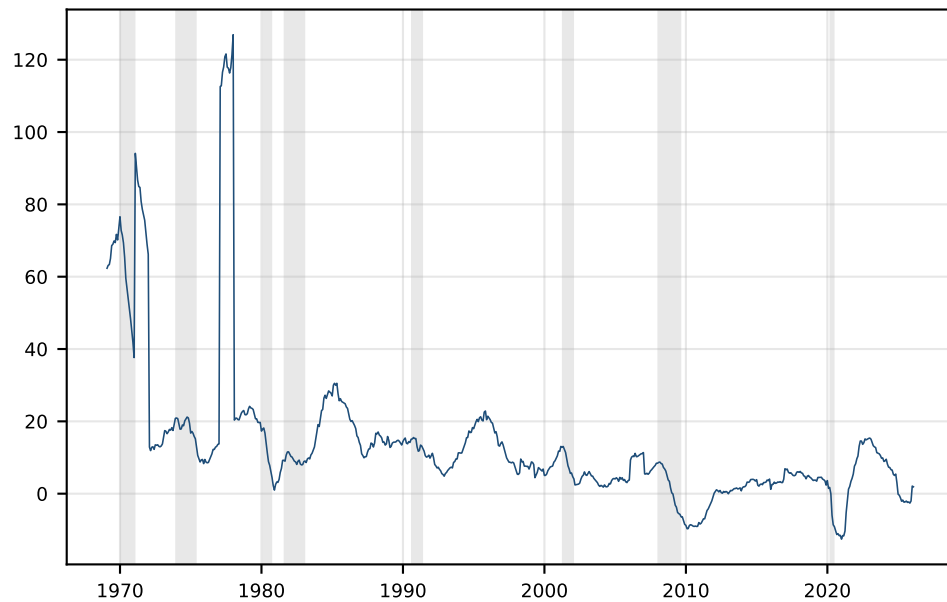
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	253	2.0%	98.0%	-1.3684
Q2	217	11.5%	88.5%	1.3307
Q3	233	16.3%	83.7%	3.5213
Q4 (highest)	234	23.9%	76.1%	11.6489

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

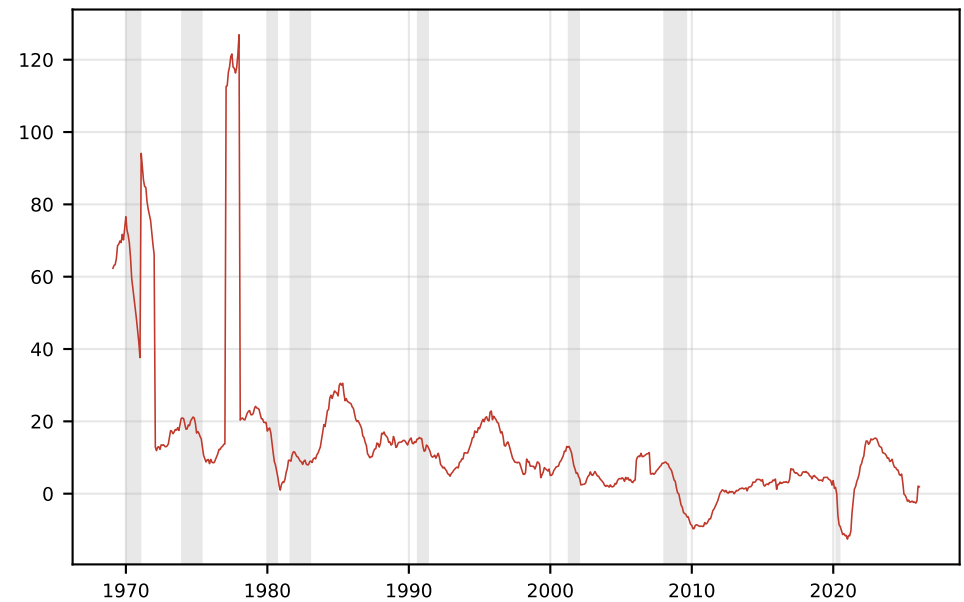
Revolving Consumer Credit YoY

Block: Selected export / YoY growth variables | Source: Derived | Col: derived_revolsl_yoy | Transform: nan

Raw Series



After: nan



SUMMARY STATISTICS

N : 1287
Valid : 685
NaN : 602
Start : 1969-01
End : 2026-01
Mean : 13.8748
Std : 21.0082
Min : -12.5812
Max : 126.9071
Skew : 3.124
Kurt : 11.175

STATIONARITY TESTS

ADF stat : -3.761
ADF p : 0.0033
ADF lags : 12
ADF : Stationary
KPSS stat : 1.906
KPSS p : 0.0100
KPSS : Non-stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

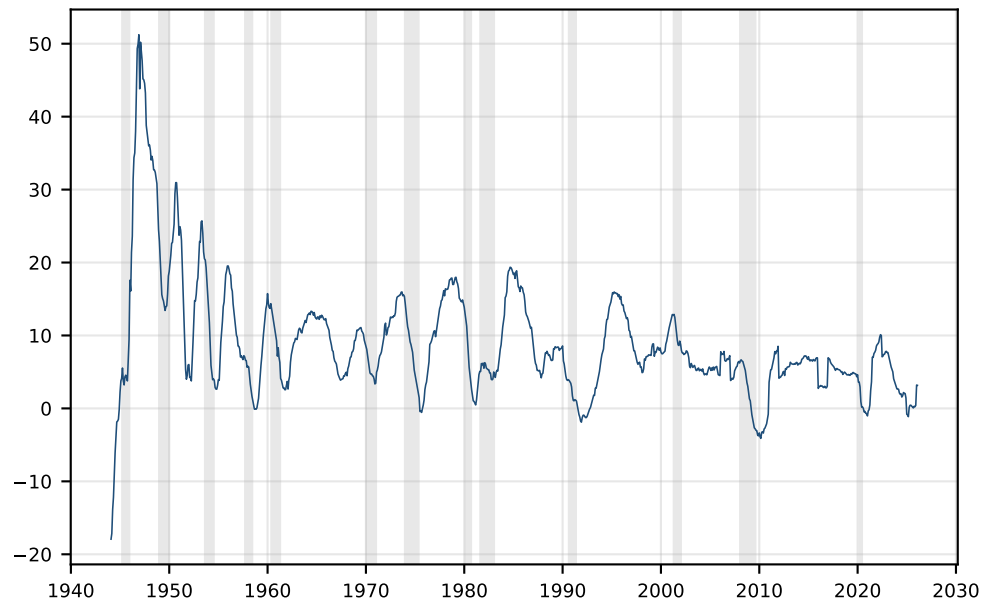
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	172	7.0%	93.0%	-0.6707
Q2	171	12.9%	87.1%	6.4852
Q3	171	12.3%	87.7%	12.2268
Q4 (highest)	171	17.5%	82.5%	37.5429

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

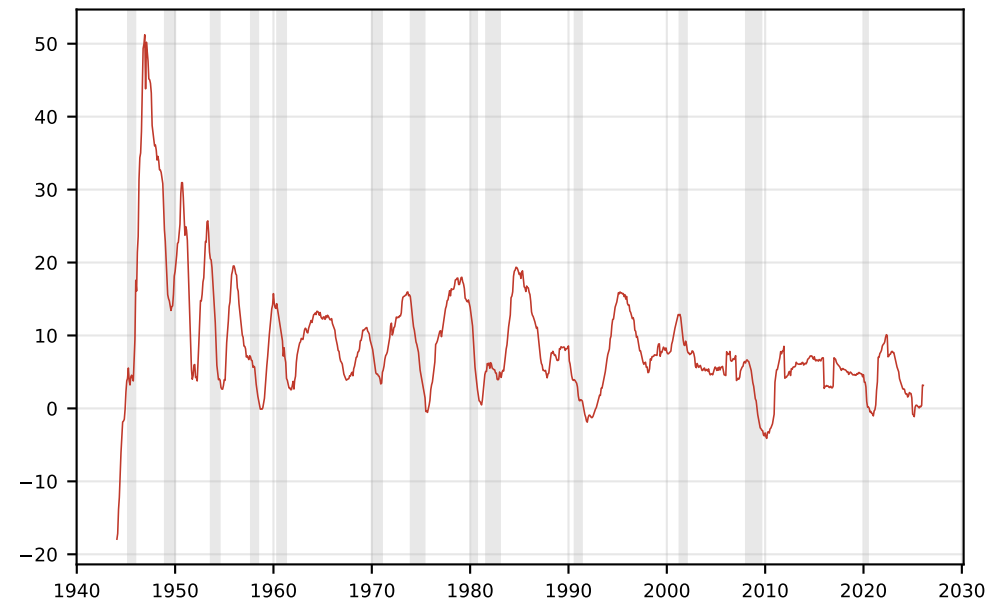
Total Consumer Credit YoY

Block: Selected export / YoY growth variables | Source: Derived | Col: derived_totalsl_yoy | Transform: nan

Raw Series



After: nan



SUMMARY STATISTICS

N : 1287
Valid : 985
NaN : 302
Start : 1944-01
End : 2026-01
Mean : 8.8544
Std : 8.0826
Min : -17.9278
Max : 51.2297
Skew : 1.899
Kurt : 6.512

STATIONARITY TESTS

ADF stat : -3.270
ADF p : 0.0163
ADF lags : 12
ADF : Stationary
KPSS stat : 1.287
KPSS p : 0.0100
KPSS : Non-stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

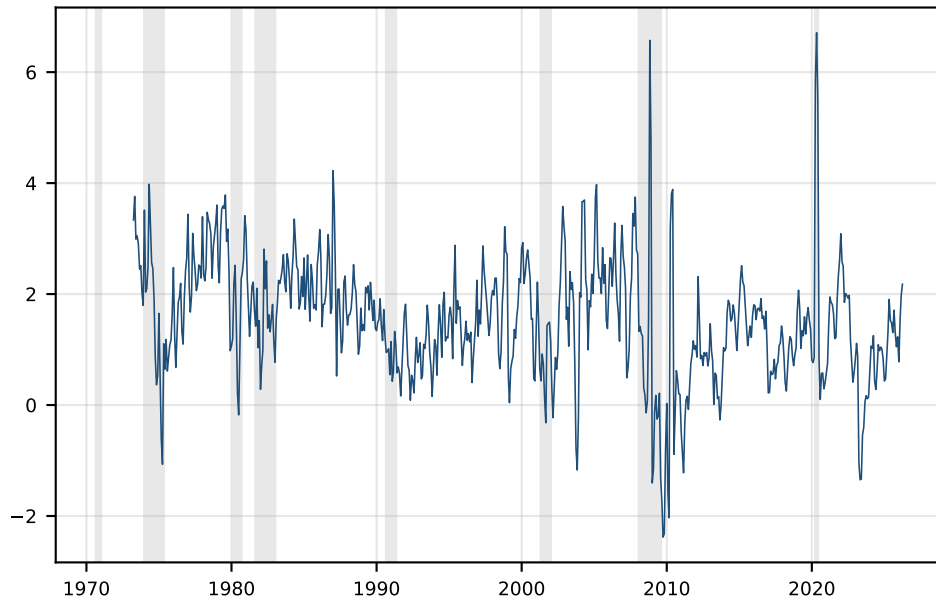
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	247	16.6%	83.4%	1.4220
Q2	246	16.3%	83.7%	5.7505
Q3	246	11.4%	88.6%	9.0709
Q4 (highest)	246	9.3%	90.7%	19.2043

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

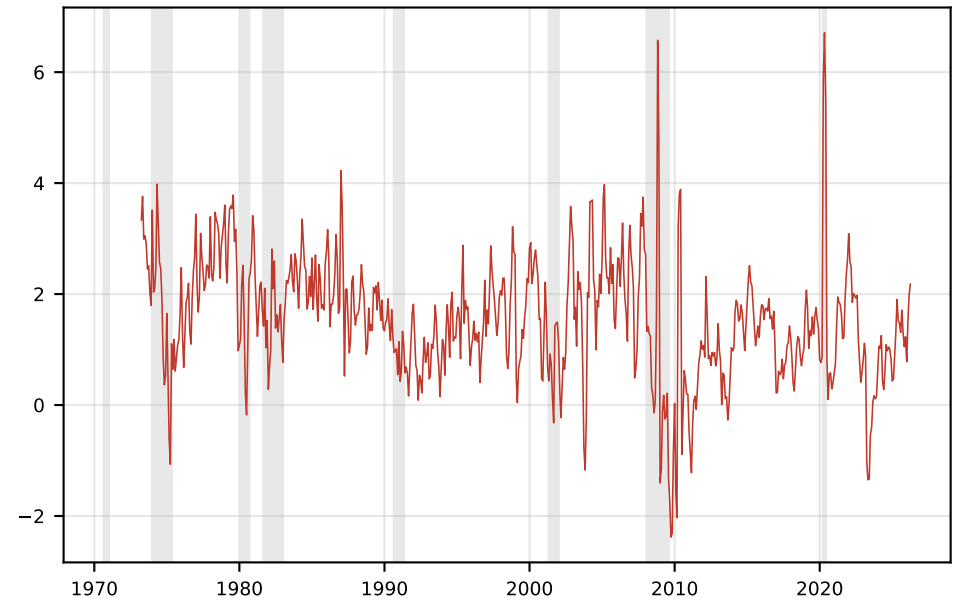
Total Bank Credit YoY

Block: Selected export / YoY growth variables | Source: Derived | Col: derived_totbkcr_yoy | Transform: nan

Raw Series



After: nan



SUMMARY STATISTICS

N : 1287
Valid : 637
NaN : 650
Start : 1973-03
End : 2026-03
Mean : 1.5368
Std : 1.0993
Min : -2.3816
Max : 6.7101
Skew : 0.190
Kurt : 2.157

STATIONARITY TESTS

ADF stat : -6.303
ADF p : 0.0000
ADF lags : 5
ADF : Stationary
KPSS stat : 1.067
KPSS p : 0.0100
KPSS : Non-stationary

QUARTILE-BUCKET ANALYSIS

(vs NBER recession/expansion)

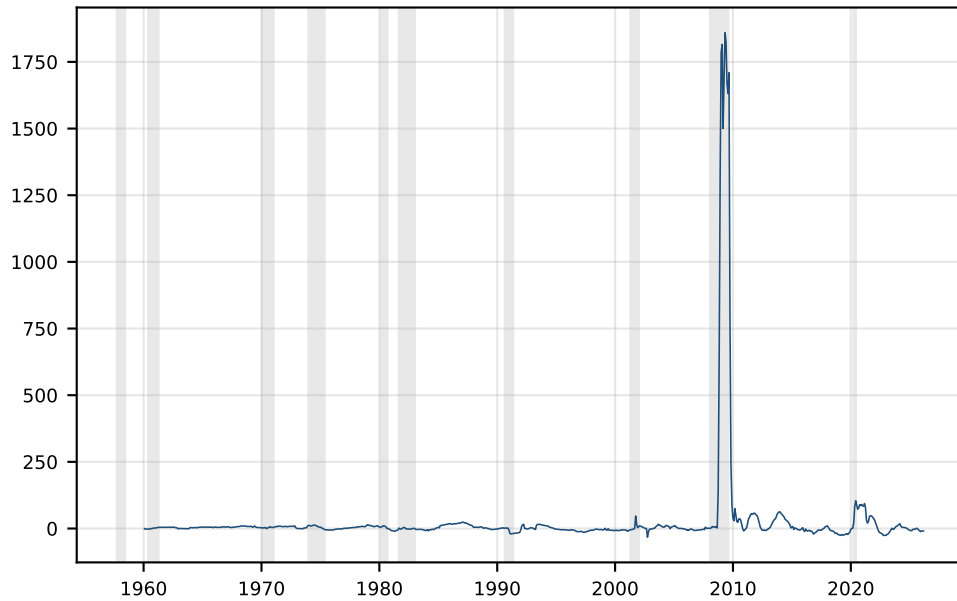
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	159	17.6%	82.4%	0.2163
Q2	159	13.2%	86.8%	1.1962
Q3	159	7.5%	92.5%	1.8462
Q4 (highest)	159	8.2%	91.8%	2.8843

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

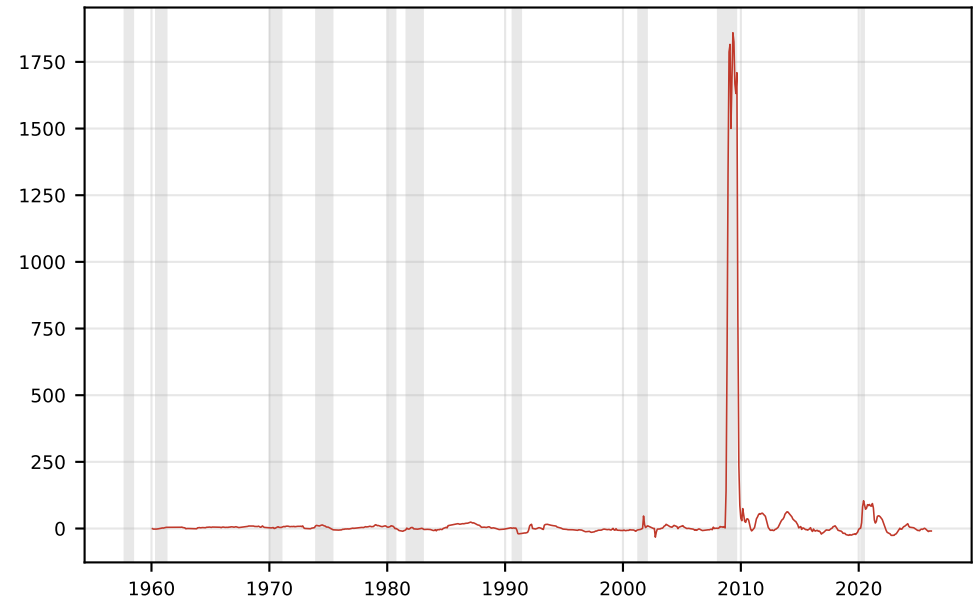
Total Reserves YoY

Block: Selected export / YoY growth variables | Source: Derived | Col: derived_totresns_yoy | Transform: nan

Raw Series



After: nan



SUMMARY STATISTICS

N : 1287
Valid : 794
NaN : 493
Start : 1960-01
End : 2026-02
Mean : 27.9753
Std : 191.9762
Min : -32.4653
Max : 1859.3333
Skew : 8.390
Kurt : 70.669

STATIONARITY TESTS

ADF stat : -4.732
ADF p : 0.0001
ADF lags : 11
ADF : Stationary
KPSS stat : 0.201
KPSS p : 0.1000
KPSS : Stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

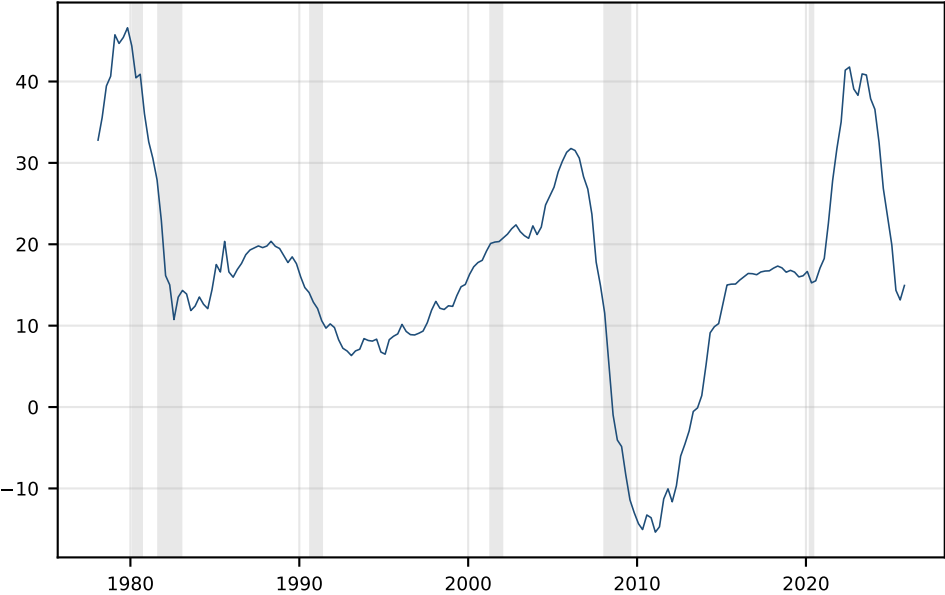
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	199	3.0%	97.0%	-9.7804
Q2	198	17.7%	82.3%	-0.7732
Q3	198	13.6%	86.4%	4.9584
Q4 (highest)	199	13.6%	86.4%	117.2361

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

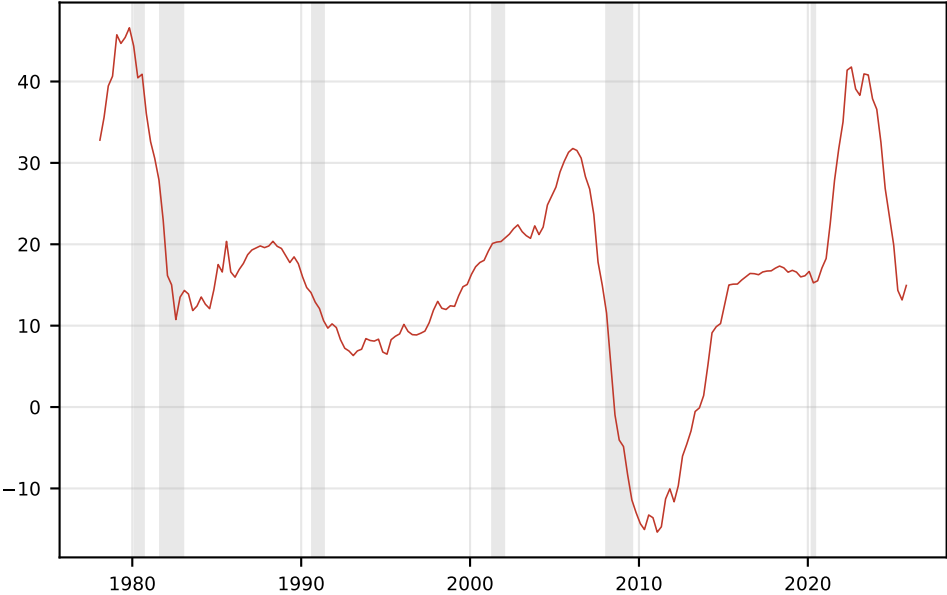
FHFA US House Price Index YoY

Block: Selected export / YoY growth variables | Source: Derived | Col: derived_ussthipi_yoy | Transform: nan

Raw Series



After: nan



SUMMARY STATISTICS

N : 1287
Valid : 192
NaN : 1095
Start : 1978-01
End : 2025-10
Mean : 16.3267
Std : 13.0375
Min : -15.3747
Max : 46.6060
Skew : -0.102
Kurt : 0.511

STATIONARITY TESTS

ADF stat : -1.713
ADF p : 0.4245
ADF lags : 12
ADF : Non-stationary
KPSS stat : 0.212
KPSS p : 0.1000
KPSS : Stationary

QUARTILE-BUCKET ANALYSIS

(vs NBER recession/expansion)

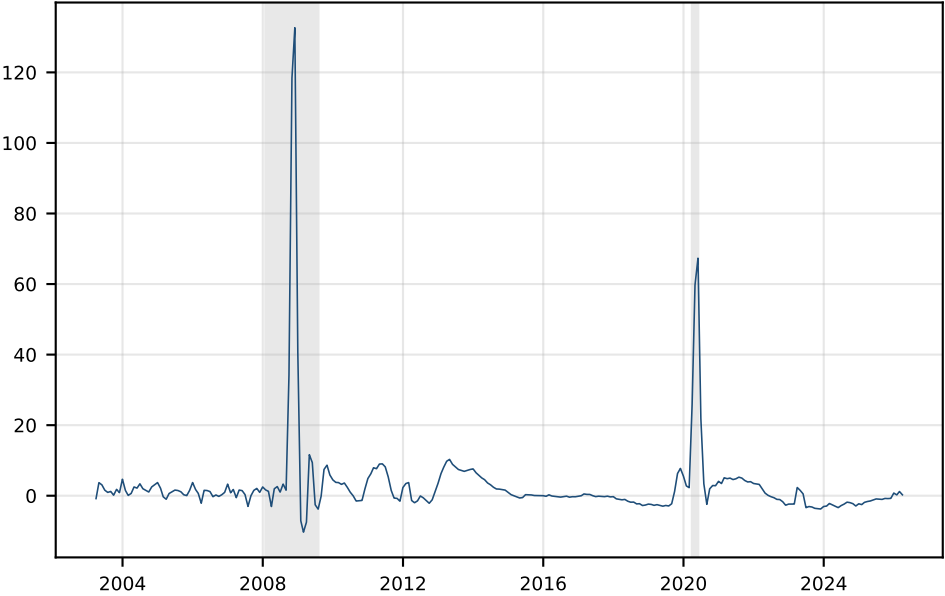
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	48	10.4%	89.6%	0.5223
Q2	48	16.7%	83.3%	13.5992
Q3	48	6.2%	93.8%	18.3868
Q4 (highest)	48	6.2%	93.8%	32.7985

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

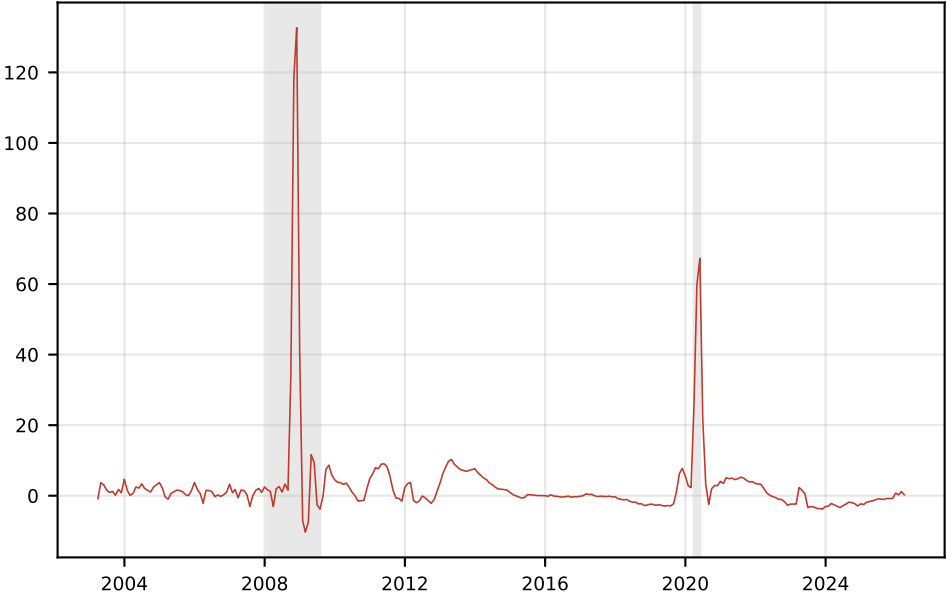
Fed Total Assets YoY

Block: Selected export / YoY growth variables | Source: Derived | Col: derived_walcl_yoy | Transform: nan

Raw Series



After: nan



SUMMARY STATISTICS

N : 1287
Valid : 277
NaN : 1010
Start : 2003-03
End : 2026-03
Mean : 2.8475
Std : 12.7169
Min : -10.3330
Max : 132.6551
Skew : 7.584
Kurt : 66.820

STATIONARITY TESTS

ADF stat : -5.857
ADF p : 0.0000
ADF lags : 5
ADF : Stationary
KPSS stat : 0.186
KPSS p : 0.1000
KPSS : Stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

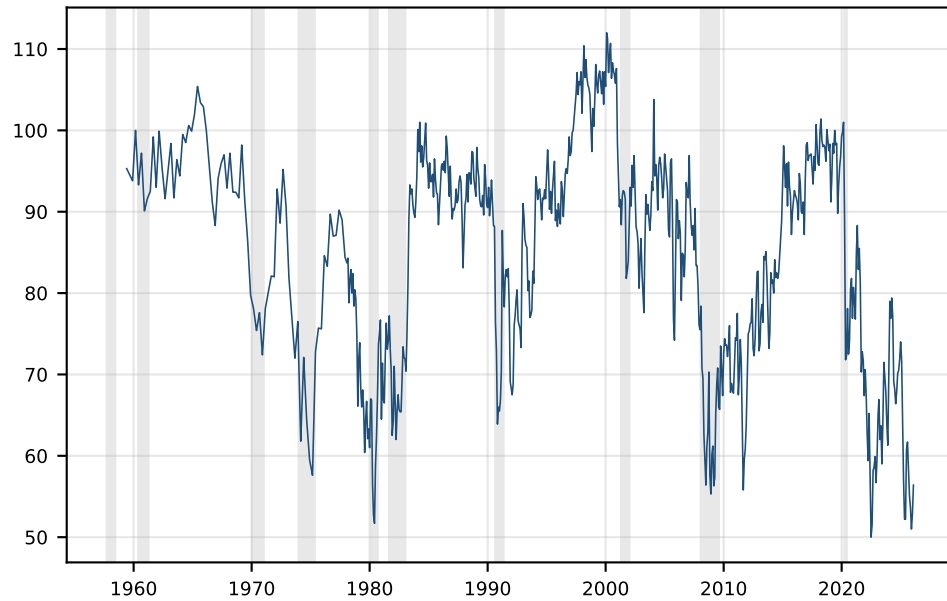
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	69	7.2%	92.8%	-2.5510
Q2	69	0.0%	100.0%	-0.1838
Q3	69	8.7%	91.3%	1.7613
Q4 (highest)	69	13.0%	87.0%	12.4013

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

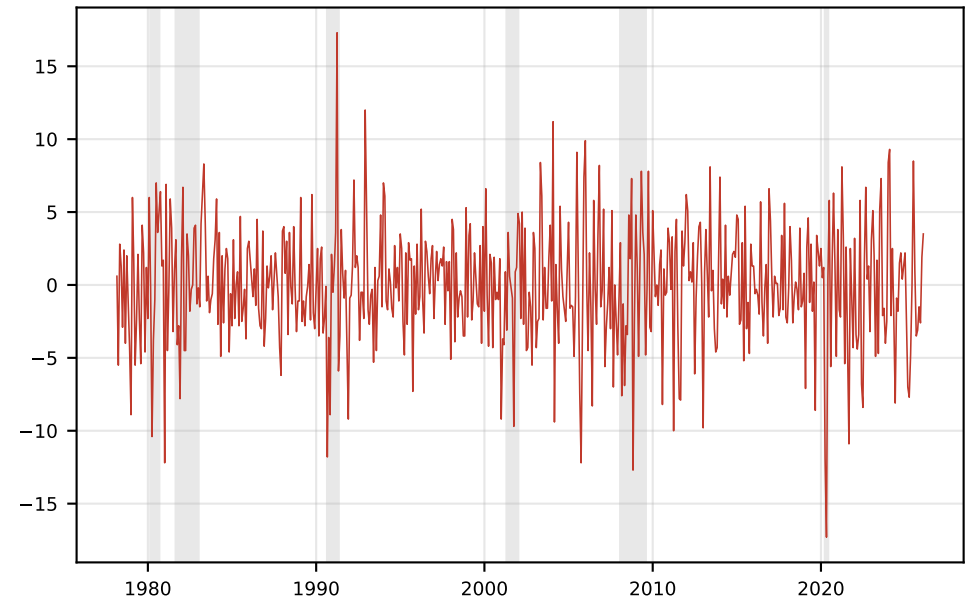
UMich Consumer Sentiment

Block: Sentiment / Surveys / Consumer | Source: FRED-MD | Col: UMCSENTx | Transform: 2: First diff

Raw Series



After: 2: First diff



SUMMARY STATISTICS

N : 1287
Valid : 651
NaN : 636
Start : 1959-05
End : 2026-01
Mean : 84.5800
Std : 13.5012
Min : 50.0000
Max : 112.0000
Skew : -0.440
Kurt : -0.645

STATIONARITY TESTS

ADF stat : -14.020
ADF p : 0.0000
ADF lags : 4
ADF : Stationary
KPSS stat : 0.084
KPSS p : 0.1000
KPSS : Stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

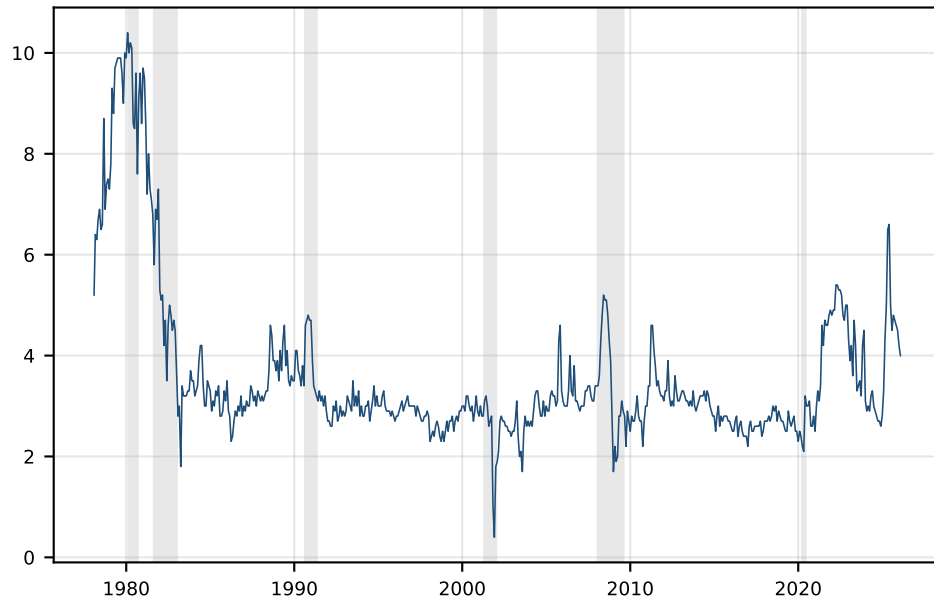
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	146	13.7%	86.3%	-5.0233
Q2	143	6.3%	93.7%	-1.2322
Q3	150	8.7%	91.3%	1.2993
Q4 (highest)	137	11.7%	88.3%	5.0175

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

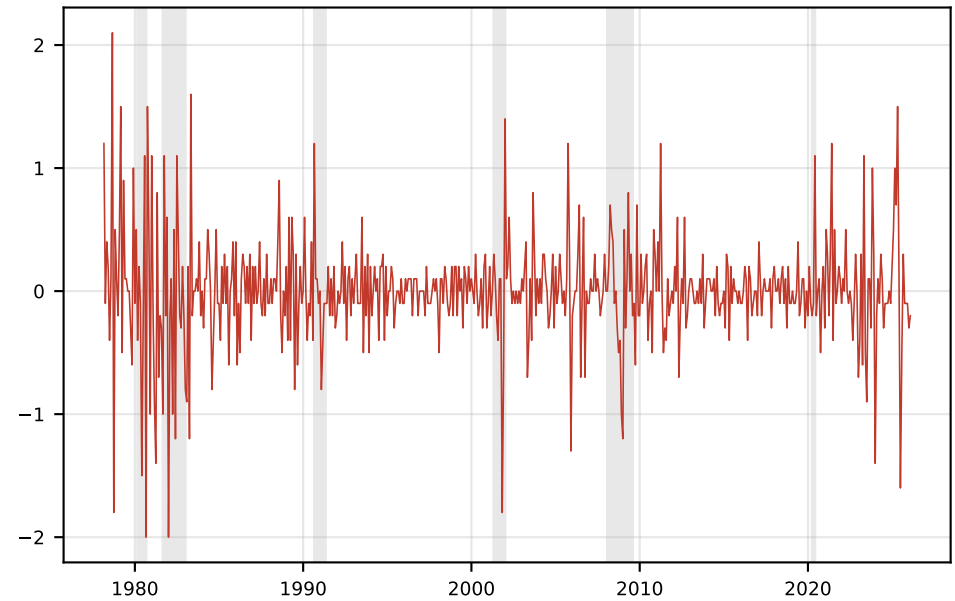
University of Michigan: Inflation Expectation

Block: Inflation / Prices / Inflation Expectations | Source: FRED API | Col: MICH | Transform: 2: First diff

Raw Series



After: 2: First diff



SUMMARY STATISTICS

```
N          : 1287
Valid       : 577
NaN         : MICH    710
MICH       709
dtype: int64
Start      : 1978-01
End        : 2026-01
Mean       : 3.6069
Std        : 1.6145
Min        : 0.4000
Max        : 10.4000
Skew       : 2.460
Kurt       : 6.095
```

STATIONARITY TESTS

```
ADF stat    : -6.202
ADF p       : 0.0000
ADF lags    : 11
ADF         : Stationary
KPSS stat   : 0.055
KPSS p      : 0.1000
KPSS        : Stationary
```

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

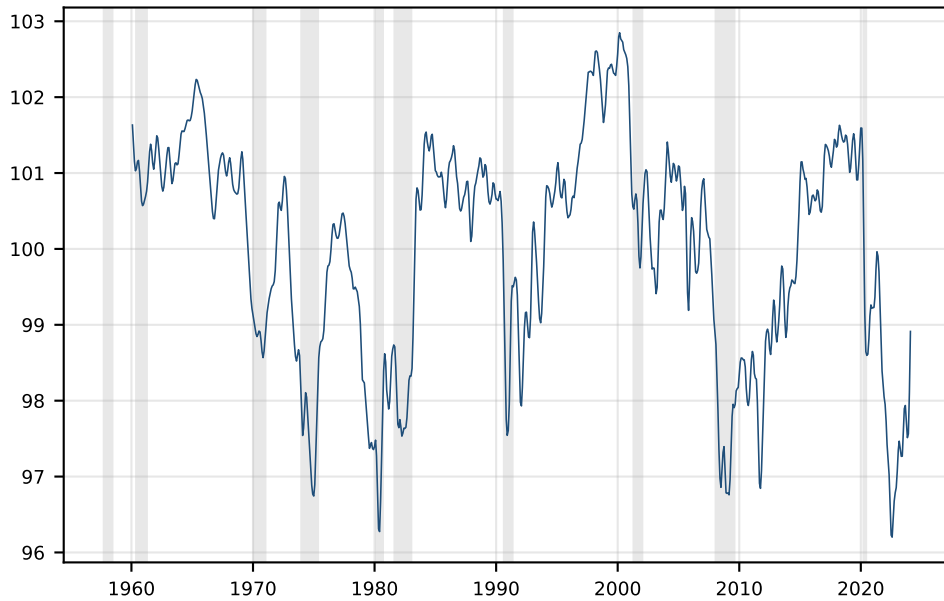
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	148	16.2%	83.8%	-0.4730
Q2	193	5.2%	94.8%	-0.0539
Q3	98	8.2%	91.8%	0.1153
Q4 (highest)	137	11.7%	88.3%	0.4956

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

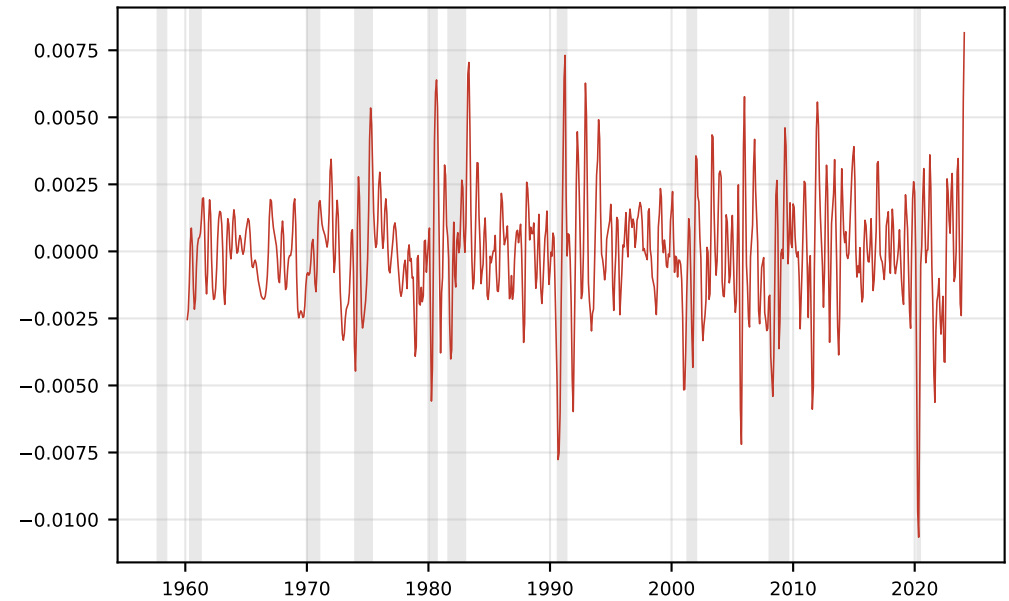
Conf Board Consumer Confidence (OECD proxy)

Block: Sentiment / Surveys / Consumer | Source: FRED API | Col: CSCICP03USM665S | Transform: 5: Log diff

Raw Series



After: 5: Log diff



SUMMARY STATISTICS

N : 1287
Valid : 769
NaN : 518
Start : 1960-01
End : 2024-01
Mean : 100.0000
Std : 1.4645
Min : 96.2013
Max : 102.8489
Skew : -0.512
Kurt : -0.573

STATIONARITY TESTS

ADF stat : -10.714
ADF p : 0.0000
ADF lags : 4
ADF : Stationary
KPSS stat : 0.029
KPSS p : 0.1000
KPSS : Stationary

QUARTILE-BUCKET ANALYSIS

(vs NBER recession/expansion)

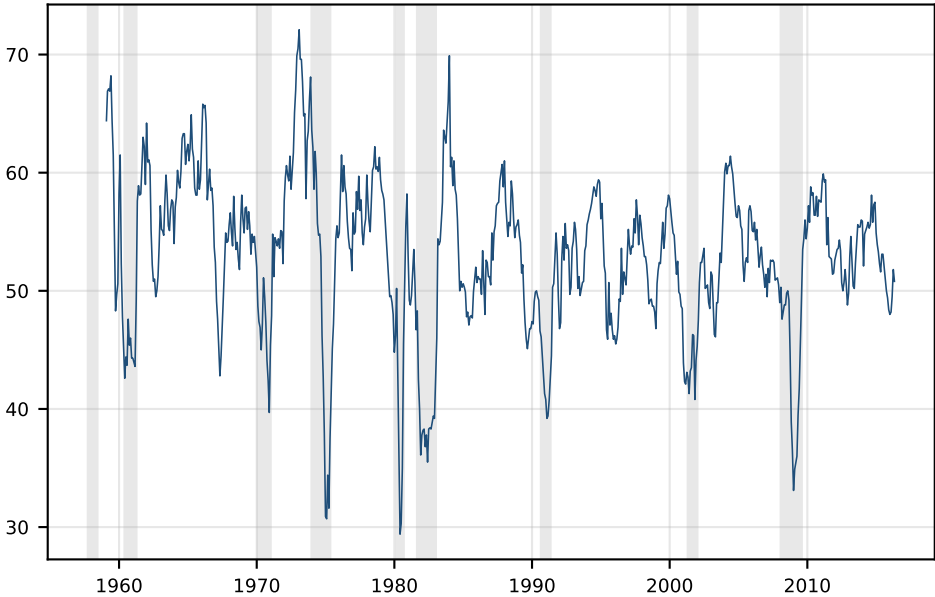
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	192	17.7%	82.3%	-0.0026
Q2	192	10.9%	89.1%	-0.0006
Q3	192	9.9%	90.1%	0.0005
Q4 (highest)	192	10.9%	89.1%	0.0025

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

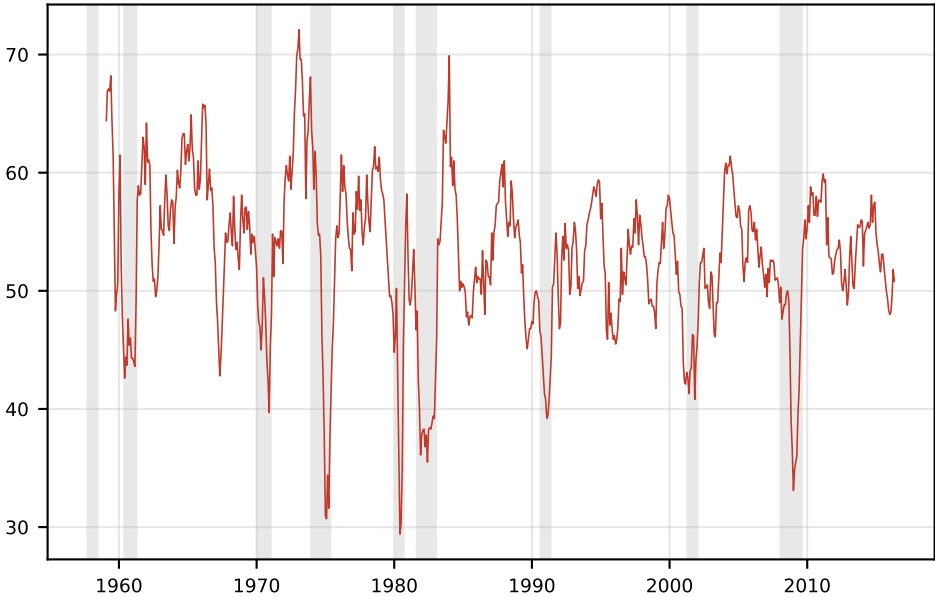
ISM Manufacturing PMI Composite

Block: Sentiment / Surveys / Business | Source: FRED-MD (hist) | Col: NAPM | Transform: 1: No transform

Raw Series



After: 1: No transform



SUMMARY STATISTICS

N : 1287
Valid : 688
NaN : 599
Start : 1959-01
End : 2016-04
Mean : 52.9850
Std : 6.6120
Min : 29.4000
Max : 72.1000
Skew : -0.481
Kurt : 0.935

STATIONARITY TESTS

ADF stat : -6.301
ADF p : 0.0000
ADF lags : 12
ADF : Stationary
KPSS stat : 0.350
KPSS p : 0.0987
KPSS : Stationary

QUARTILE-BUCKET ANALYSIS

(vs NBER recession/expansion)

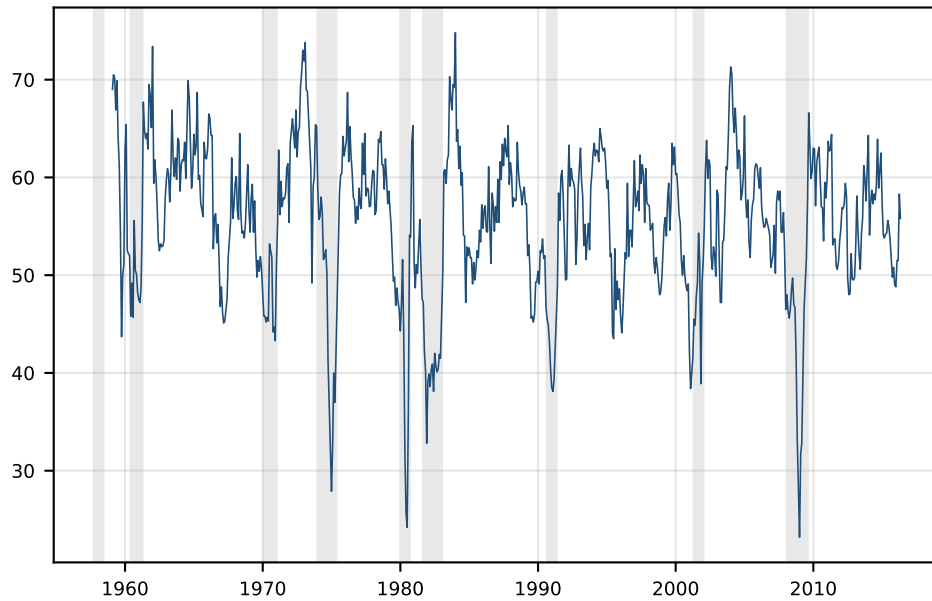
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	175	45.1%	54.9%	44.5171
Q2	176	3.4%	96.6%	51.6517
Q3	165	1.8%	98.2%	55.3188
Q4 (highest)	172	2.9%	97.1%	60.7262

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

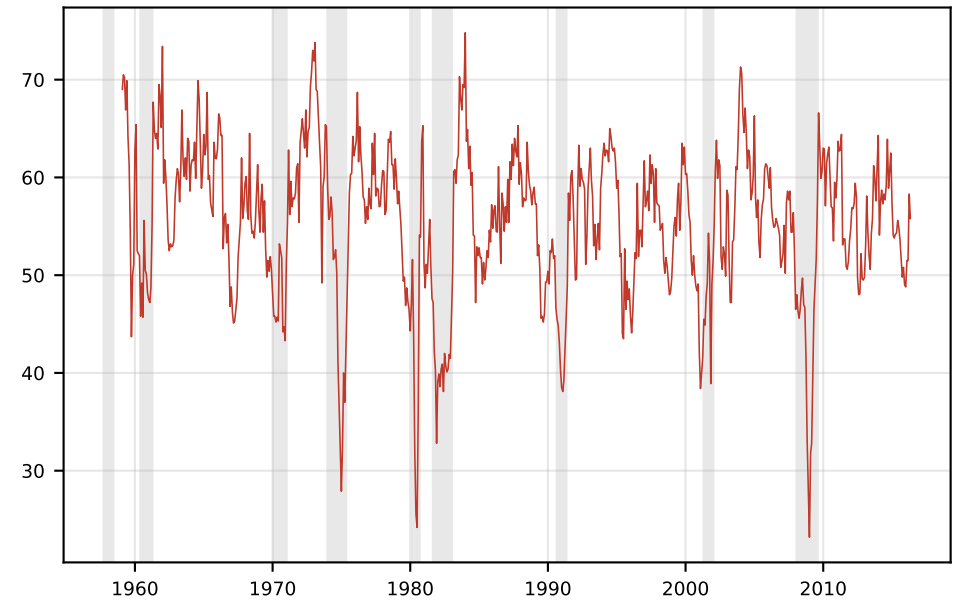
ISM Mfg New Orders

Block: Sentiment / Surveys / Business | Source: FRED-MD (hist) | Col: NAPMNOI | Transform: 1: No transform

Raw Series



After: 1: No transform



SUMMARY STATISTICS

N : 1287
Valid : 688
NaN : 599
Start : 1959-01
End : 2016-04
Mean : 55.4084
Std : 7.7683
Min : 23.2000
Max : 74.8000
Skew : -0.719
Kurt : 1.232

STATIONARITY TESTS

ADF stat : -6.862
ADF p : 0.0000
ADF lags : 10
ADF : Stationary
KPSS stat : 0.189
KPSS p : 0.1000
KPSS : Stationary

QUARTILE-BUCKET ANALYSIS

(vs NBER recession/expansion)

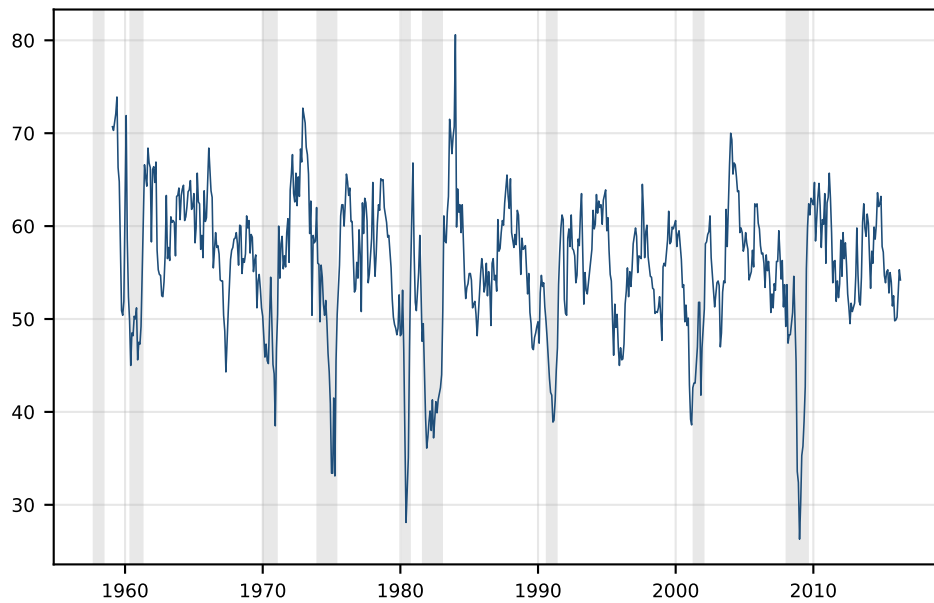
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	173	44.5%	55.5%	45.2139
Q2	171	7.6%	92.4%	53.6468
Q3	173	1.7%	98.3%	58.5694
Q4 (highest)	171	0.0%	100.0%	64.2860

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

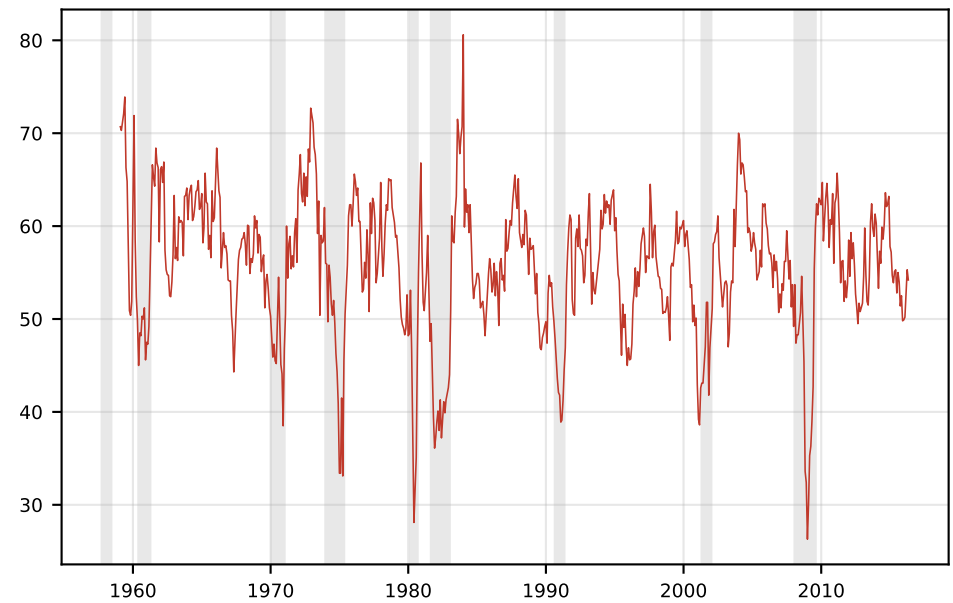
ISM Mfg Production

Block: Sentiment / Surveys / Business | Source: FRED-MD (hist) | Col: NAPMPI | Transform: 1: No transform

Raw Series



After: 1: No transform



SUMMARY STATISTICS

N : 1287
Valid : 688
NaN : 599
Start : 1959-01
End : 2016-04
Mean : 55.6547
Std : 7.3718
Min : 26.3000
Max : 80.6000
Skew : -0.631
Kurt : 1.161

STATIONARITY TESTS

ADF stat : -7.385
ADF p : 0.0000
ADF lags : 11
ADF : Stationary
KPSS stat : 0.216
KPSS p : 0.1000
KPSS : Stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

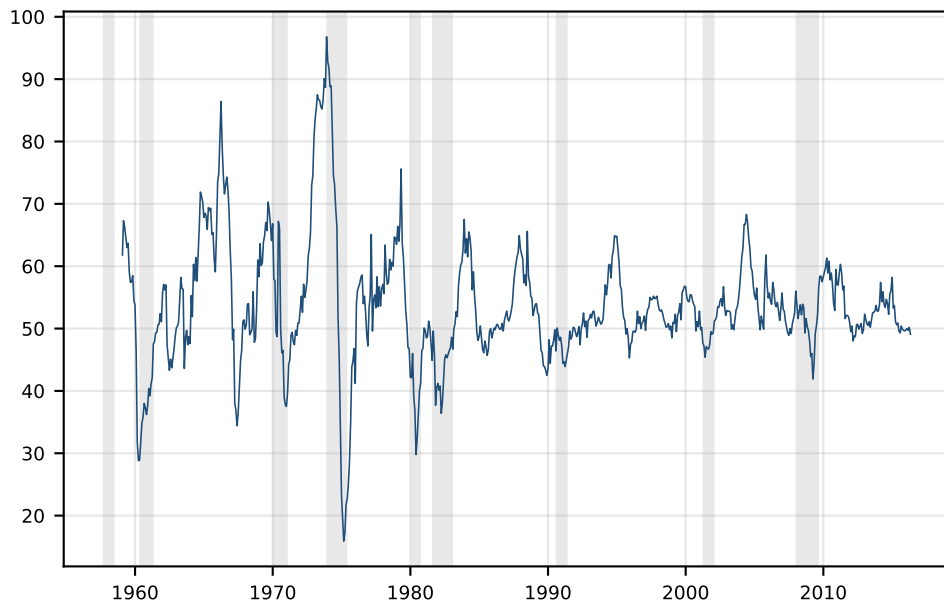
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	176	46.0%	54.0%	46.1330
Q2	168	7.1%	92.9%	54.1732
Q3	172	0.0%	100.0%	58.3628
Q4 (highest)	172	0.0%	100.0%	64.1366

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

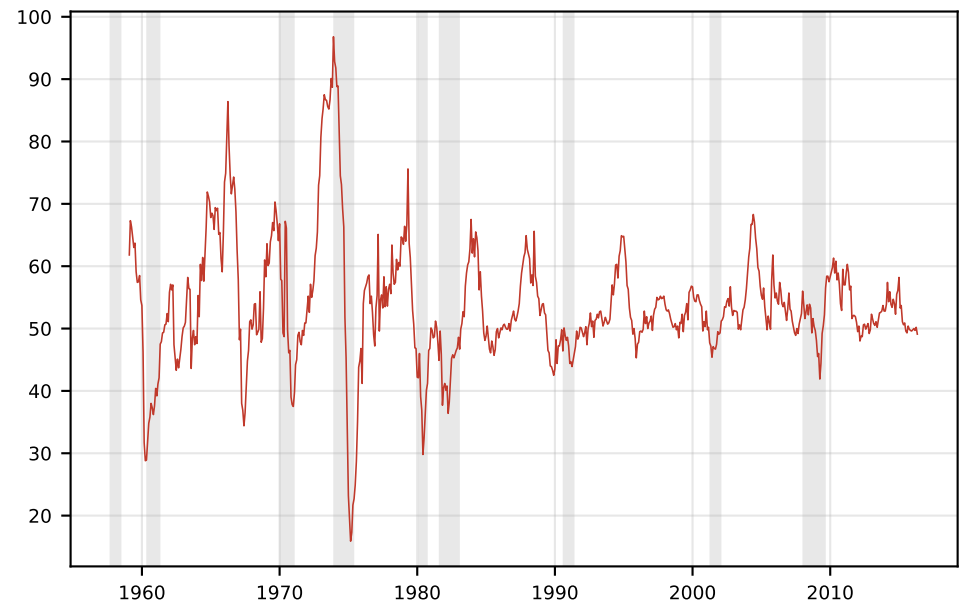
ISM Mfg Supplier Deliveries

Block: Sentiment / Surveys / Business | Source: FRED-MD (hist) | Col: NAPMSDI | Transform: 1: No transform

Raw Series



After: 1: No transform



SUMMARY STATISTICS

N : 1287
Valid : 688
NaN : 599
Start : 1959-01
End : 2016-04
Mean : 53.6624
Std : 9.9740
Min : 15.9000
Max : 96.8000
Skew : 0.697
Kurt : 3.441

STATIONARITY TESTS

ADF stat : -5.851
ADF p : 0.0000
ADF lags : 12
ADF : Stationary
KPSS stat : 0.117
KPSS p : 0.1000
KPSS : Stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

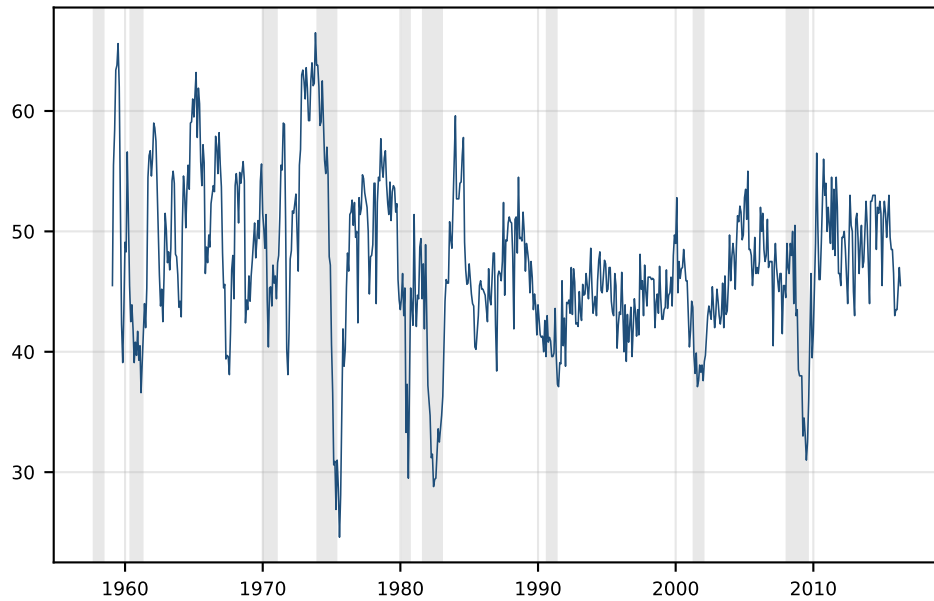
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	172	36.0%	64.0%	43.3453
Q2	172	7.6%	92.4%	50.5756
Q3	174	2.9%	97.1%	54.6109
Q4 (highest)	170	7.6%	92.4%	66.2529

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

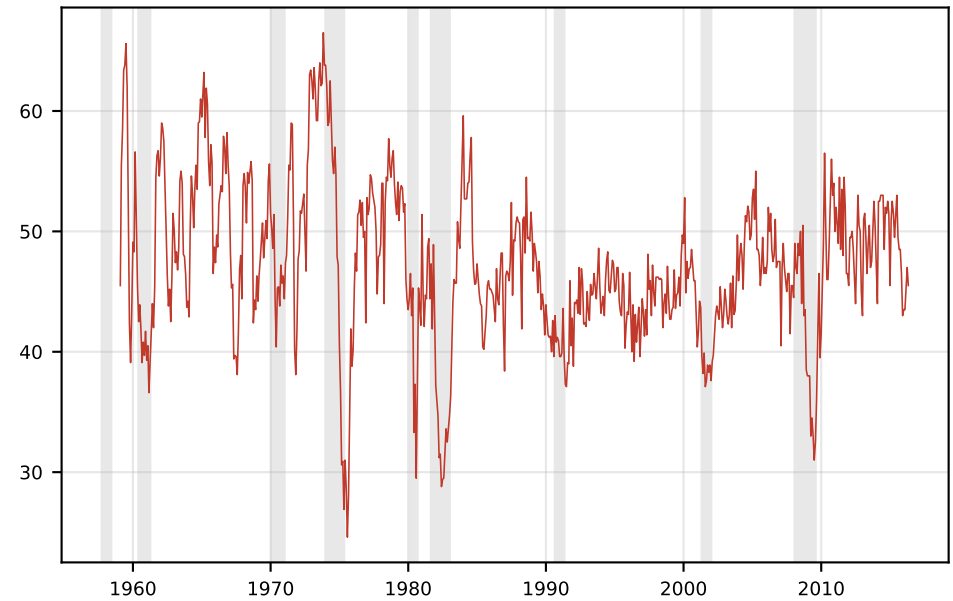
ISM Mfg Inventories

Block: Sentiment / Surveys / Business | Source: FRED-MD (hist) | Col: NAPMII | Transform: 1: No transform

Raw Series



After: 1: No transform



SUMMARY STATISTICS

N : 1287
Valid : 688
NaN : 599
Start : 1959-01
End : 2016-04
Mean : 47.2202
Std : 6.4998
Min : 24.6000
Max : 66.5000
Skew : 0.010
Kurt : 0.703

STATIONARITY TESTS

ADF stat : -5.420
ADF p : 0.0000
ADF lags : 12
ADF : Stationary
KPSS stat : 0.594
KPSS p : 0.0232
KPSS : Non-stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

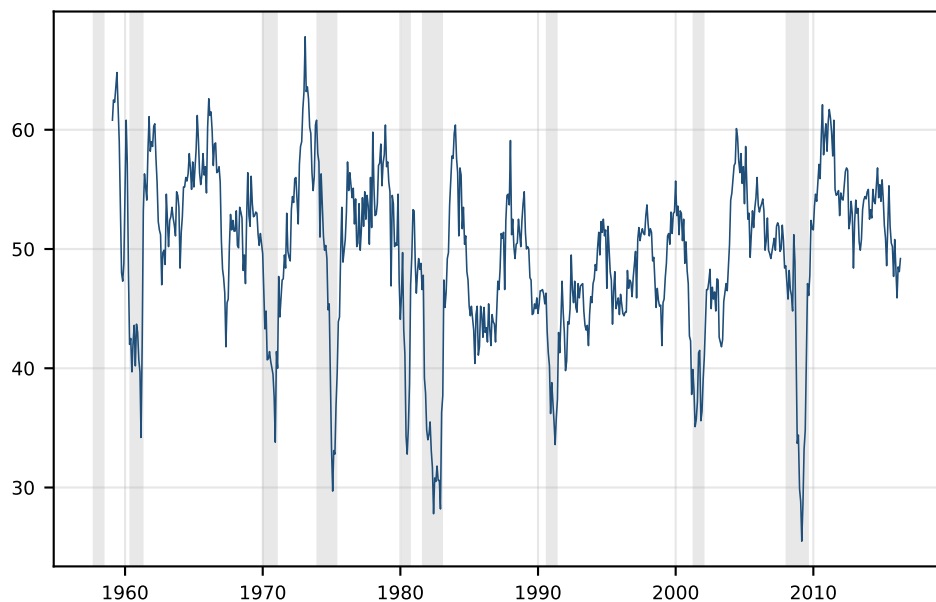
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	179	31.8%	68.2%	39.5950
Q2	167	7.2%	92.8%	45.2114
Q3	174	8.0%	92.0%	48.8609
Q4 (highest)	168	6.0%	94.0%	55.6423

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

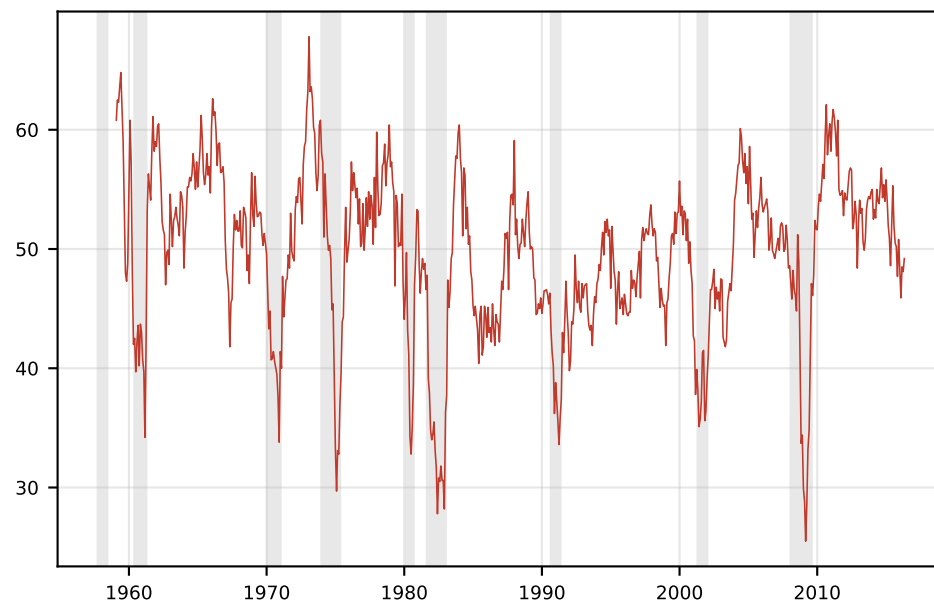
ISM Mfg Employment

Block: Sentiment / Surveys / Business | Source: FRED-MD (hist) | Col: NAPMEI | Transform: 1: No transform

Raw Series



After: 1: No transform



SUMMARY STATISTICS

N : 1287
Valid : 688
NaN : 599
Start : 1959-01
End : 2016-04
Mean : 49.6012
Std : 6.8497
Min : 25.5000
Max : 67.8000
Skew : -0.637
Kurt : 0.556

STATIONARITY TESTS

ADF stat : -6.033
ADF p : 0.0000
ADF lags : 12
ADF : Stationary
KPSS stat : 0.303
KPSS p : 0.1000
KPSS : Stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

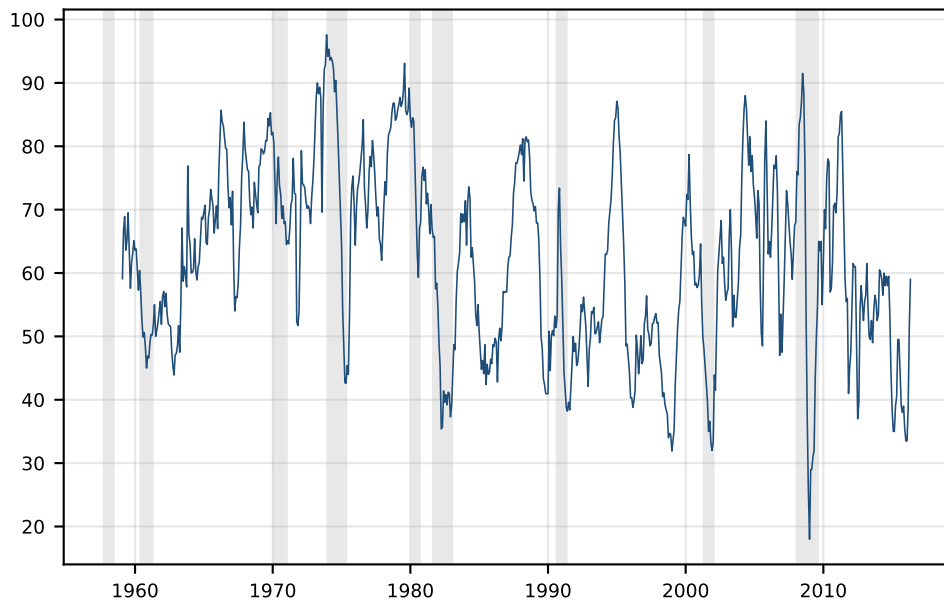
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	172	43.0%	57.0%	40.4779
Q2	174	6.9%	93.1%	48.1437
Q3	172	2.3%	97.7%	52.4012
Q4 (highest)	170	1.8%	98.2%	57.4906

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

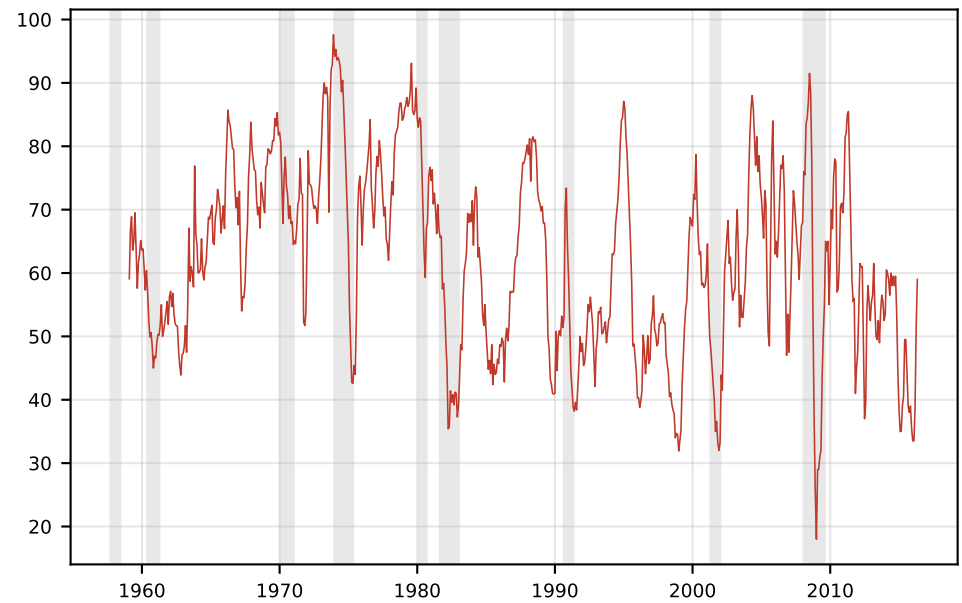
ISM Mfg Prices

Block: Sentiment / Surveys / Business | Source: FRED-MD (hist) | Col: NAPMPRI | Transform: 1: No transform

Raw Series



After: 1: No transform



SUMMARY STATISTICS

N : 1287
Valid : 688
NaN : 599
Start : 1959-01
End : 2016-04
Mean : 62.1033
Std : 14.7136
Min : 18.0000
Max : 97.6000
Skew : -0.012
Kurt : -0.696

STATIONARITY TESTS

ADF stat : -6.064
ADF p : 0.0000
ADF lags : 2
ADF : Stationary
KPSS stat : 0.739
KPSS p : 0.0100
KPSS : Non-stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

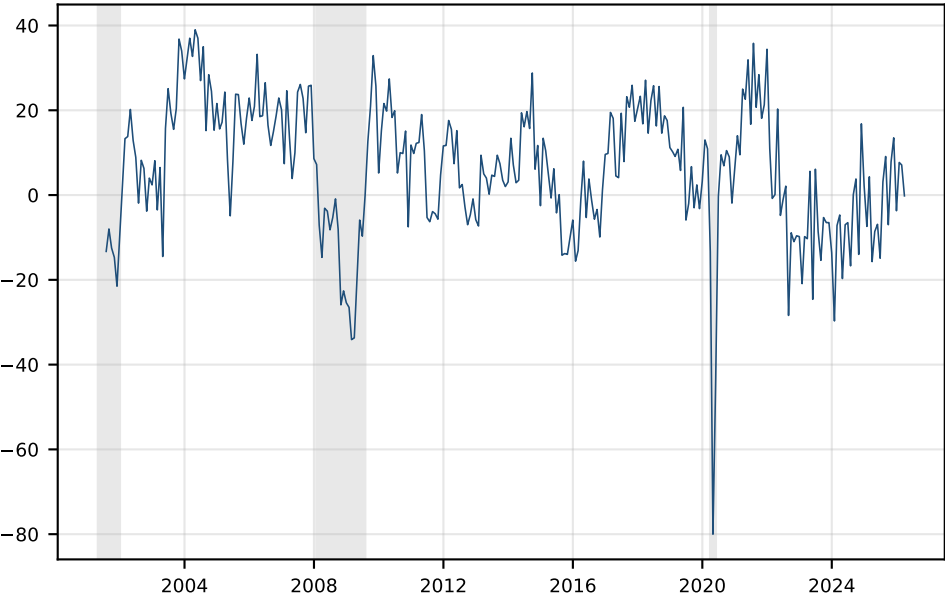
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	172	23.3%	76.7%	43.3500
Q2	174	6.3%	93.7%	56.2230
Q3	170	8.2%	91.8%	67.9576
Q4 (highest)	172	16.3%	83.7%	81.0192

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

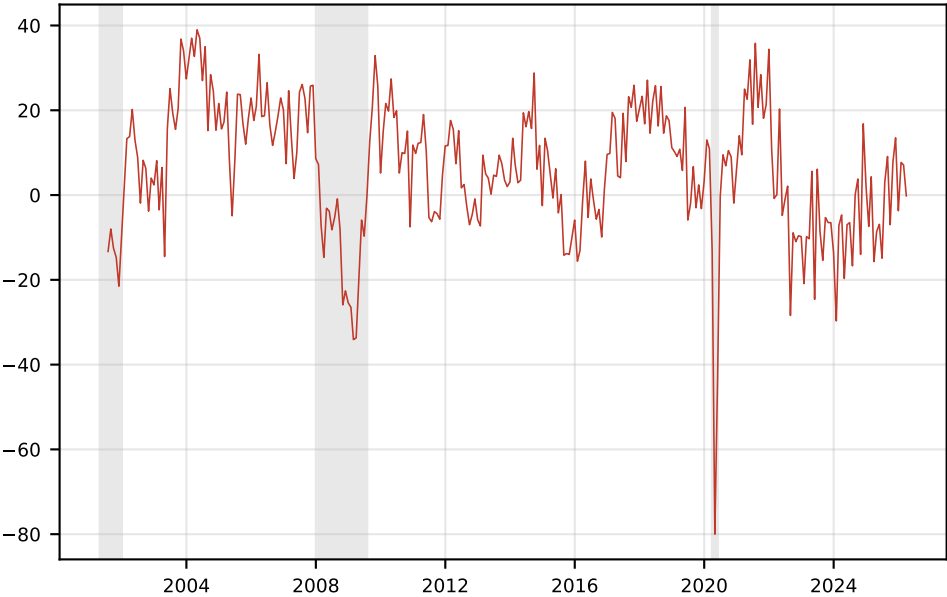
Philly/Empire State Mfg

Block: Sentiment / Surveys / Business | Source: FRED API | Col: GACDISA066MSFRBNY | Transform: 1: No transform

Raw Series



After: 1: No transform



SUMMARY STATISTICS

N : 1287
Valid : 297
NaN : 990
Start : 2001-07
End : 2026-03
Mean : 6.2721
Std : 15.5118
Min : -80.0000
Max : 39.0000
Skew : -0.794
Kurt : 2.671

STATIONARITY TESTS

ADF stat : -4.618
ADF p : 0.0001
ADF lags : 2
ADF : Stationary
KPSS stat : 0.406
KPSS p : 0.0746
KPSS : Stationary

QUARTILE-BUCKET ANALYSIS

(vs NBER recession/expansion)

Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	74	28.4%	71.6%	-13.5865
Q2	75	5.3%	94.7%	2.0147
Q3	75	0.0%	100.0%	12.4720
Q4 (highest)	72	0.0%	100.0%	24.7486

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

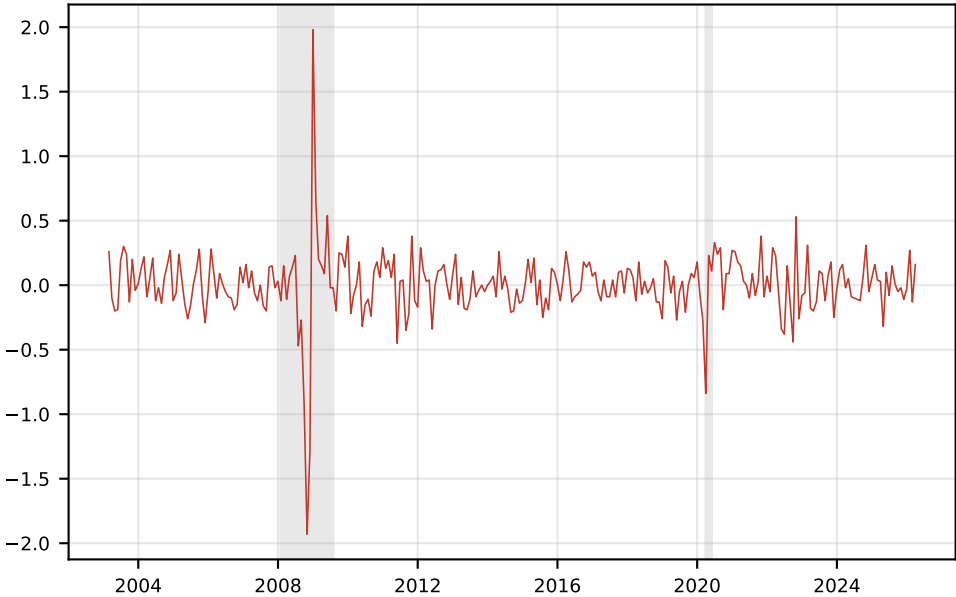
5-Year Breakeven Inflation Rate

Block: Inflation / Prices / Inflation Expectations | Source: FRED API | Col: T5YIE | Transform: 2: First diff

Raw Series



After: 2: First diff



SUMMARY STATISTICS

N : 1287
Valid : 279
NaN : T5YIE 1008
T5YIE 1008
dtype: int64
Start : 2003-01
End : 2026-03
Mean : 1.9565
Std : 0.5978
Min : -2.2400
Max : 3.3400
Skew : -1.924
Kurt : 10.163

STATIONARITY TESTS

ADF stat : -6.472
ADF p : 0.0000
ADF lags : 12
ADF : Stationary
KPSS stat : 0.028
KPSS p : 0.1000
KPSS : Stationary

QUARTILE-BUCKET ANALYSIS

(vs NBER recession/expansion)

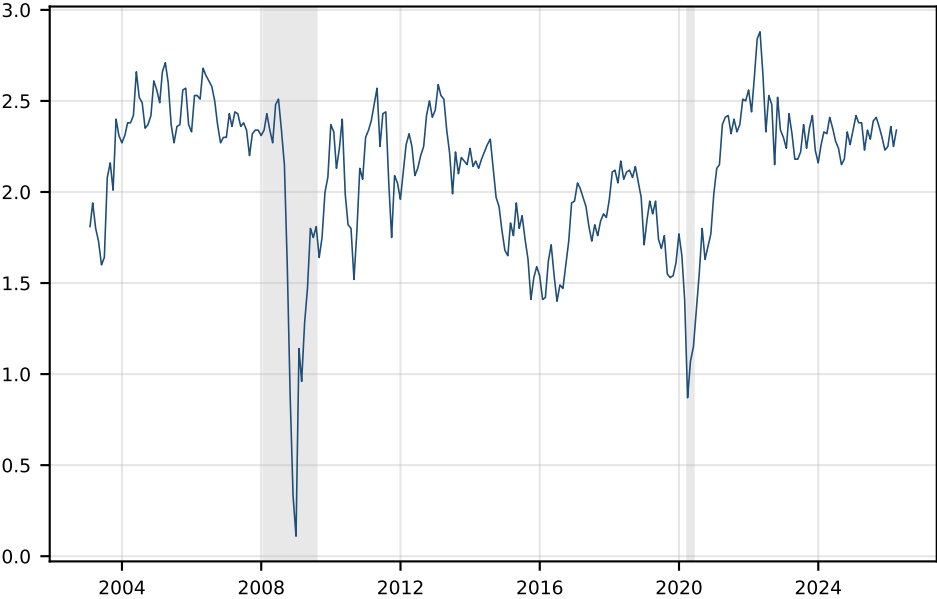
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	71	9.9%	90.1%	-0.2587
Q2	70	2.9%	97.1%	-0.0501
Q3	67	3.0%	97.0%	0.0722
Q4 (highest)	69	13.0%	87.0%	0.2601

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

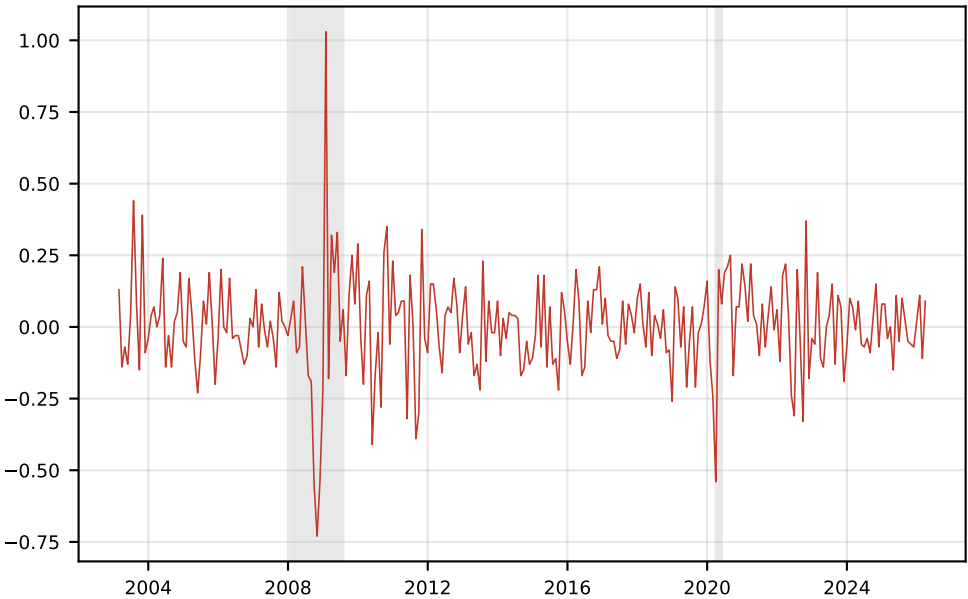
10-Year Breakeven Inflation Rate

Block: Inflation / Prices / Inflation Expectations | Source: FRED API | Col: T10YIE | Transform: 2: First diff

Raw Series



After: 2: First diff



SUMMARY STATISTICS

N : 1287
Valid : 279
NaN : T10YIE 1008
T10YIE 1008
dtype: int64
Start : 2003-01
End : 2026-03
Mean : 2.1057
Std : 0.3961
Min : 0.1100
Max : 2.8800
Skew : -1.373
Kurt : 3.270

STATIONARITY TESTS

ADF stat : -9.179
ADF p : 0.0000
ADF lags : 4
ADF : Stationary
KPSS stat : 0.024
KPSS p : 0.1000
KPSS : Stationary

QUARTILE-BUCKET ANALYSIS

(vs NBER recession/expansion)

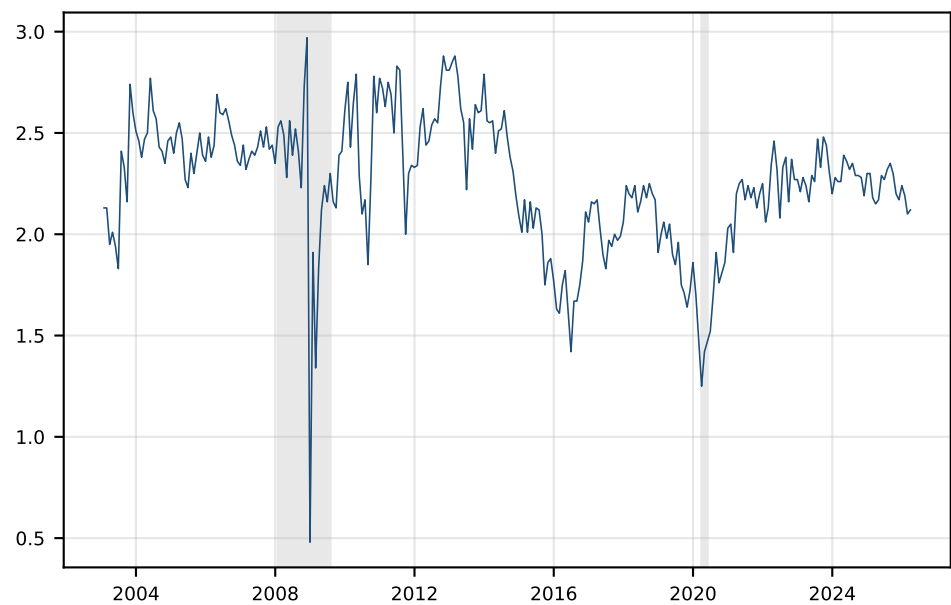
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	71	12.7%	87.3%	-0.1921
Q2	70	2.9%	97.1%	-0.0409
Q3	68	2.9%	97.1%	0.0518
Q4 (highest)	68	10.3%	89.7%	0.1974

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

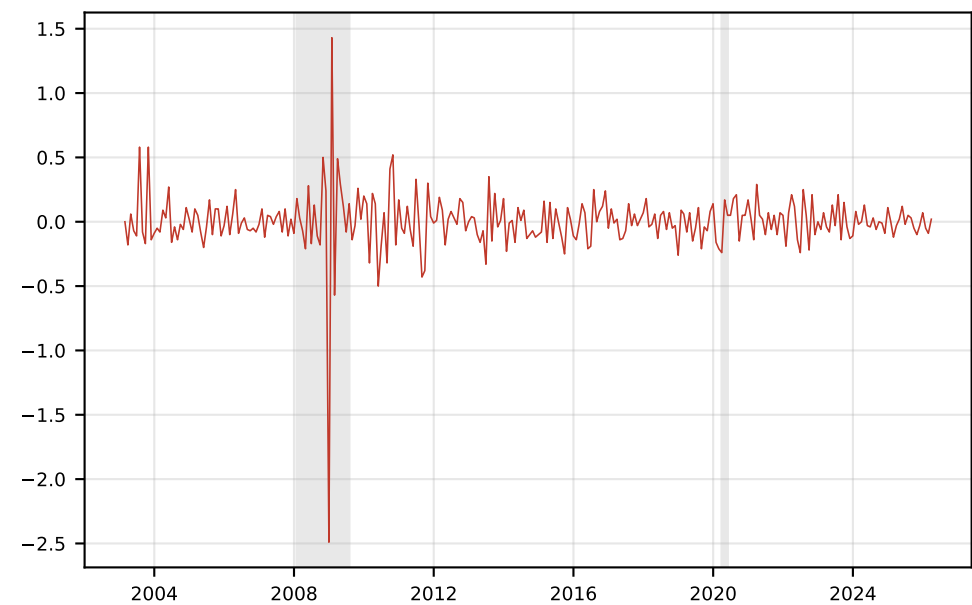
5-Year, 5-Year Forward Inflation Expectation Rate

Block: Inflation / Prices / Inflation Expectations | Source: FRED API | Col: T5YIFR | Transform: 2: First diff

Raw Series



After: 2: First diff



SUMMARY STATISTICS

N : 1287
Valid : 279
NaN : T5YIFR 1008
T5YIFR 1008
dtype: int64
Start : 2003-01
End : 2026-03
Mean : 2.2553
Std : 0.3269
Min : 0.4800
Max : 2.9700
Skew : -0.958
Kurt : 2.828

STATIONARITY TESTS

ADF stat : -8.861
ADF p : 0.0000
ADF lags : 6
ADF : Stationary
KPSS stat : 0.033
KPSS p : 0.1000
KPSS : Stationary

QUARTILE-BUCKET ANALYSIS

(vs NBER recession/expansion)

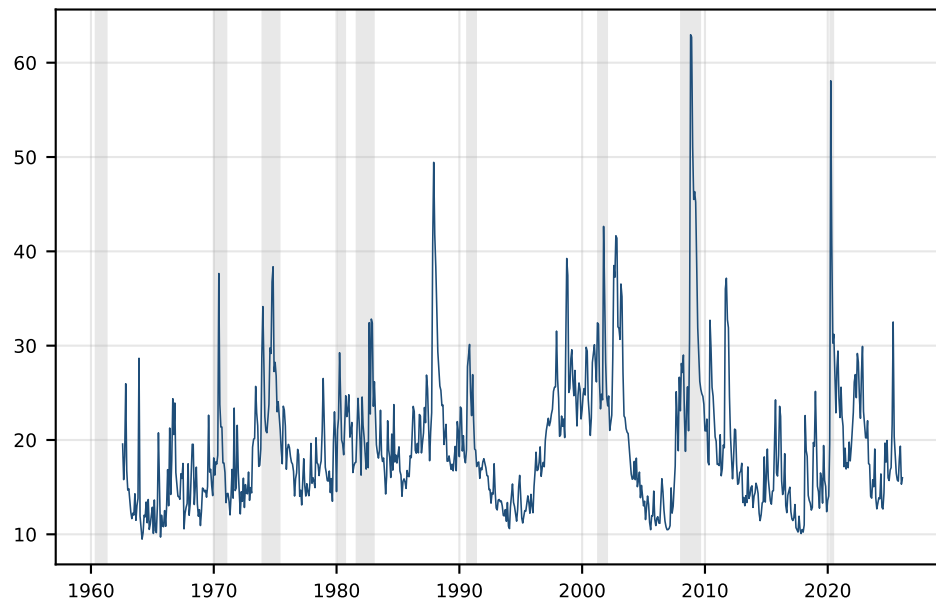
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	71	9.9%	90.1%	-0.2096
Q2	69	2.9%	97.1%	-0.0488
Q3	69	1.4%	98.6%	0.0442
Q4 (highest)	68	14.7%	85.3%	0.2231

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

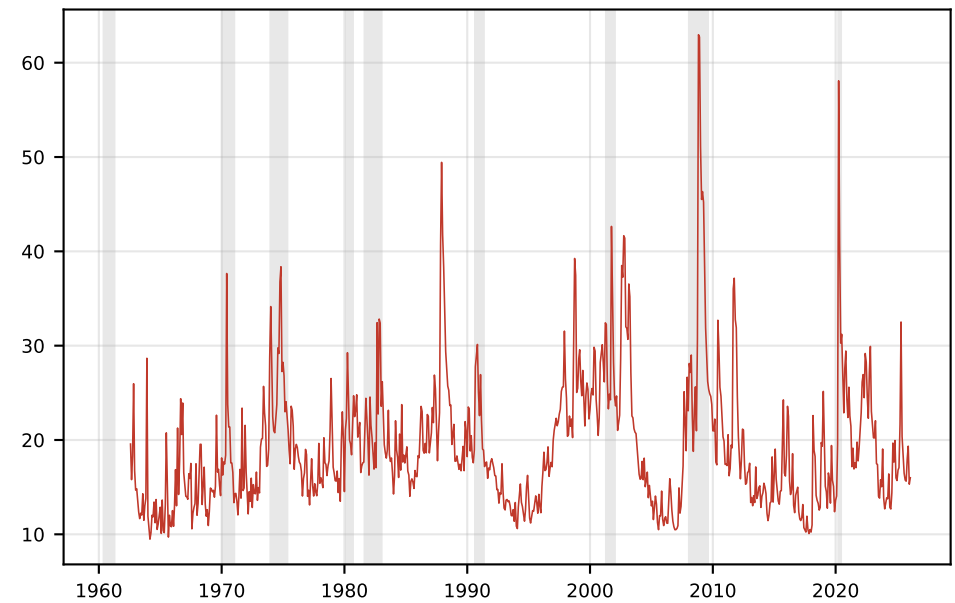
CBOE VIX

Block: Volatility / Stress / Equity volatility | Source: FRED-MD | Col: VIXCLSx | Transform: 1: No transform

Raw Series



After: 1: No transform



SUMMARY STATISTICS

N : 1287
Valid : 763
NaN : 524
Start : 1962-07
End : 2026-01
Mean : 19.2867
Std : 7.0299
Min : 9.4844
Max : 62.9648
Skew : 1.911
Kurt : 6.288

STATIONARITY TESTS

ADF stat : -5.766
ADF p : 0.0000
ADF lags : 4
ADF : Stationary
KPSS stat : 0.257
KPSS p : 0.1000
KPSS : Stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	191	0.0%	100.0%	12.5880
Q2	191	4.2%	95.8%	16.0650
Q3	190	12.1%	87.9%	19.8675
Q4 (highest)	191	28.3%	71.7%	28.6292

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

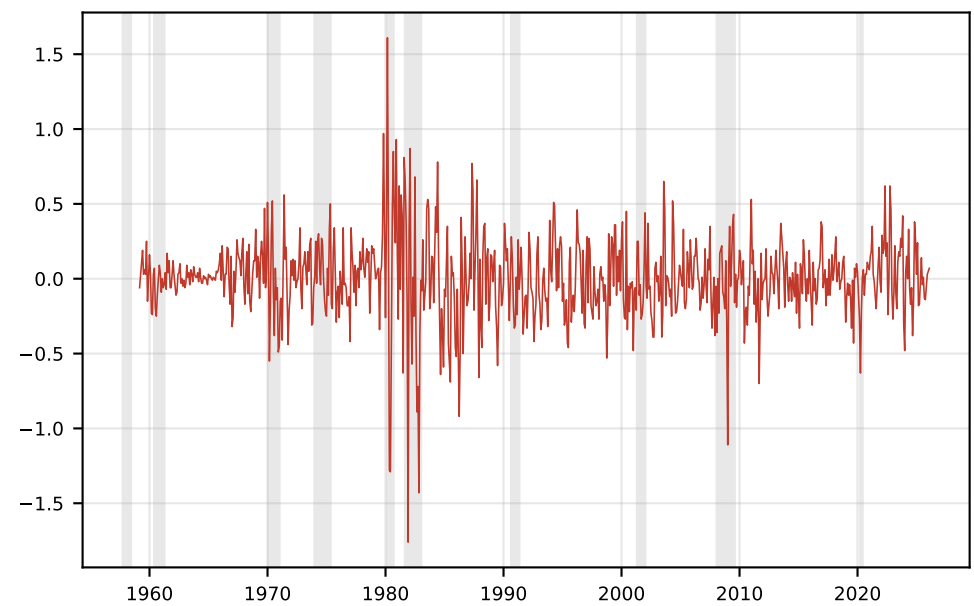
10Y Treasury Yield

Block: Volatility / Stress / Rates volatility | Source: FRED-MD | Col: GS10 | Transform: 2: First diff

Raw Series



After: 2: First diff



SUMMARY STATISTICS

```
N          : 1287
Valid       : 805
NaN         : GS10    482
GS10       412
dtype: int64
Start      : 1959-01
End        : 2026-01
Mean       : 5.7448
Std        : 2.8954
Min        : 0.6200
Max        : 15.3200
Skew       : 0.809
Kurt       : 0.499
```

STATIONARITY TESTS

```
ADF stat    : -7.506
ADF p       : 0.0000
ADF lags    : 11
ADF         : Stationary
KPSS stat   : 0.146
KPSS p      : 0.1000
KPSS        : Stationary
```

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	201	17.9%	82.1%	-0.3135
Q2	213	8.5%	91.5%	-0.0577
Q3	189	7.9%	92.1%	0.0660
Q4 (highest)	201	12.9%	87.1%	0.3136

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

St. Louis Fed Financial Stress Index

Block: Volatility / Stress / Broad uncertainty | Source: FRED API | Col: STLFSI2 | Transform: 1: No transform

Raw Series



After: 1: No transform



SUMMARY STATISTICS

N : 1287
Valid : 338
NaN : 949
Start : 1993-12
End : 2022-01
Mean : -0.0075
Std : 0.9752
Min : -1.1281
Max : 7.0442
Skew : 3.516
Kurt : 17.902

STATIONARITY TESTS

ADF stat : -4.257
ADF p : 0.0005
ADF lags : 2
ADF : Stationary
KPSS stat : 0.150
KPSS p : 0.1000
KPSS : Stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

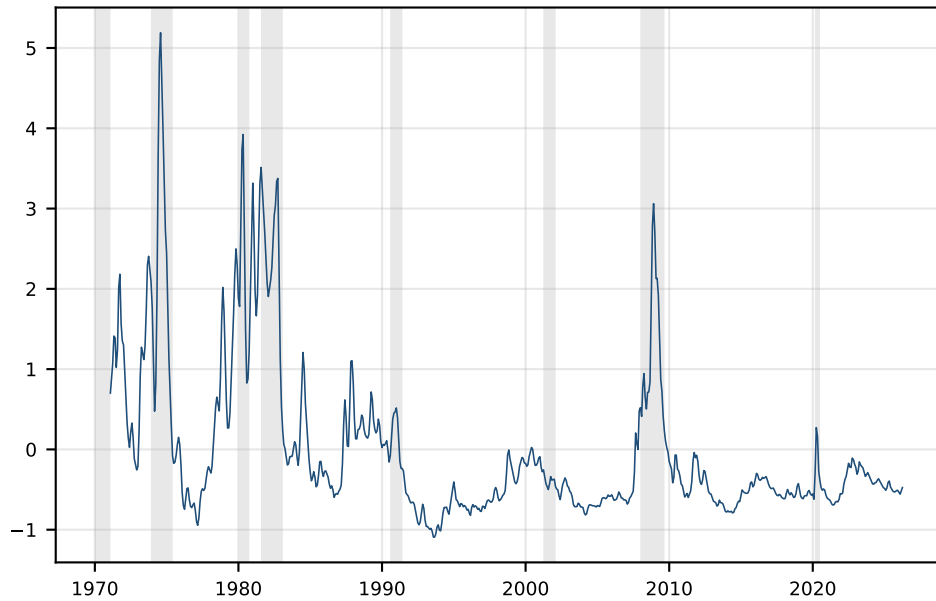
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	85	0.0%	100.0%	-0.7750
Q2	84	0.0%	100.0%	-0.3649
Q3	84	3.6%	96.4%	-0.0167
Q4 (highest)	85	29.4%	70.6%	1.1222

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

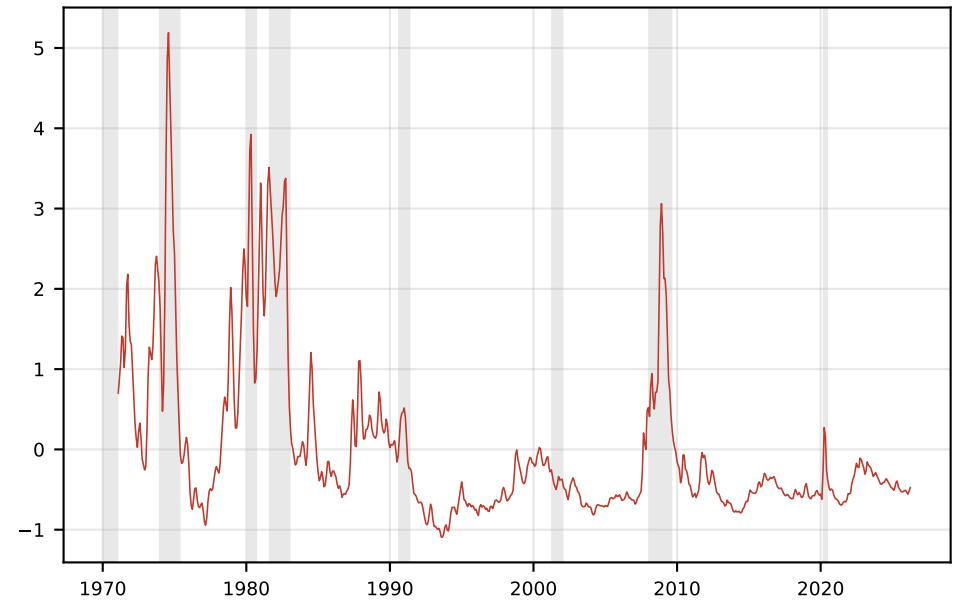
Chicago Fed NFCI

Block: Volatility / Stress / Broad uncertainty | Source: FRED API | Col: NFCI | Transform: 1: No transform

Raw Series



After: 1: No transform



SUMMARY STATISTICS

N : 1287
Valid : 663
NaN : 624
Start : 1971-01
End : 2026-03
Mean : -0.0003
Std : 1.0011
Min : -1.0931
Max : 5.1900
Skew : 2.148
Kurt : 4.769

STATIONARITY TESTS

ADF stat : -3.404
ADF p : 0.0108
ADF lags : 10
ADF : Stationary
KPSS stat : 1.317
KPSS p : 0.0100
KPSS : Non-stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

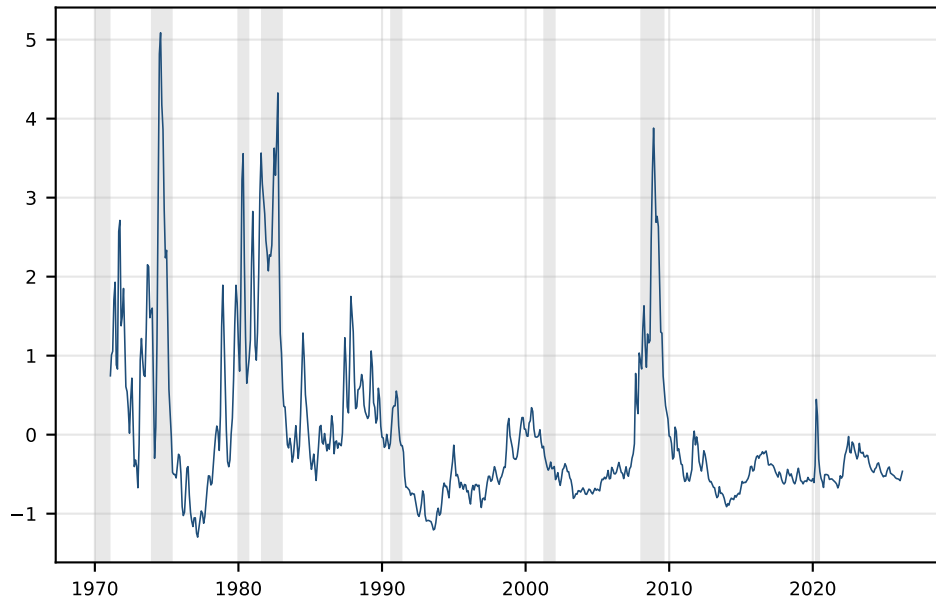
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	166	0.0%	100.0%	-0.7235
Q2	165	3.0%	97.0%	-0.5033
Q3	165	3.0%	97.0%	-0.1690
Q4 (highest)	166	38.6%	61.4%	1.3935

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

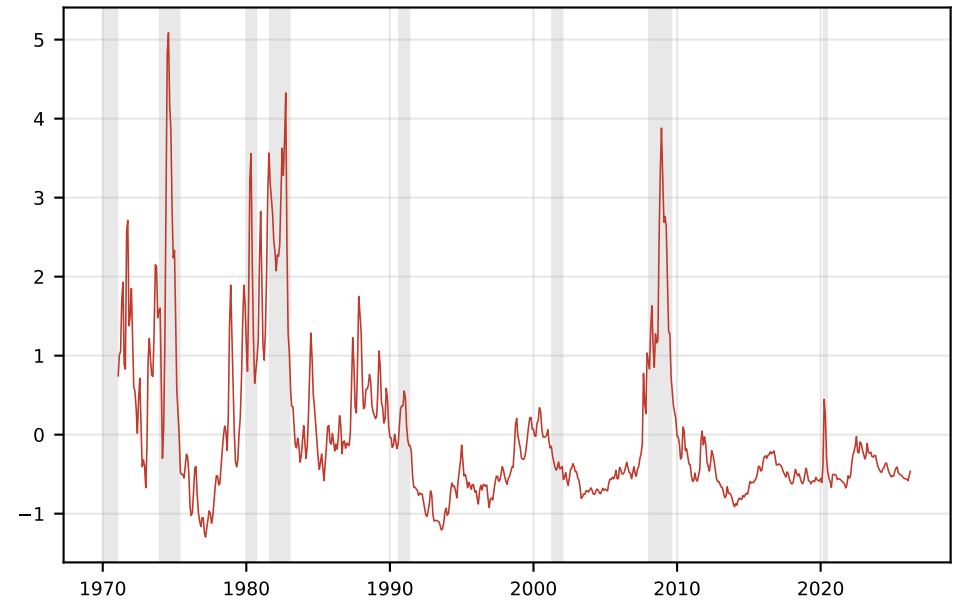
Chicago Fed Adjusted NFCI

Block: Volatility / Stress / Broad uncertainty | Source: FRED API | Col: ANFCI | Transform: 1: No transform

Raw Series



After: 1: No transform



SUMMARY STATISTICS

N : 1287
Valid : 663
NaN : 624
Start : 1971-01
End : 2026-03
Mean : 0.0004
Std : 1.0021
Min : -1.2975
Max : 5.0872
Skew : 2.067
Kurt : 4.653

STATIONARITY TESTS

ADF stat : -4.004
ADF p : 0.0014
ADF lags : 5
ADF : Stationary
KPSS stat : 0.957
KPSS p : 0.0100
KPSS : Non-stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

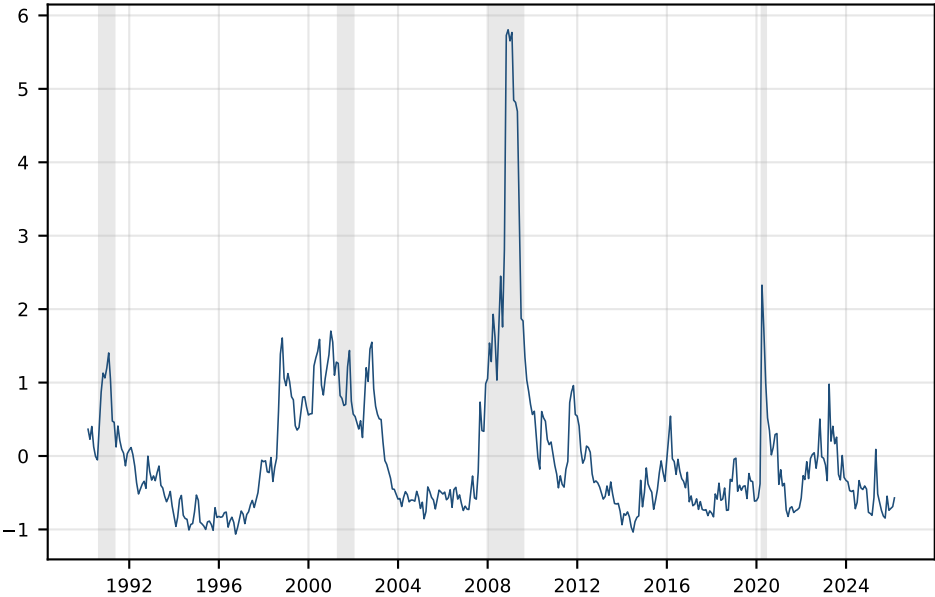
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	166	0.0%	100.0%	-0.7759
Q2	165	3.0%	97.0%	-0.4850
Q3	165	4.8%	95.2%	-0.1302
Q4 (highest)	166	36.7%	63.3%	1.3917

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

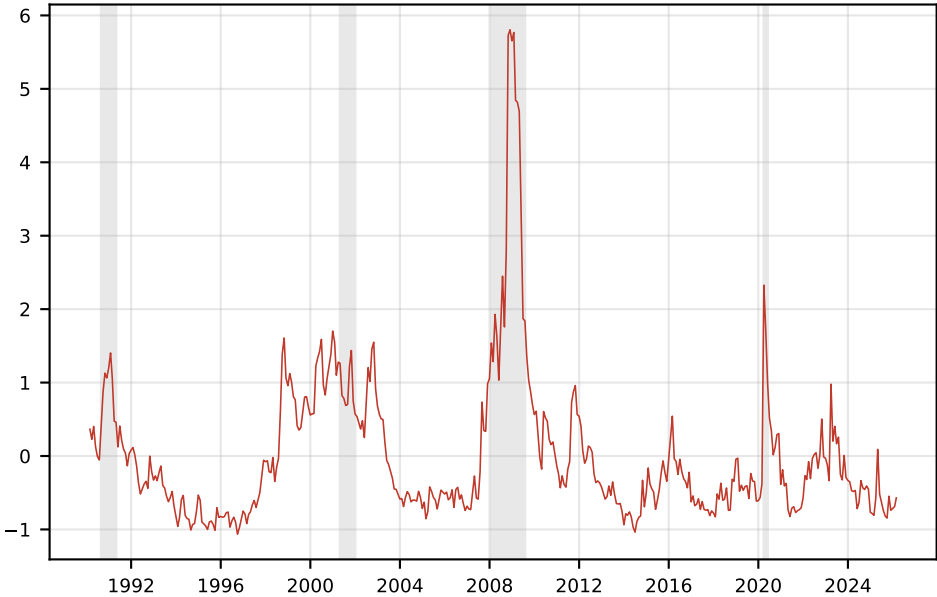
Kansas City Fed Financial Stress

Block: Volatility / Stress / Broad uncertainty | Source: FRED API | Col: KCFSI | Transform: 1: No transform

Raw Series



After: 1: No transform



SUMMARY STATISTICS

N : 1287
Valid : 433
NaN : 854
Start : 1990-02
End : 2026-02
Mean : 0.0000
Std : 1.0000
Min : -1.0645
Max : 5.8050
Skew : 2.870
Kurt : 11.978

STATIONARITY TESTS

ADF stat : -3.907
ADF p : 0.0020
ADF lags : 3
ADF : Stationary
KPSS stat : 0.192
KPSS p : 0.1000
KPSS : Stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

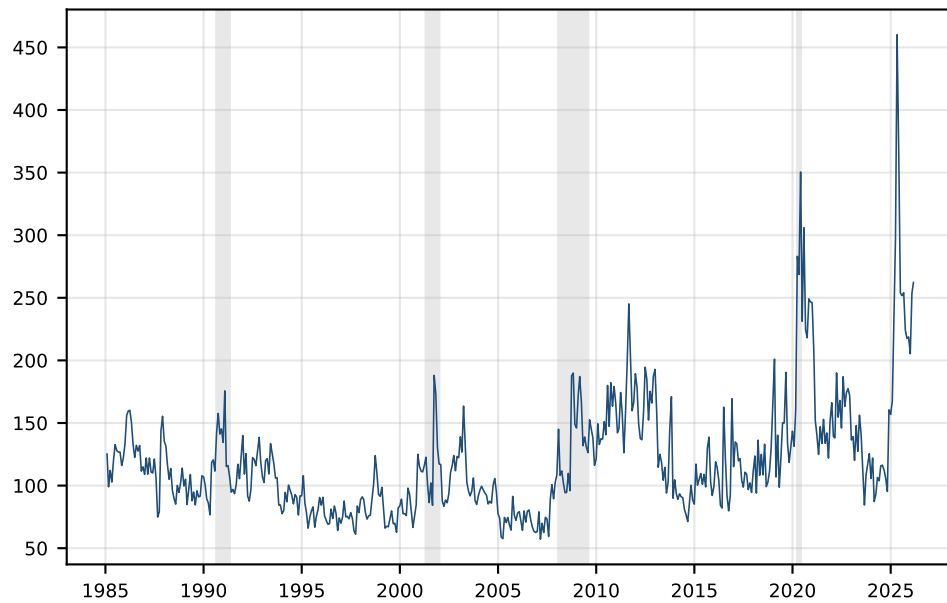
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	109	0.0%	100.0%	-0.7796
Q2	108	0.0%	100.0%	-0.4779
Q3	108	0.0%	100.0%	-0.0375
Q4 (highest)	108	33.3%	66.7%	1.3023

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

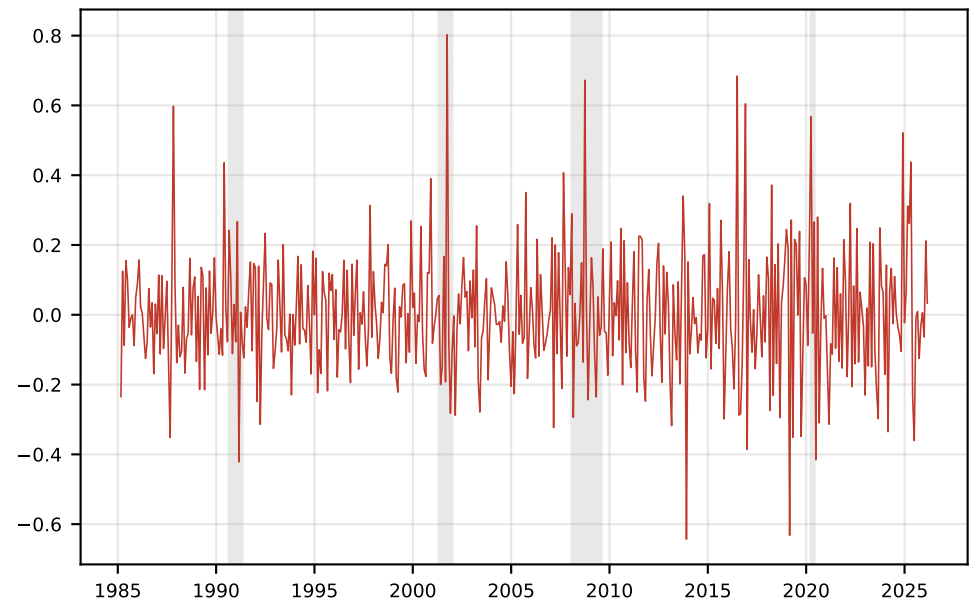
Economic Policy Uncertainty (Monthly)

Block: Volatility / Stress / Broad uncertainty | Source: FRED API | Col: USEPUINDXM | Transform: 5: Log diff

Raw Series



After: 5: Log diff



SUMMARY STATISTICS

N : 1287
Valid : 494
NaN : 793
Start : 1985-01
End : 2026-02
Mean : 119.4638
Std : 47.1001
Min : 57.2026
Max : 460.1143
Skew : 2.243
Kurt : 8.754

STATIONARITY TESTS

ADF stat : -11.356
ADF p : 0.0000
ADF lags : 6
ADF : Stationary
KPSS stat : 0.119
KPSS p : 0.1000
KPSS : Stationary

QUARTILE-BUCKET ANALYSIS

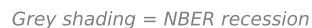
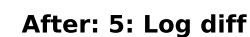
(vs NBER recession/expansion)

Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	124	8.9%	91.1%	-0.1990
Q2	123	5.7%	94.3%	-0.0567
Q3	123	6.5%	93.5%	0.0449
Q4 (highest)	123	8.1%	91.9%	0.2184

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

Block: Cross-Asset / Equity block | Source: FRED-MD | Col: S&P 500 | Transform: 5: Log diff

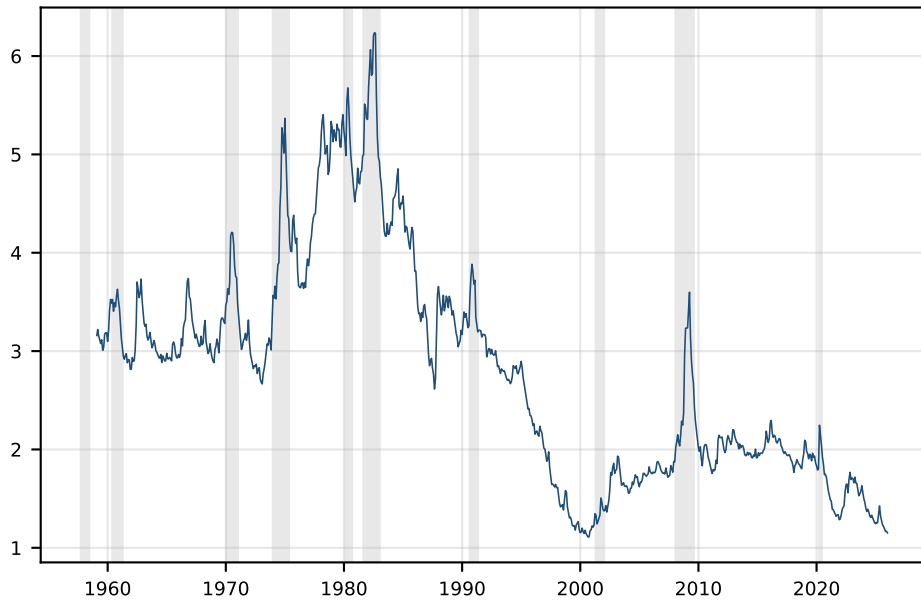
Raw Series



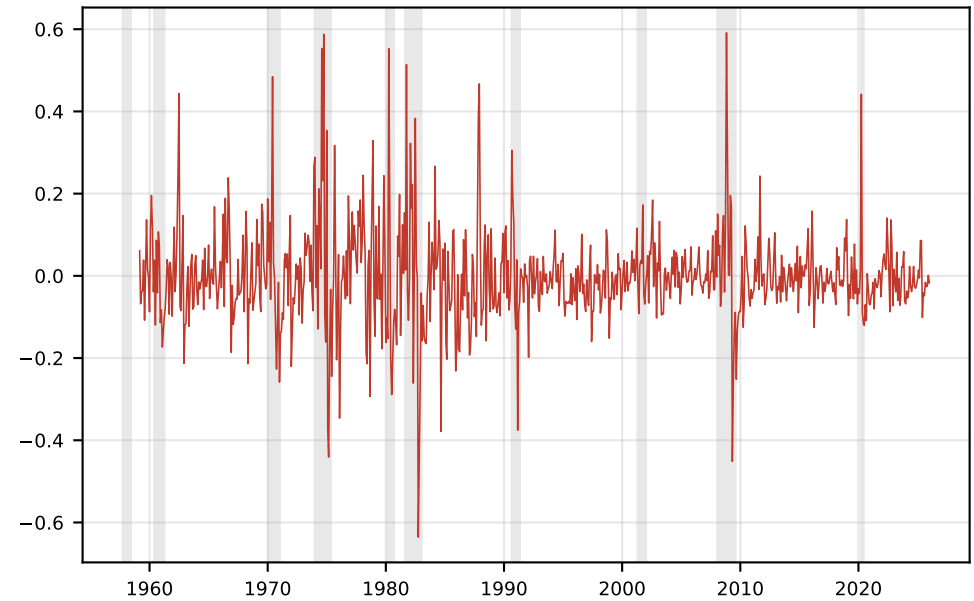
S&P 500 Dividend Yield

Block: Cross-Asset / Equity block | Source: FRED-MD | Col: S&P div yield | Transform: 2: First diff

Raw Series



After: 2: First diff



SUMMARY STATISTICS

N : 1287
Valid : 804
NaN : 483
Start : 1959-01
End : 2025-12
Mean : 2.8050
Std : 1.1441
Min : 1.1081
Max : 6.2370
Skew : 0.643
Kurt : -0.265

STATIONARITY TESTS

ADF stat : -8.570
ADF p : 0.0000
ADF lags : 11
ADF : Stationary
KPSS stat : 0.069
KPSS p : 0.1000
KPSS : Stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

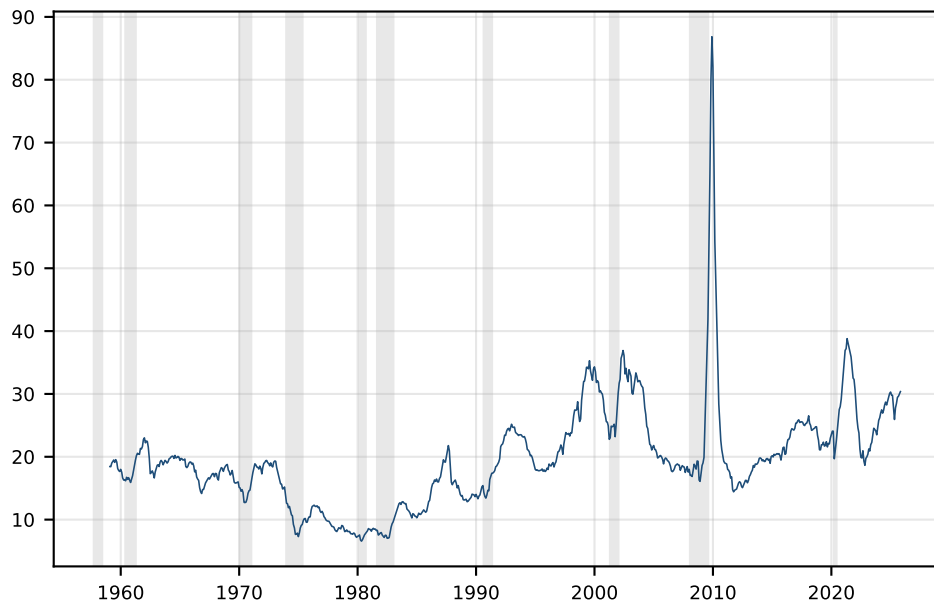
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	201	17.4%	82.6%	-0.1239
Q2	201	4.0%	96.0%	-0.0322
Q3	200	8.0%	92.0%	0.0137
Q4 (highest)	201	17.9%	82.1%	0.1325

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

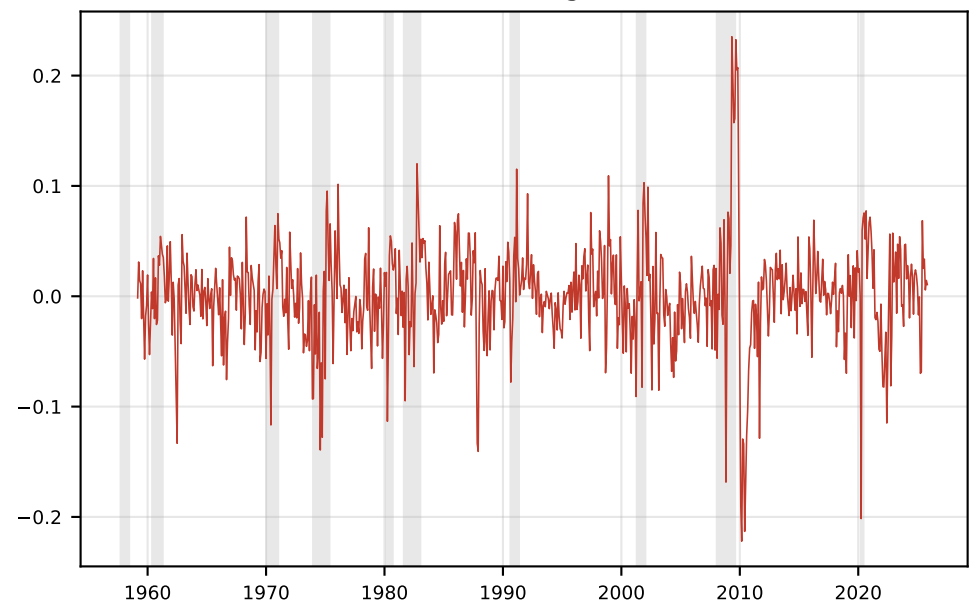
S&P 500 PE Ratio

Block: Cross-Asset / Equity block | Source: FRED-MD | Col: S&P PE ratio | Transform: 5: Log diff

Raw Series



After: 5: Log diff



SUMMARY STATISTICS

N : 1287
Valid : 802
NaN : 485
Start : 1959-01
End : 2025-10
Mean : 19.6028
Std : 8.3819
Min : 6.5696
Max : 86.8431
Skew : 2.411
Kurt : 14.014

STATIONARITY TESTS

ADF stat : -9.588
ADF p : 0.0000
ADF lags : 8
ADF : Stationary
KPSS stat : 0.040
KPSS p : 0.1000
KPSS : Stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

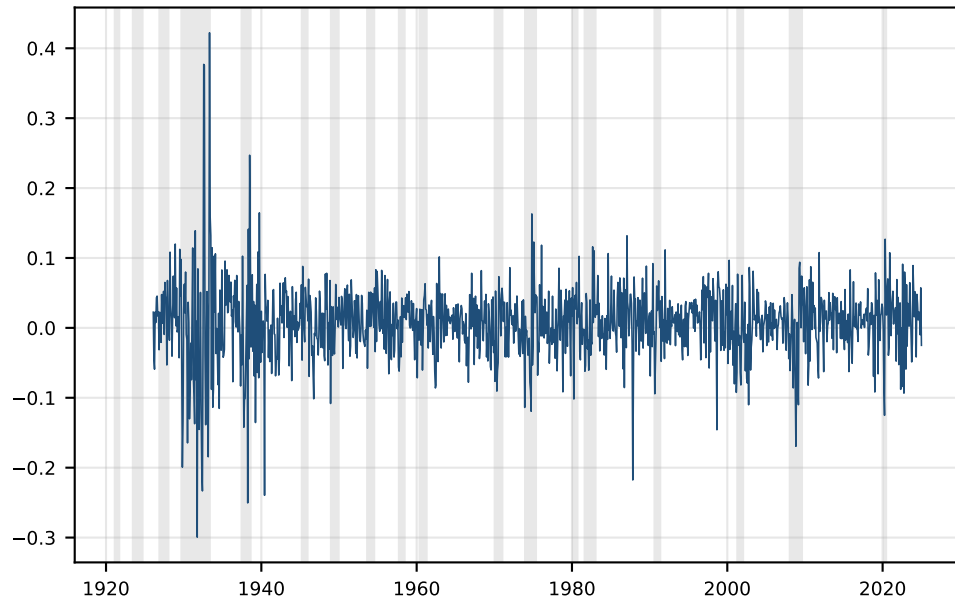
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	201	14.4%	85.6%	-0.0526
Q2	200	7.0%	93.0%	-0.0082
Q3	200	8.0%	92.0%	0.0122
Q4 (highest)	200	18.0%	82.0%	0.0513

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

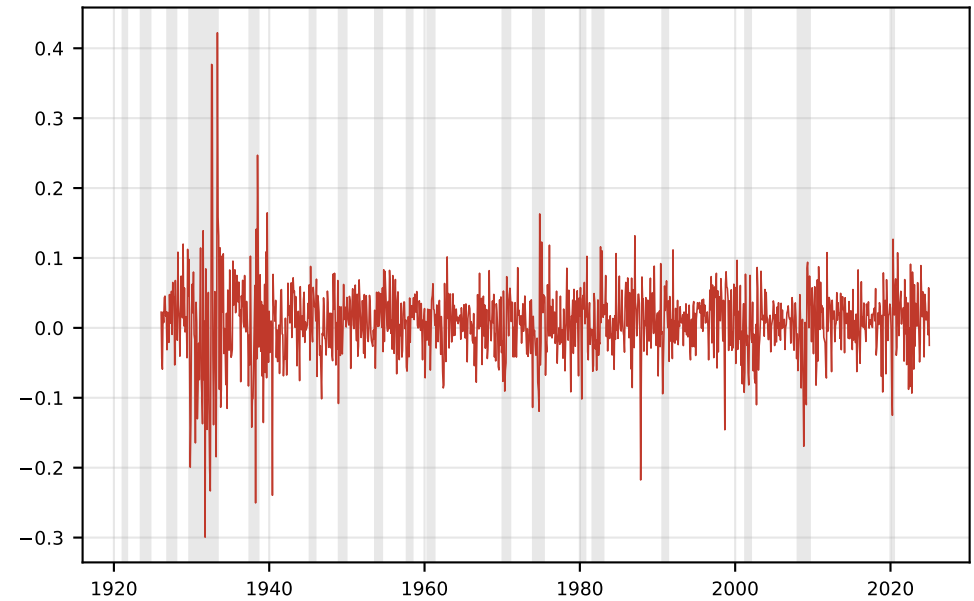
S&P 500 Total Return (monthly)

Block: Cross-Asset / Equity block | Source: WRDS (crsp.msi sprtrn, from 1925) | Col: sprtrn_sp500 | Transform: 1: No transform

Raw Series



After: 1: No transform



SUMMARY STATISTICS

N : 1287
Valid : 1187
NaN : 100
Start : 1926-01
End : 2024-12
Mean : 0.0066
Std : 0.0536
Min : -0.2994
Max : 0.4222
Skew : 0.247
Kurt : 9.116

STATIONARITY TESTS

ADF stat : -10.297
ADF p : 0.0000
ADF lags : 8
ADF : Stationary
KPSS stat : 0.074
KPSS p : 0.1000
KPSS : Stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

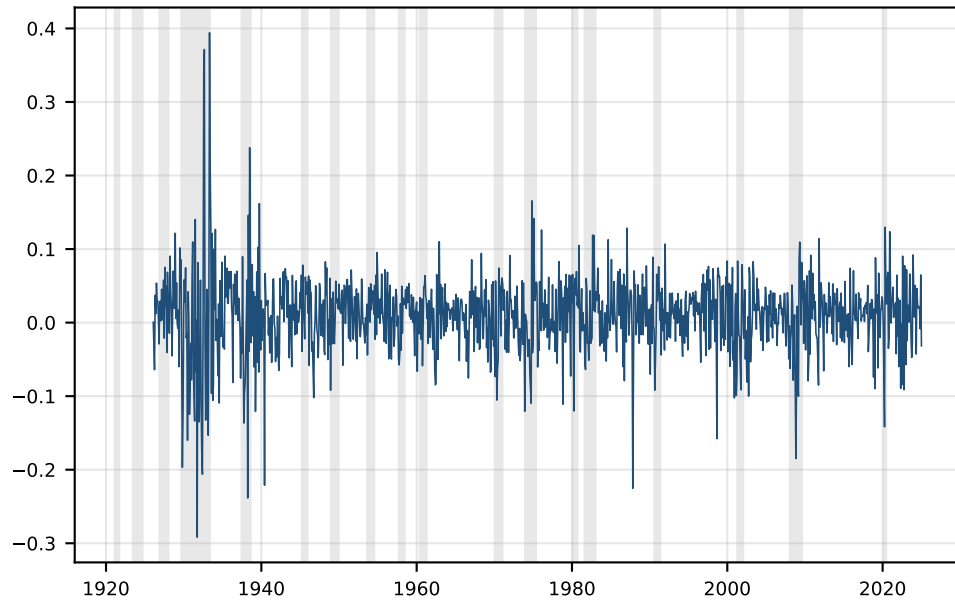
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	297	26.6%	73.4%	-0.0572
Q2	297	10.8%	89.2%	-0.0037
Q3	296	10.8%	89.2%	0.0222
Q4 (highest)	297	19.5%	80.5%	0.0653

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

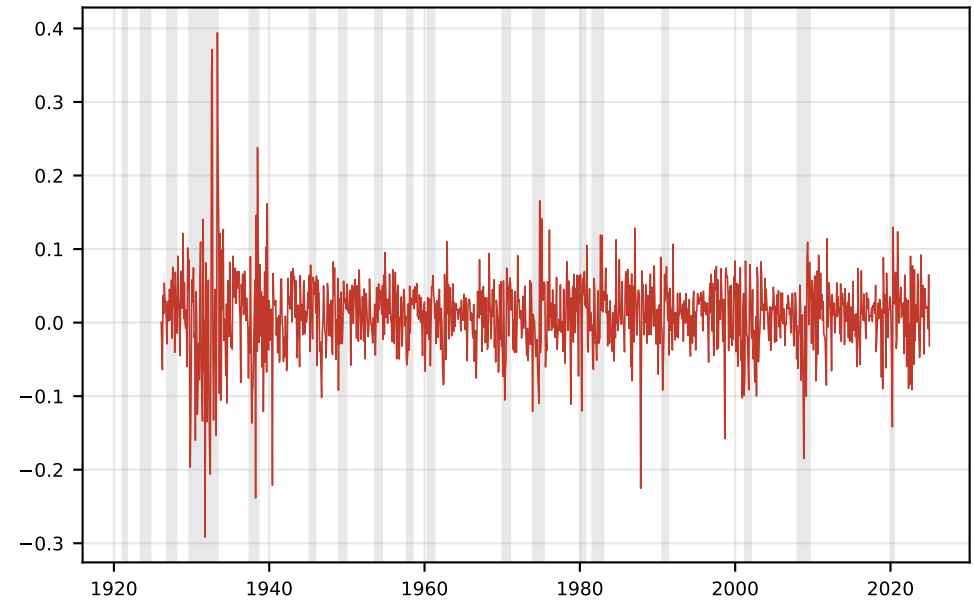
CRSP Value-Weighted Market Return

Block: Cross-Asset / Equity block | Source: Computed | Col: vwretd | Transform: 1: No transform

Raw Series



After: 1: No transform



SUMMARY STATISTICS

N : 1287
Valid : 1188
NaN : 99
Start : 1926-01
End : 2024-12
Mean : 0.0093
Std : 0.0531
Min : -0.2917
Max : 0.3941
Skew : 0.125
Kurt : 7.629

STATIONARITY TESTS

ADF stat : -11.425
ADF p : 0.0000
ADF lags : 7
ADF : Stationary
KPSS stat : 0.029
KPSS p : 0.1000
KPSS : Stationary

QUARTILE-BUCKET ANALYSIS

(vs NBER recession/expansion)

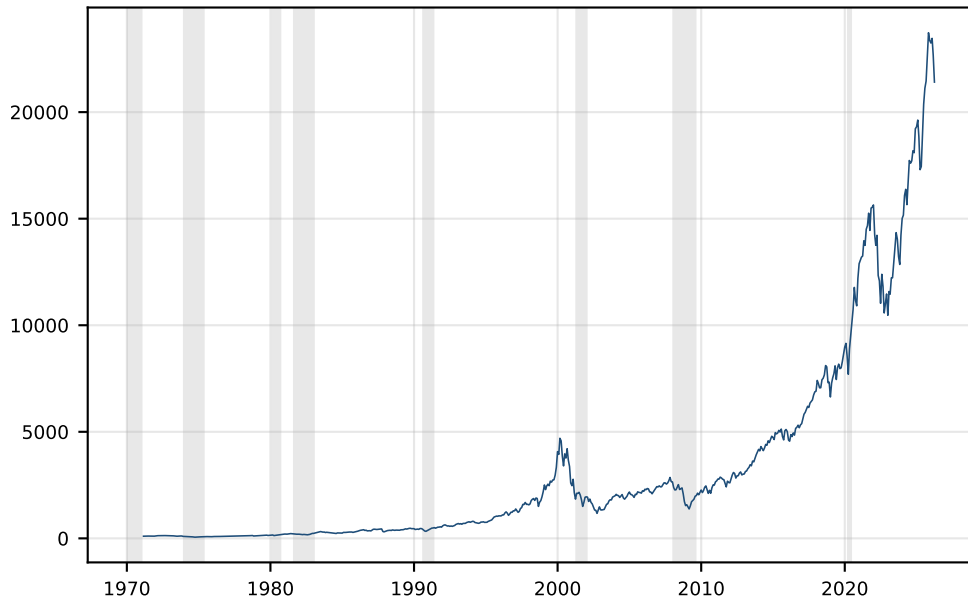
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	297	26.3%	73.7%	-0.0547
Q2	297	11.1%	88.9%	-0.0009
Q3	297	10.4%	89.6%	0.0253
Q4 (highest)	297	19.9%	80.1%	0.0677

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

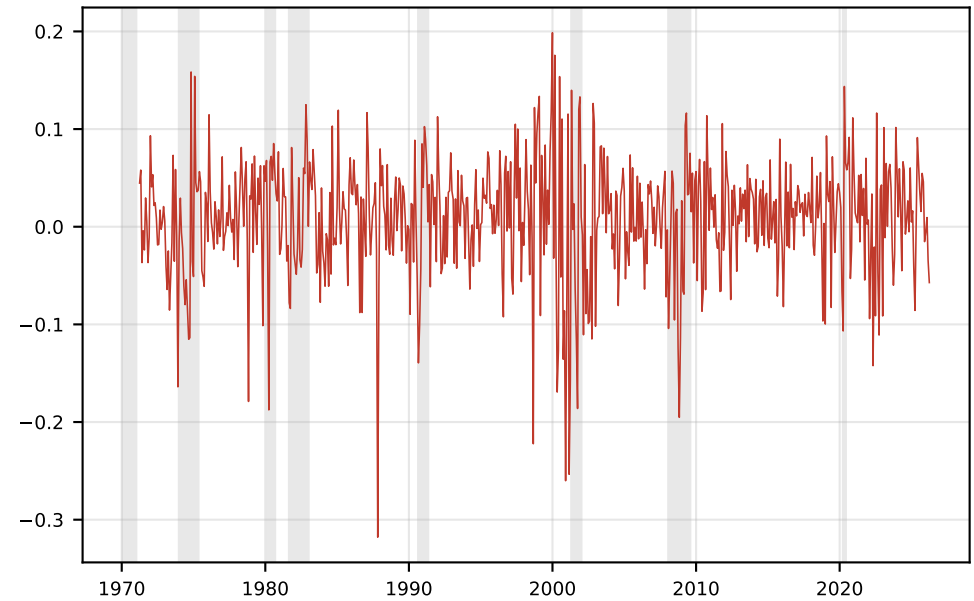
Nasdaq Composite

Block: Cross-Asset / Equity block | Source: FRED API | Col: NASDAQCOM | Transform: 5: Log diff

Raw Series



After: 5: Log diff



SUMMARY STATISTICS

N : 1287
Valid : 662
NaN : 625
Start : 1971-02
End : 2026-03
Mean : 3230.6061
Std : 4779.9135
Min : 55.6700
Max : 23724.9600
Skew : 2.232
Kurt : 4.624

STATIONARITY TESTS

ADF stat : -23.108
ADF p : 0.0000
ADF lags : 0
ADF : Stationary
KPSS stat : 0.068
KPSS p : 0.1000
KPSS : Stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

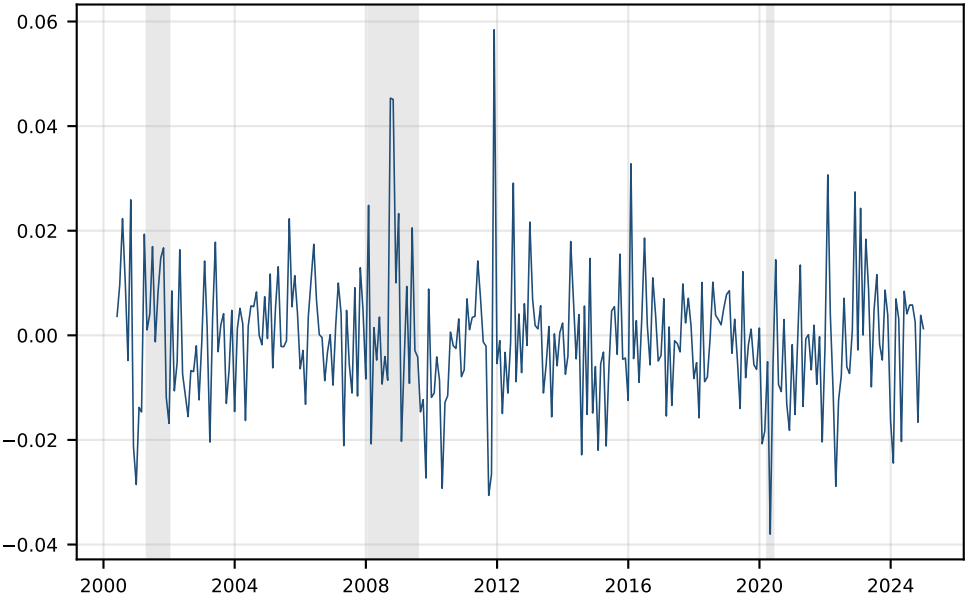
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	165	20.0%	80.0%	-0.0687
Q2	165	3.6%	96.4%	-0.0043
Q3	165	6.7%	93.3%	0.0297
Q4 (highest)	165	14.5%	85.5%	0.0761

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

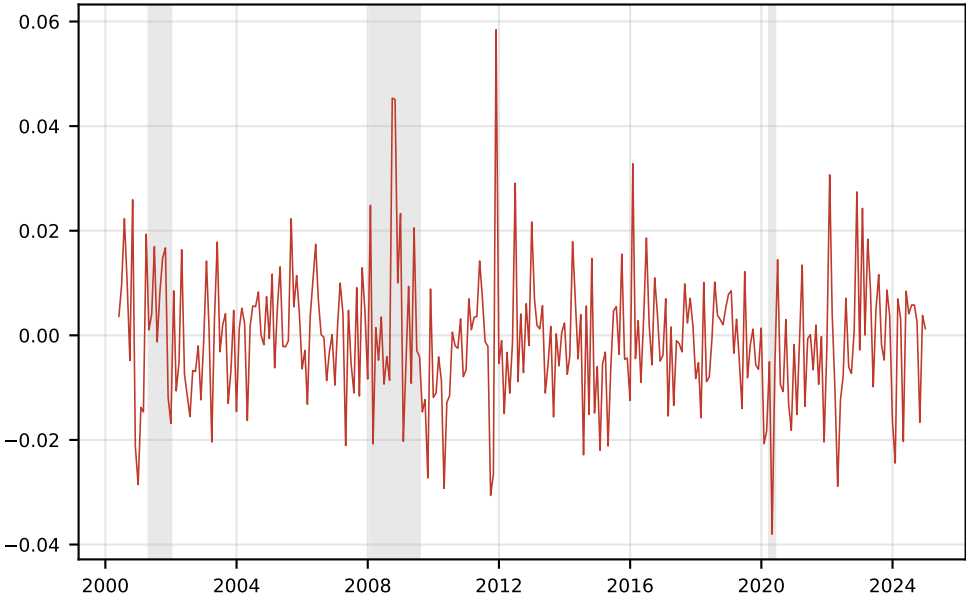
Russell 2000 (IWM ETF)

Block: Cross-Asset / Equity block | Source: WRDS (crsp.dsf IWM. Starts ~2000.) | Col: iwm_ret | Transform: 1: No transform

Raw Series



After: 1: No transform



SUMMARY STATISTICS

N : 1287
Valid : 296
NaN : 991
Start : 2000-05
End : 2024-12
Mean : -0.0007
Std : 0.0125
Min : -0.0380
Max : 0.0584
Skew : 0.622
Kurt : 2.435

STATIONARITY TESTS

ADF stat : -16.279
ADF p : 0.0000
ADF lags : 0
ADF : Stationary
KPSS stat : 0.170
KPSS p : 0.1000
KPSS : Stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

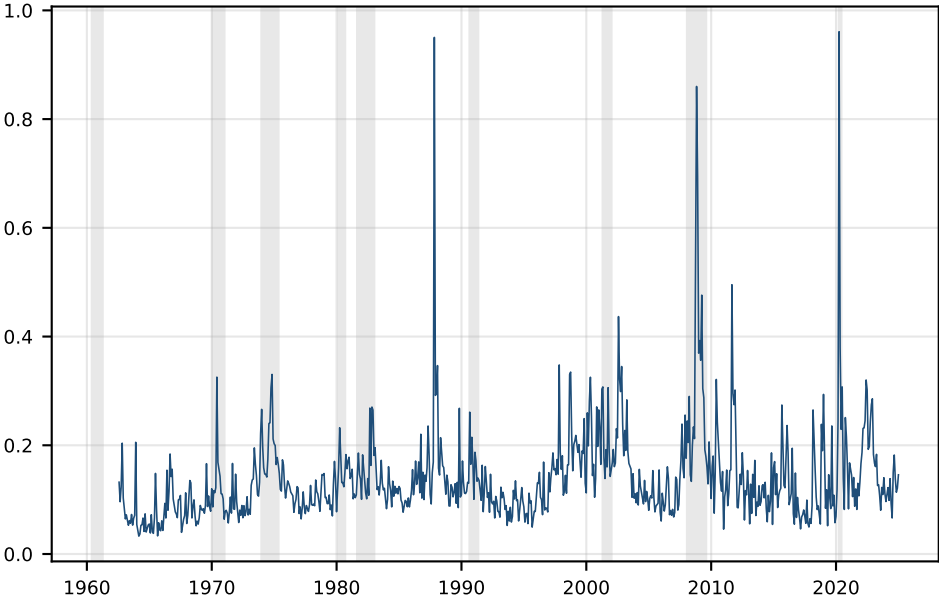
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	74	9.5%	90.5%	-0.0155
Q2	74	8.1%	91.9%	-0.0044
Q3	74	5.4%	94.6%	0.0024
Q4 (highest)	74	14.9%	85.1%	0.0146

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

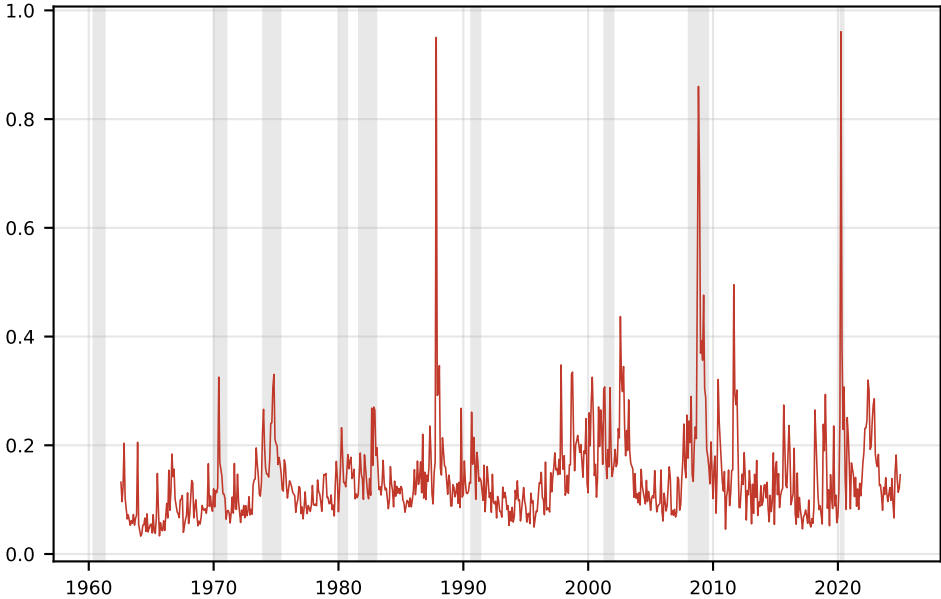
S&P 500 Realized Vol (20d)

Block: Cross-Asset / Equity block | Source: Computed | Col: rv_20d | Transform: 1: No transform

Raw Series



After: 1: No transform



SUMMARY STATISTICS

N : 1287
Valid : 750
NaN : 537
Start : 1962-07
End : 2024-12
Mean : 0.1389
Std : 0.0872
Min : 0.0328
Max : 0.9607
Skew : 4.060
Kurt : 28.998

STATIONARITY TESTS

ADF stat : -6.452
ADF p : 0.0000
ADF lags : 5
ADF : Stationary
KPSS stat : 0.812
KPSS p : 0.0100
KPSS : Non-stationary

QUARTILE-BUCKET ANALYSIS

(vs NBER recession/expansion)

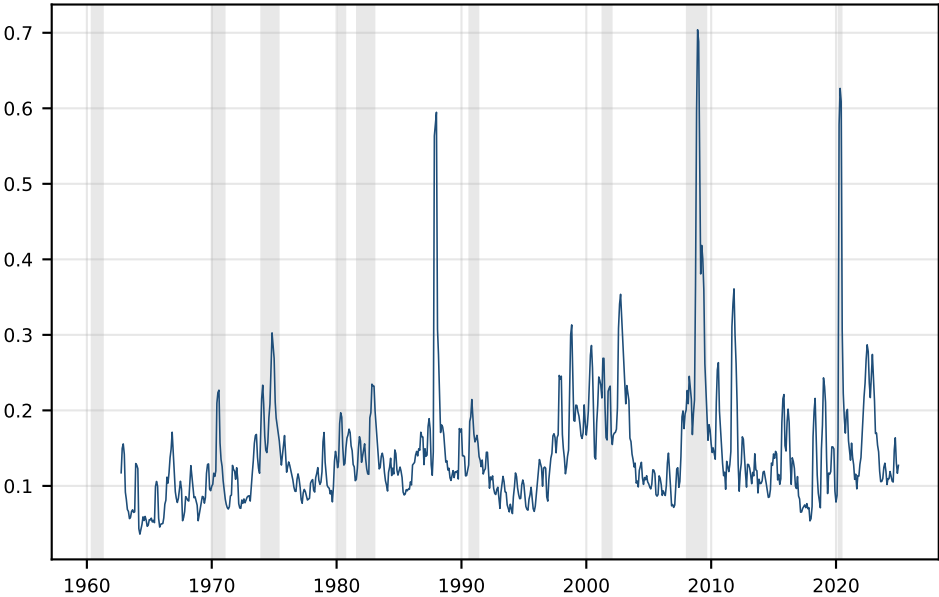
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	188	0.5%	99.5%	0.0696
Q2	187	5.3%	94.7%	0.1035
Q3	187	12.3%	87.7%	0.1402
Q4 (highest)	188	27.1%	72.9%	0.2423

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

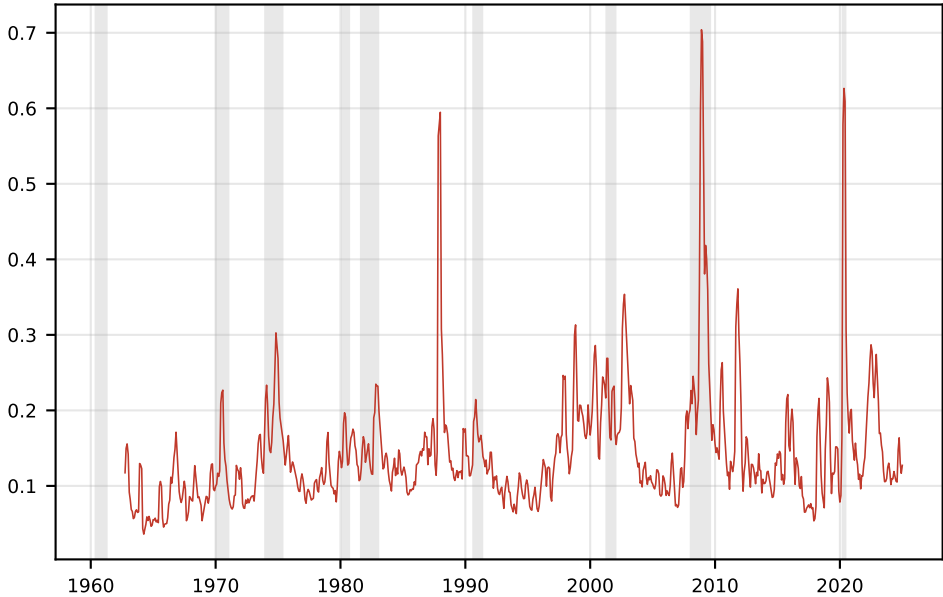
S&P 500 Realized Vol (60d)

Block: Cross-Asset / Equity block | Source: Computed | Col: rv_60d | Transform: 1: No transform

Raw Series



After: 1: No transform



SUMMARY STATISTICS

N : 1287
Valid : 748
NaN : 539
Start : 1962-09
End : 2024-12
Mean : 0.1437
Std : 0.0800
Min : 0.0361
Max : 0.7040
Skew : 3.130
Kurt : 14.860

STATIONARITY TESTS

ADF stat : -4.621
ADF p : 0.0001
ADF lags : 12
ADF : Stationary
KPSS stat : 0.776
KPSS p : 0.0100
KPSS : Non-stationary

QUARTILE-BUCKET ANALYSIS

(vs NBER recession/expansion)

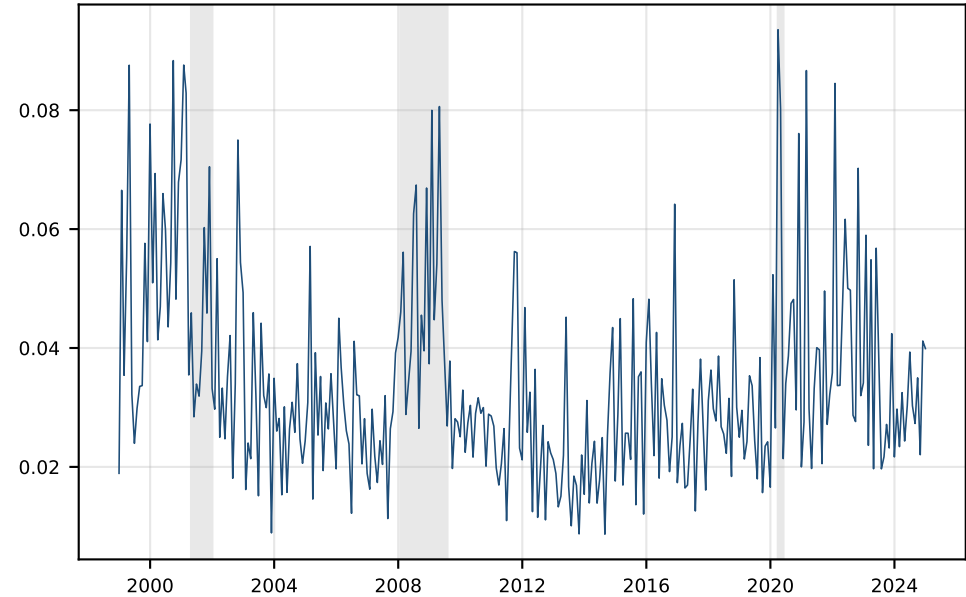
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	187	0.0%	100.0%	0.0783
Q2	187	4.8%	95.2%	0.1120
Q3	187	11.8%	88.2%	0.1423
Q4 (highest)	187	28.9%	71.1%	0.2421

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

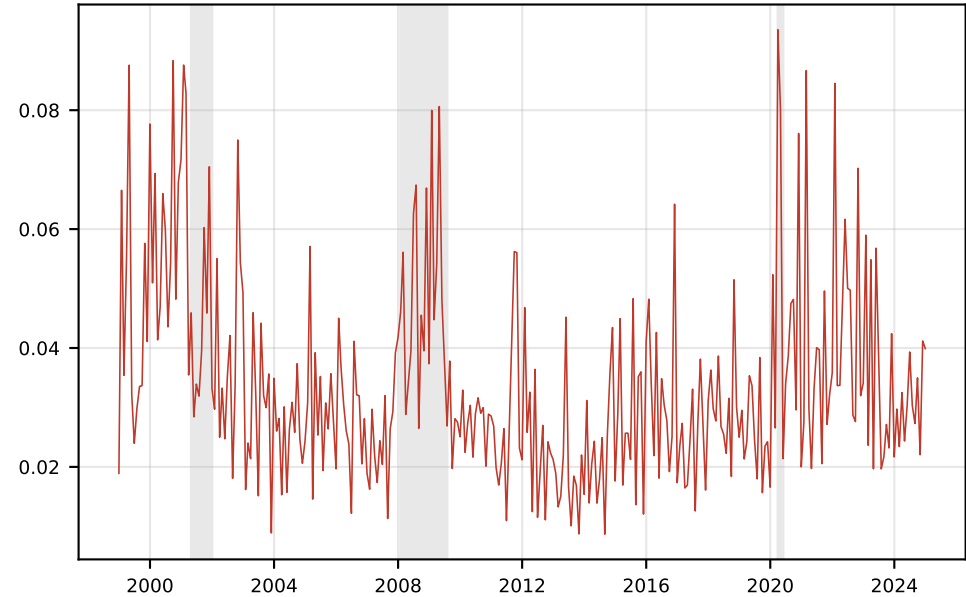
Sector Return Dispersion

Block: Cross-Asset / Equity block | Source: Computed | Col: sector_dispersion | Transform: 1: No transform

Raw Series



After: 1: No transform



SUMMARY STATISTICS

N : 1287
Valid : 313
NaN : 974
Start : 1998-12
End : 2024-12
Mean : 0.0339
Std : 0.0166
Min : 0.0087
Max : 0.0935
Skew : 1.315
Kurt : 1.690

STATIONARITY TESTS

ADF stat : -3.538
ADF p : 0.0071
ADF lags : 6
ADF : Stationary
KPSS stat : 0.468
KPSS p : 0.0489
KPSS : Non-stationary

QUARTILE-BUCKET ANALYSIS

(vs NBER recession/expansion)

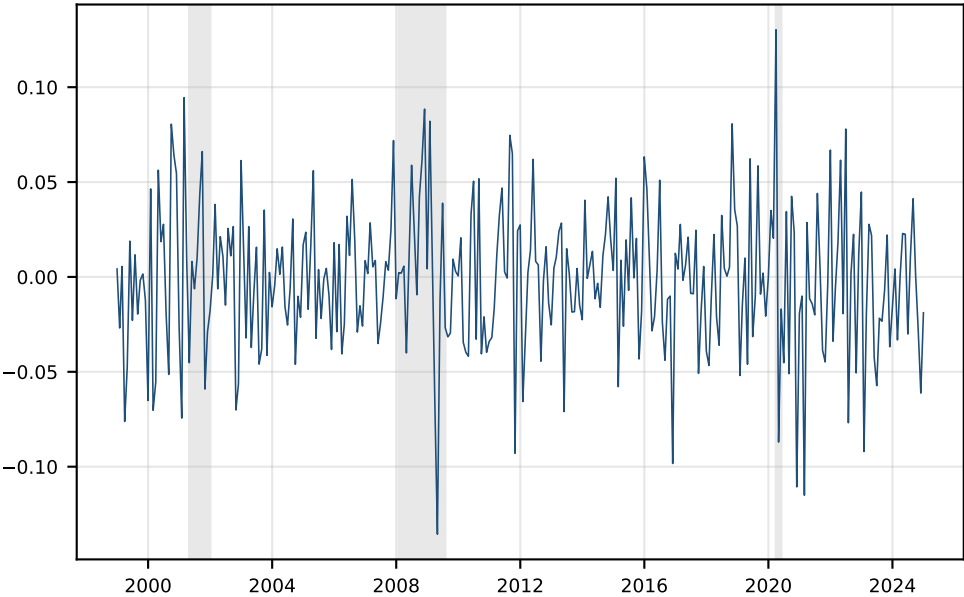
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	79	0.0%	100.0%	0.0175
Q2	78	3.8%	96.2%	0.0265
Q3	78	10.3%	89.7%	0.0344
Q4 (highest)	78	21.8%	78.2%	0.0573

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

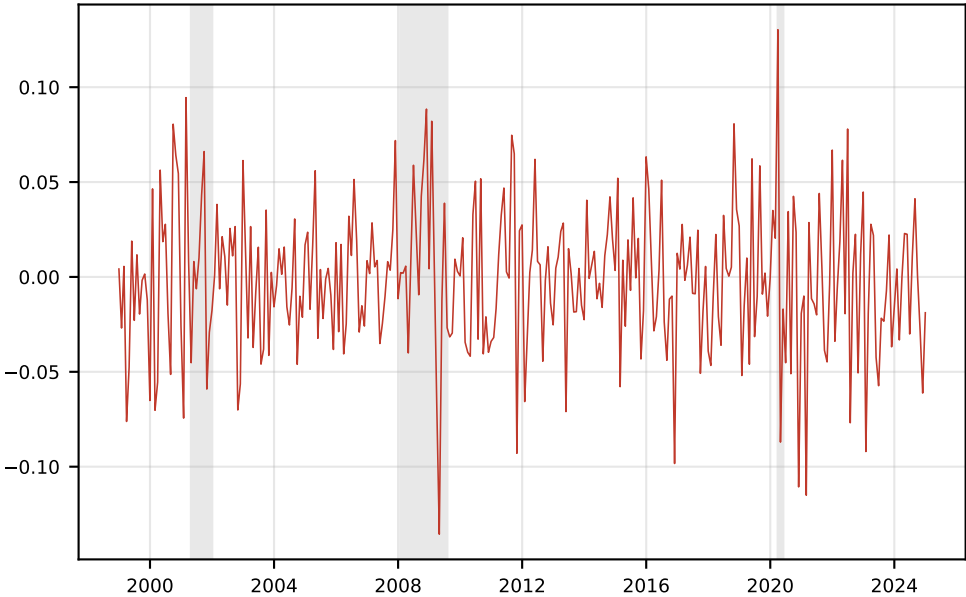
Defensive vs Cyclical Rel Perf

Block: Cross-Asset / Equity block | Source: Computed | Col: def_vs_cyc | Transform: 1: No transform

Raw Series



After: 1: No transform



SUMMARY STATISTICS

N : 1287
Valid : 313
NaN : 974
Start : 1998-12
End : 2024-12
Mean : -0.0018
Std : 0.0377
Min : -0.1356
Max : 0.1303
Skew : -0.086
Kurt : 0.796

STATIONARITY TESTS

ADF stat : -14.033
ADF p : 0.0000
ADF lags : 1
ADF : Stationary
KPSS stat : 0.097
KPSS p : 0.1000
KPSS : Stationary

QUARTILE-BUCKET ANALYSIS

(vs NBER recession/expansion)

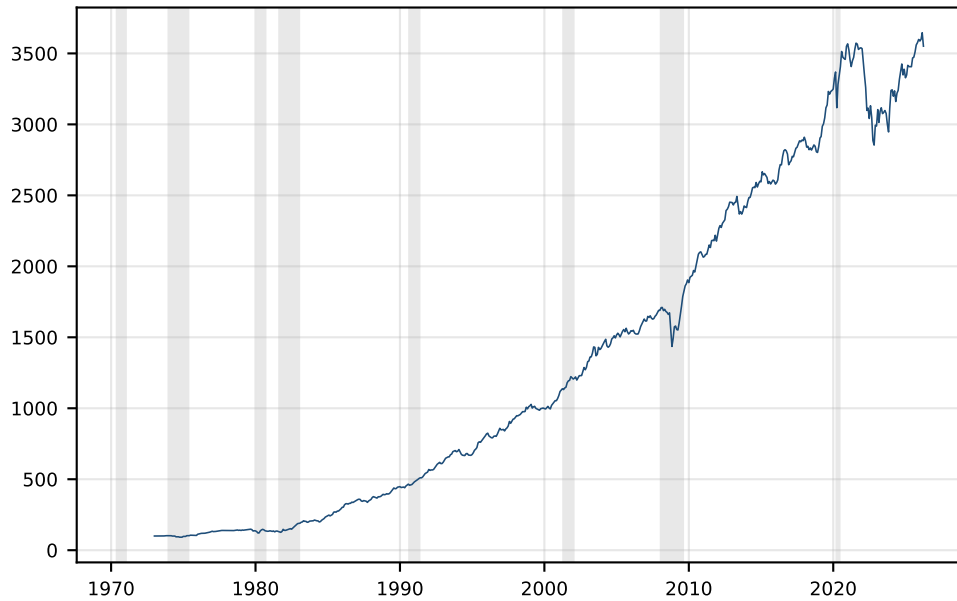
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	79	8.9%	91.1%	-0.0486
Q2	78	3.8%	96.2%	-0.0123
Q3	78	10.3%	89.7%	0.0095
Q4 (highest)	78	12.8%	87.2%	0.0449

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

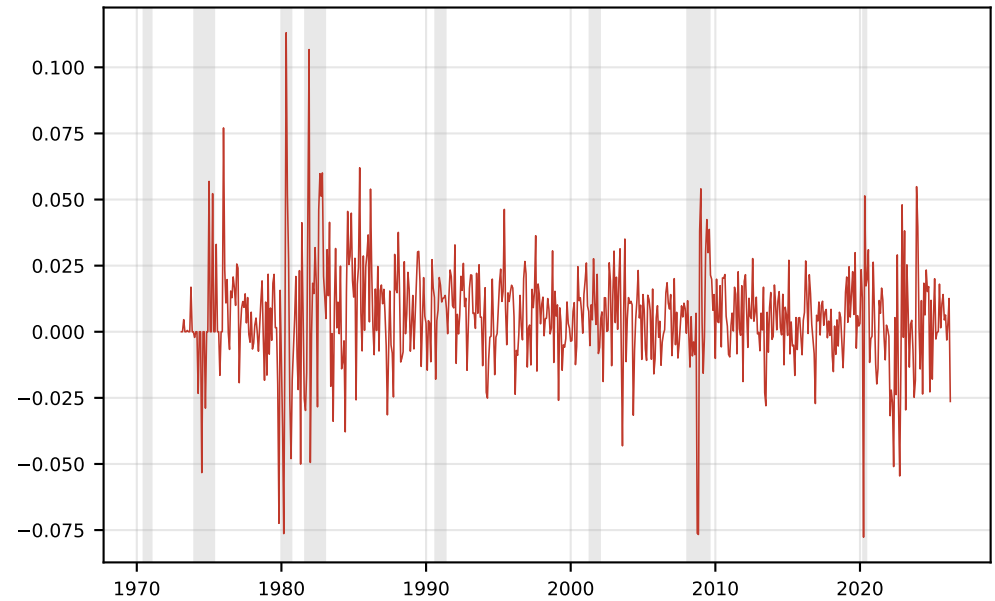
ICE BofA US Corp Total Return Index

Block: Cross-Asset / Bond block | Source: FRED API | Col: BAMLCC0A0CMTRIV | Transform: 5: Log diff

Raw Series



After: 5: Log diff



SUMMARY STATISTICS

N : 1287
Valid : 640
NaN : 647
Start : 1972-12
End : 2026-03
Mean : 1342.8317
Std : 1127.3427
Min : 91.7900
Max : 3645.6500
Skew : 0.568
Kurt : -1.067

STATIONARITY TESTS

ADF stat : -8.801
ADF p : 0.0000
ADF lags : 5
ADF : Stationary
KPSS stat : 0.510
KPSS p : 0.0394
KPSS : Non-stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

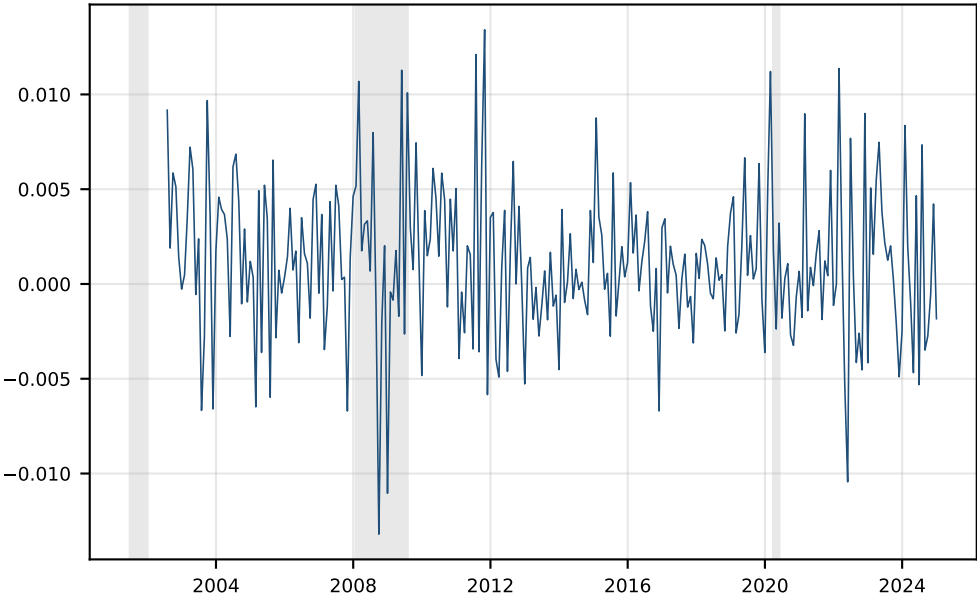
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	160	12.5%	87.5%	-0.0170
Q2	159	11.9%	88.1%	0.0012
Q3	159	6.3%	93.7%	0.0104
Q4 (highest)	160	15.6%	84.4%	0.0280

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

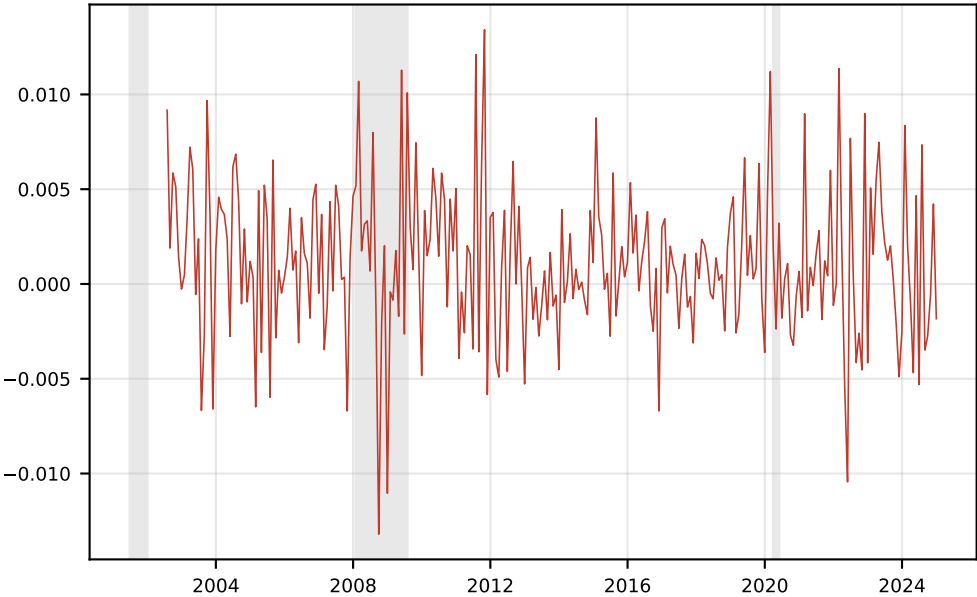
iShares 7-10Y Treasury ETF

Block: Cross-Asset / Bond block | Source: WRDS (crsp.dsf IEF. Intermediate Treasury proxy) | Col: ief_ret | Transform: 1: No transform

Raw Series



After: 1: No transform



SUMMARY STATISTICS

N : 1287
Valid : 270
NaN : 1017
Start : 2002-07
End : 2024-12
Mean : 0.0011
Std : 0.0040
Min : -0.0132
Max : 0.0134
Skew : 0.070
Kurt : 0.953

STATIONARITY TESTS

ADF stat : -7.399
ADF p : 0.0000
ADF lags : 7
ADF : Stationary
KPSS stat : 0.275
KPSS p : 0.1000
KPSS : Stationary

QUARTILE-BUCKET ANALYSIS

(vs NBER recession/expansion)

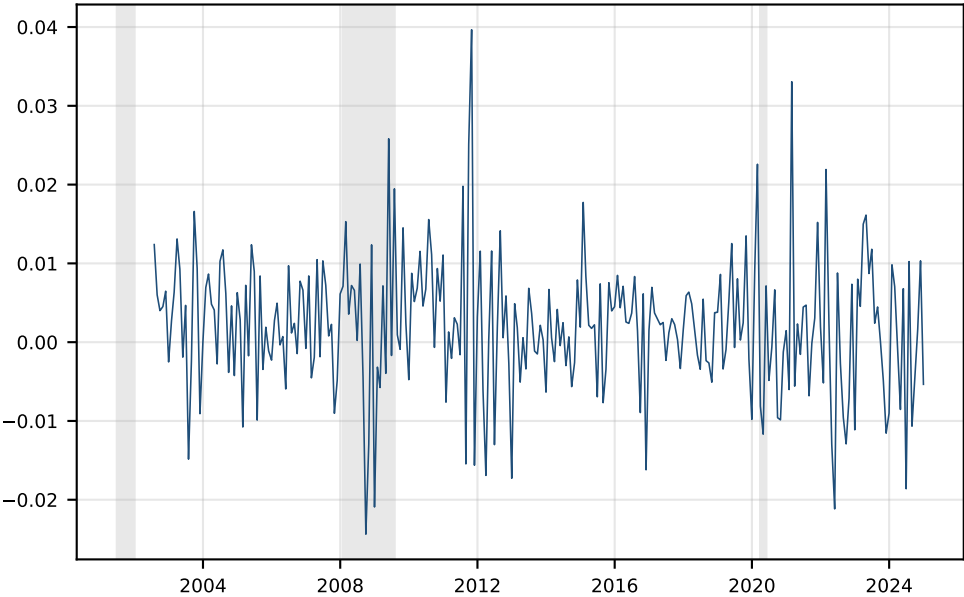
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	68	10.3%	89.7%	-0.0037
Q2	67	4.5%	95.5%	-0.0000
Q3	67	9.0%	91.0%	0.0021
Q4 (highest)	68	5.9%	94.1%	0.0062

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

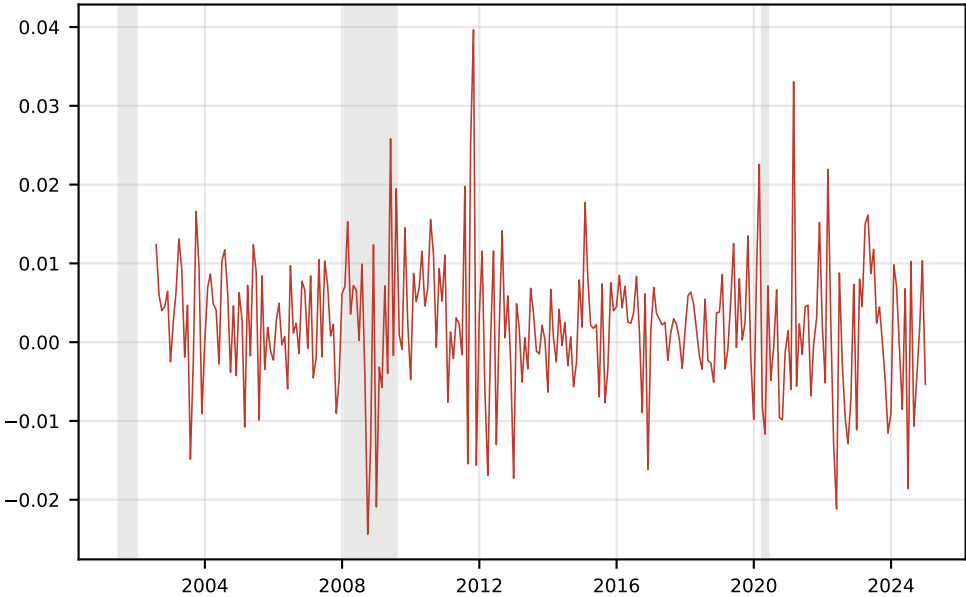
iShares 20+Y Treasury ETF

Block: Cross-Asset / Bond block | Source: WRDS (crsp.dsf TLT. Long Treasury proxy) | Col: tlt_ret | Transform: 1: No transform

Raw Series



After: 1: No transform



SUMMARY STATISTICS

N : 1287
Valid : 270
NaN : 1017
Start : 2002-07
End : 2024-12
Mean : 0.0021
Std : 0.0086
Min : -0.0244
Max : 0.0397
Skew : 0.243
Kurt : 2.011

STATIONARITY TESTS

ADF stat : -7.490
ADF p : 0.0000
ADF lags : 7
ADF : Stationary
KPSS stat : 0.275
KPSS p : 0.1000
KPSS : Stationary

QUARTILE-BUCKET ANALYSIS
(vs NBER recession/expansion)

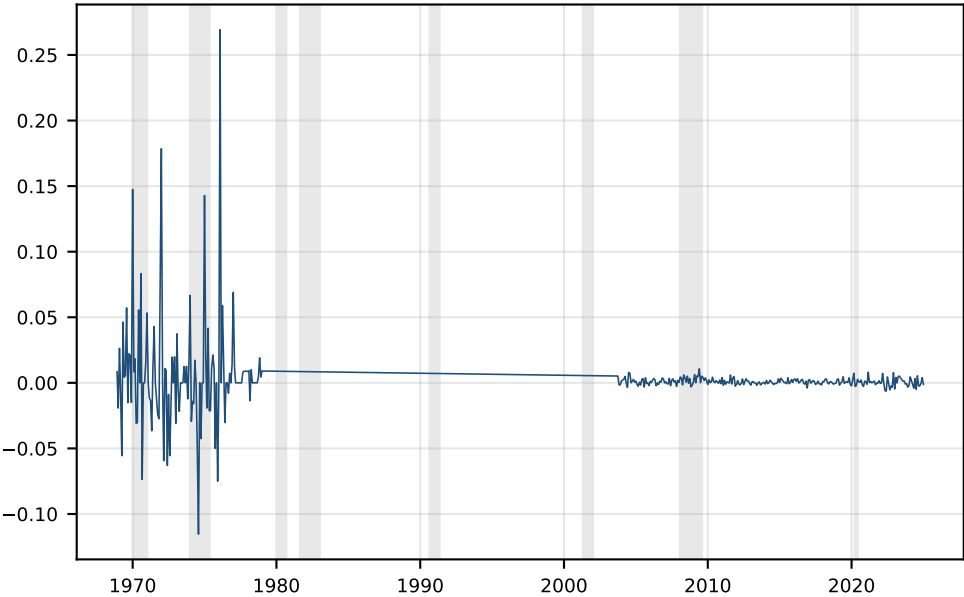
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	68	13.2%	86.8%	-0.0084
Q2	67	3.0%	97.0%	-0.0001
Q3	67	3.0%	97.0%	0.0045
Q4 (highest)	68	10.3%	89.7%	0.0122

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

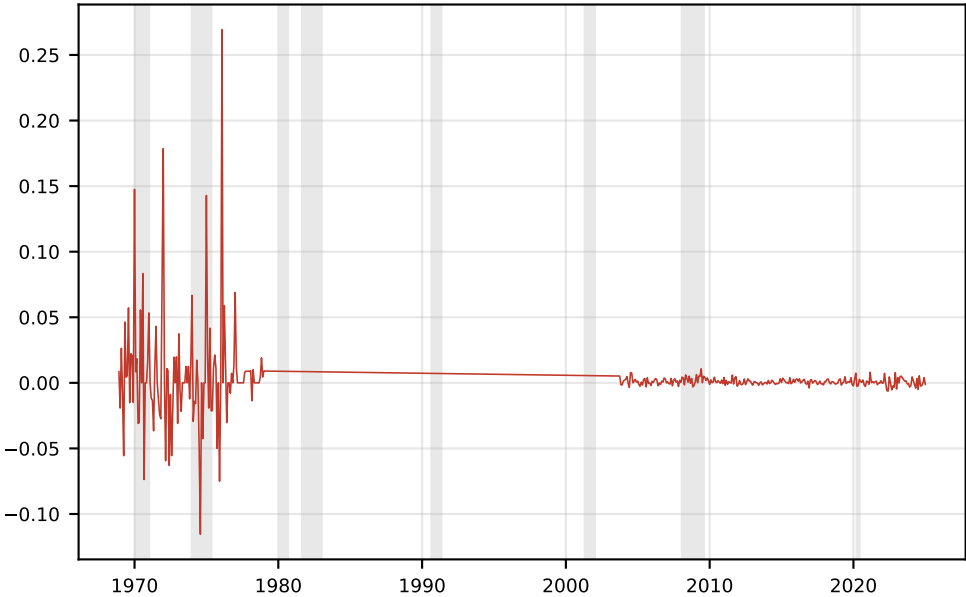
iShares US Agg Bond ETF

Block: Cross-Asset / Bond block | Source: WRDS (crsp.dsf AGG. Broad bond market proxy) | Col: agg_ret | Transform: 1: No transform

Raw Series



After: 1: No transform



SUMMARY STATISTICS

N : 1287
Valid : 378
NaN : 909
Start : 1968-11
End : 2024-12
Mean : 0.0025
Std : 0.0258
Min : -0.1154
Max : 0.2692
Skew : 4.413
Kurt : 41.798

STATIONARITY TESTS

ADF stat : -4.718
ADF p : 0.0001
ADF lags : 12
ADF : Stationary
KPSS stat : 0.334
KPSS p : 0.1000
KPSS : Stationary

QUARTILE-BUCKET ANALYSIS
(vs NBER recession/expansion)

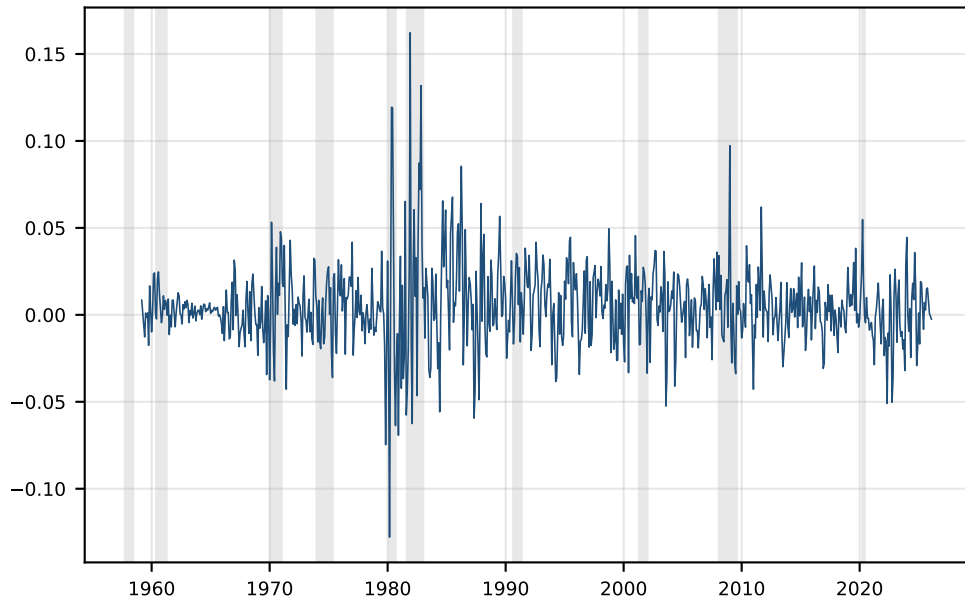
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	95	15.8%	84.2%	-0.0135
Q2	94	10.6%	89.4%	-0.0001
Q3	94	2.1%	97.9%	0.0016
Q4 (highest)	95	21.1%	78.9%	0.0221

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

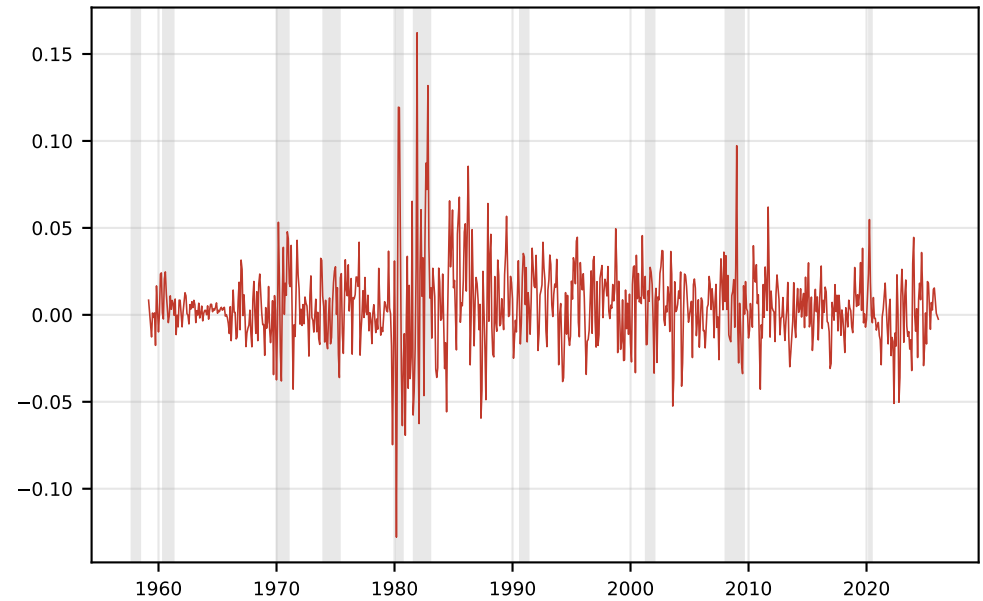
10Y Treasury Approx Total Return

Block: Cross-Asset / Bond block | Source: Computed | Col: bond_10y_ret | Transform: 1: No transform

Raw Series



After: 1: No transform



SUMMARY STATISTICS

N : 1287
Valid : 804
NaN : 483
Start : 1959-02
End : 2026-01
Mean : 0.0048
Std : 0.0233
Min : -0.1278
Max : 0.1622
Skew : 0.710
Kurt : 6.663

STATIONARITY TESTS

ADF stat : -7.181
ADF p : 0.0000
ADF lags : 11
ADF : Stationary
KPSS stat : 0.268
KPSS p : 0.1000
KPSS : Stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

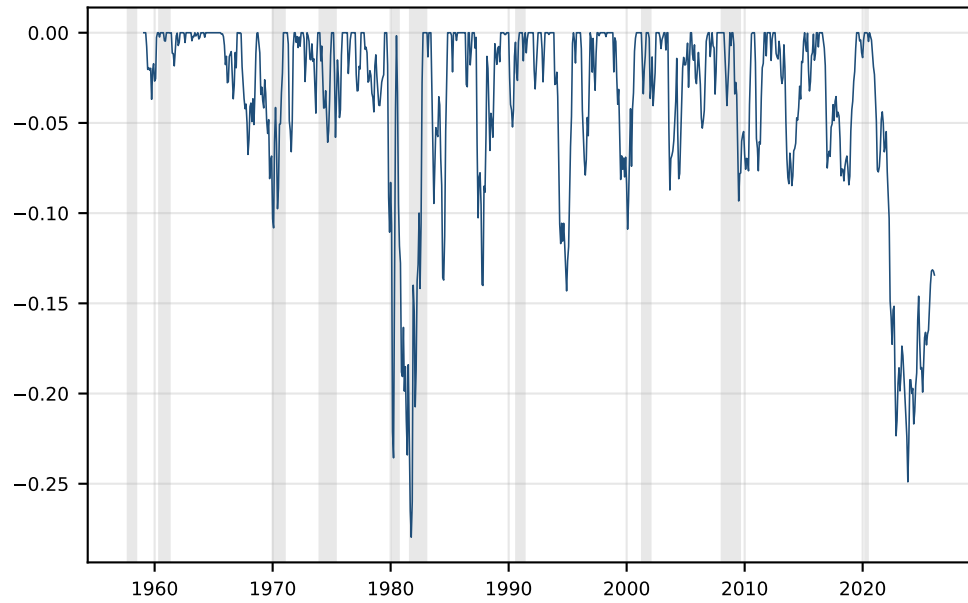
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	201	12.9%	87.1%	-0.0215
Q2	201	7.5%	92.5%	-0.0012
Q3	201	8.5%	91.5%	0.0095
Q4 (highest)	201	18.4%	81.6%	0.0323

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

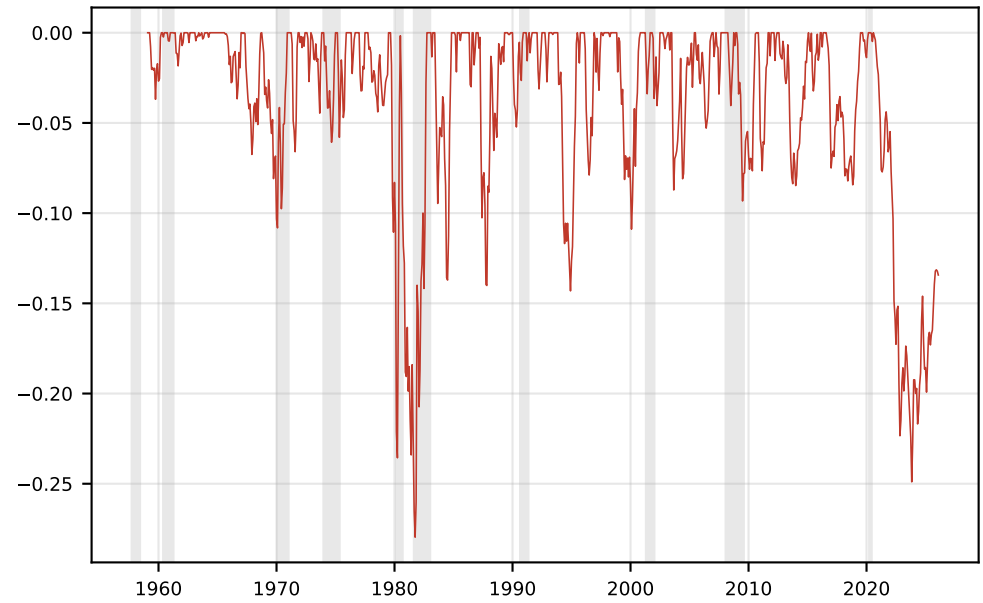
10Y Treasury Drawdown

Block: Cross-Asset / Bond block | Source: Computed | Col: bond_10y_drawdown | Transform: 1: No transform

Raw Series



After: 1: No transform



SUMMARY STATISTICS

N : 1287
Valid : 805
NaN : 482
Start : 1959-01
End : 2026-01
Mean : -0.0395
Std : 0.0544
Min : -0.2796
Max : 0.0000
Skew : -1.907
Kurt : 3.310

STATIONARITY TESTS

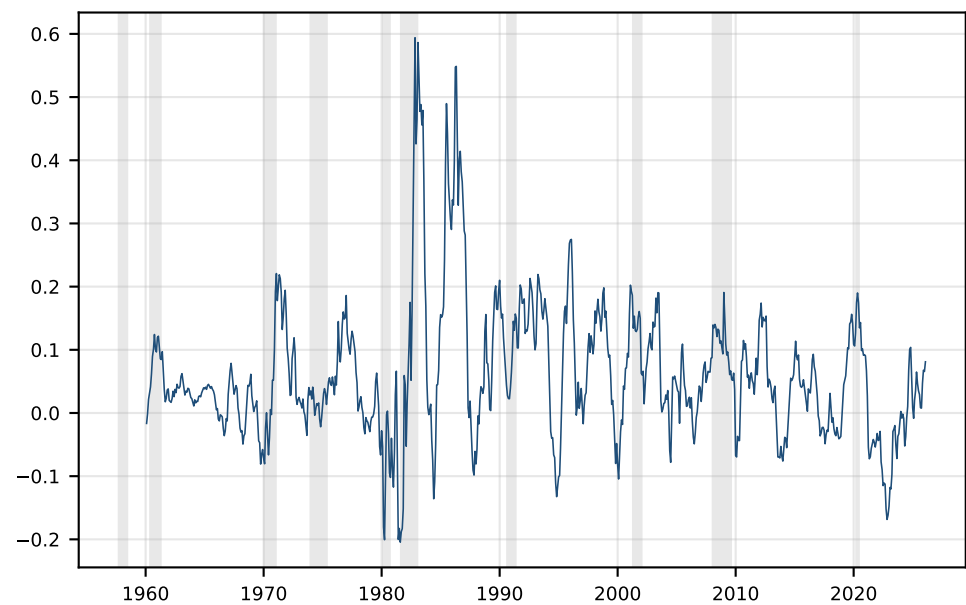
ADF stat : -3.863
ADF p : 0.0023
ADF lags : 8
ADF : Stationary
KPSS stat : 0.593
KPSS p : 0.0233
KPSS : Non-stationary

Quartile analysis not available

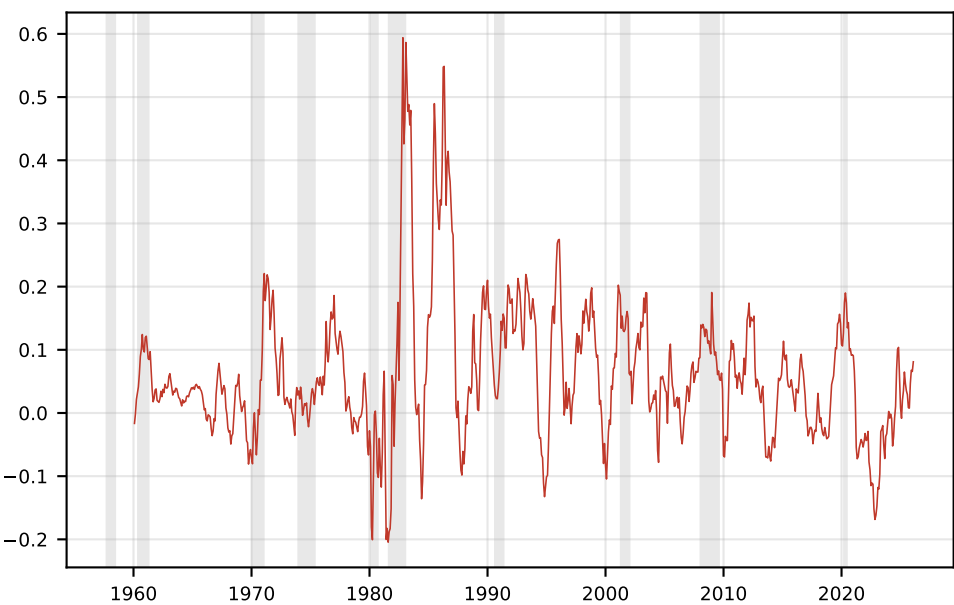
10Y Treasury Momentum (12m)

Block: Cross-Asset / Bond block | Source: Computed | Col: bond_10y_mom_12m | Transform: 1: No transform

Raw Series



After: 1: No transform



SUMMARY STATISTICS

N : 1287
Valid : 793
NaN : 494
Start : 1960-01
End : 2026-01
Mean : 0.0612
Std : 0.1095
Min : -0.2045
Max : 0.5939
Skew : 1.411
Kurt : 4.171

STATIONARITY TESTS

ADF stat : -5.038
ADF p : 0.0000
ADF lags : 12
ADF : Stationary
KPSS stat : 0.285
KPSS p : 0.1000
KPSS : Stationary

QUARTILE-BUCKET ANALYSIS

(vs NBER recession/expansion)

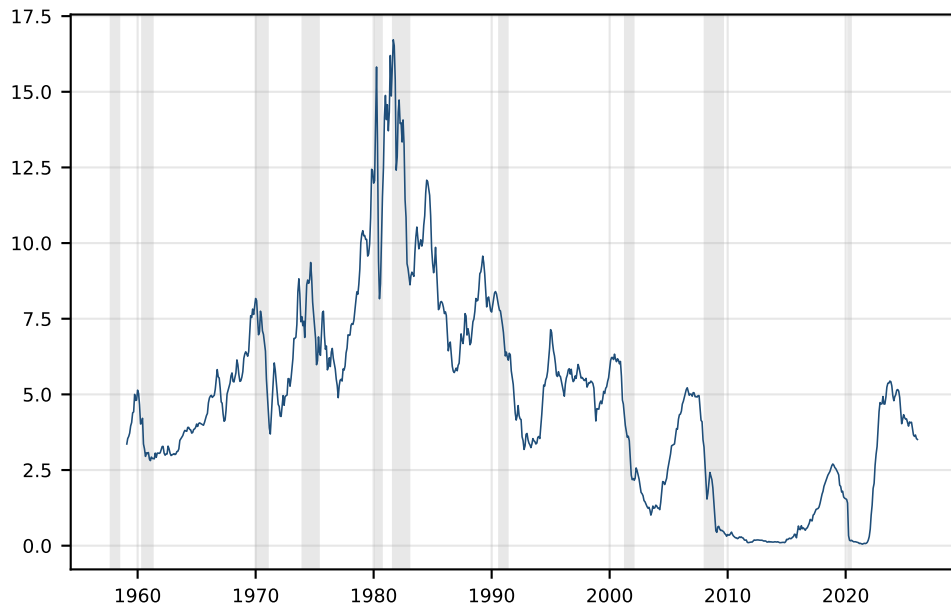
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	199	9.5%	90.5%	-0.0521
Q2	198	10.6%	89.4%	0.0231
Q3	198	11.6%	88.4%	0.0722
Q4 (highest)	198	16.2%	83.8%	0.2022

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

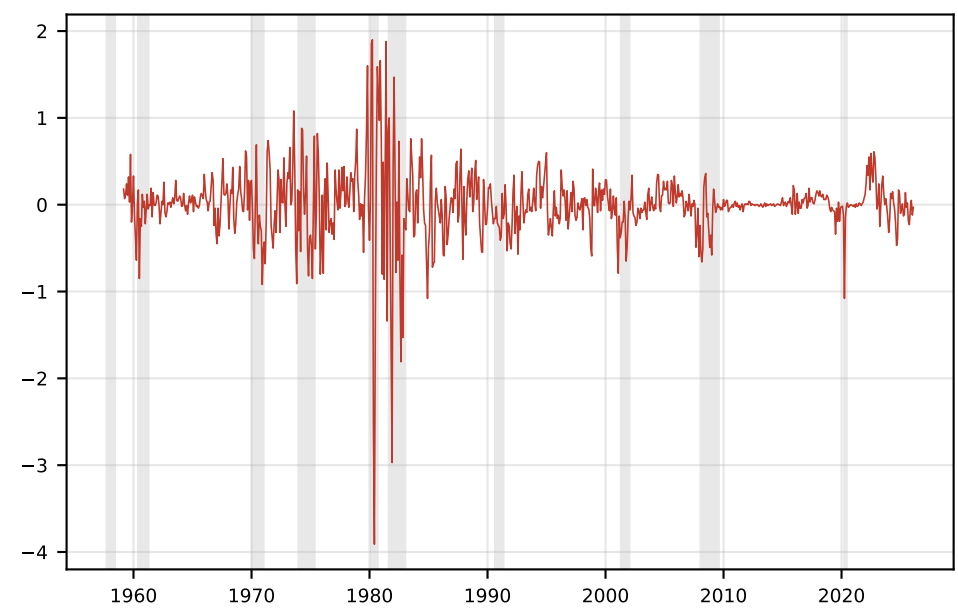
1Y Treasury Yield

Block: Cross-Asset / Rates | Source: FRED-MD | Col: GS1 | Transform: 2: First diff

Raw Series



After: 2: First diff



SUMMARY STATISTICS

N : 1287
Valid : 805
NaN : 482
Start : 1959-01
End : 2026-01
Mean : 4.8288
Std : 3.2973
Min : 0.0500
Max : 16.7200
Skew : 0.728
Kurt : 0.767

STATIONARITY TESTS

ADF stat : -6.809
ADF p : 0.0000
ADF lags : 12
ADF : Stationary
KPSS stat : 0.071
KPSS p : 0.1000
KPSS : Stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

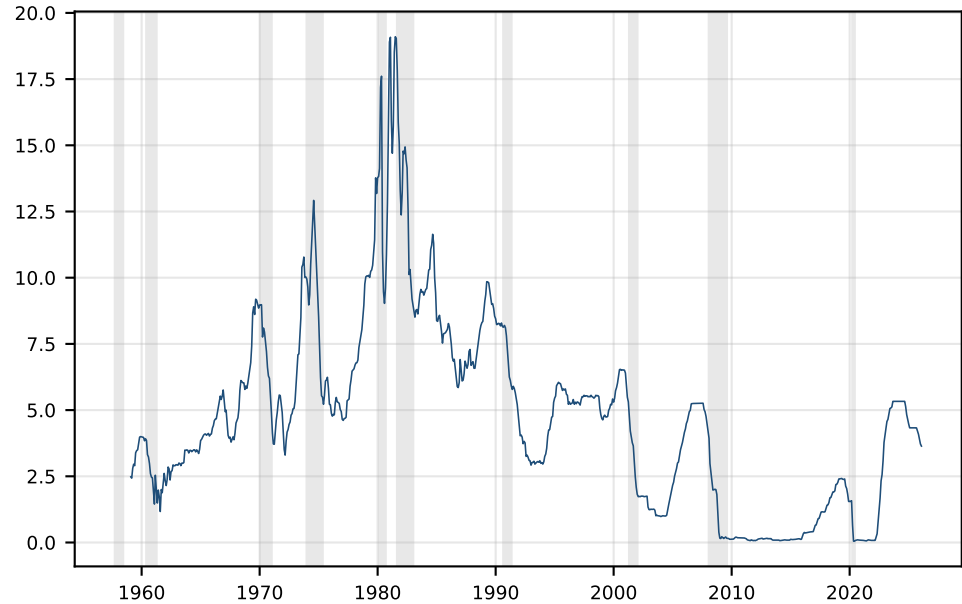
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	202	27.2%	72.8%	-0.4200
Q2	206	4.9%	95.1%	-0.0406
Q3	197	5.6%	94.4%	0.0613
Q4 (highest)	199	9.5%	90.5%	0.4085

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

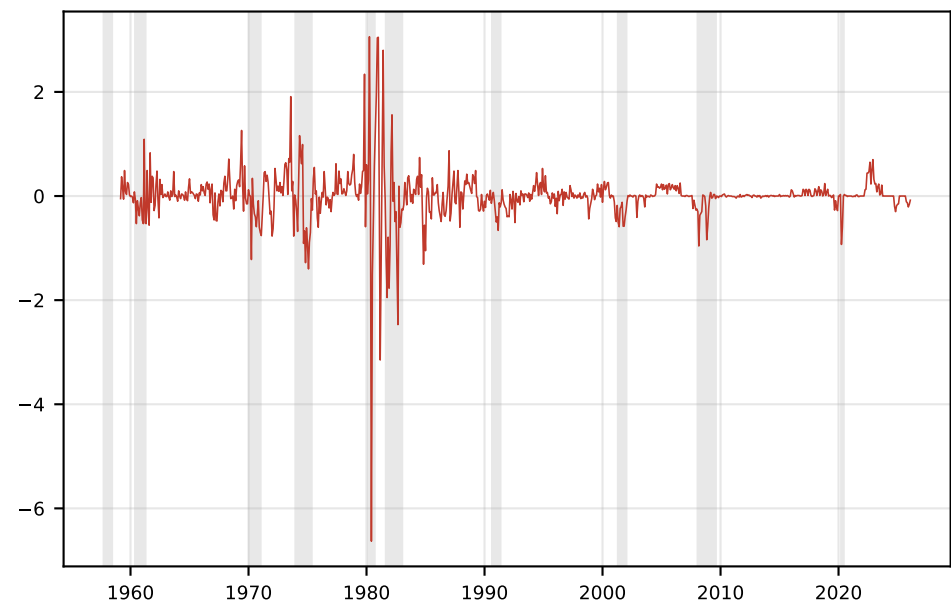
Effective Fed Funds Rate

Block: Cross-Asset / Rates | Source: FRED-MD | Col: FEDFUNDS | Transform: 2: First diff

Raw Series



After: 2: First diff



SUMMARY STATISTICS

N : 1287
Valid : 805
NaN : FEDFUNDS 482
FEDFUNDS 427
dtype: int64
Start : 1959-01
End : 2026-01
Mean : 4.7669
Std : 3.5925
Min : 0.0500
Max : 19.1000
Skew : 0.994
Kurt : 1.542

STATIONARITY TESTS

ADF stat : -6.859
ADF p : 0.0000
ADF lags : 12
ADF : Stationary
KPSS stat : 0.058
KPSS p : 0.1000
KPSS : Stationary

QUARTILE-BUCKET ANALYSIS

(vs NBER recession/expansion)

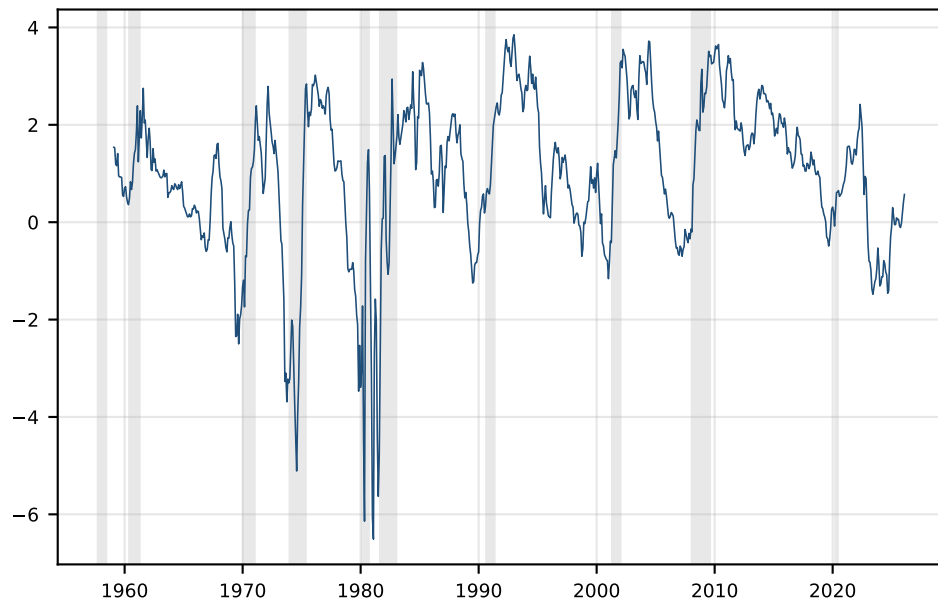
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	202	32.7%	67.3%	-0.4390
Q2	205	3.9%	96.1%	-0.0159
Q3	196	3.6%	96.4%	0.0423
Q4 (highest)	201	7.0%	93.0%	0.4218

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

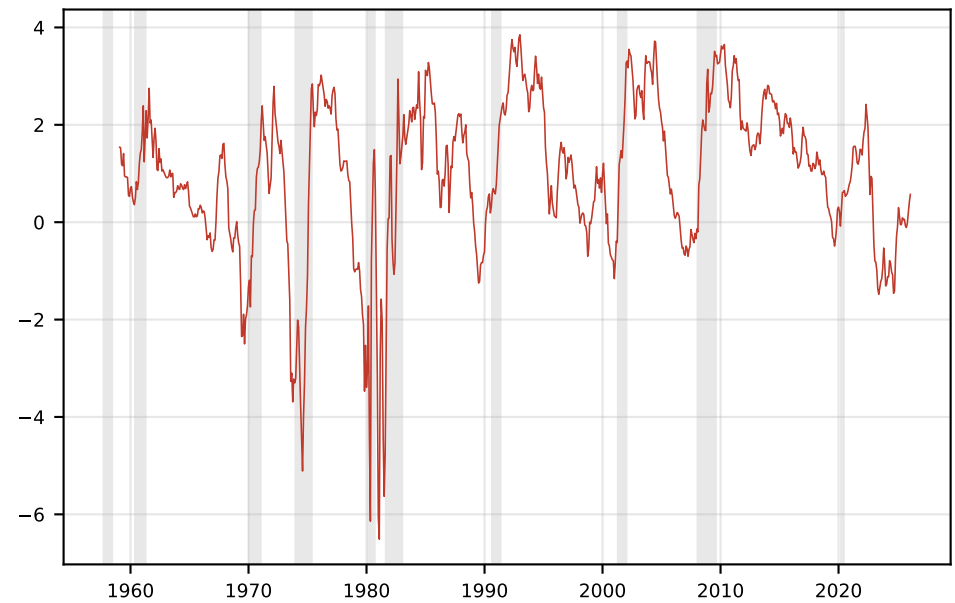
10Y Treasury minus Fed Funds Spread

Block: Cross-Asset / Rates | Source: FRED-MD | Col: T10YFFM | Transform: 1: No transform

Raw Series



After: 1: No transform



SUMMARY STATISTICS

N : 1287
Valid : 805
NaN : 482
Start : 1959-01
End : 2026-01
Mean : 0.9782
Std : 1.5905
Min : -6.5100
Max : 3.8500
Skew : -1.034
Kurt : 2.181

STATIONARITY TESTS

ADF stat : -5.096
ADF p : 0.0000
ADF lags : 9
ADF : Stationary
KPSS stat : 0.364
KPSS p : 0.0928
KPSS : Stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

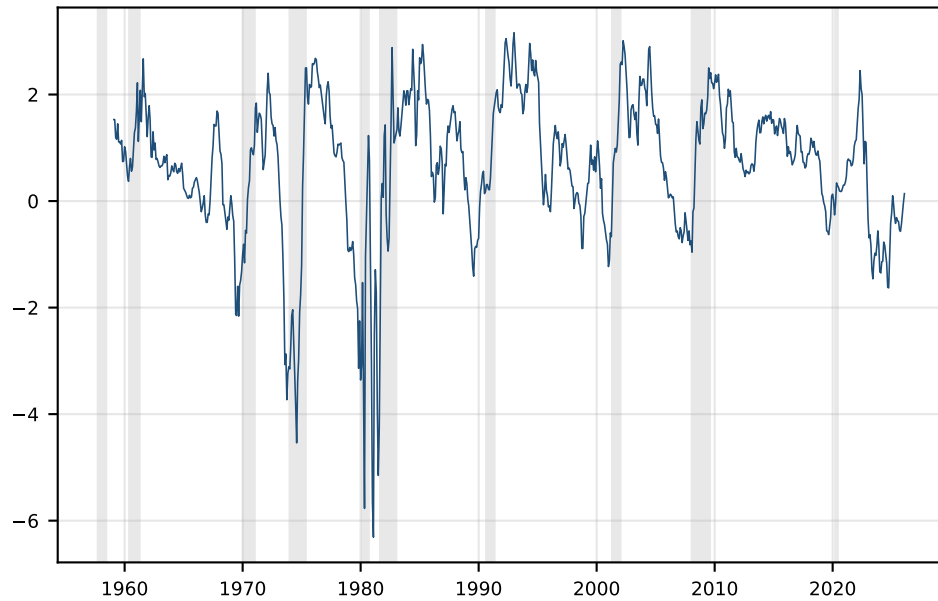
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	202	15.3%	84.7%	-1.0948
Q2	201	10.9%	89.1%	0.6506
Q3	201	14.4%	85.6%	1.5918
Q4 (highest)	201	6.5%	93.5%	2.7754

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

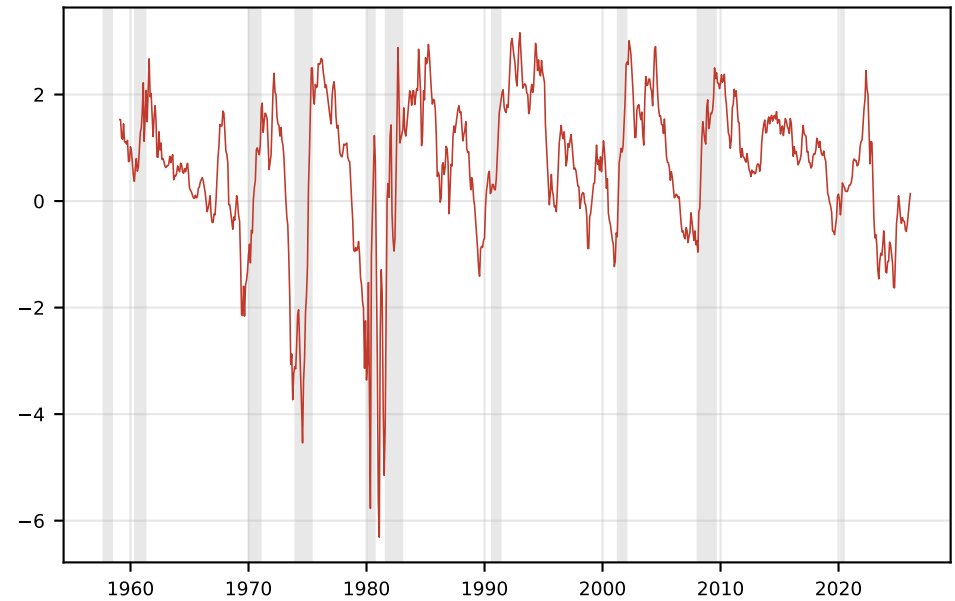
5Y Treasury minus Fed Funds Spread

Block: Cross-Asset / Rates | Source: FRED-MD | Col: T5YFFM | Transform: 1: No transform

Raw Series



After: 1: No transform



SUMMARY STATISTICS

N : 1287
Valid : 805
NaN : 482
Start : 1959-01
End : 2026-01
Mean : 0.6678
Std : 1.3276
Min : -6.3100
Max : 3.1600
Skew : -1.267
Kurt : 3.198

STATIONARITY TESTS

ADF stat : -5.336
ADF p : 0.0000
ADF lags : 9
ADF : Stationary
KPSS stat : 0.158
KPSS p : 0.1000
KPSS : Stationary

QUARTILE-BUCKET ANALYSIS

(vs NBER recession/expansion)

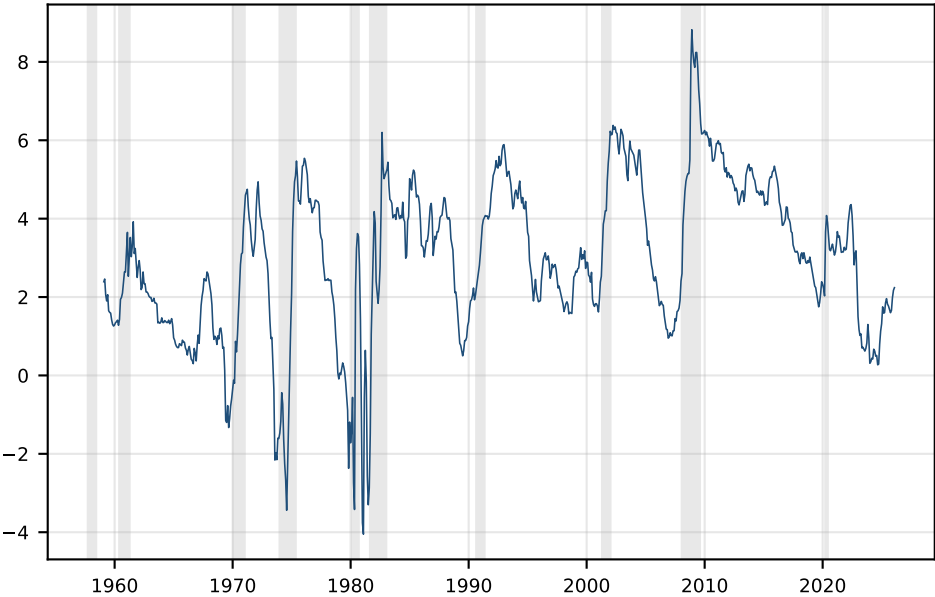
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	203	16.3%	83.7%	-1.0663
Q2	203	10.3%	89.7%	0.4746
Q3	198	13.6%	86.4%	1.1813
Q4 (highest)	201	7.0%	93.0%	2.1085

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

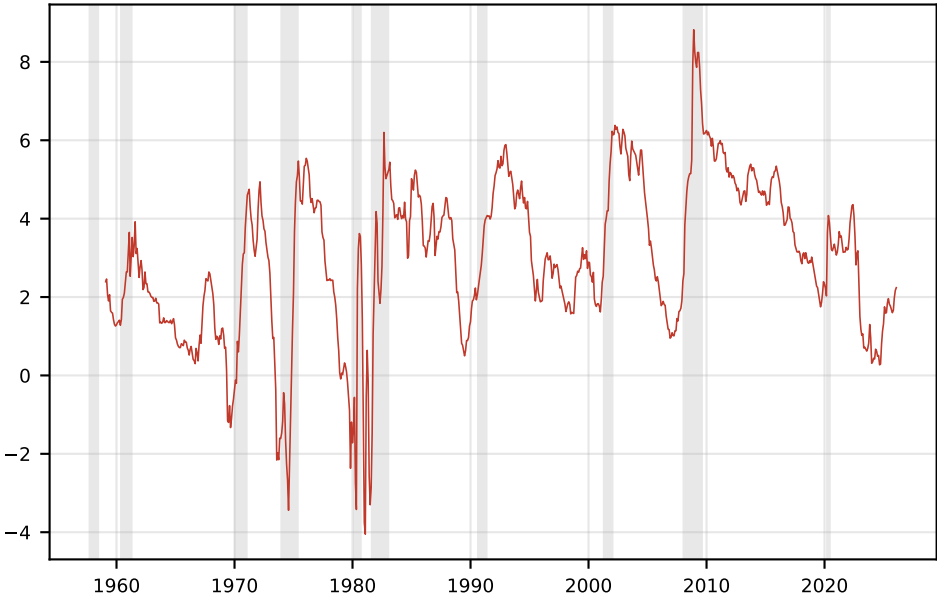
Baa Corp minus Fed Funds Spread

Block: Cross-Asset / Rates | Source: FRED-MD | Col: BAAFFM | Transform: 1: No transform

Raw Series



After: 1: No transform



SUMMARY STATISTICS

N : 1287
Valid : 805
NaN : 482
Start : 1959-01
End : 2026-01
Mean : 2.9683
Std : 1.9891
Min : -4.0500
Max : 8.8200
Skew : -0.355
Kurt : 0.371

STATIONARITY TESTS

ADF stat : -4.224
ADF p : 0.0006
ADF lags : 11
ADF : Stationary
KPSS stat : 0.922
KPSS p : 0.0100
KPSS : Non-stationary

QUARTILE-BUCKET ANALYSIS

(vs NBER recession/expansion)

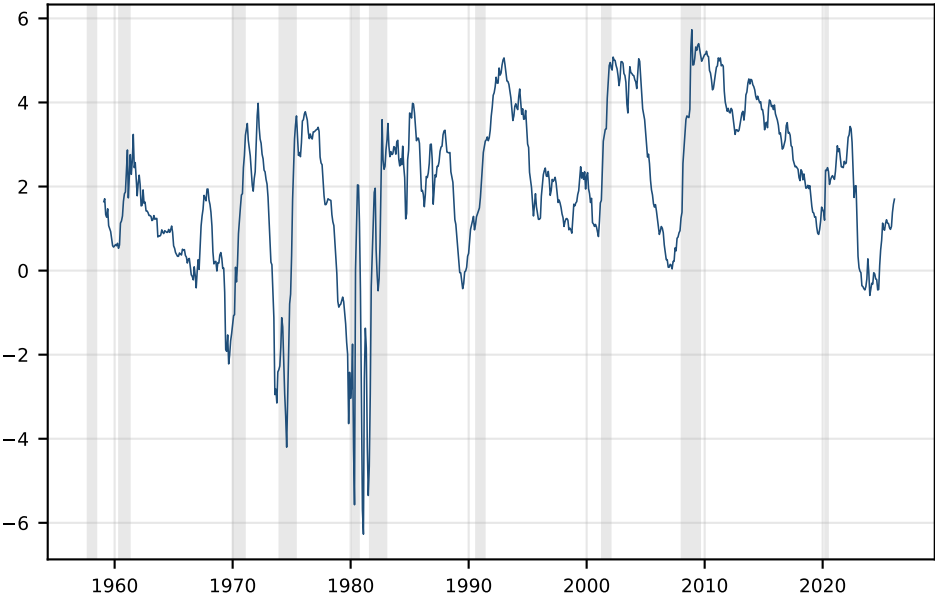
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	202	11.9%	88.1%	0.4258
Q2	201	10.9%	89.1%	2.3272
Q3	201	12.9%	87.1%	3.7510
Q4 (highest)	201	11.4%	88.6%	5.3820

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

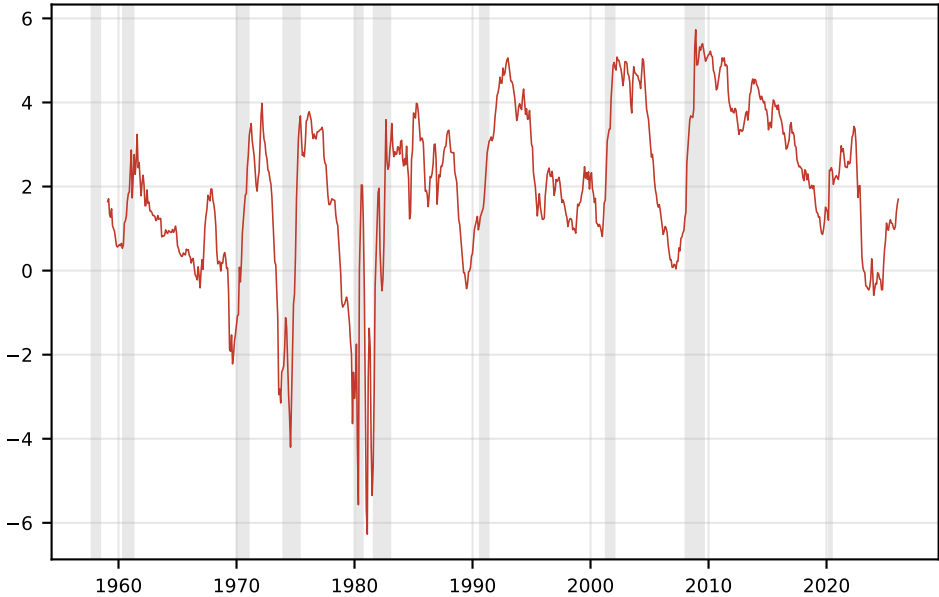
Aaa Corp minus Fed Funds Spread

Block: Cross-Asset / Rates | Source: FRED-MD | Col: AAAFFM | Transform: 1: No transform

Raw Series



After: 1: No transform



SUMMARY STATISTICS

N : 1287
Valid : 805
NaN : 482
Start : 1959-01
End : 2026-01
Mean : 1.9762
Std : 1.8875
Min : -6.2700
Max : 5.7300
Skew : -0.742
Kurt : 1.255

STATIONARITY TESTS

ADF stat : -4.025
ADF p : 0.0013
ADF lags : 11
ADF : Stationary
KPSS stat : 0.964
KPSS p : 0.0100
KPSS : Non-stationary

QUARTILE-BUCKET ANALYSIS

(vs NBER recession/expansion)

Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	203	16.3%	83.7%	-0.4403
Q2	201	11.4%	88.6%	1.4731
Q3	202	9.4%	90.6%	2.6950
Q4 (highest)	199	10.1%	89.9%	4.2199

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

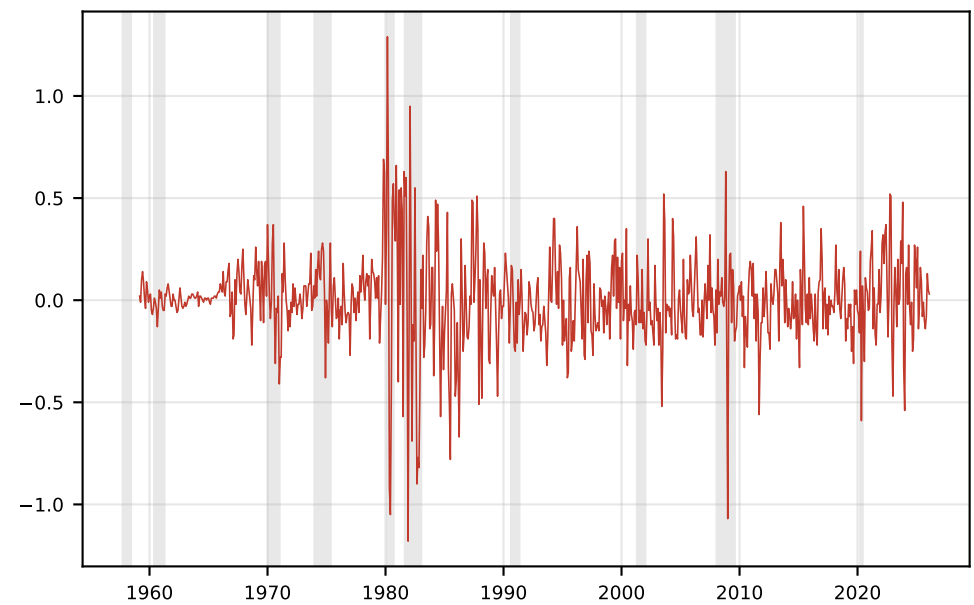
Moody Aaa Corporate Bond Yield

Block: Cross-Asset / Rates | Source: FRED-MD | Col: AAA | Transform: 2: First diff

Raw Series



After: 2: First diff



SUMMARY STATISTICS

N : 1287
Valid : 805
NaN : 482
Start : 1959-01
End : 2026-01
Mean : 6.7428
Std : 2.6708
Min : 2.1400
Max : 15.4900
Skew : 0.852
Kurt : 0.458

STATIONARITY TESTS

ADF stat : -8.931
ADF p : 0.0000
ADF lags : 7
ADF : Stationary
KPSS stat : 0.315
KPSS p : 0.1000
KPSS : Stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	202	13.4%	86.6%	-0.2452
Q2	216	9.7%	90.3%	-0.0383
Q3	188	9.6%	90.4%	0.0447
Q4 (highest)	198	14.6%	85.4%	0.2557

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

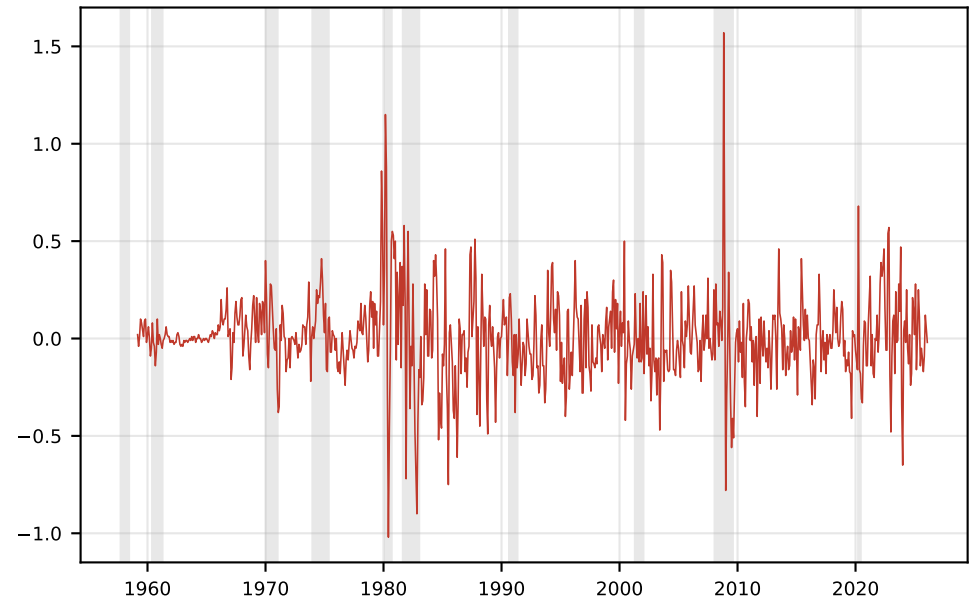
Moody Baa Corporate Bond Yield

Block: Cross-Asset / Rates | Source: FRED-MD | Col: BAA | Transform: 2: First diff

Raw Series



After: 2: First diff



SUMMARY STATISTICS

N : 1287
Valid : 805
NaN : 482
Start : 1959-01
End : 2026-01
Mean : 7.7349
Std : 2.8988
Min : 3.1600
Max : 17.1800
Skew : 0.998
Kurt : 0.800

STATIONARITY TESTS

ADF stat : -10.588
ADF p : 0.0000
ADF lags : 4
ADF : Stationary
KPSS stat : 0.215
KPSS p : 0.1000
KPSS : Stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

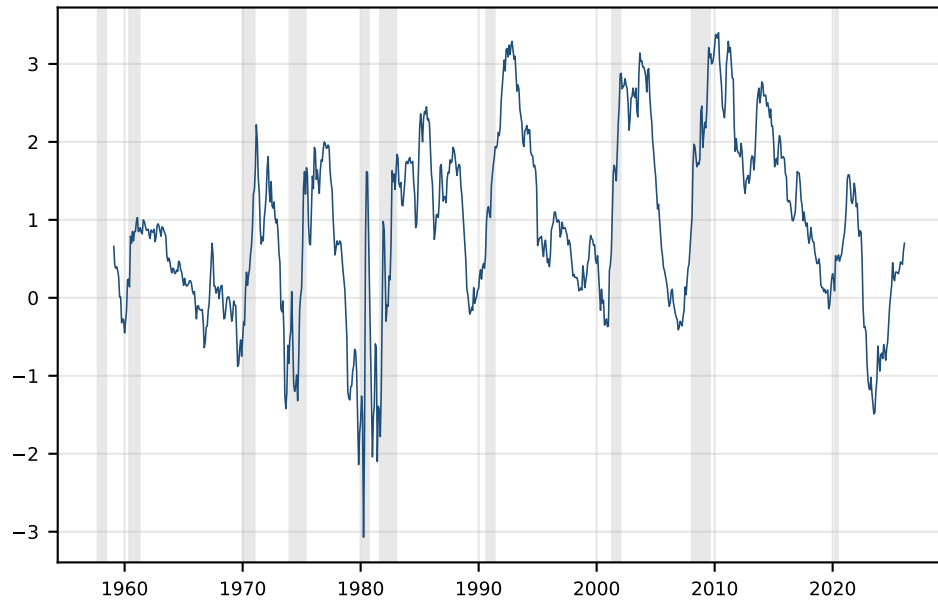
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	204	13.7%	86.3%	-0.2392
Q2	227	7.9%	92.1%	-0.0391
Q3	180	8.9%	91.1%	0.0491
Q4 (highest)	193	17.1%	82.9%	0.2582

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

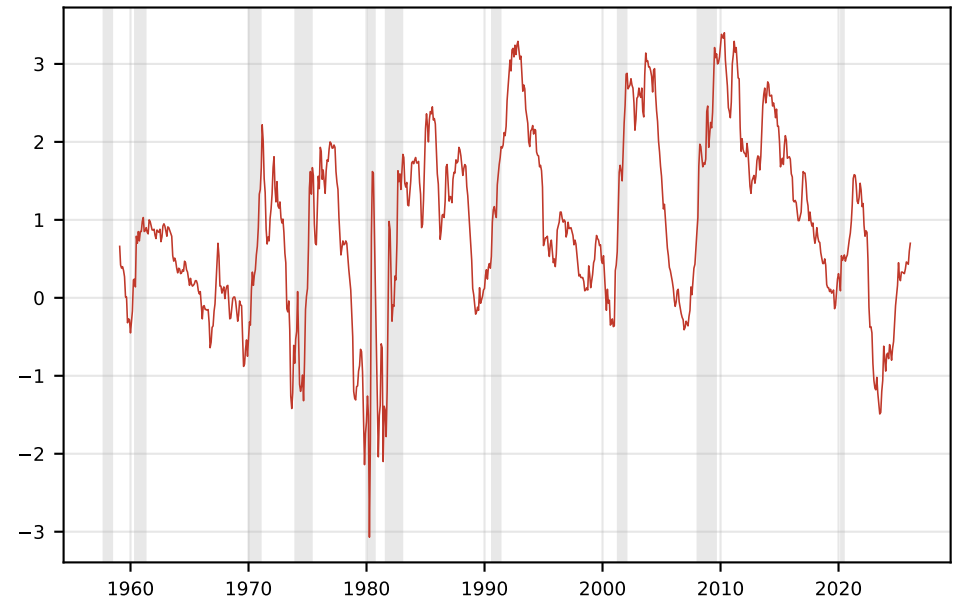
Term Spread 10Y-1Y

Block: Cross-Asset / Rates | Source: Computed | Col: term_spread_10y1y | Transform: 1: No transform

Raw Series



After: 1: No transform



SUMMARY STATISTICS

N : 1287
Valid : 805
NaN : 482
Start : 1959-01
End : 2026-01
Mean : 0.9160
Std : 1.1229
Min : -3.0700
Max : 3.4000
Skew : -0.014
Kurt : -0.271

STATIONARITY TESTS

ADF stat : -4.184
ADF p : 0.0007
ADF lags : 9
ADF : Stationary
KPSS stat : 0.571
KPSS p : 0.0256
KPSS : Non-stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

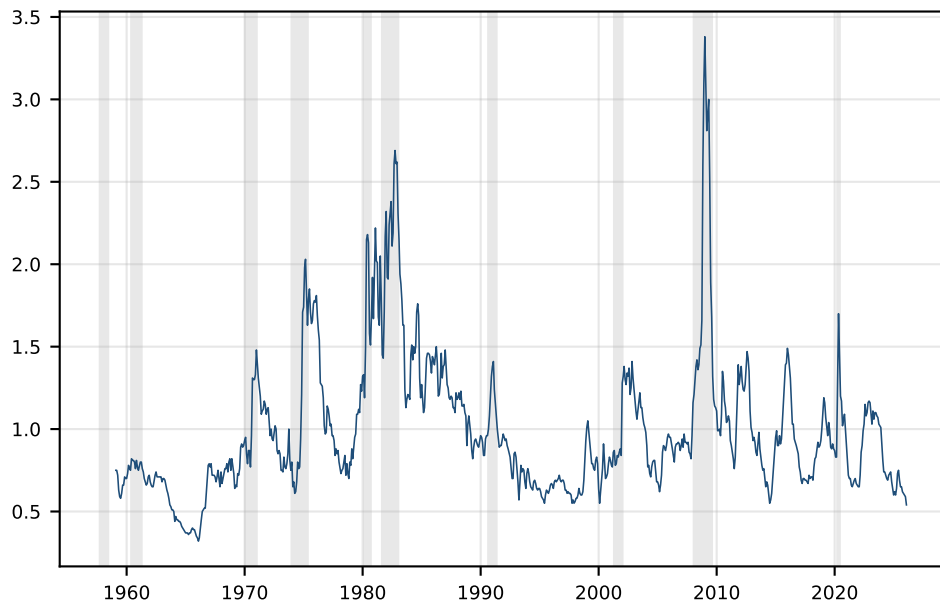
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	204	12.7%	87.3%	-0.4636
Q2	199	8.5%	91.5%	0.5009
Q3	201	17.4%	82.6%	1.2672
Q4 (highest)	201	8.5%	91.5%	2.3759

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

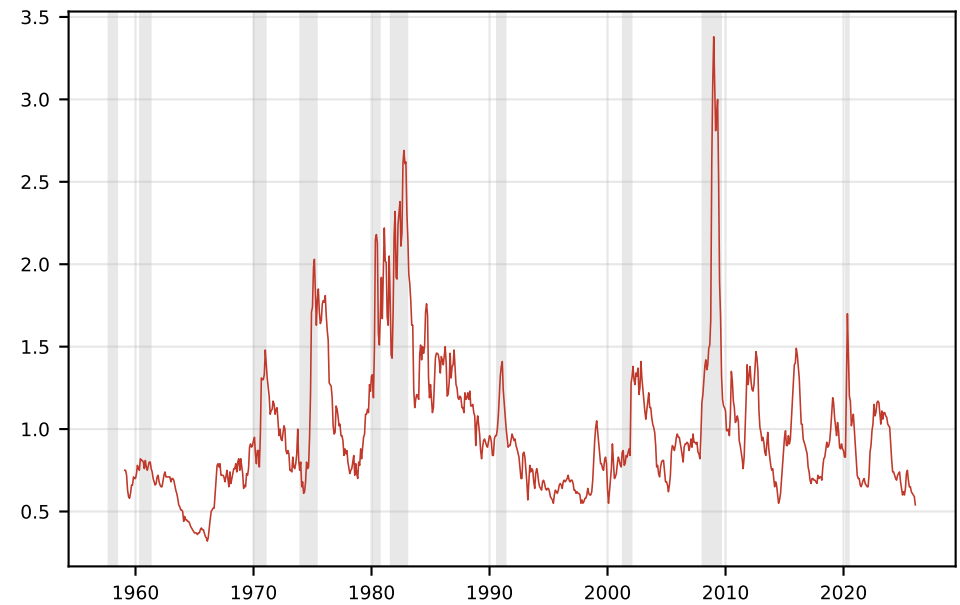
Credit Spread Baa-Aaa

Block: Cross-Asset / Rates | Source: Computed | Col: credit_spread_baa_aaa | Transform: 1: No transform

Raw Series



After: 1: No transform



SUMMARY STATISTICS

N : 1287
Valid : 805
NaN : 482
Start : 1959-01
End : 2026-01
Mean : 0.9921
Std : 0.4255
Min : 0.3200
Max : 3.3800
Skew : 1.878
Kurt : 5.115

STATIONARITY TESTS

ADF stat : -4.026
ADF p : 0.0013
ADF lags : 10
ADF : Stationary
KPSS stat : 0.265
KPSS p : 0.1000
KPSS : Stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

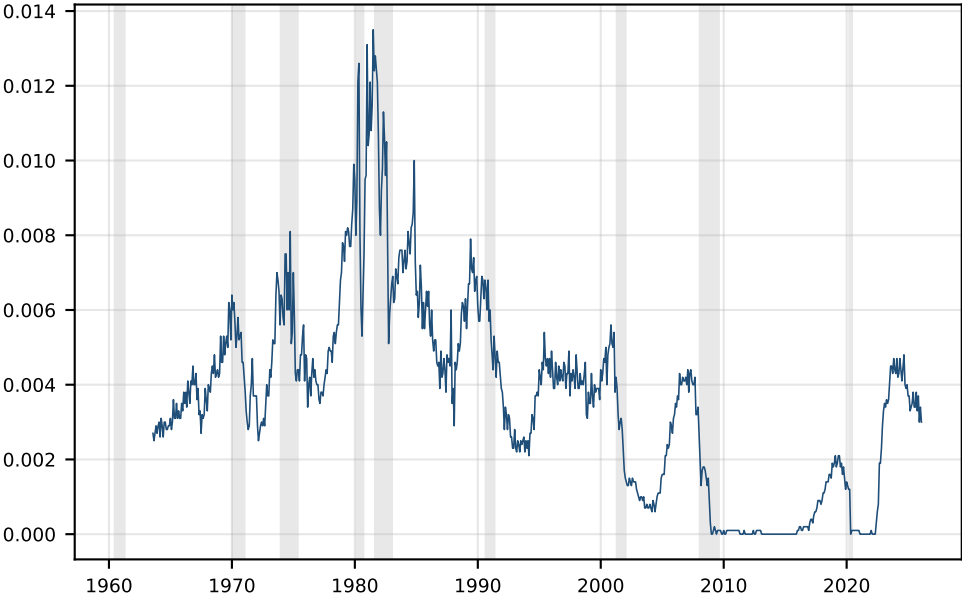
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	209	2.4%	97.6%	0.6120
Q2	202	13.4%	86.6%	0.7984
Q3	197	3.0%	97.0%	1.0124
Q4 (highest)	197	28.9%	71.1%	1.5736

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

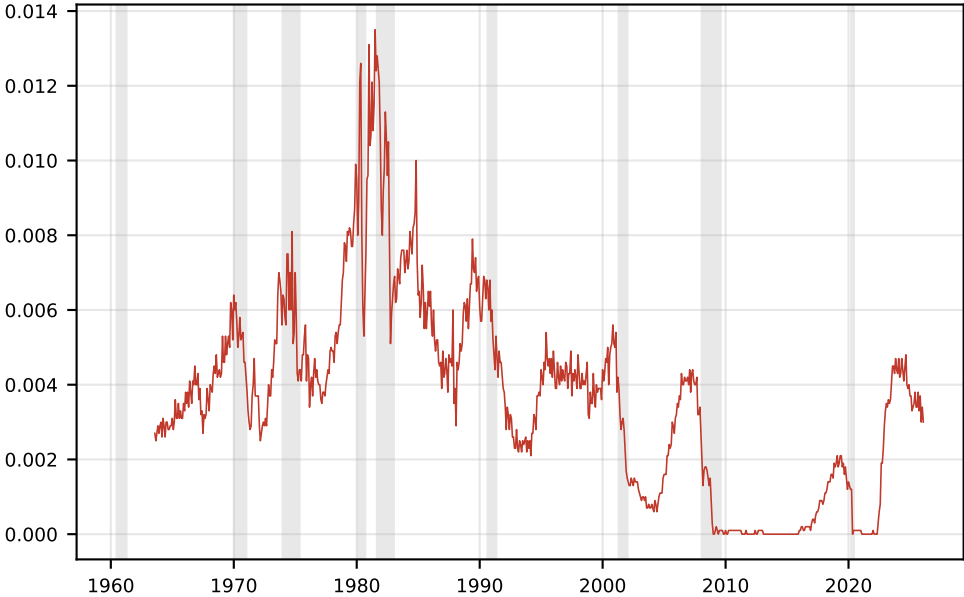
Risk-Free Rate (1m T-bill)

Block: Cross-Asset / Rates | Source: Computed | Col: rf | Transform: 1: No transform

Raw Series



After: 1: No transform



SUMMARY STATISTICS

N : 1287
Valid : 751
NaN : 536
Start : 1963-07
End : 2026-01
Mean : 0.0036
Std : 0.0026
Min : 0.0000
Max : 0.0135
Skew : 0.664
Kurt : 0.806

STATIONARITY TESTS

ADF stat : -2.589
ADF p : 0.0954
ADF lags : 12
ADF : Non-stationary
KPSS stat : 2.038
KPSS p : 0.0100
KPSS : Non-stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

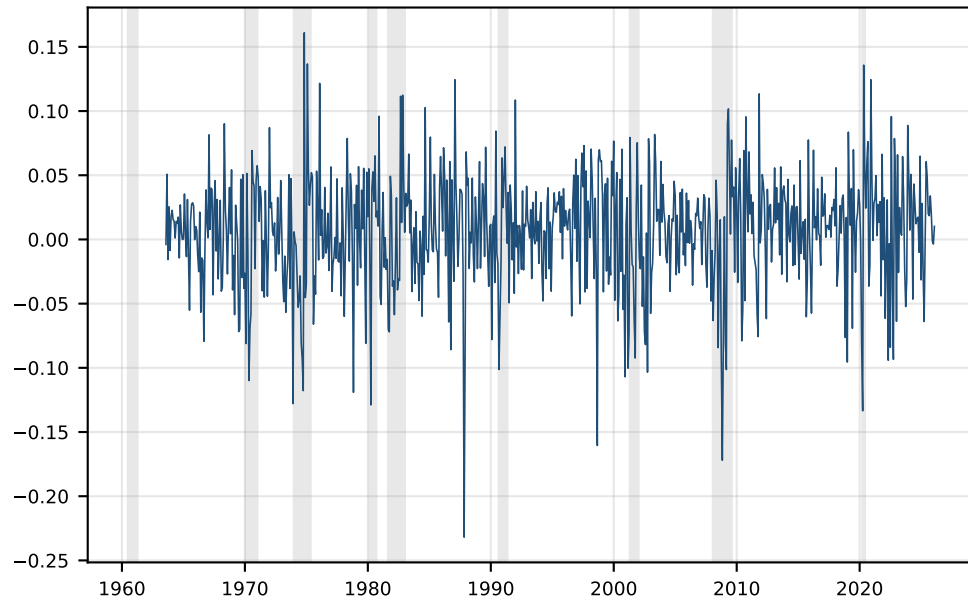
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	192	7.8%	92.2%	0.0004
Q2	196	6.1%	93.9%	0.0030
Q3	178	3.9%	96.1%	0.0043
Q4 (highest)	185	27.6%	72.4%	0.0070

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

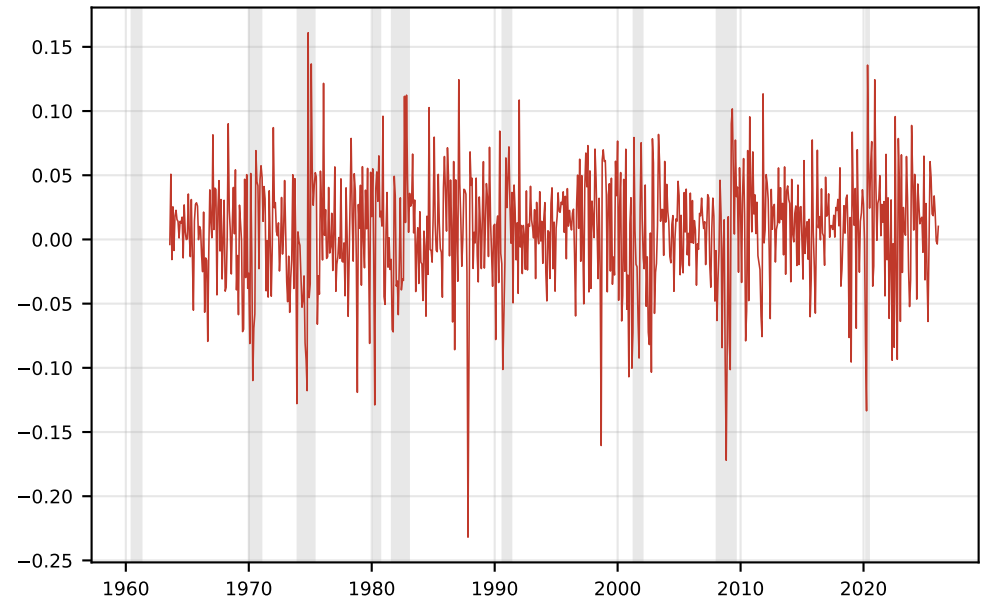
FF Market Excess Return

Block: Cross-Asset / Rates | Source: Computed | Col: mktfrf | Transform: 1: No transform

Raw Series



After: 1: No transform



SUMMARY STATISTICS

N : 1287
Valid : 751
NaN : 536
Start : 1963-07
End : 2026-01
Mean : 0.0059
Std : 0.0446
Min : -0.2319
Max : 0.1610
Skew : -0.504
Kurt : 1.752

STATIONARITY TESTS

ADF stat : -26.227
ADF p : 0.0000
ADF lags : 0
ADF : Stationary
KPSS stat : 0.232
KPSS p : 0.1000
KPSS : Stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

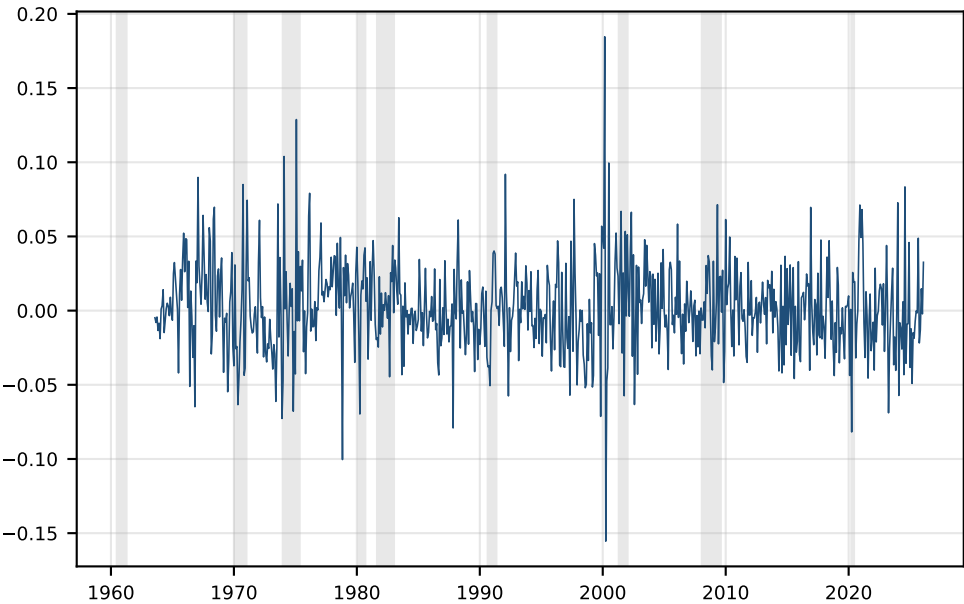
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	189	19.6%	80.4%	-0.0512
Q2	188	6.4%	93.6%	-0.0038
Q3	186	5.9%	94.1%	0.0215
Q4 (highest)	188	13.3%	86.7%	0.0578

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

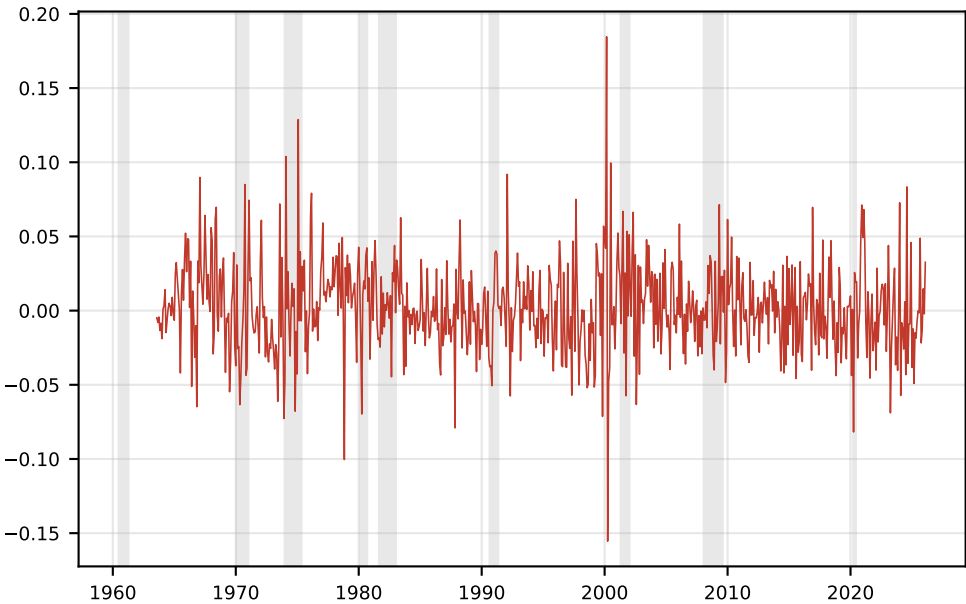
FF SMB (Small Minus Big)

Block: Cross-Asset / Rates | Source: Computed | Col: smb | Transform: 1: No transform

Raw Series



After: 1: No transform



SUMMARY STATISTICS

N : 1287
Valid : 751
NaN : 536
Start : 1963-07
End : 2026-01
Mean : 0.0018
Std : 0.0303
Min : -0.1554
Max : 0.1846
Skew : 0.360
Kurt : 3.066

STATIONARITY TESTS

ADF stat : -15.858
ADF p : 0.0000
ADF lags : 2
ADF : Stationary
KPSS stat : 0.311
KPSS p : 0.1000
KPSS : Stationary

QUARTILE-BUCKET ANALYSIS
(vs NBER recession/expansion)

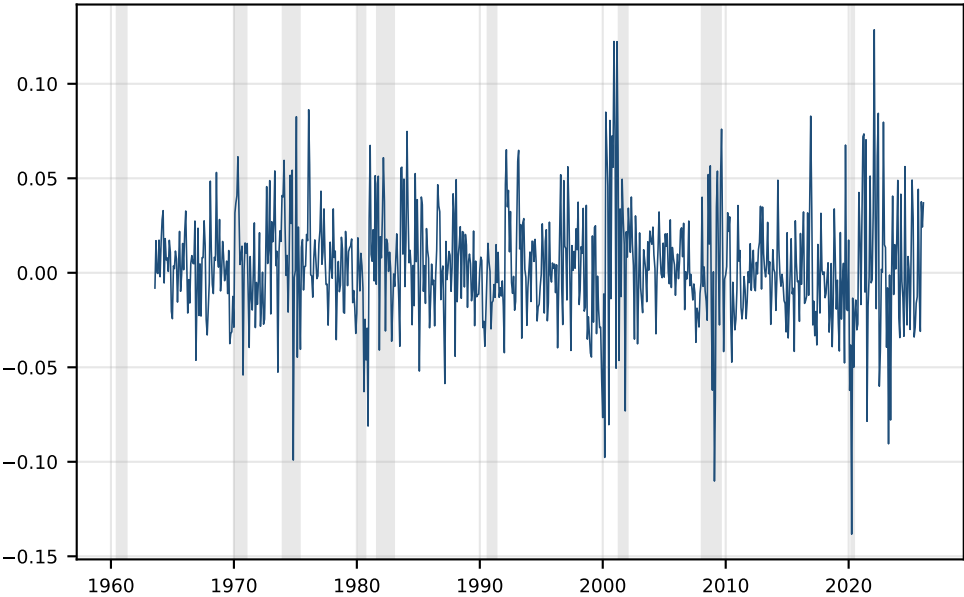
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	189	14.3%	85.7%	-0.0342
Q2	188	8.0%	92.0%	-0.0068
Q3	186	9.1%	90.9%	0.0087
Q4 (highest)	188	13.8%	86.2%	0.0399

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

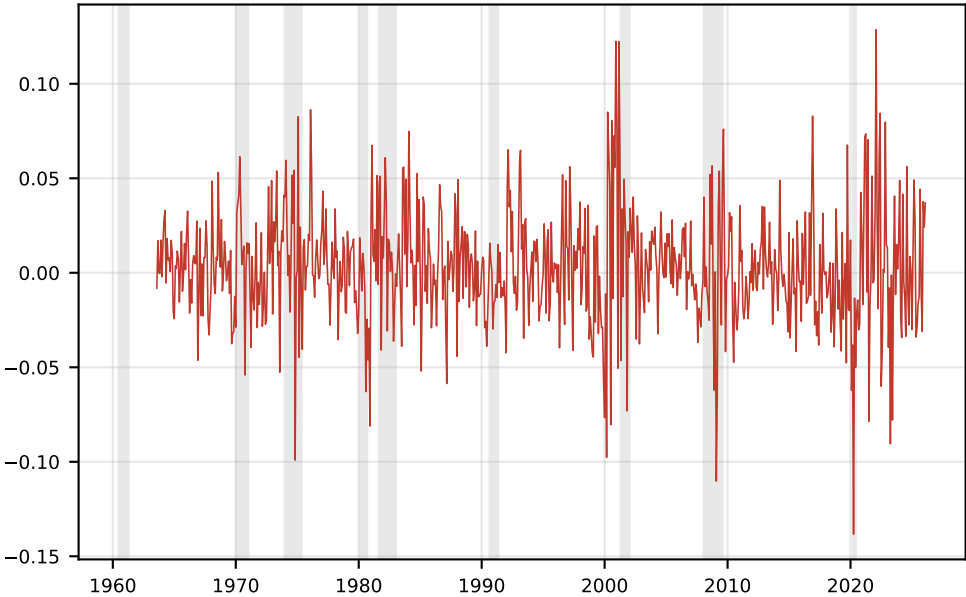
FF HML (High Minus Low)

Block: Cross-Asset / Rates | Source: Computed | Col: hml | Transform: 1: No transform

Raw Series



After: 1: No transform



SUMMARY STATISTICS

N : 1287
Valid : 751
NaN : 536
Start : 1963-07
End : 2026-01
Mean : 0.0029
Std : 0.0297
Min : -0.1383
Max : 0.1286
Skew : 0.095
Kurt : 2.185

STATIONARITY TESTS

ADF stat : -23.322
ADF p : 0.0000
ADF lags : 0
ADF : Stationary
KPSS stat : 0.258
KPSS p : 0.1000
KPSS : Stationary

QUARTILE-BUCKET ANALYSIS
(vs NBER recession/expansion)

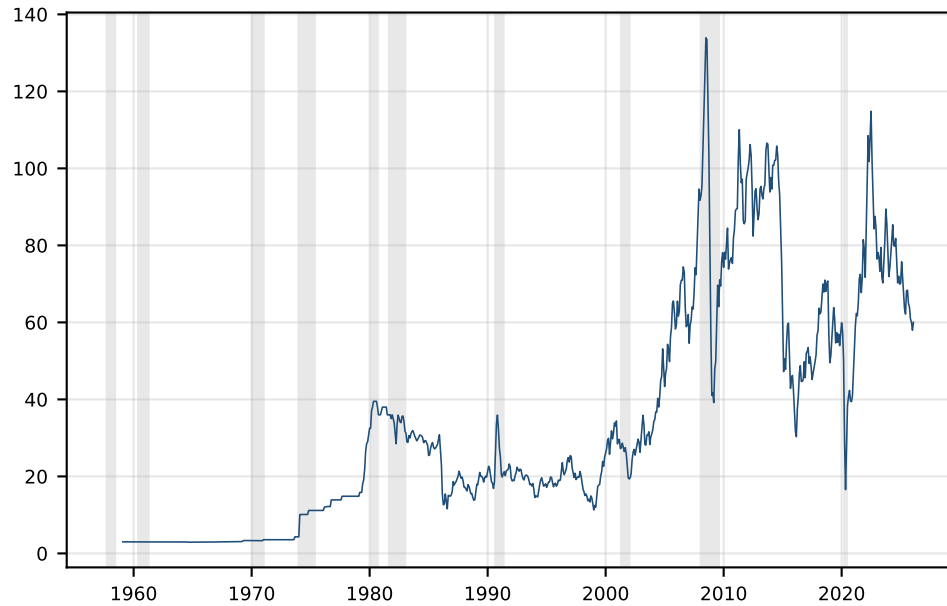
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	189	11.6%	88.4%	-0.0321
Q2	188	8.0%	92.0%	-0.0054
Q3	186	10.2%	89.8%	0.0096
Q4 (highest)	188	15.4%	84.6%	0.0398

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

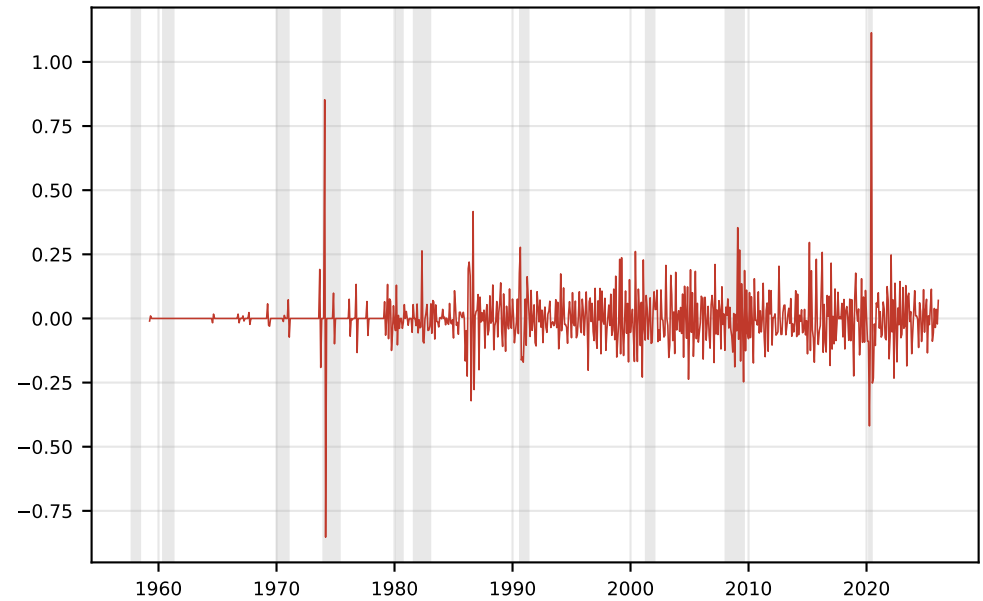
WTI Crude Oil (spliced)

Block: Commodities / Energy | Source: FRED-MD | Col: OILPRICEx | Transform: 6: Log 2nd diff

Raw Series



After: 6: Log 2nd diff



SUMMARY STATISTICS

N : 1287
Valid : 805
NaN : 482
Start : 1959-01
End : 2026-01
Mean : 33.6518
Std : 29.7382
Min : 2.9200
Max : 133.9300
Skew : 0.975
Kurt : -0.004

STATIONARITY TESTS

ADF stat : -12.991
ADF p : 0.0000
ADF lags : 12
ADF : Stationary
KPSS stat : 0.046
KPSS p : 0.1000
KPSS : Stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

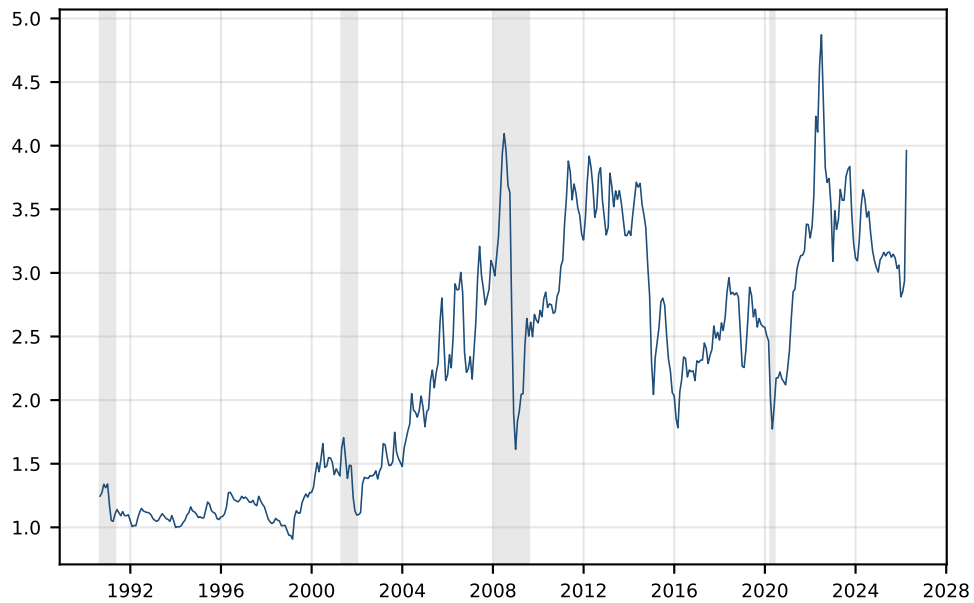
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	202	12.4%	87.6%	-0.1044
Q2	335	13.4%	86.6%	-0.0064
Q3	65	9.2%	90.8%	0.0155
Q4 (highest)	201	9.5%	90.5%	0.1107

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

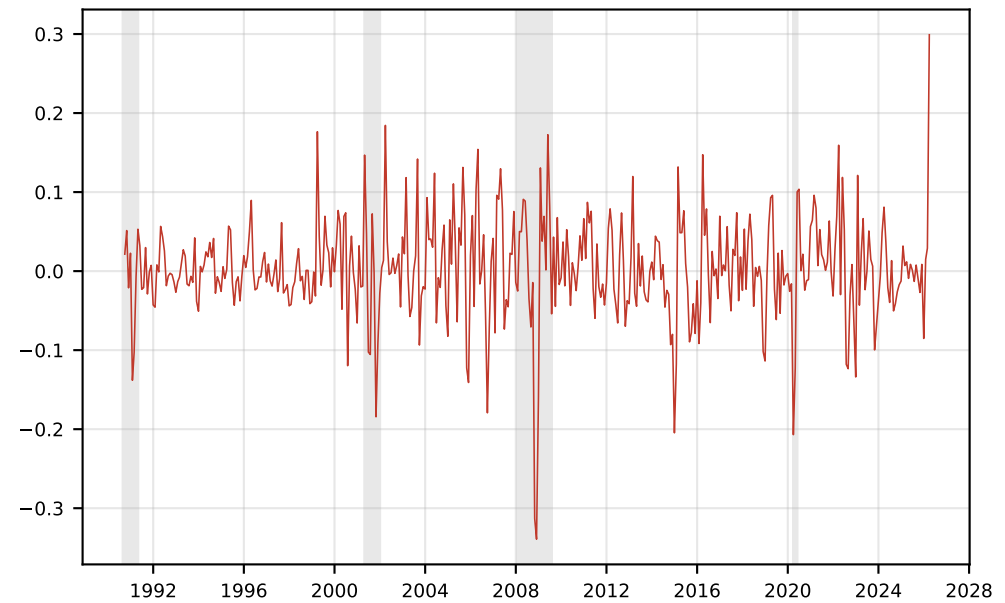
Regular Gasoline Price

Block: Commodities / Energy | Source: FRED API | Col: GASREGW | Transform: 5: Log diff

Raw Series



After: 5: Log diff



SUMMARY STATISTICS

N : 1287
Valid : 428
NaN : 859
Start : 1990-08
End : 2026-03
Mean : 2.2520
Std : 0.9504
Min : 0.9070
Max : 4.8720
Skew : 0.218
Kurt : -1.179

STATIONARITY TESTS

ADF stat : -7.513
ADF p : 0.0000
ADF lags : 9
ADF : Stationary
KPSS stat : 0.030
KPSS p : 0.1000
KPSS : Stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	107	12.1%	87.9%	-0.0705
Q2	106	4.7%	95.3%	-0.0131
Q3	106	2.8%	97.2%	0.0160
Q4 (highest)	107	13.1%	86.9%	0.0756

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

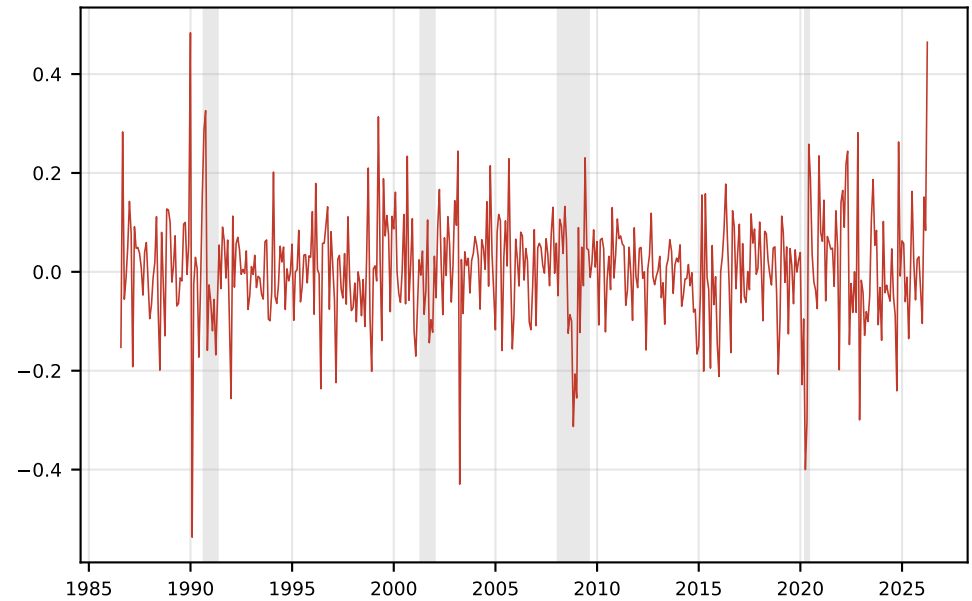
No.2 Heating Oil

Block: Commodities / Energy | Source: FRED API | Col: DHOILNYH | Transform: 5: Log diff

Raw Series



After: 5: Log diff



SUMMARY STATISTICS

N : 1287
Valid : 478
NaN : 809
Start : 1986-06
End : 2026-03
Mean : 1.4601
Std : 0.9441
Min : 0.3210
Max : 4.5290
Skew : 0.649
Kurt : -0.595

STATIONARITY TESTS

ADF stat : -21.092
ADF p : 0.0000
ADF lags : 0
ADF : Stationary
KPSS stat : 0.030
KPSS p : 0.1000
KPSS : Stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	119	15.1%	84.9%	-0.1268
Q2	119	4.2%	95.8%	-0.0215
Q3	119	3.4%	96.6%	0.0360
Q4 (highest)	119	7.6%	92.4%	0.1284

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

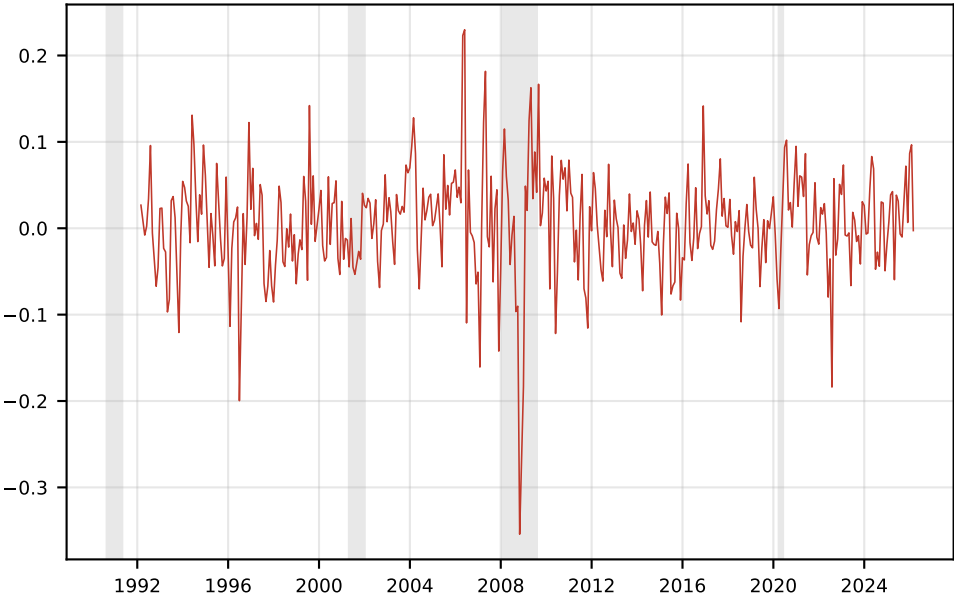
Copper (Global Price)

Block: Commodities / Metals | Source: FRED API | Col: PCOPPU5DM | Transform: 5: Log diff

Raw Series



After: 5: Log diff



SUMMARY STATISTICS

N : 1287
Valid : 410
NaN : 877
Start : 1992-01
End : 2026-02
Mean : 5214.5287
Std : 2876.7683
Min : 1377.3761
Max : 12986.6068
Skew : 0.167
Kurt : -1.241

STATIONARITY TESTS

ADF stat : -13.620
ADF p : 0.0000
ADF lags : 0
ADF : Stationary
KPSS stat : 0.064
KPSS p : 0.1000
KPSS : Stationary

QUARTILE-BUCKET ANALYSIS

(vs NBER recession/expansion)

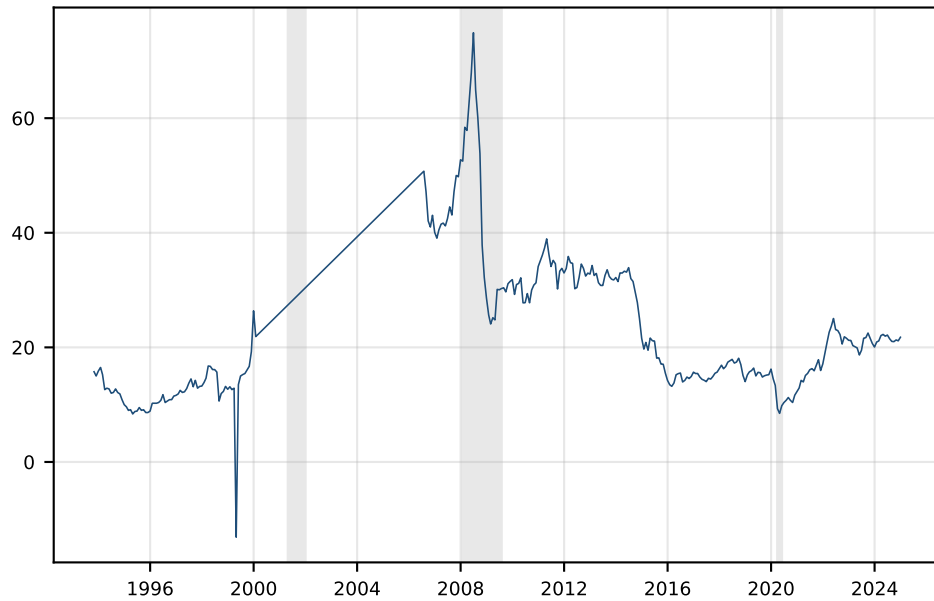
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	103	12.6%	87.4%	-0.0675
Q2	102	2.0%	98.0%	-0.0093
Q3	102	4.9%	95.1%	0.0228
Q4 (highest)	102	7.8%	92.2%	0.0723

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

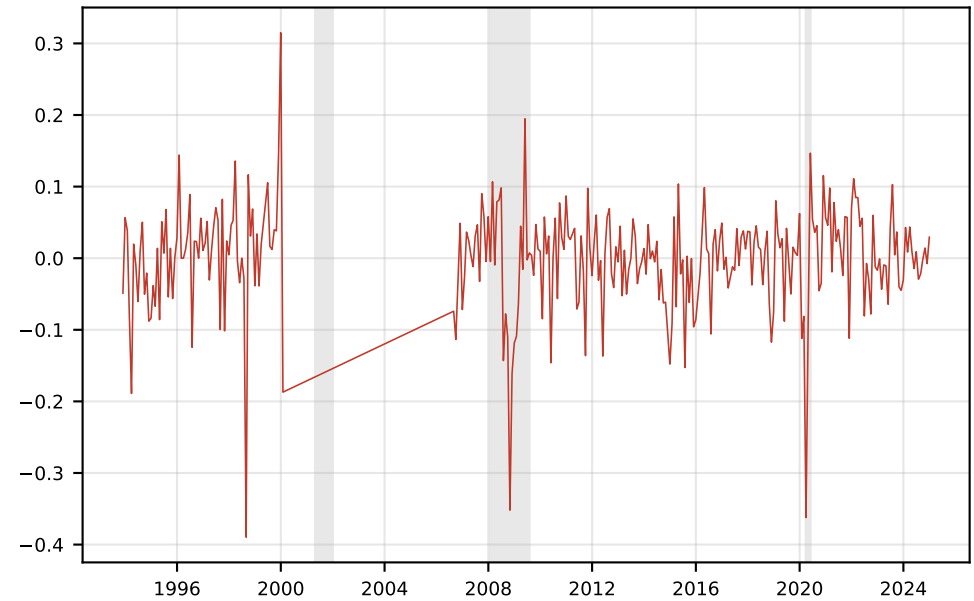
iShares GSCI Commodity ETF (GSG)

Block: Commodities / Commodity structure | Source: WRDS (GSG as proxy for GSCI. If unavailable, try DJP or DBC.) | Col: gsg_prc | Transform: 5: Log diff

Raw Series



After: 5: Log diff



SUMMARY STATISTICS

N : 1287
Valid : 298
NaN : 989
Start : 1993-10
End : 2024-12
Mean : 22.5872
Std : 12.1635
Min : -13.1250
Max : 74.9100
Skew : 1.277
Kurt : 2.192

STATIONARITY TESTS

ADF stat : -10.227
ADF p : 0.0000
ADF lags : 1
ADF : Stationary
KPSS stat : 0.059
KPSS p : 0.1000
KPSS : Stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

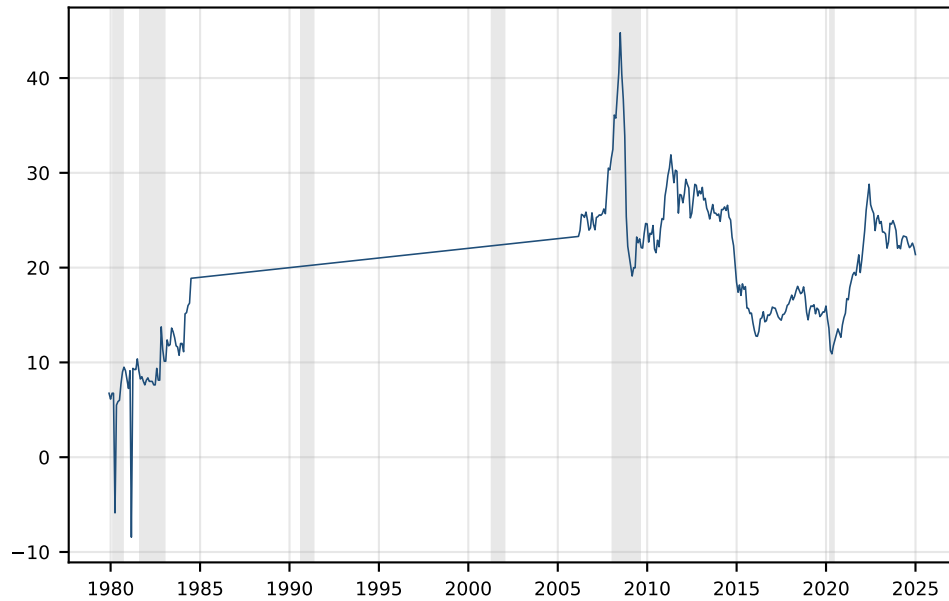
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	74	13.5%	86.5%	-0.0941
Q2	73	5.5%	94.5%	-0.0115
Q3	73	0.0%	100.0%	0.0227
Q4 (highest)	74	8.1%	91.9%	0.0755

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

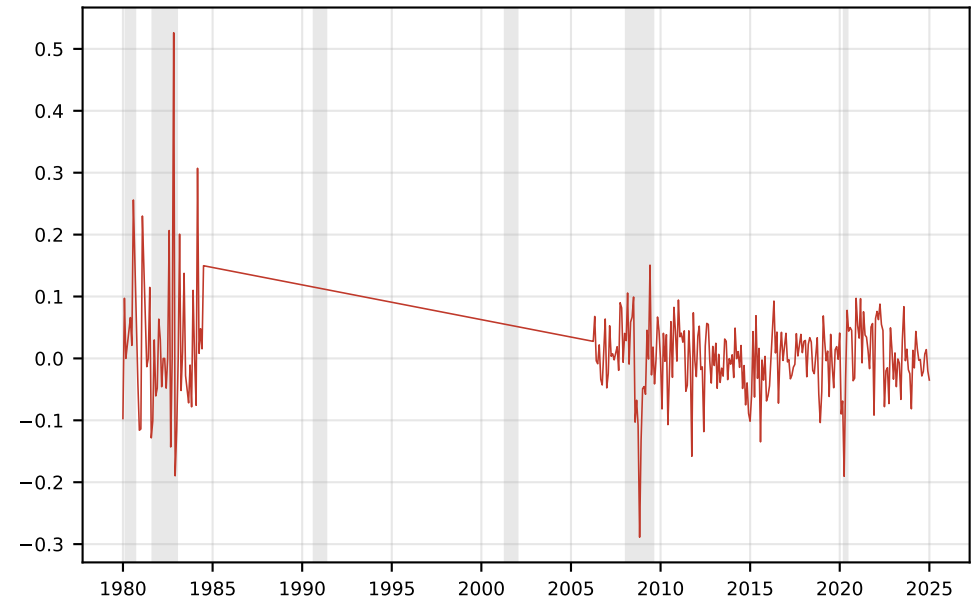
Invesco DB Commodity ETF (DBC)

Block: Commodities / Commodity structure | Source: WRDS (crsp.dsf DBC. Starts ~2006.) | Col: dbc_prc | Transform: 5: Log diff

Raw Series



After: 5: Log diff



SUMMARY STATISTICS

N : 1287
Valid : 283
NaN : 1004
Start : 1979-11
End : 2024-12
Mean : 19.6151
Std : 7.6766
Min : -8.4375
Max : 44.7800
Skew : -0.082
Kurt : 0.434

STATIONARITY TESTS

ADF stat : -7.583
ADF p : 0.0000
ADF lags : 4
ADF : Stationary
KPSS stat : 0.230
KPSS p : 0.1000
KPSS : Stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

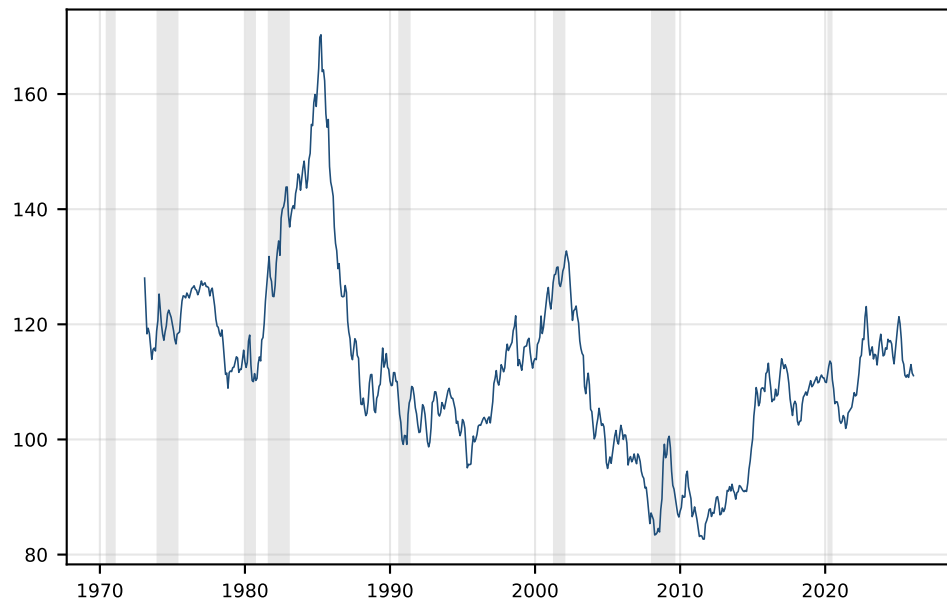
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	70	22.9%	77.1%	-0.0774
Q2	74	12.2%	87.8%	-0.0122
Q3	64	6.2%	93.8%	0.0209
Q4 (highest)	69	15.9%	84.1%	0.0884

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

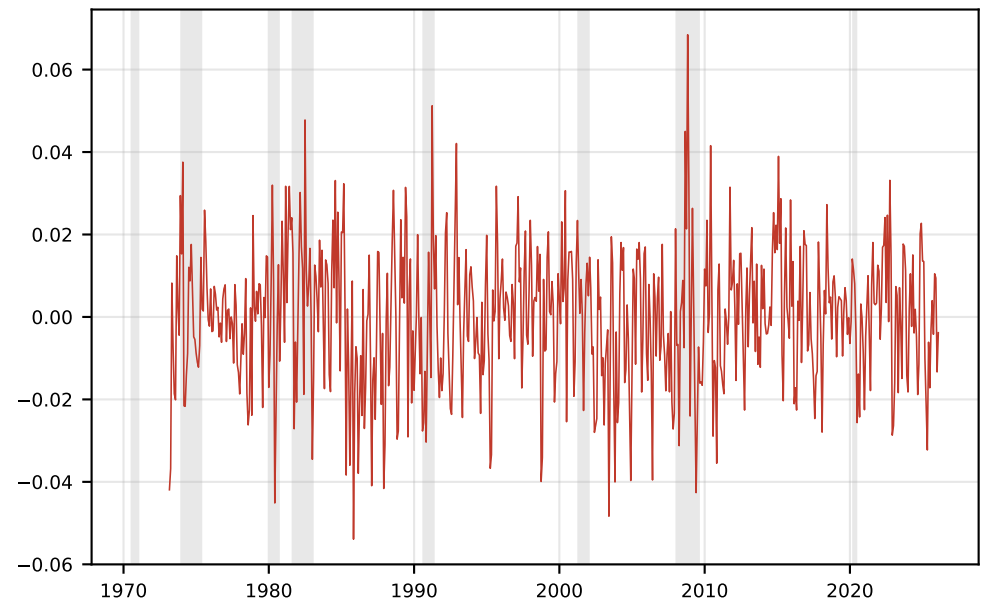
Trade-Weighted Dollar Index

Block: FX / Global Macro / FX | Source: FRED-MD | Col: TWEXAFEGSMTHx | Transform: 5: Log diff

Raw Series



After: 5: Log diff



SUMMARY STATISTICS

N : 1287
Valid : 637
NaN : 650
Start : 1973-01
End : 2026-01
Mean : 111.6945
Std : 15.4234
Min : 82.6783
Max : 170.3020
Skew : 0.805
Kurt : 1.359

STATIONARITY TESTS

ADF stat : -16.003
ADF p : 0.0000
ADF lags : 1
ADF : Stationary
KPSS stat : 0.071
KPSS p : 0.1000
KPSS : Stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

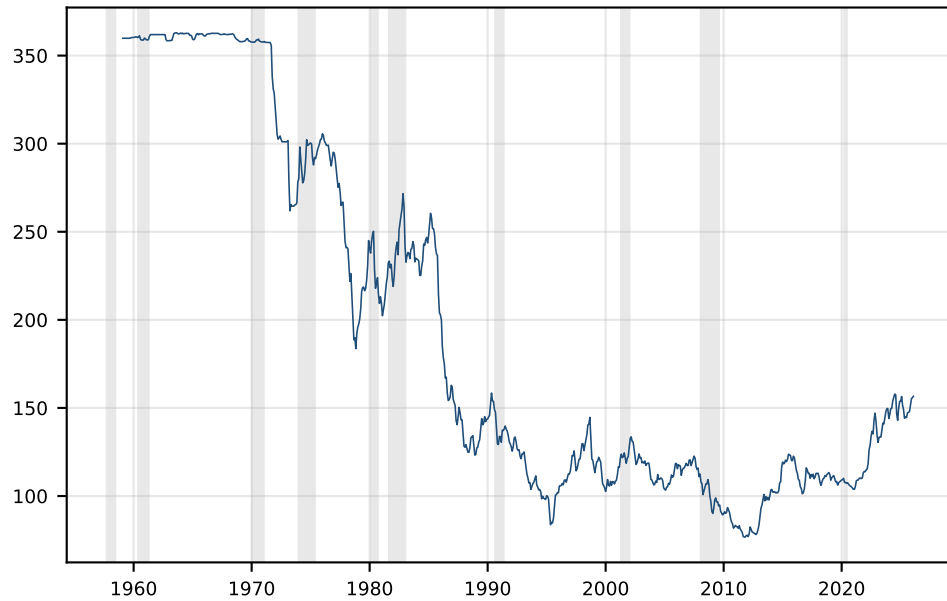
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	159	12.6%	87.4%	-0.0216
Q2	159	10.1%	89.9%	-0.0048
Q3	159	9.4%	90.6%	0.0055
Q4 (highest)	159	14.5%	85.5%	0.0200

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

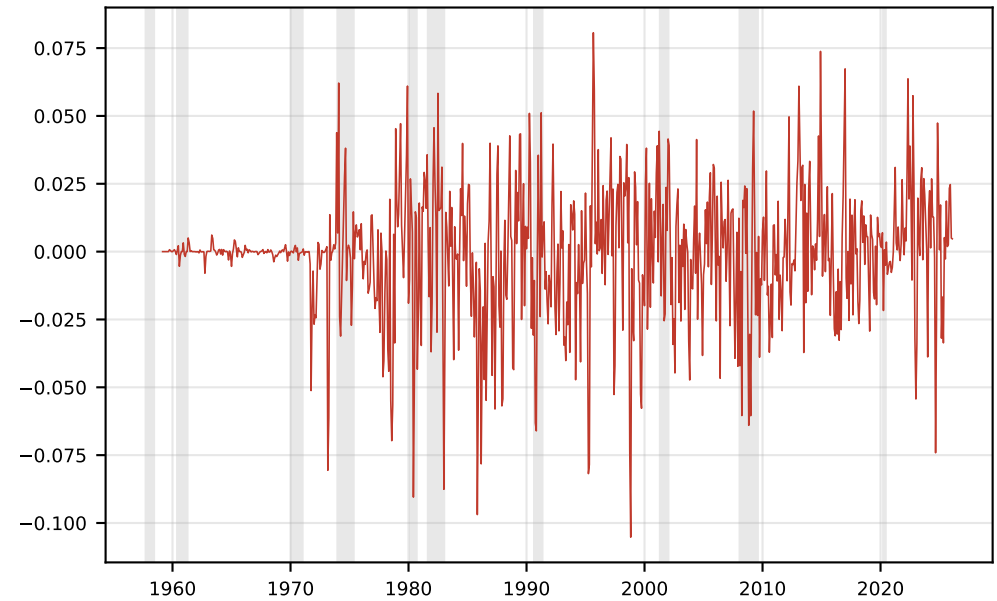
USD/JPY

Block: FX / Global Macro / FX | Source: FRED-MD | Col: EXJPUSx | Transform: 5: Log diff

Raw Series



After: 5: Log diff



SUMMARY STATISTICS

N : 1287
Valid : 805
NaN : 482
Start : 1959-01
End : 2026-01
Mean : 192.9782
Std : 100.4977
Min : 76.6430
Max : 362.9764
Skew : 0.676
Kurt : -1.162

STATIONARITY TESTS

ADF stat : -6.760
ADF p : 0.0000
ADF lags : 10
ADF : Stationary
KPSS stat : 0.295
KPSS p : 0.1000
KPSS : Stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

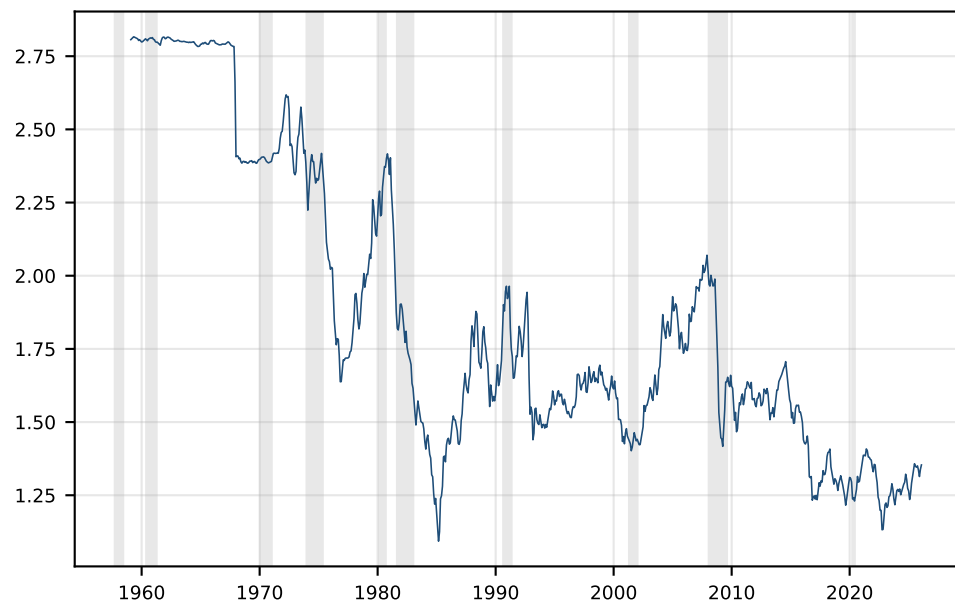
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	201	13.4%	86.6%	-0.0309
Q2	206	10.7%	89.3%	-0.0030
Q3	196	9.7%	90.3%	0.0035
Q4 (highest)	201	13.4%	86.6%	0.0263

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

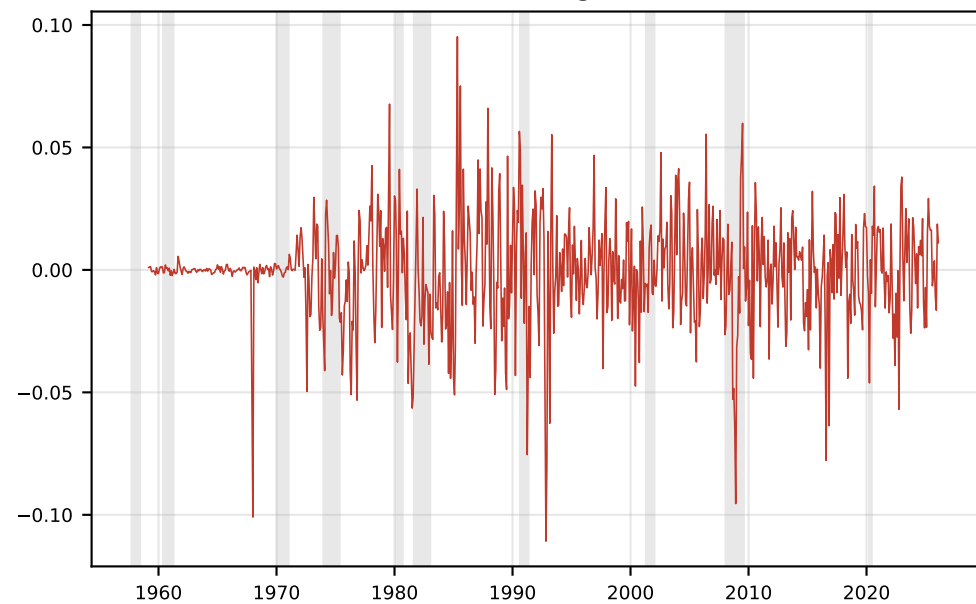
GBP/USD

Block: FX / Global Macro / FX | Source: FRED-MD | Col: EXUSUKx | Transform: 5: Log diff

Raw Series



After: 5: Log diff



SUMMARY STATISTICS

N : 1287
Valid : 805
NaN : 482
Start : 1959-01
End : 2026-01
Mean : 1.8582
Std : 0.4983
Min : 1.0931
Max : 2.8165
Skew : 0.736
Kurt : -0.727

STATIONARITY TESTS

ADF stat : -13.881
ADF p : 0.0000
ADF lags : 2
ADF : Stationary
KPSS stat : 0.043
KPSS p : 0.1000
KPSS : Stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

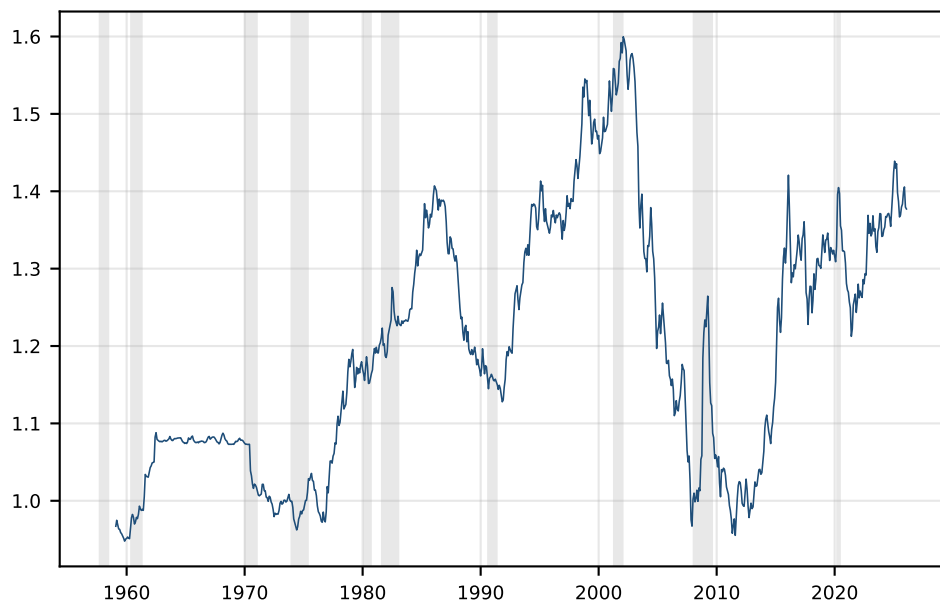
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	201	12.4%	87.6%	-0.0273
Q2	201	13.4%	86.6%	-0.0038
Q3	201	10.9%	89.1%	0.0036
Q4 (highest)	201	10.4%	89.6%	0.0238

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

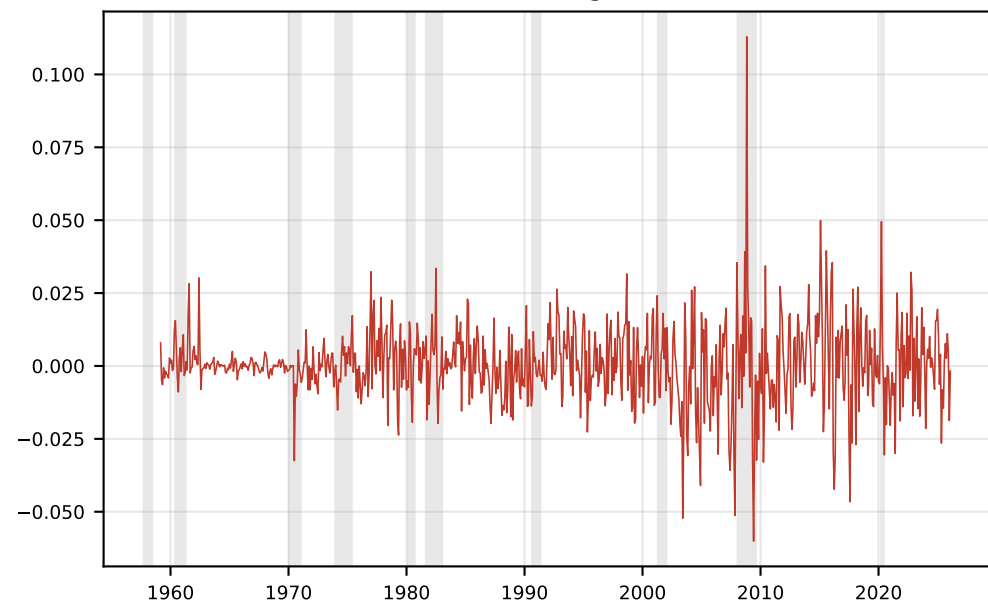
CAD/USD

Block: FX / Global Macro / FX | Source: FRED-MD | Col: EXCAUSx | Transform: 5: Log diff

Raw Series



After: 5: Log diff



SUMMARY STATISTICS

N : 1287
Valid : 805
NaN : 482
Start : 1959-01
End : 2026-01
Mean : 1.2015
Std : 0.1625
Min : 0.9478
Max : 1.5997
Skew : 0.329
Kurt : -0.876

STATIONARITY TESTS

ADF stat : -7.827
ADF p : 0.0000
ADF lags : 11
ADF : Stationary
KPSS stat : 0.071
KPSS p : 0.1000
KPSS : Stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	201	10.9%	89.1%	-0.0152
Q2	203	9.9%	90.1%	-0.0024
Q3	199	16.1%	83.9%	0.0031
Q4 (highest)	201	10.4%	89.6%	0.0164

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

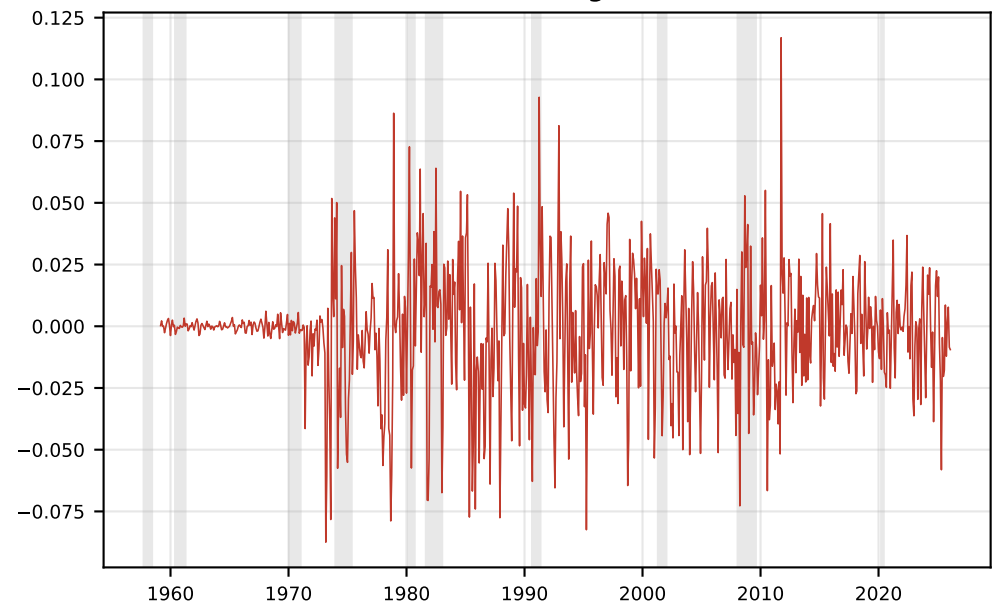
CHF/USD

Block: FX / Global Macro / FX | Source: FRED-MD | Col: EXSZUSx | Transform: 5: Log diff

Raw Series



After: 5: Log diff



SUMMARY STATISTICS

N : 1287
Valid : 805
NaN : 482
Start : 1959-01
End : 2026-01
Mean : 2.0606
Std : 1.2445
Min : 0.7800
Max : 4.3490
Skew : 0.963
Kurt : -0.648

STATIONARITY TESTS

ADF stat : -18.183
ADF p : 0.0000
ADF lags : 1
ADF : Stationary
KPSS stat : 0.053
KPSS p : 0.1000
KPSS : Stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	201	13.4%	86.6%	-0.0329
Q2	201	11.9%	88.1%	-0.0051
Q3	201	9.5%	90.5%	0.0031
Q4 (highest)	201	12.4%	87.6%	0.0264

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

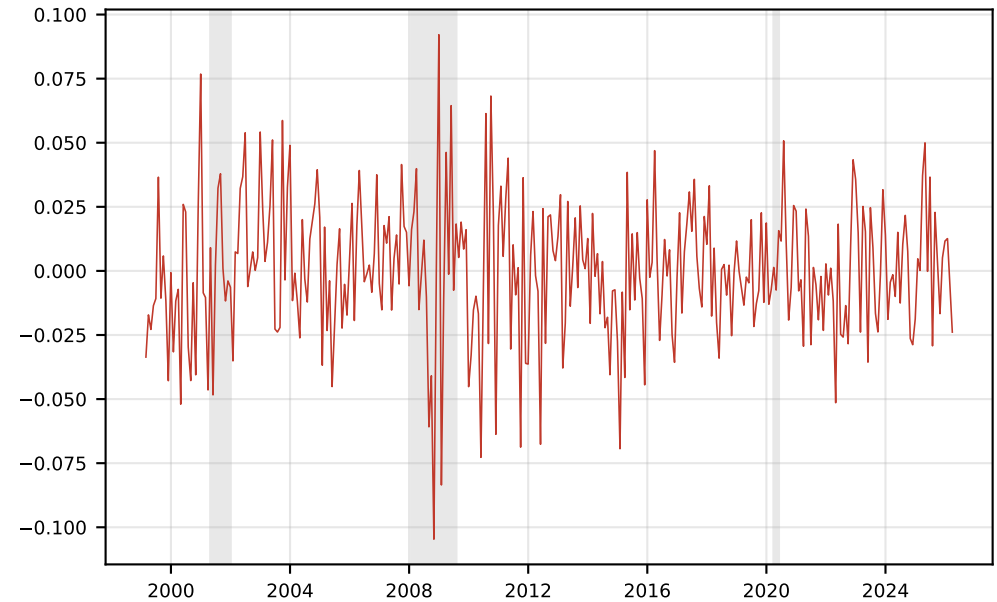
EUR/USD

Block: FX / Global Macro / FX | Source: FRED API | Col: DEXUSEU | Transform: 5: Log diff

Raw Series



After: 5: Log diff



SUMMARY STATISTICS

N : 1287
Valid : 327
NaN : 960
Start : 1999-01
End : 2026-03
Mean : 1.1828
Std : 0.1542
Min : 0.8455
Max : 1.5805
Skew : 0.115
Kurt : -0.281

STATIONARITY TESTS

ADF stat : -17.224
ADF p : 0.0000
ADF lags : 0
ADF : Stationary
KPSS stat : 0.072
KPSS p : 0.1000
KPSS : Stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

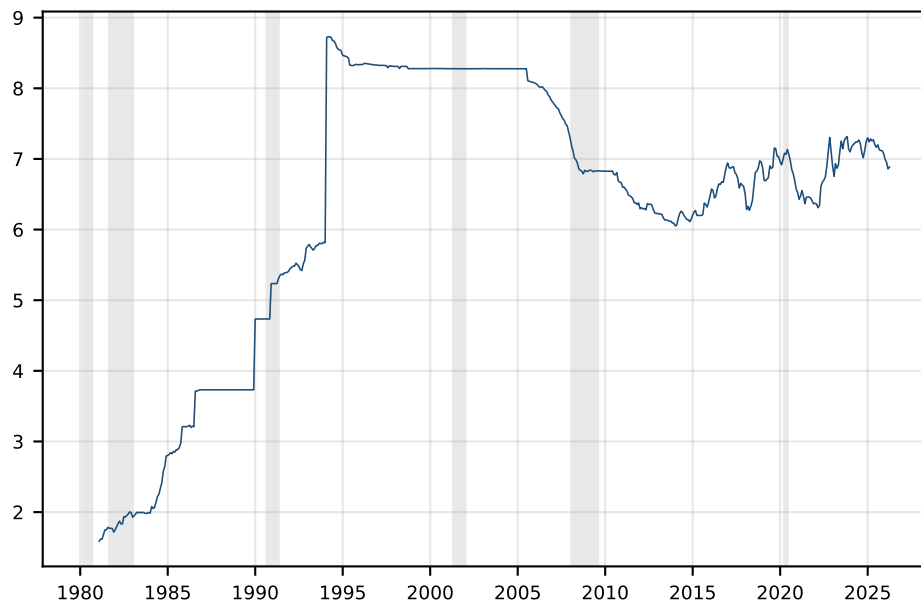
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	82	6.1%	93.9%	-0.0326
Q2	81	11.1%	88.9%	-0.0072
Q3	81	8.6%	91.4%	0.0079
Q4 (highest)	81	8.6%	91.4%	0.0328

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

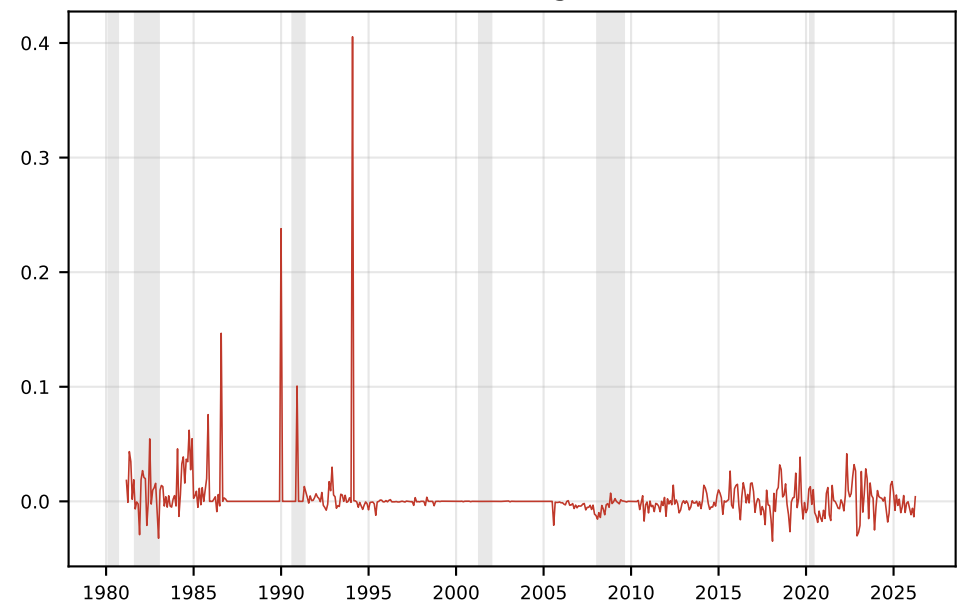
USD/CNY

Block: FX / Global Macro / FX | Source: FRED API | Col: DEXCHUS | Transform: 5: Log diff

Raw Series



After: 5: Log diff



SUMMARY STATISTICS

N : 1287
Valid : 543
NaN : 744
Start : 1981-01
End : 2026-03
Mean : 6.2876
Std : 1.9715
Min : 1.5886
Max : 8.7298
Skew : -0.936
Kurt : -0.148

STATIONARITY TESTS

ADF stat : -22.004
ADF p : 0.0000
ADF lags : 0
ADF : Stationary
KPSS stat : 1.279
KPSS p : 0.0100
KPSS : Non-stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

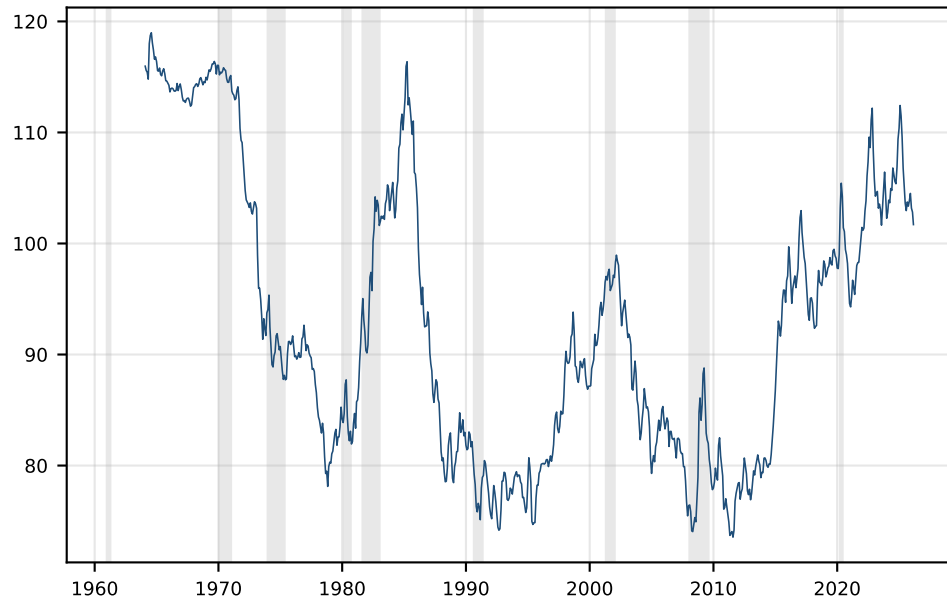
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	136	9.6%	90.4%	-0.0089
Q2	196	11.2%	88.8%	-0.0004
Q3	74	6.8%	93.2%	0.0004
Q4 (highest)	135	8.9%	91.1%	0.0202

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

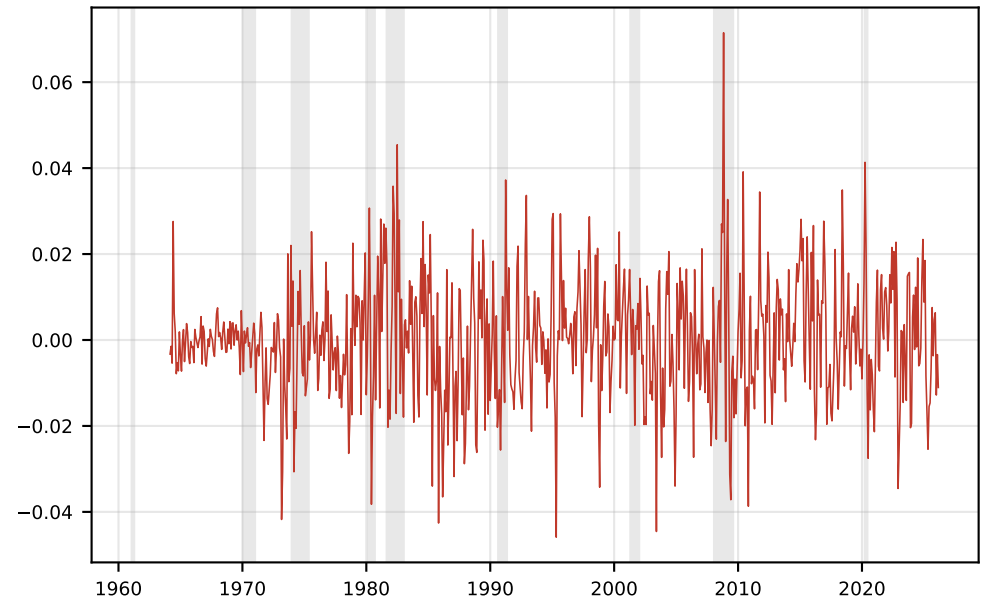
Real Effective Exchange Rate (BIS)

Block: FX / Global Macro / FX | Source: FRED API | Col: RNUSBIS | Transform: 5: Log diff

Raw Series



After: 5: Log diff



SUMMARY STATISTICS

N : 1287
Valid : 746
NaN : 541
Start : 1964-01
End : 2026-02
Mean : 92.7253
Std : 12.5466
Min : 73.5500
Max : 118.9800
Skew : 0.444
Kurt : -0.995

STATIONARITY TESTS

ADF stat : -17.356
ADF p : 0.0000
ADF lags : 1
ADF : Stationary
KPSS stat : 0.209
KPSS p : 0.1000
KPSS : Stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

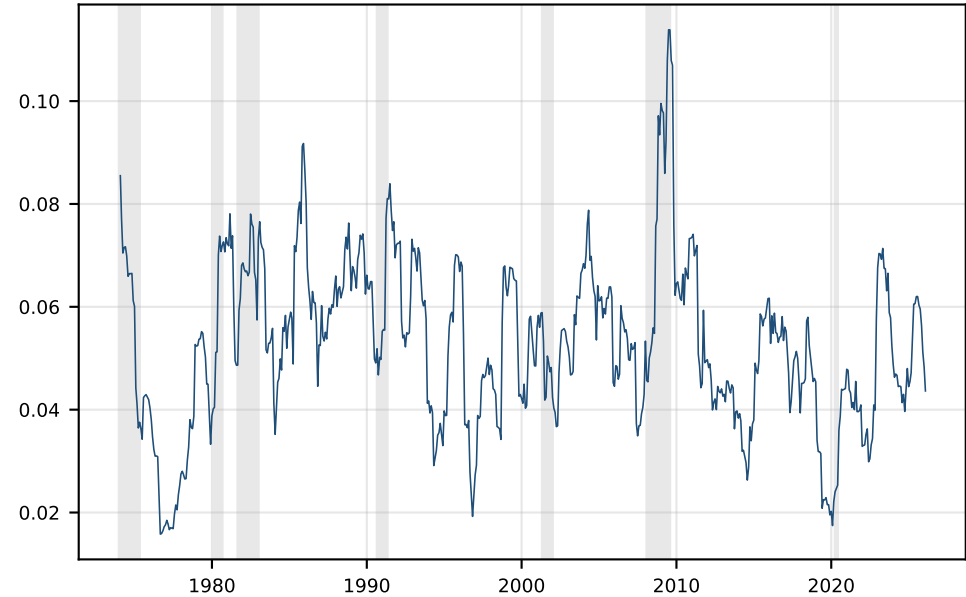
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	187	11.8%	88.2%	-0.0171
Q2	187	10.7%	89.3%	-0.0039
Q3	185	9.2%	90.8%	0.0034
Q4 (highest)	186	14.0%	86.0%	0.0171

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

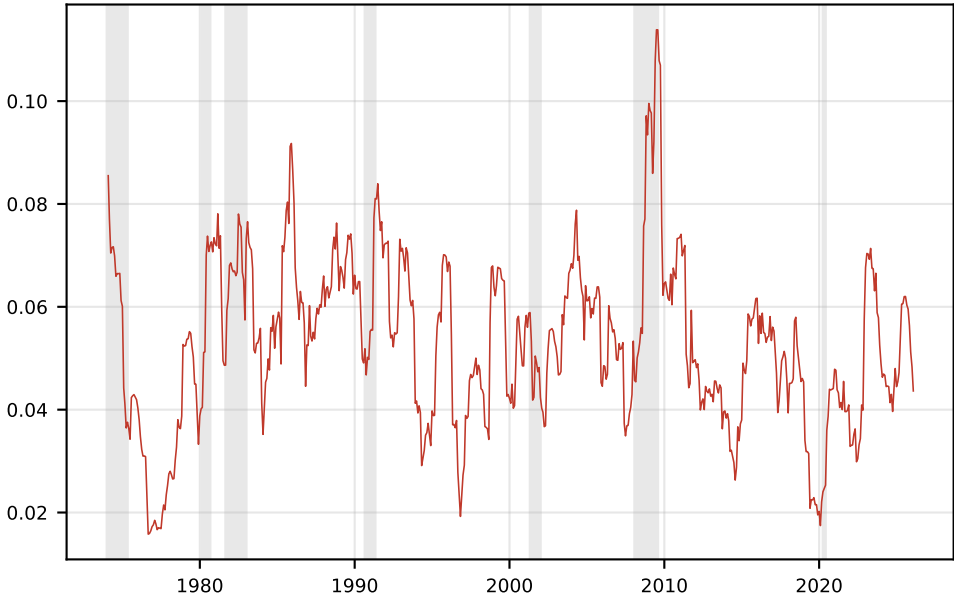
Dollar Realized Vol (12m)

Block: FX / Global Macro / FX | Source: Computed | Col: dollar_rv_12m | Transform: 1: No transform

Raw Series



After: 1: No transform



SUMMARY STATISTICS

N : 1287
Valid : 625
NaN : 662
Start : 1974-01
End : 2026-01
Mean : 0.0528
Std : 0.0162
Min : 0.0158
Max : 0.1139
Skew : 0.280
Kurt : 0.656

STATIONARITY TESTS

ADF stat : -3.716
ADF p : 0.0039
ADF lags : 12
ADF : Stationary
KPSS stat : 0.227
KPSS p : 0.1000
KPSS : Stationary

QUARTILE-BUCKET ANALYSIS

(vs NBER recession/expansion)

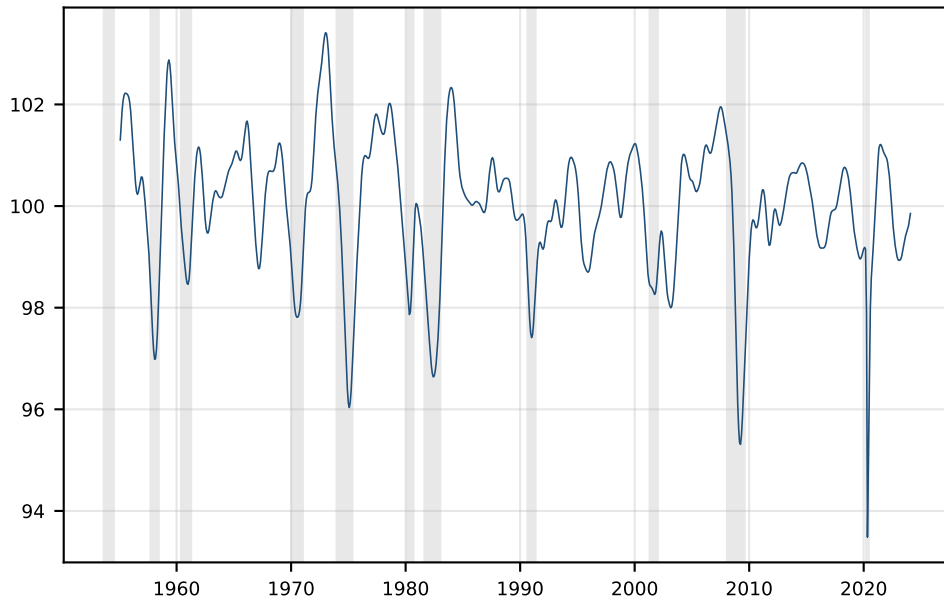
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	157	3.8%	96.2%	0.0329
Q2	156	10.9%	89.1%	0.0472
Q3	156	8.3%	91.7%	0.0580
Q4 (highest)	156	23.7%	76.3%	0.0733

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

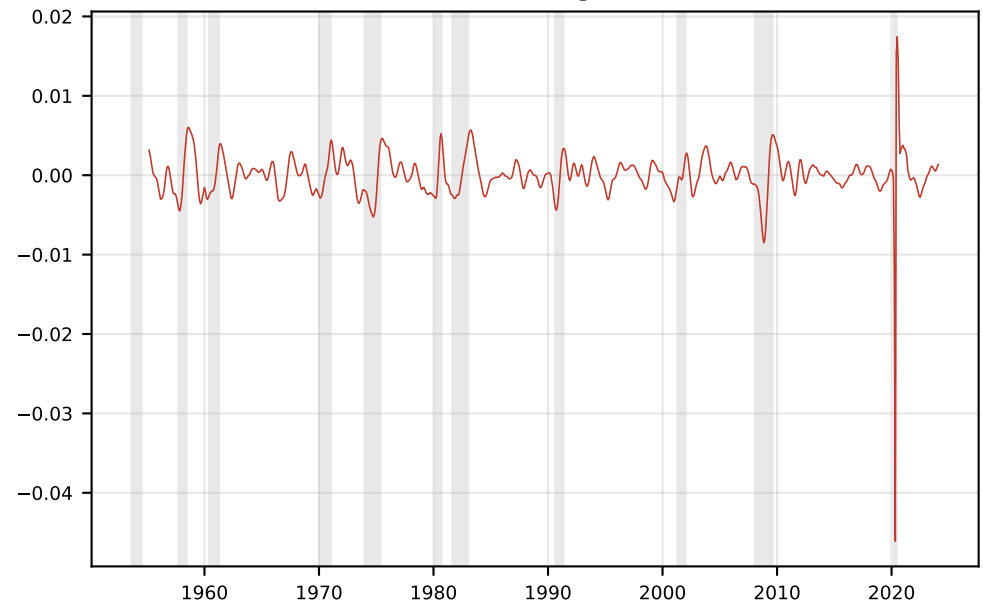
OECD CLI USA

Block: FX / Global Macro / Global cycle | Source: FRED API | Col: USALOLITONOSTSAM | Transform: 5: Log diff

Raw Series



After: 5: Log diff



SUMMARY STATISTICS

N : 1287
Valid : 829
NaN : 458
Start : 1955-01
End : 2024-01
Mean : 100.0000
Std : 1.3265
Min : 93.4837
Max : 103.4109
Skew : -0.732
Kurt : 1.535

STATIONARITY TESTS

ADF stat : -9.404
ADF p : 0.0000
ADF lags : 10
ADF : Stationary
KPSS stat : 0.010
KPSS p : 0.1000
KPSS : Stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

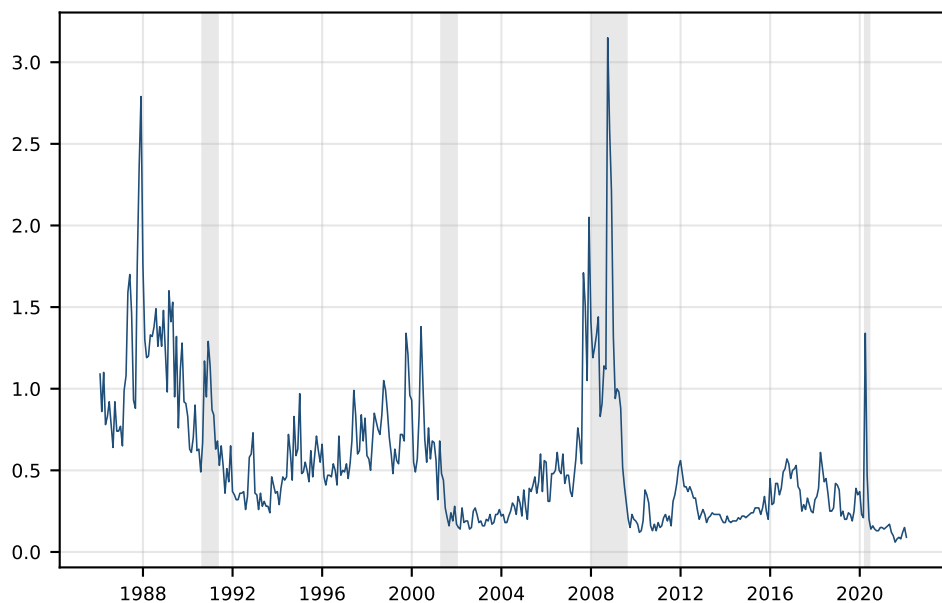
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	207	28.5%	71.5%	-0.0028
Q2	207	7.2%	92.8%	-0.0006
Q3	207	5.3%	94.7%	0.0005
Q4 (highest)	207	8.7%	91.3%	0.0028

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

TED Spread

Block: FX / Global Macro / Global funding | Source: FRED API | Col: TEDRATE | Transform: 1: No transform

Raw Series



After: 1: No transform



SUMMARY STATISTICS

N : 1287
Valid : 433
NaN : 854
Start : 1986-01
End : 2022-01
Mean : 0.5476
Std : 0.4306
Min : 0.0600
Max : 3.1500
Skew : 2.102
Kurt : 6.470

STATIONARITY TESTS

ADF stat : -3.048
ADF p : 0.0307
ADF lags : 4
ADF : Stationary
KPSS stat : 1.142
KPSS p : 0.0100
KPSS : Non-stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

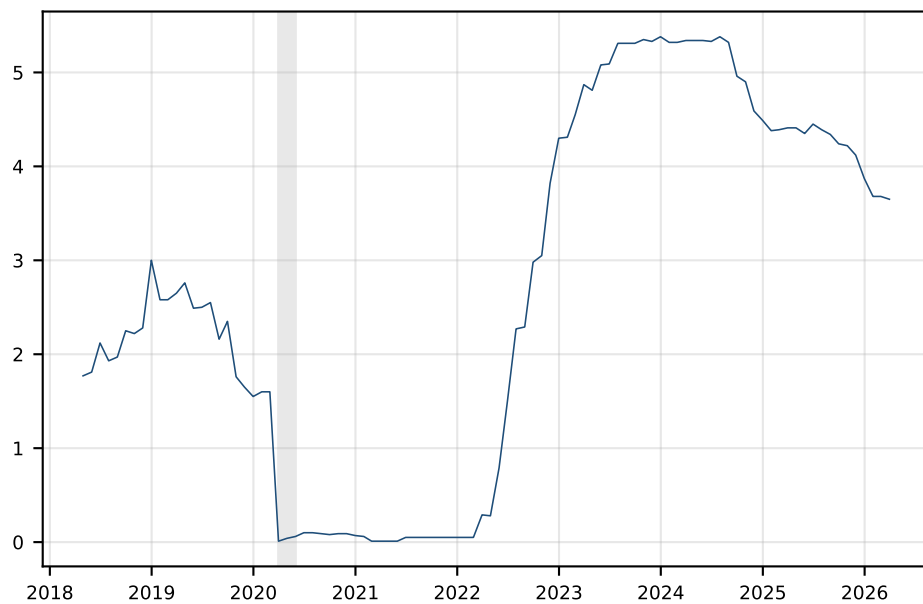
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	115	3.5%	96.5%	0.1851
Q2	105	2.9%	97.1%	0.3349
Q3	105	5.7%	94.3%	0.5471
Q4 (highest)	108	21.3%	78.7%	1.1408

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

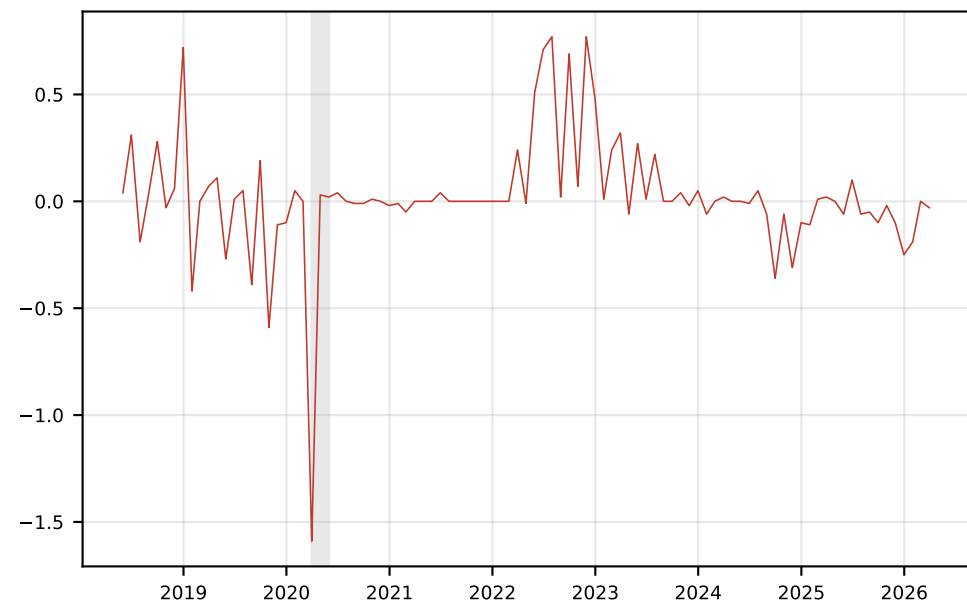
SOFR Rate

Block: FX / Global Macro / Global funding | Source: FRED API | Col: SOFR | Transform: 2: First diff

Raw Series



After: 2: First diff



SUMMARY STATISTICS

N : 1287
Valid : 96
NaN : 1191
Start : 2018-04
End : 2026-03
Mean : 2.6864
Std : 1.9829
Min : 0.0100
Max : 5.3800
Skew : -0.096
Kurt : -1.478

STATIONARITY TESTS

ADF stat : -2.147
ADF p : 0.2259
ADF lags : 4
ADF : Non-stationary
KPSS stat : 0.203
KPSS p : 0.1000
KPSS : Stationary

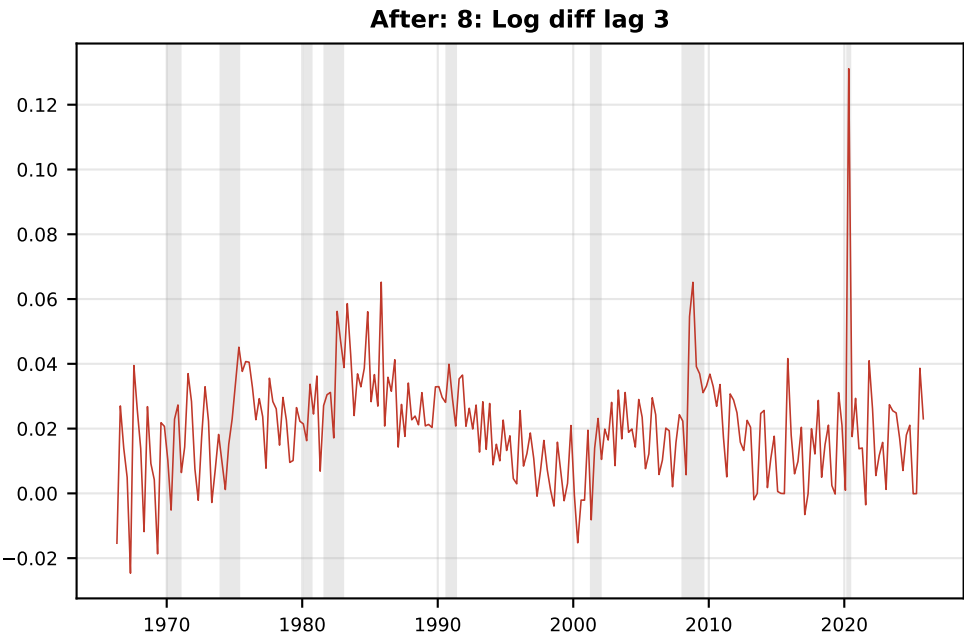
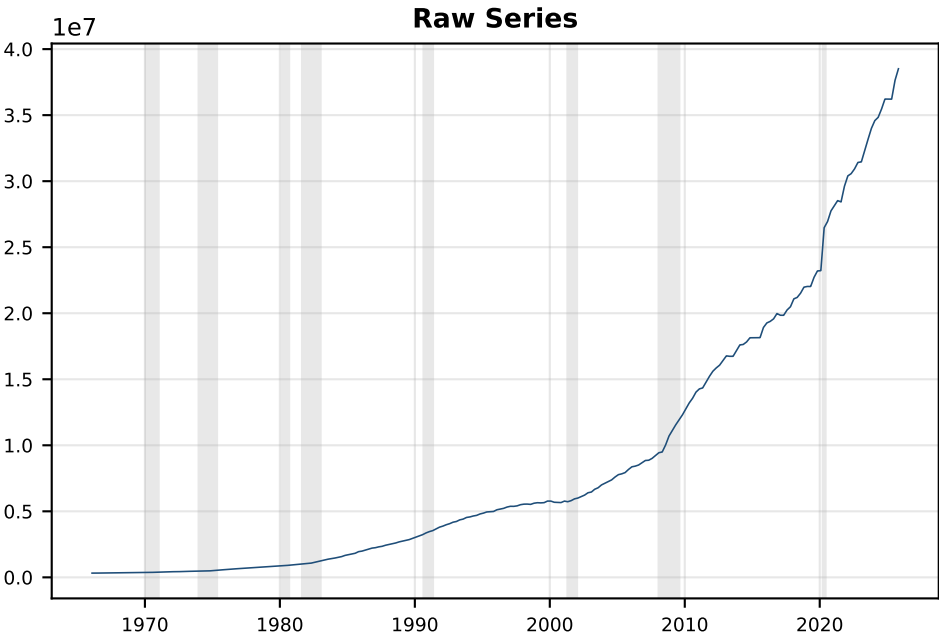
QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	24	4.2%	95.8%	-0.2350
Q2	31	0.0%	100.0%	-0.0045
Q3	15	6.7%	93.3%	0.0240
Q4 (highest)	24	0.0%	100.0%	0.3054

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

Federal Debt: Total Public Debt

Block: Debt / Debt | Source: FRED API | Col: GFDEBTN | Transform: 8: Log diff lag 3



SUMMARY STATISTICS

N : 1287
Valid : 240
NaN : 1047
Start : 1966-01
End : 2025-10
Mean : 8690938.5380
Std : 9923436.7816
Min : 316097.0000
Max : 38514009.0000
Skew : 1.359
Kurt : 0.868

STATIONARITY TESTS

ADF stat : -2.799
ADF p : 0.0584
ADF lags : 11
ADF : Non-stationary
KPSS stat : 0.209
KPSS p : 0.1000
KPSS : Stationary

QUARTILE-BUCKET ANALYSIS

(vs NBER recession/expansion)

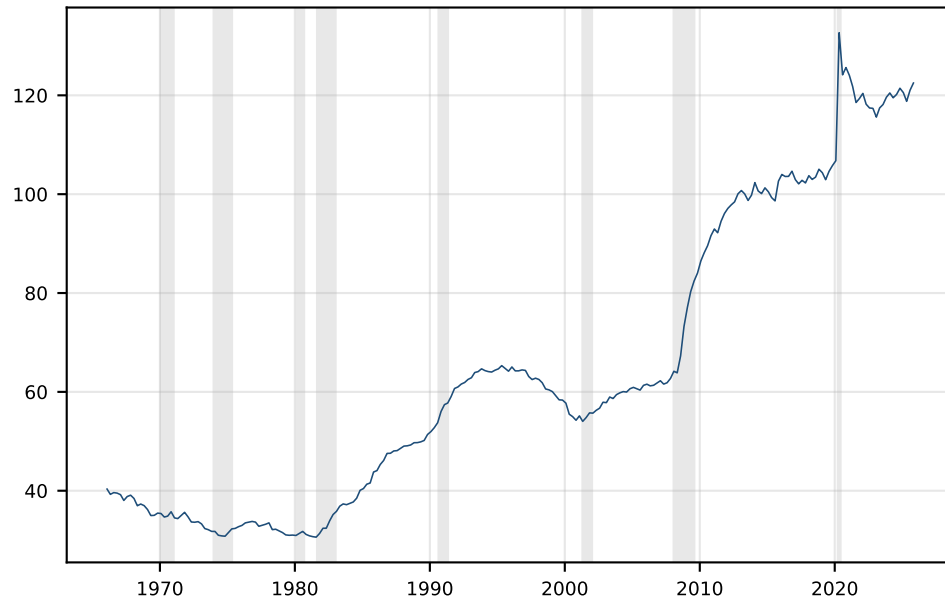
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	60	8.3%	91.7%	0.0010
Q2	60	8.3%	91.7%	0.0156
Q3	59	8.5%	91.5%	0.0245
Q4 (highest)	60	21.7%	78.3%	0.0391

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

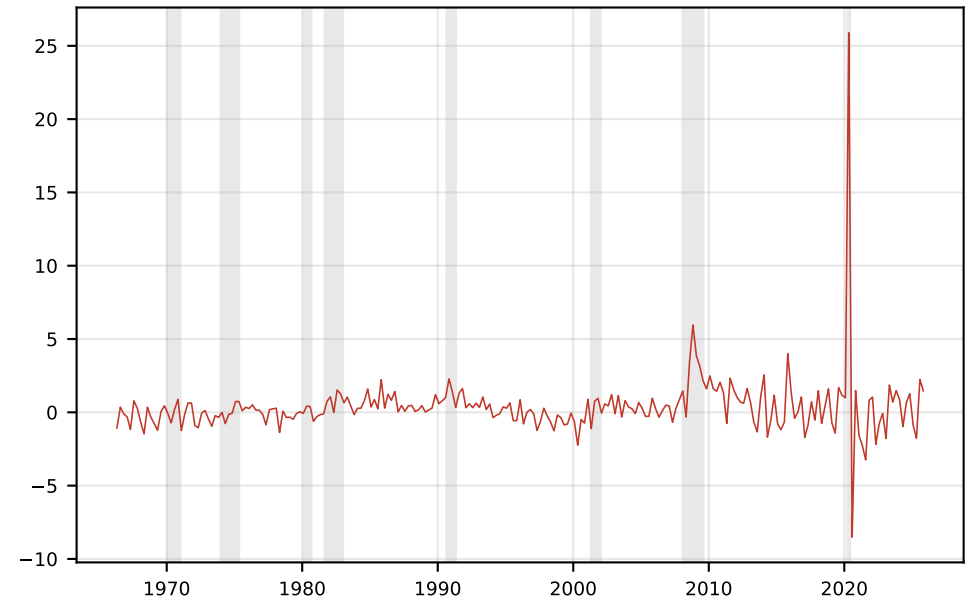
Total Public Debt as Percent of GDP

Block: Debt / Debt | Source: FRED API | Col: GFDEGDQ188S | Transform: 10: First diff with lag 3

Raw Series



After: 10: First diff with lag 3



SUMMARY STATISTICS

N : 1287
Valid : 240
NaN : 1047
Start : 1966-01
End : 2025-10
Mean : 63.9723
Std : 29.2446
Min : 30.6033
Max : 132.6629
Skew : 0.692
Kurt : -0.796

STATIONARITY TESTS

ADF stat : -7.641
ADF p : 0.0000
ADF lags : 2
ADF : Stationary
KPSS stat : 0.410
KPSS p : 0.0730
KPSS : Stationary

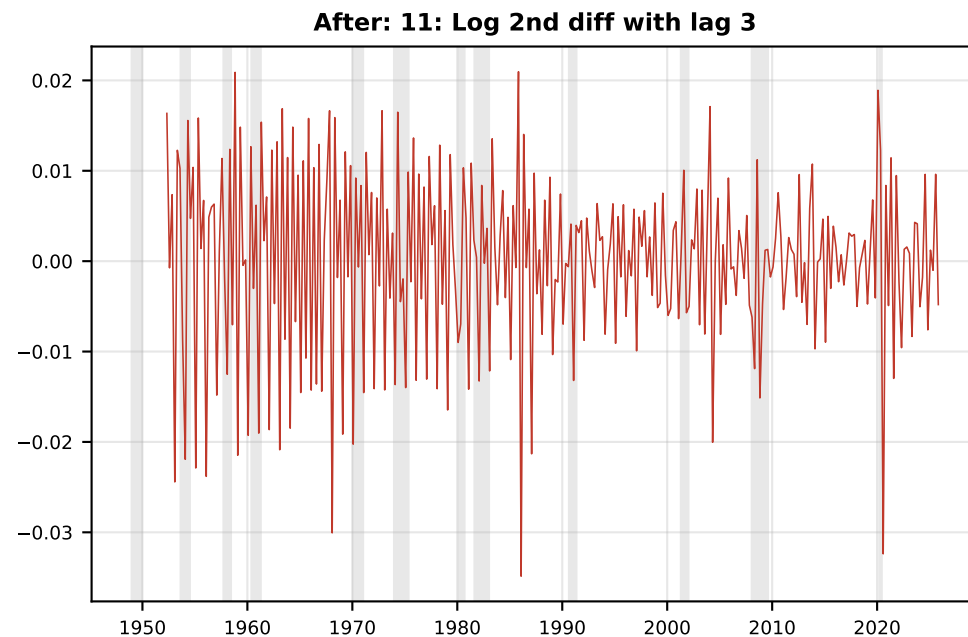
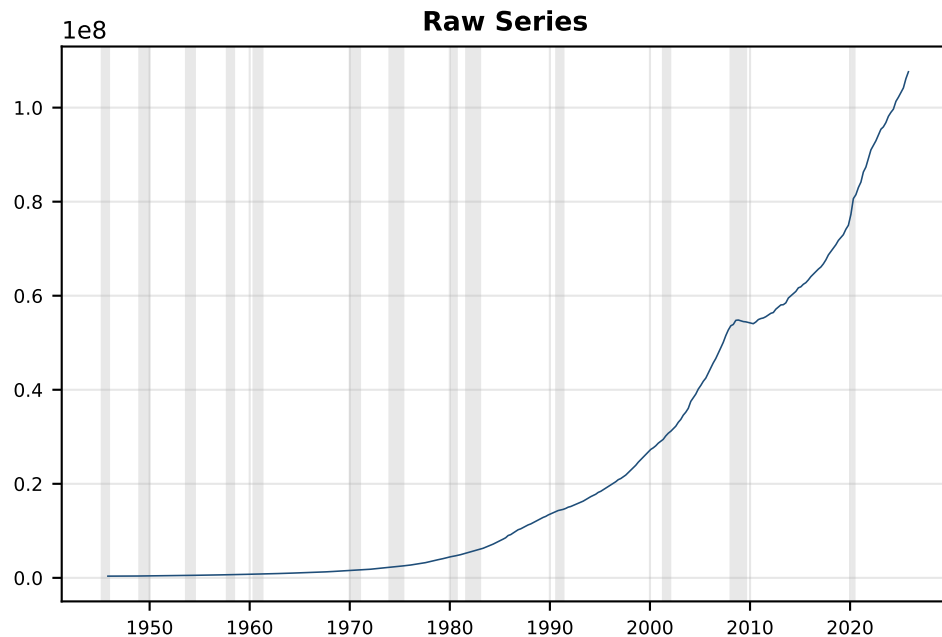
QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	60	5.0%	95.0%	-1.1354
Q2	60	11.7%	88.3%	-0.0426
Q3	59	8.5%	91.5%	0.5113
Q4 (highest)	60	21.7%	78.3%	2.0444

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

All Sectors; Debt Securities and Loans; Liability, Level

Block: Debt / Debt | Source: FRED API | Col: TCMDO | Transform: 11: Log 2nd diff with lag 3



SUMMARY STATISTICS

N : 1287
Valid : 303
NaN : 984
Start : 1945-10
End : 2025-10
Mean : 25256358.4389
Std : 29711834.0395
Min : 357032.0000
Max : 107632484.0000
Skew : 1.116
Kurt : 0.076

STATIONARITY TESTS

ADF stat : -6.960
ADF p : 0.0000
ADF lags : 10
ADF : Stationary
KPSS stat : 0.088
KPSS p : 0.1000
KPSS : Stationary

QUARTILE-BUCKET ANALYSIS

(vs NBER recession/expansion)

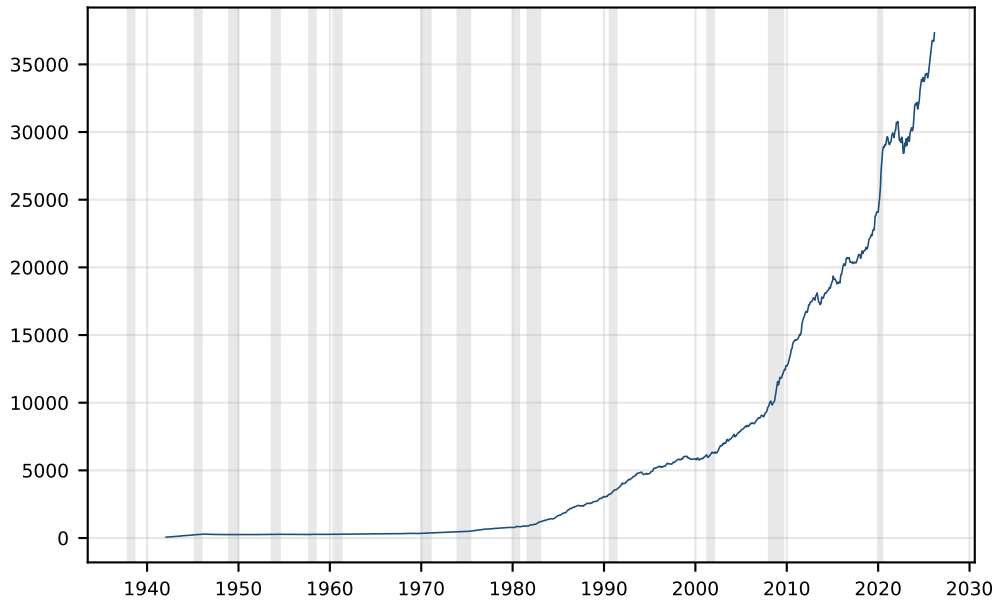
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	74	18.9%	81.1%	-0.0130
Q2	74	12.2%	87.8%	-0.0019
Q3	73	5.5%	94.5%	0.0036
Q4 (highest)	74	13.5%	86.5%	0.0115

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

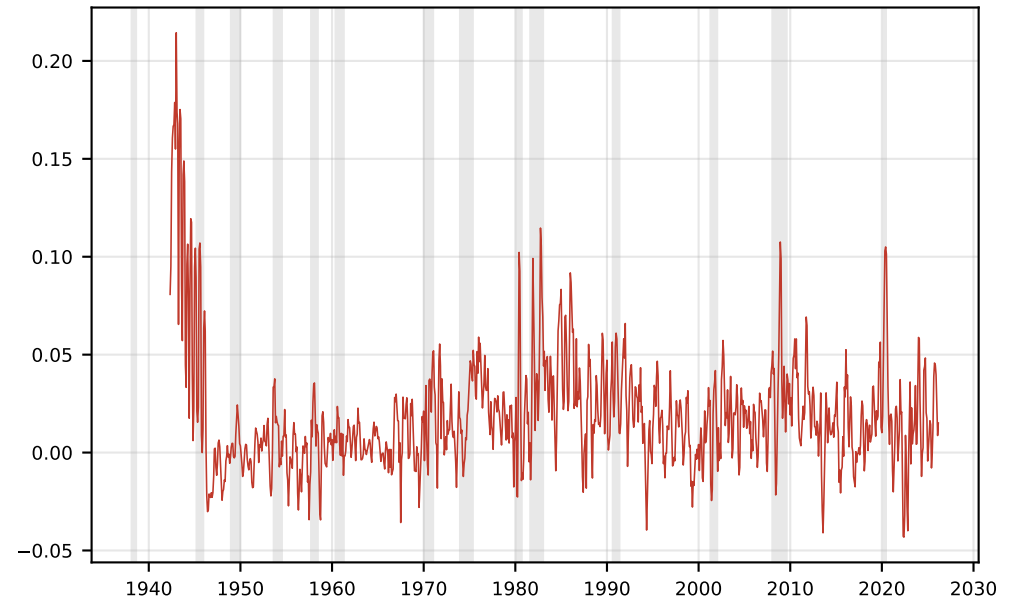
Market Value of Gross Federal DebtI

Block: Debt / Debt | Source: FRED API | Col: MVGFD027MNFBDAL | Transform: 8: Log diff lag 3

Raw Series



After: 8: Log diff lag 3



SUMMARY STATISTICS

N : 1287
Valid : 1010
NaN : 277
Start : 1942-01
End : 2026-02
Mean : 6402.5906
Std : 9177.2097
Min : 63.0000
Max : 37338.4000
Skew : 1.653
Kurt : 1.692

STATIONARITY TESTS

ADF stat : -7.079
ADF p : 0.0000
ADF lags : 12
ADF : Stationary
KPSS stat : 0.156
KPSS p : 0.1000
KPSS : Stationary

QUARTILE-BUCKET ANALYSIS

(vs NBER recession/expansion)

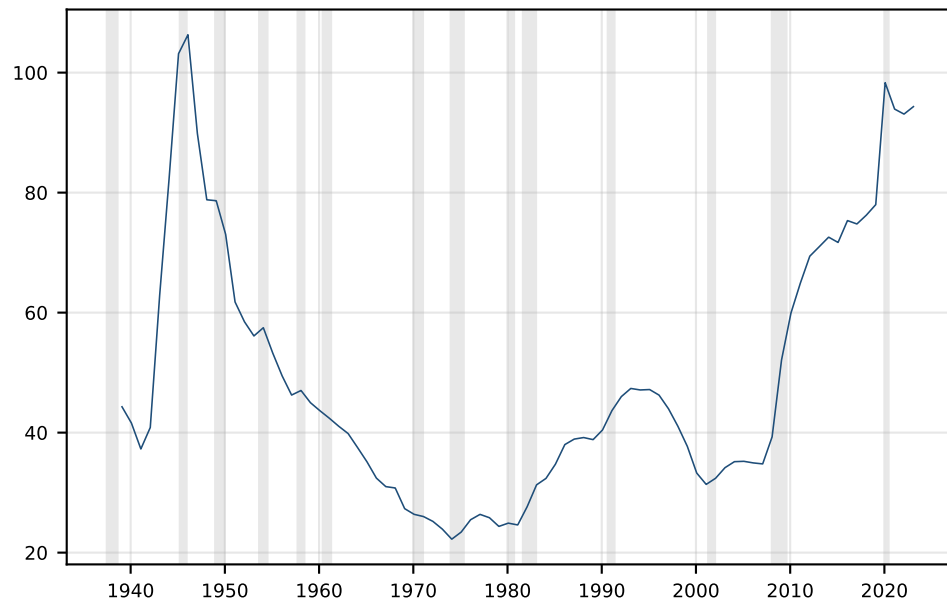
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	252	9.9%	90.1%	-0.0100
Q2	252	8.7%	91.3%	0.0074
Q3	251	13.1%	86.9%	0.0209
Q4 (highest)	252	20.6%	79.4%	0.0573

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

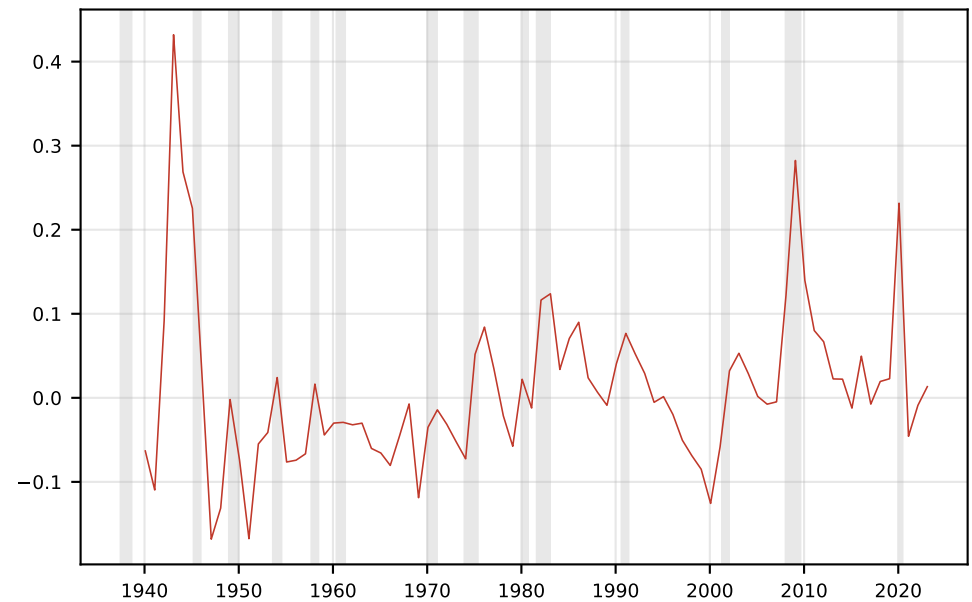
Federal Debt Held by the Public as Percent of GDP

Block: Debt / Debt | Source: FRED API | Col: FYPUGDA188S | Transform: 9: Log diff lag 12

Raw Series



After: 9: Log diff lag 12



SUMMARY STATISTICS

N : 1287
Valid : 85
NaN : 1202
Start : 1939-01
End : 2023-01
Mean : 48.8458
Std : 21.5083
Min : 22.2425
Max : 106.3133
Skew : 0.979
Kurt : 0.043

STATIONARITY TESTS

ADF stat : -4.681
ADF p : 0.0001
ADF lags : 0
ADF : Stationary
KPSS stat : 0.203
KPSS p : 0.1000
KPSS : Stationary

QUARTILE-BUCKET ANALYSIS

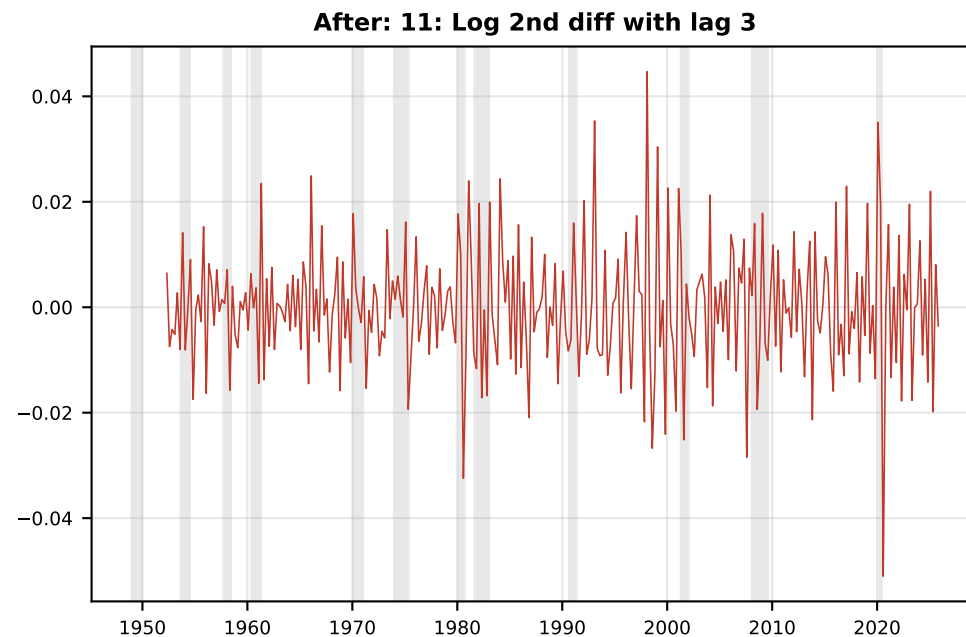
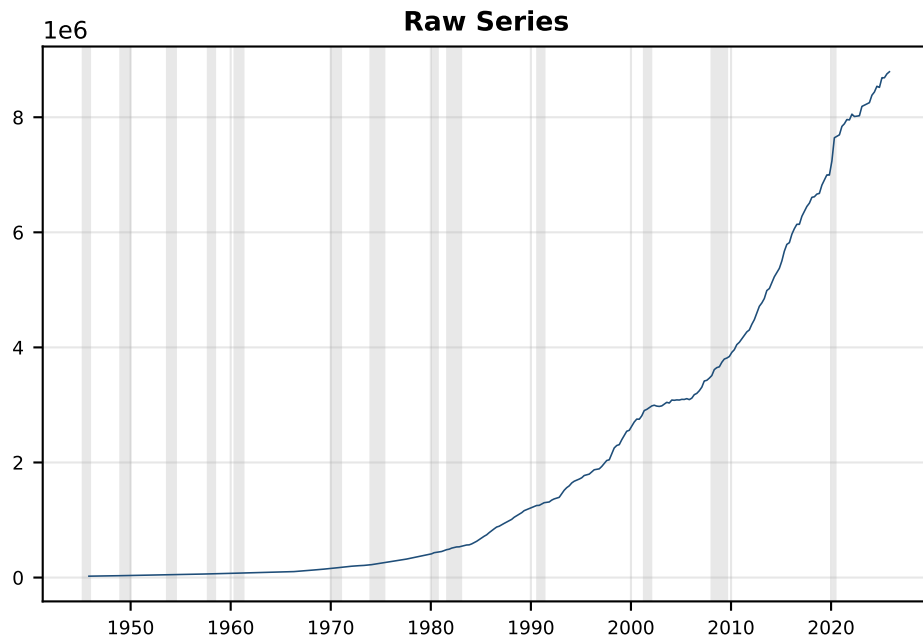
(vs NBER recession/expansion)

Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	21	4.8%	95.2%	-0.0872
Q2	21	9.5%	90.5%	-0.0255
Q3	21	14.3%	85.7%	0.0177
Q4 (highest)	21	23.8%	76.2%	0.1309

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

Nonfinancial Corporate Business; Debt Securities; Liability, Level

Block: Debt / Debt | Source: FRED API | Col: NCBDDBIQ027S | Transform: 11: Log 2nd diff with lag 3



SUMMARY STATISTICS

N : 1287
Valid : 303
NaN : 984
Start : 1945-10
End : 2025-10
Mean : 2170751.0297
Std : 2555182.7327
Min : 24000.0000
Max : 8790804.0000
Skew : 1.186
Kurt : 0.233

STATIONARITY TESTS

ADF stat : -8.292
ADF p : 0.0000
ADF lags : 10
ADF : Stationary
KPSS stat : 0.061
KPSS p : 0.1000
KPSS : Stationary

QUARTILE-BUCKET ANALYSIS

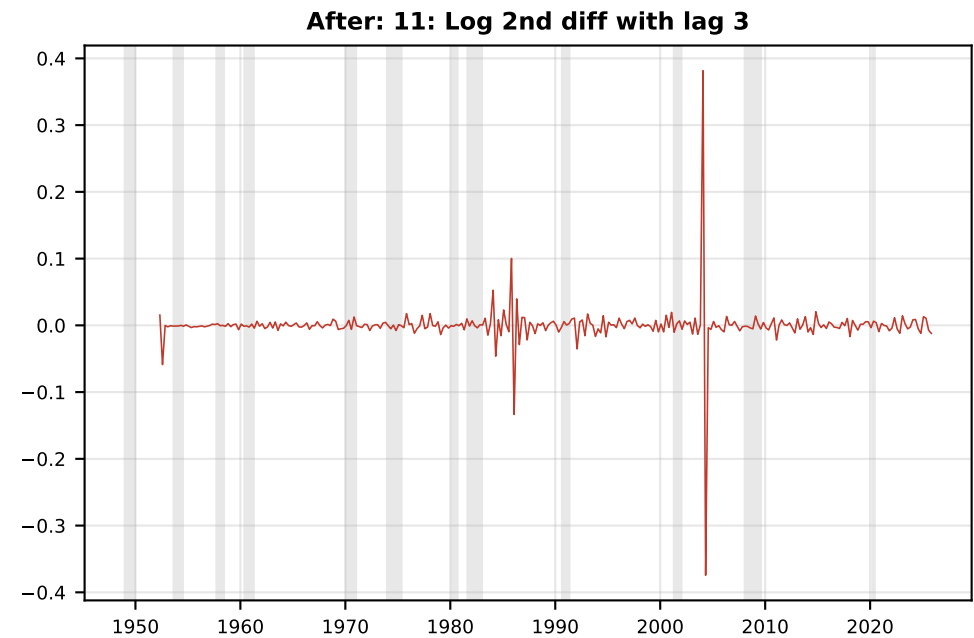
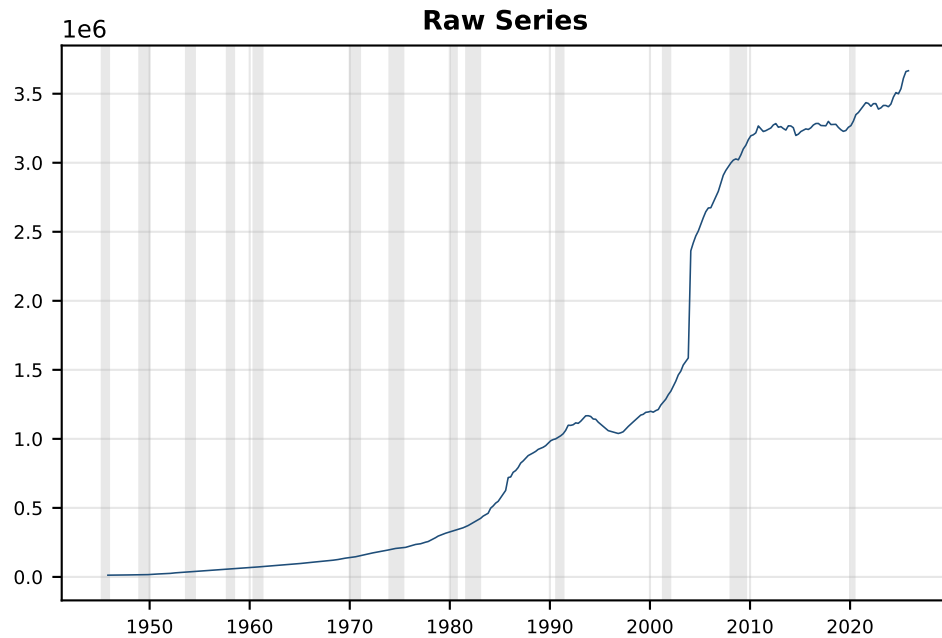
(vs NBER recession/expansion)

Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	74	12.2%	87.8%	-0.0144
Q2	74	10.8%	89.2%	-0.0037
Q3	73	12.3%	87.7%	0.0029
Q4 (highest)	74	14.9%	85.1%	0.0149

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

State and Local Governments; Debt Securities and Loans; Liability, Level

Block: Debt / Debt | Source: FRED API | Col: SLGSDODNS | Transform: 11: Log 2nd diff with lag 3



SUMMARY STATISTICS

N : 1287
Valid : 303
NaN : 984
Start : 1945-10
End : 2025-10
Mean : 1268604.1551
Std : 1296523.1695
Min : 12635.0000
Max : 3666493.0000
Skew : 0.695
Kurt : -1.190

STATIONARITY TESTS

ADF stat : -11.575
ADF p : 0.0000
ADF lags : 5
ADF : Stationary
KPSS stat : 0.186
KPSS p : 0.1000
KPSS : Stationary

QUARTILE-BUCKET ANALYSIS

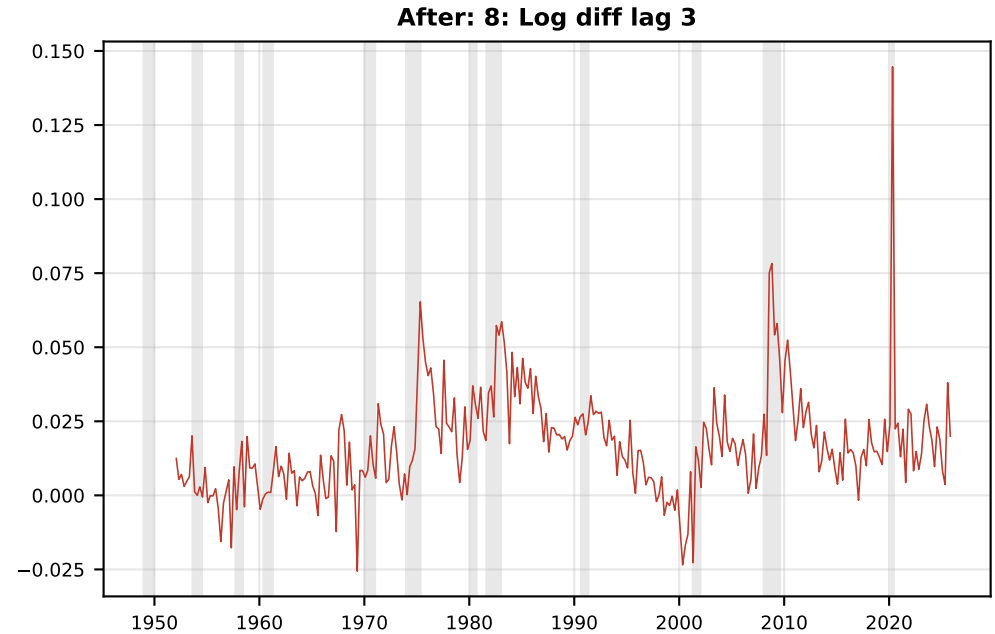
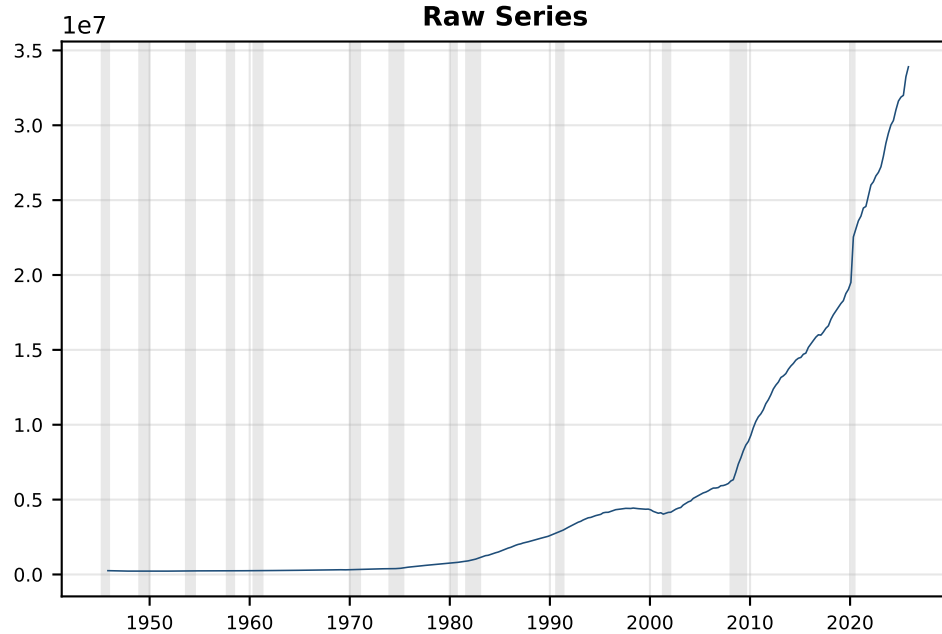
(vs NBER recession/expansion)

Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	74	8.1%	91.9%	-0.0172
Q2	74	17.6%	82.4%	-0.0018
Q3	73	15.1%	84.9%	0.0014
Q4 (highest)	74	9.5%	90.5%	0.0165

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

Federal Government; Debt Securities and Loans; Liability, Level

Block: Debt / Debt | Source: FRED API | Col: FGSDODNS | Transform: 8: Log diff lag 3



SUMMARY STATISTICS

N : 1287
Valid : 303
NaN : 984
Start : 1945-10
End : 2025-10
Mean : 5644524.3069
Std : 8011937.8591
Min : 220012.0000
Max : 33908985.0000
Skew : 1.847
Kurt : 2.615

STATIONARITY TESTS

ADF stat : -3.689
ADF p : 0.0043
ADF lags : 5
ADF : Stationary
KPSS stat : 0.491
KPSS p : 0.0438
KPSS : Non-stationary

QUARTILE-BUCKET ANALYSIS

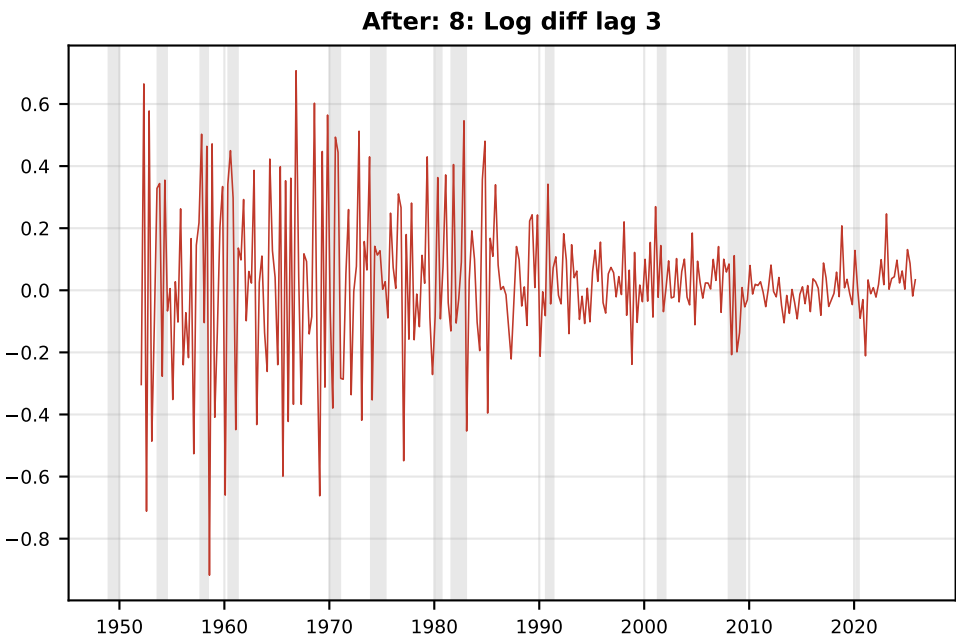
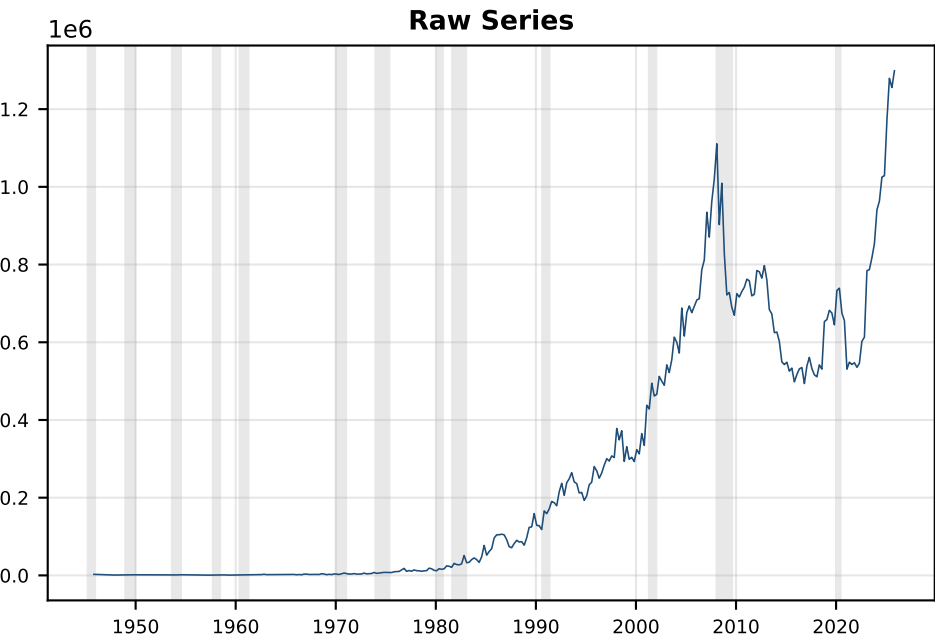
(vs NBER recession/expansion)

Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	74	13.5%	86.5%	-0.0012
Q2	74	9.5%	90.5%	0.0107
Q3	74	6.8%	93.2%	0.0197
Q4 (highest)	74	20.3%	79.7%	0.0386

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

Security Brokers and Dealers; Debt Securities; Asset, Level

Block: Debt / Debt | Source: FRED API | Col: SBDDSA | Transform: 8: Log diff lag 3



SUMMARY STATISTICS

N : 1287
Valid : 303
NaN : 984
Start : 1945-10
End : 2025-10
Mean : 275835.9340
Std : 325159.8039
Min : 570.0000
Max : 1298819.0000
Skew : 0.943
Kurt : -0.190

STATIONARITY TESTS

ADF stat : -4.960
ADF p : 0.0000
ADF lags : 11
ADF : Stationary
KPSS stat : 0.142
KPSS p : 0.1000
KPSS : Stationary

QUARTILE-BUCKET ANALYSIS

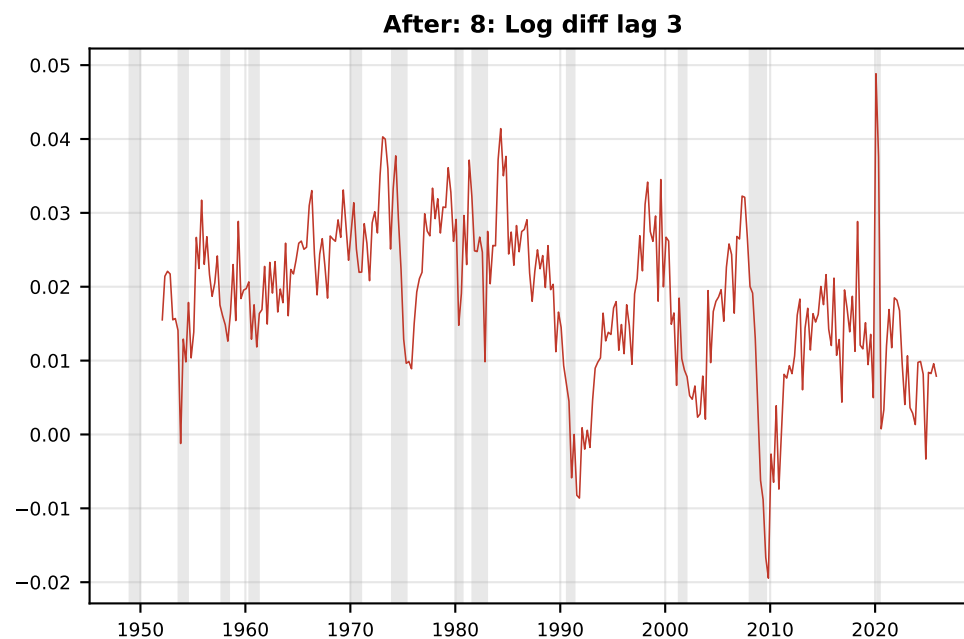
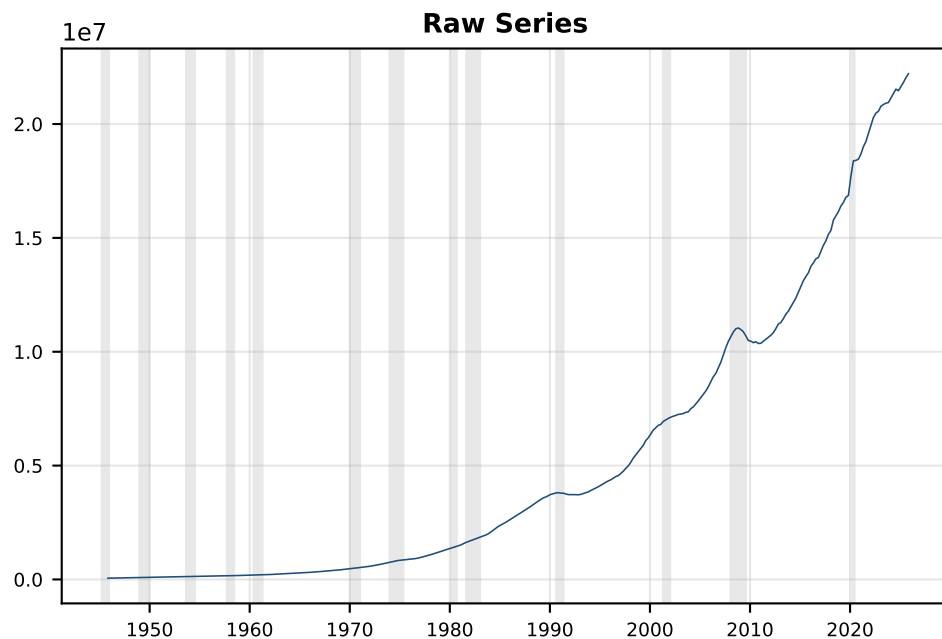
(vs NBER recession/expansion)

Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	74	13.5%	86.5%	-0.2481
Q2	74	10.8%	89.2%	-0.0215
Q3	74	4.1%	95.9%	0.0613
Q4 (highest)	74	21.6%	78.4%	0.3029

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

Nonfinancial Business; Debt Securities and Loans; Liability, Level

Block: Debt / Debt | Source: FRED API | Col: TBSDODNS | Transform: 8: Log diff lag 3



SUMMARY STATISTICS

N : 1287
Valid : 303
NaN : 984
Start : 1945-10
End : 2025-10
Mean : 5506378.9835
Std : 6234705.9547
Min : 55940.0000
Max : 22209277.0000
Skew : 1.174
Kurt : 0.309

STATIONARITY TESTS

ADF stat : -3.629
ADF p : 0.0052
ADF lags : 6
ADF : Stationary
KPSS stat : 0.865
KPSS p : 0.0100
KPSS : Non-stationary

QUARTILE-BUCKET ANALYSIS

(vs NBER recession/expansion)

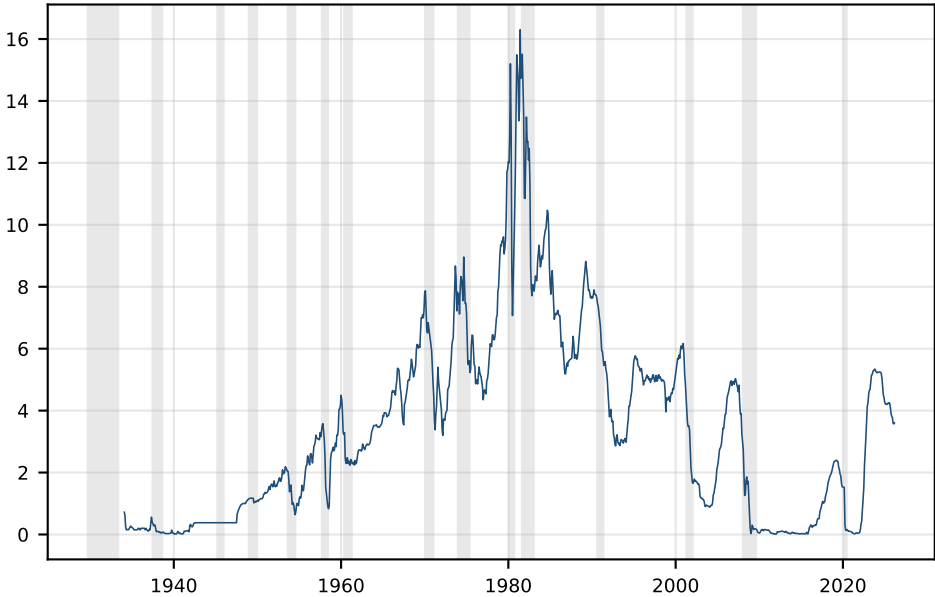
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	74	13.5%	86.5%	0.0043
Q2	74	14.9%	85.1%	0.0155
Q3	74	12.2%	87.8%	0.0222
Q4 (highest)	74	9.5%	90.5%	0.0306

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

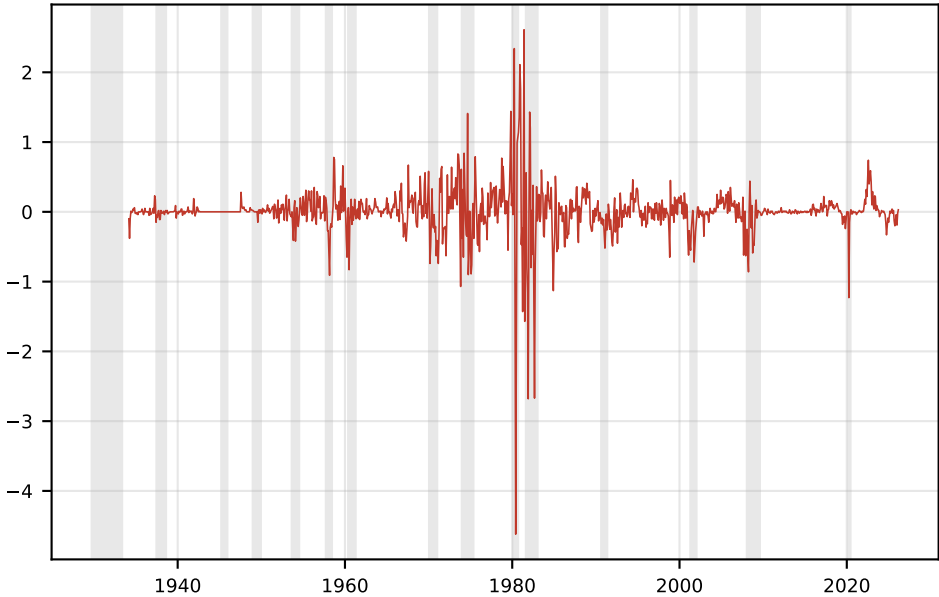
3-Month Treasury Bill Secondary Market Rate, Discount Basis

Block: Debt / Debt | Source: FRED API | Col: TB3MS | Transform: 2: First diff

Raw Series



After: 2: First diff



SUMMARY STATISTICS

N : 1287
Valid : 1106
NaN : 181
Start : 1934-01
End : 2026-02
Mean : 3.4181
Std : 3.0929
Min : 0.0100
Max : 16.3000
Skew : 1.056
Kurt : 1.250

STATIONARITY TESTS

ADF stat : -8.185
ADF p : 0.0000
ADF lags : 12
ADF : Stationary
KPSS stat : 0.059
KPSS p : 0.1000
KPSS : Stationary

QUARTILE-BUCKET ANALYSIS

(vs NBER recession/expansion)

Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	279	27.6%	72.4%	-0.3179
Q2	306	8.8%	91.2%	-0.0115
Q3	246	7.7%	92.3%	0.0370
Q4 (highest)	274	8.0%	92.0%	0.3139

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

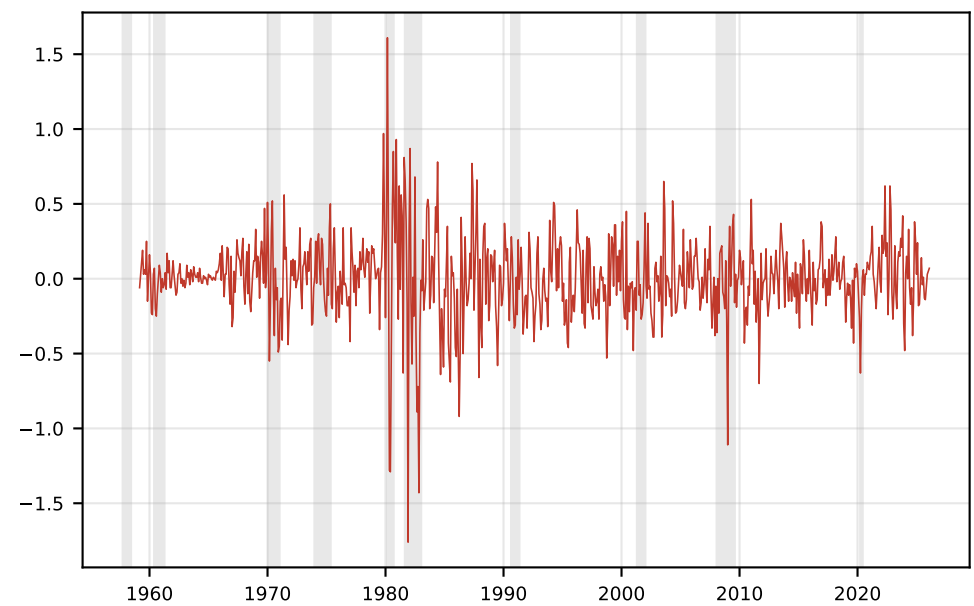
10Y Treasury Yield

Block: Volatility / Stress / Rates volatility | Source: FRED-MD | Col: GS10 | Transform: 2: First diff

Raw Series



After: 2: First diff



SUMMARY STATISTICS

```
N          : 1287
Valid       : 805
NaN         : GS10    482
GS10       412
dtype: int64
Start      : 1959-01
End        : 2026-01
Mean       : 5.7448
Std        : 2.8954
Min        : 0.6200
Max        : 15.3200
Skew       : 0.809
Kurt       : 0.499
```

STATIONARITY TESTS

```
ADF stat    : -7.506
ADF p       : 0.0000
ADF lags    : 11
ADF         : Stationary
KPSS stat   : 0.146
KPSS p      : 0.1000
KPSS        : Stationary
```

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

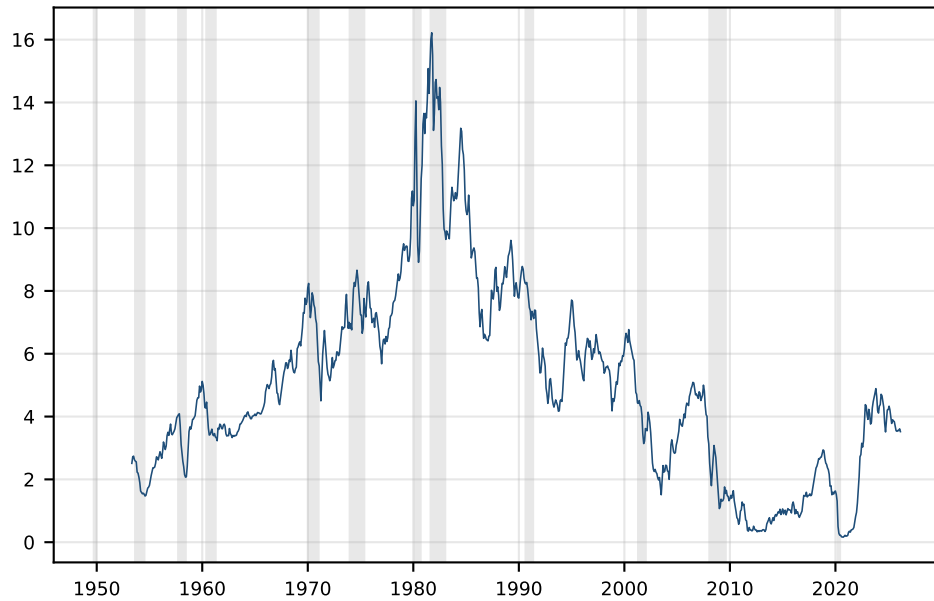
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	201	17.9%	82.1%	-0.3135
Q2	213	8.5%	91.5%	-0.0577
Q3	189	7.9%	92.1%	0.0660
Q4 (highest)	201	12.9%	87.1%	0.3136

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

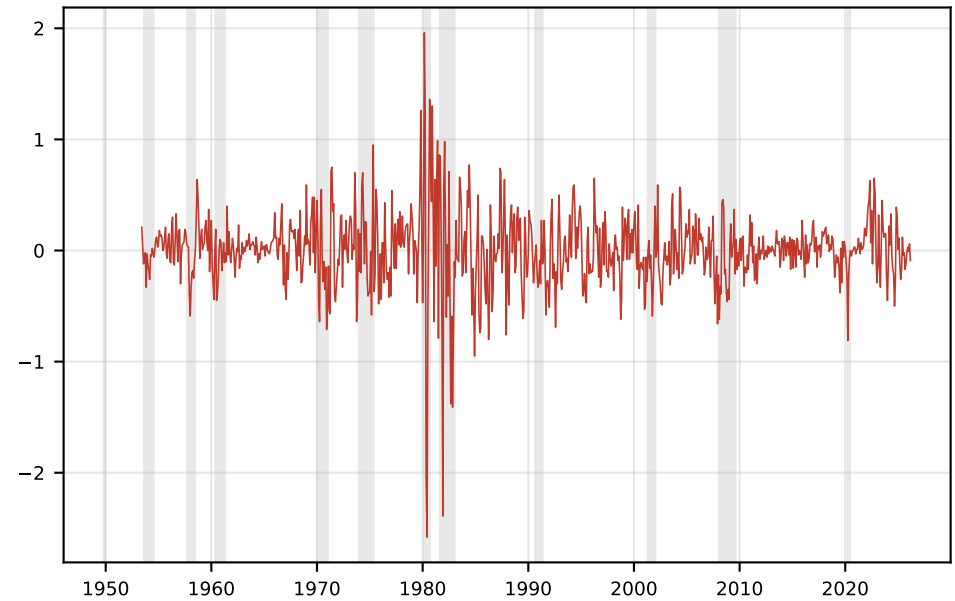
Market Yield on U.S. Treasury Securities at 3-Year Constant Maturity, Quoted on an Investment Basis

Block: Debt / Debt | Source: FRED API | Col: GS3 | Transform: 2: First diff

Raw Series



After: 2: First diff



SUMMARY STATISTICS

N : 1287
Valid : 875
NaN : 412
Start : 1953-04
End : 2026-02
Mean : 4.9999
Std : 3.1414
Min : 0.1600
Max : 16.2200
Skew : 0.790
Kurt : 0.642

STATIONARITY TESTS

ADF stat : -7.788
ADF p : 0.0000
ADF lags : 11
ADF : Stationary
KPSS stat : 0.106
KPSS p : 0.1000
KPSS : Stationary

QUARTILE-BUCKET ANALYSIS

(vs NBER recession/expansion)

Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	219	26.0%	74.0%	-0.3782
Q2	222	7.2%	92.8%	-0.0624
Q3	218	7.3%	92.7%	0.0764
Q4 (highest)	215	11.2%	88.8%	0.3768

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

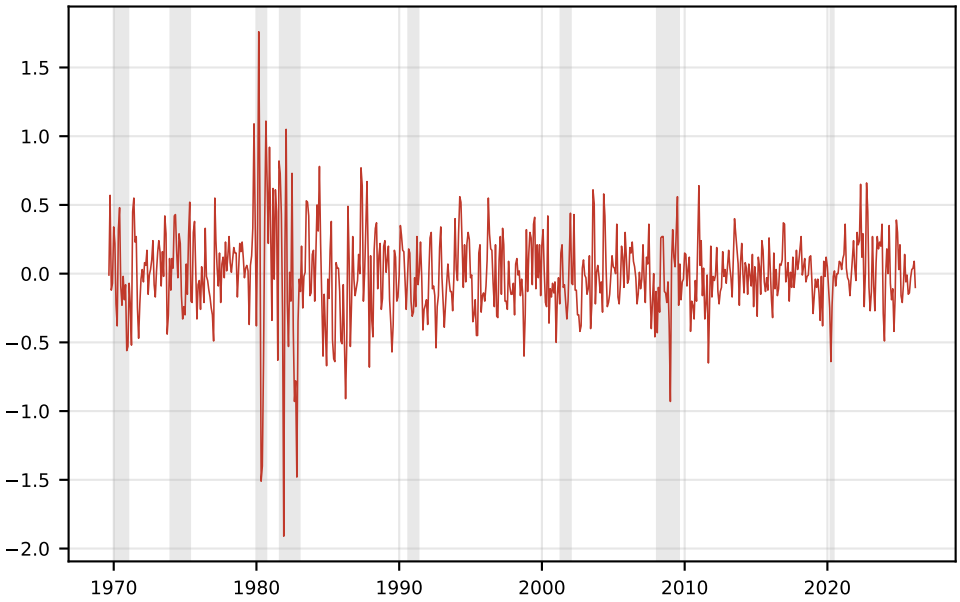
Market Yield on U.S. Treasury Securities at 7-Year Constant Maturity, Quoted on an Investment Basis

Block: Debt / Debt | Source: FRED API | Col: GS7 | Transform: 2: First diff

Raw Series



After: 2: First diff



SUMMARY STATISTICS

N : 1287
Valid : 680
NaN : 607
Start : 1969-07
End : 2026-02
Mean : 5.8179
Std : 3.1946
Min : 0.4600
Max : 15.6500
Skew : 0.564
Kurt : -0.032

STATIONARITY TESTS

ADF stat : -7.431
ADF p : 0.0000
ADF lags : 9
ADF : Stationary
KPSS stat : 0.079
KPSS p : 0.1000
KPSS : Stationary

QUARTILE-BUCKET ANALYSIS

(vs NBER recession/expansion)

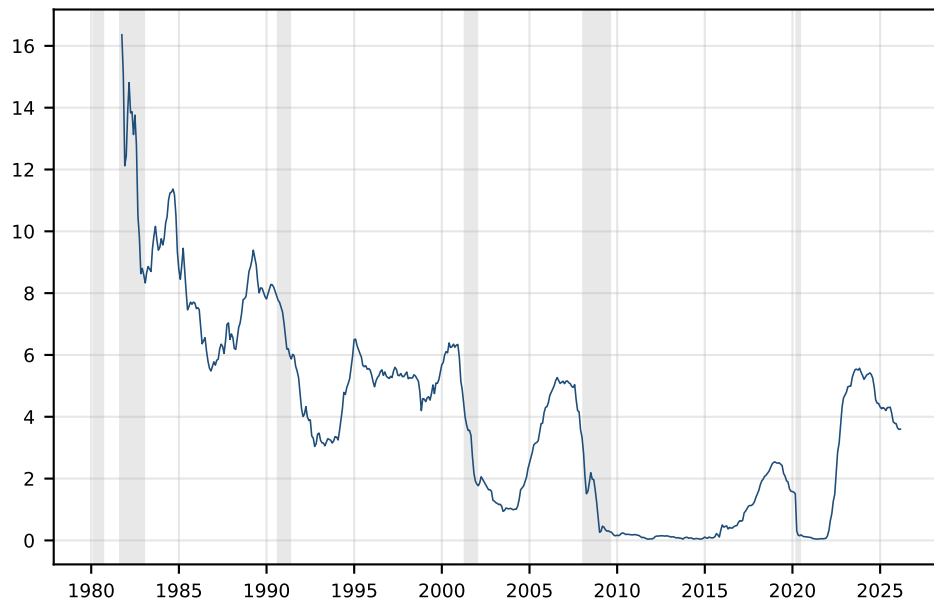
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	171	20.5%	79.5%	-0.3595
Q2	169	8.3%	91.7%	-0.0879
Q3	176	6.2%	93.8%	0.0749
Q4 (highest)	163	15.3%	84.7%	0.3695

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

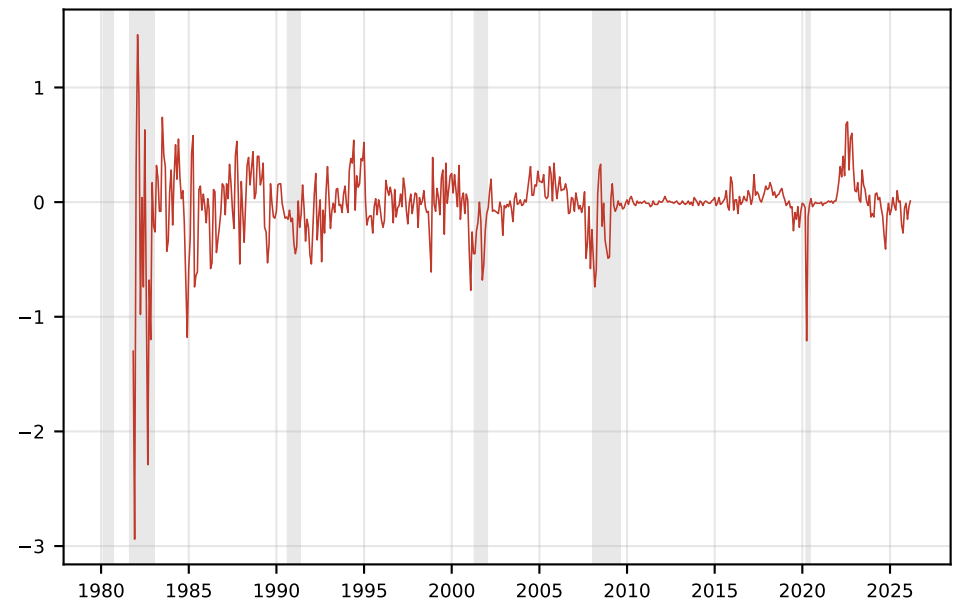
Market Yield on U.S. Treasury Securities at 6-Month Constant Maturity, Quoted on an Investment Basis

Block: Debt / Debt | Source: FRED API | Col: GS6M | Transform: 2: First diff

Raw Series



After: 2: First diff



SUMMARY STATISTICS

N : 1287
Valid : 534
NaN : 753
Start : 1981-09
End : 2026-02
Mean : 3.9968
Std : 3.2325
Min : 0.0400
Max : 16.3600
Skew : 0.706
Kurt : 0.348

STATIONARITY TESTS

ADF stat : -6.039
ADF p : 0.0000
ADF lags : 9
ADF : Stationary
KPSS stat : 0.374
KPSS p : 0.0882
KPSS : Stationary

QUARTILE-BUCKET ANALYSIS

(vs NBER recession/expansion)

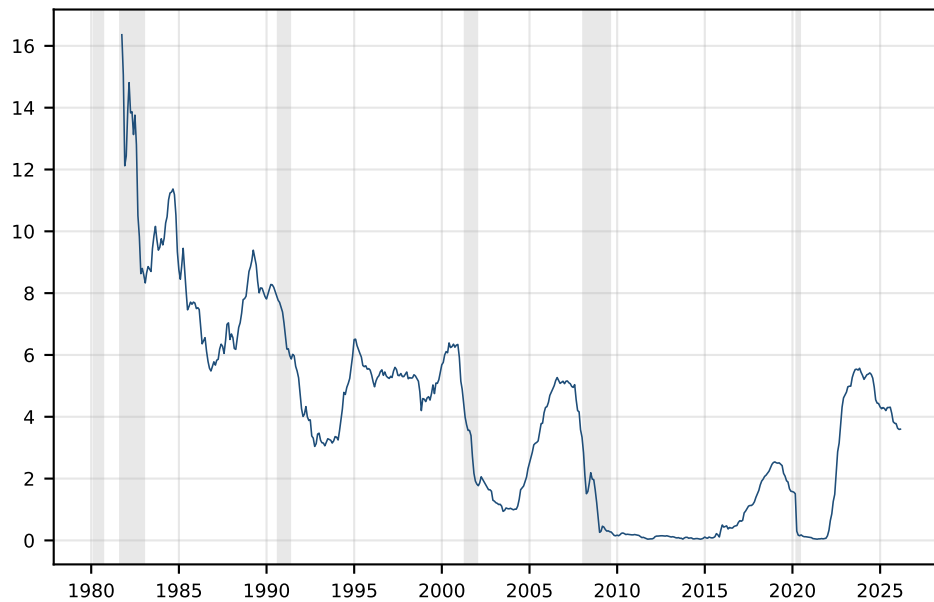
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	142	21.8%	78.2%	-0.3417
Q2	141	4.3%	95.7%	-0.0260
Q3	118	4.2%	95.8%	0.0400
Q4 (highest)	132	6.1%	93.9%	0.2630

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

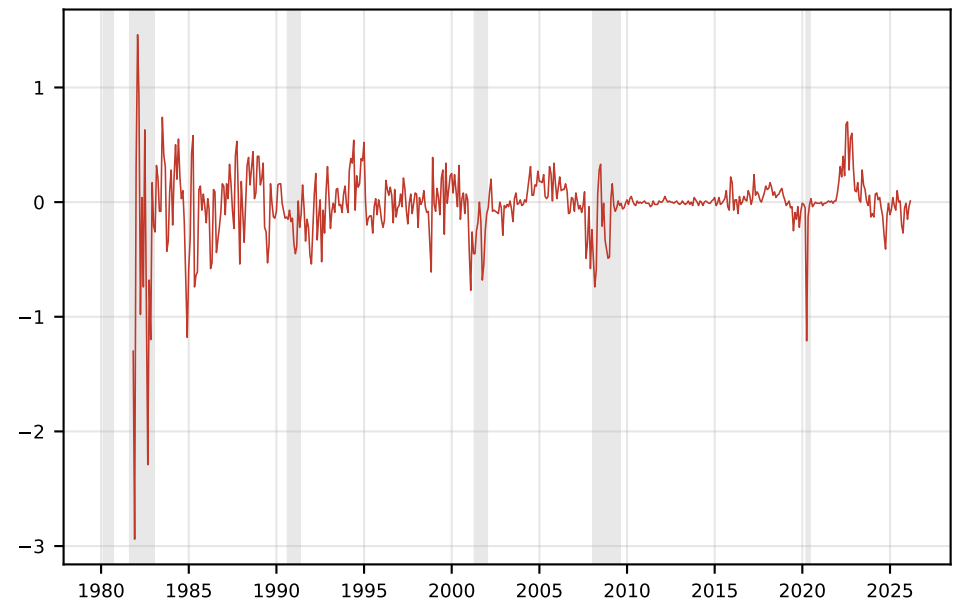
Market Yield on U.S. Treasury Securities at 6-Month Constant Maturity, Quoted on an Investment Basis

Block: Debt / Debt | Source: FRED API | Col: GS6M | Transform: 2: First diff

Raw Series



After: 2: First diff



SUMMARY STATISTICS

N : 1287
Valid : 534
NaN : 753
Start : 1981-09
End : 2026-02
Mean : 3.9968
Std : 3.2325
Min : 0.0400
Max : 16.3600
Skew : 0.706
Kurt : 0.348

STATIONARITY TESTS

ADF stat : -6.039
ADF p : 0.0000
ADF lags : 9
ADF : Stationary
KPSS stat : 0.374
KPSS p : 0.0882
KPSS : Stationary

QUARTILE-BUCKET ANALYSIS

(vs NBER recession/expansion)

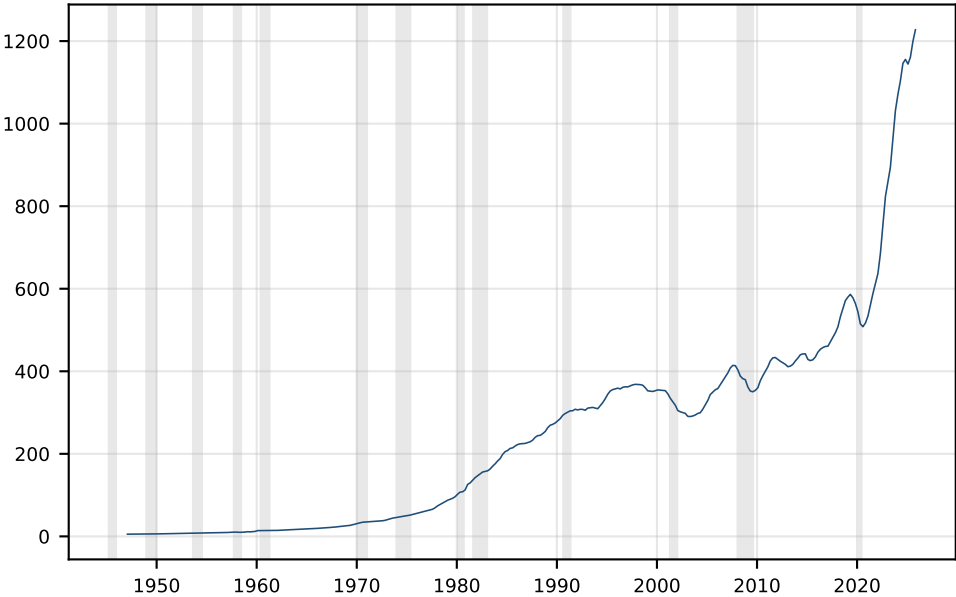
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	142	21.8%	78.2%	-0.3417
Q2	141	4.3%	95.7%	-0.0260
Q3	118	4.2%	95.8%	0.0400
Q4 (highest)	132	6.1%	93.9%	0.2630

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

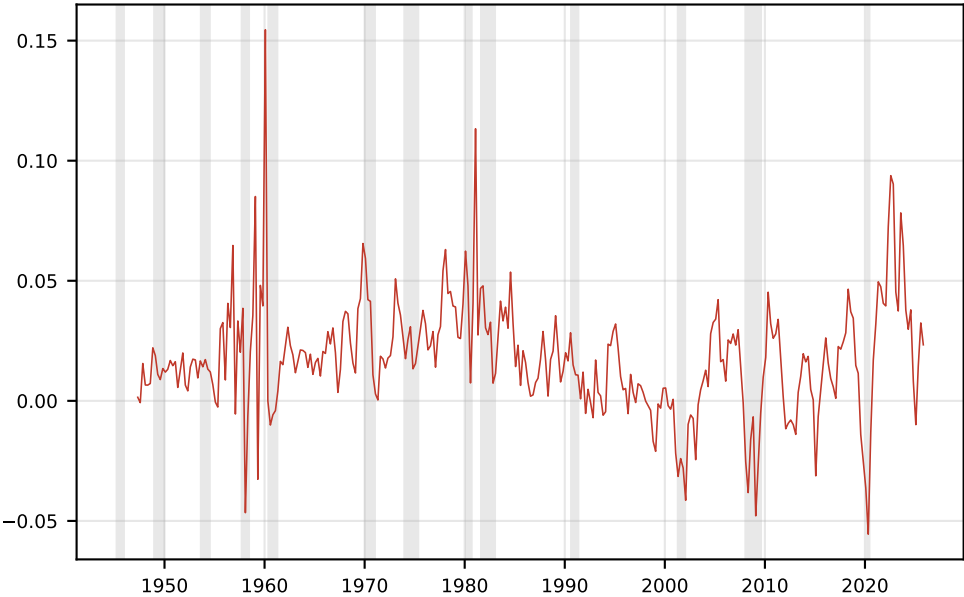
Federal government current expenditures: Interest payments

Block: Sovereign credibility / Sovereign credibility | Source: FRED API | Col: A091RC1Q027SBEA | Transform: 8: Log diff lag 3

Raw Series



After: 8: Log diff lag 3



SUMMARY STATISTICS

N : 1287
Valid : 316
NaN : 971
Start : 1947-01
End : 2025-10
Mean : 242.7559
Std : 249.2108
Min : 5.3520
Max : 1227.4950
Skew : 1.506
Kurt : 3.086

STATIONARITY TESTS

ADF stat : -4.089
ADF p : 0.0010
ADF lags : 8
ADF : Stationary
KPSS stat : 0.355
KPSS p : 0.0964
KPSS : Stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

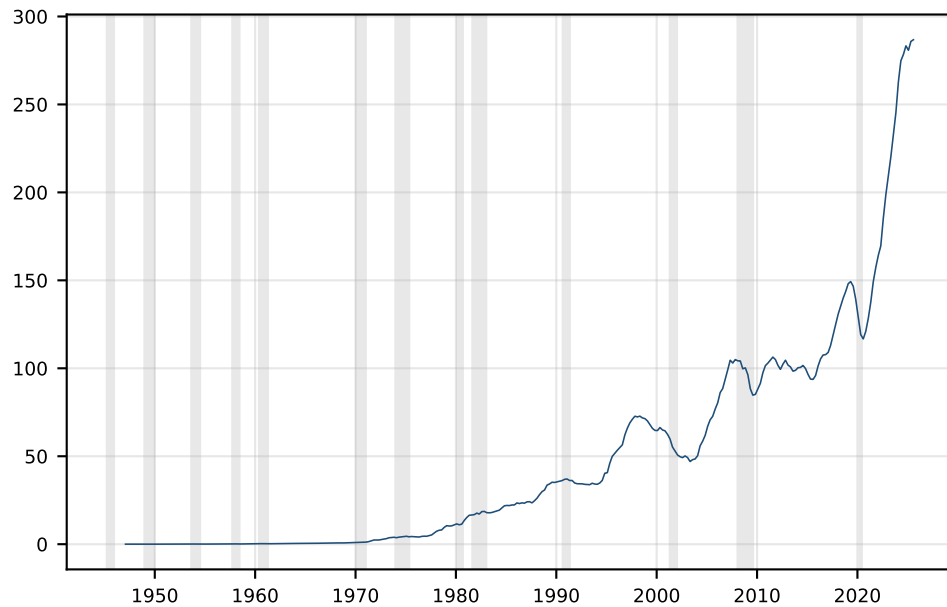
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	79	19.0%	81.0%	-0.0095
Q2	79	15.2%	84.8%	0.0110
Q3	78	6.4%	93.6%	0.0222
Q4 (highest)	79	11.4%	88.6%	0.0455

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

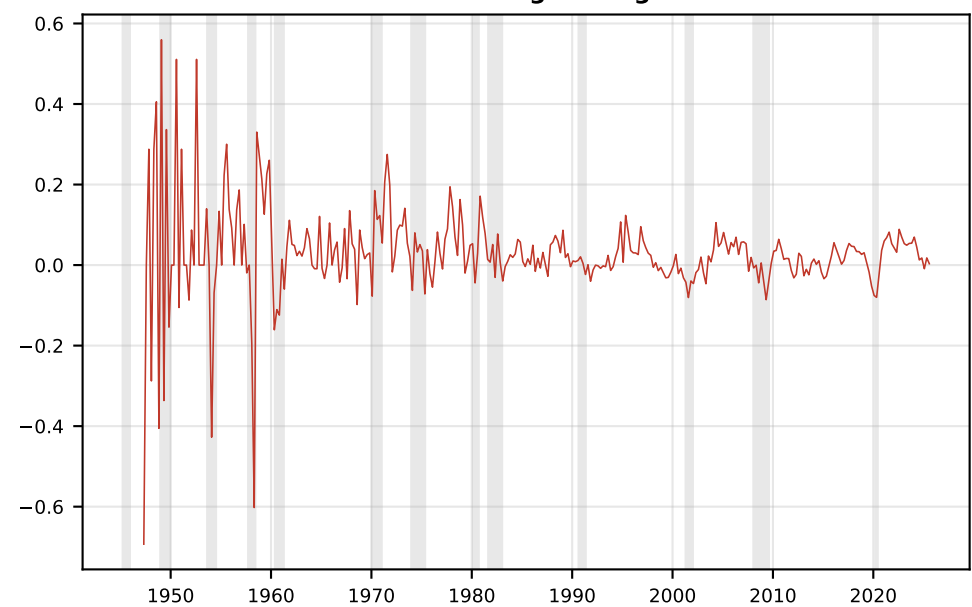
Federal government current expenditures: Interest payments: to the rest of the world

Block: Sovereign credibility / Sovereign credibility | Source: FRED API | Col: B093RC1Q027SBEA | Transform: 8: Log diff lag 3

Raw Series



After: 8: Log diff lag 3



SUMMARY STATISTICS

N : 1287
Valid : 315
NaN : 972
Start : 1947-01
End : 2025-07
Mean : 47.6213
Std : 60.8424
Min : 0.0120
Max : 286.7790
Skew : 1.787
Kurt : 3.581

STATIONARITY TESTS

ADF stat : -5.533
ADF p : 0.0000
ADF lags : 7
ADF : Stationary
KPSS stat : 0.408
KPSS p : 0.0738
KPSS : Stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

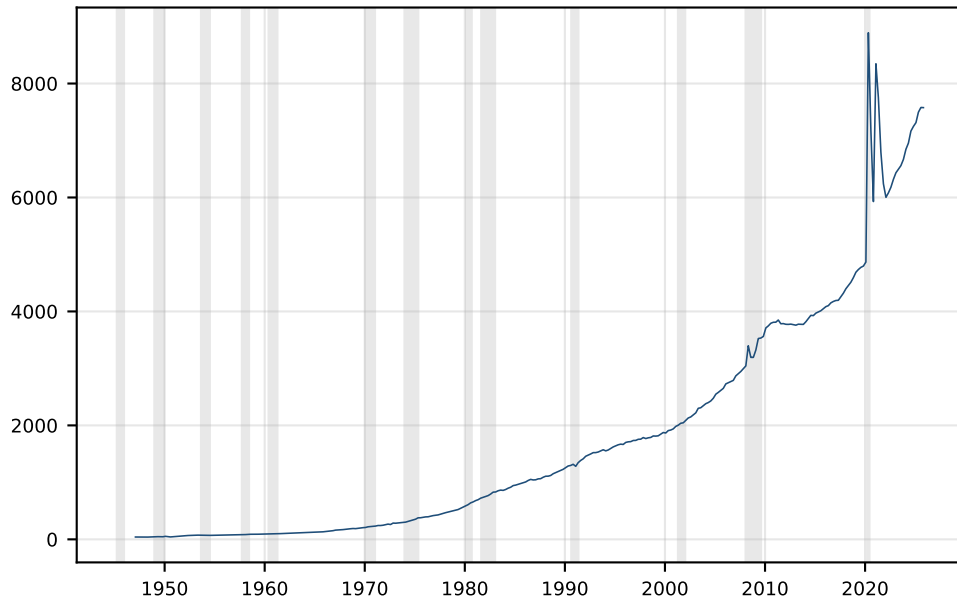
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	79	24.1%	75.9%	-0.0740
Q2	78	12.8%	87.2%	0.0062
Q3	78	6.4%	93.6%	0.0389
Q4 (highest)	79	8.9%	91.1%	0.1483

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

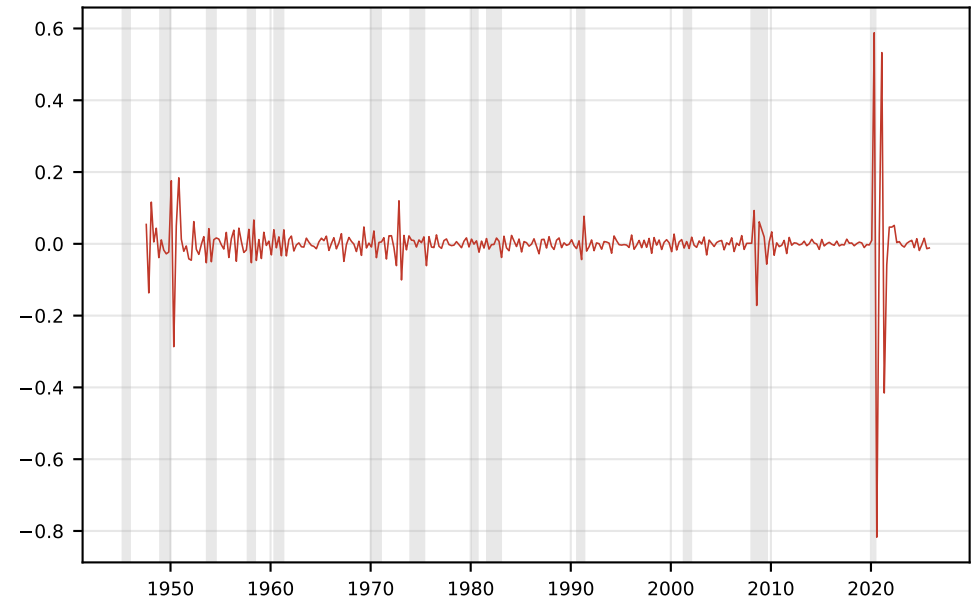
Federal Government: Current Expenditures

Block: Sovereign credibility / Sovereign credibility | Source: FRED API | Col: FGEXPND | Transform: 11: Log 2nd diff with lag 3

Raw Series



After: 11: Log 2nd diff with lag 3



SUMMARY STATISTICS

N : 1287
Valid : 316
NaN : 971
Start : 1947-01
End : 2025-10
Mean : 1755.4208
Std : 2001.4859
Min : 38.1490
Max : 8891.3730
Skew : 1.389
Kurt : 1.289

STATIONARITY TESTS

ADF stat : -10.226
ADF p : 0.0000
ADF lags : 10
ADF : Stationary
KPSS stat : 0.011
KPSS p : 0.1000
KPSS : Stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

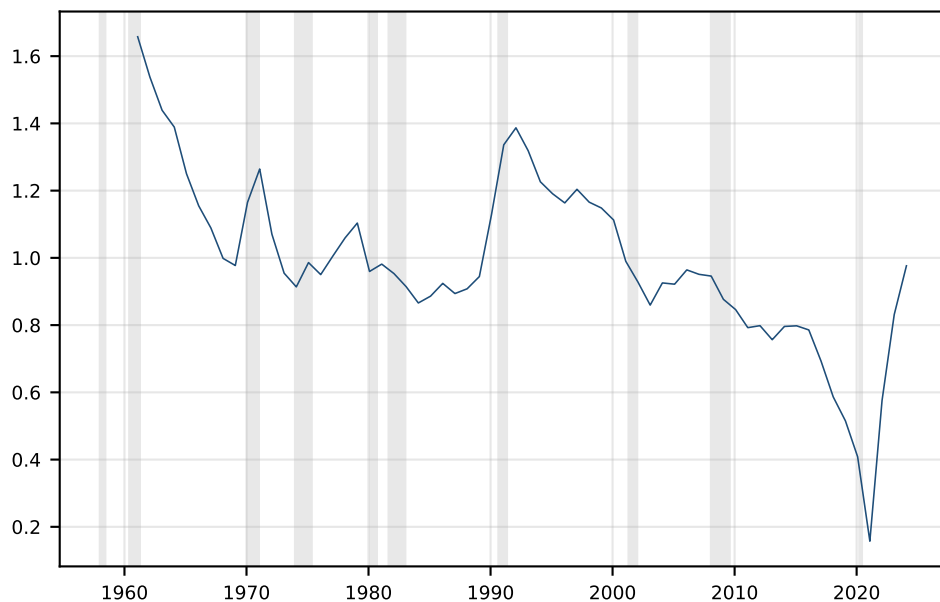
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	79	16.5%	83.5%	-0.0491
Q2	78	6.4%	93.6%	-0.0046
Q3	78	15.4%	84.6%	0.0065
Q4 (highest)	79	13.9%	86.1%	0.0474

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

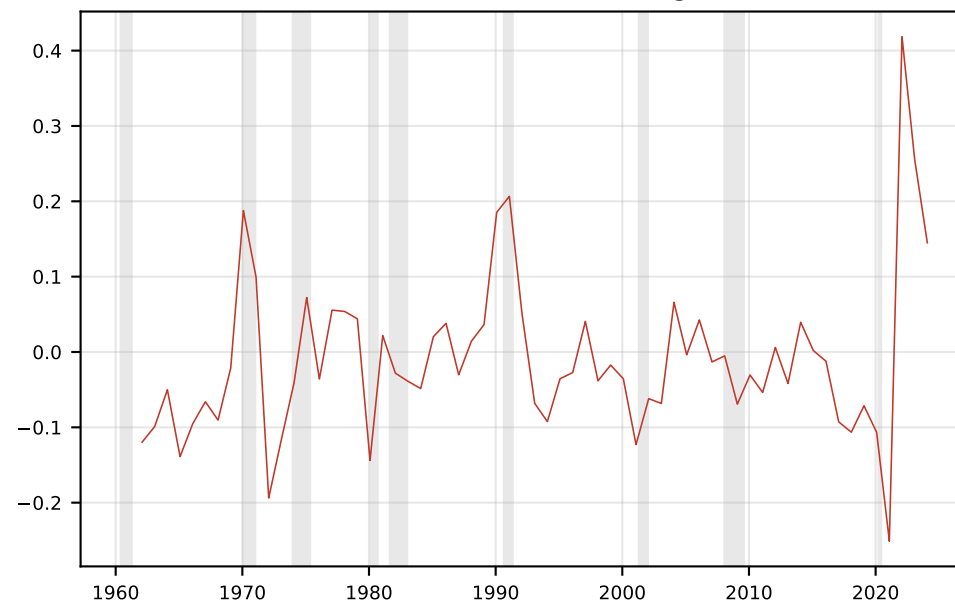
Population Growth for the United States

Block: Demographics / Demographics | Source: FRED API | Col: SPPOPGROWUSA | Transform: 12: First diff with lag 12

Raw Series



After: 12: First diff with lag 12



SUMMARY STATISTICS

N : 1287
Valid : 64
NaN : 1223
Start : 1961-01
End : 2024-01
Mean : 0.9884
Std : 0.2543
Min : 0.1573
Max : 1.6577
Skew : -0.234
Kurt : 1.647

STATIONARITY TESTS

ADF stat : -5.468
ADF p : 0.0000
ADF lags : 0
ADF : Stationary
KPSS stat : 0.148
KPSS p : 0.1000
KPSS : Stationary

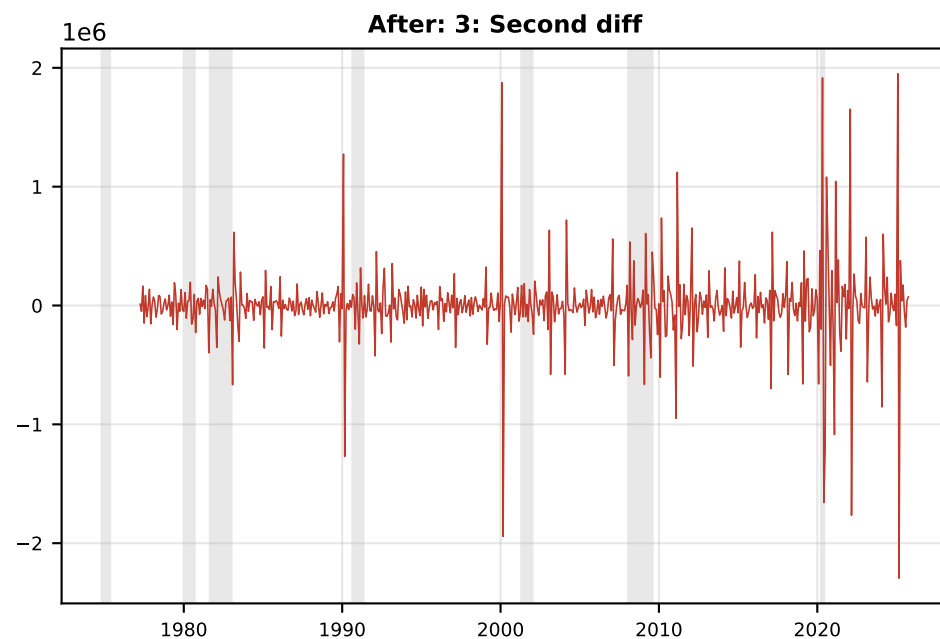
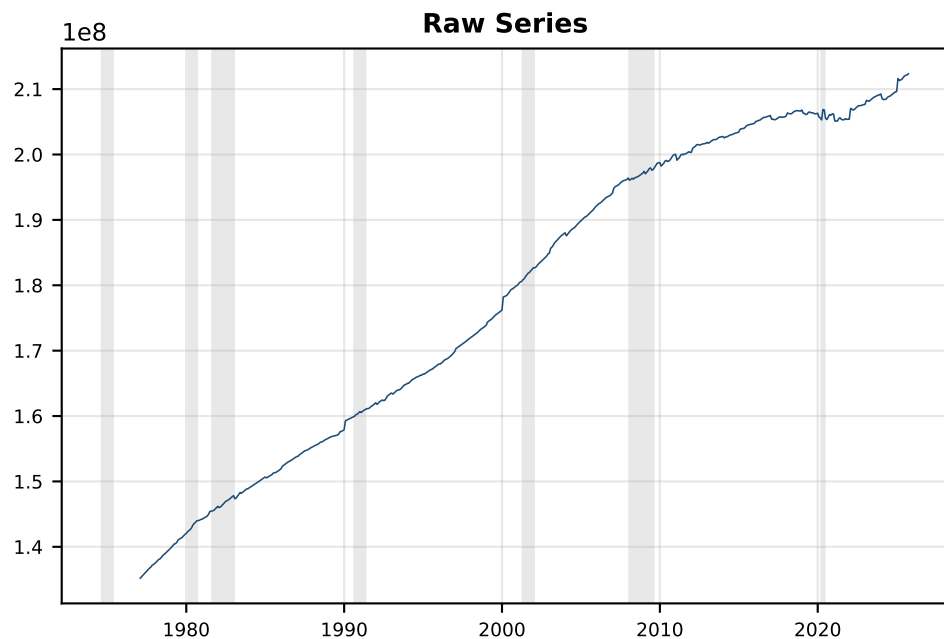
QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	16	6.2%	93.8%	-0.1194
Q2	16	6.2%	93.8%	-0.0465
Q3	15	13.3%	86.7%	0.0007
Q4 (highest)	16	18.8%	81.2%	0.1227

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

Working-Age Population Total: From 15 to 64 Years for United States

Block: Demographics / Demographics | Source: FRED API | Col: LFWA64TTUSM647S | Transform: 3: Second diff



SUMMARY STATISTICS

N : 1287
Valid : 585
NaN : 702
Start : 1977-01
End : 2025-09
Mean : 178771461.3675
Std : 23625697.9044
Min : 135204600.0000
Max : 212352300.0000
Skew : -0.215
Kurt : -1.413

STATIONARITY TESTS

ADF stat : -10.894
ADF p : 0.0000
ADF lags : 12
ADF : Stationary
KPSS stat : 0.049
KPSS p : 0.1000
KPSS : Stationary

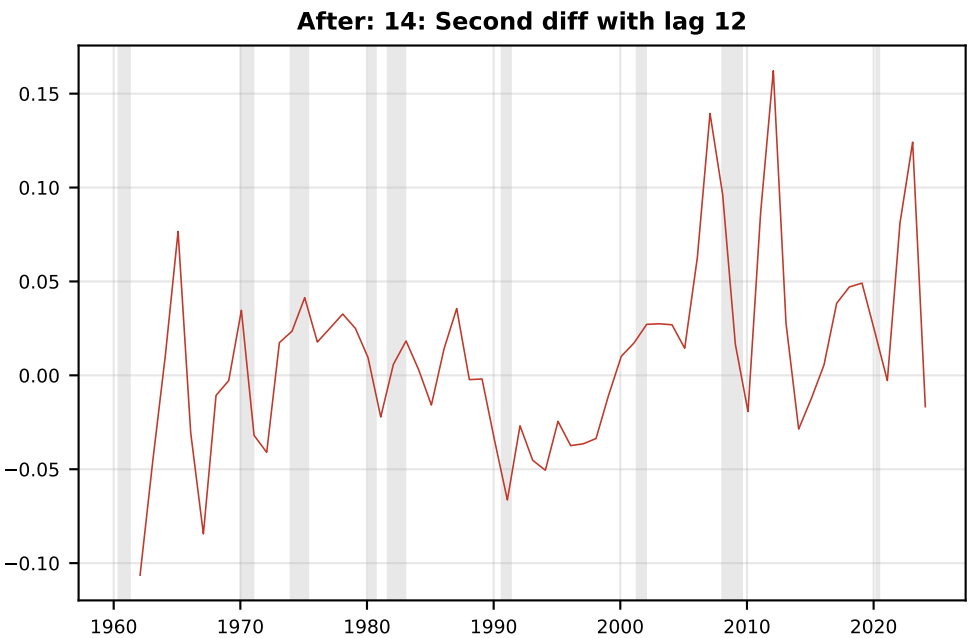
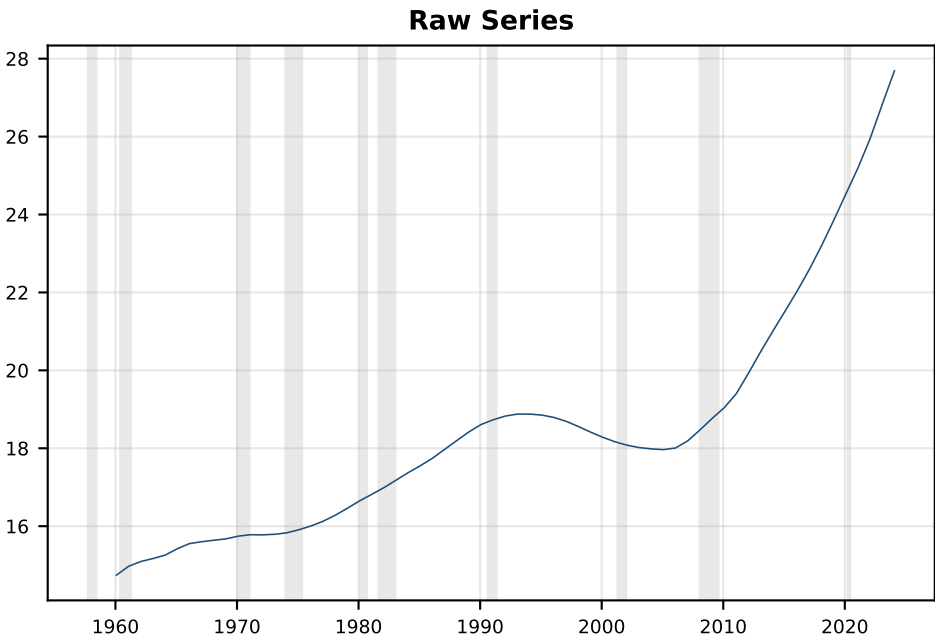
QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	146	12.3%	87.7%	-285323.2877
Q2	146	4.8%	95.2%	-30534.9315
Q3	145	11.7%	88.3%	37635.8621
Q4 (highest)	146	11.0%	89.0%	278296.5753

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

Age Dependency Ratio: Older Dependents to Working-Age Population for the United States

Block: Demographics / Demographics | Source: FRED API | Col: SPPODPNDOLUSA | Transform: 14: Second diff with lag 12



SUMMARY STATISTICS

N : 1287
Valid : 65
NaN : 1222
Start : 1960-01
End : 2024-01
Mean : 18.4638
Std : 2.9988
Min : 14.7428
Max : 27.6906
Skew : 1.348
Kurt : 1.547

STATIONARITY TESTS

ADF stat : -1.323
ADF p : 0.6184
ADF lags : 12
ADF : Non-stationary
KPSS stat : 0.656
KPSS p : 0.0176
KPSS : Non-stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

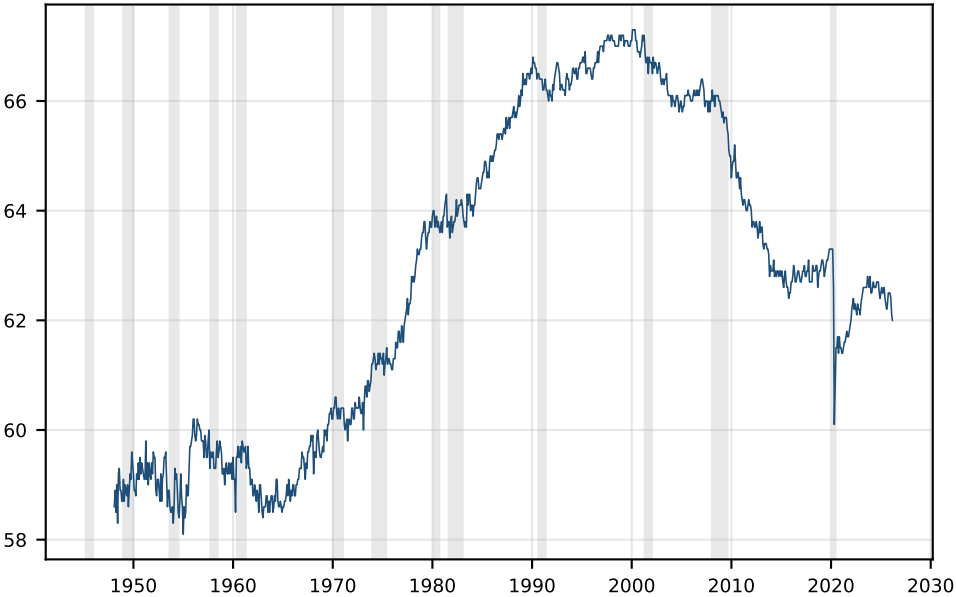
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	16	6.2%	93.8%	-0.0453
Q2	16	6.2%	93.8%	-0.0052
Q3	15	13.3%	86.7%	0.0203
Q4 (highest)	16	18.8%	81.2%	0.0710

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

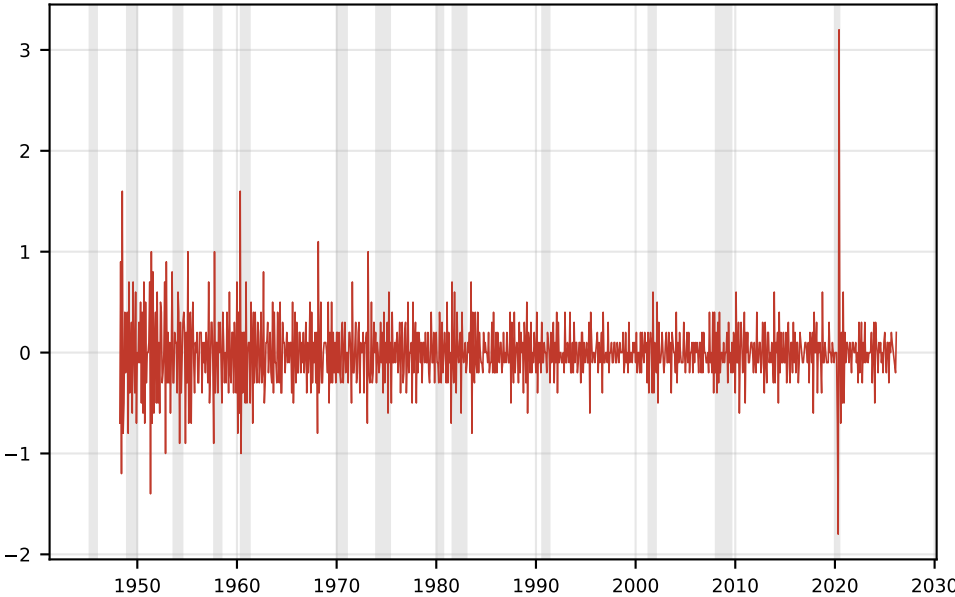
Labor Force Participation Rate

Block: Demographics / Demographics | Source: FRED API | Col: CIVPART | Transform: 3: Second diff

Raw Series



After: 3: Second diff



SUMMARY STATISTICS

N : 1287
Valid : 937
NaN : 350
Start : 1948-01
End : 2026-02
Mean : 62.8304
Std : 2.8827
Min : 58.1000
Max : 67.3000
Skew : -0.015
Kurt : -1.423

STATIONARITY TESTS

ADF stat : -16.527
ADF p : 0.0000
ADF lags : 12
ADF : Stationary
KPSS stat : 0.169
KPSS p : 0.1000
KPSS : Stationary

QUARTILE-BUCKET ANALYSIS

(vs NBER recession/expansion)

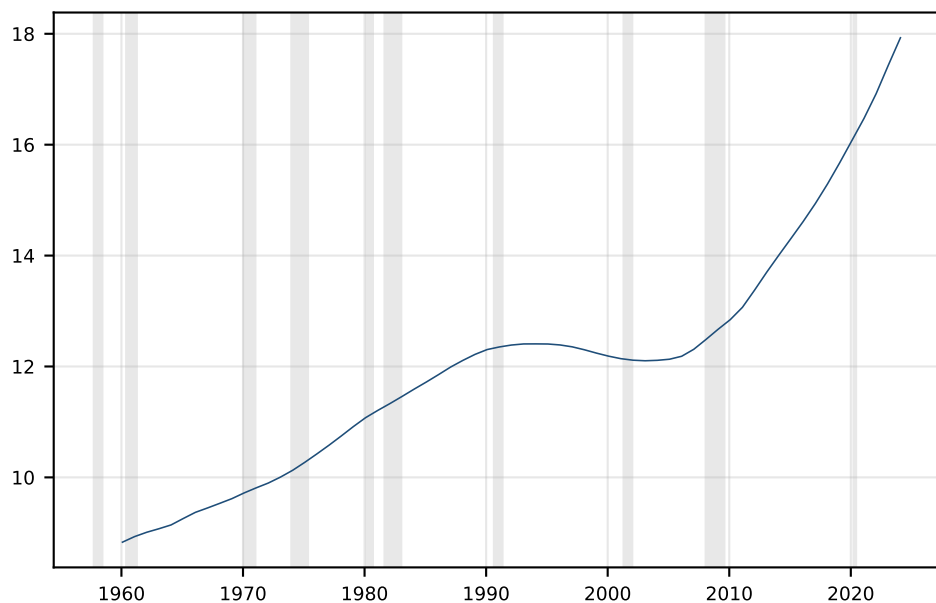
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	238	16.4%	83.6%	-0.3777
Q2	264	9.5%	90.5%	-0.0697
Q3	215	11.2%	88.8%	0.1023
Q4 (highest)	216	16.7%	83.3%	0.3991

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

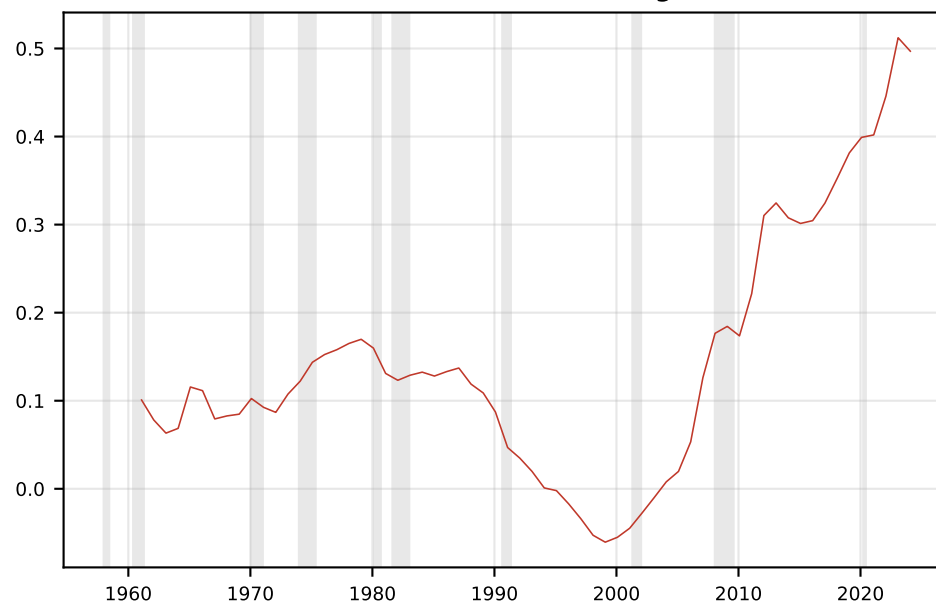
Population ages 65 and above for the United States

Block: Demographics / Demographics | Source: FRED API | Col: SPPOP65UPTOZSUSA | Transform: 12: First diff with lag 12

Raw Series



After: 12: First diff with lag 12



SUMMARY STATISTICS

N : 1287
Valid : 65
NaN : 1222
Start : 1960-01
End : 2024-01
Mean : 12.0601
Std : 2.1497
Min : 8.8314
Max : 17.9287
Skew : 0.759
Kurt : 0.451

STATIONARITY TESTS

ADF stat : -2.891
ADF p : 0.0465
ADF lags : 11
ADF : Stationary
KPSS stat : 0.459
KPSS p : 0.0516
KPSS : Stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

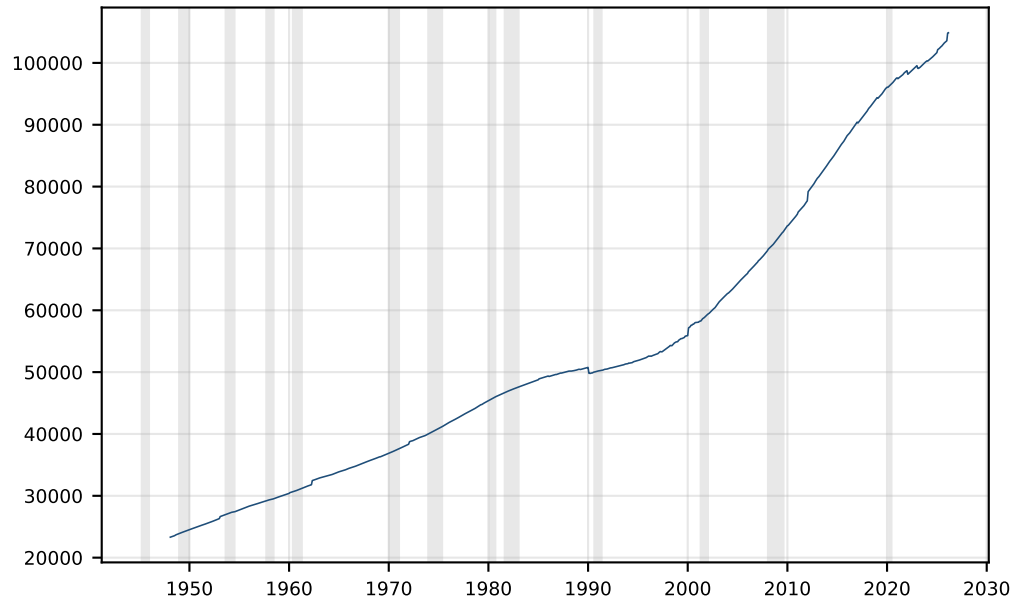
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	16	6.2%	93.8%	-0.0076
Q2	16	12.5%	87.5%	0.0931
Q3	16	18.8%	81.2%	0.1428
Q4 (highest)	16	12.5%	87.5%	0.3403

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

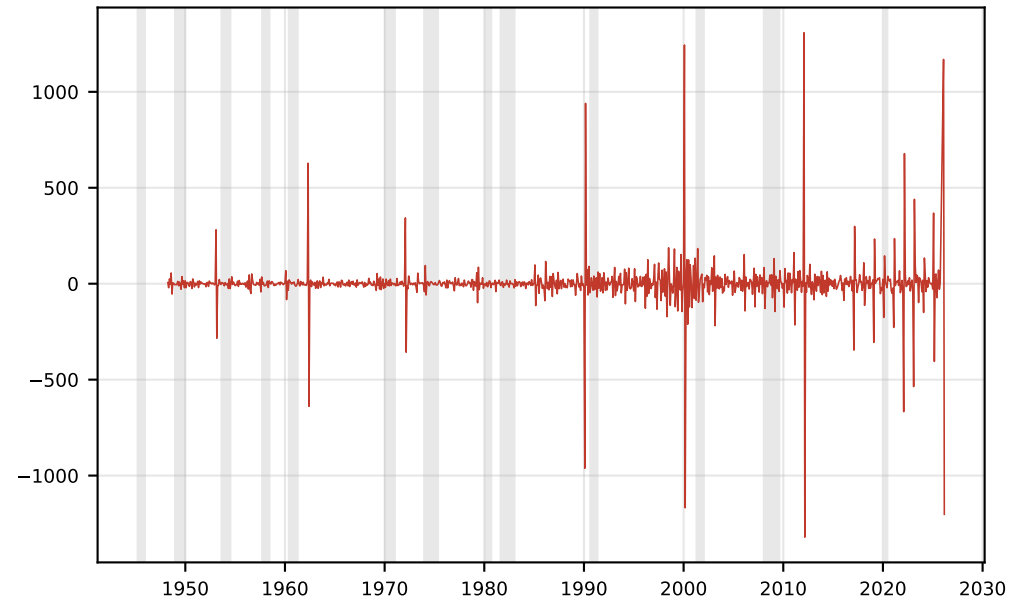
Population Level - 55 Yrs. & over

Block: Demographics / Demographics | Source: FRED API | Col: LNU00024230 | Transform: 3: Second diff

Raw Series



After: 3: Second diff



SUMMARY STATISTICS

N : 1287
Valid : 937
NaN : 350
Start : 1948-01
End : 2026-02
Mean : 53628.1057
Std : 22490.4378
Min : 23311.0000
Max : 104878.0000
Skew : 0.731
Kurt : -0.532

STATIONARITY TESTS

ADF stat : -16.193
ADF p : 0.0000
ADF lags : 11
ADF : Stationary
KPSS stat : 0.493
KPSS p : 0.0432
KPSS : Non-stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

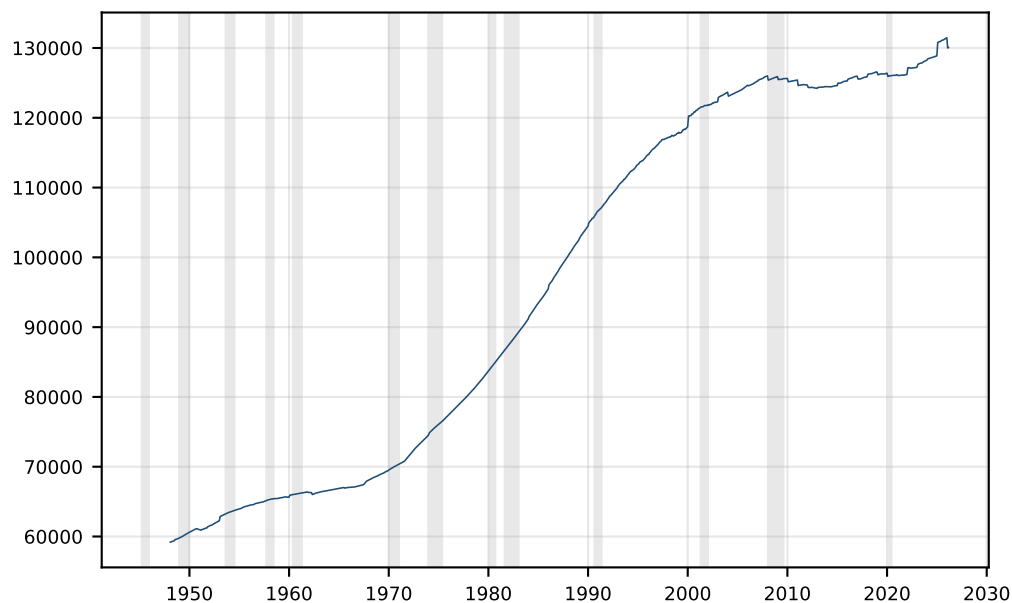
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	239	8.4%	91.6%	-78.8326
Q2	244	20.5%	79.5%	-4.8238
Q3	221	15.4%	84.6%	7.7647
Q4 (highest)	229	8.7%	91.3%	80.2620

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

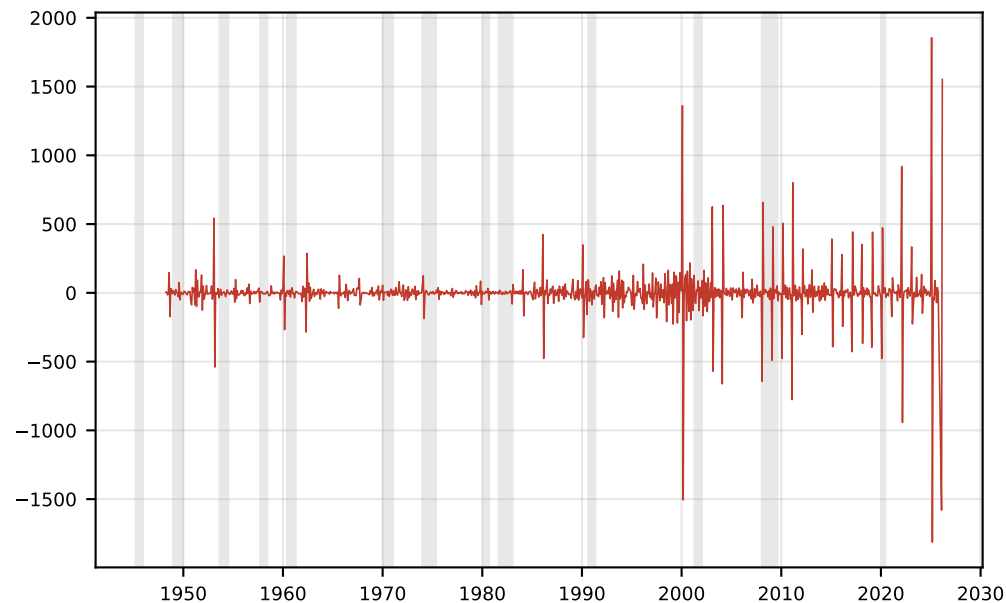
Population Level - 25-54 Yrs.

Block: Demographics / Demographics | Source: FRED API | Col: LNU00000060 | Transform: 3: Second diff

Raw Series



After: 3: Second diff



SUMMARY STATISTICS

N : 1287
Valid : 937
NaN : 350
Start : 1948-01
End : 2026-02
Mean : 96326.7001
Std : 25692.5295
Min : 59188.0000
Max : 131469.0000
Skew : -0.077
Kurt : -1.690

STATIONARITY TESTS

ADF stat : -13.876
ADF p : 0.0000
ADF lags : 12
ADF : Stationary
KPSS stat : 0.092
KPSS p : 0.1000
KPSS : Stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

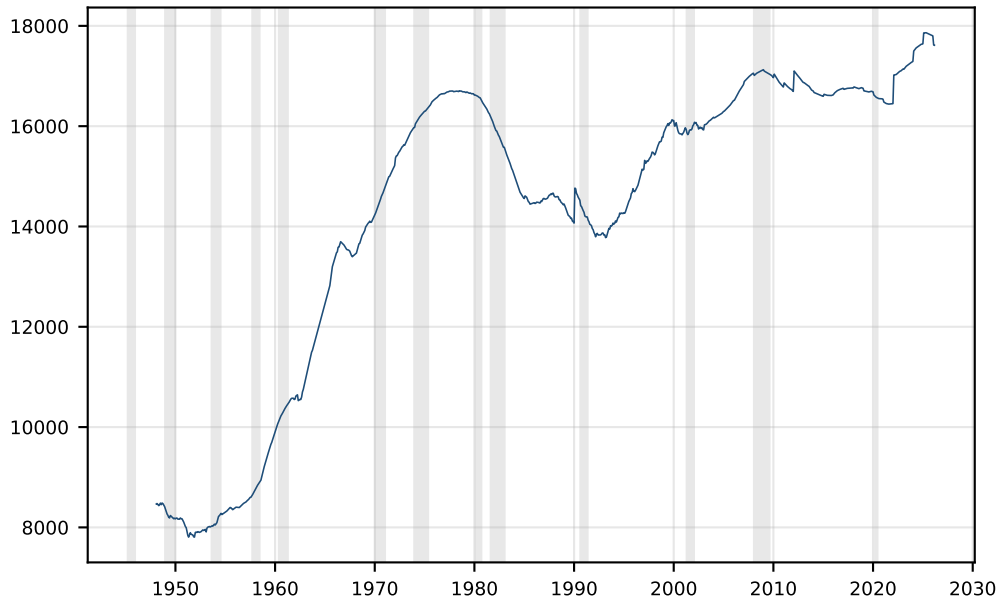
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	234	8.5%	91.5%	-113.9060
Q2	233	16.3%	83.7%	-8.2103
Q3	233	19.7%	80.3%	8.0129
Q4 (highest)	233	8.6%	91.4%	114.5665

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

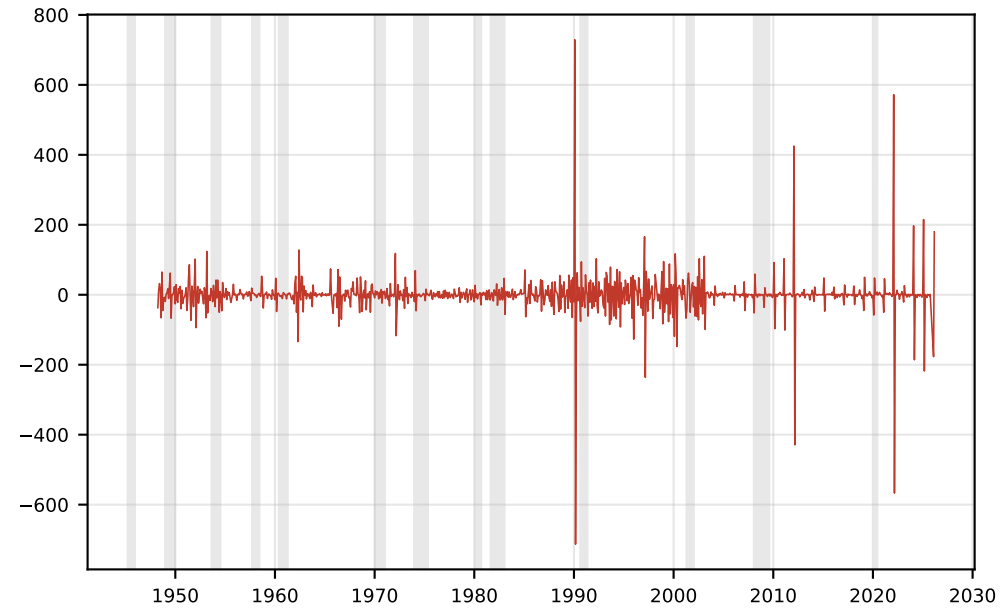
Population Level - 16-19 Yrs.

Block: Demographics / Demographics | Source: FRED API | Col: LNU00000012 | Transform: 3: Second diff

Raw Series



After: 3: Second diff



SUMMARY STATISTICS

N : 1287
Valid : 937
NaN : 350
Start : 1948-01
End : 2026-02
Mean : 14314.4728
Std : 2982.9366
Min : 7805.0000
Max : 17863.0000
Skew : -1.080
Kurt : -0.197

STATIONARITY TESTS

ADF stat : -13.785
ADF p : 0.0000
ADF lags : 12
ADF : Stationary
KPSS stat : 0.067
KPSS p : 0.1000
KPSS : Stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

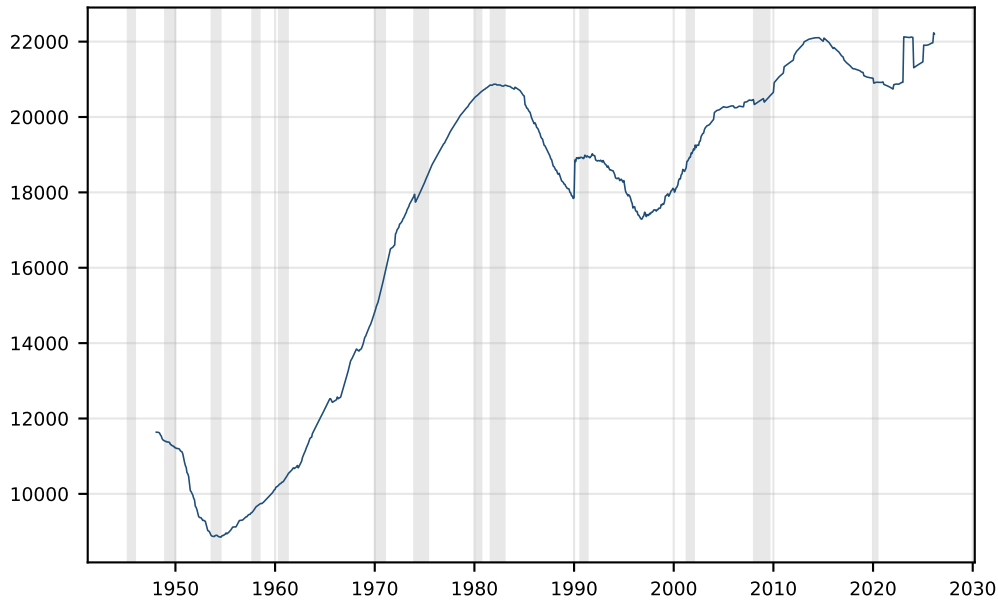
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	240	13.3%	86.7%	-42.1958
Q2	248	13.3%	86.7%	-2.0161
Q3	220	14.5%	85.5%	2.9455
Q4 (highest)	225	12.0%	88.0%	44.2800

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

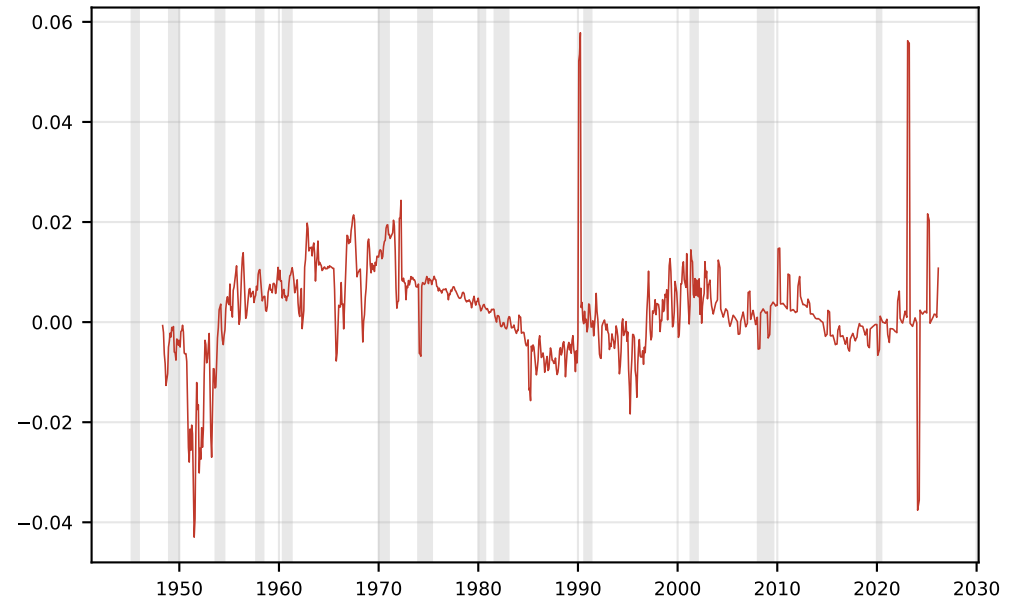
Population Level - 20-24 Yrs.

Block: Demographics / Demographics | Source: FRED API | Col: LNU00000036 | Transform: 8: Log diff lag 3

Raw Series



After: 8: Log diff lag 3



SUMMARY STATISTICS

N : 1287
Valid : 937
NaN : 350
Start : 1948-01
End : 2026-02
Mean : 17264.3362
Std : 4270.3251
Min : 8851.0000
Max : 22236.0000
Skew : -0.786
Kurt : -0.880

STATIONARITY TESTS

ADF stat : -3.708
ADF p : 0.0040
ADF lags : 12
ADF : Stationary
KPSS stat : 0.323
KPSS p : 0.1000
KPSS : Stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

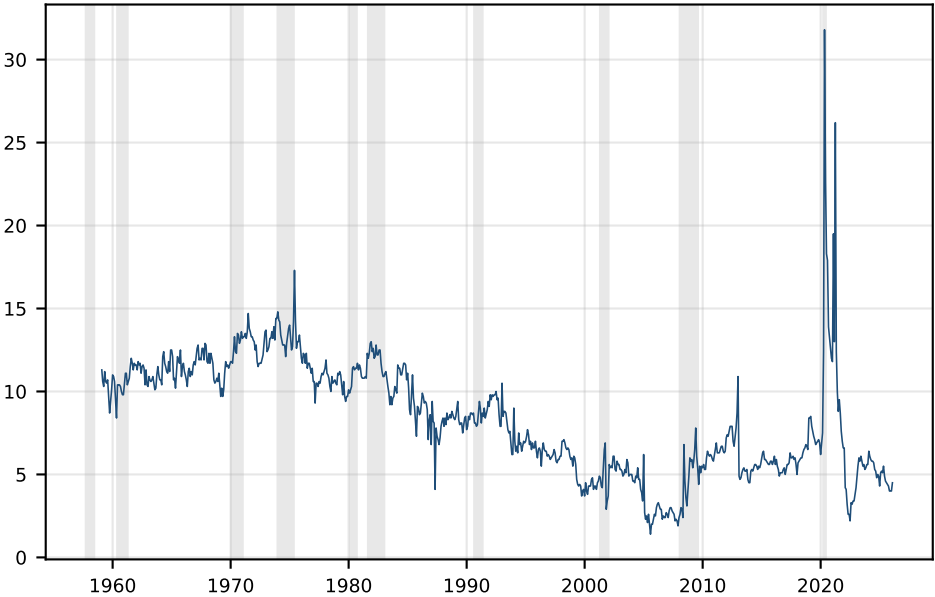
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	234	9.4%	90.6%	-0.0080
Q2	233	13.3%	86.7%	-0.0000
Q3	233	14.2%	85.8%	0.0040
Q4 (highest)	233	16.3%	83.7%	0.0123

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

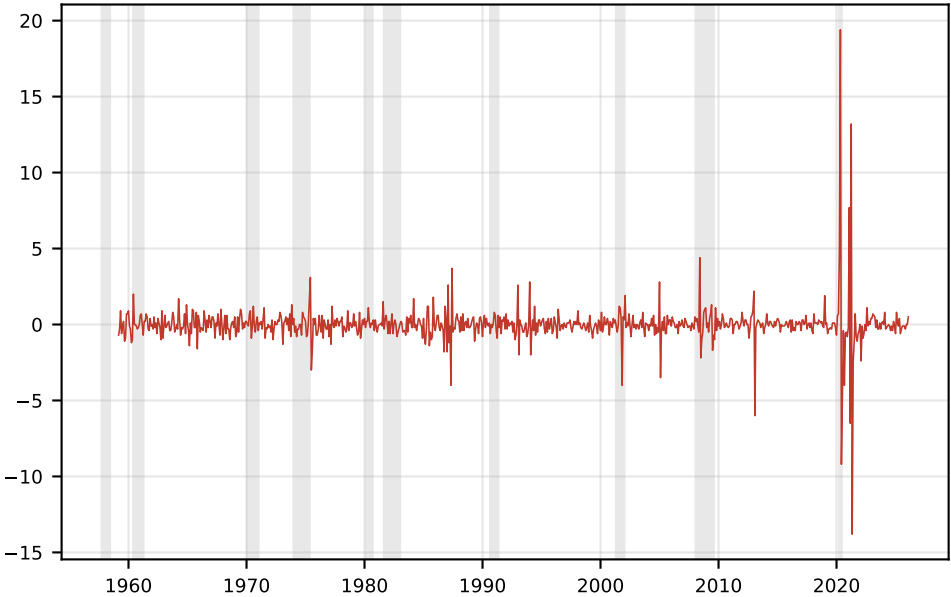
Personal Saving Rate

Block: Demographics / Demographics | Source: FRED API | Col: PSAVERT | Transform: 2: First diff

Raw Series



After: 2: First diff



SUMMARY STATISTICS

N : 1287
Valid : 805
NaN : 482
Start : 1959-01
End : 2026-01
Mean : 8.3965
Std : 3.4243
Min : 1.4000
Max : 31.8000
Skew : 0.671
Kurt : 2.735

STATIONARITY TESTS

ADF stat : -8.625
ADF p : 0.0000
ADF lags : 12
ADF : Stationary
KPSS stat : 0.041
KPSS p : 0.1000
KPSS : Stationary

QUARTILE-BUCKET ANALYSIS

(vs NBER recession/expansion)

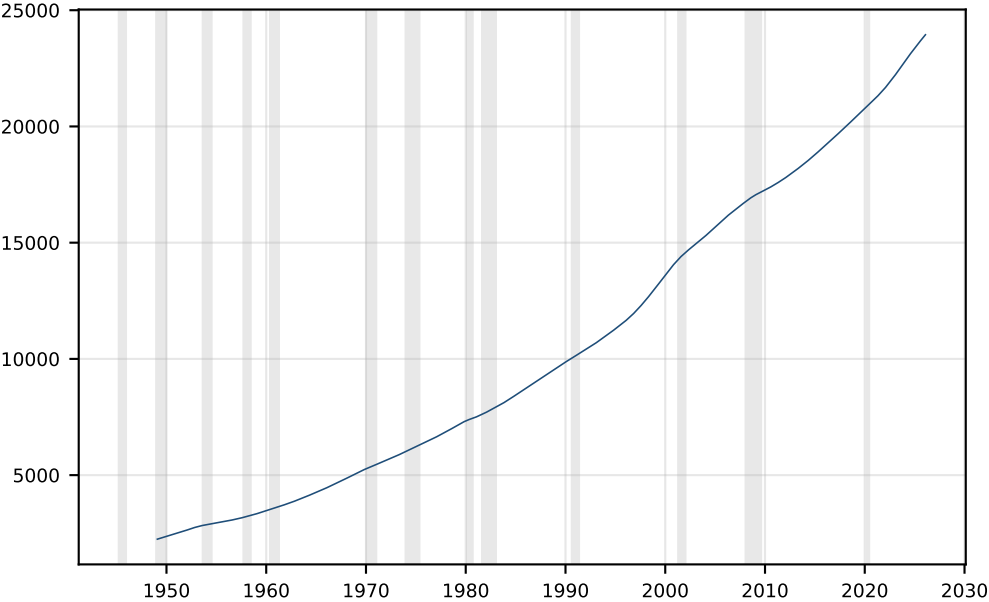
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	211	12.3%	87.7%	-0.9147
Q2	249	9.6%	90.4%	-0.1129
Q3	157	8.3%	91.7%	0.1707
Q4 (highest)	187	17.1%	82.9%	1.0027

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

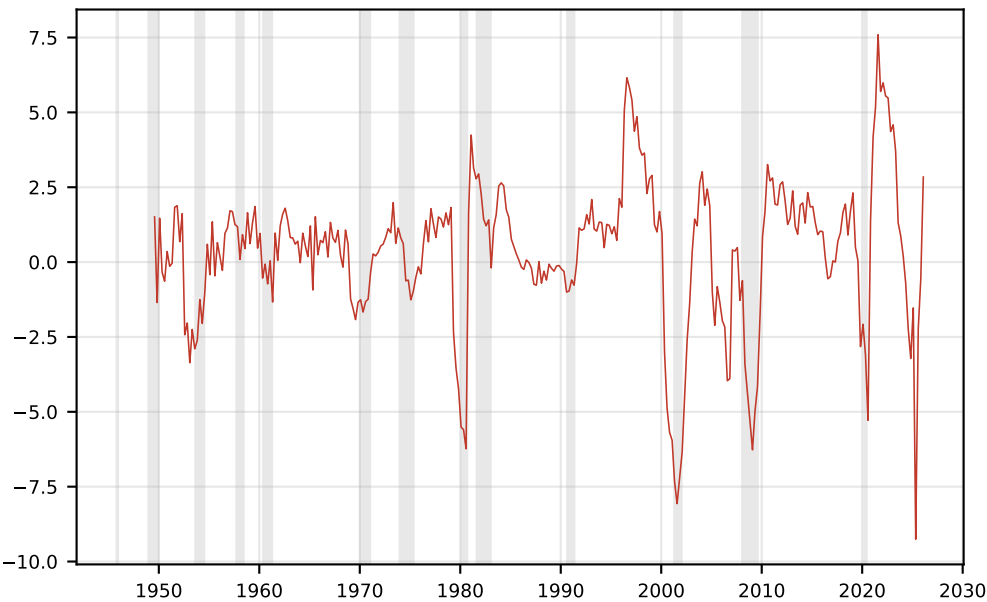
Real Potential Gross Domestic Product

Block: Productivity / Productivity | Source: FRED API | Col: GDPDOT | Transform: 13: Second diff with lag 3

Raw Series



After: 13: Second diff with lag 3



SUMMARY STATISTICS

N : 1287
Valid : 309
NaN : 978
Start : 1949-01
End : 2026-01
Mean : 10612.5324
Std : 6381.8658
Min : 2244.6967
Max : 23945.1378
Skew : 0.440
Kurt : -1.107

STATIONARITY TESTS

ADF stat : -6.821
ADF p : 0.0000
ADF lags : 3
ADF : Stationary
KPSS stat : 0.051
KPSS p : 0.1000
KPSS : Stationary

QUARTILE-BUCKET ANALYSIS

(vs NBER recession/expansion)

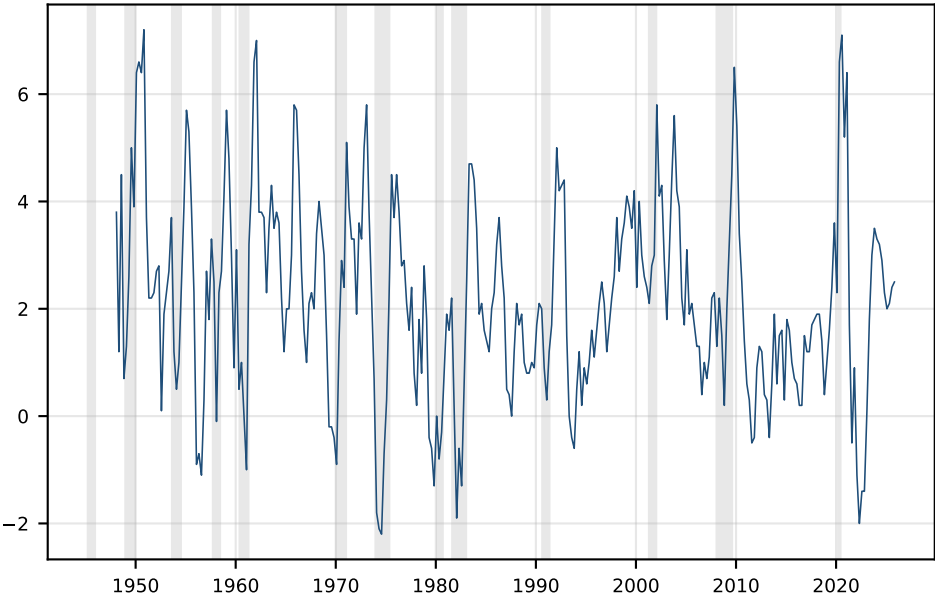
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	77	32.5%	67.5%	-2.7478
Q2	77	6.5%	93.5%	0.1018
Q3	76	9.2%	90.8%	1.1227
Q4 (highest)	77	2.6%	97.4%	2.8705

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

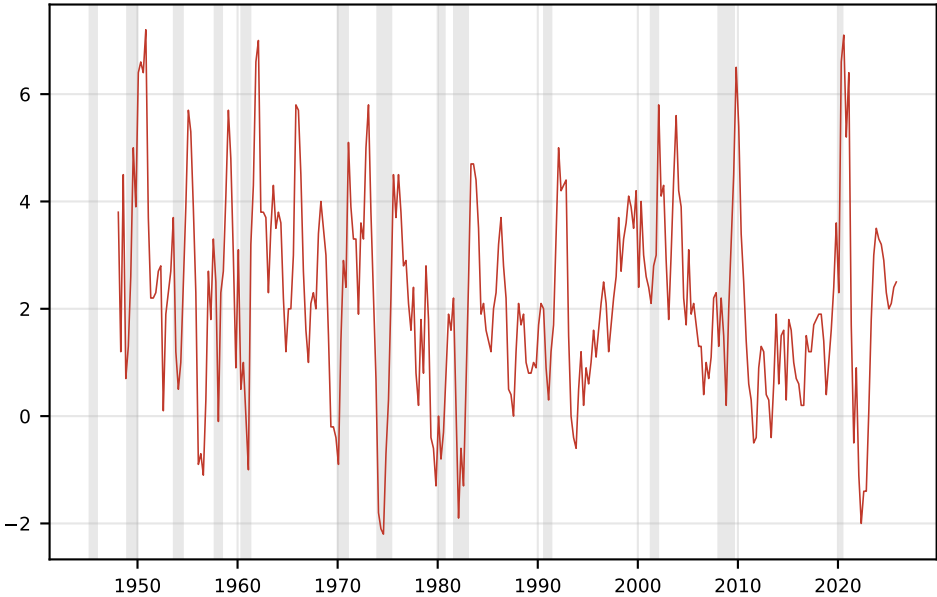
Nonfarm Business Sector: Labor Productivity (Output per Hour) for All Workers

Block: Productivity / Productivity | Source: FRED API | Col: PRS85006091 | Transform: 1: No transform

Raw Series



After: 1: No transform



SUMMARY STATISTICS

N : 1287
Valid : 312
NaN : 975
Start : 1948-01
End : 2025-10
Mean : 2.1593
Std : 1.8332
Min : -2.2000
Max : 7.2000
Skew : 0.287
Kurt : 0.063

STATIONARITY TESTS

ADF stat : -4.192
ADF p : 0.0007
ADF lags : 12
ADF : Stationary
KPSS stat : 0.326
KPSS p : 0.1000
KPSS : Stationary

QUARTILE-BUCKET ANALYSIS
(vs NBER recession/expansion)

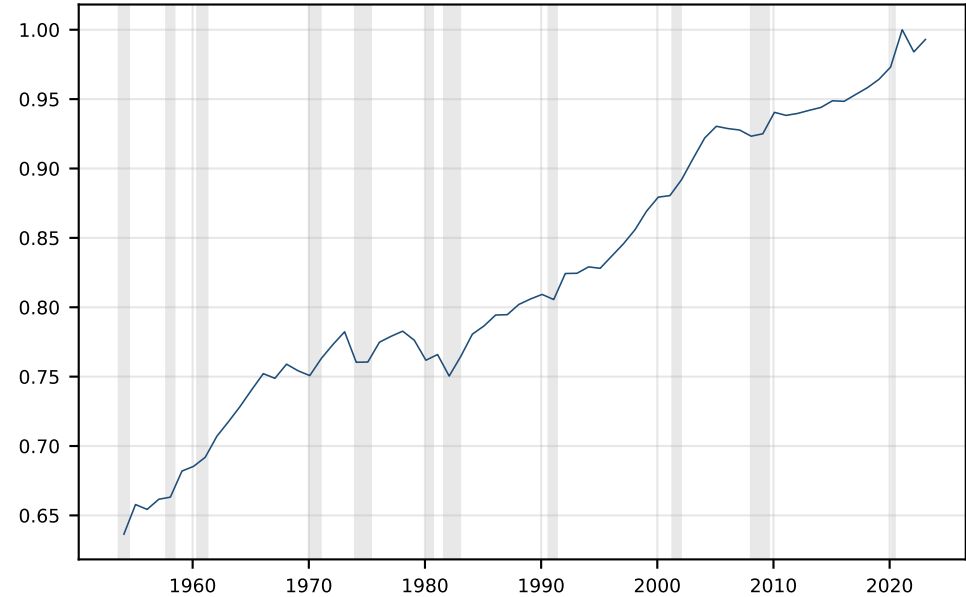
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	78	25.6%	74.4%	-0.0731
Q2	86	10.5%	89.5%	1.5895
Q3	71	12.7%	87.3%	2.6648
Q4 (highest)	77	3.9%	96.1%	4.5909

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

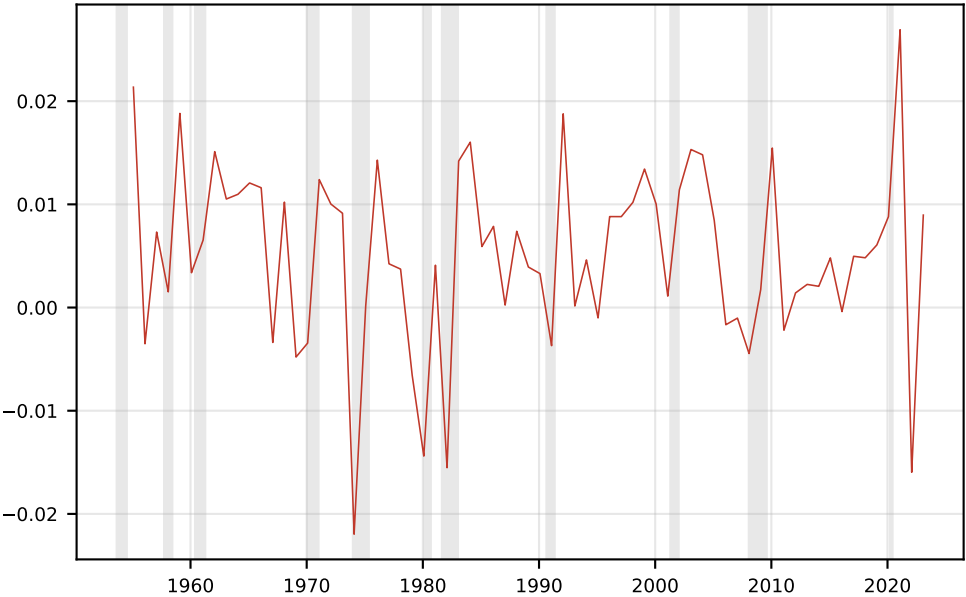
Total Factor Productivity at Constant National Prices for United States

Block: Productivity / Productivity | Source: FRED API | Col: RTFPNAUSA632NRUG | Transform: 12: First diff with lag 12

Raw Series



After: 12: First diff with lag 12



SUMMARY STATISTICS

N : 1287
Valid : 70
NaN : 1217
Start : 1954-01
End : 2023-01
Mean : 0.8232
Std : 0.0978
Min : 0.6365
Max : 1.0000
Skew : 0.085
Kurt : -1.041

STATIONARITY TESTS

ADF stat : -8.819
ADF p : 0.0000
ADF lags : 0
ADF : Stationary
KPSS stat : 0.093
KPSS p : 0.1000
KPSS : Stationary

QUARTILE-BUCKET ANALYSIS

(vs NBER recession/expansion)

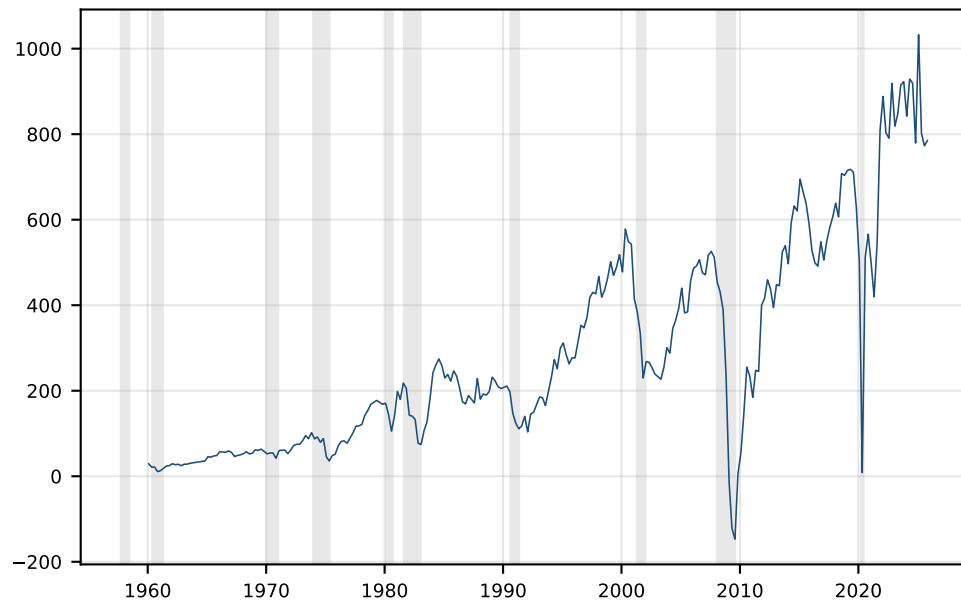
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	18	33.3%	66.7%	-0.0058
Q2	17	11.8%	88.2%	0.0031
Q3	17	5.9%	94.1%	0.0085
Q4 (highest)	17	0.0%	100.0%	0.0155

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

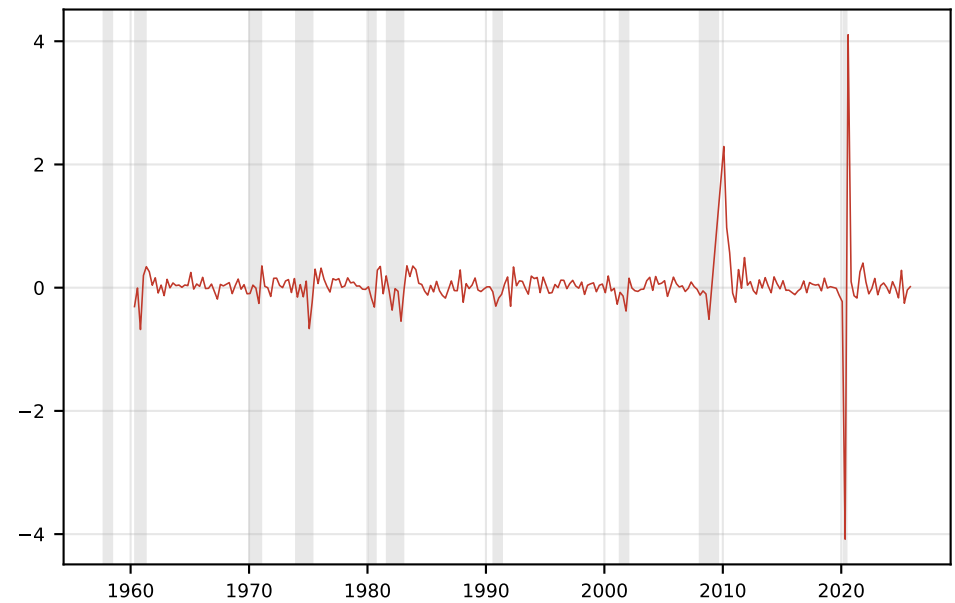
Net domestic investment: Private: Domestic business

Block: Productivity / Productivity | Source: FRED API | Col: W790RC1Q027SBEA | Transform: 8: Log diff lag 3

Raw Series



After: 8: Log diff lag 3



SUMMARY STATISTICS

N : 1287
Valid : 264
NaN : 1023
Start : 1960-01
End : 2025-10
Mean : 286.8732
Std : 242.9009
Min : -147.1210
Max : 1032.5440
Skew : 0.885
Kurt : 0.018

STATIONARITY TESTS

ADF stat : -14.537
ADF p : 0.0000
ADF lags : 1
ADF : Stationary
KPSS stat : 0.099
KPSS p : 0.1000
KPSS : Stationary

QUARTILE-BUCKET ANALYSIS

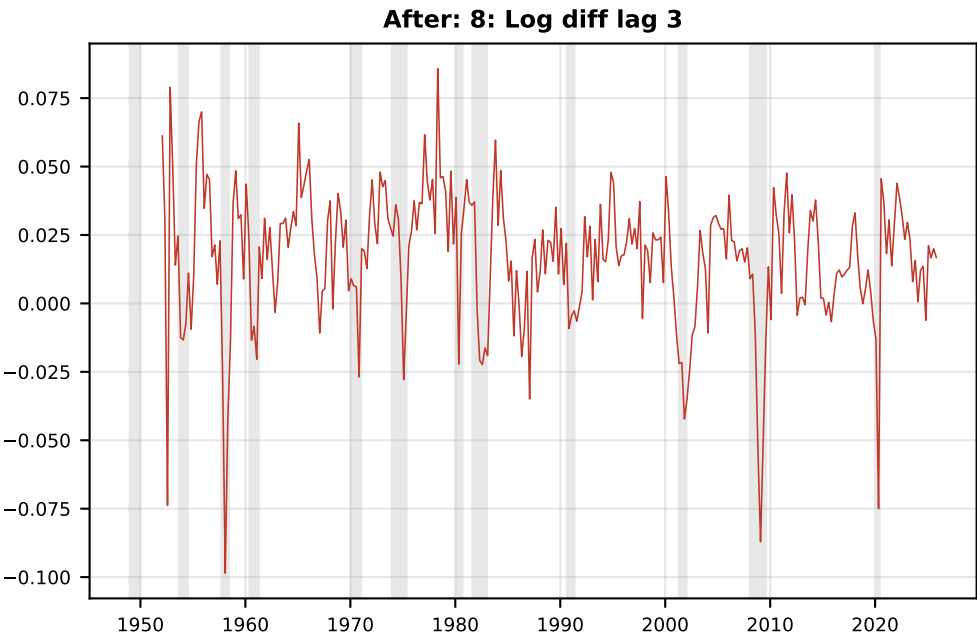
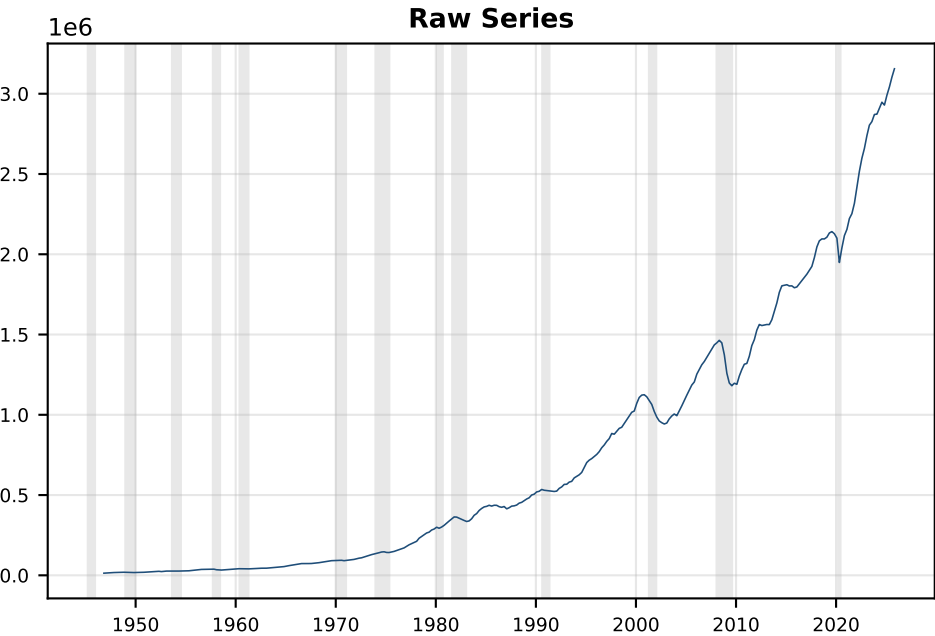
(vs NBER recession/expansion)

Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	65	29.2%	70.8%	-0.2355
Q2	65	9.2%	90.8%	-0.0201
Q3	64	3.1%	96.9%	0.0554
Q4 (highest)	65	3.1%	96.9%	0.3093

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

Nonfinancial Corporate Business; Gross Fixed Investment, Transactions

Block: Productivity / Productivity | Source: FRED API | Col: BOGZ1FA105019005Q | Transform: 8: Log diff lag 3



SUMMARY STATISTICS

N : 1287
Valid : 302
NaN : 985
Start : 1946-10
End : 2025-10
Mean : 770369.2881
Std : 803714.1396
Min : 13361.0000
Max : 3155735.0000
Skew : 1.109
Kurt : 0.365

STATIONARITY TESTS

ADF stat : -7.124
ADF p : 0.0000
ADF lags : 2
ADF : Stationary
KPSS stat : 0.316
KPSS p : 0.1000
KPSS : Stationary

QUARTILE-BUCKET ANALYSIS

(vs NBER recession/expansion)

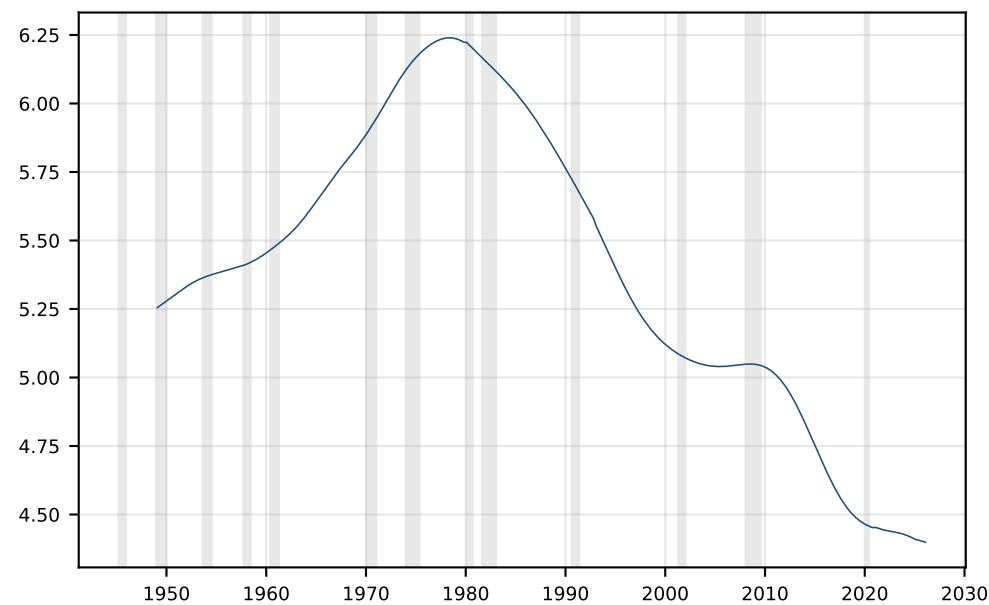
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	74	35.1%	64.9%	-0.0150
Q2	74	8.1%	91.9%	0.0128
Q3	74	4.1%	95.9%	0.0256
Q4 (highest)	74	2.7%	97.3%	0.0431

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

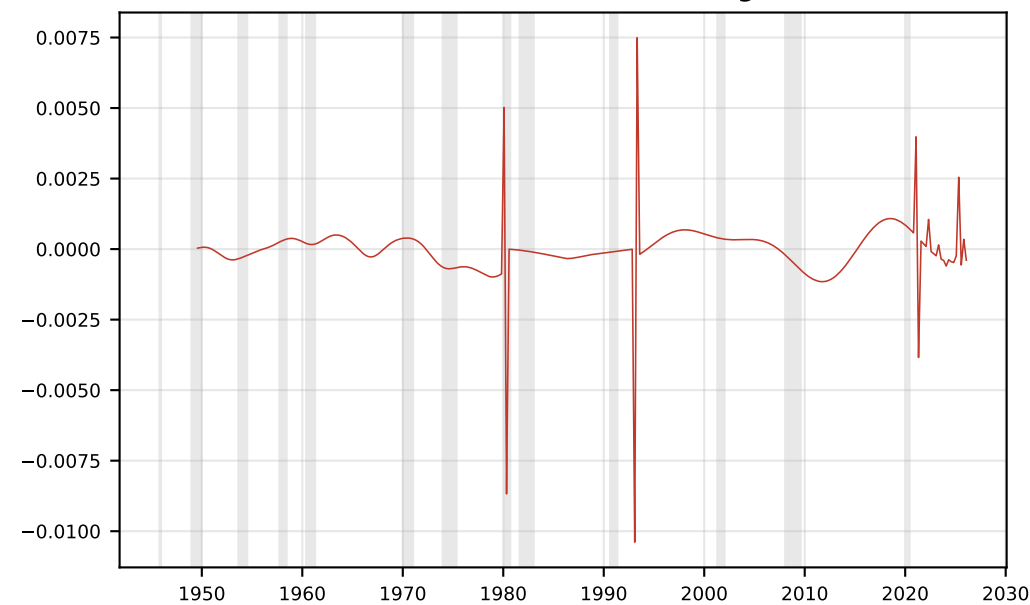
Noncyclical Rate of Unemployment

Block: Productivity / Productivity | Source: FRED API | Col: NROU | Transform: 13: Second diff with lag 3

Raw Series



After: 13: Second diff with lag 3



SUMMARY STATISTICS

N : 1287
Valid : 309
NaN : 978
Start : 1949-01
End : 2026-01
Mean : 5.3981
Std : 0.5377
Min : 4.3991
Max : 6.2399
Skew : -0.184
Kurt : -0.891

STATIONARITY TESTS

ADF stat : -2.758
ADF p : 0.0645
ADF lags : 6
ADF : Non-stationary
KPSS stat : 0.163
KPSS p : 0.1000
KPSS : Stationary

QUARTILE-BUCKET ANALYSIS

(vs NBER recession/expansion)

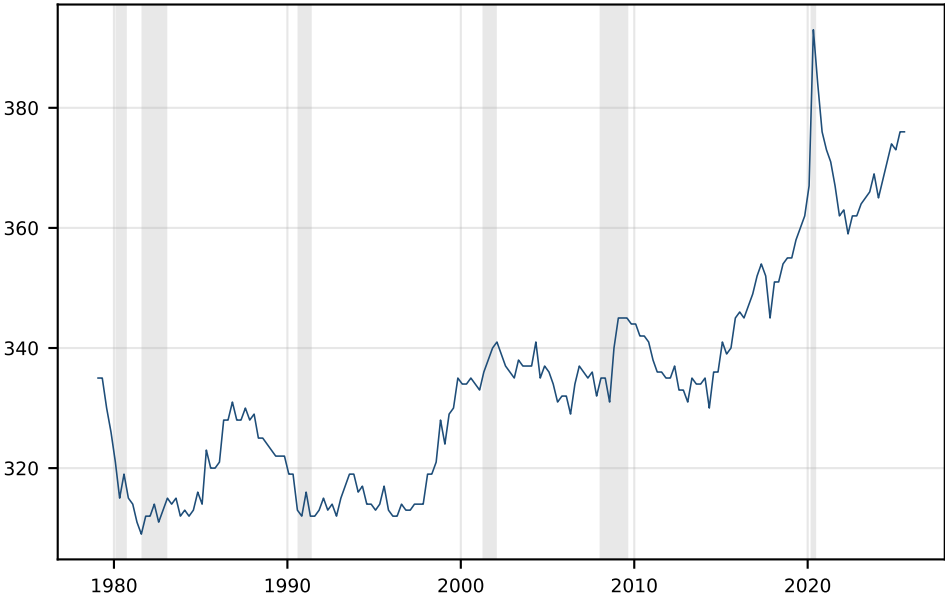
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	77	14.3%	85.7%	-0.0009
Q2	77	14.3%	85.7%	-0.0002
Q3	76	10.5%	89.5%	0.0002
Q4 (highest)	77	11.7%	88.3%	0.0008

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

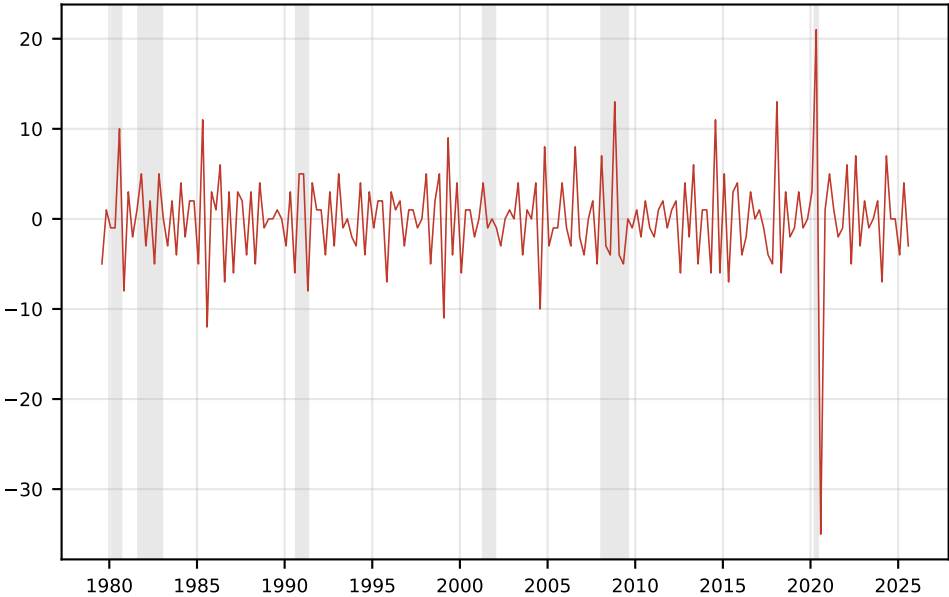
Median usual weekly real earnings: Wage and salary workers: 16 years and over

Block: Productivity / Productivity | Source: FRED API | Col: LES1252881600Q | Transform: 13: Second diff with lag 3

Raw Series



After: 13: Second diff with lag 3



SUMMARY STATISTICS

N	: 1287
Valid	: 187
NaN	: 1100
Start	: 1979-01
End	: 2025-07
Mean	: 333.9572
Std	: 18.2049
Min	: 309.0000
Max	: 393.0000
Skew	: 0.807
Kurt	: 0.112

STATIONARITY TESTS

ADF stat	: -6.786
ADF p	: 0.0000
ADF lags	: 12
ADF	: Stationary
KPSS stat	: 0.266
KPSS p	: 0.1000
KPSS	: Stationary

QUARTILE-BUCKET ANALYSIS
(vs NBER recession/expansion)

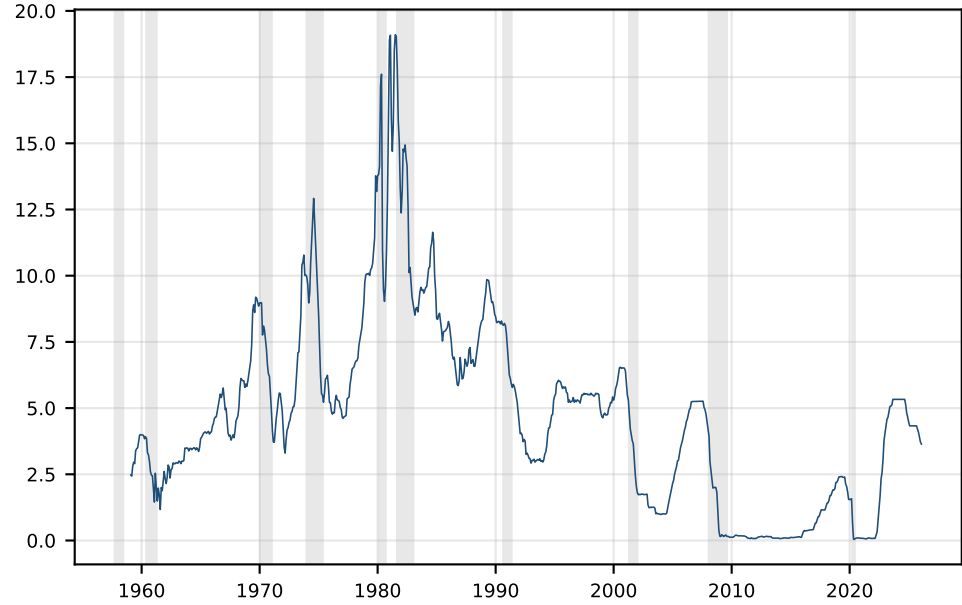
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	51	11.8%	88.2%	-5.6471
Q2	48	6.2%	93.8%	-0.8542
Q3	50	2.0%	98.0%	1.8600
Q4 (highest)	36	25.0%	75.0%	6.5556

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

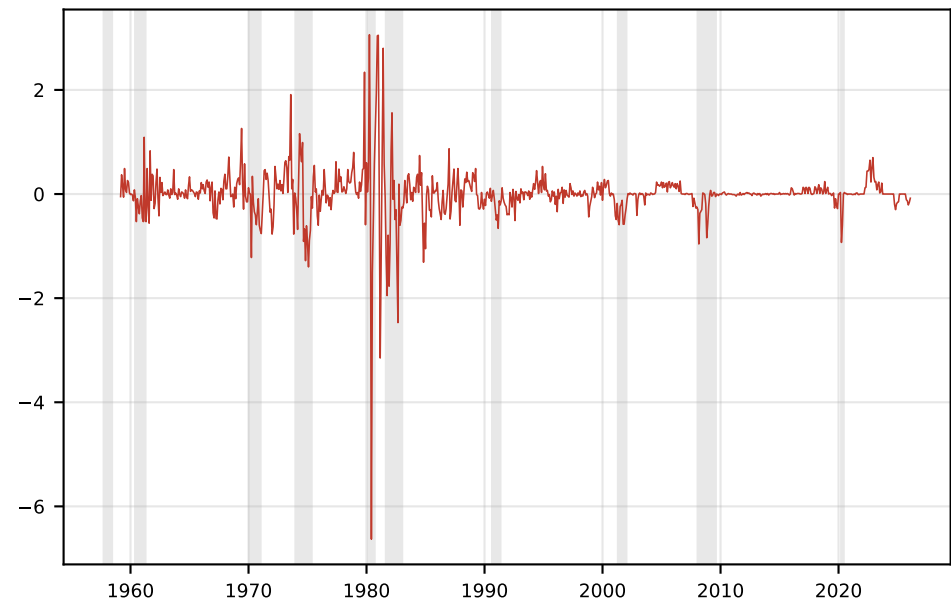
Effective Fed Funds Rate

Block: Cross-Asset / Rates | Source: FRED-MD | Col: FEDFUNDS | Transform: 2: First diff

Raw Series



After: 2: First diff



SUMMARY STATISTICS

N : 1287
Valid : 805
NaN : FEDFUNDS 482
FEDFUNDS 427
dtype: int64
Start : 1959-01
End : 2026-01
Mean : 4.7669
Std : 3.5925
Min : 0.0500
Max : 19.1000
Skew : 0.994
Kurt : 1.542

STATIONARITY TESTS

ADF stat : -6.859
ADF p : 0.0000
ADF lags : 12
ADF : Stationary
KPSS stat : 0.058
KPSS p : 0.1000
KPSS : Stationary

QUARTILE-BUCKET ANALYSIS

(vs NBER recession/expansion)

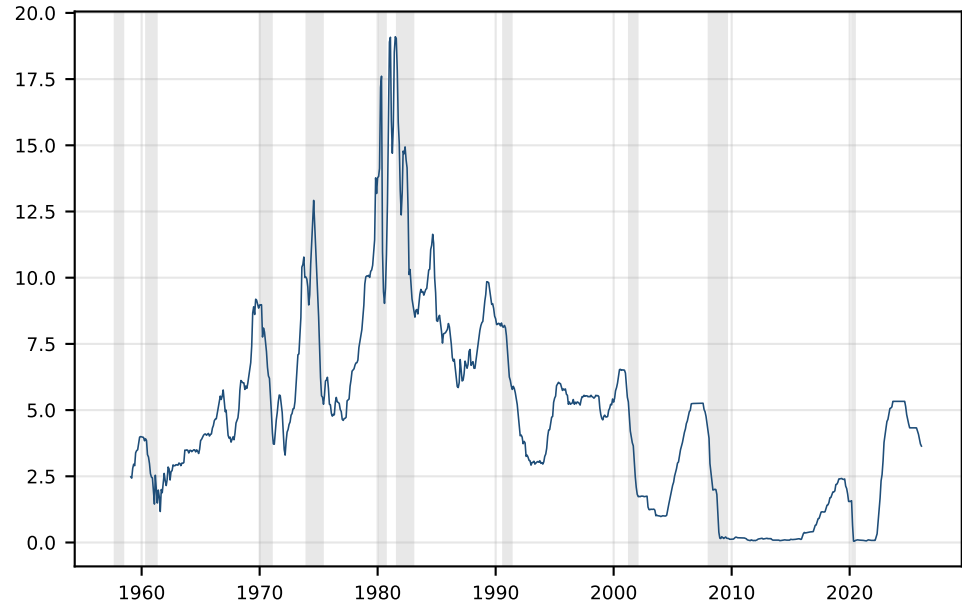
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	202	32.7%	67.3%	-0.4390
Q2	205	3.9%	96.1%	-0.0159
Q3	196	3.6%	96.4%	0.0423
Q4 (highest)	201	7.0%	93.0%	0.4218

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

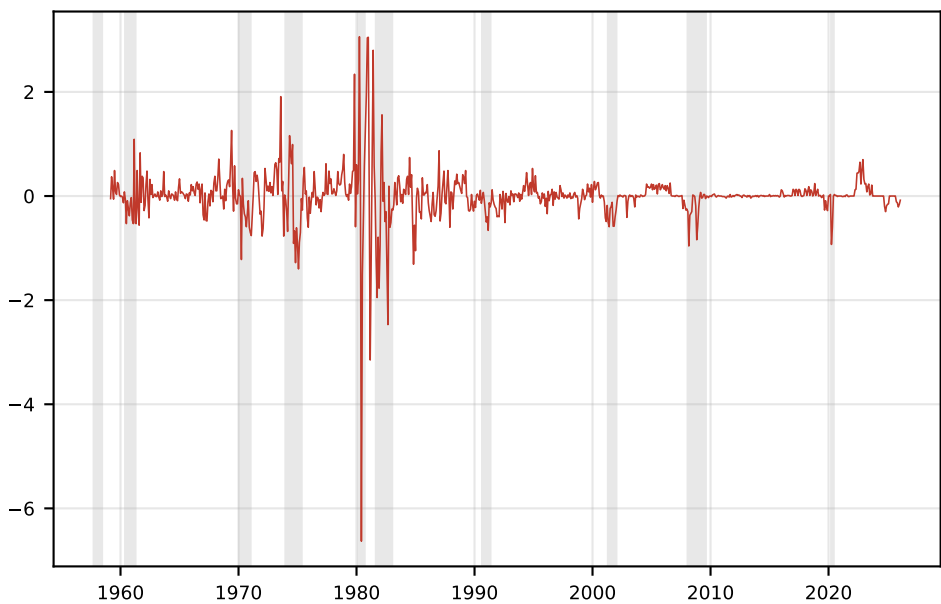
Effective Fed Funds Rate

Block: Cross-Asset / Rates | Source: FRED-MD | Col: FEDFUNDS | Transform: 2: First diff

Raw Series



After: 2: First diff



SUMMARY STATISTICS

N : 1287
Valid : 805
NaN : FEDFUNDS 482
FEDFUNDS 427
dtype: int64
Start : 1959-01
End : 2026-01
Mean : 4.7669
Std : 3.5925
Min : 0.0500
Max : 19.1000
Skew : 0.994
Kurt : 1.542

STATIONARITY TESTS

ADF stat : -6.859
ADF p : 0.0000
ADF lags : 12
ADF : Stationary
KPSS stat : 0.058
KPSS p : 0.1000
KPSS : Stationary

QUARTILE-BUCKET ANALYSIS

(vs NBER recession/expansion)

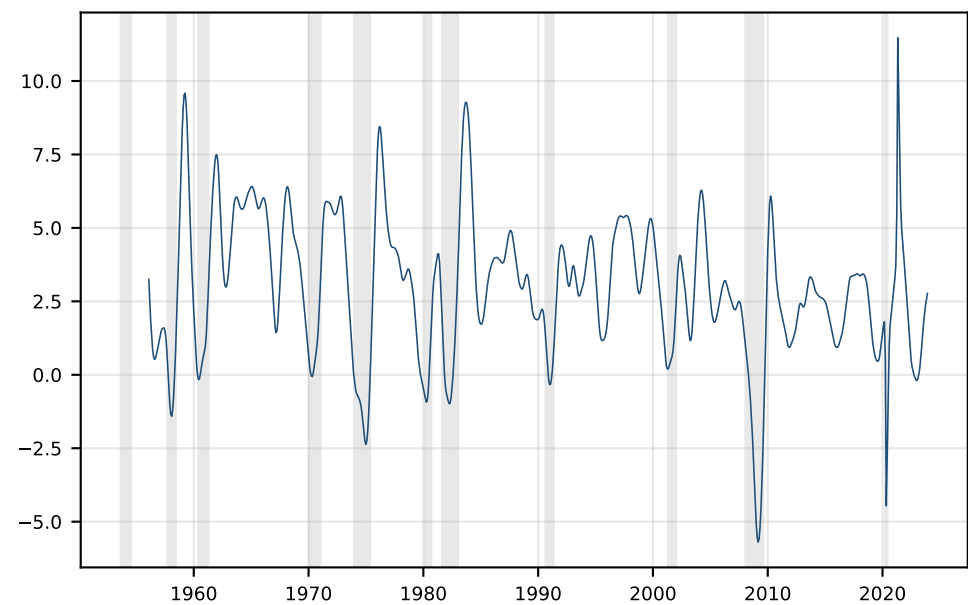
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	202	32.7%	67.3%	-0.4390
Q2	205	3.9%	96.1%	-0.0159
Q3	196	3.6%	96.4%	0.0423
Q4 (highest)	201	7.0%	93.0%	0.4218

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

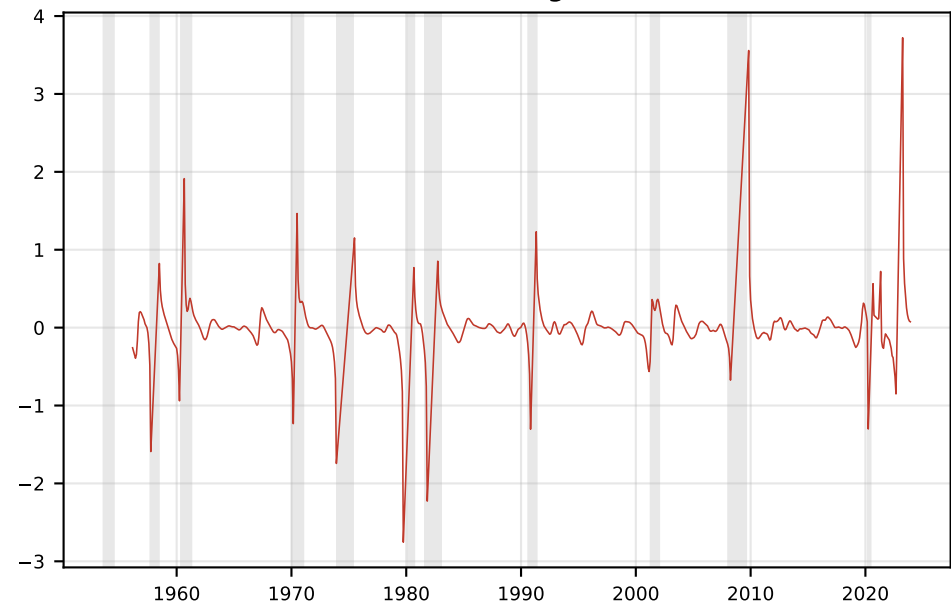
Composite Leading Indicators: Composite Leading Indicator (CLI) Trend Restored for United States

Block: Structural macro / Structural macro | Source: FRED API | Col: USALOLITOTRGYSAM | Transform: 5: Log diff

Raw Series



After: 5: Log diff



SUMMARY STATISTICS

N : 1287
Valid : 815
NaN : 472
Start : 1956-01
End : 2023-11
Mean : 2.9632
Std : 2.3857
Min : -5.6976
Max : 11.4733
Skew : -0.115
Kurt : 0.916

STATIONARITY TESTS

ADF stat : -22.127
ADF p : 0.0000
ADF lags : 0
ADF : Stationary
KPSS stat : 0.122
KPSS p : 0.1000
KPSS : Stationary

QUARTILE-BUCKET ANALYSIS
(vs NBER recession/expansion)

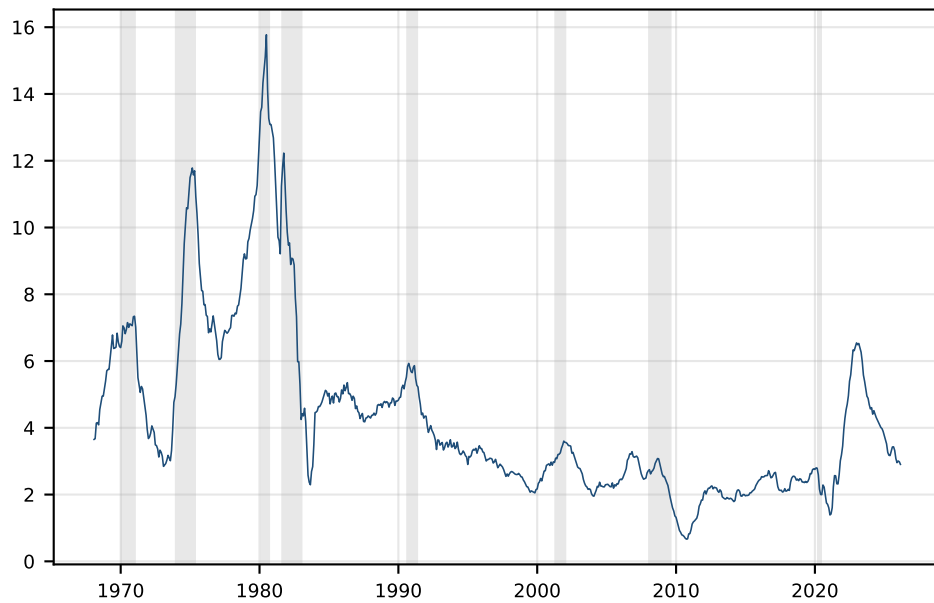
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	182	7.1%	92.9%	-0.2733
Q2	182	0.5%	99.5%	-0.0380
Q3	182	0.0%	100.0%	0.0234
Q4 (highest)	182	12.6%	87.4%	0.2754

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

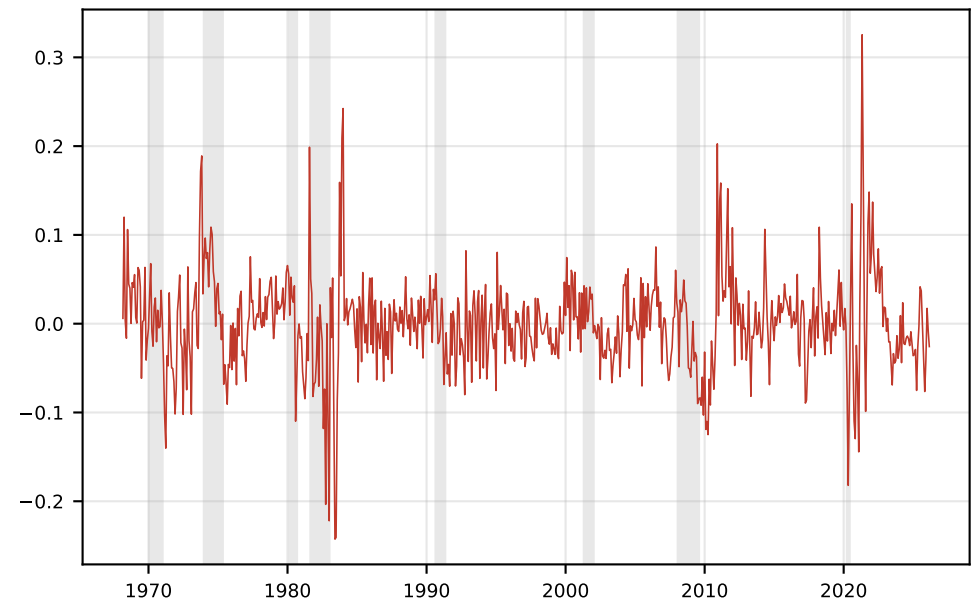
Sticky Price Consumer Price Index less Food and Energy

Block: Structural macro / Structural macro | Source: FRED API | Col: CORESTICKM159SFRBATL | Transform: 5: Log diff

Raw Series



After: 5: Log diff



SUMMARY STATISTICS

N : 1287
Valid : 698
NaN : 589
Start : 1968-01
End : 2026-02
Mean : 4.3061
Std : 2.6541
Min : 0.6639
Max : 15.7742
Skew : 1.626
Kurt : 2.654

STATIONARITY TESTS

ADF stat : -6.799
ADF p : 0.0000
ADF lags : 12
ADF : Stationary
KPSS stat : 0.041
KPSS p : 0.1000
KPSS : Stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

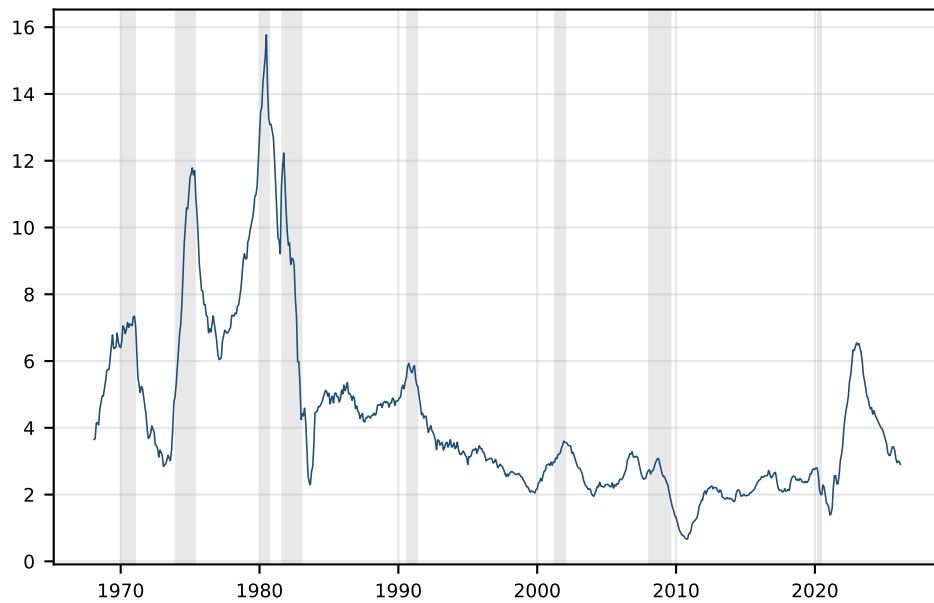
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	175	12.6%	87.4%	-0.0621
Q2	174	8.0%	92.0%	-0.0124
Q3	174	10.3%	89.7%	0.0122
Q4 (highest)	174	17.8%	82.2%	0.0613

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

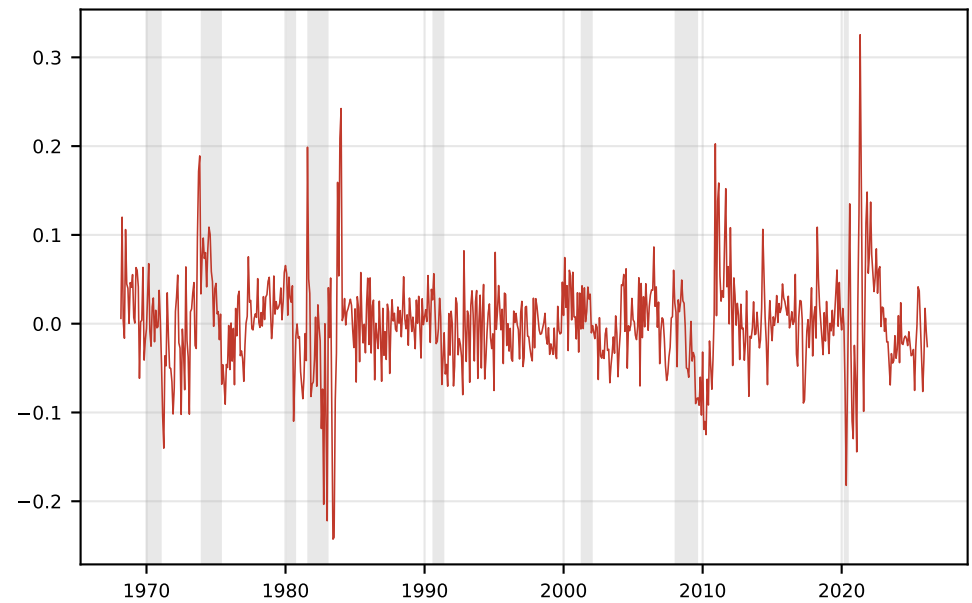
Sticky Price Consumer Price Index less Food and Energy

Block: Structural macro / Structural macro | Source: FRED API | Col: CORESTICKM159SFRBATL | Transform: 5: Log diff

Raw Series



After: 5: Log diff



SUMMARY STATISTICS

N : 1287
Valid : 698
NaN : 589
Start : 1968-01
End : 2026-02
Mean : 4.3061
Std : 2.6541
Min : 0.6639
Max : 15.7742
Skew : 1.626
Kurt : 2.654

STATIONARITY TESTS

ADF stat : -6.799
ADF p : 0.0000
ADF lags : 12
ADF : Stationary
KPSS stat : 0.041
KPSS p : 0.1000
KPSS : Stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

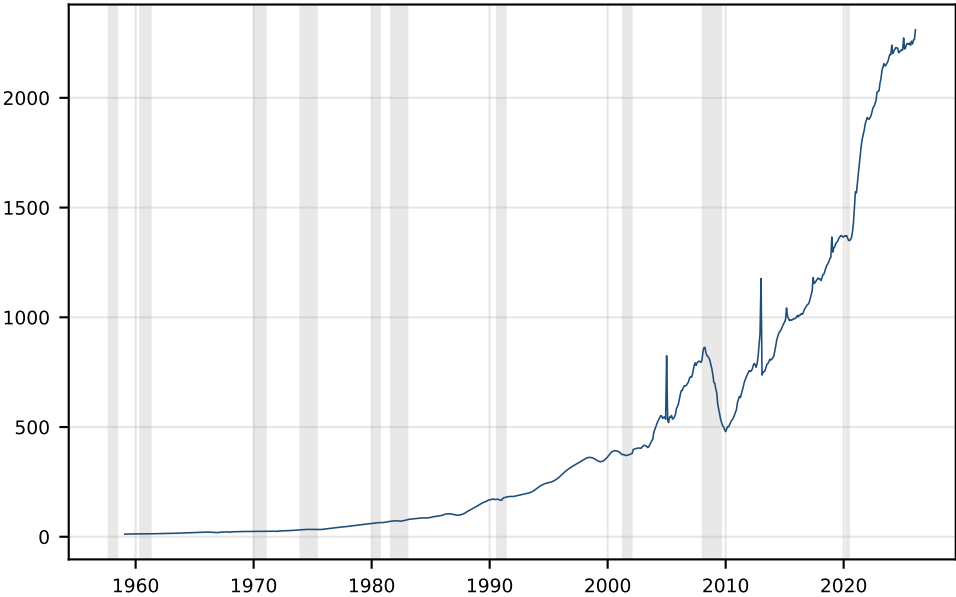
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	175	12.6%	87.4%	-0.0621
Q2	174	8.0%	92.0%	-0.0124
Q3	174	10.3%	89.7%	0.0122
Q4 (highest)	174	17.8%	82.2%	0.0613

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

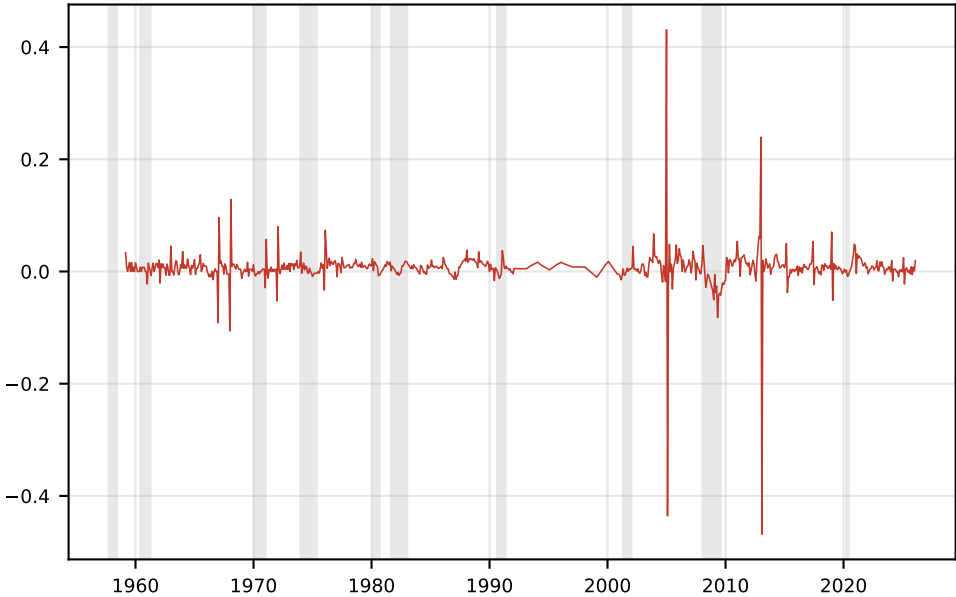
Personal Income Receipts on Assets: Personal Dividend Income

Block: Equities / Equities | Source: FRED API | Col: PDI | Transform: 5: Log diff

Raw Series



After: 5: Log diff



SUMMARY STATISTICS

N : 1287
Valid : 805
NaN : 482
Start : 1959-01
End : 2026-01
Mean : 463.6201
Std : 590.5396
Min : 11.8000
Max : 2310.3000
Skew : 1.643
Kurt : 1.942

STATIONARITY TESTS

ADF stat : -8.518
ADF p : 0.0000
ADF lags : 6
ADF : Stationary
KPSS stat : 0.049
KPSS p : 0.1000
KPSS : Stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

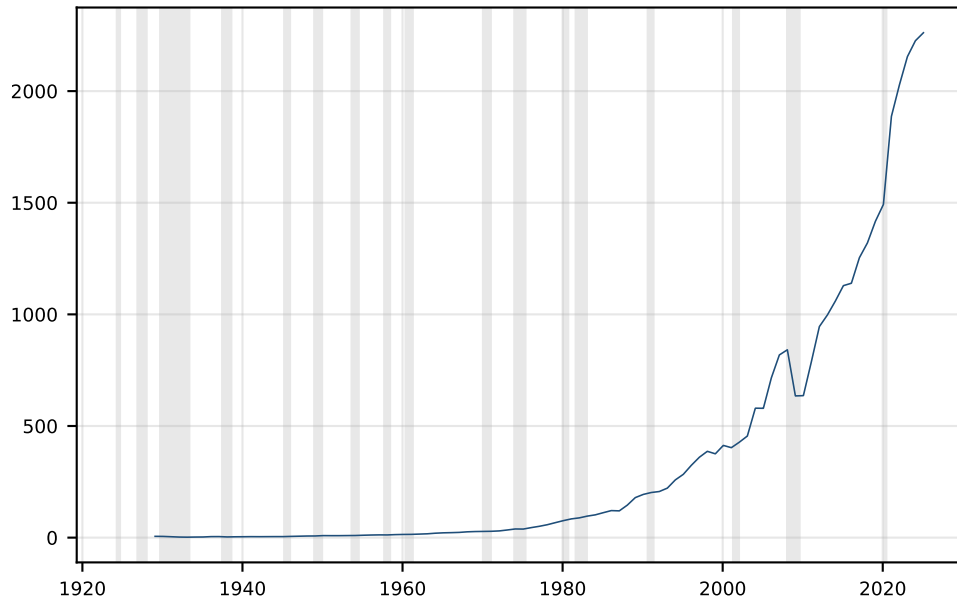
Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	219	23.3%	76.7%	-0.0130
Q2	183	9.8%	90.2%	0.0043
Q3	201	7.5%	92.5%	0.0097
Q4 (highest)	201	5.5%	94.5%	0.0268

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

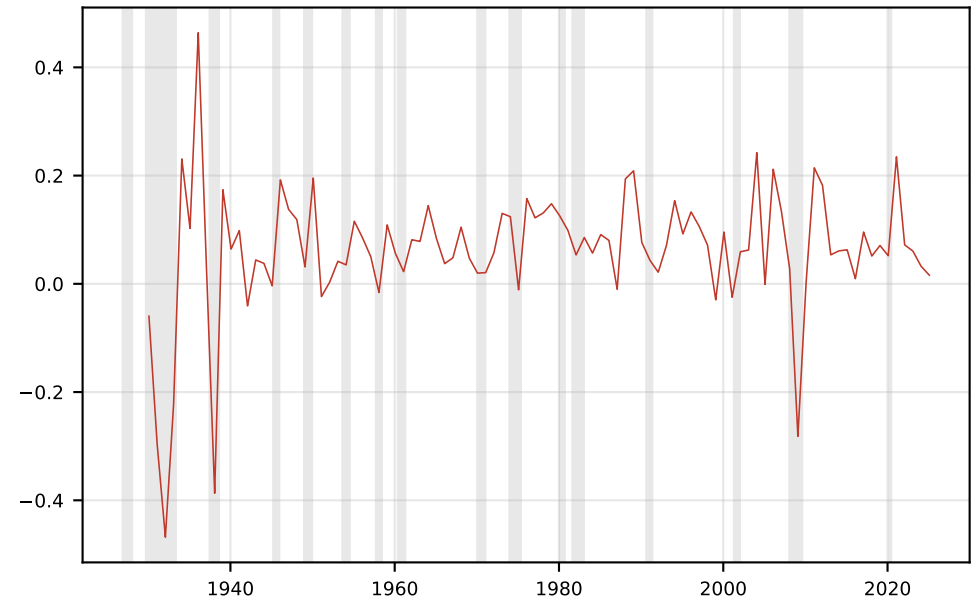
Net corporate dividend payments

Block: Equities / Equities | Source: FRED API | Col: B056RC1A027NBEA | Transform: 9: Log diff lag 12

Raw Series



After: 9: Log diff lag 12



SUMMARY STATISTICS

N : 1287
Valid : 97
NaN : 1190
Start : 1929-01
End : 2025-01
Mean : 343.9000
Std : 556.7640
Min : 2.0380
Max : 2261.4650
Skew : 2.026
Kurt : 3.522

STATIONARITY TESTS

ADF stat : -7.388
ADF p : 0.0000
ADF lags : 4
ADF : Stationary
KPSS stat : 0.308
KPSS p : 0.1000
KPSS : Stationary

QUARTILE-BUCKET ANALYSIS (vs NBER recession/expansion)

Quartile	Count	Rec%	Exp%	Avg
Q1 (lowest)	24	45.8%	54.2%	-0.0722
Q2	24	16.7%	83.3%	0.0486
Q3	24	0.0%	100.0%	0.0895
Q4 (highest)	24	4.2%	95.8%	0.1827

If Q4 high Recession% -> UP = bad
If Q4 high Expansion% -> UP = good

Data Book Summary

Series profiled: 307
Failed (not profiled): 12

By source:

source	
FRED API	149
WRDS	80
Derived	37
FRED-MD	20
Computed	15
FRED-MD (hist)	7
ABS	7

By block:

block	
Selected export	146
Cross-Asset	32
Inflation / Prices	21
Debt	16
Sentiment / Surveys	14
Commodities	13
FX / Global Macro	12
Demographics	10
Productivity	7
Volatility / Stress	7
Output / Production	5
Orders / Inventories	5
Consumption	5
Labor market	4
Income	4
Wages / labor costs	3
Sovereign credibility	3
Structural macro	3
Credit	2
Equities	2
Reference	1

Notes:

- ADF: null=non-stationary; $p < 0.05$ = stationary
- KPSS: null=stationary; $p < 0.05$ = non-stationary
- Transforms: FRED-MD t-codes where available, suggested otherwise